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Technical Guidelines on Integrated Disease Surveillance and Response in Cameroon

Extreme Nord
Makari
Kousseri
Mora
Kolofata
Maroua 2
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Meri
Maroua 1
Maroua 3
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Bafut
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Loum
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Nkongsamba
Mbanga
Yabassi
Littoral
Cité des palmiers
Edea
Pouma
Ndikinimeki
Bafia
Ntui
Yoko
Centre
Nanga Eboko
Saa
Obala
Mbandjock
Esse
Ayos
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Ngog Mapubi
Okola
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Awae
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Doume
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Mbang
Abong Mbang
Est
Ndelele
Messamena
Yokadouma
Lomie
Moloundou
Lolodorf
Ebolowa
Meyomessala
Sangmelima
Kribi
Ambam
Sud
Djoum

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MINSANTE

Organisation mondiale de la Santé

This 2021 edition of Cameroon's Technical Guide for Integrated Disease Surveillance and Response (SIMR) was developed in several stages with the participation of experts in charge of disease surveillance at all levels of the health pyramid. Ministerial departments represented in the One Health platform, WHO and development partners also participated.

This Guide is adapted from the Technical Guide for SIMR in the African Region developed by the WHO Regional Office. This document aims to update the concepts and list of priority diseases and health events and to integrate the disease surveillance aspects of the International Health Regulations (IHR) 2005.

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- Any death of a woman aged between 10 to 50; 26
- Any death of a child less than one month old; 26
- Any child under 15 years of age with fever, rash, cough/runny nose; 26
- Any child under15 years of age with sudden weakness in one or more limbs; 26
- Any person with 3 or more liquid stools in one day, and signs of dehydration (permanently cracked skin or sunken eyes) and/or death; 26
- Any person with fever accompanied by vomiting, liquid stools and unexplained bleeding (nose, mouth, skin or eyes, in urine or stool) or death; 26
- Anyone who has been bitten by a cat, dog or monkey; 26
- Anyone who has been bitten by a snake or scorpion; 26
- Two or more cases of people showing two or more similar signs/symptoms in the same community, school or workplace within a week; 26
- Unexplained death of a group (3 or more) of animals (e.g., 3 chickens) within a week; . 26
- Any public health risk event, including natural disasters or rarely observed symptoms; 26
- Two or more cases of respiratory illness in the same community or social group in the same week; 27
- Multiple cases of influenza-like syndromes (fever, cough, runny nose) in a community/school/workplace during the same week 27
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ABBREVIATIONS

AAR	After Action Review
AEFI	Adverse Effect Following Immunization
AFP	Acute Flaccid Paralysis
ARI	Acute Respiratory Infection
CBS	Community-Based Surveillance
CENAD	Cameroon Epidemiological Network for Animal Diseases
CERPLE	Regional Center for Epidemics Prevention and Control
CHW	Community Health Worker
CIR	Critical Information Requirements
CTC	Cholera Treatment Center
CTU	Cholera Treatment Unit
DHC	District Health Committee
DHIS2	District Health Information System version 2
DLMEP	Department for the Control of Disease, Epidemics and Pandemics
DMC	District Management Committee
DQA	Data Quality Assessment
EBS	Event-Based Surveillance
EMRs	Electronic Medical Records
e-IDSR	Electronic Integrated Disease Surveillance and Response
EPD	Epidemic-Prone Disease
EPI	Expanded Program on Immunization
EWAR	Early Warning Alert and Response
FLO	Forces of Law and Order
HA	Health Area
HC	Health Committee
HD	HD
HF	Health Facility
HIS	Health Information System
IAP	Incident Action Plan
IBS	Indicator-Based Surveillance
ICS	Incident Command System
ICT	Information and Communication Technology
IDSR	Integrated Disease Surveillance and Response
IHR	International Health Regulation
IMCI	Integrated Management of Childhood Illness
IMS	Incident Management System

IPC	Infection Prevention and Control
IVC	International Vaccination Center
JEE	Joint External Evaluation
KAP	Knowledge Attitude and Practices
MAR	Monthly Activity Report
MC	Management Committee
MEF	Monitoring and Evaluation Framework
MINAGRI	Ministry of Agriculture
MINEPDEP	Ministry of Environment, Nature Protection and Sustainable Development
MINEPIA	Ministry of Livestock, Fisheries and Animal Industries
MINFOF	Ministry of Forestry and Wildlife
NACC	National Aids Control Committee
NFP	National Focal Point
NGO	Non-Governmental Organization
NHDP	National Health Development Plan
NHIS	National Health Information System
NLYLBUCP	National Leprosy, Yaws, Leishmaniasis and Buruli ulcer Control Program
NMCP	National Malaria Control Program
NMP/MNCM	National Multisector Program to combat Maternal, Newborn and Child Mortality
NOCP	National Onchocerciasis Control Program
NPHL	National Public Health Laboratory
NPHO	National Public Health Observatory
NSHIPC	National Schistosomiasis Conrol Program
NTCP	National Tuberculosis Control Program
ORP	Oral Rehydratation Points
ORS	Oral Rehydration Salts
PHEIC	Public Health Emergency of International Concern
PHEMC	Public Health Emergency Management Committee
PHEOCC	Public Health Emergency Operations Coordination Center
PNPLZER	National Program for the Prevention and Control of Emerging and Re-emerging Zoonosis
PoE	Point of Entry
PPE	Personal Protective Equipment
RDPH	Regional Delegation of Public Health
RRT	Rapid Response Team
RTA	Road Traffic Accident
RVF	Rift Valley Fever
SARS	Severe Acute Respiratory Syndrome

SFP	Surveillance Focal Person
SITREP	Situation Report
SMPDR	Surveillance of Maternal, Perinatal Deaths and Response
SOP	Standard Operating Procedure
STD	Sexually Transmitted Disease
STI	Sexually Transmitted Infection
TCG/EPI	Technical Central Group /Expanded Programme on Immunization
TG	Technical Guidelines
VHF	Viral Haemorrhagic Fever
VPD	Vaccine Preventable Disease
WASH	Water, Hygiene and Sanitation
WHO	World Health Organization
WHO-AFRO	World Health Organization Regional Office for Africa

GLOSSARY

Aetiology	Refers to the cause, set of causes, or origin of a disease or condition.
Chronic	Any health condition that develops slowly or is of long duration and leads to some functional limitation and/or need for ongoing medical care.
Cluster	Aggregation of cases or health-related conditions in a given area, over a particular period, regardless of whether the number of cases is more than expected in relation to time or place or both.
Disaster	The serious disruption of the functioning of a community or a society, causing widespread human, material, economic or environmental losses exceeding the ability of the affected community or society to cope using its own resources.
Elimination	Reduction to zero or to a defined very low target rate of new cases in a defined geographic area.
Endemic	A disease or condition regularly found among particular people or in a certain area.
Epidemiology	The study of the distribution and determinants of health-related states and events in specified populations and the application of this information to decision making.
Eradication	The purposeful reduction of specific disease prevalence to the point of continued absence of transmission in the world.
Event	According to IHR 2005, an event is a manifestation of a disease or an occurrence that creates a potential for disease.
Incident	An occurrence or event, natural or human-caused that requires an emergency response to protect life, property, or the environment.
Outbreak	An increase in the number of cases of a disease or an event above what is normally expected in that population in a given area over a particular period.
Signal	Patterns of disease or other information considered by the Early Warning and Response system as representing potential acute risk to human health, such as an outbreak. Signals must be triaged and verified to determine whether or not they represent a proven event.

FOREWORD

For over 20 years, the World Health Organization Regional Office for Africa (WHO AFRO) together with its technical partners have been working to develop and implement Integrated Disease Surveillance and Response (IDSR) systems in countries of the African Region.

The first edition of the IDSR technical guidelines (2002) was widely adopted by Member States and adapted to the specificities of each country of the Region. The second edition (2010) was published to address new challenges, namely: the need to revise the International Health Regulations (IHR 2005), the emergence of new diseases, conditions and public health events, as well as the formulation of strategies for disaster risk management (DRM). There was also a need to address the increasing burden of non-communicable diseases, to reinforce community-based surveillance, in accordance with the "One Health Approach" for early detection, rapid confirmation and response to public health threats to be enhanced. This should align actions with the objectives of strengthening the overall health system.

The unprecedented Ebola virus disease (EVD) outbreak in West Africa and DRC in 2014 and 2019 respectively, the ongoing COVID-19 pandemic, the various outbreaks of emerging and re-emerging zoonotic diseases at the "Human-Animal-Environment" interface, and other health emergencies such as recent natural disasters demonstrate that health security remains a priority for health systems, Cameroon inclusive.

In Cameroon, the establishment and operationalisation of the surveillance system for human diseases, including zoonotic diseases and health events, is a key element in the 2016-2027 health sector strategy. This strategy is in line with the 2020-2030 National Development Strategy (NDS30), which aims at achieving structural transformation and inclusive development in its 4th pillar on the development of human capital and well-being.

Indeed, the implementation of robust surveillance systems is the starting point for health security. This third edition of the IDSR Technical Guidelines describes what needs to be established at each level of the health system, as well as the animal and environmental health systems, in order to early detect and respond to diseases and other public health events.
These guidelines should be used by:

- Health professionals at all levels (including surveillance officers, clinicians, laboratory personnel, and public health officers);
- Data managers;
- IHR national focal point and other sectors implementing IHR;

- Competent authorities at points of entry;
- Veterinaries and other wildlife health officers;
- Environmental health officers;
- Health training institutions;
- Supply chain officers;
- Other public health experts.

These guidelines are intended for use as:

- A general reference for surveillance activities at all levels and a stand-alone reference for level-specific responsibilities;
- A set of standard definitions for threshold levels that initiate action for responding to specific diseases;
- A useful resource for defining the boundaries of training, supervision, monitoring and evaluation of surveillance activities;
- A compass for improving early detection and response to Epidemic-prone Diseases.

Experts from the Ministry of Public Health and from the “One Health” sectors, as well as numerous technical and financial partners worked based on the IDSR Guide for the WHO African Region to produce this third edition of technical guidelines. It is the result of a realistic adaptation to our needs and priorities arising from our political and socio-economic context. In this regard, I call on all stakeholders to implement the strategy developed in this third edition within the broader context of strengthening health systems to achieve universal health coverage and better coordination between human, animal and environmental health surveillance. The purpose of this approach is to improve health security for Cameroonians, in line with the 2035 vision valued by His Excellency Paul BIYA, President of the Republic of Cameroon.

Finally, I would like to personally thank all those who were involved and participated in the development of this guide, namely: the entire team of the Ministry of Public Health, the World Health Organization, US CDC, the Focal Points of the sectors involved in the “One Health” approach in Cameroon, as well as all other technical and financial partners.

I hope that they will all work to support the dissemination and implementation of the guidelines contained in this guide, to ensure health security in our country.

Dr. MANAOUDA MALACHIE

Minister of Public Health

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We would like to thank all the sectors, technical and financial partners who have contributed to adapt this guide. We are particularly grateful to the World Health Organization (WHO) and the Centers for Disease Control and Prevention (CDC) Atlanta through its Cameroon office [GHSA] and to Metabiota for their multifaceted support in the development and adaptation process of this document.

We would also like to express our gratitude to all stakeholders at all levels of the health system who actively participated in the development of this document.

Section 0 : Introduction

This section describes:

- the concept of IDSR;
- IDSR objectives;
- the core functions of IDSR;
- IDSR and IHR implementation;
- the “One Health” approach;
- the general organisation of the IDSR Technical Guidelines

Introduction

This section introduces the concept of IDSR, which incorporates indicator-based surveillance (IBS) and event-based surveillance (EBS) as integral parts of an Early Warning Alert and Response (EWAR) system. The section also provides guidance on how IDSR works, its objectives, and how it can help to build and sustain the IHR essential capacities, thereby facilitating its implementation. The section introduces other aspects such as the "One Health" approach; the linkage between Disaster Risk Management and IDSR; essential surveillance functions; how the subnational level (for example districts) can use these guidelines to reinforce surveillance and response; the roles and responsibilities of the various actors at different levels; and the priority diseases, conditions and public health events recommended in IDSR.

It is important to emphasize that these guidelines are to help build and strengthen surveillance systems for disease, events or other IHR threats. These guidelines are NOT limited to known diseases only.

1. Surveillance in Public Health

Public Health Surveillance is the ongoing systematic identification, collection, collation, analysis and interpretation of disease occurrence and public health event data, for the purpose of timely dissemination of the resulting information for decision-making. Surveillance is also essential for planning, implementation, monitoring and evaluation of public health practice. WHO/AFRO has decided to achieve its public health surveillance objectives through the implementation of the IDSR strategy.

1.1. Definition of the different types /approaches to surveillance in Public Health

Passive Surveillance. A system whereby a health institution receives routine reports submitted from HF, such as hospitals, clinics and public health units, the community or other sources. There is no active search for cases. This is the most common form of surveillance, which includes the surveillance of diseases and other public health events using routine surveillance, routine health management and information system or any other public health information system.

Active Surveillance. It involves an ongoing search for cases in the community or HF (HF). This may involve regular contacts with key reporting sources, by making telephone calls to health care workers at a facility or laboratory or physically moving to the source and carrying out record review of data. Examples include active case search of measles and polio during outbreaks.

Specific Disease Surveillance. This approach aims at collecting health data for a particular disease to conduct vertical surveillance. Tuberculosis, malaria and HIV surveillance systems are some examples.

Integrated Disease Surveillance. This approach aims at collecting health data for multiple diseases and public health events, using standardised tools.

Indicator-Based Surveillance (IBS). It is the systematic identification, collection, monitoring, analysis and interpretation of structured data, such as indicators produced by well-identified (mostly HF) formal sources.

Community-Based Surveillance (CBS). It is defined as the systematic detection and reporting of events of public health importance within the community by its members. Community surveillance incorporates both indicator-based and event-based surveillance methods. As part of this surveillance, resource persons are designated to report cases or events to the Surveillance Focal Person of the nearest health facility. Community-based surveillance strategies focus on two approaches to collect community information.

- The first one relies on identifying and reporting events based on agreed indicators (community case definitions). For example, trusted community members and Community Health Workers (CHWs) are trained to identify diseases such as measles, cholera, polio and dracunculiasis, using community case definitions and the use of standardised reporting system to the next level.
- The second strategy relies on reporting of unusual events (signals) which can alert on the early stages of an outbreak or any other public health threat in the community. For example, CHWs or any other representative from the community, who have been trained to detect events, such as unusual animal deaths and report them to the next level. Often, CHWs would link the patient identified through any of the strategies, to a nearby HFand can help identify contacts.

Event-Based Surveillance (EBS). Event-based surveillance is the organized and rapid capture of information about events that are of potential risk to public health. Information is initially captured as a signal, considered by the early warning and response system (EWAR) as representing a potential acute risk (such as an outbreak). All signals may not necessarily become real events, and as such, need to be triaged and verified before a response is initiated. Signals which may signify potential risks include:

- Occurrence of disease in humans, such as unexplained clustered cases of a disease or syndromes, unusual disease trends or unexpected deaths, as recognized by health workers and/or other key informants in the community;
- Events related to potential exposure for humans, such as diseases and deaths in animals, contaminated food products or water, and environmental hazards, including chemical and radio-nuclear events;
- Potential exposure of human beings to biological, chemical or radiological and nuclear hazards, or occurrence of natural or man-made disasters.

The event-based surveillance system is very sensitive, and information received through it should be synchronised with IBS and rapidly assessed for the risk the event poses to public health and responded to appropriately (***illustrated in Figure 1***).

Intersection of IBS and EBS: All events detected in the EBS system that are investigated and meet the standard case definition should be captured in the IBS system and reported to the next level of the health care system.

1.2 Event-based surveillance and indicator-based surveillance as backbone to the IDSR strategy

EBS and IBS are components of the early warning, response and epidemic intelligence, incorporated into the IDSR strategy. EBS and IBS complement each other, albeit with separate roles and purposes. EBS is most likely to generate alerts to detect small outbreaks early, while IBS is better at monitoring disease trends overtime, and useful for signalling the start of regular seasonal outbreaks of endemic diseases, using alert and outbreak thresholds. EBS is also better at generating alerts indicating outbreaks in areas where access to healthcare is limited.

In order to ensure effective warning and rapid response, the IDSR data collection and analysis system relies on two main channels of information or signal generation: Indicator Based Surveillance (IBS) and Event Based Surveillance (EBS)

Unlike indicator-based surveillance, event-based surveillance is not based on routine monitoring of indicators and action thresholds, but rather on the analysis of all available information to detect any event occurring in the community (unusual illness or death in humans or animals, and unusual occurrence of clusters of cases of diseases, events or conditions in the community including environmental incidents).

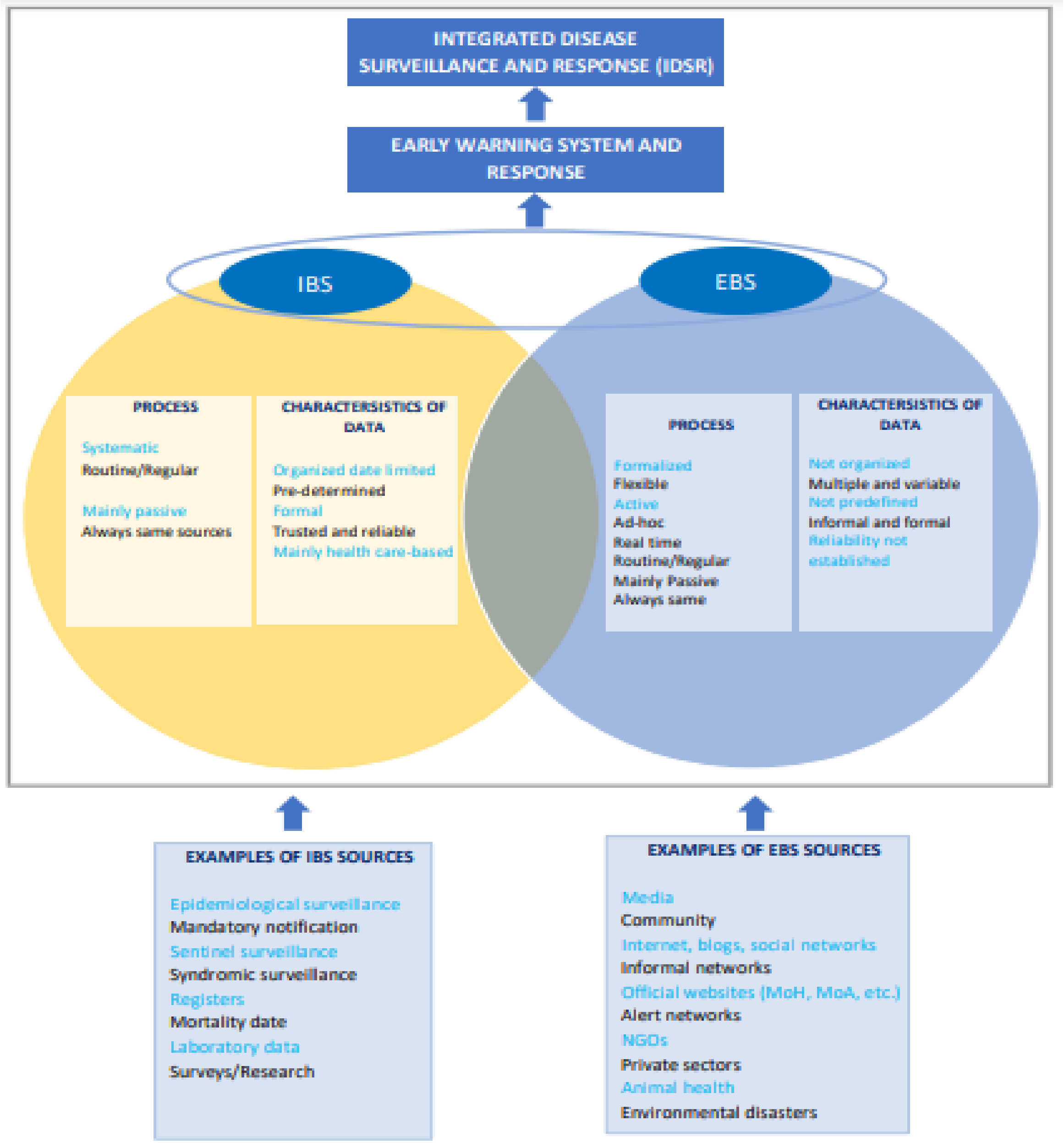
INTEGRATED DISEASE SURVEILLANCE AND RESPONSE (IDSR)
EARLY WARNING SYSTEM AND RESPONSE
IBS
EBS
PROCESS
Systematic
Routine/Regular
Mainly passive
Always same sources
CHARACTERSISTICS OF DATA
Organized date limited
Pre-determined
Formal
Trusted and reliable
Mainly health care-based
PROCESS
Formalized
Flexible
Active
Ad-hoc
Real time
Routine/Regular
Mainly Passive
Always same
CHARACTERSISTICS OF DATA
Not organized
Multiple and variable
Not predefined
Informal and formal
Reliability not established
EXAMPLES OF IBS SOURCES
Epidemiological surveillance
Mandatory notification
Sentinel surveillance
Syndromic surveillance
Registers
Mortality date
Laboratory data
Surveys/Research
EXAMPLES OF EBS SOURCES
Media
Community
Internet, blogs, social networks
Informal networks
Official websites (MoH, MoA, etc.)
Alert networks
NGOs
Private sectors
Animal health
Environmental disasters

Figure 1: Indicator-based and event-based surveillance for Early Warning Alert and Response System (EWARS) for IDSR Strategy

In the context of the IDSR strategy, the flow of information from the EBS follows the same information channels as the IBS, i.e. from the community to the Health District and from the region to the central level. Regardless of the type of surveillance, it is essential that the health data collected be used for public health actions.

1.3 What are common methods of surveillance?

Facility-Based Surveillance. All reporting units, such as HF, are required to report on a weekly, monthly, quarterly or annual basis to the next level, based on the categories of the diseases, conditions and public health events. Additionally, they are required to report any epidemic-prone disease to the next level immediately.

Case-Based Surveillance. This involves the ongoing and rapid identification of cases for the purpose of case follow-up. It is the type of surveillance used for diseases targeted for elimination or eradication or during confirmed outbreaks. In these scenarios, every individual case identified is reported immediately to the next level, using a case-based form.

Sentinel surveillance. This type of surveillance is done for specific conditions in a specific cohort, such as a geographical area or population subgroup, to estimate trends in a larger population. A given number of HF or reporting sites are usually designated as sentinel sites. It is aimed at monitoring the rate of occurrence of priority events such as pandemic or outbreak and other public health events. These sites act as early warning sites. Examples of sentinel surveillance include sentinel surveillance for human influenza, rotavirus, paediatric bacterial meningitis, and environmental sewage sampling for polio.

Syndromic Surveillance. This is an active or passive system that uses Standard Case Definitions, based entirely on clinical features, without any laboratory diagnosis. Examples of these are: Acute Flaccid Paralysis (AFP) as an alert for polio; acute watery diarrhoea among persons aged two years and above as an alert for cholera; “rash illness” as an alert for measles; acute haemorrhagic fever as an alert for viral haemorrhagic diseases, or severe acute respiratory infection or influenza-like illness as alerts for influenza. Because of the lack of specificity of this system, reports require more investigation from higher levels.

Laboratory surveillance. This consists of surveillance conducted at laboratories to detect events or trends, which may not be seen as a problem at other locations or originate from laboratory testing, mainly done routinely or used when conducting sentinel surveillance. Laboratories can be the source of an initial alert for a specific outbreak or public health event that necessitates in-depth epidemiological investigations. For example, the laboratory may be the first to detect the emergence of resistant strains, such as multi-drug resistant tuberculosis in the community, virological surveillance for influenza and bacteriological surveillance under the antimicrobial resistance surveillance system.

2. Integrated Disease Surveillance and Response

The Integrated Disease Surveillance and Response (IDSR) strategy was adopted by the Member States of the WHO Regional Office for Africa in September 1998 as an approach to improve public health surveillance and response to priority diseases, conditions and public health events at the community, health facility, district, regional and national levels. IDSR promotes the rational use of resources by integrating and channelling routine surveillance activities. The IDSR strategy makes surveillance and laboratory data more usable and helps public health managers and policy makers to improve on the detection and response to major causes of disease, deaths and the condition of their populations. As part of improvement to the health system, the IDSR strategy also assisted countries to better monitor and evaluate time-bound objectives.

Surveillance activities for different diseases involve similar functions (detection, specimen collection, reporting, analysis and interpretation, feedback, and action), and often use the same structures, processes and personnel. As such, the principles of surveillance are the same whether applied to a single or multiple disease (s), condition or public health event. The only difference is the question of whether the objective of surveillance is elimination or eradication, which may require time-limited intensive efforts aimed at proving the absence of disease or event.

2.1 What takes place in an integrated system?

- All surveillance activities are coordinated and streamlined. Rather than using scarce resources to maintain multiple surveillance systems with separate vertical activities, resources are combined to collect, manage and analyse information at a single surveillance focal point at each level.

- Several activities are combined into one integrated activity, and take advantage of similar surveillance functions, skills, resources and target populations. For example, surveillance activities for Acute Flaccid Paralysis (AFP) often address surveillance for neonatal tetanus, measles and other Vaccine Preventable Diseases (VPDs) or any unusual events. Thus, health workers who routinely visit HF to search for AFP cases also, review HFrecords for information about other priority diseases in the area. Community focal persons interact with their community members on a regular basis and ask about a range of diseases, conditions and public health events. Communities know they can bring anything unusual to the attention of their focal persons.

- The district level is the place where all the surveillance functions are integrated. It is the operational level in the local health system. It has dedicated staff for all aspects of public health, such as planning, supporting implementation of the National Health Strategy Plan, monitoring health events in the HFand the community, mobilising community action, seeking assistance at the national level, and accessing regional resources for protection at the district level. Similar functions also occur at various administrative levels.

- Surveillance focal points at the district, regional and national levels collaborate with Public health emergency management committees (PHEMC) at each level to plan relevant public health response actions and actively seek opportunities for combining resources.

- The focus is on the creation of a comprehensive public health surveillance system with sufficient capacity for detecting, confirming and responding to diseases, priority conditions and public health events. IDSR ensures that the information flow is bi-directional (horizontal and vertical), so that each level is informed promptly of potential outbreaks and response interventions. Information flow should also reach communities and neighbouring districts.

Integration refers to the efficient use of human resources, and harmonising different methods, software, data collection forms, standards and case definitions in order to prevent inconsistent information and maximize efforts among all disease prevention and control programs and stakeholders. Where possible, countries use a common reporting form, a single data entry system for multiple diseases, and common communication channels. Training and supervision are integrated. A common feedback bulletin is used, and other resources, such as computers and vehicles are shared. IDSR involves full-time coordination of surveillance activities and joint actions (planning, implementation, monitoring and evaluation), whenever possible and useful.

Coordination: refers to working or acting together effectively for the rational and efficient use of available, but limited resources, such as the Health Information System and various disease programs. Coordination involves developing standards, information sharing, joint planning, monitoring and evaluation to provide accurate, consistent and relevant data and information to policy-makers and stakeholders at district, regional and national levels.

2.2 Objectives of Integrated Disease Surveillance and Response

IDSR objectives are as follows:

- Strengthen the capacity of countries to conduct effective surveillance activities:
    - Train personnel at all levels;
    - Develop and carry out plans of action;
    - Advocate and mobilise resources;

- Increase involvement of clinicians and other health workforce in surveillance activities;
- Integrate multiple surveillance systems so that tools, personnel and resources are used more efficiently;
- Improve the triangulation and use of information to detect changes in trends in order to:
    - Conduct a rapid response to suspected and confirmed outbreaks;
    - Monitor the impact of interventions (for example, declining incidence, spread, and case fatality);
    - And facilitate evidence-based response to public health events, health policy design, planning and management:
- Improve the flow of surveillance information between and within levels of the health system using electronic tools;
- Build strong laboratory systems and networks at national, regional and district levels, to:
    - Confirm pathogens and other hazards;
    - Monitor drug sensitivity;
    - Increase efficacy of point-of-care tests;
- Trigger epidemiological investigations of reported public health problems and implementation of effective public health interventions;
- Mount an effective response to public health emergencies;
- Emphasize community participation in detection, reporting and response to public health problems, including surveillance and response to confirmed cases and public health event, and risk communication in line with IHR.

3 IDSR and IHR (2005)

The IHR (2005) is a binding and legal instrument, which urges all States parties to develop minimum essential public health capacities.

3.1 IHR (2005) purpose and goal

The purpose of the IHR (2005) is to prevent, protect against, control and provide public health response to the international spread of disease in ways that are relevant and restricted to public health risks, and which avoid unnecessary interference with international traffic and trade.

The scope of IHR has been expanded from three diseases (cholera, plague and yellow fever) to all PHEIC. They include those caused by infectious diseases, chemical agents, radioactive materials and contaminated food. Since the goal of IDSR is to strengthen the overall national system for the surveillance of diseases, particularly at the district level, and ensure continuous and timely provision and use of information for public health decision-making, IDSR provides the following resources for the implementation of IHR (2005):

- An infrastructure for surveillance, investigation, confirmation, reporting and response.
- Skilled human resources.
- A defined implementation process (sensitization, assessment, plan of action, implementation, monitoring and supervision).
- Generic guides for assessment, plan of action development, technical guidelines training materials, tools and Standard Operating Procedures that incorporate IHR (2005) components.

Member States in the African Region have thus recommended that IHR (2005) should be implemented in the context of IDSR. IHR (2005) is therefore not a separate surveillance system, but rather, one that requires countries to put in place a "sensitive, reliable and flexible surveillance system that meets international standards". IDSR is a strategy, which will ensure a reliable supply of information to the national level to fulfil IHR requirements. The IHR (2005) provides an opportunity to address the threat to international public health security and trade caused by emerging and re-emerging infectious diseases, including PHEICs. It also provides an excellent opportunity for strengthening surveillance and response systems and acting as a potent driver for IDSR implementation.

IDSR and IHR (2005) share common functions, as described in figure 3 below (detection, reporting, verification and confirmation, and timely response).

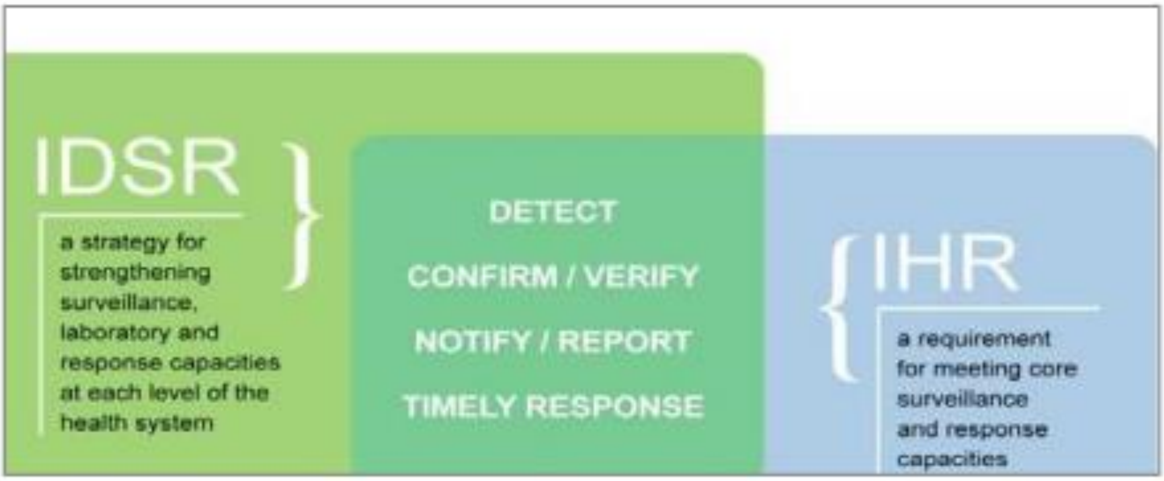
IDSR
a strategy for strengthening surveillance, laboratory and response capacities at each level of the health system
DETECT
CONFIRM / VERIFY
NOTIFY / REPORT
TIMELY RESPONSE
IHR
a requirement for meeting core surveillance and response capacities

Figure 2: Implementing IHR through IDSR[1]

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[1] *A guide for assessment teams. International Health Regulations (2005): Protocol for assessing national surveillance and response capacities for the International Health Regulations in accordance with Appendix 1A of the regulations.* February 2009.

The IHR (2005) guidelines have practical implications for IDSR. In the IHR (2005) guidelines, all public health emergencies should be detected, assessed and responded to promptly, using an adapted response to the situation. The IHR (2005) guidelines include control at points of entry (airports, seaports and land border posts) and containment at source of public health events. The IHR (2005) guidelines also include capturing signals (rumours) of “clusters of unexplained illness” as an event category for reporting from lower levels. Because of the major role IHR (2005) plays for timely detection and verification of suspected public health emergencies and events, event-based surveillance is now part of IDSR and the IHR.

Note: the notification process of events to WHO under IHR (2005) requires the use of “the decision-making instrument” which implies the implementation of the IDSR core functions: detection and confirmation of cases in the laboratory, analysis and-interpretation of data from results and reporting (***Appendix 2C***). A summary of the events required to be reported by the IHR (2005) is provided below.

The three main categories of public health events likely to be notified under the IHR are:

- Four diseases should immediately be reported to WHO. They are: smallpox, poliomyelitis due to wild-type poliovirus, human influenza caused by a new subtype, and SARS (see decision algorithm in ***Appendix 2*** of the IHR (2005)); These technical guidelines cover these four diseases;
- Other diseases and public health events may require notification if they are considered potential health events of international concern. The assessment will normally be conducted at the district level or at a higher level (regional, national), as decided by the national authorities using the decision-making instrument in ***Appendix 2*** of the IHR (***see Appendix to Section 2)***. Diseases classified in this category by the IHR include cholera, plague, yellow fever, viral haemorrhagic fevers, West Nile fever, and other diseases of national or regional importance such as dengue, rift valley fever, and meningitis. These pathologies are fully covered in these Technical Guidelines;
- “All public health emergencies of international concern including those of unknown cause or origin, and those involving events or diseases other” than those mentioned in the previous two paragraphs should be assessed using the IHR ***Appendix 2*** decision-making instrument ***(Appendix to Section 2)***. A list of these events is given in section 2.

These events are not given in detail in this Technical Guidelines. For more information, consult related documentation[2].

3.2 Monitoring and evaluating the core functional capacity for implementation of IHR (2005)

Following the Ebola outbreak experience in 2015, several IHR (2005) review committees and various expert panels have recommended the use of other tools to monitor and evaluate IHR (2005) implementation, to complement its annual monitoring. Consequently, since 2016, WHO Member States and partners have adopted the combined approach to the IHR (2005) monitoring and evaluation process. The four components of the IHR (2005) monitoring and evaluation framework are:

(a) Mandatory annual reporting to the World Health Assembly (WHA)
(b) Joint External Evaluation (JEE)
(c) After Action Review (AAR)
(d) Simulation exercises

The four components highlight a more functional approach to assessing IHR (2005) capacities and foster transparency and mutual accountability. This is illustrated in figure 4 below.

Protocol for assessing national surveillance and response for communicable diseases for the International Health Regulations (IHR). Regulations for assessment teams in accordance with Appendix 1A of the Regulations. February 2009.

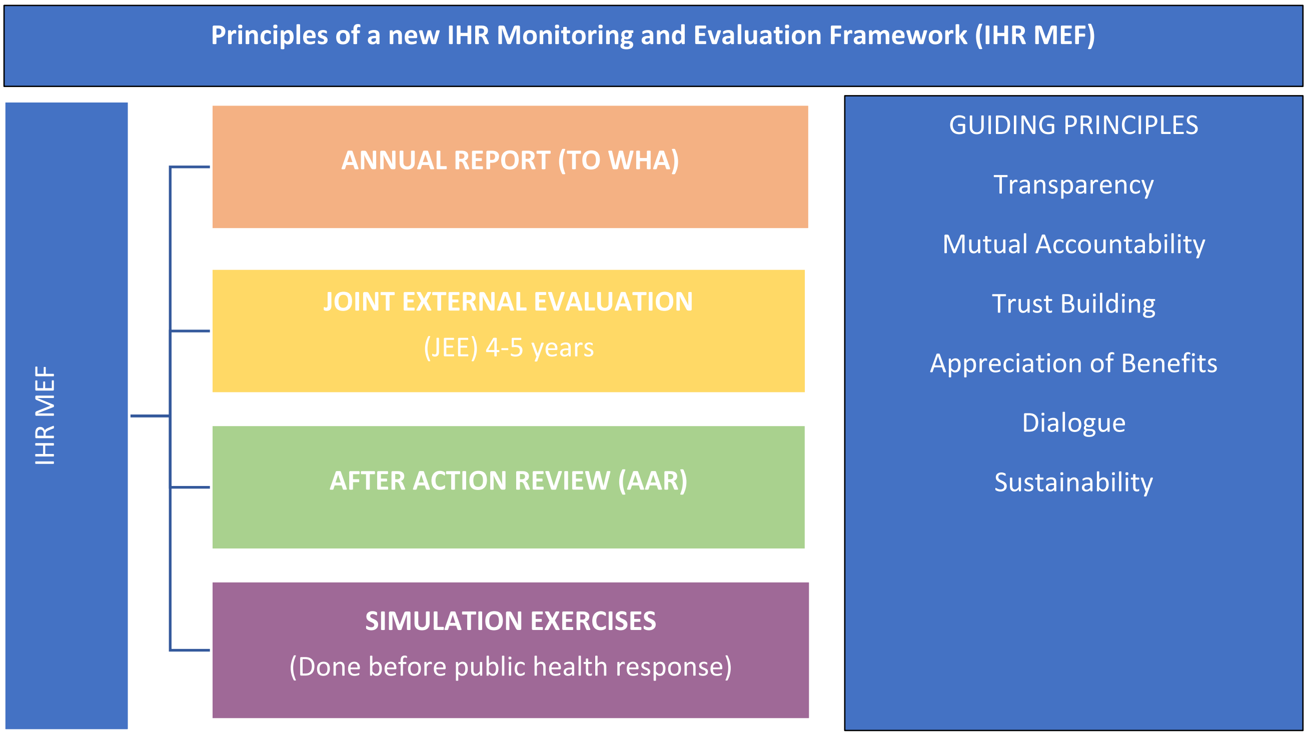

Principles of a new IHR Monitoring and Evaluation Framework (IHR MEF)
IHR MEF
ANNUAL REPORT (TO WHA)
JOINT EXTERNAL EVALUATION
(JEE) 4-5 years
AFTER ACTION REVIEW (AAR)
SIMULATION EXERCISES
(Done before public health response)
GUIDING PRINCIPLES
Transparency
Mutual Accountability
Trust Building
Appreciation of Benefits
Dialogue
Sustainability

Figure 3: IHR monitoring and evaluation framework

4 “One Health” approach

One Health is an approach to address a shared health threat at the human-animal-environment interface. This approach is based on collaboration, communication and coordination across all relevant sectors and disciplines with the ultimate goal of achieving optimal health outcomes for humans and animals alike and minimise the impact of different interactions on the environment. The “One Health” approach applies to the local, regional, national, and global levels. Humans and animals (domestic and wildlife) share the same eco-system and opportunities for spillover of diseases are increasing with modern trends in globalization, growing population pressures, climate change, economic development, mass urbanization and increasing demand for animal-sourced foods. The “One Health” approach is intrinsic to the IHR (2005), IDSR strategy, as well as other global health frameworks. It is enhanced by the latter.

It is meant to improve indicator- and event-based surveillance, which is the cornerstone of the early warning function of the IDSR. Animal and human health workers as well as other relevant partners should be engaged at various levels, as information sources for IDSR, to further facilitate information sharing and joint rapid response activities. The “One Health” approach offers a comprehensive framework for IHR (2005) implementation and helps to address PHEIC of all sources. The key principles of the “One Health” approach include prevention and control of emerging infectious diseases and support to health national services relying on existing structures (reference to IHR [2005] and the World Organization for Animal Health “OIE” international standards).

The “One Health” approach principle also considers the role of changing environments, with regard to infectious and chronic disease risks affecting humans and animals. By using data, expertise and management approaches in the environment, environmental health practitioners can assist in enhancing the understanding of the root causes of diseases, and better account for the complexity of environmental factors.

A strong functional IDSR thus requires improved communication, coordination and collaboration from all sectors, for the implementation of an effective “One Health” framework. A national “One Health” strategy document was elaborated on this purpose.

5 IDSR and disaster risk management

Disaster is defined as the serious disruption of the functioning of a community or society, causing widespread human, material, economic or environmental losses, exceeding the ability of the affected community or society to cope, using its own resources. At its sixty-second session held in November 2012 in Luanda, the Regional Committee for Africa adopted a paper entitled “Disaster Risk Management (DRM): a strategy for the health sector in the African Region”, in an effort to adopt a comprehensive approach to tackling DRM.

Disaster risk management is defined as the systematic process of using administrative and organizational directives, operational skills and capacities to implement strategies, policies and improved coping capacities, thereby lessening the adverse impact of hazards and possibility of disaster. In disaster risk management, a hazard analysis is conducted, followed by an assessment of the level of vulnerability and available coping capacity. The ultimate objective of DRM is to lower risk by reducing vulnerability or improving the capacity to mitigate the impact of a hazard. IDSR is an important tool in the DRM, as it provides early warning information, which is crucial for risk assessment and ultimately, risk reduction. IDSR assists in identification of hazards, assessment, risk communication and monitoring of disaster risks, thereby enhancing the early warning component. A disaster risk preparation and control national strategy document was elaborated for that purpose.

6 Implementing cross-border activities within the IDSR framework

Given the ecological distribution of communicable diseases and the porosity of international borders, it is imperative that countries in the Region work together to control and contain the spread of these diseases. The free movement of people and goods across the Region's borders provides opportunities for cross-border spread of diseases. In addition, at urban centres located at border points, a disaster on one side of the border can easily affect the health of a large number of people on both sides of the border. It is therefore logical that countries in the Region would coordinate and synchronize interventions, in an effort to control the spread of communicable diseases. In cross-border health activities management, Cameroon designed The National Guidelines for the Detection, Management and Transfer of travellers at points of entry, the Operational Guidelines for cross-border response to boost priority cross-border activities for disease control, including the following aspects: disease surveillance, outbreak preparedness and control, as well as building core capacities to ensure compliance with IHR (2005). The implementation of surveillance activities necessitates that:

- Countries, in collaboration with WHO, should establish a cross-border surveillance and response framework with neighbouring countries, using the existing IDSR systems in the respective countries;
- Countries should establish procedures for data sharing within the framework of IDSR;
- When outbreaks are detected through the IDSR system, the neighbouring cross-border regions and districts should be notified, using the IDSR reporting tools. If they are reporting a similar outbreak, coordinate response efforts with the IDSR response structures as described in Sections 4, 5 and 6 of the third edition of the IDSR Technical Guidelines;
- Ensure cross-border (district-district) coordination and collaboration on surveillance issues and provide notification of any outbreaks in the neighbouring district. International or cross-border notification should also be given if needed;
- Develop and organize simulation exercises with cross-border district teams;
- Organise regular cross-border meetings;
- Cross-border district surveillance and response initiatives should be facilitated.

A national strategy document for cross-border surveillance at the points of entry was elaborated for that purpose.

7 Implementing of antimicrobial resistance (AMR) within the IDSR framework

Anti-Microbial Resistance (AMR) is a major concern in human, animal and environmental health. It is generally accepted that self-medication, inappropriate use of agrochemical products, inadequate waste management and excessive consumption of antimicrobials in agriculture, human and veterinary medicine are the main factors contributing to the emergence of AMR. All these practices are at the origin of the appearance of antimicrobials and/or their residues in foodstuffs and in the environment, leading to serious direct consequences, in particular difficulties with disease management (lengthening of the duration of treatment, prolongation of stays in health facilities), increase in mortality, and in the costs of care.

AMR is a public health problem in Cameroon. Efforts have been made by the country to improve its detection capacities, but important challenges remain to strengthen its surveillance. AMR surveillance in Cameroon integrates human, animal and environmental health sectors. It is initially a sentinel surveillance with variations in the collection method which can be passive and/or active depending on the sector. In accordance with the WHO (GLASS) and WOAH guidelines, cases are detected using antimicrobial susceptibility testing.

The establishment of an integrated AMR surveillance system refers to the harmonisation of the different surveillance methods, data collection tools and a common platform for sharing information. This platform is located within the National AMR Coordination Centre (NCC) (reference to the National Public Health Laboratory) which receives data from the different sentinel sites of the three main sectors, while ensuring the sharing of information with the departments concerned. The AMR surveillance cycle respects the eight functions of Integrated Disease Surveillance and Response (IDSR) which go from detection to follow-up evaluation for each sector.

In human health, detection takes place at the various surveillance sites identified through case definitions. In animal health, the animal samples come from the main sectors concerned by surveillance. This includes Ruminants (especially cattle), which are mainly extensive, and poultry and pigs (broilers, layers and pigs), which are mainly intensive in certain regions. In environmental health, AMR surveillance began initially at human and animal surveillance sites before being extended. A national AMR surveillance strategy document has been elaborated for this purpose.

8 Electronic IDSR (e-IDSR) as a platform for enhancing real-time surveillance

The application of e-tools in the health sector has the potential to provide real-time validated data for public health surveillance, investigation and prompt outbreak response. Electronic IDSR provides new opportunities for accelerating the achievement of the IHR (2005) core capacities. Electronic IDSR is the application of electronic tools to the principles of IDSR, to facilitate prevention, prediction, detection, reporting and response. It is based on:

- Standardised interoperable and interconnected information systems administered within the national context;
- Rapid collection, analysis, reporting and use of disease/events data in real-time for appropriate public health action.

While paper-based tools can also provide timely information, countries should aim to have electronic tools to facilitate timely data transmission and response to public health threats. Countries implement e-IDSR to:

- Fulfil the regional committee recommendations of WHO AFRO on the use of information technology, which is essential to the achievement of IHR (2005) requirements by countries;
- Assist in standardization of data;
- Assist in improving timeliness and completeness of reporting;
- Assist in early detection, investigation, and response to outbreak or public health events;
- Reduce manual data entry, as it is prone to errors;
- Ensure systematic information sharing across levels and sectors;
- Enable better data transmission and management including data storage and easy access;
- Improve the capacity of electronic surveillance of diseases, near real-time notification;
- Improve data quality;
- Reduce system costs and easily generate automated alerts.

Section 9 describes in detail the key guiding principles for establishing e-IDSR

9 Description of surveillance functions as described in these Guidelines

The guidelines assume that all levels of the health system are involved in conducting surveillance activities for detecting and responding to priority diseases, conditions and public health events (even though the different levels do not perform identical functions). These activities include the following core functions:

Step 1 – Identify and record cases, conditions and events. Standard case definition is used for health service delivery points (human, animal and environment); simplified case definition is used at the community level to identify priority diseases, conditions, and signals that may generate alerts for emerging public health events. Additionally, case identification can be done through other health service delivery points (animal and environment) using public and private HF or community structures. Case definitions and a functioning alert and verification system are vital for detecting cases and outbreaks. After identification, all events must be recorded in the applicable register (HFregister, line list).

Step 2 – Report suspected cases, conditions or public health events to the next level. Notification deals with informing higher level of detected cases or public health events for taking adequate measures. If this is an epidemic-prone disease, a potential PHEIC or a disease targeted for elimination or eradication, respond immediately by investigating the case or public health event, collecting the necessary diagnostic sample, and submitting a detailed report. For events to be notified under IHR to WHO, the national focal point is required to use the decision-making instrument (***Appendix 2*** of IHR) to identify any potential PHEIC.

Step 3 – Analyse (person, place and time) and interpret findings. Surveillance data should be compiled, analysed for trends, compared with data from previous periods and interpreted for use in public health actions.

Step 4 – Investigate and confirm suspected cases, outbreaks or public health events. Case/outbreak confirmation involves the epidemiological investigation of suspected cases and capacity of the laboratory to make confirmation. Take action to ensure that the case and contacts, and the outbreak or public health events are investigated, and confirmed in the laboratory. The capacity for case confirmation is enhanced through improved referral systems, networking and partnerships. Gather evidence about what may have caused the outbreak or public health event, by including animal and environmental health sources and using this to select appropriate control and prevention strategies. Social, gender and behavioural factors should also be collected and used to produce locally appropriate responses and risk communication.

Step 5 – Prepare. Preparedness refers to the availability of public health emergency preparedness and response plans, taking into account acquisition and mobilisation of multiform resources:

- Relevant personnel;
- Finances;
- Material: stockpiling (vaccines, drugs and laboratory reagents), personal protective equipments, designation of isolation facilities, care centers, etc., rolling stock needed for outbreak interventions.

A mechanism for coordinating response measures set during preparedness is essential for effective response. Establishing pre-positioned 'outbreak response contracts and memorandums of understanding between United Nations agencies and Non-Governmental Organizations (NGOs) or civil society speeds up the process of sending logistical support at all levels.

Step 6 – Respond. When an outbreak, acute public health event or condition is detected, an investigation should take place to determine the cause and extent of the problem, the results of the investigation should guide the response. An assessment should be made to identify gaps and vulnerabilities, coordinate and mobilise resources to implement the appropriate public health response.

The MoH has adopted the Incident Management System (IMS) as its emergency management model and has a Public Health Emergency Operations Coordination Centre (PHEOCC) and regional response coordination platforms. According to the IMS principles, the event related analysis of risks provides an opportunity to decide whether the IMS should be activated under the supervision of the Minister of Public Health.

Step 7 – Risk communication. Risk communication is an essential element for all surveillance systems, during preparedness and response to public health events and emergencies. It is the real-time exchange of information, advice and opinions between experts, community leaders, or officials and people who are at risk. It encourages future cooperation by communicating with all levels, including communities.

Step 8 – Monitor, evaluate, supervise and provide feedback to improve the surveillance system. To improve the surveillance system, it is necessary to monitor, evaluate and supervise all related activities and provide feedback. Evaluating the effectiveness of surveillance and response systems requires an assessment of its preparedness, information quality, responsiveness, and overall performance. Feedback helps to motivate and reinforce health workers' efforts to participate in the surveillance system. Monitoring, evaluation and supervision allow for taking action to correct problems and make improvements to response. Different evaluation procedures such as the JEE, Intra-Action Reviews (IARs), After Action Reviews (AARs) and simulation exercises may be used.

Community representatives, the private sector and NGOs should be included in these evaluation activities.

9.1 Different levels where surveillance is performed

The levels of the health system are defined as follows:

Peripheral:

- Community. Represented by basic community-level services such as trained birth attendants, community health workers, or similar care providers, village or community leaders (religious, traditional or political), school teachers, health extension workers, locally identified community-based surveillance volunteers, veterinarians, drug vendors and traditional practitioners;
- Health facility. All institutions (public, private, NGOs or faith-based organisations) with outpatient and/or inpatient facilities are defined as a health facility;
- Health Area. A geographic unit consisting of several neighbourhoods and/or villages, serving a population of 5,000 to 15,000.
- HD. A geographic unit generally serving a population of 50,000 to 300,000.

At the intermediate level (Regional). Represented by the Regional Delegation of Public Health (RDPH), coordinates health actions in line with policies and strategies defined by the central level. The main dialogue structure at the intermediate level is the Regional Fund for Health Promotion (RFHP).

At the central level (national) The elaboration, monitoring and evaluation of the government policies related to public health are made by the MoH that defines strategies and allocates resources.

In an integrated system, some laboratory services should be available at each level described above.

9.2 Roles of the different levels in implementing surveillance

Each level of the health system plays a specific role in each of the surveillance functions (see ***Appendix A***).

9.3 How Districts can strengthen surveillance and response

Most countries have assessed their surveillance systems using the standard protocol for evaluating the surveillance system developed by WHO AFRO (Protocol for the Assessment of National Communicable Disease Surveillance and Response Systems WHO/CDS/CSR/ISR/2001.2).

Districts can also use a matrix of IDSR functions and skills to describe their role in the surveillance system. Such a matrix describes a complete system in which all the skills and activities are in place. Each level supports activities of lower levels and reinforces the opportunity for successful decision-making at corresponding levels. In an IDSR system under development, the matrix provides a systematic framework for improving and strengthening the system.

Practical uses of the IDSR matrix include:

- Ensuring that all core functions and capacities have been identified;
- Establishing accountability to provide a basis for assigning functions to appropriate levels and determining what capacities should be present;
- Organising activities and training for human resource development;
- Managing, monitoring and evaluating programs;
- Strengthening district laboratory capacity, including laboratory information system(national laboratory network);
- Planning for resources (human, material/supplies and financial).

The IDSR matrix also illustrates several key assumptions that need to occur for the core functions of the surveillance system. If one or more of the elements at each level is not present or is being performed poorly, the risk of failure increases for the achievement of surveillance and control objectives. An effective system will be supported at each level from the levels above and below. A complete system minimises delay in taking public health actions.

The functions of detection, reporting, analysis, investigation, response, risk communication, monitoring, evaluation, and providing feedback are interdependent and should always be linked. The IDSR matrix in ***Appendix A***, defines the surveillance functions, and how they are achieved at each level of the health system including the role of WHO in relation to IDSR core functions.

10 Efforts by WHO in the African Region to strengthen IDSR

WHO/AFRO provides technical support for implementation of surveillance and response at every level of the health system, including:

- The development of comprehensive technical guidelines for each level;
- A protocol for adapting the guidelines to every level within each country;
- Training of human resources involved in surveillance and response system;
- Advocacy for resources mobilization;
- Coordinating the monitoring, detection and control of diseases, conditions and public health events, outbreaks and public health emergencies across countries;
- Sharing public health information and promoting documentation of best practices.

11 Content of the Guidelines

11.1 New orientations

The previous edition of the guidelines has been revised in order to incorporate:

- Lessons learnt from previous outbreaks;
- New frameworks or strategies, such as the regional strategy for health security and emergencies;
- The revised IHR monitoring and evaluation framework;
- The initiatives for enhancing prevention, detection and response to public health events (Global Health Security Agenda, “One Health”, etc.);
- Key strategies for the African Region;
- The rising of non-communicable disease threats and road traffic injuries in the context of development of resilient health systems;
- The implementation of the IHR (2005) requirements and capacities for surveillance and response.

These guidelines should be adapted to reflect national priorities, policies and public health structures, and used in conjunction with other similar guidelines/strategies or initiatives. Overall, the revised guidelines will incorporate the following:

- Strengthening indicator-based surveillance with better analysis, reporting and use of routine data for decision making;
- Strengthening event-based surveillance;
- Improving community-based surveillance;
- Improving cross-border surveillance and response;
- Scaling up e-IDSR implementation;
- Improving reporting and information sharing platforms;
- Sharing improved data between sectors;
- Tailoring IDSR to emergency or fragile health system contexts.

The guidelines are intended for use as:

- A general reference for surveillance activities across all levels;
- A set of definitions for thresholds that trigger some action for responding to specific diseases or conditions;
- A stand-alone reference for level-specific guidelines;
- A resource for developing training, supervision and evaluation of surveillance activities;
- A guide for improving early detection and preparedness for outbreak response.

11.2 Key users

These guidelines are to be used by health care workers at the Primary Health Care level (public, private and religious), where disease is expected to be first reported. Additionally, these guidelines will be used by:

- Disease surveillance managers and officers at all levels;
- IHR national focal points;
- All the persons in charge of disease control programs (NMCP, CTG/EPI, NACC, NOCP/Blindness, National Program for the Control of Schistosomiasis and soil-transmitted Helminthiasis, NCP for Buruli Ulcer, NTCP, National Multisector Program to combat Maternal, Newborn and Child Mortality in Cameroon (PLMI), National Program for the Prevention and Control of Emerging and Re-emerging Zoonosis (PNPLZER), etc.;

- Health authority at points of entry;
- Hospital managers;
- Clinicians;
- Pharmacists;
- National Public Health Laboratory (NPHL);
- Veterinary and wildlife health officers;
- Environmental health and Sanitation officers;
- District health management teams;
- Private and religious HF;
- Heads of International Vaccination Centers (IVC);
- Officers in charge of epidemiological surveillance at the Ministry of Livestock, Fisheries and Animal Industries (MINEPIA);
- Laboratory personnel, community leaders, quarter heads, councillors and district/regional political officers;
- Other public health experts and practitioners in specialised institutions;
- Public health training institutions;
- Other health partners including NGOs;
- Other ministries;
- The person in charge of surveillance at the Ministry of Forestry and Wildlife;
- Regional Delegates of Public Health;
- Coordinators of the Regional Centers for Outbreak Prevention and Control (CERPLE);
- Environmental Health Officers and Health Engineers (MINEPDED);
- District Medical Officers (DMO);
- Surveillance Focal Persons (SFP) of HDs;
- Heads of Integrated Health Centers (IHC).

12 Priority diseases, conditions and public health events

The WHO/AFRO suggests the following communicable and non-communicable diseases and conditions or events as priorities for integrated disease surveillance in the African Region (see Table 1 for the priority diseases, conditions and public health events). The diseases or conditions are recommended because they are:

- Required internationally under IHR (for example, smallpox, poliomyelitis due to wild poliovirus, human influenza caused by a new subtype, SARS);

- Epidemic-prone diseases which have a serious public health impact due to their ability to spread rapidly internationally (for example, cholera, plague, yellow fever, viral haemorrhagic fever);
- Main causes of morbidity and mortality due to communicable diseases and conditions in the African Region (for example, malaria, pneumonia, diarrhoeal diseases, tuberculosis, HIV/AIDS, maternal deaths and injuries);
- Priority non-communicable diseases or conditions in the Region (high blood pressure, diabetes mellitus, mental health and malnutrition).
- Effective control and prevention interventions are available for addressing the public health problems they pose (for example onchocerciasis, trypanosomiasis).

Intervention programs supported by WHO for prevention and control, eradication or elimination of the diseases exist. These include EPI, PLMI, PNPLZER.

The adaptation of the IDSR AFRO Guidelines has made it possible to update the list of priority diseases and conditions for Cameroon (See. **Table 1**)

Table I: Priority Diseases, Conditions and public health events for IDSR in Cameroon

Epidemic-prone diseases, conditions or events which require immediate reporting	Diseases targeted for eradication or elimination	Other major diseases, events or conditions of public health importance
1. Chikungunya	1. Bacterial meningitis	1. Assisted delivery
2. Cholera	2. Malaria	2. Others envenoming
3. COVID-19 and Severe Acute Respiratory Infections	3. Dracunculiasis (Guinea worm disease)	3. Maternal deaths
4. Dengue fever	4. Lymphatic Filariasis	4. Perinatal deaths
5. Dracunculiasis	5. Leprosy	5. Diabetes mellitus (new cases)
6. Anthrax	6. Noma	6. Diarrhoea with dehydration in children less than 5 years of age
7. Acute haemorrhagic fever	7. Onchocerciasis	7. Diarrhoea with blood (Shigella)
8. Yellow fever	8. Yaws and endemic syphilis or Bejel	8. Sickle Cell Anaemia
9. Bacterial meningitis	9. Wild-type poliomyelitis	9. Epilepsy
10. Poliomyelitis (acute flaccid paralysis)	10. Human rabies of canine origin	10. Typhoid fever
11. Plague	11. Measles	11. Lymphatic Filariasis
12. Human rabies	12. Neonatal tetanus	12. Helminthiasis
13. Measles	13. Trachoma	13. Acute and chronic viral hepatitis
14. Neonatal tetanus	14. Buruli Ulcer	14. Hypertension (new cases)
15. Monkeypox		15. Sexually transmitted infections
16. Human Smallpox		16. Leishmaniosis
**Ebola, Marburg, Rift Valley, Lassa, Crimean-Congo, West*		17. Leprosy
		18. Malnutrition in children under 5 years of age
		19. Adverse events following immunization (AEFI)
		20. Dog bite
		21. Snake bite

Nile, Dengue Zika virus disease		22. Malaria 23. Yaws 24. Severe pneumonia in children less than 5 years of age 25. Intestinal and urinary schistosomiasis 26. Influenza-like illness 27. Syphilis 28. Neonatal tetanus 29. Trachoma 30. Injuries (road traffic accidents) 31. Trypanosomiasis 32. Tuberculosis 33. Buruli ulcer 34. HIV/AIDS (new cases)
	** Disease specified by IHR (2005) for immediate notification Human influenza due to a new subtype *** Wild-type poliomyelitis Severe acute respiratory syndrome (SARS) Smallpox*** Any public health event of international or national concern (infectious, zoonosis, food borne, chemical, radio nuclear, or due to unknown condition).	

Table II: List of EBS signals

- Any death of a woman aged between 10 to 50;
- Any death of a child less than one month old;
- Any child under 15 years of age with fever, rash, cough/runny nose;
- Any child under15 years of age with sudden weakness in one or more limbs;
- Any person with 3 or more liquid stools in one day, and signs of dehydration (permanently cracked skin or sunken eyes) and/or death;
- Any person with fever accompanied by vomiting, liquid stools and unexplained bleeding (nose, mouth, skin or eyes, in urine or stool) or death;
- Anyone who has been bitten by a cat, dog or monkey;
- Anyone who has been bitten by a snake or scorpion;
- Two or more cases of people showing two or more similar signs/symptoms in the same community, school or workplace within a week;
- Unexplained death of a group (3 or more) of animals (e.g., 3 chickens) within a week;
- Any public health risk event, including natural disasters or rarely observed symptoms;

- Two or more cases of respiratory illness in the same community or social group in the same week;
- Multiple cases of influenza-like syndromes (fever, cough, runny nose) in a community/school/workplace during the same week
- Severe acute respiratory illness in a traveller from a region or area of COVID-19 transmission.

13 IDSR Technical Guidelines organisation

The Technical Guidelines for IDSR presents a comprehensive vision for a disease surveillance and response system. In the IDSR, all levels of the health system are involved in surveillance activities to respond to priority diseases, conditions and public health events. The sections of these Guidelines correspond to the core functions of the IDSR and describe the following monitoring activities:

Section 1.	Identifying/detecting and recording cases of priority diseases, conditions and public health events
Section 2.	Reporting on priority diseases, conditions and public health events
Section 3.	Analysing and interpreting data
Section 4.	Investigating suspected outbreaks and other health events
Section 5.	Preparing response to outbreaks and other health events
Section 6.	Responding to suspected outbreaks and other health events
Section 7.	Communicating on risks
Section 8.	Monitoring, Evaluating and Supervising surveillance improvement and Response
Section 9.	Electronic system for Integrated Disease Surveillance and Response (e-IDSR)
Section 10.	Tailoring IDSR to emergency or fragile health system contexts
Section 11.	Summary of guidelines concerning diseases, priority conditions and public health events

Every section incorporates appendices that refer to the core functions of the Guidelines. All levels of the health system are addressed in each section and an overview of how Cameroon can perform each function to achieve the required level of surveillance and response is provided. In addition, a section on integrated electronic disease surveillance and online response has been added to summarise and enable the implementation of the e-IDSR system.

14 Appendices of the introductory section

Appendix A	IDSR Matrix: core functions and activities per level of the health system
Appendix B	Assessment tool for surveillance and response at the HD level
Appendix C	IHR (2005) Decision-Making Instrument
Appendix D	Potential emergency situations of public health of international concern to be reported under the IHR (2005)
Appendix E	Guide for establishing a community-based surveillance and response system
Appendix F	Core surveillance and response capacities as described in the IHR (2005)
Appendix G	Roles and responsibilities of the various IDSR actors
Appendix H	Guide to implementing point of entry surveillance and response systems
Appendix I	List of actors involved in IDSR implementation per level in Cameroun

APPENDIX A: IDSR Matrix: core functions and activities per level of the health system

Levels	Identification / Detection	Reporting	Analysis and Interpretation	Investigation and Confirmation	Preparedness	Response	Communication on risks	Monitoring, evaluation, supervision and feedback towards improvement
Community-based	Use alert triggers to identify priority diseases, conditions and public health events or other risks in the community. Support the community in detection and encourage the use of alert triggers	Report signals to appropriate authorities (community leaders, heads of health areas, SFPs of the HF, HD, Region, central level)	Involve local leaders in observation, description and definition of cases and outbreak thresholds of diseases. Assist in mapping out/locating cases	Participate in investigation activities on rumours or unusual events reported by community leaders and members. Ensure feedback to the community on monitoring actions	Participate in community health and emergency preparedness committees. Participate in identification of potential diseases, conditions and public health events. Participate in training and simulation exercises	Implement response activities. Promote community engagement. Raise community awareness on hygiene measures and good practices. Participate in activities on basic infections prevention and control measures and social distancing. Conduct active case finding in the community (suspected case and contact case), and provide community health education for behaviour change using outreach communication methods	Identify the persons who can take charge of the communication process. Build a relationship to establish functional bonds between the community and the HF and for coordinating information. Include communication with animal and environmental sectors to define a "One Health" approach at community level	Conduct situational analysis. Capture community reaction to public health action. Inform community members on reported cases, public health events and prevention activities. Conduct After Action Review. Participate in a posteriori evaluation

Levels	Identification / Detection	Reporting	Analysis and Interpretation	Investigation and Confirmation	Preparedness	Response	Communication on risks	Monitoring, evaluation, supervision and feedback towards improvement
Health facility	Use standard case definitions to detect, laboratory confirm and record priority diseases or conditions. Set up alert triggers in hospitals to identify public health events in the HF. Collect and transport specimens for laboratory confirmation. Verify alert triggers from community. Ensure the appropriate storage of surveillance material.	Report information on case for immediately reportable diseases. Send reporting forms and activity reports to the next level.	Prepare and periodically update graphs, tables and cards describing reported diseases, public health events and conditions based on time, place and person. Immediately report any disease, public health event or condition which: • Crossed the alert threshold or occurred in unusual places where it had not been seen previously • Presence of unusual trends or characteristics.	Take part in investigation of reported outbreaks. Collect, package, store and transport specimens for laboratory confirmation during investigation.	Participate in emergency preparedness and response committees' meetings. Participate in training and simulation exercises. Control and maintain the rapid intervention equipment and the emergency stock.	Participate in response activities involving case management and active case finding based on standardized guidelines. Take appropriate additional control measures. Participate as member of the rapid response team.	Ensure that the communication system is related to the leading structure in the community. Communicate with community members about the results of prevention and response activities, and maintain close contact with the community. Establish a listening unit and organise listening sessions and meetings with community workers/volunteers in surveillance.	Evaluate community engagement. Conduct self-evaluation of surveillance and response activities. Monitor and evaluate prevention activities. Provide feedback at community level. Make laboratory results available to HFs.

HD	Support the HF in risk evaluation of events reported by the community. Collect surveillance data from the HF and the community, and check their quality. Ensure that reporting sites have reliable tools for data collection and reporting. Make available to sites the necessary tools for data identification. Ensure that all care facilities have supply for laboratory specimen collection and transportation.	Ensure that HF and community surveillance agents know and use standard case definitions for reporting priority diseases, conditions and public health events. Draw and update a list of reporting sites. Report important risk events to the Region. Timely report data to the surveillance focal point at regional level.	Compile data from HFs. Use and fine-tune rate denominators. Analyse data by time, location, and individual characteristics. Assist the HF in updating weekly graphs, tables and charts describing priority diseases, conditions and public health events. Include epidemiological and laboratory data for a better analysis. Compare data and draw conclusions on trends and thresholds.	Organise and conduct investigation of reported diseases or outbreaks. Update the line list of suspected cases. Assist the HF in sampling, packaging and safely transporting laboratory specimens for confirmation. Receive laboratory results from the Region, then forward them to the HF. Send the preliminary investigation report to the Region.	Establish and ensure the functioning of emergency preparedness and response committees. Participate in risk mapping and evaluation. Organise, set up and ensure the functioning of the District RRTs. Train the HF in response and support the community. Develop an up-to-date communication plan on risks and try it during a real emergency or a drill.	In collaboration with the Region, select and set up appropriate response measures on public health. Timely plan information and education activities in the community. Document response activities. In an outbreak situation, send a daily situation report of the district.	Establish systems and a structure on risks communication. Ensure involvement of risk communication actors at the regional level. Develop and build networks of relevant actors and organisations to improve information flow. Ensure that risk communication is part of emergency interventions Alert and inform communities on outbreaks or public health events.	Conduct regular supervision of HFs. Provide feedback to the HF and the community on surveillance activities and priority public health events. Provide regular feedback to the HF and the community on routine outbreak prevention and control activities. Monitor and evaluate communication of HFs complete programs to the district. Monitor and evaluate communication during response to outbreaks. Collect information from affected communities on their needs. Organise evaluation meetings on surveillance involving all stakeholders in the district. Gather information to assess response impact to affected communities.

Regional	Ensure the coordination of detection activities in respective Districts so as to supervise and support operational surveillance services. Ensure that reporting sites have reliable tools (case definitions, reliable posters for data collection and reporting). Ensure that sampling material in the laboratory are available and specimens transportation is effective.	Ensure that the districts know and use standard case definitions for notification and verification of priority diseases, conditions and public health events. Instruct the Districts to ensure the reporting of priority diseases, conditions and public health events. Receive regular data from the surveillance focal person in the District and review the quality of data in the report.	Ensure accuracy of denominators for the Region. Compile data from the district surveillance focal person. Analyse data by time, location, and individual characteristics. Prepare up-to-date graphs, tables and charts describing the priority diseases, conditions and public health events on a weekly basis. Calculate rates and thresholds and compare current data with previous data at the same period to draw conclusions.	Organise and support investigation on reported diseases, conditions and public health events. Receive and interpret laboratory results. Compile line lists in the event of confirmed outbreak at District level. Report confirmed outbreak to the national level. Ensure that specimen collection kits for investigation activities are available.	Convene meetings of the emergency preparedness and management committee. Develop and run emergency plans. Organise training and simulation exercises for the staff. Conduct periodic review of risks for public health events. Organise and support RRTs. Develop an up-to-date regional communication plan on risks and try it during a real emergency or a simulation exercise.	select the appropriate response strategy. Activate the emergency preparedness and response committee. Train the staff in emergency management. Plan awareness raising activities. Communicate messages on behaviour change. Send daily situation reports during outbreaks.	Establish systems and a structure on risks communication. Ensure involvement of risk communication actors at the regional level. Develop standard operating procedures (SOPs) on the authorization and dissemination of information on public health emergencies. Ensure that the sources of information are updated and accessible to the media for diffusion. Ensure that the accessible and relevant information, education and communication material is tailored to populations needs. Rapidly disseminate information in a transparent manner. Assess the relevant of information before considering risk communication. Ensure population involvement to facilitate peer communication and raise knowledge on the situation. Monitor and react to rumours and public reactions to facilitate local response. Ensure that risk communication is part of emergency interventions. Ensure that a risk communication expert is found at all levels.	Monitor and evaluate program objectives and indicators to assess the quality of the surveillance system in districts and HFs. Provide feedback to Districts on surveillance and data quality. Provide feedback to the district on routine outbreak prevention and control activities. Publish the region epidemiological bulletin monthly. Conduct regular assessment of staffing needs to implement IDSR and inform the next level. Conduct regular supervision. Monitor and evaluate communication on response to public health events. Assess the acceptability of response provided to the community and improve where necessary. Trigger involvement of partners in monitoring surveillance and response activities. Organise monitoring and evaluation meetings on surveillance involving all stakeholders in the region

Levels	Identification / Detection	Reporting	Analysis and Interpretation	Investigation and Confirmation	Preparedness	Response	Communication on risks	Monitoring, evaluation, supervision and feedback towards improvement
							Alert neighbouring areas, regions and districts to the outbreak, including cross-border areas.	

National	Define actions, update them and ensure compliance with national guidelines and policies. Implement policies and procedures for reference laboratories networks including quality assurance systems. If necessary, refer to reference laboratories for confirmation and specialised tests. Collect and transport specimens for further analysis to the WHO collaborating centre (CC), where necessary.	Train, inform and support lower levels on surveillance and response. Compile reports on immediately reportable diseases and events at the regional level. Timely report other priority diseases, conditions and public health events to programs and partners concerned. Include all concerned laboratories in the reporting network. Use the IHR decision-making instrument (***Appendix* 2**) to determine the risks of priority diseases, conditions and public health events. Inform WHO under IHR (2005)	Define policies and develop procedures for data analysis and interpretation. Define denominator and ensure accuracy. Analyse and interpret data in a view to take action at the national level. Calculate national rates and compare data with previous periods. Describe risk factors for priority diseases, conditions and public health events. Organise regular meetings with the technical coordination committee to review analysed and interpreted data before dissemination. Conduct specific analyses to predict the magnitude and trend of priority events.	Ensure that standard operational guidelines and procedures for outbreak investigation are available at all levels. Dispatch rapid intervention team for investigating outbreaks and response. Coordinate and collaborate with international authorities during investigation, if necessary. Coordinate investigations with health teams at the region and the district during response, where necessary. Alert laboratories and encourage their involvement. Provide logistic support for field investigation. Share information with international networks about confirmed outbreaks. Process specimens from the investigation and send results in time.	Implement policies, procedures and training per level. Map out risks Prepare and disseminate emergency preparedness and response plans. Develop a national communication plan on risks including messages for community education. Train and support national rapid response teams (RRT) in public health emergency situations. Develop policies and procedures for response to outbreaks, priority conditions and public health events. Organise simulation exercises. Develop and run emergency plans. Set up a PHEOCC. Establish and maintain a national command centre and follow up the operational preparedness status using the checklist which is the reference tool.	Identify and support response activities. Coordinate response with health teams at the region and the district levels. Support response activities, e.g dispatching of rapid response teams for emergency public health situations. Follow up and adapt guidelines for risk communication and community engagement.	Establish systems and a structure on risks communication. Ensure the involvement of all risk communication actors. Update the risk communication plan and try it during a real emergency or a simulation exercise. Update standard operating procedures, policies and guidelines on the authorization and dissemination of information during a public health emergency situation. Ensure that the sources of information are updated and accessible to the media for diffusion. Ensure that the accessible and relevant information, education and communication material is tailored to populations needs. Rapidly disseminate information in a transparent manner. Assess the relevance of information before considering risk communication. Target and use networks of relevant partners and organizations to improve information flow. Build public support to facilitate peer-to-peer communication, rumour awareness and public response to facilitate local intervention. Ensure that risk communication is part of the emergency response systems.	Regularly follow up IDSR and laboratory key indicators. Provide regular feedback to the regions. Share epidemiological data and reports, including information on response to outbreaks in neighbouring countries. Develop and distribute the national epidemiological and public health bulletin periodically. Organise regular meetings on IDSR evaluation. Conduct regular supervision. Ensure involvement of partners in monitoring surveillance and response activities. After action review including lessons learned from investigation and response. Support annual controls of core IHR capacities. Update the work plan and the budget line on the implementation of IDSR activities. Provide the document on appropriate feedback in due time.

Levels	Identification / Detection	Reporting	Analysis and Interpretation	Investigation and Confirmation	Preparedness	Response	Communication on risks	Monitoring, evaluation, supervision and feedback towards improvement
							Ensure that staff trained on risk communication are found at all levels.	

WHO Country Office and AFRO	Develop and disseminate generic guidelines on surveillance. Encourage sharing of documents and best IDSR practices. Provide technical support at the national level for the detection and confirmation of priority diseases, conditions and public health events. Coordinate international support from the reference laboratory network, including pools of excellence.	Gather and compile reports on outbreaks and diseases and events that can be reported at the international level. Create regional annual profiles or situation reports per priority disease, conditions public health and events.	Provide guidelines for a better analysis of data and development of information bulletins/products. For each priority disease, condition and event, develop and disseminate best testing practices. Provide technical support at the national level to improve analyses capacities.	Disseminate up-to-date guidelines and tools on specific diseases. Provide assistance, at the request of the country, to conduct evaluations or investigations on priority diseases and public health events. Provide assistance for coordinating the participation of laboratories during investigations. Provide assistance for risk evaluation using the IHR decision-making tool.	Mobilize resources for training, logistics and supervision. Establish a network of experts in IDSR training and implementation. Develop, update or revise guidelines on disaster or risk management. Maintain and update a panel of experts for rapid response teams. Develop, update/revise training modules for the IDSR and IHR to support the IMS.	Coordinate and support response activities (strategic center of health operations, technical experts, standard operating procedures, guidelines, etc.) Mobilize resources and facilitate partnerships. Support the activation of the IMS. Activate the IMS.	Disseminate guidelines, manuals, training modules and other orientation elements on risk communication. Help in the coordination of partnership and share information with partners and stakeholders.	Provide feedback to assist in collaboration with national and regional teams. Upload on the WHO site and share relevant links with all persons and partners concerned. Use reports from the regions to evaluate the IDSR systems and advocate for their improvement. Develop, update or revise guidelines and tools for IDSR/IHR monitoring and evaluation. Develop and disseminate the regional bulletin on surveillance. Promote, guide and support operational research. Ensure the functioning of the IDSR task force. Regularly monitor key IDSR/IHR performance indicators and standards as part of the revised emergency response.

Levels	Identification / Detection	Reporting	Analysis and Interpretation	Investigation and Confirmation	Preparedness	Response	Communication on risks	Monitoring, evaluation, supervision and feedback towards improvement
Point of Entry	Use case definitions or signals to identify suspicious passengers or public health events related to the transportation of persons and goods. Provide assistance to the community for case finding.	Immediately report to the IHR focal point and to the district/regional and national levels. Send summary reports monthly to the national level and at the same time to share with the districts and the regions.	Prepare and update periodically the database of detected cases/public health events.	Participate in evaluation of potentially exposed/infected travellers in an isolation/treatment center. Organize the investigation of suspected passengers and contacts. Check rumours or unusual events reported by community leaders and members.	Participate in the works of emergency preparedness and response committees at the point of entry. Participate in the emergency preparedness plan at the point of entry. Participate in training and simulation exercises. Participate in cross-border meetings.	Refer the sick passenger to the appropriate health facility. Liaise with emergency and preparedness committee in response activities. Assist in case and contact finding. Implement the best prevention and control measure against basic infections.	Build relationships, communicate and ensure coordination for information sharing with the various stakeholders (IHR focal points, civil aviation authorities/port authorities, ICAO). Establish communication with vessel operators and the naval industry about the authorization and maritime health declaration. Build relationship with surveillance agents at all levels and with the national IHR focal point.	Monitor and evaluate prevention activities and modify them where necessary. Organize simulation exercises periodically.

Appendix B: Assessment tool for surveillance and response at the HD level

Most countries used an evaluation tool developed by WHO Office for the African Region to assess their national systems for surveillance, preparedness and response to outbreaks in the event of an outbreak and to determine the necessary improvements. Others have used new tools such as the JEE to evaluate the capacity of countries to prevent, detect and respond to public health events. Evaluation provides results that can be used to solve issues regarding resources, the quality and timeliness of surveillance data and the use of information. The national strategic plan may equally serve as reference tool when preparing a specific action plan at the HD for other countries, which did the JEE, the national action plan for food safety may also be used.

The IDSR does not propose the setting up of a new system; it instead seeks to give advice on how to prepare for surveillance and response activities. Yet, if the HD has the necessary competences and resources to conduct an evaluation to document the situation of surveillance and response activities in the district, it may use the checklist below after adapting it to the local context. This tool may guide HDs in identifying activities to improve their performance and IDSR capacity.

Identification of cases and events

1. Evaluate the availability and knowledge of standard case definitions of reportable priority diseases and conditions, including public health events of concern.
2. Define the sources of information (formal and informal) on health events in the HD, including points of contact of the community with health services. For example, list the sources of information based on reporting sites of the HD:
    - (a) Health structures and hospitals
    - (b) Laboratories (including private laboratories: for profit, military, NGOs, denominational)
    - (c) Point of Entry
    - (d) Community health workers (including community animal health workers)
    - (e) Volunteers or community focal persons (traders, hairdressers, farmers, etc.)
    - (f) Traditional birth attendants and healers

(g) Leaders of rural communities informed about the health events in the community (for example, village elders, traditional healers, school teachers and leaders of faith-based schools)

(h) Health personnel

(i) Private sector practitioners

(j) Public security agents of fire, rescue or police departments.

(k) Veterinary and animal health structures and services

(l) Laboratories of the sector, food safety and environmental health

(m) The Media, internet sites, and research applications on health information

(n) Others, including NGOs.

It is equally important to keep and update a register of rumours/signals to report events and a feedback loop to confirm or dispel rumours.

3. Identify focal points for each source of information. Determine and specify the possibility to involve the community in the surveillance of health events.

Reporting

4. Specify priority diseases, conditions and public health events considered in the national policy and targeted for surveillance in the HD such as:

   (a) Epidemic-prone diseases or events, an unexplained number of diseases or deaths in a locality which require immediate reporting;

   (b) Diseases targeted for eradication and elimination;

   (c) Diseases of public health concern, including non-communicable diseases.

5. For each priority disease, condition or public health event, review the minimum data to be declared by health structures and other sources. Indicate when they should be signalled, to whom and how. In other words, establish and set up a reporting circuit within the HD. Indicate information that ought to be collected from patients (inpatients and outpatients). For example, a minimum requirement should be to notify all cases and all deaths due to selected diseases and conditions.

   (a) Indicate the diseases or conditions to be reported immediately and communicate this list to HF in the district.

   - (b) Define the communication channels of district data (by telephone, hard copy, soft copy of the form, face-to-face). If reporting is online, are all structures equipped with computers and modems? Specify how online reporting should be done and if paper-based forms will be used for data collection, how transcription will be done from paper-based form to electronic form.
   - (c) Define the frequency at which required data should be communicated.
   - (d) Define a feedback mechanism to HFs and information to higher levels (regional and national levels).

6. Define the management tools available in the HD and how to use them in an integrated system. Define the frequency at which tools should be used to report diseases, conditions or public health events. Tools may include:
   - (a) Case-based reporting forms;
   - (b) Case investigation forms and surveillance reports on lab specimens;
   - (c) Specimens follow up forms (in the laboratory) and specimens transfer forms;
   - (d) Line lists of cases to be used during outbreaks, while ensuring that all variable from non-human sectors is entered;
   - (e) Contact finding forms (contact listing form);
   - (f) Summary tables with calculation formulae;
     - (i) Weekly routine report forms;
     - (ii) Monthly routine report forms;
     - (iii) Quarterly routine report forms;
     - (iv) Graphs for data analysis within a given period;
     - (v) Maps for data analysis on locations;
     - (vi) Graphs for data analysis of individual characteristics.

7. Ensure the availability of the necessary supplies for surveillance at each reporting site. (Note: If a reporting site can produce digital reports, the digital format should be compatible with the methods used at the district, regional and national levels). In most countries applying e-IDSR, DHIS 2 has often been used as data system. In the absence of an electronic report system, ensure that the focal points in charge of data management have a reliable stock of data collection forms, paper, colour pencils, graph papers and logbooks.

8. Ensure the effective and ongoing coaching of HFs by the district to guarantee that data are collected within the prescribed deadlines, and set up an accountability mechanism if the reports are not sent on time.

Data Analysis

9. Define requirements for data management for each reporting site. For instance, develop and disseminate procedures, including deadlines so that reporting sites are aware that they must report for each reporting period (e.g., every month)

    (a) Calculate, compile and communicate summary totals

    (b) Periodically check data quality and clean up where necessary

    (c) Analyse data (produce weekly, monthly, quarterly or annual summaries as tables, graphs or maps)

    (d) Provide interpretation for the next higher level

    (e) Forward data to the next level (SMS, e-mail, fax/case-based forms, line lists)

    (f) Archive and secure data backups

    (g) Provide feedback and recommendations to community focal points, all reporting sites concerned and community leaders, and follow up the implementation of recommendations

10. Decide whether current forms include current IDSR priorities. For instance, do current forms provide the necessary information to detect problems and initiate a response to priority diseases targeted for surveillance?

11. Gather and present relevant data on your district likely to be used to ask for additional resources in order to improve surveillance and response activities. (For instance, ealth workers are able to document increase in malaria cases; they know that an effective response is available with insecticide-treated mosquito nets. The district surveillance focal person has used data to illustrate the expected reduction in malaria cases if part of the cost of mosquito nets for the community is taken care of by local companies).

Investigation and confirmation of cases, outbreaks or public health events

12. Describe the reference diagnostic and laboratory network for the confirmation of priority diseases and conditions in the district. For example, give the list of:

(h) Public, private structures or NGOs of the district having reliable laboratory services for confirming priority diseases;

(i) Special prevention, control or surveillance activities in the district having access to a laboratory (e.g. HIV sentinel surveillance sites in the district)

13. Describe the methods or the mechanism for active case finding, and where necessary, procedures for contact finding.

Preparedness of the response to outbreaks and other public health events

14. Organise and update the list of the rapid response team (RRT) of the district so that evaluation of RRT preparedness become a routine activity of the HD. See Section 4 for the composition of the RRT in the event of a public health.

15. Identify a coordination mechanism to supervise preparedness and response meetings. See Section 5 to know how to develop a coordination mechanism and the composition of the team that will direct the intervention and planning process of meetings. Specify and disseminate times for:

    (a) Meeting to regularly evaluate the preparedness level to public health emergencies and discuss current issues or activities. Establish mechanisms such as reminders to ensure that meetings are held as scheduled;

    (b) Meetings to discuss outbreak response, including the review of main recommendations and measures, and the progress of the implementation.

16. For each selected priority disease, condition or public health event, indicate the public intervention activity to be carried out, and develop an emergency plan for such disease, condition or public health event. Identify possible activities and interventions for which the district will need external assistance. See Sections 4, 5, 6 and 9 to know the key standard elements for preparedness and intervention activities.

17. For each disease or condition to which the district is responding, specify the target and alert threshold, or analyse results that will trigger an action.

Communication and feedback

18. Define methods to inform and support health workers in implementing IDSR, by:

    (a) Listing current training opportunities of health workers in surveillance, intervention or data management in the district.

    (b) Coordinating training opportunities among disease control programs taking advantage of overlapping competencies such as supervision, preparation of reports, budgeting, data analysis and data use for defining priorities.

(c) Defining training needs for each category of health workers, based on supervision or as a response to a specific event. Deciding whether this will be initial training on surveillance and response techniques or recycling training on how to integrate surveillance activities.

(d) Establishing performance indicators (quality management) for health workers and by evaluating their performance regularly.

19. Describing how communication on surveillance and response is done among the district and focal points and other focal points of the animal health sector and other relevant key sectors. Clarifying who is in charge of preparing periodic reports at each level. Including methods such as monthly meetings, information bulletins and telephone calls.

20. Examining and updating procedures and feedback methods among the district, health structures and the community, and between the district and higher levels. Specifying feedback methods and updating, where necessary;

21. Bulletins summarising the data sent to the district by HF.

22. Periodic meetings to examine public health issues and recent activities.

23. Supervision.

24. Describing available communication mechanisms, including protocols and guidelines for risks communication. Identifying a speaker and ensuring that training has been offered on required protocols. Establishing a liaison mechanism between the community, health structures and preparedness and response committee, which may be activated during an outbreak and/or for routine activities. See Sections 6 and 7 of the Guide on key elements on risk communications before, during and after an outbreak.

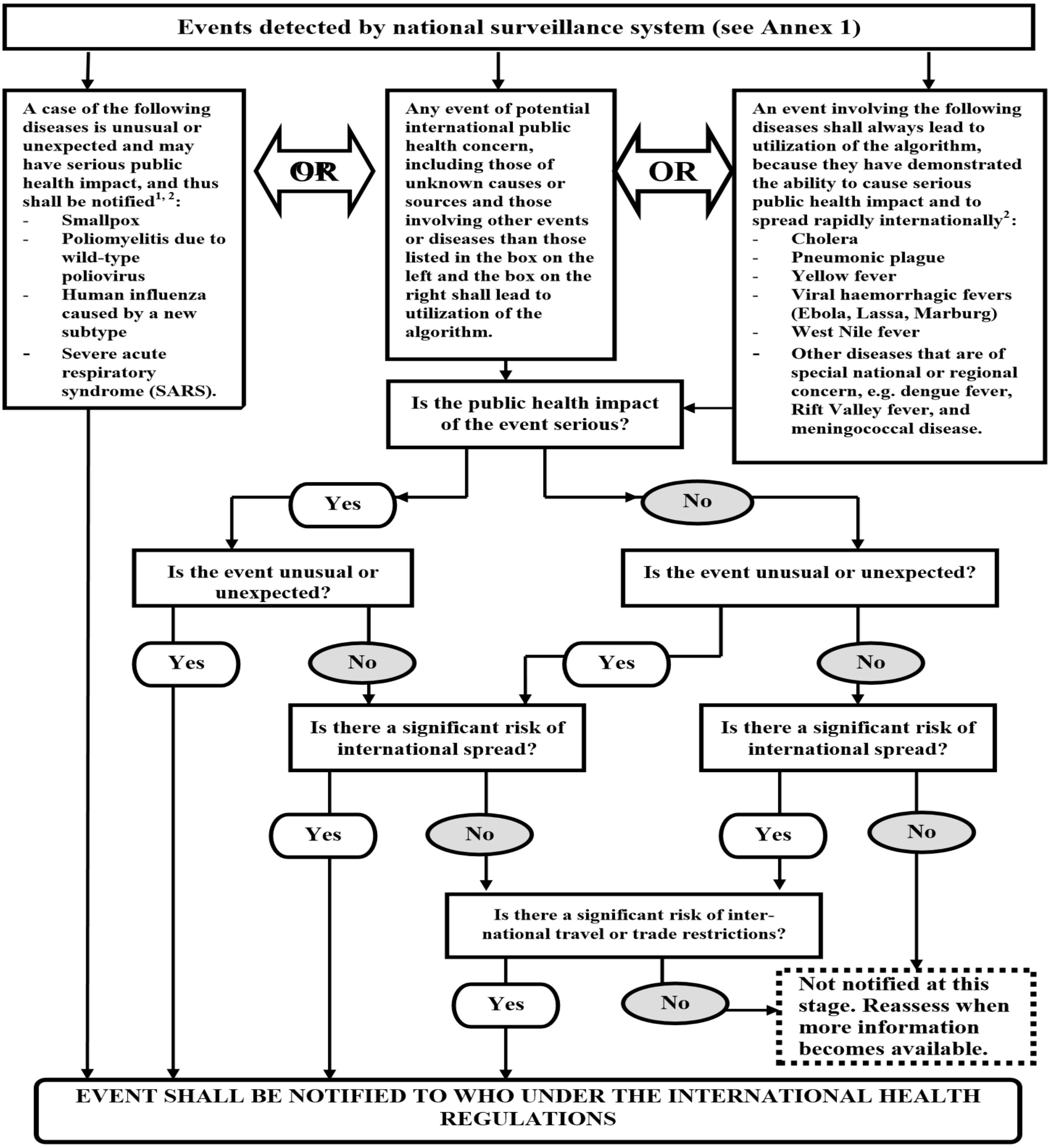
Evaluation and improvement of the surveillance system

25. Decide whether other indicators will be evaluated and plan how to monitor and evaluate the timeliness and completeness of reports.

26. State three or more goals to achieve in order to improve surveillance in the district during the following year, based on relevant data.

Appendix C. IHR (2005) Decision-making Instrument

DECISION INSTRUMENT FOR THE ASSESSMENT AND NOTIFICATION OF EVENTS THAT MAY CONSTITUTE A PUBLIC HEALTH EMERGENCY OF INTERNATIONAL CONCERN


Events detected by national surveillance system (see Annex 1)
A case of the following diseases is unusual or unexpected and may have serious public health impact, and thus shall be notified[1, 2]:
- Smallpox
- Poliomyelitis due to wild-type poliovirus
- Human influenza caused by a new subtype
- Severe acute respiratory syndrome (SARS).
OR
Any event of potential international public health concern, including those of unknown causes or sources and those involving other events or diseases than those listed in the box on the left and the box on the right shall lead to utilization of the algorithm.
OR
An event involving the following diseases shall always lead to utilization of the algorithm, because they have demonstrated the ability to cause serious public health impact and to spread rapidly internationally[2]:
- Cholera
- Pneumonic plague
- Yellow fever
- Viral haemorrhagic fevers (Ebola, Lassa, Marburg)
- West Nile fever
- Other diseases that are of special national or regional concern, e.g. dengue fever, Rift Valley fever, and meningococcal disease.
Is the public health impact of the event serious?
Yes
No
Is the event unusual or unexpected?
Is the event unusual or unexpected?
Yes
No
Yes
No
Is there a significant risk of international spread?
Is there a significant risk of international spread?
Yes
No
Yes
No
Is there a significant risk of international travel or trade restrictions?
Yes
No
Not notified at this stage. Reassess when more information becomes available.
EVENT SHALL BE NOTIFIED TO WHO UNDER THE INTERNATIONAL HEALTH REGULATIONS

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[1] As per WHO case definitions.

[2] The disease list shall be used only for the purposes of these Regulations

Appendix D. Potential emergencies of public health concern to be reported under the International Health Regulations (2005)

Surveillance of specific risks

One of the best ways to protect international public health is to control or minimize known health risks. Known risks represent a large majority of events likely to become public health emergency situations under the IHR (2005). There are programs for the control of infectious diseases, food safety and environmental health which significantly contribute in the global alert and response system of WHO;

Environmental risks include:

- Chemicals
- Food items
- Ionizing radiations
- Non-ionizing radiations

It is possible to receive technical information on these risks from various sources (see references at the end of the document). Areas of interest for building capacities in integrated surveillance should include partnerships to deal with the issues below:

1. Emergency situations on environmental health;
    (a) Natural events;
    (b) Technological incidents;
    (c) Complex emergencies;
    (d) Events caused by deliberate acts.
2. Chemical risks in food; acute or chronic food exposure (environmental or deliberate pollution);
3. Zoonosis;
    (a) Emerging zoonosis;
    (b) Neglected zoonosis.

THEMES ON THE SURVEILLANCE OF SPECIFIC HAZARDS

1. Infectious diseases hazards

Threats posed by known, new or unknown infectious diseases

2. Zoonotic diseases events

Using the "One Health" approach is key in establishing a link between human health and animal health as an interface between man, animal and environment. Coordination, collaboration and communication between sectors and partners in the "One Health" approach enables the maximization of resources while ensuring better health to people and animals sharing the same environment. It is important to detect diseases affecting animals as they may pose a risk for human health, thereby saving lives.

3. Events related to food safety

Food-borne or water-borne diarrhoeal diseases are the main causes of diseases and deaths in the poorest countries, with about 1.8 million deaths yearly, mostly children. Identifying the source of an outbreak and containing it are essential under IHR.

4. Events related to chemicals

Detecting and controlling events related to chemical, toxic products or environmental pollution are essential in enforcing IHR.

5. Radiological or nuclear events

A radiological or nuclear emergency may be caused by incidental or deliberate leakages. It may also be detected during a clinical analysis of irradiated patients admitted in a care structure, even if the exposure source has not been confirmed.

Source: *A guide for assessment teams. International Health Regulations (2005): Protocol for assessing national surveillance and response capacities for the International Health Regulations (IHR) in accordance with Appendix 1A of the regulations.* February 2009

Appendix E. Guide for establishing a community-based surveillance and response system

Community surveillance is a simple public health initiative, adaptable and cheap, managed by communities under the coordination of official surveillance structures. Designated communities and community focal persons are trained and equipped to be aware of potential health risks, including emerging events that could point out to a new health risk. They must equally ensure close surveillance of immediately reportable diseases and seasonal diseases or signs of the emergence of an existing disease. For a trained health professional, an event which seems “unusual, strange or unexplained” to the community may be an early warning sign of a more serious and significant health risk or public health event.

Two different strategies on community surveillance may be used to gather information from the community:

(a) **Event-based community surveillance (EBCS)** builds on the reporting of unusual events and is designed for quickly identifying problems in the community. The information may be incomplete, unconfirmed or simply a rumour. The definition of an “unusual event” varies from one community to the other, and may be defined in each context. This may be an event or a cluster of events that may be unusual to a specific community or at a given period in the year. For instance, an unusual event may be a “cluster of deaths from unknown cause in the same household or in adjacent households.”

(b) **Indicator-based community surveillance (IBCS).** This type of surveillance is used to identify/report events, based on agreed indicators (community case definitions). Information from the community may be provided by people, including volunteers for community-based surveillance who have already been briefed on indicators.

IBCS is based on the declaration of a suspected case or the trend of specific diseases using the community case definition. The community-based definition of a case consists of two or three easily identifiable symptoms in relation with a specific disease. This is a most basic form of syndromic declaration (symptoms) used by health professionals in surveillance systems of diseases at the national level, IDSR and others. For example, influenza whose community case definition is as follows “sudden illness, fever, cough and difficulty in breathing”, and acute flaccid paralysis (AFP) which is defined as the sudden onset of paralysis/weakness in one part of children under 15 years of age.

Both systems should be set up to ensure that all information from the community is captured and quickly communicated for follow up to a resource period appointed for surveillance at the next level These two elements of surveillance should equally be integrated at the community level.

Steps in establishing community surveillance

A crucial step in establishing community surveillance consist in ensuring the adherence of national and sub-national authorities. This will enable the official acknowledgement of the community surveillance system as part of the national surveillance system, and people will then be appointed. Therefore, a HFofficial or appointed surveillance worker in charge of coordinating community surveillance activities should:

(a) Determine, within the structure, the availability and knowledge of standard community case definitions for reporting priority diseases and conditions and public health events of concern.

(b) Raise awareness of community leaders, elders and other persons of influence on the relevance of community surveillance, necessary information, how information is used, proposed process, the characteristics of the resource persons in community surveillance, financial support or human resources provided by district and the benefits the community derives from participating.

(c) Define the sources of information on health events in the community, including points of contact between the community and the health services. A key informant selected among these sources may form community networks enabling community-based surveillance resource persons to detect signals early (e.g., by raising awareness of persons who regularly go to grain mills or tea drinking spots). Information sources include:

  (i) Home visits - during which community surveillance resource persons are supposed to regularly visit homes in their catchment area to inquire about priority diseases and deaths that may have occurred since their last visit.

  (ii) Gathering places - another method for gathering information on priority events will involve community surveillance to frequently go to gathering places or meeting parlours in villages. This will not replace home visits but will rather constitute another way of ensuring that priority events are identified on time. Gathering places or meeting parlours in the community are places where people meet to discuss and share news for mouth-to-ear. For example, community wells, boreholes or rivers where women meet to fetch potable water or wash clothes. While they are working, women share news about their family and their neighbourhood.

(iii) Grain milling places - in some communities, women gather to grind, crush or pound grains into flour. They share news about their family and their neighbourhood while working.

(iv) Beer, palm wine or tea parlours - in some communities, men gather daily in beer parlours, other drinking places, in a house or a shop, or under a specific tree, to drink and socialise. While drinking, they sometimes share news about their family, friends and neighbours.

(v) In the market - information and news are shared. People who go to the market and spend part of their time buying or selling things and the rest of the time chatting.

(vi) In churches, mosques or temples - Sometimes, religious leaders make announcements before and after the service to inform people about events in the neighbourhood. Moreover, people who go to church or to the mosque often talk together before or after the service to share news about their family, friends and neighbours.

(vii) At the village Chief house or where elders meet - the chief of the village and elders are generally informed of whatever happens in their community. They often gather to share information about the community or to discuss issues and take decisions.

(viii) In schools and schoolyards - teachers and students often share information and news about their family and friends when they meet in school or while playing in the schoolyard.

(d) Identify surveillance correspondents for each source of information, in collaboration with the community. Determine and specify possibilities for community involvement in the surveillance of health events, and the role of community-based surveillance resource persons. Resource persons should be trusted people in the community, and who are committed to signalling “zero case”. They must be reassured that they will not run into trouble if they report bad news so, they do not need to forge data.

(e) Clearly indicate alerts, events, diseases and conditions for surveillance within a controlled area, as well as those prescribed in the national policy; equally specify trigger mechanisms.

(f) Draw a list of epidemic-prone diseases, diseases targeted for eradication and elimination, and other priority public health diseases, including non- communicable diseases.

(g) Define information and support methods to focal persons in implementing community surveillance. These may include monthly meetings or phone calls. List the current training opportunities for surveillance and response liaison officers.

(h) Define training needs. Design and pre-test illustrated/simplified training material intended for surveillance and reports preparation, for illiterate/semi-literate people; design illustrated teaching aids based on games, and illustrated daily or weekly timetables.

(i) Train community surveillance correspondents in surveillance and response techniques, and in improving interpersonal relationships through interactive training, adult learning practices and role-plays. Using a cellular phone to show MP3 files or other video clips may be useful during training and in the community.

(j) Describe how communication on surveillance and response is done, and how it is followed up between the health structure/surveillance worker and contact persons in community surveillance. For community-based surveillance resource persons who are literate, design simple alert forms (see Appendix 2B) and show them how to enter information. For those who are illiterates, develop mechanisms for obtaining information on events. Think about such mechanisms as identifying a family member who can help with writing;

(k) Include methods such as monthly meetings and telephone calls to ensure the follow up of correspondents in community surveillance.

(l) Examine and update supervision and feedback procedures and methods between the health structure and community liaison officers. Regular training for updating knowledge should also be held to ensure that community focal points understand the types of signals to be reported and how to do this.

(m) Describe communication links between community focal points and health structures and the outbreak management committee, which may be activated during an outbreak and for routine activities.

(n) Develop general, illustrated and social mobilization documents to raise community, youth or schools' awareness;

(o) Organize periodic meetings between surveillance coordinators in HF, coordinators of community surveillance and community leaders to discuss progress, problems and concern, and provide reciprocal feedback;

(p) State three or more goals you would like to achieve in order to improve surveillance in the community during the following year, based on relevant data.

Official framework of community surveillance

Community surveillance should be implemented within an official framework where participants have a sound knowledge of what an unusual event is (a signal) to be reported (unexplained number of similar serious diseases, increased absenteeism at school) and how and when to report (e.g., by message or telephone calls). The framework should be supported by a trained staff of a health structure or a HD staff, which is regularly evaluated.

Community representatives who are part of the community surveillance team

Community members who are trusted by their community may be resource persons in community surveillance. They should be selected by the community where they reside for greater empowerment and ownership of community surveillance. Representatives may be staff of basic services at the village level namely: community health volunteers, community health workers, trained birth attendants or similar care providers, chiefs of villages (religious, traditional or political leaders), teachers, veterinarians, health workers, drug vendors, traditional healers and other communities, a respected person who does not belong to the health sector such as a hairdresser, shopkeeper or grandmother always discussing with members of the community are effective focal persons.

Once they have been selected, community-based surveillance resource persons should undergo training and play their role on how to recognize some diseases or conditions in order to signal suspected cases.

Supervision of community surveillance

Supervision goal is to improve timeliness in reporting, better understand case definitions and enhance person-to-person communication skills. It is important to conduct supervision through evidence-based approaches to know which area of surveillance requires improvement. All implementation activities for community surveillance should be coordinated by a surveillance officer or an official from a HFin the locality. As such, he will be in charge of:

(a) Preparing a list of priority diseases, events or conditions to be integrated in community surveillance based on technical guidelines adapted from IDSR;

(b) Sharing, as agreed, a list of simplified community case definitions to facilitate case detection, events detection and follow up;

(c) Designing pre-test and provide training material and illustrated aide-memoire;

(d) Designing an interactive training module;

(e) Building capacities of community surveillance correspondents in all areas of surveillance and response;

(f) Regularly upgrading skills and practices of focal points in all appropriate aspects of surveillance and investigations, especially treatment and dissemination of data;

(g) Establishing feedback loops, which is an essential measure to ensure the community surveillance is still working; Ensuring that suportive supervision and supervision of posts are conducted, where they are commended for the good job done; and identifying areas of improvement;

(h) Disseminating simplified case definitions and signals using posters or any recent intervention method (banners, flyers, etc.) which have proven their worth, in relevant places in the community, and adequately;

(i) Following up surveillance and response activities including the timeliness and completeness of reports;

(j) Supervising the activities of community surveillance focal persons, including development of case definitions; where community surveillance focal persons are used for contact tracing, ensure that the person working in the healthcare structure leads the process, in collaboration with the person living in the community;

(k) Identifying and mapping out key health determinants in the regions;

(l) Providing community surveillance teams with regular and timely feedback and ensuring bidirectional feedback process, in order to build an atmosphere of trust between community surveillance and the HFworker;

Note: In some countries, there may be too many community surveillance correspondents, and coordination may become problematic. A district may then appoint a focal supervisor for community surveillance in a given community (possibly among community surveillance correspondents) to supervise a specific number of community surveillance correspondents. The roles of community supervisors should be clearly defined to avoid situations where they are the ones to decide whether something poses a health hazard based only on reporting.

Sources of information for community surveillance

Operational community surveillance should build relationships with the main sources of information. These include, without restriction, living sources:

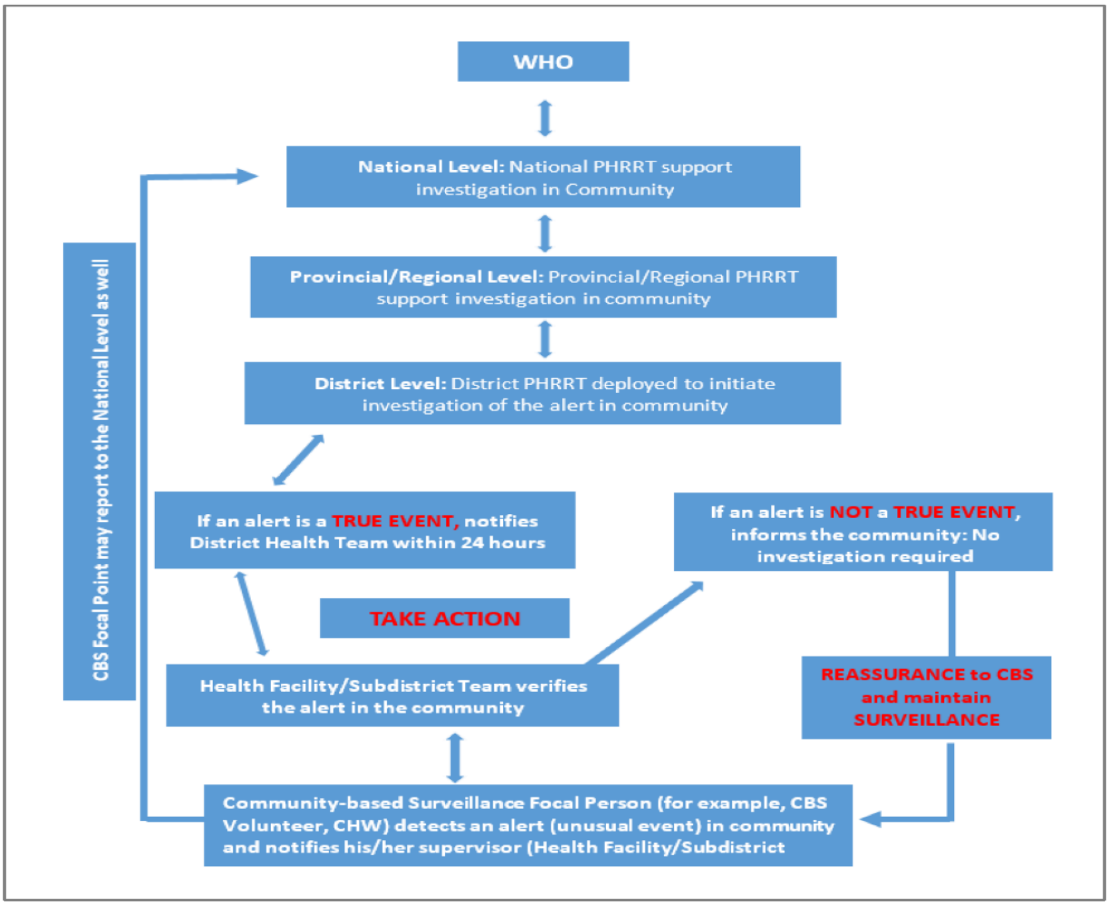
(a) All community health workers, community volunteers including traditional birth attendants, teachers, pharmacists who maintain a relation of trust with the community; They are often found in areas where access to primary health care is rare; Families often share information with a trusted and known health worker.

(b) Community leaders, traditional or religious leaders, young people and members of the civil society: these persons and groups can provide informal reports on unusual health hazards they have witnessed in their communities.

(c) Local, national and international media are important sources of information for community surveillance. Events such as clusters of human cases, outbreaks or unexpected and unusual deaths may be covered by local newspapers (printed or online), or radio reports before they are detected and signalled by local health services.

(d) Traditional medicine, traditional healers and practitioners and shrine keepers. In some African countries, a large portion of the population rely on traditional medicine for primary health care. Traditional medicine has been used for thousand of years and traditional practitioners may constitute a valuable source of information. Families whose members are sick often seek spiritual advice in shrines known to be places of healing.

(e) Alternative medicine (herbalists, for instance), complementary medicine and non-conventional medicine, including health care practices which are not integrated in the dominant healthcare system.


WHO
National Level: National PHRRT support investigation in Community
Provincial/Regional Level: Provincial/Regional PHRRT support investigation in community
District Level: District PHRRT deployed to initiate investigation of the alert in community
If an alert is a TRUE EVENT, notifies District Health Team within 24 hours
If an alert is NOT a TRUE EVENT, informs the community: No investigation required
TAKE ACTION
Health Facility/Subdistrict Team verifies the alert in the community
REASSURANCE to CBS and maintain SURVEILLANCE
Community-based Surveillance Focal Person (for example, CBS Volunteer, CHW) detects an alert (unusual event) in community and notifies his/her supervisor (Health Facility/Subdistrict
CBS Focal Point may report to the National Level as well

Note: Additional reference elements for community surveillance are found in the Integrated Disease Surveillance and Response in the African Region: *WHO Guide for establishing the Community-Based Surveillance and Response Programme* (August 2014) and the International Federation of the Red Cross and Red Crescent Societies (IFRC): Community-Based Surveillance: basic principles March 2017

Appendix F. Surveillance and response core capacities as described in the IHR (2005)

According to IHR, members States will use existing national structures and resources to meet their needs in core capacities. These needs include surveillance capacity, report writing, reporting, verification, response and collaboration. Each party must evaluate existing national structures and resource capacities in meeting minimum needs. Building on this evaluation result, each Member State shall develop and implement an action plan to ensure that core capacities are present and operational across the country.

Appendix 1, Part A of the IHR (2005) defines the required surveillance and response core capacities. The regulations acknowledge three levels in the health care system:

(a) Response at the primary public health or community level

(b) Response at the intermediate public health level

(c) National level

Response at the level of the local community or primary health care

At the local community or primary public health level, the capacities are as follows:

(a) Detect events causing diseases or deaths above expected thresholds at a given time and place in all regions of the country;

(b) Immediately signal all key information available to the higher level (within 24 hours). At the community level, CHWs will send their reports to the HF in their respective catchment areas. At the level of HDs, reports should be sent to the regional or national level. According to this Guide, key information include:

- Clinical case descriptions.
- Laboratory results.
- Sources and types of risks.
- Number of human cases and deaths.
- Conditions on the spread of the disease, include environmental issues such as water and sanitation, personal history of travels and that of neighbours, behavioural issues such as burial practices, distance to cover between house and HF to receive care, efforts to receive care before detection, weather conditions and accessibility, floods, insecurity, migrants/internally displaced persons/refugees. Public health measures are implemented, including adopted regulations and hygiene measures.

Intermediate level of response

The need of core capacities for public health response at the intermediate level should be adapted to each district context. Many countries have more than one intermediate level (health area, district and region) while others which are smaller may have only one (district or national)

The needs in core capacities and the health system functions may vary among countries. For example, while in countries with large federal States intermediate level functions may be closer to the core capacity needs described under “national level”, in smaller countries with a single level, intermediate level functions may be closer to those of the community level or the primary public health response level.

The core capacities at the intermediate level are:

(a) Confirming the situation of signalled events and supporting or enforcing additional control measures

(b) Immediately evaluating signalled events and, in case of emergency, communicate all core information to the national level within 24 to 48 hours. In this appendix, applicable criteria to emergency events include serious impact on public health or the unusual or unexpected nature presenting high risk of spreading.

National level Evaluation and Reporting

At the national level, response has two functions: evaluation and reporting

(a) Ensuring coordination with the coordinator of the World Organisation for Animal Health (OIE) and the coordinator of the international Food Safety Authorities Network and other sectors to ensure the coordination of evaluation and reporting of events.

(b) Evaluate all reports on emergency events within 48 hours.

(c) Immediately notifying WHO through the national IHR focal point when evaluation indicates that the event should be notified as per paragraph 1 of Article 6 of the IHR and the decision-making instrument to evaluate and report public health events likely to constitute a PHEIC in Appendix 2 of the IHR, and inform WHO where necessary, as per Article 7 and paragraph 2 of article 9 of these Regulations.

At the national level, public health response requires the capacity to:

(a) Ensure coordination by establishing a coordination mechanism which may include the creation of a public health emergency operations center or a similar coordination structure and the activation of the IMS (see sections 5 and 6 for further details);

(b) Quickly decide on the necessary control measures to prevent/limit national and international spread;

(c) Provide support through specialised personnel in laboratory analysis of specimens (at the national level or through collaborating centers), and a logistical assistance (material, supplies and transportation);

(d) Provide on-site assistance, where necessary, to complement local investigations;

(e) Ensure an operational direct link with high health officials and other civil servants for rapid approval;

(f) Implement lockdown and control measures;

(g) Ensure direct link with other ministries concerned;

(h) Ensure, through the most effective communication channels available, liaisons with hospitals, clinics, airports, seaports, road crossings, laboratories and other essential operational areas for disseminating information and recommendations received from WHO about events that occurred in the Member State's territory and in other Member States territories;

(i) Establish, implement and update a national plan for response to public health emergencies, including the setting up of a "One Health" team to intervene in case of events likely to constitute a PHEIC;

(j) Provide what precedes 24/7;

During various consultations at international level, the core capacities were summarised into eight components: legislation, policy and coordination, surveillance, preparedness, response, risk communication, laboratory and human resources. These eight components are equally important for IDSR.

APPENDIX G. Roles and responsibilities of the various IDSR stakeholders

Roles and responsibilities of a community-based surveillance resource person (community health worker)

Using simplified case definitions to identify diseases, events, conditions or other priority hazards in the community, the Community Health Worker:

(a) Conducts regular home visits;

(b) Regularly meets key informants;

(c) Attends local ceremonies and events and monitors every unusual event, such as an expected person who has not shown up;

(d) Records diseases, conditions or health-related unusual events in forms and reporting tools (logbook) and reports immediately within 24 hours;

(e) Participates in verbal autopsies by answering interview questions prepared by the health facility’s supervisor;

(f) Promptly notifies the nearest HFand other relevant sectors of unexpected or unusual cases of disease or death in humans and animals for immediate verification and investigation in accordance with the International Health Regulations and the IDSR strategy (within 24 hours);

(g) Involve local leaders in describing disease events and trends in the community;

(h) Educates the community on the need to report priority diseases, conditions and unusual events in order to obtain care for these diseases;

(i) Supports health workers during case or outbreak investigation and contact finding;

(j) Mobilizes local authorities and community members to support response activities;

(k) Participates in the mapping of potential risks, in training and in simulation exercises;

(l) Participates in isolation and response activities in coordination with the district;

(m) Participates in response activities, which may include home-based care, social or behavioural change of traditional practices, logistics for the distribution of medicines, vaccines or other supplies. Providing quality health education during crisis is a valuable contribution;

(n) Provides feedback about reported cases, events and prevention activities to community members;

(o) Verifies that public health interventions have taken place as planned, with the participation of the community. Participates in meetings organized by health area, district and higher-level authorities.

Roles and responsibilities of HFstaff at the point of entry

The health facility's staff:

(a) Identifies priority disease cases using standard case definitions;

(b) Records case-based information and reports immediately notifiable diseases, conditions and public health events to the next level;

(c) Liaises with the district on how to conduct an immediate laboratory investigation of suspected cases;

(d) Manages case treatment/referrals;

(e) Prepares for and participates in the investigation and response to outbreaks and treatment of cases;

(f) Communicates summary and case-based data (weekly report) within the time frame provided to the next level;

(g) Conducts simple data analysis (graphs, tables, charts) at point of collection;

(h) Communicates the diagnosis of epidemic-prone diseases to the district or community;

(i) Convenes the district Rapid Response Team;

(j) Determines the resources (human, financial, material and telephone) and the deployment schedule.

Roles and responsibilities of the surveillance officer at the district level

The role of the surveillance officer is to:

(a) Investigate and verify potential outbreaks, collect specimens, diagnose, advise on treatment or prevention protocols;

(b) Prepare and analyse weekly surveillance reports and submit them promptly to higher authorities;

(c) Ensure that surveillance sites properly maintain surveillance reports and registries or logbooks;

(d) Keep an up-to-date list of all notification sites;

(e) Establish and keep an up-to-date database of all trained and registered health workers who can serve as surveillance focal persons at reporting sites as well as other community surveillance focal persons;

(f) Ensure that reporting sites have an adequate supply of data collection and reporting tools;

(g) Ensure that standardized IDSR case definitions for all priority diseases are understood and used by health workers at the site. Provide on-site training if necessary;

(h) Monitor IDSR performance indicators (such as completeness and promptness);

(i) Periodically update graphs, tables and diagrams and compare current data with past data, over semesters and quarters, even weeks or years (which is important for seasonal events), and make recommendations;

(j) Personally provide weekly or monthly feedback on IDSR implementation to surveillance reporting sites;

(k) Call the reporting sites to ensure that they report accurate data on time;

(l) Regularly carry out support and supervision to surveillance sites, including health centers, border posts, and communities, and build their capacity to analyse and interpret their data to guide decision-making;

(m) Sign and date registers or phone calls to document the supervisor's visit, and also write recommendations for improvement;

(n) Assist the health department in verifying community signals;

(o) Organize and direct investigations of verified cases or outbreaks;

(p) Keep an updated list of suspected cases;

(q) Help HF collect, package, store and safely transport laboratory specimens for confirmation;

(r) Receive laboratory results from the region and forward them to the HF;

(s) Organize training sessions for surveillance sites with new personnel;

(t) Check the quality of surveillance data from time to time by carrying out audits and propose appropriate measures to improve the quality of such data;

(u) Keep a rumour logbook to record events regarding the surveillance sites;

(v) Ensure cross-border (district to district) coordination and collaboration on surveillance issues and report any outbreaks in the neighbouring district. International or cross-border notification should also be done if necessary;

(w) Document the added value of the IDSR and urge the management team to support IDSR activities;

(x) Participate in outbreak investigations and ensure that a registry/line list is maintained.

Roles and Responsibilities of the district management team

The district management team's role is to:

(a) Ensure liaison, through the District Medical Officer (DMO), on overall surveillance activities and plans;

(b) Help the district surveillance officer implement planned activities;

(c) Ensure that surveillance activities are included in district annual work plan;

(d) Ensure liaison with district officials to raise funds for surveillance activities;

(e) Ensure timely release of funds for surveillance activities;

(f) Monitor performance and data analysis results using an IDSR monitoring tool;

(g) Participate in the mapping of district risks and the development of an action plan, based on the results obtained;

(h) During outbreaks, assist the PHEMC in organizing and running the RRTs (see Section 5 for more details);

(i) Forward the findings of the initial investigation to the region;

(j) Participate in risk mapping and community assessment;

(k) Participate in the establishment of emergency preparedness and response committees and ensure their functioning;

(l) Design, work out and implement community health education programs;

(m) Support training of HF and the community in the response in collaboration with the region;

(n) Identify appropriate public health actions and implement the response;

(o) Plan timely community outreach and education activities;

(p) Document response activities;

(q) In the event of an outbreak, send a daily district situation report.

Roles and responsibilities of the other political leaders at the district level

Political leaders such as those in the village, neighbourhood, or district level are very important people who help foster behaviour change in disease surveillance. They can play the following roles:

(a) Support any declaration of a public health emergency;

(b) Make a local inventory to identify human, financial or logistics support. A quick response can often prevent the spread;

(c) Ensure that hygiene and sanitation principles are respected (cleanliness of the environment, availability and use of latrines, advocacy for safe water consumption, personal hygiene and sanitation measures, including hand washing);

(d) Report clusters of illnesses or deaths to a neighbouring health facility;

(e) Compel enforcement of regulations to strengthen health and sanitation measures;

(f) Actively participate in the sensitization of community members in order to promote, maintain and support good health;

(g) Facilitate community-based planning, implementation, and evaluation of the neighbourhood's surveillance activities and health programs;

(h) Monitor outbreaks in collaboration with health care providers and other extension workers in the neighbourhoods;

(i) Provide administrative support to health care providers in the neighbourhoods and villages;

(j) Support the implementation of relevant legislation to prevent or control the occurrence of infectious diseases;

(k) Supervise subordinates by ensuring that the principles of hygiene and sanitation are respected;

(l) Ensure regular convening of meetings (or organization of meetings) of the public health care committee during an outbreak;

(m) Discuss disease characteristics and implications for action in regular meetings with the DMO;

(n) Ensure that various committees are established and provided with the necessary resources to carry out their activities;

(o) Leverage resources from a variety of sources to respond to disasters, including disease outbreaks;

(p) Conduct advocacy on health issues as part of various campaigns in the district.

Roles and Responsibilities of the Regional Team

(a) Liaise with the national level on overall surveillance activities and plans under the supervision of the Regional Delegate of Public Health;

(b) Help the regional surveillance officer and district level officials implement planned activities in their various districts;

(c) Ensure that surveillance activities and district plans are included in the regional annual work plan;

(d) The Regional Delegate of Public Health liaises with regional officials to mobilize funds for surveillance activities and ensures that these funds are released on time;

(e) Monitor the performance and results of the IDSR at the district level;

(f) Participate in district risk mapping and assist districts in working out an action plan based on the results obtained;

(g) During outbreaks, assist the PHEMC in organizing and operating regional RRTs for public health emergencies at both the regional and district levels (see Section 5 for details);

(h) Communicate the findings of the initial survey at the national level;

(i) Participate in the establishment and operation of emergency preparedness and response committees;

(j) Help districts perform risk mapping and situational analysis;

(k) Help districts design and implement community health education programs;

(l) Support district training in response activities;

(m) Help districts identify appropriate public health actions for the response and facilitate cross-border surveillance and response.

Role of the national level

(a) Establish a PHEOCC or similar coordination mechanism to coordinate preparedness and response activities in case of a public health event (see Section 5 for more details);

(b) Designate a spokesperson and present a risk communication plan, including media engagement, to share information before, during and after a public health emergency;

(c) Define standards, policies and guidelines for the IDSR and update emergency preparedness and response plans based on simulations and post evaluation;

(d) Evaluate the national level’s capacity and improve it accordingly;

(e) Identify and mobilize national resources and coordinate external support for the implementation of the IDSR;

(f) Ensure the supervision, monitoring and evaluation of the IDSR's activities;

(g) Produce and distribute epidemiological reports;

(h) Monitor the implementation of international, national and regional agreements and protocols;

(i) Support investigations of suspected outbreaks;

(j) Provide data management and analysis support at the national level.

Role of WHO and other partners

(a) Contribute to the establishment of standards and the elaboration of guidelines;

(b) Provide technical, material and financial assistance to strengthen in-country disease surveillance, health information and laboratory systems;

(c) Help the Ministry of Public Health mobilize resources for surveillance and response activities;

(d) Support the supervision, monitoring and evaluation of IDSR;

(e) Provide management support (e.g., writing funding proposals or requests);

(f) Support capacity building, training, equipment, etc;

(g) In case of a health emergency, support with technical experts and back-up personnel (if necessary during the response) and provide mobile laboratories, other equipment and vaccines.

N.B. WHO's role is to facilitate coordination with other partners and other UN agencies.

APPENDIX H. Guide to implementing point of entry surveillance and response systems

Object

The purpose of the IHR 2005 is to prevent, protect against, control and respond to the international spread of disease through public health action that is proportionate and limited to the risks to public health, while avoiding unnecessary interference with international traffic and trade. It calls for strengthening national surveillance and monitoring capacity, including sites such as: sea ports, airports and land crossings; prevention, alert and response to international public health emergencies; global partnerships and international collaboration. In addition to the IHR, it is critical that border health activities be sustainable and in line with other surveillance activities conducted under the IDSR.

A system of detection, reporting and appropriate response for ill travellers is necessary. The long-term strategy is to ensure that formal points of entry (PoEs) are fully compliant with the IHR, while ensuring that points of entry also have emergency plans. All designated points of entry must have routine surveillance and response capabilities.

Key partners

The main partners accompanying the Ministry of Public Health include: the related sectors; the descentralised territorial authorities, the actors of the " one health" platform, the air, maritime and port authorities; the UN agencies; US CDC and Africa CDC and other key partners.

Key surveillance and intervention areas at points of entry

(a) Routine measures should be put in place at points of entry for detection of ill travellers, reporting to health authorities, rapid public health assessment, and access to care for severely ill travellers or those whose symptoms pose a public health risk, including safe transport from the point of entry to a health facility.

(b) Detection of ill travellers should include at least the following:

- Reporting ill or dead travellers from aircraft, ships or land/ border crossings;
- Maritime health officers or immigration officers at certain points of entry should be trained to recognize ill travellers they encounter during their routine assessments; and should also conduct an initial assessment to determine whether or not the illness is a potential public health risk.

(c) Arrangements for the initial response to an ill traveler, if the disease is detected at a point of entry, should include, at a minimum, the following:

- The ability to quickly isolate the ill traveller from others to prevent the potential spread of the disease.in question
- Standby health teams should be available, either in person or remotely by telephone, to conduct a rapid assessment of ill travellers detected at points of entry to determine if a communicable disease of public health concern is suspected.
- A HFlocated near the point of entry should be designated to provide medical care, as needed, to seriously ill travellers or those suspected of having a communicable disease of public health concern. The designated HFshould have adequate infection prevention and control capacity to prevent the spread of disease to staff or other patients, as well as diagnostic capacity, including access to laboratory diagnostics.
- Ambulance services or other safe means of transportation should be available to facilitate the transportation of ill travellers from the point of entry to the designated health facility.

(d) If necessary, in case of a declared public health emergency affecting international travellers or posing a risk of international disease spread, there should also be the ability to implement traveller screening or other border health measures quickly, as recommended by the WHO.

Role of competent authorities

(a) Immediately report all events and illnesses with outbreak potential detected at points of entry to the immediate higher level. Notification should also be made at the same time at the national level, with a copy of the report to the IHR National Focal Point, to evaluate the use of the decision-making algorithm. If yellow fever is suspected, include vaccination for cases from endemic or risk areas.

(b) If a traveller is a suspected case, fill in the passenger location form/alert notification form immediately. Ensure that the traveller/suspect is separated from others, including family members, and transferred to the nearest waiting room.

(c) If a suspected traveller is recognized and may not be symptomatic during the trip, be sure to take the appropriate details and transfer this information to a nearby health center for close surveillance. The HFwill liaise with the community focal person (CFP) for close monitoring.

(d) Monitor luggage, cargo, containers, conveyances, goods, postal parcels, and human remains departing from and arriving to affected areas so that they are maintained in such a condition that they are free of sources of infection or contamination, including vectors and reservoirs.

(e) Ensure, as far as possible, that facilities used by travellers at points of entry are maintained in a safe condition and free of sources of infection or contamination, including vectors and reservoirs.

(f) Supervise the pest control, disinfection or decontamination of luggage, cargo, containers, conveyances, goods, postal parcels and human remains or sanitary measures for persons, if any, under these regulations.

(g) Inform conveyance operators, as early as possible, of their intention to apply control measures to a conveyance and provide, where appropriate, written information on the methods to be used.

(h) Report suspected cases to the health center as soon as possible so that transportation can be arranged.

(i) Ensure that all filled in forms are kept in a safe place. Create a database for events, if a computer is available. Keep a record or log of all events.

When responding to an emergency or outbreak, those responsible for cross-border coordination should follow these steps:

(a) Organize a meeting with partners as soon as the outbreak or event is confirmed;

(b) Assess the need for the establishment of a regional or national Emergency Preparedness and Response Committee or RRTs, and if necessary, request their support;

(c) Meet regularly to assess the situation of the outbreak as appropriate;

(d) Share surveillance data regularly, take into account the number of cases (including zero cases, if applicable) and contact tracing status (if necessary);

(e) Exchange information on the movement history of the persons concerned and, on the persons, to contact to facilitate the coordination of interventions on both sides of the border;

(f) Regularly evaluate the response to the outbreak and take steps to improve its control as indicated above; document and communicate the response to the outbreak by escalating notifications as needed.

APPENDIX I. List of actors involved in IDSR implementation per level in Cameroon

Various levels	Actors
National	Technical departments, NPHO, reference laboratories
Regional	RDPH, focal points, communication FPs, head of the health information bureau (CBIS), CERPLE
HD	District medical officer, district health management team, district data manager, district surveillance fp, district communication FP, DHC, DMC
Health Area	Head of the health area, health area surveillance fp, heads OF HFs, health area communication FP, health committee, HMC
Community	CBO, CHW, community leaders, traditional healers and traditional birth attendants, state and council agents

Section 1: Detecting and recording cases of priority diseases, conditions and public health events

This section describes:

- how to integrate both Indicator-Based surveillance (IBS) and Event-Based Surveillance (EBS) to early detection of priority diseases, conditions and public health events;
- how to use standard case definitions to detect priority diseases, conditions and public health events;
- how to establish EBS and using this approach for signals detection, triage and verification to detect public health events;
- procedures which need to be followed when planning surveillance and response activities at the district level and how to update the map and reporting/guideline tools;
- how to use networks and emphasize the role of the laboratory to improve surveillance and response activities to confirm suspected outbreak.

Introduction

Health staff (human, animal, and environmental) conduct surveillance activities at all levels of the health system (public and private) so they can detect public health problems of concern in their communities.

Community members also play an important role in surveillance by actively participating in detecting, reporting, responding to and monitoring health events related to humans or animals in their catchment area.

Various public health events and/or risks may also occur at Points of Entry (PoE) and they can be recognized before, during or after travel, even when travellers have already left the Point of Entry. Staff at Points of Entry must be vigilant in ensuring that these events are identified and reported on time to facilitate response.

Surveillance priorities may be communicable and non-communicable diseases, conditions or public health events that include national or local priorities such as epidemic-prone diseases, maternal deaths or public health events associated with human and/or animal health, which might have direct consequences to human health.

An essential function of a public health surveillance system is to be vigilant in its capacity to detect not only known public health threats with established case definitions and formal reporting channels but also events or potential hazards that are not specifically included in the formal reporting system. These may be events such as clusters of disease patterns or rumours of unexplained deaths. These diseases, conditions and public health events may come to the attention of the health system in several ways through community, hospital, media or call centre channels.

For example:

- a person falls sick and seeks treatment from a health facility;
- high rate of hospital admission for the same diseases or symptoms;
- community members report unusual circumstances or events at the local level;
- community members report unusual occurrences of deaths or unusual disease pattern to the health facility;
- a pharmacy may also report a large increase in purchase of a particular drug or treatment;

- a school might report unusual absences due to similar signs and symptoms such as an influenza-like illness (ILI);
- health staff who conduct routine record reviews observe that cases of another priority disease have not been reported. For example, an officer who normally reviews the clinic register of a HF for cases of Acute Flaccid Paralysis (AFP) also sees that a case of cholera has recently been recorded in that clinic register. The register also keeps records of cases of non-communicable diseases such as high blood pressure and diabetes;
- health staff conduct routine record reviews of the laboratory register and observe recorded confirmed cases of priority diseases such as malaria and typhoid fever;
- radio, television, newspapers, or social media (WhatsApp, Facebook, etc.) report a rumour of rare or unexplained events in the area with potential exposure for humans;
- civil registration records show an increase in maternal deaths;
- unusual reports of illness among health care workers;
- a HFhas reported adult deaths due to bloody diarrhoea. During analysis of the routine reports from all the facilities in the area, the district medical officer notices adult deaths due to bloody diarrhoea which might signify that there might be an outbreak of *Bacillary dysenteriae* or *Escherichia coli*;
- an unusual death or number of deaths among animals, such as livestock, birds or rodent species, or an unusually high number of sick animals presenting the same signs;
- environmental officers observed during assessment of water bodies, contamination that might be due to chemicals like lead, or due to other related chemicals from mining activities, which might be an early trigger for public health interventions.

1.1. IBS and EBS to detect and record cases of priority diseases, conditions and public health events

The IDSR strategy integrates both IBS and EBS approaches to early detection of priority diseases, conditions and public health events.

As part of efforts to increase the sensitivity of the surveillance system, all countries should also establish EBS system alongside the IBS at all levels of the health system, that is, at the national, regional, district, HFand community levels. The IBS involves the use of standard case definitions to identify diseases, priority conditions, and public health events, while EBS uses signal detection, triage and verification to detect events. In contrast with case definitions that are narrow and disease-specific, EBS requires the detection and immediate reporting of signals, which are broad and indicate the possibility of a serious public health event. Signals that are verified are classified as events according to the result of this verification.

IBS and EBS are an integral component of the routine IDSR activities of the surveillance staff. Both IBS and EBS should use existing resources and infrastructure set aside for routine IDSR strategy.

1.2. Use of standard case definitions

A standard case definition is an agreed-upon set of criteria used to decide if a person has a particular suspected disease or condition. The definition specifies clinical criteria, laboratory diagnosis and specifications on time, place and person.

Why do we need case definition?

- to help decide if a person has a presumed disease or condition or public health event, or to exclude other potential disease diagnosis;
- to ensure that every case is diagnosed in the same way, regardless of where or when it occurred, or who identified it;
- to initiate action for reporting and investigating quickly if the clinical diagnosis takes longer to confirm;
- to compare the number of cases of the diseases, conditions or public health events that occurred in one time or place with the number occurring in another time or place.

Using standard case definitions throughout the national system helps in detecting priority diseases and use thresholds or signals to take necessary action. However, when HF and districts use different case definitions, it is difficult to identify the trend of a particular disease or event. It is then impossible to intervene urgently and investigate the causes of the change in trend. This is because health workers analysing data based on a particular standard case definition will not be able to know whether the trends observed in another region have similar or different causes.

Using standard case definitions is also important in implementing the IHR (2005). For this reason, it is important that health staff at all levels of the health system, including community, must be aware of case definitions of diseases, conditions or public health events that may afflict not only the local community but also have the potential for spreading across geographical boundaries.

The process of reporting events to WHO under the IHR requires the use of the decision-making instrument based on the IDSR standard case definitions, confirmation, data analysis and interpretation of results. The IHR (2005) decision-making instrument is included in ***Appendix 2C, Section 2***.

In describing standard case definitions, for HFlevel, a three-tiered classification system is normally used – Suspected, Probable, Confirmed:

(a) **Suspected case**. Indicative clinical picture, that is, patient will have typical clinical signs or symptoms that could be a disease, priority condition or public health event;
(b) **Probable case**. Suspected case linked epidemiologically to a confirmed case, but has not been confirmed yet by the laboratory or for which a laboratory sample cannot be taken because the case is lost or dead or a sample had been taken but was not available for laboratory testing or was not viable for sufficient laboratory testing;
(c) **Confirmed case**. A suspected case or any person confirmed by a laboratory test.

The classification might vary according to the epidemiology of the individual diseases. In all outbreak scenarios, a more sensitive case definition to identify all suspected cases should always be used. Identification of cases in these scenarios will use the syndromic surveillance approach where case detection will be based on clinical features without any laboratory diagnosis (See the introductory section for the description of syndromic surveillance). If in the middle of an outbreak, the cause of the agent has been established, cases may continue to be classified as either suspected cases or confirmed cases. An additional tier classification, that is, “Probable case definition”, may be added if officials feel that conducting laboratory tests on every patient with a consistent clinical picture and a history of exposure (for example, measles) is unnecessary.

Case definitions at the community level are usually simplified and are used to facilitate early detection of priority diseases, public health events and conditions or other hazards in the community. Case definitions at this level use key signs and symptoms to help the community to recognize when they should refer a person with these signs and symptoms to a CHW or a HF(HF).

Examples of how key signs and symptoms of community case definitions may be described are in ***Appendix 1B***.

All cases (suspected, probable and confirmed) should always be recorded in a recognized facility register and the IDSR reporting forms.

1.2.1. “One Health” approach in identification of public health events

“One Health” aims at applying a holistic approach in jointly detecting events and conducting risk assessment in responding to possible public health events occurring at the human-animal environment interface. Detection of events under the “One Health” approach thus requires all levels from community, district, and region to national to strengthen collaboration across sectors, and jointly share responsibility of detecting events, which might have an impact on the health of humans, animals and their shared environment.

Examples of the “One Health” approach include detection of a rabid animal or reports of animal illness from the veterinary sector, which can facilitate investigations of human cases of disease or reports of human diseases, which can be traced through exposure to chemical hazards within the environment.

Detection of events at PoE also requires a “One Health” approach and this requires involvement of all relevant sectors such as ministries responsible for human health, animal, health environmental health and other sectors responsible for the security of goods and persons. All events detected should be shared with relevant sectors as part of the “One Health” approach.

1.2.2. Distribute standard case definitions and registers to Health Facilities

Make sure that HFpersonnel at all levels including PoE(s) know and have available standard case definitions (including those for reporting unusual events, disease patterns, or unexplained deaths) specified by the national level.

Some countries have prepared and disseminated case definitions for diseases under surveillance in the form of a poster or as a small pocket-sized booklet. These tools reinforce the use of standard case definitions for detecting and reporting priority diseases, conditions and public health events that could be risky.

Ensure that HF personnel know the process for recording and notification, including reporting sites. They should also be trained in detection, and use of surveillance tools. Also, ensure that HF record rumours. The registers, which are normally used in most countries, are the outpatient department or inpatient department registers. Surveillance officers should always liaise with the health information focal person to extract the priority disease of IDSR from the register.

Proposed case definitions based on established disease-specific programs are found in ***Appendix 1A*** and are available in Section 11 of these guidelines.

1.2.3. Distribute community level case definitions using key signs and symptoms

Provide information to community health workers (CHW), traditional healers, birth attendants and community leaders on how to recognize and report priority diseases, conditions or public health events to the health facility. The case definitions for community level should be simpler than those used in HF. A list of examples of case definitions for use at the community level is in ***Appendix 1B*** of this section.

At the same time, emphasize the need to refer people with the suspected disease or condition to a CHW or HF.

Provide them with procedures for reporting, including when and who to report to (CHWs, Heads of HAs, community leaders, heads of HFs, etc.). Also, ensure that the necessary surveillance tools are made available to them. It is important to design simple community signal forms for detecting priority diseases, conditions and public health events, as well as other reporting tools (see ***Appendix 2B***) to enable them to refer a suspected case. It is important to show them how to fill information and for those who are illiterate, develop mechanisms of capturing information of events from them. Think of mechanisms like identifying someone from the family member who can assist with actual writing. Also, provide information to the community on priority diseases using posters, newsletters and announcements during meetings. Also, provide feedback methods and how timely information will be made available to the community, considering that this will encourage community members to participate in surveillance and response activities.

1.3. Establish Event-Based Surveillance (EBS) at all levels

All countries should ensure that the EBS system is established at all levels of the health system alongside the IBS system. EBS implementation involves taking into consideration the functions of EBS as illustrated in ***figure 4***.

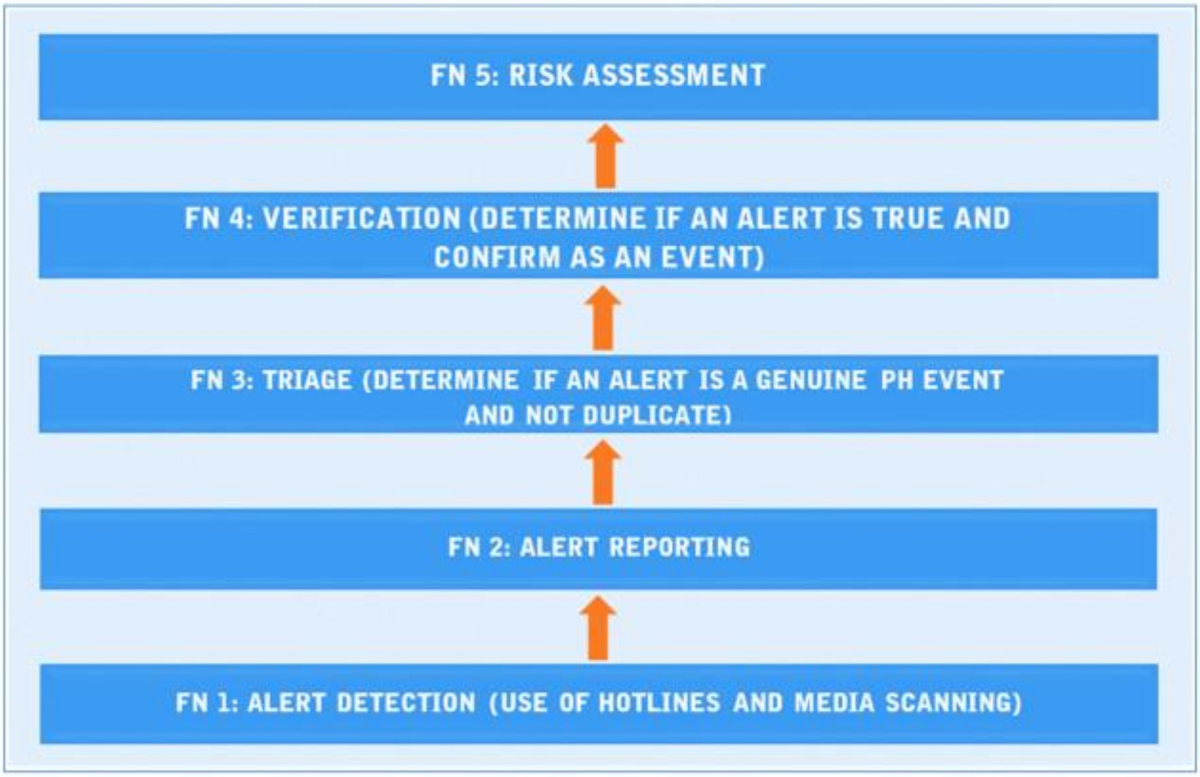

FN 5: RISK ASSESSMENT
FN 4: VERIFICATION (DETERMINE IF AN ALERT IS TRUE AND CONFIRM AS AN EVENT)
FN 3: TRIAGE (DETERMINE IF AN ALERT IS A GENUINE PH EVENT AND NOT DUPLICATE)
FN 2: ALERT REPORTING
FN 1: ALERT DETECTION (USE OF HOTLINES AND MEDIA SCANNING)

Figure 4: Functions of EBS at all levels of the health system

The steps below are followed in establishing and monitoring an EBS system:

- Step 1: Establish mechanisms to use the four sources of EBS (call centre, media screening, HF, community) to detect signals
- Step 2: Detect signals
- Step 3: Register and report signals
- Step 4: Conduct triage of signals
- Step 5: Conduct verification of signals and include true signals (event) in the IDSR
- Step 6: Conduct risk assessment and characterisation of signals changed as events

The steps for establishing EBS at the national, regional, district and HFlevels are described in ***Appendix 1C*** of this section.

1.4. Update district procedures for surveillance and response

Each year, national, regional and district health officials should work together to update and adjust procedures for surveillance and response accordingly. Results of the assessment should be used to update surveillance procedures and adjust action plans according to priorities.

1.4.1. Update the mapping of the catchment area

At least annually, update information about the catchment area (HF, PoE, laboratories). This activity should be part of the health planning at the district, regional and national levels. Make sure there is a description on local population characteristics in the catchment area, what activities are happening, what risks should be accounted for, and what
surveillance assets and gaps exist.

Risk mapping should extend to all public health hazards as specified by IHR (2005), including chemical, zoonotic, radiological and nuclear hazards. WHO has developed an integrated risk profiling tool for assessment of public health threats, and this can be used within the broader framework of disaster risk management. *(Strategic Tool for Assessing Risk STAR, WHO, July 2017)*.

This information is used to define the measures to be taken to reduce these risks and avoid associated diseases and deaths. Examples of potential risks include: sources of contaminated water, lack of urgent transportation to a referral facility for women in childbirth, or potential hazards such as inadequate safety precautions in mining or occupational sites or slums where there is a public health risk, especially during heavy rains or poor latrine coverage.

To update the catchment area description, make sure you have current information about:

- the size of key target populations at all levels such as children between 0-11 months, from 12-59 months, from 5-14 years, from 15 years and above, women of childbearing age, pregnant women, elderly, refugees, internally displaced persons, out of school youth, and other vulnerable groups, etc;
- major public health activities in the area including public, private, and NGO, immunization activities, clean water projects, family planning clinics, nutritional centers for malnourished children, refugee camp health activities, information related to risk factors for non-communicable diseases etc.

In updating the district profile, you can use several methods among which is the creation of a forum with key health stakeholders at all levels, where there will be discussion on surveillance and response activities related to priority health events at the district level, and this can facilitate getting updates from stakeholders on various key areas in surveillance and response in which they are involved. This could be done through a monthly or quarterly meeting. Take the opportunity also to provide feedback about surveillance data that is reported from their institutions to the district. Involve officials from other relevant sectors in the forum in order to address health issues following the “One Health” approach.

1.4.2. Update the list of reporting sites and the directory of surveillance focal persons in the Health District

Updating the list of reporting sites and the directory of focal surveillance officers in the district requires:

- identifying all HF, Points of Entry, and any other location in the country including community focal points required to report unusual surveillance data or events;
- building relationships with private facilities, other stakeholders of the "One Health" approach and NGOs, the district, and involving them in surveillance activities;
- recording (updating as needed) HFand Points of Entry locations and names of staff who are responsible for surveillance activities. Also updating the records for community focal persons.

Ensure that telephone and e-mail contact information of focal persons, which may include community health workers, trained birth attendants, community leaders, public safety officials, etc., is updated and recorded.

Ensure that also in recording or updating the focal persons, identification is done whether the focal persons have been trained in surveillance or not in order to plan for either new training or orientation to update their skills. A sample worksheet for listing the reporting sites and the contact focal person at each site is in ***Appendix 1C***.

1.4.3. Identify potential community representatives that can be engaged in community-based surveillance

Any community member known and accepted by the community can be a community-based surveillance (CBS) focal person. They should be selected by the communities they live in so as to increase empowerment and ownership of CBS. Representation could be from basic community-level services such as trained birth attendants, community or village health workers, or similar health care providers, village leaders (religious, traditional or political), school teachers, veterinarians, health extension workers, drug vendors and traditional healers and in many communities, a respected non-health person such as the barber, shop keeper, security personnel or grandmother who regularly talks to community members, are effective focal persons.

Keep an updated list of the selected people with their contact information, including the corresponding health facility. Ensure they have a list of simplified community case definitions to facilitate case detection and reporting. A sample worksheet for listing the reporting sites and the contact focal person at each site is in ***Appendix 1C***.

1.4.4. Distribute updated data collection forms, reporting tools and technical guidelines

As you conduct updates of the catchment area description, check to see that all reporting sites have an adequate supply of surveillance reporting tools, forms, line list, registers or other means for reporting surveillance data (telephone and/or e-mail). Monthly coordination meetings with HF and other reporting sites provide opportunities to distribute updated versions of forms and procedures for notification, investigation and response to public health events.

1.5. Role of the laboratory in surveillance and response

There are several diseases or conditions with signs and symptoms that are the same or similar as other diseases or conditions. For example, a child with fever and rash over the entire body might be diagnosed with measles, even though there could be several causes for the child's clinical presentation (for example, scarlet fever, rubella). Laboratory confirmation of diagnosis of diseases, events under surveillance is essential to accurately confirm the diagnosis in an individual patient, and verify the cause (or aetiology) and/or the source of an outbreak.

Laboratories should be used as early warning alerts to detect pathogens and other hazards that have potential to spread. For example, emergence of resistant strains in the hospital or the community (for example, multi-drug resistant tuberculosis, hospital-acquired infections).

1.5.1. Specimens' collection, storage and transportation

The type of specimens collected, its packaging (storage media) and its shipment depends on the suspected disease. Specimens should be collected in adequate quantity into appropriate containers, taking into consideration the risk assessment at the health care facility level or, if necessary, in the field during an outbreak investigation. All specimens must be triple packaged and labelled correctly and accompanied by the correct laboratory forms and supporting documentation (autorisations, agreements, etc.) in order to arrive at the laboratory in good condition, and provide reliable results.

Ensure that HF have trained personnel, equipment as well as adequate reagents and consumables to enable specimen collection. A clearly defined transportation process is required to enable HF to understand where to send specimen according to defined protocols. Equally, minimize time delay between collection, shipment and analysis of the specimen.

Many factors can affect the reliability of interpretation of laboratory test results. Thus, it is difficult to interpret results when:

- a specimen has not been properly taken (e.g., haemolysed blood specimen);
- delay in transportation and/or processing may result in bacterial contamination in a collected specimen such as urine or cerebrospinal fluid;
- use of wrong transport or storage media or container may cause reduced viability of the suspected organism;
- giving antibiotics before specimens for cultures are collected;
- wrong temperature is used for storage of specimen.

The disease-specific reference tables in ***section 9*** lists recommended laboratory procedures for confirming priority diseases and conditions including:

- The diagnostic test for confirming the disease or condition;
- The specimen to be collected;
- When to collect the specimen;
- How to collect, store and transport the specimen;
- When to expect the results;
- Sources for additional information.

It is necessary to initiate public health measures even before laboratory confirmation has been received. It should be noted that the patient should be isolated based on signs and symptoms, and case management should be initiated immediately even prior to laboratory results such as in the case of viral haemorrhagic fevers.

1.5.2. Update the list of facilities belonging to the laboratory network

Appendix 1D of this section describes the functions of laboratories according to the level of the health system.

The surveillance Focal Person (FP) and the laboratory focal persons at each level of the health system should maintain an updated list of the laboratories that have the capacity to perform required laboratory testing. ***Appendix 1F*** provides a sample worksheet for listing national laboratories for confirming priority diseases and conditions. It provides to all HFs information about the methods for transporting specimens including how to prepare, handle, transport and store the specimens. Make sure to disseminate information about packaging and transporting infectious material as directed by the national policy.

At health facility, district and regional levels, the focus is on safe collection, handling, transportation and processing of specimens. The surveillance focal person (HA, HD, region) should establish or strengthen routine communication with identified laboratories that receive specimens from your HFor district. The purpose of this routine contact is to strengthen communication between the HF in the district that will be sending specimens, and the laboratory that will be receiving them. Ensure that the procedures for specimen collection, transportation, confirming the disease or condition through laboratory testing and reporting the results are clear and can be reliably carried out.

To support regional or district level laboratories within the network, the national level health authority will establish a memorandum of understanding with laboratories outside the area or network that have the capacity for specific diagnostic procedures not available locally. The national level should also support the laboratory through advocacy with high decision makers to procure the necessary supplies to collect, handle, store, and transport specimens safely throughout the network.

In addition, it is also crucial to improve collaboration between human, veterinary and other relevant public health laboratories in line with the "One Health" approach.

1.5.3. Update the inventory of supplies, reagents and equipment used for confirmation of diseases by laboratories performing the test

Surveillance activities should actively work with the laboratories regarding supplies, reagents and equipment to avoid duplication and maintain an updated list of supplies, reagents and equipment available in each laboratory. This should be done especially in public HF. In addition, an attempt should be made to obtain a comprehensive inventory from private facilities. The inventory should also consist of telephone numbers of the laboratory focal persons. An inventory of hazardous biological agents and toxins identified through surveillance should also be made.

1.5.4. Describe laboratory procedures for confirming priority diseases and conditions

The national level should make sure that laboratory protocols and guidelines are established and known at all levels. A laboratory focal person should be identified at all levels. Each laboratory focal person should make sure that laboratory protocols, guidelines and procedures are followed at their assigned level. (See ***Appendix 1F)***

When designated, the HD Laboratory officer must ensure that the national laboratory confirmation procedures are known and followed at the district. The appointed staff must:

- ensure pre-positioning of specimen collection and transport equipment (immediate availability) at the district laboratory. Rapid Diagnostic Tests (RDTs) or serological tests to detect priority diseases and hazards (e.g., chemical) must be available for timely use;
- assist the HF to collect appropriate specimen for confirmation of suspected cases;
- coordinate with the laboratory, if needed, to identify appropriate specimen collection and any special procedures or difficulties;
- collect and package specimen properly or assist the facility in doing so;
- ensure safe and reliable transport of the specimen from the facility to the selected laboratory;
- receive laboratory results and quickly send them to the HF and national authorities;
- also send results to the patient's physician;
- based on the laboratory results, take measures with the HFaccording to national guidelines.

1.5.5. Establish a laboratory quality control and assurance program

A quality assurance program (internal and external quality control) is the backbone of good laboratory performance. Laboratory quality control and quality assurance are important for building confidence in the results obtained. Establishing or strengthening the laboratory quality assurance programs will allow improvement of the reliability and reproducibility of laboratory results. Coordinate with regional or national laboratory authorities to establish activities for ensuring quality results from laboratories.

Standard Operating Procedures (SOPs) are among the most important documents in a diagnostic laboratory. Ensure that each laboratory has up-to-date written SOPs for all techniques performed in the laboratory. These procedures should be the same throughout a country’s laboratory network so that each laboratory is performing tests in the same manner. These SOPs should also incorporate internal quality controls. In addition, laboratories should participate in external quality evalauation programs and corrective actions implemented based on sub-standard/poor results, in order to maintain excellence in the laboratory. Laboratories should be encouraged to engage in the WHO Stepwise Laboratory Improvement Process Towards Accreditation (SLIPTA) if not yet accredited. Refer to WHO Stepwise Laboratory Quality Improvement Process towards Accreditation (SLIPTA) [Checklist Version 2:2015 for

Clinical and Public Health Laboratories for how to do SLIPTA assessment and *http://apps.who.int/iris/handle/10665/204423*].

1.6. Appendices of Section 1

APPENDIX 1A	WHO/AFRO standard case definitions for reporting suspected priority diseases, conditions and public health events from the HFto the district
APPENDIX 1B	Community level case definitions using key signs and symptoms
APPENDIX 1C	Guide for establishing Event-Based Surveillance (EBS) at the national, regional, district and HFlevels
APPENDIX 1D	List of district reporting sites
APPENDIX 1E	Laboratory functions by health system level
APPENDIX 1F	Responsibilities of laboratory focal persons at all levels
APPENDIX 1G	List of national laboratories of human and animal health for confirming priority diseases, conditions and public health events

APPENDIX 1A. Standard case definitions recommended by the WHO AFRO for reporting suspected cases of priority diseases, conditions or public health events by HF at the district level.

The WHO AFRO advises HF (HFs) to use the following standard case definitions to report suspected cases of priority diseases, conditions and public health events to the HD. For additional information on each priority disease targeted for surveillance by WHO AFRO, see specific guidelines on ***section 11*** that equally integrates the actions to be taken for alerts or crossing outbreak thresholds.

Priority Diseases or Conditions	
Diseases or Condition	**Standard case definition**
Acute Haemorrhagic Fever Syndrome	**Suspected case**: sudden onset of fever lasting less than three weeks in a seriously sick patient or a deceased person **AND** two of the following signs: haemorrhage or purpura; epistaxis (nasal bleeding); hematemesis (vomiting blood); haemoptysis (presence of blood in sputum); blood in stool; other haemorrhagic manifestations without known predisposing factor to haemorrhagic phenomena **OR** clinical suspicion of one of the viral diseases. **Confirmed case**: A suspected case with laboratory confirmation Note: During an outbreak, case definitions may be changed to correspond to the local event. It is important to note that during outbreaks, most cases might not show haemorrhagic manifestation, a proper history taking is crucial
Acute and Chronic Viral Hepatitis	**(a) Acute viral Hepatitis:** **Suspected Case:** Any person with discrete onset of an acute illness with signs/symptoms of: (i) Acute infectious illness (e.g: fever, malaise, fatigue) (ii) Liver damage (e.g.: anorexia, nausea, jaundice, dark coloured urine, right upper quadrant tenderness of body), **OR** (iii) Raised alanine aminotransferase (ALT) levels more than ten times the upper limit of normal values **Confirmed Case:** A suspected case that is laboratory confirmed by virus specific biomarkers • **Acute Hepatitis A:** IgM anti-HAV positive or positive RNA HAV; • **Acute Hepatitis B:** detection of the surface antigen of the VHB virus (HBsAg) **AND** presence of the essential antigen of the VHB virus (anti-HBc) class IgM, detection of the DNA of the hepatitis B virus; • **Acute Hepatitis C:** presence of VHC RNA (viral load), presence of the care antigen of the hepatitis C virus (if available) and detection of anti-VHC IgM. **Absence of acute hepatitis A markers (anti-VHA IgM) and Hepatitis E (IgM anti-VHE);**

	• **Acute Hepatitis D:** VHc RNA positive (Viral Load) and anti-VHc negative and presence of anti-VHD (generally IgM), and detection of VHD virus ARN (infection with VHD occurs ONLY as co-infection or superinfection of hepatitis B); • **Acute Hepatitis E:** presence of IgM anti-VHE positive. **(b) Chronic Viral Hepatitis Case definition (VHB and VHC):** **Chronic Hepatitis B:** • HBsAg is the first serological marker to appear. Persistence of HBsAg for at least 6 months indicates chronic infection • Positive anti-HBc (generally IgG); **Chronic Hepatitis C:** • Hepatitis C virus RNA present in a person with antibodies against hepatitis C (Anti-VHC positive) VHC RNA positive **OR** VHC core antigen positive • Positive VHC ARN **OR** positive core VHC antigen **Note**: Antibody detection (i.e., HCV Ab positive) cannot differentiate between acute, chronic infection and past infection.
Adverse Events Following Immunization (AEFI)	A medical incident that takes place after immunization, causes concern and is believed to be caused by the immunization. The adverse event may equally be a negative or unintentional sign, any abnormal lab result, a symptom or a disease.
Anthrax	**Suspected case:** Any person with acute onset characterized by several clinical forms which are: **1. Cutaneous form:** Any person with skin lesion evolving over 1 to 6 days from a papular through a vesicular stage, to a depressed black eschar invariably accompanied by oedema that may be mild to extensive. **2. Gastro-intestinal**: Any person with abdominal distress characterized by nausea, vomiting, anorexia and followed by fever **3. Pulmonary (inhalation):** any person with brief prodrome resembling acute viral respiratory illness, followed by rapid onset of hypoxia, dyspnoea and high temperature, with X-ray evidence of mediastinal widening **4. Meningeal**: Any person with acute onset of high fever possibly with convulsions, loss of consciousness, meningeal signs and symptoms; commonly noted in all systemic infections, but may present without any other clinical symptoms of anthrax. **Confirmed Case:**

	A confirmed case of anthrax in a human can be defined as a clinically compatible case of cutaneous, inhalational or gastrointestinal illness that is laboratory-confirmed by: 1. isolation of *B. anthracis* from an affected tissue or site; or 2. Other laboratory evidence of *B. anthracis* infection based on at least two supportive laboratory tests. **Note:** *it may not be possible to demonstrate B. anthracis in clinical specimens if the patient has been treated with antimicrobial agents.*
Buruli Ulcer (*Mycobacterium ulcerans Disease*)	**Suspected case**: A person presenting a painless skin nodule, plaque or ulcer, living or having visited a BU endemic area If a suspected case is confirmed (Not applicable to BU) **Confirmed Case:** A suspected case confirmed by at least one laboratory test (ZN for AFB, PCR, culture or histology).
Chikungunya	**Suspected case**: a person presenting with sudden high fever onset >38.5 °C, along with serious pain in joints or arthritis unexplained by other clinical conditions. **Confirmed case**: A suspected case with laboratory confirmation
Cholera	**Suspected case**: In areas where a cholera outbreak has not been declared: Any patient aged two years and older presenting with acute watery diarrhoea and severe dehydration **OR** who died from acute watery diarrhoea. In areas where a cholera outbreak is declared: any person presenting with or dying from acute watery diarrhoea. **Confirmed case:** A suspected case with Vibrio cholera O1 or O139 confirmed by culture or polymerase chain reaction (PCR) and, in countries where cholera is not present or has been eliminated, the Vibrio cholera O1 or O139 strain is demonstrated to be toxigenic.
Dengue Fever	Any person with acute febrile illness of 2-7 days duration with 2 or more of the following: headache, retro-orbital pain, myalgia, arthralgia, rash, haemorrhagic manifestations, leukopenia. **Suspected case:** A suspected case with laboratory confirmation (positive IgM antibody, four times or more rise in IgG antibody titres in paired serum specimens (acute and convalescent), positive PCR or viral isolation of Dengue by cellular culture). **Dengue Fever**: A probable or confirmed case of dengue with bleeding tendencies as evidenced by one or more of the following: positive tourniquet test; petechieae, ecchymosis or purpura; bleeding: mucosa, gastrointestinal tract, injection sites or other; hematemesis or melena; and thrombocytopenia (100 000 cells or less per mm3) and evidence of plasma leakage due to increased vascular permeability, manifested by one or more of the following: 20 % rise in average haematocrit for age and sex, 20 % drop in haematocrit following volume replacement therapy compared to baseline, signs of plasma leakage (pleural effusion, ascites, hypo-proteinaemia).

	Dengue Shock Syndrome: All the above criteria, plus evidence of circulatory failure manifested by rapid and weak pulse, and narrow pulse pressure (≤ 20 mm Hg) or hypotension for age, cold, clammy skin and altered mental status.
Diabetes	**Suspected new case**: Any person presenting with the following symptoms: (a) Increased thirst (polydipsia); (b) Increased hunger (polyphagia); (c) Frequent urination (polyuria) **Confirmed new case:** Any person with a non-fasting glucose ≥ 11,1 nnik>k (200 venous plasma glucose measurement of ≥ 11.1mmol/L (200 mg/dl) or capillary mg/dl) **OR** Any person with a fasting glucose 6.1 mmol/L (110 mg/dl) venous plasma glucose measurement of ≥ 7 mmol/L (126 mg/dl) or capillary glucose ≥11.1 mmol/L (200mg/dl)
Diarrhoea with Blood (Dysentery)	**Suspected case**: A person with abdominal pain and diarrhoea with visible blood in stool. **Confirmed case**: Suspected case with stool culture positive for Shigella dysenteriae type1.
Diarrhoea with dehydration in children below five years old	**Suspected case:** Passage of at least three loose or watery stools in the past 24 hours, with or without dehydration **AND**: *Moderate dehydration* - at least two of the following signs: restlessness, irritability, sunken eyes, thirst, skin that relaxes slowly after pinching, **OR** *Severe dehydration* - two or more of the following signs: lethargy or loss of consciousness, sunken eyes, inability or difficulty in drinking, skin that relaxes very slowly after pinching **Confirmed case:** Suspected case confirmed by stool culture revealing the presence of a known enteric pathogen. *Note*: Laboratory confirmation of the specific pathogen causing the outbreak is not routinely recommended for surveillance purposes.
Draconculiasis	**Rumour:** Information about the occurrence of Dracunculiasis from any source. **Suspected Case:** Anyone with an itchy or blistering skin lesion living in a Guinea worm endemic region or areas, and Guinea worm emergence.

	Confirmed case: someone with a skin lesion with guinea worm emergence, and when the laboratory confirms that the worm is *D. medinensis*. The person is considered a case only once during the calendar year, when the first worm emerges. Worm specimens should be collected from each patient for laboratory confirmation and sent to the U.S. Centers for Disease Control and Prevention (CDC). All cases should be monitored at least twice a month for the remainder of the calendar year for early detection of the possible emergence of other guinea worms.
Ebola or Marburg Virus Disease **Ebola or Marburg Virus Disease**	**Routine surveillance:** **Suspected case**: Anyone with a high fever that does not respond to any treatment of the usual causes of fever in the area, and who has at least one of the following signs: bloody diarrhoea, gum bleeding, skin bleeding (purpura), conjunctival injection and blood in the urine. **Confirmed case**: Laboratory-confirmed suspect case (positive IgM serology, positive polymerase chain reaction (PCR), or virus isolation, or with an epidemiological link to confirmed cases or an outbreak **Note: During an outbreak, case definitions may be adapted to the new clinical picture(s) or to the different modes of transmission related to the local event.** **Suspected Case:** • **Suspected case**: Any person, alive or dead, suffering or having suffered from a sudden onset of high fever and having had contact with: a suspected, probable or confirmed Ebola or Marburg case; a dead or sick animal (for Ebola); a mine (for Marburg); • Any person with sudden onset of high fever and at least three of the following symptoms: headaches - lethargy - anorexia / loss of appetite - aching muscles or joints - stomach pain - difficulty swallowing - vomiting - difficulty breathing - diarrhoea - hiccups; **OR** • Any person with inexplicable bleeding; **OR** • Any sudden, inexplicable death ; **Probable case:** • Any suspected case evaluated by a clinician; **OR** • Any deceased suspected case (where it was not possible to collect specimens for laboratory confirmation) with an epidemiological link to a confirmed case.
Epilepsy	**Suspected case:** Any person with one epileptic seizure (**Note**: For a new suspected case, report only the first diagnostic) **Confirmed case**: Any person with recurrence of, at least, two epileptic seizures. Any positive response to treatment with an antiepileptic drug reinforces the hypothesis of a confirmed case. Seizures can last from 30 seconds to three minutes. When occur continuously without a pause, they can lead to *status epilepticus*.

Influenza caused by a new sub-type	**Suspected case:** any person with unexplained acute disease of the lower respiratory tract, with fever (>38°C), cough, shortness of breath **OR** difficulty breathing **AND** with one or more of the following exposures during the seven days prior to symptom onset: (a) Close contact (within one meter, while taking care, talking to or touching) with a person who is a suspected, probable or confirmed case of H5N1 while; (b) exposure (for instance handling, slaughtering, plucking, cutting, preparing for food) to poultry, wild birds, their remains or environments contaminated by their dunk in an area where cases of non-seasonal H5N1 influenza infections in animals or humans have been suspected or confirmed in the past month; (c) consumption of raw or undercooked bird products in an area where there have been suspected or confirmed cases of H5N1 influenza infection in animals or humans in the past month; (d) Close contact with an animal other that poultry or wild birds with confirmed H5N1 influenza infection; (e) handling of specimens (animal or human) suspected of containing non-seasonal H5N1 influenza virus in a laboratory or other environment **Confirmed case**: a person fulfilling the criteria of a suspected case **AND** with laboratory confirmation by a laboratory whose tests results for H5N1 are accepted by WHO as being a confirmation. Note: ***Include IHR case definition for reporting human influenza caused by a new subtype***
High blood pressure	**New suspected case at first visit**: Anyone with a resting blood pressure (measured over an average of three readings, five minutes apart) ≥140 mm Hg for systolic pressure, or ≥90 mm Hg for diastolic pressure. **Confirmed case**: Anyone who, on at least two occasions, has a resting blood pressure (three readings) ≥140 mm Hg for systolic pressure, or ≥90 mm Hg for diastolic pressure.
Influenza-like illness	**Suspected Case:** An acute respiratory infection in the child or adult with sudden onset of fever (≥38ºC) **AND** cough; which occurred within the last ten days. **Confirmed case**: case corresponding to the clinical case definition and confirmed in the laboratory (laboratory results should reveal the presence of the influenza virus).
Injuries (Road traffic accident)	**Road traffic accident**: any person suffering from a road traffic injury, seen in consultation for the first time. **Traffic accident fatality**: Any person killed immediately or dying within 30 days as a result of a road traffic accident.
Lassa and Crimean-Congo Haemorrhagic Fevers	**Suspected CCHF case**: Anyone with sudden onset of fever, malaise, generalized weakness, irritability, headache, severe pain in the limbs and lumbar region, and marked anorexia. Early congestion of the face and chest, injected conjunctiva, haemorrhagic enanthema of the soft palate, uvula, and pharynx, and very often a rash of fine petechiae extending from the chest and abdomen to the rest of the body, sometimes with large purpuric areas. **Confirmed CCHF case**: Laboratory-confirmed suspect case (positive IgM serology, positive PCR, virus isolation or IgG seroconversion by ELISA or IFA) or with an epidemiological link to confirmed cases or an outbreak

	Suspected case of Lassa fever: Any person with a progressive illness with at least one of the following: malaise, fever, headache, sore throat, cough, nausea, vomiting, diarrhoea, myalgia, chest pain, hearing loss; and a history of contact with rodent excreta or with a confirmed case of Lassa fever **Confirmed case of Lassa fever**: Laboratory-confirmed suspect case (positive IgM serology, positive PCR, virus isolation) or epidemiologically linked to a laboratory-confirmed case
Leprosy	**Suspected case**: A person showing one of three cardinal signs of leprosy: hypo-pigmented or reddish skin lesion, loss or decrease of sensations in skin patch, enlargement or peripheral nerve. **Confirmed case**: A person showing at least two cardinal signs of leprosy and who has not completed a full course of treatment with multidrug therapy (MDT).
Lymphatic filariasis	**Suspected case**: Resident of an endemic area with a clinical sign of hydrocoele or lymphoedema for which other causes of these findings have been excluded. **Confirmed case**: A person with positive laboratory diagnosis of microfilaremia in blood smear, filarial antigenaemia or positive ultrasound test.
Malaria	**Uncomplicated malaria**: any person living in a malaria-risk area with fever or a history of fever within the previous 24 hours, without signs of severe malaria (malfunctioning of vital organs) is clinically diagnosed as having malaria. **Confirmed case of uncomplicated malaria:** any person having fever or with a history of fever within the previous 24 hours, with laboratory confirmation: microscopic blood smear or any other diagnostic test for malaria parasites. **Suspected case of severe malaria**: any person living in a malaria-risk area, hospitalized with high fever and whose clinical diagnosis shows signs of vital organ dysfunction. **Confirmed case of severe malaria**: a patient hospitalized with laboratory confirmed *P. falciparum* infection (asexual forms), accompanied by signs and symptoms of the severe form of the disease (vital organs dysfunction) diagnosed by a laboratory.
Malnutrition	**Low birth weight**: any new born with a birth weight less than 2500 grams (or 5.5 lbs) **Malnutrition in children:** (a) Children under 59 months who are underweight (indicator: weight for age<-2 Z score) (b) Children 6 to 59 months with Middle Upper Arm Circumference (MUAC)<11.5 cm (high risk of mortality) (c) Bilateral pitting oedema **Malnutrition in pregnant women**: Pregnant women giving birth to low-birth-weight babies (birth weight < 2.5 Kg) Poor nutritional and health status of the women, can predict which population groups may benefit from improved antenatal care of women and neonatal care for infants.
Maternal deaths	Death of a woman while pregnant or within 42 days of the delivery or termination of the pregnancy, irrespective of the duration and site of the pregnancy, from any cause related to or aggravated by the pregnancy or its management but not from accidental or incidental cause.

Measles	**Suspected case**: Any person with fever and maculopapular (non-vesicular) generalized rash and cough, flu, or conjunctivitis (red eyes) or any person in whom a clinician suspects measles. **Confirmed case:** A suspected case with laboratory confirmation (positive IgM antibody) or epidemiological link to confirmed cases in an outbreak.
Coronavirus of the Middle East respiratory syndrome (MERS-CoV)	**Note: Many case definitions are possible depending on whether the person resides or not in the Middle East. For more information, see Section 11.** **Suspected Case:** Any person with acute respiratory infection, with a history of fever and cough, and signs of pulmonary parenchymal disease (e.g., Pneumonia or acute respiratory distress syndrome), based on evident clinical or radiological data, and who travelled to the Middle East[2] within the 14 days before onset of the disease, or in countries where the presence of MERS-CoV is known to be circulating in dromedary camels or where human infections occurred recently. A person with an acute respiratory disease, regardless of the severity, who in the 14 days prior to the onset was exposed to one of the following situations: (a) close contact with a confirmed or probable case of MERS-CoV infection, while that patient was ill; (b) visit to a healthcare facility in a country where hospital-associated MERS-CoV infections have been reported; (c) direct contact with dromedary camels or consumption or exposure to dromedary camel products (raw meat, unpasteurized milk, urine) in countries where MERS-CoV is known to be circulating in dromedary camel populations or where human infections occurred as a result of presumed zoonotic transmission. **Confirmed case**: A person with laboratory confirmation of MERS-CoV infection1, irrespective of clinical signs and symptoms, who has been exposed to either one of the following situations (***note***: see Section on the Recommendations for testing)
Coronavirus disease 19	**Suspected case A:** any person compatible with the clinical **AND** epidemiological criteria: Clinical criteria (a) Acute onset of fever **AND** cough **OR** (b) Acute onset of at least **THREE** of the following signs and symptoms: fever, cough, disorders, weakness/tiredness, headaches, myalgia, sore throat, coryza, dyspnoea, anorexia/nausea/vomiting, diarrhoea, altered mental state Epidemiological criteria (a) Living or working in an area where there is high-risk transmission of the virus: for instance, a close residential area and humanitarian settings, such as internally displaced persons (IDP) settlements, any time within 14 days before the onset of symptoms **OR** (b) Residing or having travelled to an area where there is community transmission of the virus any time within 14 days before the onset of symptoms **OR** (c) Having worked in the health sector including HF and households within 14 days before the onset of symptoms **OR** **Suspected case B:** A patient with severe acute respiratory disease (SARI: acute respiratory infection: fever + cough which occurred within 10 days and requiring hospitalisation) **Probable case**

	Probable case A: A patient corresponding to the clinical criteria below (suspected case) **AND** who is the contact of a probable or confirmed case, or has epidemiological link to a cluster of cases in which at least one confirmed case has been identified. **Probable case B**: A person living or who has travelled to an outbreak-stricken area a presenting with recent anosmia (loss of smelling) or agueusia (loss of taste) without any identified cause. **Probable case C**: Unexplained death in an adult who has suffered from respiratory distress **AND**, who was the contact of a confirmed case or had epidemiological link to a cluster of cases in which at least one confirmed case has been identified. **Confirmed Case:** Any person with a positive diagnostic test (antigen RDT or PCR) to the COVID-19 infection, irrespective of the clinical signs and symptoms.
Bacterial Meningitis	**Suspected case**: any person with sudden onset of fever (>38.5 °C rectal or 38.0 °C axillary) and one of the following signs: neck stiffness, altered consciousness or other meningeal signs including bulging fontanelle in toddlers. **Probable case**: any suspected case with macroscopic aspect of CSF turbid, cloudy or purulent; or with a CSF leukocyte count >10 cells/mm3; or with bacteria identified by Gram stain in CSF; or antigen detection (for instance, by latex agglutination) in CSF. **In infants**: CSF leucocyte count >100 cells/mm3; or CSF leucocyte count 10–100 cells/ mm3 AND either an elevated protein (>100 mg/dl) or decreased glucose (<40 mg/dl) level **Confirmed case**: any suspected or probable case that is laboratory confirmed by culturing or identifying (i.e., by polymerase chain reaction,) a bacterial agent of *Neisseria meningitidis*, *Streptococcus pneumoniae* or *Haemophilus influenzae* type b in the CSF or blood
Monkeypox	**Suspected case**: An acute illness with fever > 38.3°C (101 F), intense headache, lymphadenopathy, back pain, myalgia, and intense asthenia followed one to three days later by a progressively developing rash often beginning on the face (most dense) and then spreading elsewhere on the body, including soles of feet and palms of hand. **Probable case:** A case that meets the clinical case definition, is not laboratory confirmed, but has an epidemiological link to a confirmed or probable case. **Confirmed case:** A clinically compatible case that is laboratory confirmed. **Differential diagnosis**: Alternative causes of clinical symptoms that must be considered include other rash illnesses, such as, smallpox, chickenpox, measles, bacterial skin infections, scabies, syphilis, and medication-associated allergies.
Neonatal tetanus	**Suspected Case:** *Neonatal tetanus* - Any newborn with a normal ability to suck and cry during the first two days of life, and who, between the 3rd and 28th day of age, cannot suck normally, and becomes stiff or has spasms. **Or** both. **Confirmed case**: No laboratory confirmation recommended.
Non-neonatal tetanus	**Suspected case**: any person aged above 28 days with acute onset of one of the following symptoms: trismus, continuous spasms of facial muscles or generalized muscular spasms.

	Confirmed case: No laboratory confirmation recommended.
New HIV Cases	A positive ELISA for confirming HIV and a rapid test for confirming the positive results are sufficient for an epidemiological case definition for HIV infection (WHO Regional Office for Africa recommends that countries use either Bangui or Abidjan HIV/IDSR case definitions).
Noma	**Suspected new case**: any child with a mouth ulcer and other warning signs such as malnutrition, poor hygiene, recent illness (measles, persistent diarrhoea, or malaria). **Confirmed new case**: any person with a gangrenous disease, which starts as gingival ulceration and spreads rapidly through the tissues of the mouth and face, destroying the soft and hard tissues.
Onchocerciasis	**Suspected case**: any person with fibrous nodules in subcutaneous tissues (in an endemic area). **Confirmed case**: A suspected case that is laboratory confirmed by presence of one or more of the following: microfilariae in skin snips, adult worms in excised nodules, or typical ocular manifestations (such as slit-lamp observations of microfilariae in the cornea, the anterior chamber of the eye, or the vitreous body).
Plague	**Suspected Case:** (a) Clinical presentation (sudden fever, headaches, serious malaise, prostration and very painful swelling of the lymph nodes or cough with blood in sputum, chest pain and difficult breathing) **AND** (b) Consistent with epidemiological features, such as exposure to infected animals or humans and/or evidence of fleabites and/or residence in or stay in a known endemic area within the past 10 days. **Confirmed case:** Any suspected case confirmed by isolation of *Yersinia pestis* in blood or by bubonic aspiration, or by specific seroconversion or a rapid diagnostic test detecting F1 antigen in endemic areas.
Poliomyelitis (Acute Flaccid Paralysis)	**Suspected case**: Any child under 15 years of age with loss or reduced muscle strength and/or muscle tone (paresia, weakness of the limb, difficulty in walking or standing, difficulty in moving a limb, hypotonia, etc...) of rapid onset (24-72 days) and involving one or many members including trauma of the nerve by injection **OR** any person in whom the clinician suspects poliomyelitis. **Confirmed case**: A suspected case with virus isolation in stool.
Perinatal deaths	**Perinatal death** includes the death of a foetus from 28 weeks of gestation to the first seven days after birth. Perinatal death is defined as any death of live newborn occurring up to 28 days after birth.
Human Rabies	**Suspected case**: A person suspected to have been in contact with a rabid animal and with one or more of the following: headache, neck pain, nausea, fever, fear of water, anxiety, agitation, abnormal tingling sensations or pain at the wound site,

	Confirmed case: A suspected case that is laboratory confirmed
Rift Valley Fever (RVF)	**Suspected case** **Early stage of the disease**: acute febrile illness (axillary temperature >37.5ºC or oral >38ºC) of more than 48 hours that does not respond to antibiotic or antimalarial therapy, and is associated with: (a) Direct contact with sick or dead animal or its products **OR** (b) Recent travel (withing the last week) to, or living in an area where, after heavy rains, livestock die or abort, and where RVF virus activity is suspected/confirmed **OR**: (c) Abrupt onset of at least 1 of the following signs: exhaustion, backache, muscle pains, headache (often severe), discomfort when exposed to light, and nausea/vomiting **OR**: (d) Nausea/vomiting, diarrhoea **OR** abdominal pain with 1 or more of the following: • Pale skin (or Hb < 8mg/dl) • Low platelets (thrombocytopenia) as evidence by presence of small skin and mucous membrane haemorrhages (petechiae) (or platelet count < 100x109 / dl.) • evidence of kidney failure (oedema, reduced urine output) (or creatinine > 150 mol/L) **OR**: • cutaneous bleeding into skin, bleeding from puncture wounds, from mucous membranes or nose, from gastrointestinal tract and unnatural bleeding from vagina **OR**: • jaundice (transaminase level three times higher that normal) **Late stages of diseases or complications (two to three weeks after onset of the first symptoms)**: Patients who have experienced, in the preceding month a flu-like illness, with clinical criteria, who additionally develop the following: (a) CNS disorders which resemble meningo-encephalitis **OR** (b) Unexplained visual loss **OR** (c) Unexplained death following sudden onset of acute flu-like illness with haemorrhage, meningoencephalitis, or visual loss during the preceding month. **Confirmed case**: any patient who, after clinical screening, has a positive anti-RVF IgM ELISA or quantitative reverse transcriptase polymerase chain reaction (RT-PCR) tests (characteristic appearance of antibodies between 4 and 6 days after the onset of symptoms).
Severe Acute Respiratory Infections (SARI)	**Severe Acute Respiratory Infection**: any seriously ill person aged five years or above presenting the signs of an acute respiratory infection of the lower tract with: (a) Sudden onset of fever (≥ 38ºC)

	(b) Cough or sore throat (c) Shortness of breath or difficulty in breathing (d) With or without clinical or radiologic evidence of pneumonia **OR** Any person who died from an unexplained respiratory condition
Severe Acute Respiratory Syndrome (SARS)	**Suspected case**: any person presenting (a) a history of a febrile flare-up, or a documented fever ≥38°C **AND** (b) One or more symptoms of lower respiratory tract illness (cough, difficulty breathing, shortness of breath) (c) Radiographic evidence of lung infiltrates consistent with pneumonia or ARDS or autopsy findings consistent with the pathology of pneumonia or ARDS without an identifiable cause **AND** (d) No alternative diagnosis can fully explain the illness **Confirmed case**: An individual who tests positive for SARS-CoV infection by the WHO recommended testing procedures.
Severe Pneumonia in Children under 5 years of age in	**Clinical case definition (IMCI) for pneumonia:** A child presenting with cough or difficult breathing and: (a) 50 or more breaths per minute for infant age 2 months up to 1 year (b) 40 or more breaths per minute for young child 1 year up to 5 years. ***Note:*** *A young infant age 0 up to 2 months with cough and fast breathing is classified in IMCI as "serious bacterial infection" and is referred for further evaluation.* **Clinical case definition (IMCI) for severe pneumonia:** A child presenting with cough or difficult breathing and any general danger sign, or chest indrawing or stridor in a calm child. General danger signs for children 2 months to 5 years are unable to drink or breastfeed, vomits everything, convulsions, lethargy, or unconsciousness. **Confirmed case:** radiographic or laboratory confirmation of pneumonia will not be feasible in most districts.
Sexually Transmitted Infections	**Genital ulcer syndrome (non-vesicular) :** Suspected case: Any male with an ulcer on the penis, scrotum, or rectum, with or without inguinal adenopathy, or any female with ulcer on labia, vagina, or rectum, with or without inguinal adenopathy. **Confirmed case**: Any suspected case confirmed by a laboratory test.

	Urethral discharge syndrome: **Suspected case**: Any male with urethral discharge with or without dysuria. **Confirmed case**: A suspected case confirmed by a laboratory test (for example Gram stain showing intracellular Gram-negative diplococci).
Smallpox	**Suspected case**: An acute illness with sudden onset of high fever >38.3°C (101 F) followed by a characteristic rash (macules, vesicles, pustules, crusts) with centrifugal distribution in the same stage of development without other apparent cause. **Probable case**: A case that meets the clinical case definition, is not laboratory confirmed, but has an epidemiological link to a confirmed or probable case. **Confirmed case**: A clinically compatible case that is laboratory confirmed.
Trachoma	**Suspected case**: any patient with red sticky eyes who complains of pain and itchiness of the eyes. **Confirmed case**: a suspected case in whom eye examination confirms one of the stages of Chlamydia trachomatis infection according to the **WHO Simplified Trachoma Grading System**.
Trypanosomiasis	**Suspected Case:** ***Early age***: A painful chancre originating as a papule and then evolving into a nodule at the primary fly bite site. There may be fever, intense headache, insomnia, painless lymphadenopathy, anaemia, local oedema and rash. ***Late stage***: Cachexia, somnolence, and central nervous system signs **Confirmed case**: a suspected case confirmed by card agglutination trypanosomal test (CATT) or by isolation of trypanosomes in blood lymph nodes or cerebrospinal fluid.
Tuberculosis	**Suspected case**: any person with a cough of at least 3 weeks. **Confirmed case:** ***Smear-positive pulmonary TB:*** (a) a suspected patient with at least 2 sputum specimens positive for acid-fast bacilli (AFB), or (b) one sputum specimen positive for AFB by microscopy and radiographic abnormalities consistent with active PTB as determined by the physician, **OR** (c) one positive sputum smear by microscopy and one sputum specimen positive on culture for AFB (d) Smear negative PTB: a patient who fulfils all the following criteria: (e) two sets taken at least 2 weeks apart of at least two sputum specimens negative for AFB on microscopy, radiographic abnormalities consistent with PTB and a lack of clinical response despite one week of a broad-spectrum antibiotic, a decision by a physician to treat with a full course of anti-TB chemotherapy,

	(f) a patient who is severely ill, at least two sputum specimens negative for AFB by microscopy, radiographic abnormalities consistent with extensive pulmonary TB (interstitial and miliary), a decision by a physician to treat with a full course of anti-TB chemotherapy, **OR** (g) a patient whose initial sputum smears were negative, who had sputum sent for culture initially, and whose subsequent sputum culture result is positive.
Typhoid Fever	**Suspected case**: Any person with gradual onset of steadily increasing and then persistently high fever, chills, malaise, headache, sore throat, cough, and, sometimes, abdominal pain and constipation or diarrhoea. **Confirmed case**: suspected case confirmed by isolation of Salmonella typhi from blood, bone marrow, bowel fluid or stool.
West Nile Fever	**Suspected case**: a hospitalized case of encephalitis due to unknown cause **Confirmed case**: Confirmation of West Nile Fever is through laboratory diagnostics to identify WNV-specific IgM
Yaws and Endemic Syphilis or Bejel	**Suspected case**: a person with a history of residence in an endemic area (past or present) who presents with clinically active (visible) yaws lesions **Confirmed case**: a suspected case with a positive serological test (rapid treponemal test for syphilis confirmed by DPP test) **Imported case**: a person who presents with clinically active yaws serologically confirmed in an area where yaws is not known to be endemic **Index case**: first case of yaws which is detected in a community **Contact of a case:** a person who has close, frequent contact with the infected person. A contact for the purpose of yaws eradication is the household, classmates or close playmates as identified by the contact
Yellow fever	**Suspected case** Any person with acute onset of fever, with jaundice appearing within 14 days of onset of the first symptoms. **Probable case:** A suspected case with positive IgM (confirmed by the national reference laboratory of the African Region) **Confirmed case:** A probable case + positive seroneutralization (confirmed by the national reference laboratory of the African Region)
Zika Virus Infection	**Suspected Case:** A person presenting with rash and/or fever and at least one of the following signs or symptoms: arthralgia, arthritis; or conjunctivitis (non-purulent/hyperaemic). **Probable case**: A suspected case with presence of IgM antibody against Zika virus and an epidemiological link (with no evidence of infection with other flaviviruses). **Confirmed case**: a person with laboratory confirmation of recent Zika virus infection, presence of Zika virus RNA or antigen in serum or other specimens (e.g., saliva, urine, tissue, whole blood); or IgM antibody against Zika virus positive (ELISA available on the market)

NB: These case definitions may change based on new knowledge.

APPENDIX 1B: Community level; case definitions using key signs and symptoms

Inform community leaders, community health workers, traditional healers, traditional birth attendants and health workers on the field in remote areas about priority diseases and conditions targeted for surveillance in your region. For community case definitions, use the main signs and symptoms describing them with simple words easily understandable than the IDSR case definitions in HF. Here are some examples of case definitions to be used to help the community in recognising cases of disease and referring the persons presenting the signs described for treatment and notify the health facility.

How to describe the main signs and symptoms in case definitions at community level - Some examples	
Acute Flaccid Paralysis (AFP)	Any child under 15 years of age with sudden weakness or unable to use his hands or legs.
Acute Watery Diarrhoea	Any person who passed at least three watery stools in the past 24 hours
Acute Haemorrhagic Fever Syndrome	Any person with unexplained disease accompanied by fever and bleeding or any person who died from a severe unexplained disease accompanied by fever and bleeding
Adverse Events Following Immunization (AEFI)	Unusual manifestation following immunization
Severe Diarrhoea in Children under 5 years of age	Any child aged below five years who passed at least three loose or watery stool in the previous 24 hours, with or without dehydration.
Diarrhoea with Blood (Dysentery)	Any person with diarrhoea, abdominal pain and visible blood in stool.
Dracunculiasis (Guinea worm Disease)	Any person with a skin lesion living in a Guinea worm endemic region, and Guinea worm emergence
Hepatitis	Any person with fever and yellow eyes
Animal bites (potential rabies)	Any person bitten by a dog or other mammals
Influenza-like illness	Any person with fever, cough, sore throat or flu
Leprosy	Any person with skin lesions and loss of sensitivity

Malaria	**[In an endemic country]:** any person with fever or who had fever in the previous 24 hours or with pale (white) palms in young children **[In a non endemic area]:** any person who was exposed to mosquito bites and had fever in the previous three days
Measles	Any person with fever and rash
Meningitis	Any person with fever and stiff neck
Maternal deaths	Death of a woman during pregnancy or within 42 days after childbirth
Neonatal death	Death of a newborn occurring before 28 full days of life
Neonatal tetanus	Any newborn normal at birth, then after two days of life becomes stiff, cannot suck, has convulsions or spasms
Onchocerciasis	In endemic areas, any person with fibrous nodules in subcutaneous tissues.
Plague	Any person with painful swelling under the armpits or in the groin area. In a region where plague is rife, any person with cough, chest pain and fever
Pneumonia	Any child below five years of age with cough, shortness of breath or difficulty in breathing
Human Rabies	Any person suffering from anxiety, fever, malaise and abnormal sensations most often at the wound site where the person was bitten by an animal Frequent symptom are agitation and fear of water.
Sexually Transmitted Infections (STI)	Any man or woman with urethral or vaginal discharge respectively, or genital pain or ulcer.
Tuberculosis	Any person having cough for at least two weeks or more
Typhoid Fever	Any person with persistent fever for at least three weeks
Viral Haemorrhagic Fever	Any person with fever and at least two other symptoms (headache, vomiting, yellow eyes, diarrhoea, fatigue), or who died after a severe illness accompanied by fever or bleeding.
Yellow Fever	Any person with fever and at least two other symptoms (headache, vomiting, yellow eyes, diarrhoea, fatigue), or who died after a severe illness accompanied by fever or bleeding.

Unusual health events	• Two persons or more with a similar severe disease in the same environment (household, workplace, school, street, for instance) in one week; • two persons or more who die in the same community during the same week; • Increased number of diseases or deaths in animals, including poultry during one week; • Any human or animal death following exposure to animals or animal products, including poultry (e.g., Consumption or handling); • Any person who was bitten, scratched or whose wound was licked by a dog or other animal; • Two persons or more with watery stool or who vomit after eating or drinking in a specific place (for e.g., a wedding party, funerals, festival, canteen, street food vendors, etc.); • An unexpected number of children absent from school due to the same disease; • Any community event causing anxiety in the population

APPENDIX 1C: Guide for the establishment of event-based surveillance at national, regional, district and HFlevels

Event-based surveillance is the organized and timely collection of information about events that may represent a risk to public health. Information is first collected in the form of signals that are considered by the early warning and response system as alerts representing a potential acute risk to human health, such as an outbreak. Not all signals may become events and therefore all must be triaged, verified and evaluated before the response is launched.

Event-based surveillance allows for early detection of events and thus timely intervention. It is therefore mandatory that all countries strive to implement event-based surveillance in addition to indicator-based surveillance at all levels of the health system, i.e., national, regional, district, facility and community levels.

The following is a description of the steps involved in implementing event-based surveillance.

Stages in the implementation of event-based surveillance at the national and regional levels (National or regional call centres and media watch teams)

Stage 1: Set up a hotline and a media watch system

This stage comprises two main activities as described below:

A. Establish a hotline

- a hotline is a telephone line that the public can use to obtain information from or provide information to an organization. It is a short number to receive direct phone calls or information from social networks such as WhatsApp, Facebook or Twitter.
- it should be free of charge (the cost of notifying public health authorities of alerts should be zero).

- it is recommended to have a single number easy to remember that can be used as telephone hotline for notifications. The same number can be used for support calls, text messages and social networks to avoid confusion. For example, if the hotline number is 499, the same number should be used to send SMS or messages through Facebook Messenger.
- community members should be motivated to self-report events that may affect population health, including emerging public health events or outbreaks.
- disseminate the hotline number through advocacy, health authorities, community health workers, NGOs, religious and other leaders, or schools, and advertise in local languages on television, radio, and in newspapers.
- develop a partnership with communications companies that can disseminate the hotline number through text messages to their customers. Messages sent should include the purpose of event-based surveillance, the importance of immediate notification, and the notification circuit.
- form a team to run the 24/7 event-based surveillance hotline to answer calls or request information from the community.

Call Methodology

- The responder should begin by thanking the caller for taking the initiative to report potential public health events of concern.
- Next, the employee must ask a series of prepared questions, modelled on the questions asked in the signal logbook.
- The call should be ended by thanking the caller for their availability, patience and proactive attitude.
- The employee must record directly in the appropriate logbook those signals that match the predefined list of signals.
- Informants should be called back as soon as possible in case a call is interrupted or disconnected, or if calls are made while the person answering is busy; this will ensure that all siganls are collected.

Messaging Methodology

- Once a message is received via SMS or social networks, an automatic instant message should greet the sender, thanking him/her and indicating that an operator will contact him/her.
- Questions or autoresponders can be used to collect information from the sender.
- The data should be recorded directly in the signal register according to the predefined list of signals in the country.
- Sender information should be collected for further communication and details of notified signals. A direct call to the sender may be necessary if more information is required.

Note: A hotline should be set up at the national, regional, and district PHEOCCs.

B. Establish a media scanning center

- The media are channels of general communication within a population and serve as collection tools used to store and distribute information or data. Examples include newspapers, magazines, television, radio, newsletters and other printed media, as well as electronic or online sources.
- Media scanning is an active process that must be carried out using different means.
- It is recommended that the media scanning be carried out at the national level.
- Health care staff should be trained in regular media scanning, such as daily analysis.
- Media scanning sources can be both official and non-official.
    - Official media sources:
        - NB: signals detected from official sources are reliable and do not require further verification.

      Examples of official media sources are:

        - Public sector websites, including those of the Ministries of Health, Agriculture, Environment, Foreign Affairs, etc.
        - Websites of official organizations such as universities and internationally recognized research centres;

  - Official WHO early warning websites, e.g., WHO Event Information Site for IHR National Focal Points, a secure platform accessible only to National Focal Points;
  - WHO Outbreak Information Bulletin;
  - Websites of WHO regional offices, e.g., AFRO, EMRO, EURO, SEARO, WPRO, PAHO;
  - Disease-specific websites, such as the Global Influenza Surveillance and Response System.

- Non official media sources:
  - NB: signals detected by these sources are not reliable and must be verified.

  Examples of non-official media sources are:

  - newspapers and magazines;
  - online content from TV and radio stations;
  - social media for example Facebook, Twitter etc.;
  - non-official websites such as ProMED, the Global Public Health Information network, HealthMap, MEDISYS, etc.

Online media scanning methods

The scanning of online information can be done manually and automatically.

Manual scanning stages

- make a checklist for periodic review of online sources (e.g., daily);
- make a list of priority signals regarding the country's strategies, capacities and resources;
- make a list of keywords related to prioritysignals, including diseases, syndromes or public health events;
- visit all pre-determined websites in the online source checklist to search for keywords.

Automated analysis

(a) there are many automated technology tools that can be used to analyse online information from predefined sources.

(b) these tools can help save time and effort, and promote early detection of public health threats.

(c) examples of automated analysis are:

- news feeds (or RSS feeds) are standard software tools that track predefined websites and inform the user with updates;
- contributory sources are based on information sharing between health professionals, in which individuals collect information accessible through shared feeds, e.g., ProMed;
- automated information feeds or services developed by governments or international organizations that collect health information from multiple sources and thus can reduce the time spent searching for individual sources. They are data aggregators.

Stage 2: Alert detection

(a) alert detection is the process of gathering information about potential public health events reported to the hotline.

(b) the public can contact the service hotline by phone, SMS, instant messaging on social networks or online chat.

(c) the hotline service team must filter the notifications received from callers to determine which signals are valid.

(d) a list of signals prepared by national public health authorities should be provided to hotline operators or public responders so that they can continue to record signals.

(e) the operator shall record valid alerts in a register provided for that purpose.

(f) signals can also be detected by manual or automated media scanning.

Examples of pre-determined signals:

Code	Signals to be notified
01	Two or more people with a similar serious illness in the same setting (Household, workplace, school, street, for example) in one week
02	Large number of unexplained deaths of poultry, livestock, other domestic animals or wildlife

03	Severe illness of a health worker after exposure to patients with similar symptoms
04	One or more hospitalized patients with unexplained severe illness, including lack of response to standard treatment.

Stage 3: Signals recording

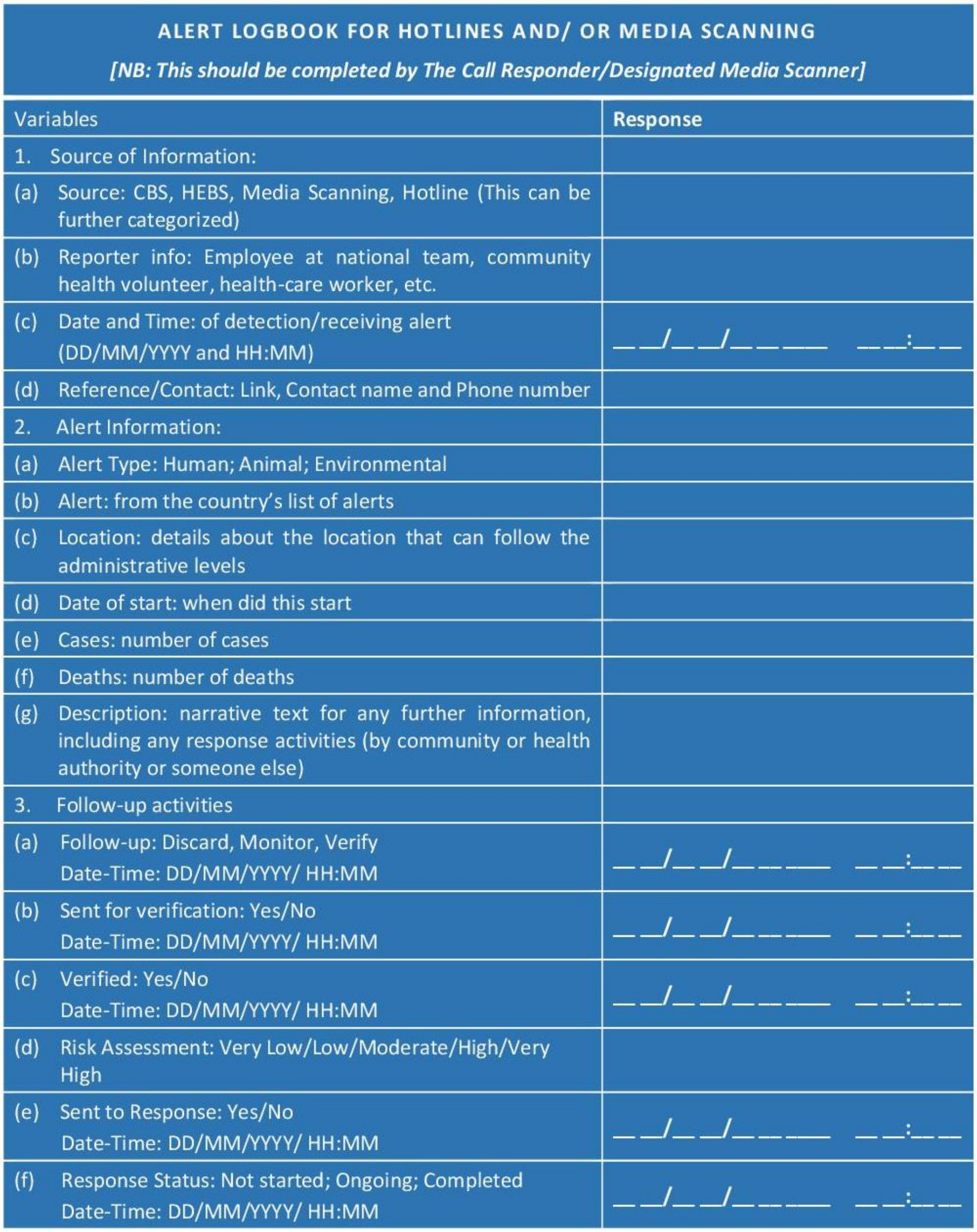
(a) signals that are collected from media and hotline, and that match the predefined list of signals, must be recorded in a signal logbook. See a sample signal logbook for hotline or media scanning on the next page.

(b) each signal collected should include data on the detection, sorting, and verification of the signal through to response.

(c) signal recording must include the minimum data set for signal monitoring, such as:

- source or informant: name, phone number of contact person, date and time of call or detection;
- signal: when it happened, who was affected (cases, deaths) and where it starts and spreads;
- signal monitoring: sorting, verification, risk assessment and response.

Sample signal logbook for hotline or media scanning



ALERT LOGBOOK FOR HOTLINES AND/ OR MEDIA SCANNING

[NB: This should be completed by The Call Responder/Designated Media Scanner]

Variables	**Response**
1. Source of Information:	
(a) Source: CBS, HEBS, Media Scanning, Hotline (This can be further categorized)	
(b) Reporter info: Employee at national team, community health volunteer, health-care worker, etc.	
(c) Date and Time: of detection/receiving alert (DD/MM/YYYY and HH:MM)	__ __/__ __/__ __ __ __ __ __:__ __
(d) Reference/Contact: Link, Contact name and Phone number	
2. Alert Information:	
(a) Alert Type: Human; Animal; Environmental	
(b) Alert: from the country's list of alerts	
(c) Location: details about the location that can follow the administrative levels	
(d) Date of start: when did this start	
(e) Cases: number of cases	
(f) Deaths: number of deaths	
(g) Description: narrative text for any further information, including any response activities (by community or health authority or someone else)	
3. Follow-up activities	
(a) Follow-up: Discard, Monitor, Verify Date-Time: DD/MM/YYYY/ HH:MM	__ __/__ __/__ __ __ __ __ __:__ __
(b) Sent for verification: Yes/No Date-Time: DD/MM/YYYY/ HH:MM	__ __/__ __/__ __ __ __ __ __:__ __
(c) Verified: Yes/No Date-Time: DD/MM/YYYY/ HH:MM	__ __/__ __/__ __ __ __ __ __:__ __
(d) Risk Assessment: Very Low/Low/Moderate/High/Very High	
(e) Sent to Response: Yes/No Date-Time: DD/MM/YYYY/ HH:MM	__ __/__ __/__ __ __ __ __ __:__ __
(f) Response Status: Not started; Ongoing; Completed Date-Time: DD/MM/YYYY/ HH:MM	__ __/__ __/__ __ __ __ __ __:__ __

Stage 4: Triage

Perform signal evaluation for verification

(a)If the signal matches one of the priority signals for the country and has not been previously reported, it should be verified immediately.

(b)If the signal is defined broadly, such as an unusual event that may pose a threat to public health, a trained public health specialist or team leader should assess it to decide whether to reject it or proceed with verification.

Stage 5: Verification

(a) Verification is an essential step in confirming the validity of the signal collected and should be performed by experts in the field, such as public health specialists.

(b) Verification should be done at the local level closest to the location of the signal.

(c) If the signal is detected at the national level, it is notified to the focal point in the region (regional health team) where it was detected, by phone call, SMS, e-mail, etc.

(d) The regional health team then notifies the relevant district management team.

(e) A trained HD team with support from regional or national experts should perform signal verification.

(f) All signals must be verified within 24 hours.

(g) Once a signal is verified, and requires action, it is considered an event.

(h) The HD team must carry out a risk assessment of this event in order to define the actions to be taken, and with support from regional or national experts, should promptly begin investigations (when necessary) by collecting more information in the field (physical tests, laboratory specimens, etc.) using existing case or event investigation forms from the IDSR.

(i) Confirmed events that meet the standard case definition should be recorded by the relevant HD team in the indicator-based surveillance system and reported to the next level of the health system, i.e., using existing data collection tools and following the IDSR reporting procedures (***see Section 2 of the IDSR Technical Guide, Third Edition***).

Stage 6: Risk assessment and characterization

(a) Once a signal is verified and confirmed as an event, risk assessment begins.

- *Risk assessment is a systematic and ongoing process of collecting, evaluating, and documenting information that serves as the basis for actions to manage and reduce the negative consequences of an acute public health event.*

(b) The first risk assessment of an event should occur within 48 hours of the detection of one or more signals.

(c) The national team should lead the risk assessment with the appropriate regional and district teams.

(d) Each assessment is a process by which available information about an actual event is analysed and a judgment is made as to whether the event poses an immediate risk to public health. In this case, a full risk assessment is conducted ***(see Section 4 of the Third Edition of the Technical Guide for the IDSR).***

- For signals that have been proven to be true, but do not represent an immediate threat to the population, the team should monitor the event and undertake risk assessments as new information becomes available.

Stages for implementing event-based surveillance at the district level

(a) the stages for implementing event-based surveillance at the district level are those to implement at the community and fealth facility levels.

(b) It involves setting up a network of key informants who will interact with community health workers at the community level or with the surveillance focal point of the health facility at the hospital level.

(c) Signals are detected by community members and sensitized key informants, as well as by CHWs who notify the SFP of the health area either directly or indirectly through the SFP of the nearest health facility. This notification can be done through logbooks, electronic forms or by phone.

(d) Once the signal is notified, it is recorded and processed by the health area SFP and only the alerts transformed into events are sent to the health district level and more or less integrated into the IDSR

(e) Record oral or written information from HF and communities about suspected outbreaks, rumours, unexplained events or signals transformed into events in the district's register of suspected outbreaks ***(see Section 4, Appendix 4A of the Third Edition of the Technical Guide for IDSR).***

(f) the health area surveillance focal point should perform the following functions: recording, triaging, and verification of signals / alerts transformed in event within 24 hours.

(g) The health district management team must conduct a risk assessment of the notified events within a timeframe of 24h.

(h) Signal / alert triaging

- when the focal surveillance point of the health area / health district receives information about a reported signal, it should triage by asking the following questions:
    - does the information provided justify an early warning? (In other words, could this signal be a real public health event?
    - has this signal already been reported? (In other words, is it a duplicate?)
- triaging can be personnally done in the field, by text message, or by phone.
- after triaging:
    - if the notification is not relevant or is a duplicate, it may be rejected. No further action is required.
    - if the signal is to be rejected, communicate the following information to the health facility surveillance focal points who notified the signal:
    - that they continue to monitor the situation and inform in case the situation changes and a threshold is crossed;
    - if they reported an alert that turned out to be false, it is okay; they should continue to report alerts when they detect them.
    - If the signal is relevant and not duplicated, it should be verified by the surveillance focal point that received the information.

- The district management team shall ensure that the signal registers are properly maintained and completed and that the signals notified are traceable.

(i) verify signals

- the HA /HD surveillance focal point should verify all triaged signals that fall under event-based surveillance. Before they be considered as events.
- verification involves determining whether a signal is valid (i.e., not a false alarm or rumour), reliable, and matches at least one of the predefined signals for event-based surveillance implementation.
- verification criteria may include asking questions to those who reported the signal to ensure that they understood it, that the signal was confirmed or not confirmed by at least two different sources, or that it was reported by a person with medical authority (e.g., a veterinarian, physician, or laboratory assistant).
- to conduct the verification, the HA /HD surveillance focal point will ask questions to the person who reported the signal, and possibly to other persons. This may include the patient, his or her family and friends, or other community members.
- verification can be done in person, by site visit, or by telephone.
- use the event-based surveillance verification tool; see an example on the next page.
- the result of the verification is confirmation wether the signal is true or false. Once the signal is verified and found valid and of public health interest, it becomes an event.
  - after verification: If the signal is considered a public health event, it is immediately notified to the health distric management team, which will conduct a risk assessment of this event.
  - if the signal is not considered a public health event, the situation will be monitored to ensure that it does not become one.
  - record confirmed events in existing IDSR data collection tools and platforms and notify the next level ***(see Section 2 of the Third Edition of the IDSR Technical Guide).***

(j) perform a risk assessment in compliance with national guidance.

The District Management Team must carry out a risk assessment of all events notified by the lower level. This assessment can be done with the support of the regional and/or national level. This assessment should lead to the characterization of the risk and define the course of action. A possible response plan must be drawn up if the risk is considerable. After the risk assessment, any event posing a significant risk (moderate to very high) should be immediately notified to the higher level.

Example of event-based monitoring: verification tool

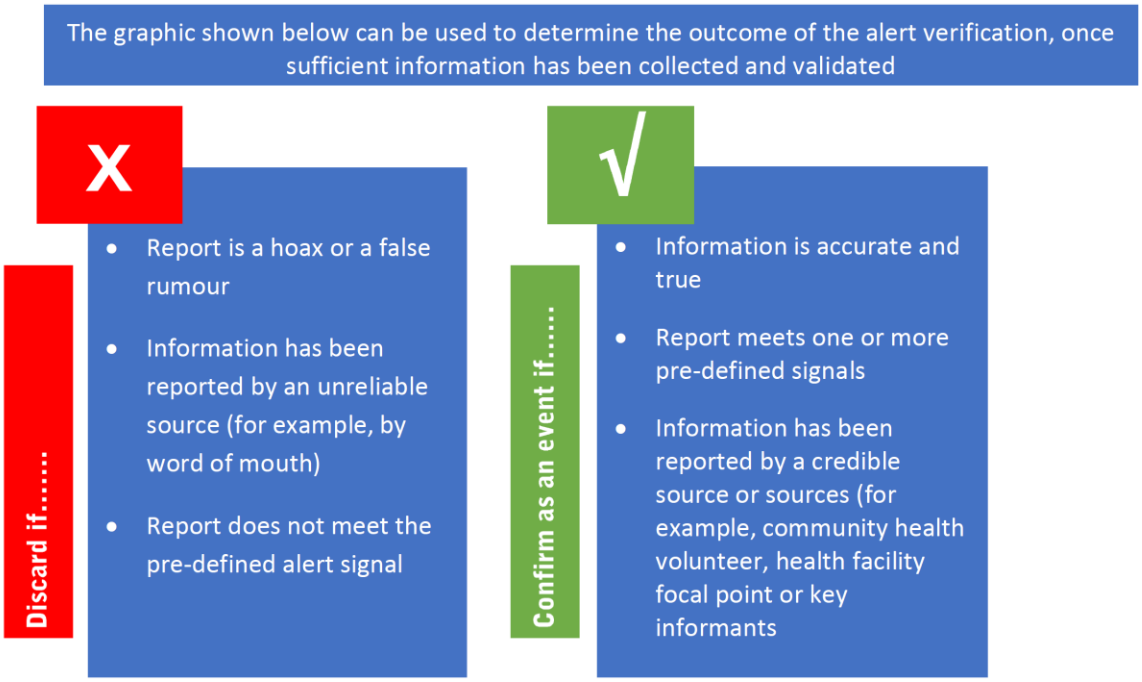
When a signal is reported by a community surveillance focal persont or health facility, the district management team should use this tool to verify whether the signal is TRUE or FALSE.

The signal verification process should answer three main questions:

(a) is the notification accurate (true)?

(b) was the information reported by a reliable source(s)?

(c) does the notification meet the criteria for one or more signals?



The graphic shown below can be used to determine the outcome of the alert verification, once sufficient information has been collected and validated

X

Discard if……

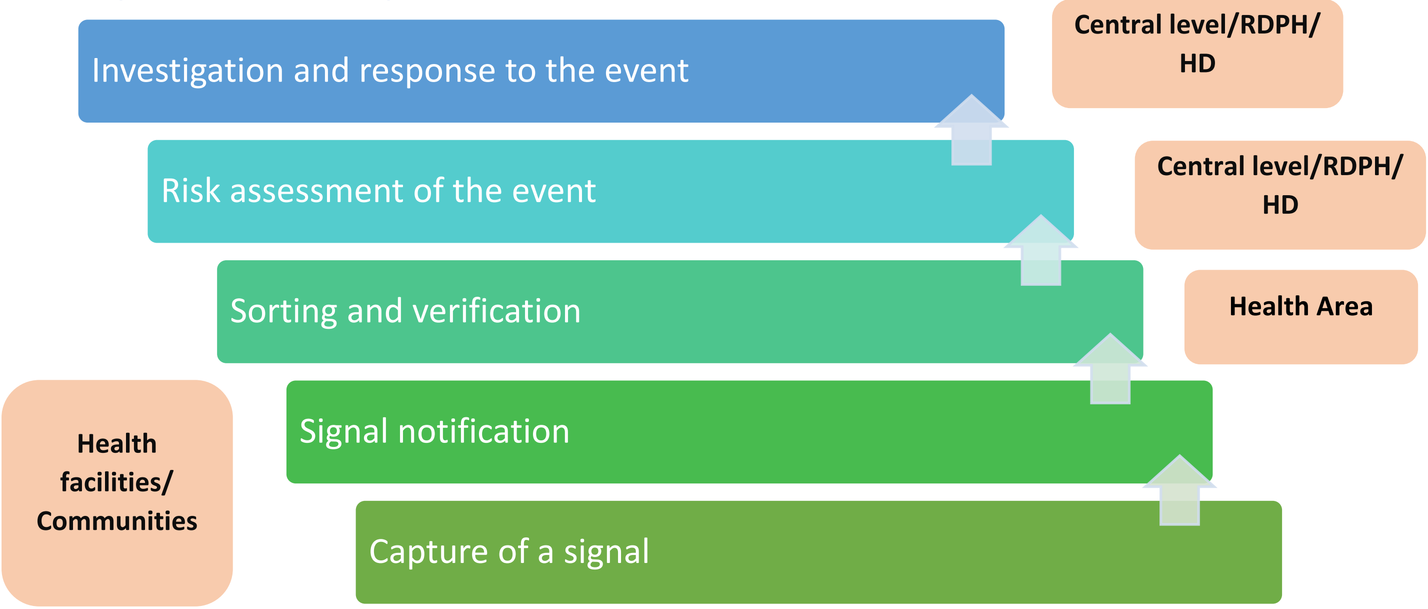
- Report is a hoax or a false rumour
- Information has been reported by an unreliable source (for example, by word of mouth)
- Report does not meet the pre-defined alert signal

√

Confirm as an event if……

- Information is accurate and true
- Report meets one or more pre-defined signals
- Information has been reported by a credible source or sources (for example, community health volunteer, health facility focal point or key informants

Stages of EBS reporting and role at each level


Investigation and response to the event
Central level/RDPH/ HD
Risk assessment of the event
Central level/RDPH/ HD
Sorting and verification
Health Area
Signal notification
Health facilities/ Communities
Capture of a signal

III. Stages of Event-Based Surveillance at the HFLevel

The stages involve consideration of the following important elements:

(a) indicator-based surveillance at the HFlevel includes immediate, weekly, or monthly reporting of a predetermined list of diseases based on case definitions.

(b) event-based surveillance in HF train clinicians, nurses, and other relevant health professionals to report signals of disease patterns, such as a cluster of diseases, and is not disease-specific.

(c) event-based surveillance has the potential to detect emerging or re-emerging public health threats because it is not disease-specific, requires immediate notification, and is highly sensitive and broad.

(d) in addition, because reporting does not require laboratory results and relies on the experience of clinicians, event-based surveillance is more practical and relatively simple to implement and maintain.

(e) HF should participate in both indicator-based and event-based surveillance, as the two complement each other and allow for early detection of diseases, conditions and public health events.

Stages of Event-Based Surveillance at the HFLevel

Stage 1: Signal detection

(a) identify and train facility-based event-based surveillance focal persons: Facility-based surveillance focal persons can be trained for this role.

(b) facility-based event-based surveillance focal persons should ask other staff to inform them immediately when they see or hear of a signal in their workplace.

(c) health professionals, including clinicians, nurses, infection control workers, security/reception and mortuary staff, if any should be made aware of how to recognize signals and report them immediately to the HF surveillance focal point.

(d) Detecting a signal means identifying or suspecting the presence of a signalt predetermined by national public health authorities.

Examples of event-based surveillance signals in HF:

Code	Notifiable event-based surveillance signals at HF
01	Any serious illness observed in health personnel after they have treated a patient with a similar illness
02	Sudden significant increase of inpatients for any serious illness of the same type
03	Any serious, unusual and unexplainable illness, including lack of response to standard treatment
04	Increased use of a particular medicine.

Stage 2: Signal notification

(a) notification of signals involves contacting one or more event-based surveillance focal persons in the HF who intend to notify the district team immediately.

(b) this notification can be done by phone, SMS, or in person, but it must be done immediately: the same day and as soon as possible.

Stage 3: triage and verification

(a) upon receiving notification of signals, the district management team should triage and verify all signals within 24 hours of detection using the verification tool.

(b) In case of an actual event, investigations and response measures are immediately implemented according to the existing IDSR structures.

(c) the district management team should keep the reporting HF regularly informed of the situation.

APPENDIX 1D: List of district notification sites

Note the contact information for health personnel, community health workers, or point-of-entry workers who provide surveillance and detection data on outbreaks and public health events to the district. Please include, for example, the names and contact information of community health workers, skilled birth attendants, community leaders, and public safety officials. This list should be updated regularly, adding new sites and removing those that are no longer in service or not participating.

Example:

Name of the HFor patient contact point with the health department	Address or location of the facility or point of contact	Name of designated focal person for surveillance and response	Telephone or email address (or other contact information)
Lima Health Center	*PO Box 123* *lima area*	*Dr. Moyo*	*Tel.: 123-458 or send a message through the daily railway contact with Mlima station*

APPENDIX 1E: Laboratory functions at various levels of the health system

Laboratory functions at various levels of the health system			
Level	Collect	Confirm	Report
HEALTH FACILITY	Use standard case definitions to determine when to collect specimens; Help the front-line laboratory to collect specimens as per agreed guidelines; Document specimens with clinical history; Transport specimens to the front-line laboratory or referral laboratory, in compliance with the agreed guidelines, along with the IDSR case-based reporting form by the laboratory.	Use standard case definitions to start or request appropriate analyses for disease confirmation; Handle specimens according to the standardized operating rules and approved guidelines;	Note down data on specimen collection and transportation; Receive laboratory results and provide feedback.
DISTRICT OR REGION	Communicate the sampling principles and procedures to healthcare providers; Ask for additional sampling material, where necessary; Store specimens in appropriate conditions while waiting for shipment or further analyses; Request for additional specimens, where necessary, with regard to disease investigation; Organise the transportation of specimens to the front-line laboratory	Perform laboratory tests for presumptive diagnosis, if necessary; Store representative specimens for transportation according to guidelines; Conduct systematic analysis of laboratory results; Observe changes in trend during systematic analysis of laboratory results;	Record, store and save laboratory results and details of laboratory tests, including all the tests conducted and the opportunity to test; Communicate results to the health personnel and patients; Ensure to receive laboratory results regularly at the national level; Update the list of laboratory results and follow up any missing result at the laboratory which did the test; Notify results and information on delays to the next level; Report any changes in trend during systematic analysis of laboratory results to the national level;

	or referral laboratory, in compliance with the agreed guidelines, along with the IDSR case-based reporting form by the laboratory.		Use information systems to investigate outbreaks;
National referral laboratories (some laboratories operate both as front-line and referral laboratories);	Establish guidelines, policies and procedures for specimen collection with national authorities; Distribute the appropriate specimen collection and transportation kits for epidemic-prone diseases; Ask the laboratory or healthcare providers for additional specimens depending on the needs; Store specimens in appropriate conditions while waiting for shipment or further analyses;	Establish principles and procedures for specimen collection with national authorities; Perform appropriate confirmatory laboratory analyses; • Microscopic, culture, antibiogramme, serotyping, serological test, molecular detection and identification, genomic sequencing. Store representative isolates of the outbreak, where necessary.	Record, store and save laboratory results and details of laboratory tests, including all the tests conducted and the opportunity to test; Notify results to regional or district health teams and to all stakeholders at the national, regional or district levels for transmission to the requesting HFor laboratory; Report data on cases and summary data in conformity with the agreed protocol; Report laboratory results on testing of sentinel populations living in target sites; Conduct systematic review of tests, data and laboratory results and scrutinize trend changes.
Global reference	Establish guidelines, policies and procedures for specimen collection with national authorities; Request additional sampling, where necessary.	Conduct further tests on specimens or isolates sent, where necessary.	Record, store and save laboratory results and details of laboratory tests, including all the tests conducted and the opportunity to test; Report laboratory results to the national reference laboratory or to the national laboratory Coordination team in charge of upcoming dissemination

APPENDIX 1F: Responsibilities of laboratory focal persons at all levels

Laboratory focal person at the national level

(a) coordinate all laboratory activities in support of disease preparedness, surveillance and response;

(b) establish and support collaboration with epidemiologists or surveillance officers;

(c) define laboratory testing capabilities in-country and those that serve as international referrals and share this information with all stakeholders;

(d) keep a list of laboratories that perform the required analyses;

(e) keep and update the inventory list of supplies, reagents and equipment for all laboratories;

(f) establish and coordinate agreements with international laboratories for the provision of diagnostics or confirmation of priority diseases not yet available in the country;

(g) support the laboratory through advocacy at higher levels to ensure that it has access to the infrastructure, equipment, and supplies needed to collect, handle, test, store, and safely ship specimens;

(h) ensure that there is a framework for transporting specimens in and out of the country to facilitate the shipment of specimens;

(i) ensure that laboratory results are communicated timely to all relevant stakeholders and used appropriately to inform public health measures and clinical management of patients;

(j) ensure that laboratory results are recorded appropriately;

(k) ensure that laboratories have a quality assurance program to improve the reliability and reproducibility of laboratory results.

Regional Laboratory Focal Person

(a) keep a list of laboratories that will perform the required analyses;

(b) provide information to all HF for the appropriate transportation of specimens;

(c) keep and update the inventory list of supplies, reagents and equipment for all laboratories in the region;

(d) ensure that laboratory confirmation procedures established at the national level are known and followed in the region and districts;

(e) ensure that specimen collection, transport equipment, and laboratory diagnostic tests are available for rapid detection of priority diseases;

(f) coordinate with the HF and the laboratory for the collection, safe packaging and reliable transportation of the appropriate specimen for confirmation of the suspected case;

(g) receive laboratory results and communicate them promptly, according to national procedures, to all those who need them for public health action and clinical patient care;

(h) ensure that laboratory results are recorded appropriately;

(i) communicate with reference and national laboratory focal persons as needed;

(j) ensure that laboratories have a quality assurance program to improve the reliability and reproducibility of laboratory results.

District Laboratory Focal Person

(a) establish or strengthen systematic communication with identified laboratories that receive specimens and the HF or districts that send them;

(b) keep and update the stock list of supplies, reagents and equipment for all the district HF and laboratories;

(c) ensure clear and reliable procedures for both specimen collection and shipment, confirmation of disease/conditions and reporting of results;

(d) communicate with the regional laboratory focal person;

(e) communicate with the national reference laboratory, if applicable;

(f) ensure that laboratory results are recorded appropriately;

(g) ensure that laboratories have a quality assurance program to improve the reliability and reproducibility of their results.

Laboratory Focal Person of the health facility

(a) keep and update the stock list of supplies, reagents and HFequipment;

(b) ensure availability and adherence to standard operating procedures for both sample collection and shipment, confirmation of disease cases and reporting of results;

(c) communicate with district and national laboratory focal persons as needed;

(d) ensure that laboratory results are recorded appropriately;

(e) ensure that laboratories have a quality (internal and external quality control) assurance program to improve the reliability and reproducibility of their results.

APPENDIX 1G: List of national laboratories responsible for confirming priority diseases and conditions

Periodically update the list of laboratories in your district or those designated at the national level to confirm priority diseases and conditions. Include on this list the person to contact for assistance. The list below is only an example.

Example:

Priority diseases, conditions and public health events	Focal person, laboratory name, address, phone number, e-mail address
Poliomyelitis	***Example: John Zimbe, National Laboratory, 145 Kenyatta Road, Pretoria, South Africa, 234-701342555***
Cholera	
HIV	
Tuberculosis	
Measles	
Plague	
Human influenza caused by a new subtype	
Rift Valley Fever	
Dengue fever	
Public health events of national or international concern.	
Anthrax	
Chikungunya	
Typhoid fever	

Section 2: Reporting priority diseases, conditions and public health events

This section describes how to

- report immediately reportable diseases, conditions and public health events;
- transmit weekly and monthly summary data on priority diseases, conditions and public health events;
- improve regular reporting practices.

Introduction

Integrated Disease Surveillance and Response (IDSR) is a system with the potential to ensure a reliable supply of epidemiological information to the national level in order to fulfil IHR (2005) requirements. Ensuring with reliable reporting of surveillance data throughout the system is important. Reliable reporting provides information for surveillance officers, the national IHR focal point, competent authority at Point of Entry (PoE) the WHO contact point and other health staff to:

- identify emerging problems or conditions and plan appropriate responses, including informing relevant staff or levels.
- take action in a timely way.
- monitor disease trends in the area.
- evaluate the effectiveness of the response

This section describes how to report priority diseases, conditions and public health events within the required timelines. In IDSR, data collection and data reporting follow different timelines for different purposes:

a) immediate reporting of case-based information allows for early detection of unexpected or highly pathogenic/lethal public health events. All the diseases and conditions under immediate reporting should also be reported under aggregated weekly report in the IDSR Weekly Summary Reporting Form.
b) weekly aggregated reporting provides data for monitoring trends of diseases, conditions or public health events to early detect outbreaks.
c) monthly aggregated reporting provides data for monitoring the health status of the population and impact of disease specific programs, and for planning allocation of resources.

National policy will determine reporting requirements and the public health events to be reported and whether data from the districts and HF are to be reported immediately, weekly or monthly. Recommended frequency will depend on the specific disease control activities in the country. Diseases targeted by the IDSR are public health priorities requiring rapid response. HDs should also detect and report unusual events that may affect human health.

Paper-based tools are the most commonly used tools for reporting these diseases, condition and public health events. While paper-based tools can provide timely information, countries should aim to have electronic tools to facilitate rapid transmission of data to enable timely response to public health threats (eIDSR). The potential benefits of using electronic reporting tools for eIDSR include more timely reporting, investigation, and response to outbreaks. Electronic reporting may also improve data quality; enhance virtual, near real-time disease and public health events monitoring capability; may lead to reduced system costs and easily generate automated signals. In addition, information can be more easily stored and accessed. See ***Section 9*** for electronic IDSR (e-IDSR) System Guide for countries.

The targeted public health workforce for IDSR are primarily staff at all levels of the health system (both human and animal), data management personnel who will oversee the Information Communication Technology (ICT) aspect of the system, supervisory and disease-specific program personnel, that is, district, regional and decision-makers at the national level. It is important that countries aim at having an interoperable approach of strengthening e-IDSR by creating systematic linkages and information-sharing platforms. This can be done by formalising agreements between the Ministries of Public Health (MoH), Agriculture (MINAGRI), Livestock (MINEPIA), Wildlife (MINFOF) and Environment (MINEPDED) Units, Nationl Health Information System (NHIS), IDSR, Maternal, Perinatal and Response (MPDSR), Cameroon Epidemiological Network for Animal Diseases (CENAD).

2.1. Immediately reportable priority diseases, conditions and public health events

Immediate reporting is indicated when an epidemic-prone disease or other potential Public Health Emergency of International Concern (PHEIC) is suspected or is otherwise required under the International Health Regulations (2005). Immediate reporting should also be done for diseases and events considered priorities at the national level which may not necessarily be PHEICs. The diseases, conditions and public health events requiring immediate reporting to the next level are listed in ***Table 2.*** Immediate reporting allows timely action to be taken to prevent the re-emergence or rapid transmission of epidemic-prone diseases or events or their propagation, especially those due to highly virulent infectious, chemical, biological or radio nuclear agents.

Information that is reported immediately, such as single cases or clusters of reportable events (See examples below), will generate a siganl and initiate a case-based reporting system. This means that, specific information about that suspected case, or, if it is a cluster, specific information of each of the cases identified, will be collected thoroughly and reported to the next level. At the same time, an initial investigation will be initiated. For events reported at PoE, information is reported to the next level (district in which the PoE is situated) as well as simultaneously to the IHR NFP. Reporting units with no diagnostic capacity will use the suspected case definition given to identify and report diseases, conditions and public health events. Additionally, information of contacts will be collected. ***Section 4*** describes how to conduct contact tracing and how to report contacts.

For conditions like maternal and perinatal deaths, the circumstances leading to the death need to be gathered and analysed and health providers should use the Maternal Perinatal Death Surveillance and Response (MPDSR) in consultation with the relevant focal persons.

In IDSR, there are two types of thresholds used to initiate response: an alert threshold and an outbreak threshold. These thresholds are normally expressed in terms of the number (or proportion) of cases of a disease and the critical point (threshold) beyond which action must be taken. Trained health-care personnel should always determine the alert and outbreak thresholds. Thresholds for alerts and outbreak for epidemic-prone diseases, conditions or public health events are shown in ***section 11***.

Please refer to ***Section 11*** for disease-specific information including surveillance case definitions, suspected events.

Table III*: Immediately reportable priority diseases, conditions and public health events (source: MAPE notification form, Cameroon, 2023)*

1. Acute haemorrhagic fever syndrome (Ebola, Marburg, Lassa, Rift Valley, Crimean-Congo fever) 2. post-vaccine adverse events (PVA) 3. Anthrax 4. Bacterial meningitis 5. Chikungunya 6. Cholera 7. Dengue fever 8. Bloody diarrhoea (shigellosis) 9. Dracunculiasis (Guinea worm disease) 10. Influenza due to a new viral subtype 11. Listeriosis 12. Maternal deaths 13. Measles 14. Middle East Respiratory Syndrome (MERS) 15. Simian orthopoxvirosis	16. Neonatal tetanus 17. Perinatal deaths 18. Leprosy 19. Poliomyelitis (acute flaccid paralysis) 20. Rabies (Humanà 21. SARI 22. SARS 23. Typhoid fever 25. Pian, endemic syphilis or bejel 26. Yellow fever 27. Zika virus disease 28. Group of unexplained human or animal/avian diseases/deaths*. 29. Any public health event of international concern (infection, zoonosis, foodborne infection, chemical or radioactive contamination or event of unknown origin)

1. Chikungunya 2. Cholera 3. COVID-19 and Severe Acute Respiratory Infections 4. Dengue 5. Dracunculose 6. Anthrax 7. Acute hemorrhagic fever* ** Ebola, Marburg, Rift Valley, Lassa, Crimean-Congo, West Nile, Dengue fever Zika virus disease*	1. Yellow fever 2. Bacterial meningitis 3. Poliomyelitis (acute flaccid paralysis) 4. Across 5. Human rabies 6. Measles 7. Neonatal tetanus 8. Monkeypox 9. Human smallpox

Examples of clusters can be:

- any cluster of illness or deaths among people living in the same community within a specific time period (for example, one week);
- cluster of deaths of animals within a specific time period (for example, one week);
- illness or death among people after exposure to animals;
- health-care worker illness after exposure to patients with similar illnesses;
- unexpected increase in admission to health care facilities of persons with similar severe symptoms;
- sudden illness in a person who has travelled out of the country in the past 14 days;
- any unusual illness or sudden death in the community within a specific time period (for example, one week);
- unexpected large numbers of children absent from school due to the same illness in the same seven-day period;
- unexpected large numbers of sales at pharmacies of many people buying medicines for the same kind of illness.

Note Ensure that adequate information is collected for events which are reported. Some of the events might have a link with the Agricultural or Livestock/Wildlife sector or Food or Environment or other sectors, ensure information is also sought from these sectors.

2.1.1. Reporting of events from community sources

Any suspected event occurring in the community, including maternal and neonatal events should be reported immediately. The trigger mechanisms of reporting must be clearly defined and the CHW must immediately notified the surveillance focal person of the health area or the head of facility. Minimum information collected should include:

- date of onset of event and date of report;
- suspected disease, condition, or event;
- what happened?
- when did this happen (day, month, year)?
- where did this happen (exact location, village, district, Region)?

- who is affected (age, gender, occupation, etc.)?
- how many have been affected?
- has anyone died? If yes, how many?
- is the event ongoing?
- are there any animal deaths/exposures?
- recent history of travel to an affected area;
- name and contact number of the person reporting;
- any action taken;
- other information you have.

See ***Appendix 2C*** for a reporting format when an event is identified, ***Appendix 2D*** for reporting format for community base surveillance and ***Appendix 2E*** for reporting structure for community alert and verification.

2.1.2. Case-based Reporting to the higher level

If an immediately reportable disease, condition or other public health event is suspected, the HFmust report case-based information to the next level within 24 hours. Information obtained through preliminary investigation of suspected or probable case includes:

- patient's geographical location;
- HFthat handled the patient or referred the patient;
- patient's identification and demographic information;
- Information about signs and symptoms, including date of onset, history of vaccination (where applicable) and information about any relevant risk factors including contacts;
- laboratory results (if available);
- history of travel;
- history of contacts (human or animal);

Any maternal or perinatal death, once it occurs, should also be reported immediately. A sample reporting form for both is given in ***Appendix K***.

In the event of a suspected disease or other immediately reportable public health event:

- make the initial report by the fastest means possible (telephone, text message, e-mail, walkie-talkie or paper). The HFshould contact the district health authority immediately and provide information about the patient;
- follow up the initial verbal report with a written report using a standardised case-based report form. A sample case-based reporting form for recording case-based information is in ***Appendix 2A*** at the end of this section. If a computer or other electronic device is available for surveillance or case management, complete and submit the form electronically to the next level;
- if a laboratory specimen is requested at this time, make sure that the patient's information on the specimen, the laboratory investigation form, and the case-based reporting form all match. Ensure proper packaging to ensure reliable results;
- ensure also that a copy of the case-based form accompanies the laboratory form. A sample laboratory form is included in ***Appendix 2B***;
- use forms to begin gathering initial information for case investigation. Disease-specific case-based reporting forms for particular diseases and conditions of concern (for example, cholera, VHF, maternal death, and MDR/XDR TB) are in the appendix at the end of ***Section 9***.

Note: *Some epidemic-prone diseases or conditions like Maternal or Perinatal deaths have specific reporting requirements, depending on national or regional policies. Please refer to disease-specific conditions and requirements in* ***Section 11*** *of this guide.*

- ensure that adequate information is available for events which are reported, as some events might have a link with the Agricultural or Livestock, Wildlife sector or Food or environment or other sectors including the community. Such information sharing is crucial and should start at the community level, HFand subsequently at the district and region. At the National level, the IHR (2005) Focal Point (FP) should notify WHO of an event that is a potential public health emergency of international concern (PHEIC) using the decision instrument in the IHR (2005) (***Appendix 2A***);
- for all events, establish a line listing of suspected cases or events or conditions reported as part of initial and ongoing investigation and ensure it is always updated, while at the same time maintaining the link with appropriate sectors, depending on a particular disease or event. The line list should be established in case of suspected outbreak at the level of the care/isolation unit

a consolidated by the different higher levels (SA, districts, Regions, Central). Refer to ***Appendix 4E*** for a line list sample;

- if a potential Public Health Emergency of International Concern (PHEIC) is detected at Point of Entry, immediate reporting should also be made to the higher level and a copy of the notification forwarded by the national IHR FP to the central level for evaluation using decision algorithm. For cases from endemic or risk areas for yellow fever, include information on vaccination status.

2.1.3. Weekly reporting of cases of disease or public health events

Weekly reporting provides data for monitoring trends of diseases or conditions to early detect outbreaks. It is important to ensure that the weekly reporting format is adhered to across all HF and districts to facilitate comparison within and between the facilities and districts.

After immediately reporting to the next level about instances of notifiable diseases, conditions or public health events, collect and report weekly summary information, as listed in ***Table 2***. See ***Appendix 2F*** for format for developing a weekly summary form which is an aggregate of case-based forms.

At the district level, weekly data analysis includes verification of the quality of the data coming from the reporting sites and the completeness and timeliness of these reports. The incidence and case-fatality rates should be compared with the set alert and outbreak thresholds. Outbreak curves should be updated and help for in-depth analysis of individual case data received from reporting sites. Laboratory data are analysed in the same manner as individual cases: for action-oriented analysis, quality of the specimens, the pathogens detected, their resistance to recommended drugs, the location and the individual characteristics (socio-demographic data) of cases sampled are of primary interest.

Districts which have computers are encouraged to store the information electronically and forward the surveillance data sets to the regional level pending the delivery of physical copies duly signed by appropriate staff.

With eIDSR (see ***section 9***), this will be updated automatically in the database, while in countries using paper-based reporting; this will be done manually and entered into a computer. This aggregation is important to understand the trend of the immediately reportable diseases and plan for effective intervention. For early detection of outbreaks via weekly aggregated reporting, it is recommended to keep the number of variables at a minimum, ideally reporting only the number of cases and deaths, to avoid unnecessary burden on the health care facilities and maximize reporting efficiency.

Based on epidemiological evidence, the countries may decide to include additional diseases, conditions and public health events in diseases for weekly reporting, for example, malaria, MDR-TB, diarrhoea with severe dehydration in children under five years of age, severe malnutrition, and neonatal deaths. Only diseases or conditions or public health events which could result in public health action should be considered for entry on the list of aggregated weekly reporting. Some rare but high-risk public health events should be removed from routine aggregated reporting to be reported immediately.

2.1.4. Monthly and quarterly reporting of summarized data for priority diseases, conditions and public health events

Some priority diseases are reported monthly or quarterly. Data from these diseases should be synthesized and reported to higher level, taking into account the frequency of reporting. These data are crucial for priority health programs and can be used in monitoring the progress of prevention and control activities and in detecting any emerging, unexplained or unusual events and trends in these diseases.

The total number of cases and deaths recorded during a specific period (e.g., monthly) for diseases of public health importance should be reported. All public and private HF, including referral, regional and teaching hospitals, should report all cases and deaths of monthly reportable diseases to the district under which they operate. Districts will compile the information coming from all reporting sites and forward the summaries to the regional level and from there to the central level. Each level of the health system will thus be able to detect the increase or the occurrence of unusual health events, during these monthly summaries analysis.

The results of this analysis will make it possible to monitor progress towards disease control objectives, measure progress in prevention activities in the HD and identify outbreaks or other public health events that have not been detected, so that early response measures can be taken.

Table IV: Priority diseases, conditions and public health events to be reported monthly (MAR)

1. Neurological conditions (epilepsy)
2. Maternal deaths
3. Diabetes
4. Diarrhoea with acute dehydration in children under 5 years of age
5. Sickle Cell Anaemia
6. Lymphatic Filariasis
7. Intestinal Helminthiases
8. Acute viral hepatitis
9. High blood pressure
10. Sexually transmissible infections
11. Low birth weight in neonates (birth weight less than 2500 g)
12. Leprosy
13. Malnutrition in children under 5 years of age
14. Noma
15. Onchocerciasis
16. Malaria
17. Severe pneumonia in Children under 5 years of age
18. Intestinal schistosomiasis
19. Urinary schistosomiasis
20. Influenza-like Illness
21. Trachoma
22. Injuries (road traffic accidents)
23. Trypanosomiasis
24. Tuberculosis
25. Buruli ulcer
26. HIV (new cases)

Each month, each facility calculates the total number of cases and deaths due to priority diseases and public health events recorded. Totals are calculated separately for outpatient and inpatient cases. A summary form recording the aggregated data totals (see ***Appendix 2G***) is sent to the district level. The district summarizes aggregated data from all reporting sites coming from the Health Areas and forward it to the regional level, which in turn forwards it to the central level.

Every effort should be made to obtain from the Health Information System the total number of outpatients and the total number of patients hospitalized for various health problems (including those absent from the IDSR list) during the period under consideration.

If a computer is available for surveillance or data management, analysis of the medical records will help produce weekly or monthly reporting. This information is important to produce situation reports (SITREPs) at the peripheral, regional and central levels. All data should be shared with the health authorities, with a copy for the corresponding prevention and control program. This is important not only for coordination at the central level, but also for creating or enriching a national IDSR database.

Depending on the level of laboratory services, laboratory data should be recorded in a register in order to calculate accumulations of the number of monthly cases of diseases under surveillance. During outbreaks, events on the types of specimens processed should also be included in the weekly summaries; their results should help to complete the variables in the descriptive register. Efforts should be made to update the laboratory component of the IDSR data and to link these data to epidemiological/clinical data. Monthly summaries may include baseline tests performed for which the country has selected pathogens as an indicator based on the main PHEICs, which is important since the analysis may produce trends likely to justify in-depth investigations.

2.1.5. Zero reporting

If no cases of disease or event have been diagnosed during the epidemiological week or current month, record a zero (0) on the reporting form for that disease. If the space is left blank, the staff that receives the report will not be able to develop information from a blank space. Submitting a zero report when no cases of reportable disease were detected during the week tells the staff at the next level that a complete report has been filled.

2.1.6. Reporting of a Public Health Emergency of International Concern (PHEIC) according to the IHR (2005)

If a PHEIC as defined in ***Appendix 2*** of the IHR (2005) is suspected, the District FP should immediately report to the IHR (2005) Focal Point and the Regional Surveillance Focal Point using the fastest means of communication. If a PHEIC is detected at the PoE, the district and region should also be notified immediately (see the reporting framework in ***Appendix 2B***).

The process for reporting events to WHO according to the IHR (2005) includes the use of the Decision Instrument contained in the IHR. This is a country-level function coordinated by the IHR (2005) Focal Point with support from relevant experts, depending on the degree of urgency.

2.2 Compilation of Immediately reportable priority diseases, conditions and public health events

After detecting a first case of a suspected or confirmed outbreak, summary data are important for analysis and monitoring. For example, at the HFor district level, the surveillance FP should draw an outbreak curve to verify whether and when the outbreak thresholds of the diseases under surveillance have been crossed. In addition, data from outbreak investigations can be used to verify whether the case-fatality rate is below, equal to, or above the target. Analysis of data from a suspected or confirmed outbreak (Time, Place, Person) can highlight possible risk factors such as age, sex, place of residence, exposure during social events such as funerals, occupational hazards, bush meat consumption, consumption of contaminated food or drink.

At the district level, weekly data analysis includes verification of the quality of the data coming from the reporting sites and the completeness and timeliness of these reports. For eIDSR, a person should be responsible for ensuring that data verification is performed and approved before transmission to a higher level. In addition, a thorough analysis of individual case records received from reporting sites will also be conducted, besides weekly summary data.
Incidence and case-fatality rates should be calculated and compared with the alert and outbreak thresholds to assess the trend of the outbreak. Epicurves should be updated regularly in order to monitor the trends and evolution of an outbreak in the district.

Districts that have digital tools (computers, digital tablets, and smartphones) are encouraged to store information in electronic format and to transmit all surveillance data to the higher level in this format.

2.3 Improve routine reporting practices

In some HF, more than one person may be responsible for recording information about patients seen in the facility. For example, the clinician records the patient's name and diagnosis in a clinic register. Later in the day, a nurse tallies the number of cases and deaths seen in an outpatient service. The ward nurse tallies the number of admitted cases.

Each week or month, a records clerk or statistician calculates summaries for all the diseases and records them in a standard form. In case the HFis equipped with computers, individual patient records should be entered, from which the IDSR priority diseases or conditions subset will be extracted and analysed to get the required weekly, monthly or quarterly compilations.

In outbreak scenarios, isolation units that are separate from HF can be opened, and they will use a different register to record diseases or events. It is important that this information be captured in the overall IDSR weekly or monthly summaries.

2.3.1 Review the flow of information at the reporting site

During supervisory visits to reporting sites, ensure that:

- the surveillance Focal Person is identified and active;
- all reporting sites including secondary and tertiary hospitals in the catchment area of your district are visited;
- clinicians record legibly information in the patient registers using the recommended case definitions so that health workers who tally the cases at the end of the day can reliably record the required diagnoses on the tally sheet;
- clinicians, ward nurses or other responsible staff should complete the case-based reporting form preferably while the patient is still present;
- clinicians record laboratory results in the patient registers;
- records clerks or statisticians have summary forms that contain spaces for recording cases and deaths due to priority diseases or conditions according to the standard case definitions;
- health staff review the weekly, monthly and quarterly IDSR data summary totals and provide comments on the forms about results seen during data analysis. (See ***section 3***);
- health workers record the summary totals on a recommended weekly or monthly IDSR summary reporting form (See ***Appendix 2G***).

2.3.2 Keeping records and procedures for managing reporting forms

Keep a record of IDSR forms, notifications and reports received at your level. The record you keep is an essential data source for calculating indicators for your country's IHR report and for monitoring performance of the IDSR indicators. A sample IDSR Reports and Data Sharing Log Book form is in ***Appendix 2E***.

Periodically check with reporting sites that you supervise (community, HFand district) to ensure that the correct forms and procedures are available to staff so they can record and report the required cases of priority diseases and conditions:

- take steps to ensure that all health workers know or have access to the standard case definitions recommended by national policy. Establish or modify existing procedures so that all health workers are able to apply the standard case definitions in detecting and reporting priority diseases, conditions, outbreaks or events;
- sensitize staff on diseases or conditions that require immediate reporting for case-based surveillance, including potential PHEIC and other priority diseases or public health events of national and regional concern. For example, all the health staff should be aware of epidemic-prone diseases for which a single suspected or probable case is a suspected outbreak requiring immediate action, and of any unusual or unexplained event with potential for affecting human health;
- review with health staff the role that case-based data plays in determining risk factors and the means of disease transmission or exposure to health risks in a public health event. Make sure the staff has access to a standardized form for reporting case-based information;
- ensure that the surveillance unit has access to fast communication means (telephone, text message, electronic mail, walkie-talkie, personal messages, or other rapid communication means). For the district, specify how the district should notify the regional or national levels and who should be contacted at these levels.

2.3.3 Perform periodic checks on data quality

While each data manager may have some preferred methods for filling in forms, describing diseases, or abbreviating terms, it is important for every level of reporting (facility, district, region or national) to use a standard approach to recording and reporting, as data that are not comparable, will lead to inappropriate decisions.

Some of the examples of factors which may affect data quality that needs to be periodically checked include:

- poorly completed forms (missing values, etc.);
- incomplete forms (for example, blank spaces);
- under-reporting or over-reporting of cases;
- duplicate reporting;
- unsystematic data collection and reporting;
- untruthful reporting, (for example, reporting zero, while there is an ongoing outbreak of epidemic-prone diseases).
- inconsistent reporting formats (forms);
- late submission or reporting;
- inconsistent reporting periods;
- calculation errors on aggregate reports;
- lack of documentation and source data or files are lost.

During supervision, stress the importance of data quality and surveillance; that correct data will lead to analysis, interpretation, and the information that will be communicated will lead to action and evaluation. It is recommended that countries conduct regular data quality audits at the reporting sites. (See ***Appendix 2H*** for checklist on key elements to assess in data quality audits).

2.3.4 Enhance linkages to strengthen community-based surveillance

A community-based surveillance system relies on the community members' capacity to identify and report public health problems to the nearest HFor to the district health office. In this system, CBS focal persons identify and report events in the community that have public health significance. CBS focal persons act as community informants, and they report to the health facility, or in the case of a serious event, directly to the district authorities.

Any community member acceptable by the community can be a CBS focal person. Representation could be from basic village-level services such as trained birth attendants, community or village health agents, or similar care providers, community health workers or volunteers, village leaders (religious, traditional or political) or school teachers, veterinarians, health extension workers, drug vendors, and traditional healers. Once selected, the CBS focal persons should receive training and carry out supportive supervision on how to recognize certain diseases or health conditions for reporting suspect cases.

Example: The *CHW hears of several cases of acute watery diarrhoea with vomiting in the community. The informant suspects cholera and reports the signal to the local HFand to the district medical officer by text message. Members of the public health emergency rapid response team (RRT) travel to the community to verify and investigate the possible outbreak, and, based on the investigation results, implement control and prevention measures. The outbreak is quickly contained. Thanks to the early warning from the community-based surveillance liaison.*

District staff may identify key informants in the community who can provide information about the community's health status. Examples of key informants include:

- pharmacists
- school teachers
- village leaders
- religious leaders
- traditional healers
- birth attendants
- community based organizations
- other societal leaders
- veterinary health workers
- any individuals involved in neighbourhood watch or other active surveillance approaches
- other community resource persons

Depending on the event, resource availability and the context, the country may choose its source of information. The district can organize community-based surveillance by:

- working with community leaders to identify members of the community to receive relevant training;
- provide necessary information to identifying priority diseases and public health conditions or events to community health informants. Give enough information about the disease so that these

persons can refer cases to the health facility, or notify the HFwhen unusual or unexplained health events occur in the community;

- involve CBS focal persons in risk mapping, emergency simulation exercises and risk communication during outbreaks;
- ensure that the CBS gives regular and timely feedback of diseases/events reported from the community level. Districts need to ensure that there is sustained commitment by CBS and hence to continuously collaborate with them;
- disseminate information on alert and outbreak thresholds.

Please refer to the list in ***Appendix 1B*** of key signs and symptoms to use in case definitions for community surveillance.

2.3.5 Strengthen linkages between laboratory and surveillance data

Public health laboratory system complements the syndromic disease surveillance.

(a) In case of a public health event, the laboratory where confirmation took place should report the laboratory results to the respective HFand simultaneously to the central level. The surveillance focal person at the HFshould report to the head of the health authority, who reports to the supervisor (FP or district medical officer) for action and concomitant transmission of information to the regional level, which in turn forwards it to the central level.
(b) To strengthen the linkages between epidemiological and laboratory data, the case reported and the laboratory samples should have the same unique ID.
(c) Submission of the weekly summaries of the samples processed, and the types of samples, as well as the results, should be done whenever there is an outbreak, to assist in completion of the variables in the line list register.
(d) During supervision at reporting sites, ensure that the laboratorians correctly record data for diseases under surveillance and that there is an established register.
(e) Make sure that the test results are linked with IDSR data at all levels.
(f) The laboratory should provide a weekly and monthly summary reporting Forms for laboratory results of specimens received.

(g) Liaise with the animal sector, to gather a comprehensive report also from the veterinary laboratory, especially if they have recorded any animal information which might have risks to public health.

2.3.6 Promote a multisectoral "One Health" approach

Ensure implementation of the "One Health" approach to improve reporting of public health risks across all levels, with emphasis also at the community level. Lay emphasis on strengthening the technical and community capacities of staff for all relevant sectors (including human physicians/nurses, veterinarians for livestock or wildlife) and environmental inspectors.

Interoperable and interconnected platforms with emphasis on strengthening information systems within and between the human, animal, and environmental sectors would be ideal in enhancing real time information sharing. There should be a conscious effort to formalize the system of sharing information with other sectors, environmental health, etc. The other multisectoral key actors to foster collaboration in reporting and assessment of public health risks include private sector, civil society, religious organizations, defence and security forces, prisons, internally displaced persons and refugee camps, technical and financial partners, academic and research institutions. Ensure that they are also included to strengthen routine reporting and analysis of public health hazards and events.

2.4 Data protection and security to protect patients' confidentiality

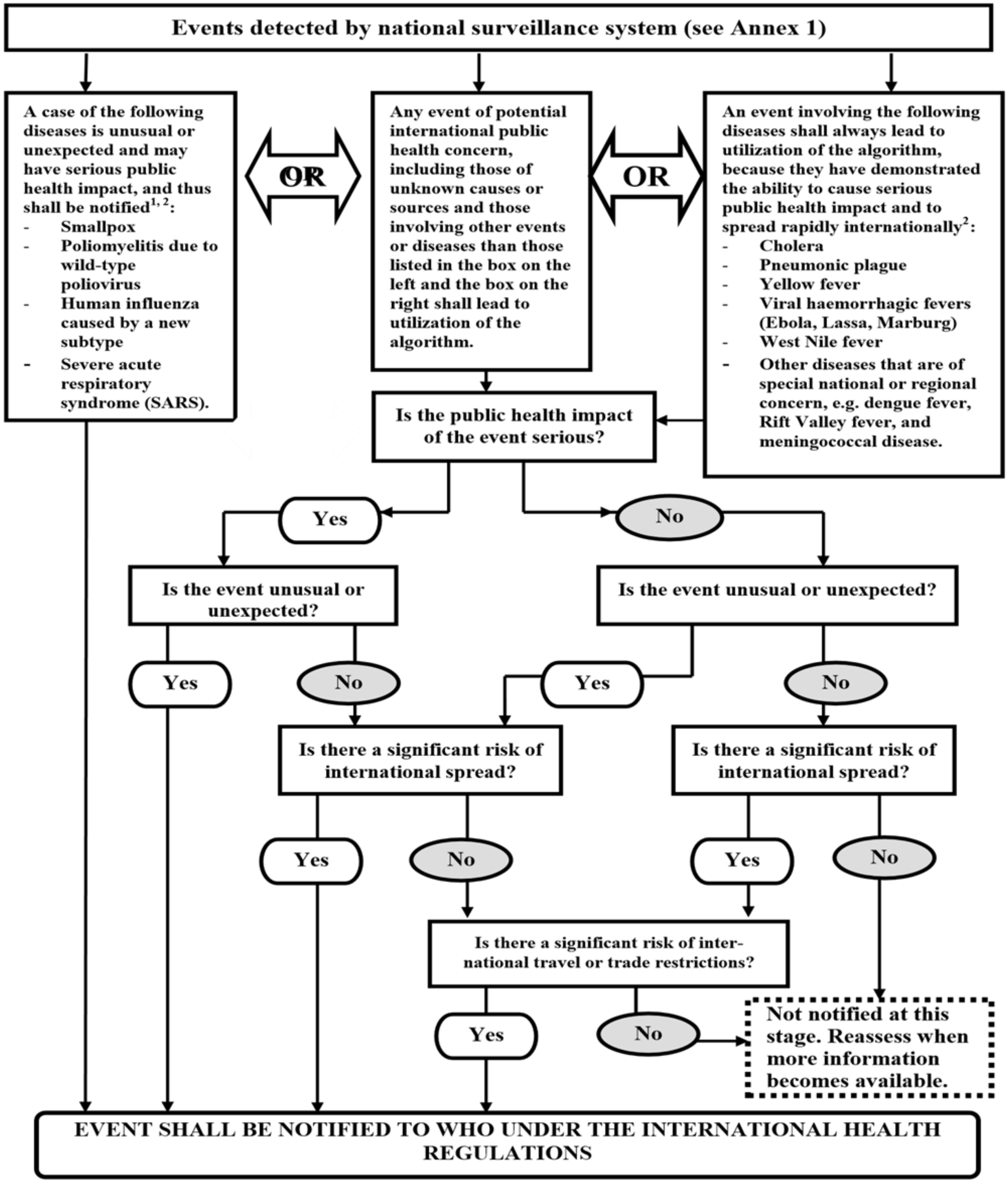
The public health community recognizes that there might be risks to both individuals and communities, if one uses name-based reporting of private health-related information. To ensure protection of patient confidentiality and privacy, when reporting, use unique identifiers such as numbers instead of names and this will prevent identities from being inadvertently disclosed. The identifiable data should be however maintained where public health surveillance interventions occur and it is usually at the HFlevel. Districts need to have guidelines on privacy and security of health data, which should be guided by the national level guidelines.

Note: Use of names may be required during an outbreak of infectious diseases for contact tracing. Refer to ***section 4*** on case tracing and recording.

2.5 Appendices of Section 2

APPENDIX 2A	IHR (2005) decision instrument
APPENDIX 2B	Algorithm of reporting immediately reportable diseases, priority conditions and public health events
APPENDIX 2C	Community-based signals reporting form
APPENDIX 2D	Reporting and verification circuit of signals received in the community or health facility
APPENDIX 2E	Case-based reporting form of other immediately reportable diseases
APPENDIX 2F	Weekly notification form of diseases and other public health events in Cameroon
APPENDIX 2G	Monthly notification form of diseases and other public health events in Cameroon
APPENDIX 2H	Maternal deaths reporting form
APPENDIX 2I	Perinatal deaths reporting form,
APPENDIX 2J	Epidemiological week cut-off format, 2023

APPENDIX 2A: IHR (2005) Decision instrument

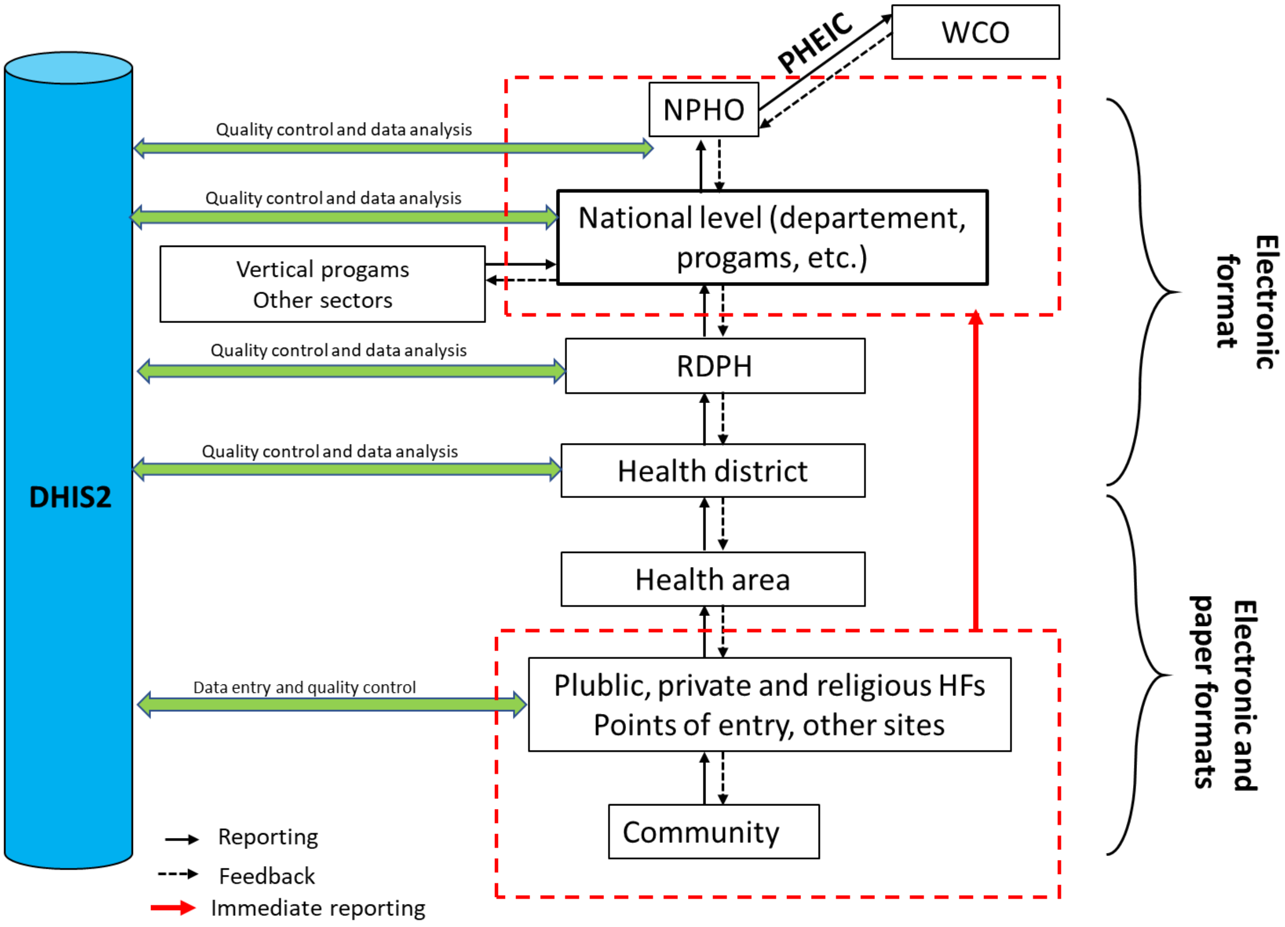

Events detected by national surveillance system (see Annex 1)
A case of the following diseases is unusual or unexpected and may have serious public health impact, and thus shall be notified1, 2:
- Smallpox
- Poliomyelitis due to wild-type poliovirus
- Human influenza caused by a new subtype
- Severe acute respiratory syndrome (SARS).
OR
Any event of potential international public health concern, including those of unknown causes or sources and those involving other events or diseases than those listed in the box on the left and the box on the right shall lead to utilization of the algorithm.
OR
An event involving the following diseases shall always lead to utilization of the algorithm, because they have demonstrated the ability to cause serious public health impact and to spread rapidly internationally2:
- Cholera
- Pneumonic plague
- Yellow fever
- Viral haemorrhagic fevers (Ebola, Lassa, Marburg)
- West Nile fever
- Other diseases that are of special national or regional concern, e.g. dengue fever, Rift Valley fever, and meningococcal disease.
Is the public health impact of the event serious?
Yes
No
Is the event unusual or unexpected?
Is the event unusual or unexpected?
Yes
No
Yes
No
Is there a significant risk of international spread?
Is there a significant risk of international spread?
Yes
No
Yes
No
Is there a significant risk of international travel or trade restrictions?
Yes
No
Not notified at this stage. Reassess when more information becomes available.
EVENT SHALL BE NOTIFIED TO WHO UNDER THE INTERNATIONAL HEALTH REGULATIONS

---

[1] As per WHO case definitions.

[2]

APPENDIX 2B. Algorithm of reporting immediately reportable diseases, conditions and public health events


PHEIC
WCO
NPHO
Quality control and data analysis
Quality control and data analysis
National level (departement, progams, etc.)
Vertical progams
Other sectors
Quality control and data analysis
RDPH
Quality control and data analysis
Health district
Health area
DHIS2
Data entry and quality control
Plublic, private and religious HFs
Points of entry, other sites
Community
Reporting
Feedback
Immediate reporting
Electronic format
Electronic and paper formats

APPENDIX 2C. Community-based signals reporting form

Send this form immediately to your supervisor or nearby health facility

Instructions: This form is completed by the CHW in the community and submitted immediately to nearest HFsurveillance focal person when he or she identifies disease (s) or public health event as per the community case definition for community surveillance. It is also completed for unusual health events/signals that are not captured by the given case definition.

COMMUNITY-BASED SIGNALS REPORTING FORM

Date of onset of signal: ____

Date and hour of signal detection: ____

Localisation of signal: ____

Description of signal: ____

____

____

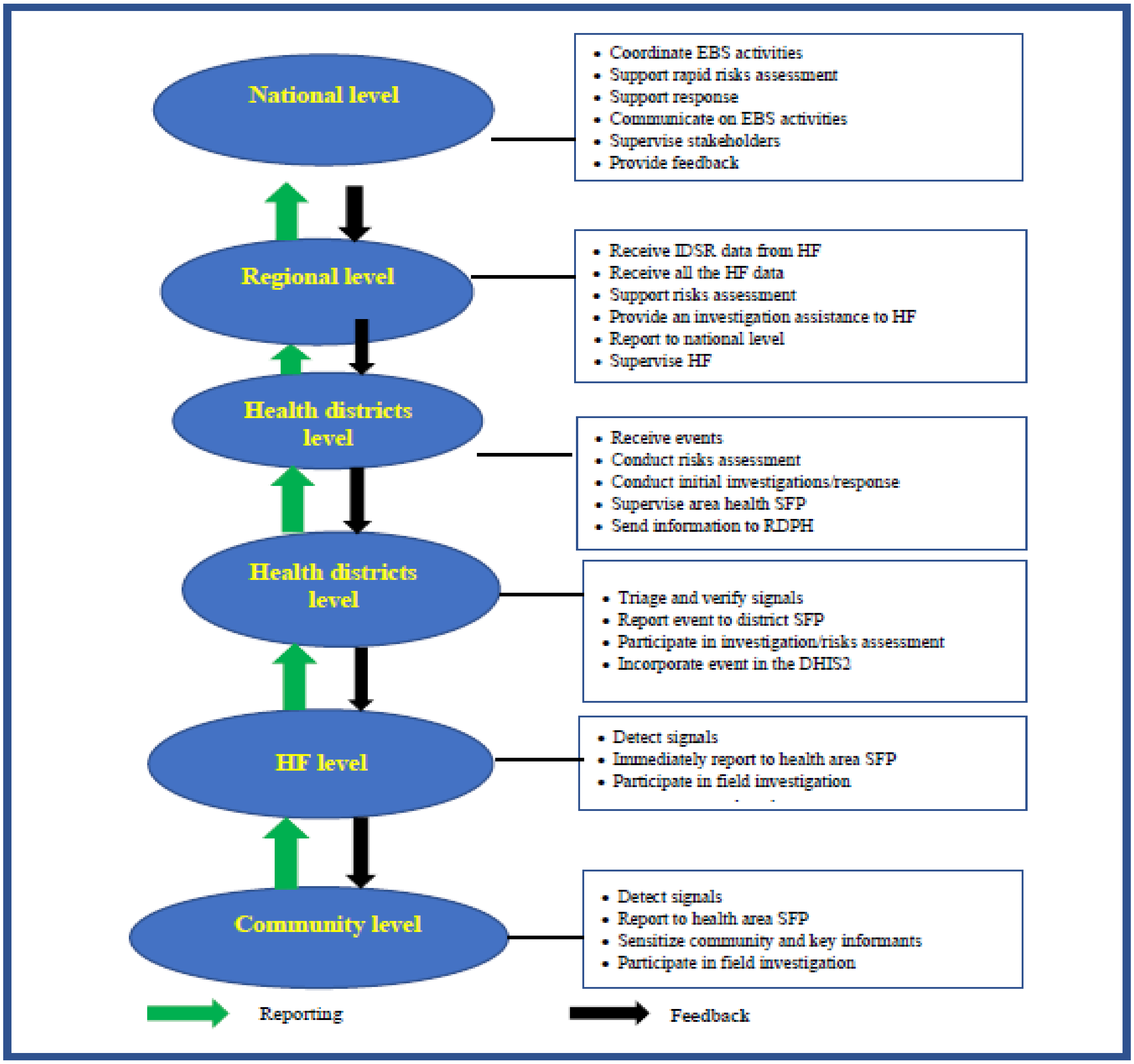
____

____

____

Name and signature of the CHW

APPENDIX 2D. Reporting and verification circuit of signals received in the community or health facility



National level

- Coordinate EBS activities
- Support rapid risks assessment
- Support response
- Communicate on EBS activities
- Supervise stakeholders
- Provide feedback

Regional level

- Receive IDSR data from HF
- Receive all the HF data
- Support risks assessment
- Provide an investigation assistance to HF
- Report to national level
- Supervise HF

Health districts level

- Receive events
- Conduct risks assessment
- Conduct initial investigations/response
- Supervise area health SFP
- Send information to RDPH

Health districts level

- Triage and verify signals
- Report event to district SFP
- Participate in investigation/risks assessment
- Incorporate event in the DHIS2

HF level

- Detect signals
- Immediately report to health area SFP
- Participate in field investigation

Community level

- Detect signals
- Report to health area SFP
- Sensitize community and key informants
- Participate in field investigation

APPENDIX 2E: Case-based reporting form of other immediate notifiable diseases

Case-based reporting form of other immediate notifiable diseases		
	Variables or questions	**Answer – Case number**
X	Report unique identifier (YEAR-WEEK-CMR-REGION-DS-Case no)	
1	Country	
2	Region	
3	HD	
4	Reporting site (HF, community)	
5	Disease or event (diagnostic): *	
6	Outpatient or inpatient	
7	Date of consultation in the HF (day/month/ year)	\___\___\___\\
8	Patient(s) name(s)	
9	Date of birth (day/ month/ year)	\___\___\___\\
10	Age (years/months/days)	
11	Sex: M=male F=female	
12	Patient's residence village/neighbourhood	
13	Name of HA	
14	Name of HD	
15	Occupation	
16	Phone number (if necessary, name of mother or father if neonate or infant)	
17	Date of onset of symptoms (day/ month/ year)	\___\___\___\\
18	Travel history (yes or no), if yes, state destination	
19	Number of vaccine doses received in the past against the disease being reported	
20	Date of last vaccination	\___\___\___\\
21	Date specimen collected	
22	Date specimen sent to the lab	
23	Laboratory results	
24	Outcome (Alive, Dead, transferred out, Lost to follow-up or unknown)	
25	Final classification: Confirmed, probable, compatible, discarded	
26	Date facility notified district (day/month/year)	\\ _ _ _ \\ _ _ _ \\ _ _ _ \\
27	Date form send to the HD (day/month/year)	\___\___\___\\
28	Person completing form: name, function, signature	

Other diseases/events: Any disease that does not have specific reporting form including unknown events

APPENDIX 2F: Weekly reporting form of diseases and other public health events

REPUBLIQUE DU CAMEROUN / REPUBLIQUE OF CAMEROON

MINISTERE DE LA SANTE PUBLIQUE / MINISTRY OF PUBLIC HEALTH

FICHE DE NOTIFICATION HEBDOMADAIRE DES MALADIES A POTENTIEL EPIDEMIQUE, AFFECTIONS PRIORITAIRES ET AUTRES EVENEMENTS DE SANTE PUBLIQUE

FORM FOR WEEKLY NOTIFICATION OF SURVEILLANCE OF EPIDEMIC PRONE DISEASES, PRIORITY CONDITIONS AND OTHER PUBLIC HEALTH EVENTS

(Fiche niveau de la Formation Sanitaire/ Health Unit's Form)

REGION

District

Aire de Santé / Health Area

Nom de la Formation Sanitaire
Name of the Health Unit

Nombre de DS ayant notifié
Nombre de DS ayant notifié à temps
Nombre de FS ayant notifié
Nombre de FS ayant notifié à temps

Année
Year

semaine épidémiologique Du
Epidemiological WeekFrom

Au
To

N°	**MALADIES / EVENEMENTS**	**DISEASES / EVENTS**	**NOMBRE/NUMBER**		
			CAS/CASES	**DECES/ DEATHS**	**CAS CONFIRMÉS/ CONFIRMED CASES**
1	**Chikungunya***	**Chikungunya***			
2	**Choléra***	**Cholera***			
3	**Dengue***	**Dengue Fever***			
4	Diarrhée avec déshydration chez les enfants de moins de cinq ans	Diarrhea with dehydration in children under five years of age			
5	Diarrhée sanglante	Bloody diarrhea			
6	**Dracunculose* (Ver de Guinée)**	**Dracunculiasis* (Guinea worm)**			
7	**Fièvre Charboneuse* (Anthrax)**	**Anthrax***			
8	**Fièvre hémorragique virale* (Ebola, Marburg, Vallée du Rift, Lassa, Crimée-Congo)**	**Viral hemorrhagic fever* (Ebola, Marburg, Rift Valley, Lassa, Crimean-Congo)**			
9	**Fièvre Jaune***	**Yellow Fever***			
10	Fièvre Typhoïde	Typhoid Fever			
11	Méningite	Meningitis			
12	Morsure de chien	Dog bite			
13	**Rage***	**Rabies***			
14	Morsure de serpent	Snake bite			
15	Envenimation	Envenomation			
16	Paludisme	Malaria			
17	**Paralysie Flasque Aiguë* (PFA)**	**Acute Flaccid Paralysis***			
18	**Peste***	**Plague***			

19	**Rougeole***	**Measles***			
20	**SRAS/IRAS* (Autre que COVID-19)**	**SARS/SARI* (Other than COVID-19)**			
21	**COVID-19* (Cas confirmé)**	**COVID-19* (Confirmed case)**			
22	Syndrome grippal / IRA (Infection Respiratoire Aiguë)	Flu-like syndrome/ARI (Acute Respiratory Infection)			
23	**Tétanos Néonatal* (TNN)**	**Neonatal Tetanus* (NNT)**			
24	**Variole du Singe***	**Monkey Pox* (MPox)**			
25	**Variole***	**Smallpox***			
26	Accouchement assisté	Assisted delivery			
27	Décès maternel	Maternel death			
28	Décès périnatal	Perinatal death			
29	Décès Néonatal	Neonatal death			
30	Autres évènements (spécifier):	Other events (specify):			
	INDICATEURS DE NUTRITION				
31	Malnutrition aigue sévère chez les enfants admis	Severe acute malnutrition in children admitted			
32	Malnutrition aigue sévère avec complications médicales chez les enfants admis	Severe acute malnutrition with medical complications in admitted children			
33	Nombre d'enfants malnutris aigues sévères ayant bénéficié d'un kit minimum WASH	Number of severely malnourished children who received a minimum WASH kit			
34	Quantité RUTF-plumpy nut consommée dans la semaine (en sachets)	Quantity of RUTF-plumpy nut consumed during the week (in sachets)			
35	Quantité RUTF-plumpy nut restante dans la semaine (en sachets)	Quantity of RUTF-plumpy nut remaining in the week (in bags)			

***** Maladie à notification immédiate / immediate notifiable diseases**

Nombre de FS avec disponibilité d'un point d'eau potable fonctionnel ☐

Number of HF with a functional drinking water point ☐

Nombre de FS avec disponibilité des produits de potabilisation de l'eau de boisson

Number of HF with availability of drinking water treatment products

Date d'envoi / Date of Submision : ……………………………... **Date de Réception / Date of Reception: …………**

Nom, Signature et cachet / Name, Signature and Round Stamp

APPENDIX 2G: Monthly reporting form of diseases and other public health events

XV. SURVEILLANCE DES MALADIES A POTENTIEL EPIDEMIQUE ET AUTRES EVENEMENTS DE SANTE PUBLIQUE / SURVEILLANCE OF EPIDEMIC-PRONE DISEASES AND OTHER PUBLIC HEALTH EVENTS					
Maladies / Diseases	Nouveaux Cas / New Cases	Décès / Deaths	Maladies / Diseases	Nouveaux Cas / New Cases	Décès / Deaths
348. ANTHRAX			349. CHIKUNGUNYA		
348. ANTHRAX			349. CHIKUNGUNYA		
350. CHOLERA			351. DENGUE		
352. DIARRHEE SANGLANTE BLOODY DIARRHOEA			353. GASTROENTERITE AIGUE SEVERE ACUTE SEVERE GASTROENTERITIS		
354. FIEVRE TYPHOIDE354. FIEVRE TYPHOIDE TYPHOID FEVER			355. FIEVRE HEMORRAGIQUE AIGUE VIRAL HAEMORRHAGIC FEVER*		
356. FIEVRE JAUNE YELLOW FEVER			357. MENINGITE MENINGITIS		
358. PESTE PLAGUE			359. ROUGEOLE MEASLES		
360. DRACUNCULOSE DRACUNCULIASIS			361. PFA AFP		
362. TETANOS NEONATAL NEONATAL TETANUS			363. DECES MATERNELS MATERNAL DEATHS		
364. MALNUTRITION MALNUTRITION			365. MAPI EPD		
366. PALUDISME MALARIA			367. MORSURE DE CHIEN DOG BITE		
368. RAGE RABIES			369. ENVENIMATION ENVENOMIZATION		
370. MORSURE DE SERPENT SNAKE BITE			371. VARIOLE SMALLPOX		
372. VARICELLE CHICKEN POX			373. DIARRHEE+DESHYDRATATION CHEZ LES MOINS DE 5 ANS DIARRHOEA WITH DEHYDRATION IN CHILDREN UNDER 5		
374. HELMINTHIASE INTESTINALE INTESTINAL HELMINTHIASIS			375. INFECTION RESPIRATOIRE AIGUE ACUTE RESPIRATORY INFECTION		
376. FIEVRE ERUPTIVE ERUPTIVE FEVER			377. FILARIOSE LYMPHATIQUE LYMPHATIC FILARIASIS		
378. ULCERE DE BURULI BURULI ULCER			379. EPILEPSIE EPILEPSY		
380. INSUFFISANCE RENALE KIDNEY FAILURE			381. AVC CVA		
382. DREPANOCYTOSE SICKLE-CELL ANAEMIA			383. NOUVEAU CAS HEPATITE VIRALE B		

			NEW CASES OF VIRAL HEPATITIS B		
384. NOUVEAU CAS HEPATITE VIRALE C NEW CASES OF VIRAL HEPATITIS C			385. NOUVEAU CAS HTA NEW CASES OF HBP		
386. IST STI			387. SCHISTOSOMIASE SCHISTOSOMIASIS		
388. TRACHOME TRACHOMA			389. AVP		
390. TUBERCULOSE RESISTANTE RESISTANT TUBERCULOSIS			391. NOUVEAU CAS VIH NEW HIV CASES		
392. TRYPANOSOMIASE TRYPANOSOMIASIS			393. RESISTANCE ANTI MICROBIENNE ANTI-MICROBIAL RESISTANCE		
394. TABAGISME SMOKING			395. ALCOOLISME ALCOHOLISM		
396. TOXICOMANIE SUBSTANCE ADDICTION			397. DESORDRE PSYCHIQUE PSYCHOTIC DISORDER		
398. TOXI INFECTION ALIMENTAIRE FOODBORNE EPIDEMIC			399. SYNDROME GRIPPAL FLU SYNDROME		
400. RHUMATISME RHEUMATISM			401. COVID-19		

APPENDIX 2H: Maternal deaths reporting form

MATERNAL DEATHS REPORTING FORM

Region______________ District___________________ Health area __________________

Report case unique identifier: ---/--- ---/---/ ---/---/

Maternal death place:

If HF, name of structure ---

If not indicate the place: ---

1.2 Maternal death place: --------/------------/----------------

1.3 Cause of death

Bleeding	
Pre-eclampsia/eclampsia	
Abortion	
Malaria	
HIV/AIDS	
Severe anaemia	
Other circumstances	

Tick the box corresponding to the probable cause of death

Name and signature

Person who reported ---

Head of the HF --

Head of the HA --

District Medical Officer --

DATE--

APPENDIX 2I: Perinatal deaths reporting form

NEONATAL AND PERINATAL DEATHS REPORTING FORM

Region_____________ District_____________ Health area ____________________

Case unique identifier: ---/--- ---/---/ ---/---/

Maternal death place:

If HF, name of structure ---

If not, indicate the place: ---

1.2 Maternal death place: --------/------------/----------------

1.3 Cause of death

Complications of preterm birth		Diarrhoea	
Congenital condition		Malaria	
Infections: Tetanus		Asphyxia	
Septicaemia		Antepartum complications	
Pneumonia		Per partum complications	
Meningitis		Others	
Syphilis			

Tick the box corresponding to the probable cause of death

Name and signature
Person who reported ---

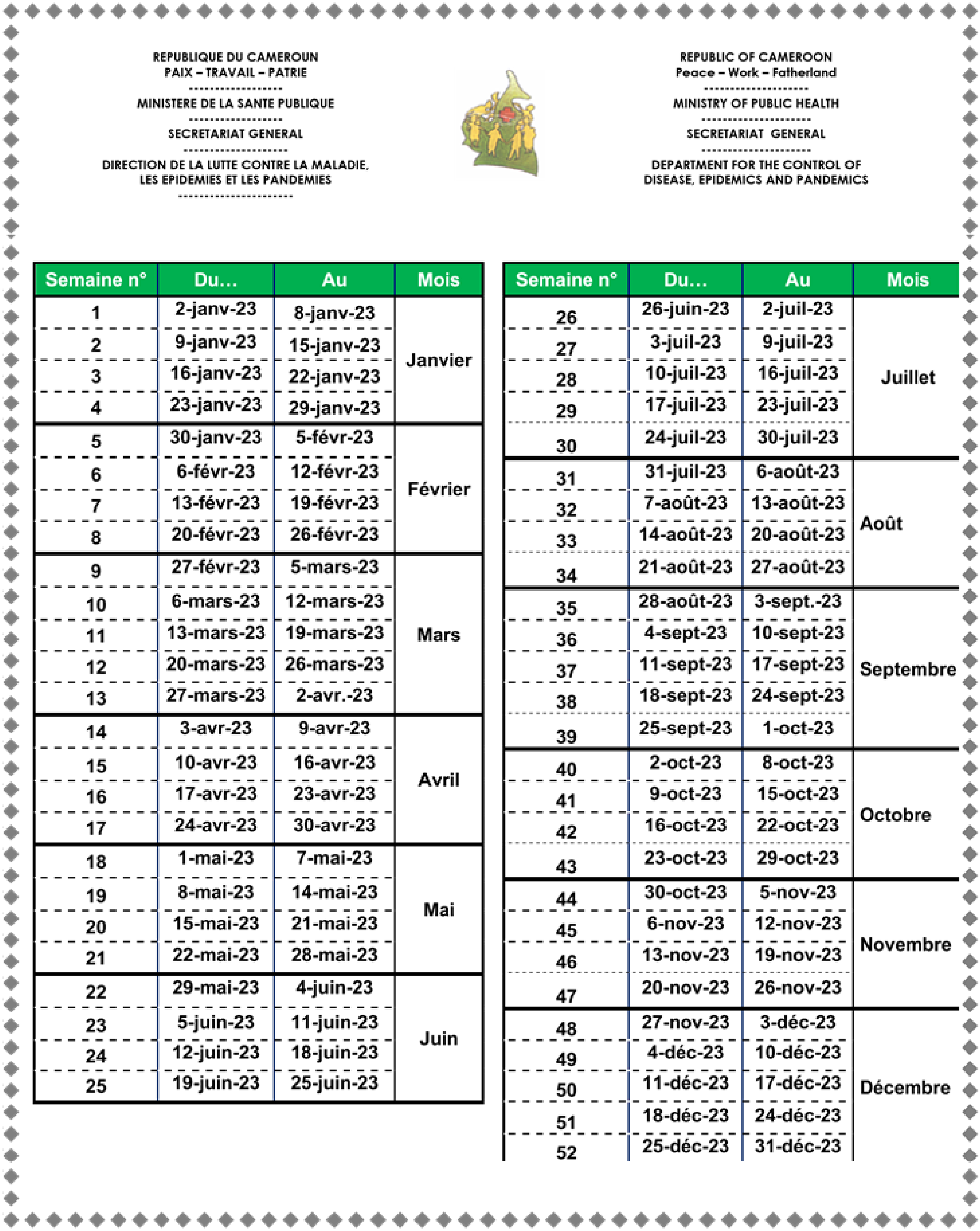
Head of the HF ---

Head of the HA --

District Medical Officer ---

DATE --

APPENDIX 2J: Epidemiological week cut-off format, 2023



REPUBLIQUE DU CAMEROUN
PAIX – TRAVAIL – PATRIE

MINISTERE DE LA SANTE PUBLIQUE

SECRETARIAT GENERAL

DIRECTION DE LA LUTTE CONTRE LA MALADIE,
LES EPIDEMIES ET LES PANDEMIES

REPUBLIC OF CAMEROON
Peace – Work – Fatherland

MINISTRY OF PUBLIC HEALTH

SECRETARIAT GENERAL

DEPARTMENT FOR THE CONTROL OF
DISEASE, EPIDEMICS AND PANDEMICS

Semaine n°	Du...	Au	Mois
1	2-janv-23	8-janv-23	Janvier
2	9-janv-23	15-janv-23	
3	16-janv-23	22-janv-23	
4	23-janv-23	29-janv-23	
5	30-janv-23	5-févr-23	Février
6	6-févr-23	12-févr-23	
7	13-févr-23	19-févr-23	
8	20-févr-23	26-févr-23	
9	27-févr-23	5-mars-23	Mars
10	6-mars-23	12-mars-23	
11	13-mars-23	19-mars-23	
12	20-mars-23	26-mars-23	
13	27-mars-23	2-avr.-23	
14	3-avr-23	9-avr-23	Avril
15	10-avr-23	16-avr-23	
16	17-avr-23	23-avr-23	
17	24-avr-23	30-avr-23	
18	1-mai-23	7-mai-23	Mai
19	8-mai-23	14-mai-23	
20	15-mai-23	21-mai-23	
21	22-mai-23	28-mai-23	
22	29-mai-23	4-juin-23	Juin
23	5-juin-23	11-juin-23	
24	12-juin-23	18-juin-23	
25	19-juin-23	25-juin-23	

Semaine n°	Du...	Au	Mois
26	26-juin-23	2-juil-23	Juillet
27	3-juil-23	9-juil-23	
28	10-juil-23	16-juil-23	
29	17-juil-23	23-juil-23	
30	24-juil-23	30-juil-23	
31	31-juil-23	6-août-23	Août
32	7-août-23	13-août-23	
33	14-août-23	20-août-23	
34	21-août-23	27-août-23	
35	28-août-23	3-sept.-23	Septembre
36	4-sept-23	10-sept-23	
37	11-sept-23	17-sept-23	
38	18-sept-23	24-sept-23	
39	25-sept-23	1-oct-23	
40	2-oct-23	8-oct-23	Octobre
41	9-oct-23	15-oct-23	
42	16-oct-23	22-oct-23	
43	23-oct-23	29-oct-23	
44	30-oct-23	5-nov-23	Novembre
45	6-nov-23	12-nov-23	
46	13-nov-23	19-nov-23	
47	20-nov-23	26-nov-23	
48	27-nov-23	3-déc-23	Décembre
49	4-déc-23	10-déc-23	
50	11-déc-23	17-déc-23	
51	18-déc-23	24-déc-23	
52	25-déc-23	31-déc-23	

Section 3: Analyze and interpret data

This section describes how to:

- receive, process and store data from notification sites;
- analyze data based on time, place, and person;
- analyze data according to thresholds to decide on Public Health actions;
- interpret the results of the analysis;
- synthesize and use the results of the analysis to improve Public Health actions.

Introduction

Data preparation and analysis is an important part of monitoring. It is not enough to only collect, record and report information about illness, death and disability from the catchment area; they also need to be analysed at each level from the point of collection. Organizing and analysing data are for surveillance. Analysing data provides the information that is needed to take relevant, timely and appropriate public health action. Analysis of surveillance data allows for:

- observing trends over time and alerting health staff and relevant surveillance stakeholders about emergent events or unusual patterns;
- identifying areas at risk;
- determining personal variables such as age, gender or occupation that help identify people at risk for a disease or public health event;
- monitoring and evaluating public health interventions.

In general, analysing routine surveillance data (time, place, and person) should address the following questions:

- have any diseases, priority conditions or public health events been detected during the reporting period (e.g., this week)? Is an outbreak or unusual public health event suspected?
- how many of the cases, deaths or public health events detected, have been confirmed? where did they occur?
- what is the current situation compared to the same period of the previous year or two previous periods of the current year? For example, compared to the beginning of the reporting period, has the problem worsened?
- are the trends stable, improving or worsening?
- are the reported surveillance data sufficiently representative of the reporting site's catchment area? Of all the sites that were supposed to report, what proportion actually reported?
- how timely and how complete were the data received from the reporting sites?
- what period (seasonality) is the event occurring?
- who is affected? Which groups are most at risk?

Each site that receives or collects data must prepare and follow a routine monitoring data analysis plan (refer to ***APPENDIX 3A*** of this section).

This analysis can be done manually or through a computerized system. This section describes the different methods used for the analysis, as well as the steps for interpreting and summarizing the results. The information in this section must apply to all levels of the health pyramid.

3.1. Receiving, handling and storing data from reporting sites

Surveillance data usually flow from reporting sites (HF, entry points, etc.) up to the national level. A reporting site is a site which reports data to the next level. This definition includes all HF (public, private, parapublic and religious), medical laboratories and Points of Entry. Some reporting sites also have reports on signals/events from community-based surveillance.

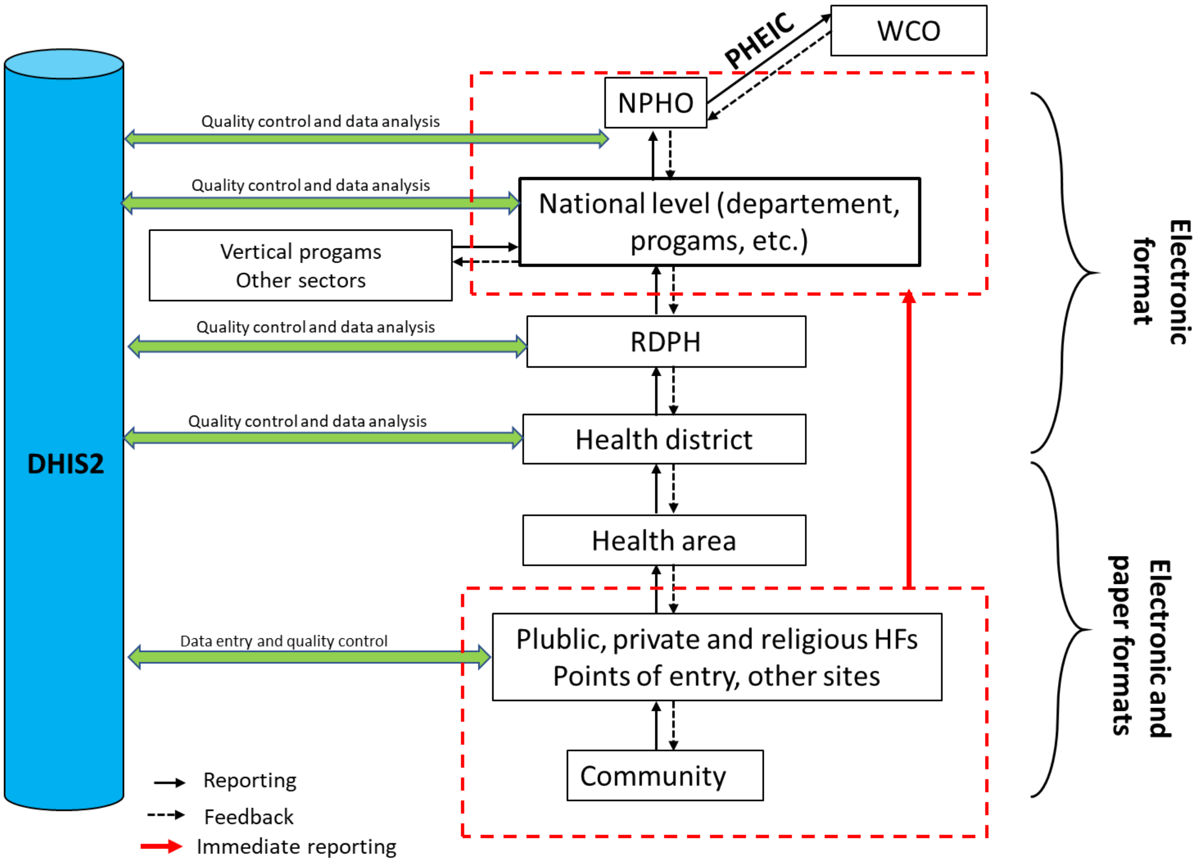
At the HF(HF) level, the outpatient and inpatient services are surveillance sites. The information collected there is compiled in standard forms, processed and forwarded to the HD (HD). In areas where electronic surveillance already exists, data are entered using a cell phone or computer.

At the community level, CHWs, using the community case definitions, notify any situation that may constitute a public health emergency to the surveillance focal point of the health area or health facility. As part of the EBS currently implemented in several regions (Centre, East, Far North, Littoral, North, North West, South and South West), signals are reported by CHWs through logbooks to the HA surveillance focal point. The latter record the signals in the signal logbook and process them by checking and sorting them. Signals that have been transformed into events are reported to the HDs. At this level, the data is compiled, analysed and transmitted to the corresponding Regional Delegation of Public Health, which in turn compiles, analyses and transmits it to the central level. In insecure areas, CHWs report signals through the EWARS application. The District Management Team (DMT) can access the compiled information from a computer (see ***section 9*** on electronic IDSR).

In summary, a Health Area (HA) team collects data from communities and HF in its catchment area, compiles them, and forwards them to the district management team. At this level, the data are compiled, analysed and forwarded to the corresponding Regional Delegation of Public Health (RDPH), which in turn compiles, analyses and forwards them to the national level. The central level in turn compiles, analyses, produces epidemiological bulletins (weekly, monthly depending on the case) and shares them with the various stakeholders. However, some events require immediate notification as shown in Figure 6 below. In case of an emergency situation, the situation report should be sent to the next level with a copy to the higher levels.

Adequate data protection and security must be ensured. Documents containing personal information (health-related and notifiable conditions) should not be left on the desk or in any other location where they can be accessed by unauthorized persons. Hard copies of forms containing personal information must be kept in locked cabinets and in a secure location. Data stored in a computer should be password protected and with restricted access. Network hardware and any back-up copies of notifiable condition data must be password protected or stored in a secure location.

The diagram below illustrates the routine monitoring data flow:



WCO
PHEIC
NPHO
Quality control and data analysis
Quality control and data analysis
National level (departement, progams, etc.)
Vertical progams
Other sectors
Quality control and data analysis
RDPH
Quality control and data analysis
Health district
DHIS2
Health area
Data entry and quality control
Plublic, private and religious HFs
Points of entry, other sites
Community
Reporting
Feedback
Immediate reporting
Electronic format
Electronic and paper formats

Figure 5 : Data flow within the IDSR

In the event of irregularities in the transmission of data or in particular situations, feedback must be given from the higher level to the lower level so that action can be taken or for information (retro-information). Similarly, in particular emergencies (events representing a significant risk to public health), a rapid data exchange system should be established in accordance with the IHR (2005). In case of an emergency, the situation report should be sent to the next level with copies to higher levels. However, the Ministry of Public Health is committed to sharing this status report and information with WHO in accordance with Article 7 of the IHR (2005). This article states: “*If a State Party has evidence of an unexpected or unusual public health event within its territory, irrespective of origin or source, which may* *constitute a public health emergency of international concern, it shall provide to WHO all relevant public health information".*

3.1.1. Data reception

Make a careful record of all received data. The surveillance team at each level of the health pyramid or site receiving the data shall:

- acknowledge receipt of the data or the report.
- record into an appropriate logbook the date and time the data was received, its purpose and the sender;
- record into an appropriate logbook any data set or surveillance report received from any reporting site (see **Appendix 2G**, **Section 2**);
- verify the completeness and timeliness of data or reports;
- review data quality;
- check that the form (paper or computer file) is filled accurately and correctly;
- ensure that the form is fully completed (no blank space);
- verify that the form does not contain any inconsistent data. Verify this data with the notification site (by phone call or message, email...) and correct any discrepancies;
- compile the data and store them in a database

For aspects related to electronic monitoring, see ***Section 9***.

3.1.2. Data entry and cleaning

At each level of the health pyramid, the surveillance team must ensure, in liaison with the Surveillance Focal Person, that data from the various reporting sites on diseases, priority conditions and public health events within the framework of the IDSR have been extracted from the registers and correctly entered into an appropriate form. It is important to identify problems and clean the data before starting the analysis. This is an essential part of data processing. Indeed, disease trends and mapping will be inaccurate without accurate information about the number of cases, the timing of their occurrence, or their geographic location. Every supervision should therefore be an opportunity to educate health care workers about the importance of properly recording all patient information in logbooks or notification forms. Emphasize that patient registries are sources of data for reporting public health information and are therefore likely to play an important role in detecting an unusual event or public health problem.

Ensure that HF staff have mastered the notification process. It is also important to ensure that records, including rumour records, are available. Registries frequently used in most countries are outpatient and inpatient registries. The surveillance manager should always work with the Surveillance Focal Point to extract data on diseases, priority conditions and public health events from the registers.

Data from surveillance can be entered and compiled manually or electronically. Regardless of the method used, the following should be done:

- update the summary totals for each week or month, ensuring that they include only cases or deaths actually reported that week or month. Late notifications from previous weeks or months should be entered in the corresponding week or month and the sums recalculated;
- record a zero when no cases were reported. If a space that should have been filled is left blank or filled in with a dash or "Nil", the higher level may get the wrong idea of the situation, as they will not be able to tell if it is missing data or if no cases have been reported. A “zero” reporting allows the next level to know that surveillance did not detect a case of the particular disease, priority condition and public health event;
- avoid duplication by using the unique identifier on the file or case notification form. This identifier makes it possible to check and prevent the same data from being entered several times;
- establish frequent contacts with the reporting sites in order to address issues of missing information or errors and address inconsistencies detected in the report.
- ensure consistency and harmonization of data.
- ensure that update of information on laboratory results is done by linking to the respective case record unique identifier.

Once surveillance data has been received and entered into the aggregate forms, review it carefully to ensure that no mistakes were made during entry. Indeed, as these data influence decisions about disease prevention and control actions, any error or delay in their capture and processing can have significant ethical, social and economic repercussions. During an outbreak, ensure that data is collected using a line list.

3.2. Analysing data by time, location and person (personal characteristics)

Findings from data analysis may trigger investigations and subsequent response to an outbreak, condition, or public health event. Data should be analysed by time, place and person (***refer to Table V below).***

Table V: Types of analysis, objectives, tools and methods

Type of analysis	Objectives	Analysis methods	Presentation of results
Time	Detect sudden or gradual changes in the occurrence of a disease or unusual event; determine its frequency, seasonality, and the time between exposure and onset of symptoms.	Compare the number of cases reported in the current period with the number of cases reported in a previous period (week, month, season or year).	Present results in a **table, line graph**, **histogram or sequence chart**
Location	Determine the location of cases (e.g., identify a HD or populations at risk for the disease, priority condition or public health event).	Plot the cases on a map and identify clusters of cases or geographic links between the location of the cases and the public health event under investigation.	Plot the cases on a **detailed** map of the HD or area affected by an outbreak. Point density analysis can also be used to represent the number of cases by location. **Note**: Information can also be presented in a table or diagram, but locating cases on a map will enable quick assessment and response.
Person (personal characteristics)	Describe the reasons for changes in the occurrence of the disease, priority condition, or public health event and how they occurred, the profiles of those most at risk, and potential risk factors.	Depending on the disease, priority condition, or public health event, characterize cases according to the data reported for case-based surveillance: age, sex, place of work, vaccination status, schooling, and other known risk factors for the disease, priority condition or public health event.	Extract specific data about the affected populations and summarize it in **a table, bar graph or pie chart**

3.2.1. Data Analysis by time

Data from this type of analysis is usually shown on a graph. The number or rate of cases or deaths is placed on the vertical or y-axis. The period being evaluated is placed along the horizontal or x-axis. Events that occurred that might affect the particular disease being analysed can also be noted on the graph.

The graphs show the number of cases and deaths over a given period. It is easier to observe variations in the number of cases and deaths on a graph, especially when the number of cases and deaths is high or to observe their evolution over time.

The number of cases is measured according to time by making graphs either in the form of curves (trend line) or bars (bar chart or histogram). Below are some examples of graphs (**Appendix 3B** of this section explains how to draw a graph).

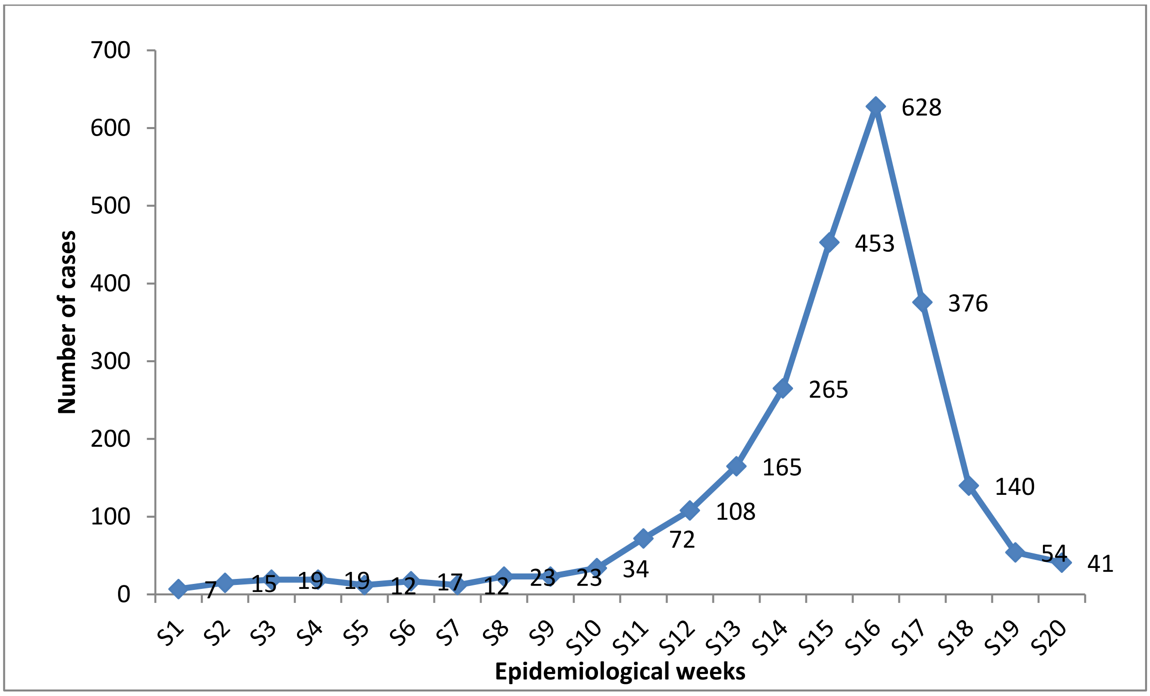


Figure 6 : Cerebrospinal meningitis cases reported in Cameroon, epidemiological weeks 1 to 20, 2011

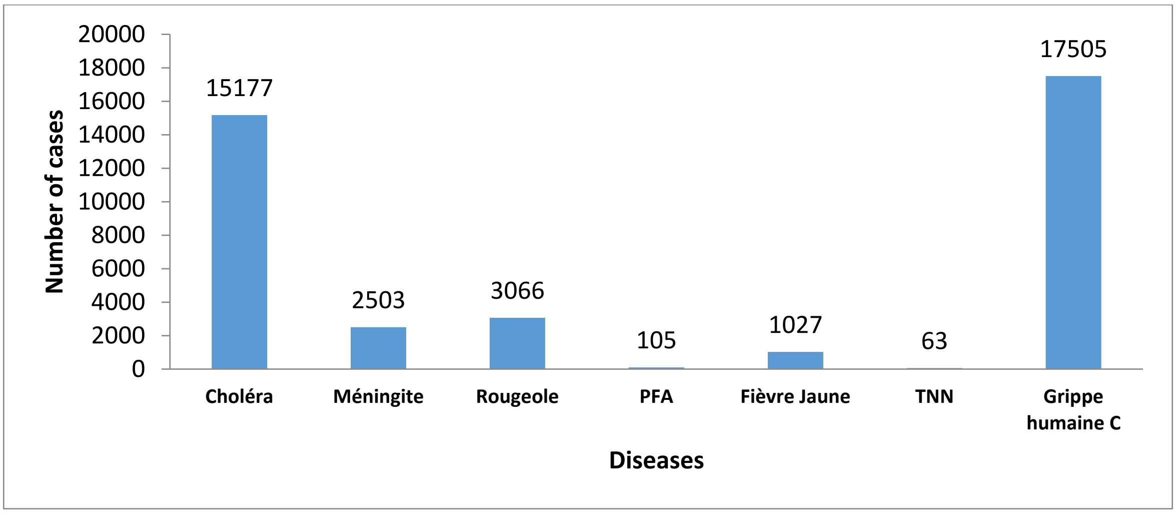


Figure 7 : Distribution of disease cases reported in Cameroon, epidemiological weeks 1 to 31, 2011

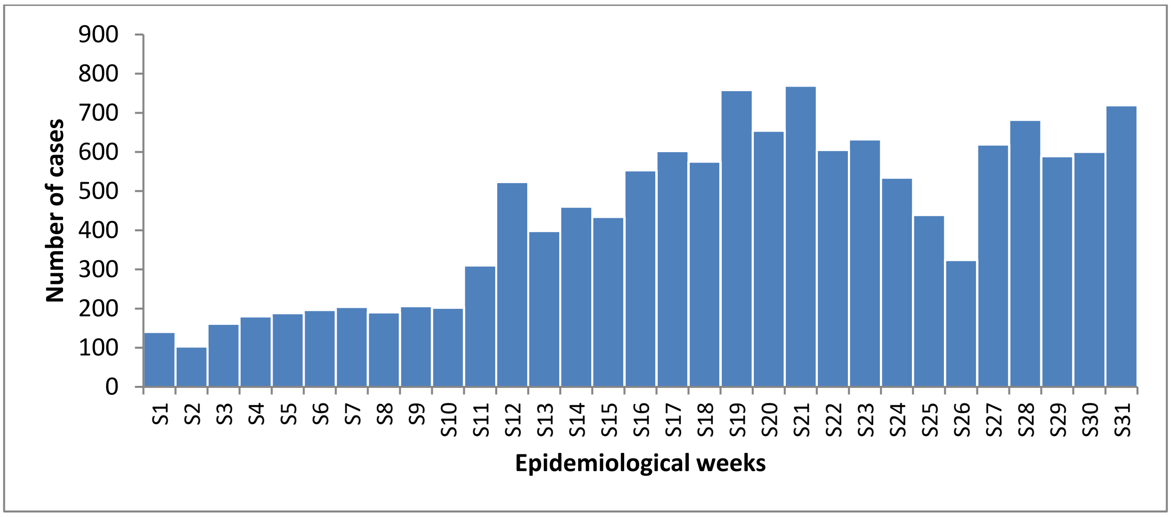


Figure 8: Distribution of cholera cases detected in Cameroon in 2011, epidemiological weeks 1 to 31, 2011

Note: Whatever the type of graph, the following essential elements must be included: the title of the graph, the titles of the different axes (vertical and horizontal) and the legend.

Drawing a histogram

A histogram can be obtained with data from notification forms or line lists. Using symbols, plot each case on the histogram according to its date of occurrence. An outbreak curve appears gradually. The title of the histogram should include the geographic area represented by this curve, which may be the entire HD or only the perimeter served by the HFwhere the cases occurred.

Indicate with arrows the important events on the histogram. For example, from the register of reported outbreaks, locate the following dates:

- onset of the first (or index) case;
- notification to the HD by the HF;
- the first case seen at the HF;
- the beginning of case investigation at the HD level;
- laboratory confirmation of the outbreak;
- the beginning of the response;
- reporting to the higher level by the HD.

The results of this time-dependent analysis allow users of these data to look back at the outbreak and answer questions about which patients were exposed to the disease and the length of the incubation period, the source, the time between detection and confirmation of the outbreak, the mode of transmission of the disease, and the likely timing of exposure to the etiologic agent.

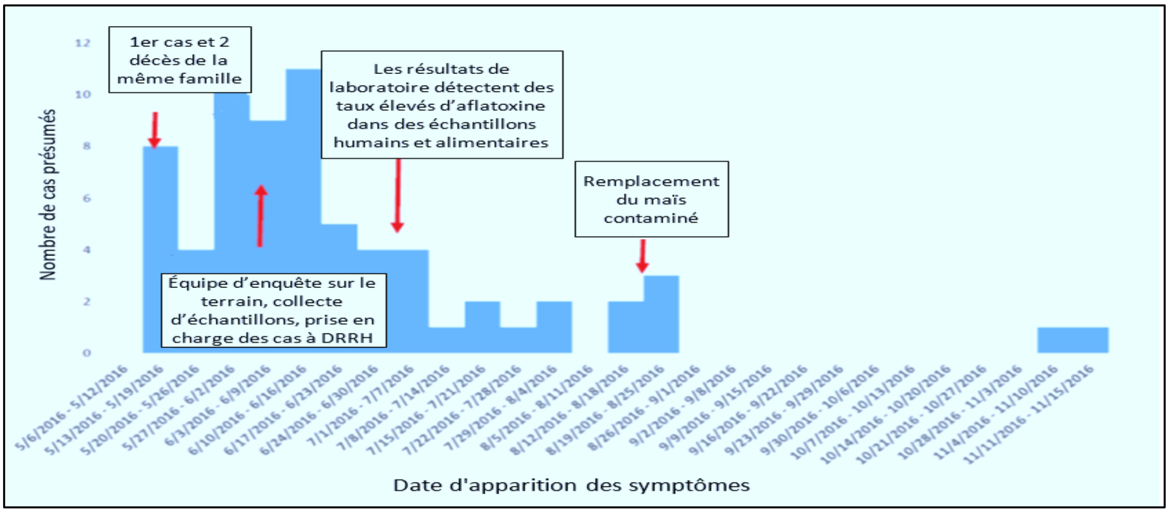


Figure 9 : Cases of aflatoxicosis by date of onset of symptoms, Nyama and Ngémé areas, Cameroon, 2016

3.2.2. Data Analysis by location

Analysing data by place provides information on the location of the disease, the priority condition, or the public health event. Drawing up a detailed map of cases for certain diseases, priority condition or public health event and updating it on a regular basis gives an idea of the answers to the following questions: where, how and why is the disease spreading?

Use the place of residence on the case reporting forms or line list to plot and describe:

- clusters of cases occurring in a particular area;
- travel patterns related to the mode of transmission of the disease;
- sources of infection common to all cases.

Use manual methods or mapping software to produce a map for use in the systematic analysis of disease surveillance data. On a map of the area where the cases occurred, indicate:

- roads, water sources, location of specific populations and other factors associated with the transmission of the disease risk, priority condition or public health event under investigation. For example, for neonatal tetanus, indicate on the map the places where traditional birth attendants work, as well as the health care facilities where mothers come to give birth;
- the most relevant patient locations or geographic characteristics for that disease, priority condition or public health event. For example, in the case of an outbreak of meningitis in young patients, remember to indicate on the map the school the patients attend;

- other relevant places depending on the disease or condition being investigated. Refer to the specific guidelines for diseases, priority conditions, or public health events for specific recommendations for data analysis by place.

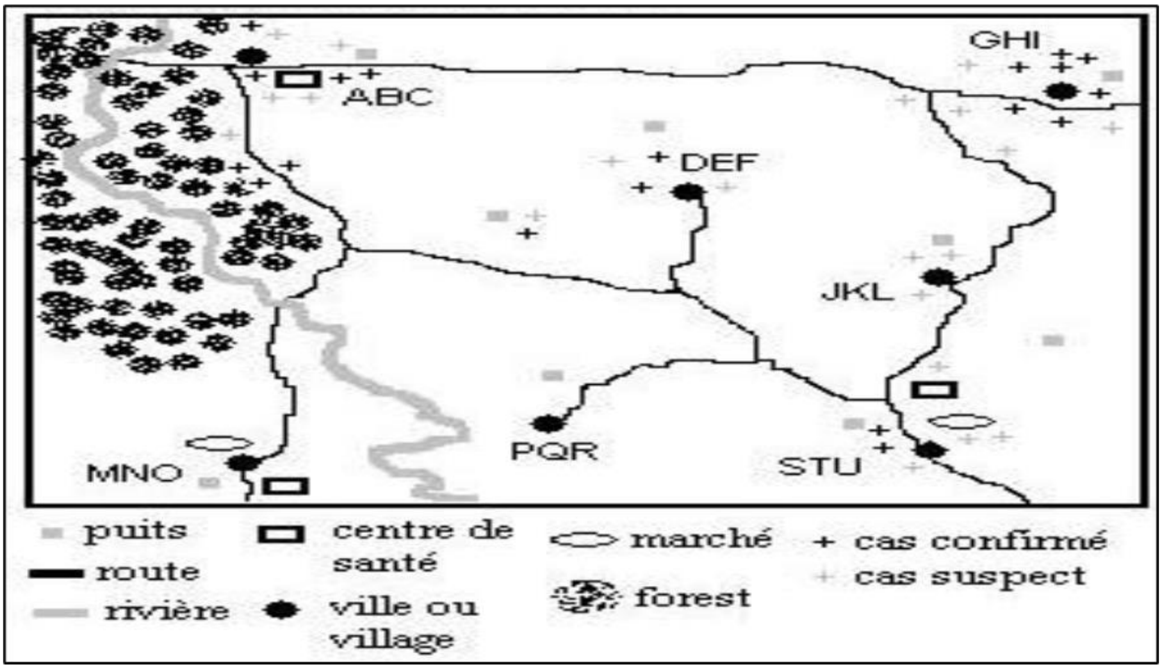

GHI
ABC
DEF
JKL
PQR
STU
MNO
puits
centre de santé
marché
cas confirmé
route
cas suspect
rivière
ville ou village
forest

Figure 10: Distribution of suspected and confirmed cases of disease Y in a HD X at a time T (manual map)

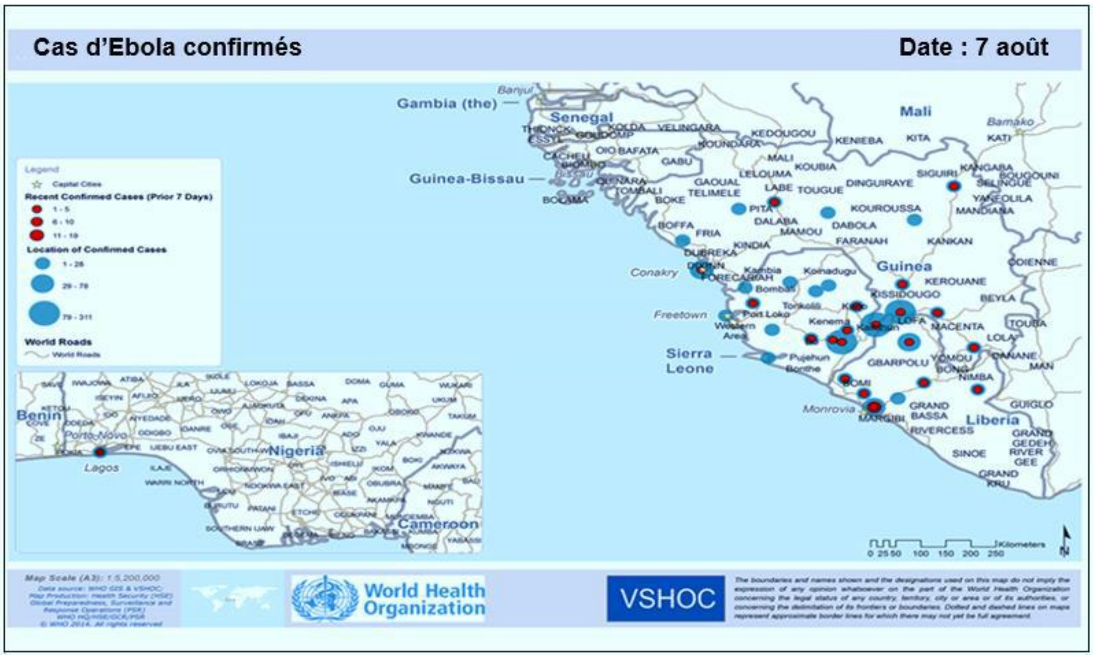

Cas d'Ebola confirmés
Date : 7 août
Gambia (the)
Senegal
Mali
Guinea-Bissau
Guinea
Conakry
Freetown
Sierra Leone
Monrovia
Liberia
Nigeria
Benin
Cameroon
Lagos
Bamako
Recent Confirmed Cases (Prior 7 Days)
Location of Confirmed Cases
World Roads
World Health Organization
VSHOC

Figure 11: Distribution of confirmed Ebola cases in Sierra Leone and Liberia; August 7 2014 (GIS generated map)

3.2.3. Data Analysis by person (personal characteristics)

Analysis by personal characteristics describes both the population affected and/or at risk of contracting the disease, or being exposed to the factors associated with it. These factors can provide insights into the understanding of the disease, the cause of its occurrence and the strategy to control it, thus preventing its further spread.

Cases should be broken down according to each individual variable on the notification form. For example, compare the total number and proportion of suspected and confirmed cases by:

- age group;
- gender;
- trade or profession;
- urban or rural residence;
- vaccination status;
- clinical manifestations;
- risk factors;
- disease outcome;
- final classification.

Use disease-specific information to decide which variables to compare. For example, if data were collected on a malaria outbreak, specify the targets of the National Malaria Control Program (NMCP), compare the number of cases detected by target (children under 5 years of age, people 5 years and older, and pregnant women).

This analysis of data by personal characteristics is generally recommended for describing a population at risk. This analysis is easier when case-based data is available. In this analysis, it is important to:

(a) identify numerators and denominators

A simple case count does not provide all the information needed to understand the impact of a disease at the community, HF, or HD level. Percentages and rates must be calculated to compare the information reported to the HD.

The first step in analysing individual data is to identify the numerator and denominator in order to calculate percentages and rates.

- the *numerator* is the number of specific events being measured, such as the number of cases or deaths attributable to a particular disease (e.g., the number of measles cases that occurred during the year in the population);
- the *denominator* is the total number of events being measured or the size of the population to which it relates, such as the number of people in the population who experienced cases or deaths from a particular disease, or the exposed population.

(b) use simple percentages

The calculation of simple percentages makes it possible to compare information for populations of different sizes. Here are some examples:

Table VI : Confirmed cases of measles in north Cameroon, year 2010

Regions	Confirmed case
Adamawa	34
Far North	38
North	10
TOTAL	82

Looking only at the number of reported cases, it appears that the Far North Region is the most affected by measles. However, if we relate the number of cases reported by each region to the total population, the situation is different.

Table VII : Total population of Northern Cameroon, Year 2010

Regions	Total population
Adamawa	1,015,622
Far North	3,480,414
North	2,050,229
TOTAL	6,546,265

By calculating the percentage of measles cases in the last 12 months in the population, we can compare the incidence of the disease in each region. The numerator is the number of new measles cases in the population, and the denominator is the total population. In this example, the incidence rate of the disease is ultimately higher in the Adamawa Region.

Table VIII: Measles incidence rate in the three Northern regions of Cameroon, Year 2010

Regions	Incidence rate (per 100,000 population)
Adamawa	3.35
Far North	1.09
North	0.49
TOTAL	1.25

Make a table to analyse data by person (personal characteristics)

For each priority disease or condition under surveillance, a table should be drawn to study the characteristics of patients who contract the disease or condition. A table is a set of data presented in a simplified form of columns and rows. This representation is used in disease surveillance to indicate the number of cases and deaths attributable to a given disease, over a given period of time and at a given location.

To make a table, you need:

- choose the type of information to present. Consider the age-specific analysis of measles cases and associated deaths;
- determine the number of columns and rows needed. There should be an extra row at the bottom of the table and an extra column on the right to show the totals. In the example chosen, we will need one row per age group and one column for each variable (age group, number of cases, number of deaths);
- put a title on each row and column, including time estimates;
- Report the total number of cases and deaths, making sure that each number corresponds to the correct row and column;
- give the table a title.

Example of a table: Distribution of Measles cases by age group in a District X at a period T

Age group	Number of reported cases	Number of deaths	Total
0 - 4 years	40	4	44
5-14 years	9	1	10
More than 15 years	1	0	1
Unknown age	28	0	28
Total	78	5	83

3.2.3.1. Calculate the percentage of cases occurring within a given age group

After summing up each age group, we can calculate the percentage of cases that occurred in a given age group. This requires:

- identify the total number of notified cases in each age group from summary data for which temporal and personal characteristics are available (e.g.;40 cases of measles in children aged 0-4 years);
- calculate the total number of cases for the time period or individual characteristic studied (for example, 78 notified measles cases in the table above);
- divide the total number of reported cases in each age group by the total number of reported cases in all age groups (e.g., for the 0-4 age group, divide 40 by 78. Result: 0.51):
- multiply the answer by 100 to calculate the percent. (Multiply 0.51x 100=51%)

Age group	Number of reported cases	Proportion of notified cases in each age group (%)
0-4 years	40	51
5-14 years	9	12
More than 15 years	1	1
Unknown age	28	36
Total	78	100

3.2.3.2. Calculate attack rate

The attack rate measures the morbidity or rate of spread of a disease, priority condition or public health event in a population at risk. An attack rate describes the risk of contracting the disease over a specified period of time, such as the duration of an outbreak. The attack rate is defined as the frequency of occurrence of an event (such as a new disease case) in a population at risk over a specified period of time, and is usually calculated in an outbreak scenario. It is expressed by the number of people at risk; for example: 4.5/100,000 population.

$$\frac{Number\ of\ new\ cases\ over\ a\ given\ period}{Size\ of\ the\ population\ at\ risk\ at\ the\ begining\ of\ the\ period}\ x\ constant\ (such\ as\ 100\ or\ 1000)$$

Example:

16 cases of cholera occurred in a village with a population of 800 inhabitants within the month of July 2020

Attack rate =16/800 =0.02

0.02 x 100 = 2.0, that is, 2 cases per 100 population = 2.0%

During an outbreak, this data will need to be frequently updated (often daily) to see if the information being received changes the hypotheses regarding the causes of the outbreak.

3.2.3.3. Calculate the case-fatality rate

A fatality rate helps to:

- indicate whether a case is identified and managed promptly;
- indicate any problems with case management once the disease has been diagnosed;
- identify a more virulent, new or drug-resistant pathogen;
- indicate poor quality of care or no medical care;
- compare the quality of case management between different catchment areas and HDs;
- assess health-seeking behaviours;
- identify conditions that predispose to severe forms of the disease (e.g., immune deficiency)

Public health programs can reduce case fatality rates by ensuring early case detection and good quality case management. Some disease-specific control recommendations use case fatality reduction as a measure of the effectiveness of the response to an outbreak

To calculate the case fatality rate, we need to:

- calculate the total number of deaths (let's use the previous example of measles: 5 deaths across all age groups);
- divide the total number of deaths (5) by the total number of reported cases (78): 5 ÷ 78 = 0.06;
- multiply the answer by 100: 0.06 X 100 equals 6%.

Age group	Number of reported cases	Number of deaths	Case fatality rate (%)
0-4 years	40	4	10
5-14 years	9	1	11
More than 15 years	1	0	0
Unknown age	28	0	0
Total	78	5	6

Note: See the disease-specific guidelines in ***Section 9*** for recommendations about the essential variables to compare for each disease.

Analysing results according to thresholds to decide on public health actions

Thresholds are markers that indicate when intervention is required. They facilitate monitoring and help program managers answer the question: "When and how to intervene?”

The thresholds are determined based on information from two different sources:

- situation analysis answers the following questions: who is at risk for the disease? What are the risks? When to intervene to prevent the spread of the outbreak? What measures should be implemented? And where do cases occur most often?
- international recommendations made by technical experts and disease control program specialists.

This document discusses two types of thresholds: the alert threshold and the outbreak threshold (or action threshold). For some diseases or conditions, no threshold is established. Nevertheless, the problem must be notified and measures must be taken.

The alert level indicates to health care personnel and the surveillance team that further investigation is necessary. Depending on the disease or condition, the alert threshold may be reached upon notification of a single suspected case (as for a disease with outbreak potential or subject to elimination or eradication measures), or when an unexplained increase in cases or unusual morbidity is observed in weekly or monthly reports over a certain period of time.

The **outbreak** or **intervention threshold** triggers a response. It indicates the existence of specific data or an investigative finding that indicates the need for intervention beyond simple confirmation or clarification of the problem. This may include:

- forwarding the laboratory confirmation to the relevant HFs;
- seting up an emergency response (e.g, vaccination);
- conducting a community awareness campaign;
- improving IPC in HFs.

Depending on surveillance results, several action levels can be proposed. For diseases that are infrequent or subject to eradication measures, the detection of a single case suggests an outbreak. Indeed, in such circumstances, the occurrence of a case is unusual and therefore constitutes a serious public health event, as these rare or eradication-targeted diseases have the potential for rapid spread or high case fatality rates.

In other situations, it will take a number of cases to trigger a response. For example, in countries of the meningitis belt including Cameroon, the alert threshold is defined as 3 suspected cases per 100,000 inhabitants in one week for a population of 30,000 to 100,000 inhabitants. For a population of less than 30,000 inhabitants, this is an incidence of 2 suspected cases in one week or an increase in the number of cases compared to previous non-epidemic years. For a population between 30,000 and 100,000, an attack rate of 10 suspected cases per 100,000 population per week is the epidemic threshold. For a population of less than 30,000 inhabitants, an incidence of 5 suspected cases in one week or a doubling of the number of cases in 3 consecutive weeks, e.g. Week 1: 1 case, Week 2: 2 cases, Week 3: 4 cases. The epidemic threshold is set at 10 suspected cases per 30,000 to 100,000 population per week and less than 30,000 population for 5 suspected cases in one week or doubling of the number of cases in three weeks (at least two cases per week) and the alert threshold is set at 3 suspected cases per 30,000 to 100,000 population per week and less than 30,000 population for three suspected cases per week or incidence compared to previous non-outbreak years.

(Source: *National guidelines for surveillance, preparedness and response to meningitis epidemics*)

The outbreak threshold for malaria in some countries is set at the 3rd quartile of confirmed malaria cases in the last five years; the alert threshold at the 2nd quartile/median of confirmed malaria cases.

In practice, the national level is responsible for communicating action levels for priority diseases to all reporting sites in the health system. This makes it easier to use surveillance data to take action at all levels where it is collected. These action levels are periodically evaluated and revised at the national or international level, based on observed trends in diseases, priority conditions or public health events under surveillance. ***Section 11*** presents suggested action levels for priority diseases and public health events.

Drawing conclusions from the findings to generate information

After data analysis, it is important to generate relevant information for decision-making. This requires:

- routinely (weekly, monthly, or quarterly) collect or present graphs, maps, and charts, and meet with the HD team or relevant stakeholders to review the results of the analysis and discuss the findings;
- systematically review the findings according to the HD analysis plan (see **APPENDIX 3a**), if available;
- correlate your analysis with other sources of data, especially those from animals (domestic or wild) or the environment for a proper interpretation of your findings. For example, if you have diagnosed cases of human rabies, it is important to obtain information from the animal health sector on the status of any bite investigations, animals placed under observation or vaccinated dogs;
- consider the data quality when interpreting results for example:
    - missing data values (completeness per month, per event);
    - inconsistencies (between linked data elements —validation);
    - arithmetic errors (in correlation and aggregation);
    - obvious fluctuations (sharp increase or decrease per month, per event).

In a system where eIDSR is being established it is important to ensure that there are features to improve data quality and these might include:

- data input validation;
- maximum and minimum values;
- validation rules at least, review the findings to:
    - assess whether the situation is improving or not;
    - compare observed data to the expected data;
    - consider possible explanations for an apparent increase in cases;
    - has there been a change in the number of HFs reporting information?

- has there been a change in reporting procedures or surveillance system?
- has there been any change in the case definition that is being used to report the disease or condition?
- is the increase or decrease a seasonal variation?
- has there been a change in screening or treatment programs, in community outreach or health education activities that would result in more people seeking care?
- has there been a recent immigration or emigration to the area or an increase in refugee populations?
- has there been any change in the quality of services being offered at the HF (for example, lines are shorter, health staff are more helpful, drugs are available, and clinic fees are charged)?
- is there an increase or improvement in laboratory testing or diagnostic procedure?
- is there an increased awareness of disease in the public? For example, mass vaccination campaign and awareness of a particular disease will lead to an increase of cases seen in the HF
- are there accumulated cases notified which should have been notified earlier?

3.3. Summarising and using analysis results to improve public health actions

Based on the surveillance results, prepare a concise, action-oriented synthesis and communicate it to all interested parties who need the information. Use tables, graphs, and maps to present results, describe them clearly and concisely, interpret them, and make comments and recommendations.

Formulate conclusions drawn from the results of surveillance data analysis and use them to:

- carry out an investigation to determine the reasons for the increase or decrease in number of cases;
- collaborate with specific disease control programs to intensify surveillance when an alert threshold is crossed;
- mobilize political leaders and the community to obtain more resources, if the increase in the number of cases is linked to a lack of resources.

The information exchange is an important part of surveillance as well as a powerful coordination mechanism, as it motivates the staff who send out the notifications and creates a spirit of partnership through the transparency displayed. It is therefore important to share the results of the analysis and provide timely feedback. See ***Section 7 and 8*** of this document for information and examples related to communication and feedback.

3.4. Appendices to Section 3

APPENDIX 3A Making a plan for routine analysis of surveillance information.

APPENDIX 3B How to manually plot a graph

APPENDIX 3A. Making a plan for routine analysis of surveillance information.

A surveillance data analysis plan must include tables, maps, and graphs. To develop an analysis plan it is necessary to:

1. assess completeness and timeliness of notification. An essential first step in the analysis of monitoring data is to check whether monitoring reports are received on time and whether all reporting sites have reported. This allows the district (or other level) surveillance team to identify silent areas (areas where health events may occur without being reported) or reporting sites that need help with reporting. This step determines whether the data are sufficiently representative of the level concerned.
2. calculate weekly (or monthly) totals for each district (or other level). Update the number of cases and deaths reported from the beginning of the year to the date of the last notification. This summary information makes it easier to describe what happened in the given reporting period.
3. calculate summary totals of cases, deaths, and case fatality rates, since the beginning of the reporting period;
4. use geographic variables (hospitals, place of residence, notification site, neighbourhoods, villages, etc.) to study the distribution of cases by location. This information will facilitate identification of risk areas;
5. analyse trends, at least for priority diseases in the district. Track changes in the number of cases, deaths, and case fatality rates to identify any increases or unusual patterns of disease;
6. validate and analyse data quality. Create a data validation team at all levels. Meetings shall be held periodically to review notifications. The consistency of all notifications sent with the various sources must be checked.

The following page shows an example of a surveillance data analysis plan.

Example of data analysed for cholera in Country A, 2010

Distribution by time

Onset (week)	Outcome		Total number of cases C = (A + B)	Case-fatality rate (B/C) X 100
	Alive (A)	Dead (B)		
26	16	7	**23**	30
27	92	5	**97**	5
28	87	1	**88**	1
29	19	2	**21**	10
32	11	0	**11**	0
33	9	2	**11**	18
Total	234	17	**251**	7

Distribution by place

District	Issue		Total number of cases C= (A + B)	Case-fatality rate (B/C) X 100
	Alive (A)	Dead (B)		
District 1	1	0	**1**	0
District 2	86	6	**92**	7
District 3	147	11	**158**	7
Total	234	17	**251**	7

District	Population (A)	Cases (B)	Attack rate per 100 000 (B /A) X 100 000
District 1	179 888	92	**51**
District 2	78 524	158	**201**

Distribution by individual characteristics

Age group	Outcome		Total number of cases C= A+B	Case-fatality rate (B/C) X 100
	Alive (A)	Dead (B)		
0-4 ans	35	2	**37**	5
05-9 ans	50	5	**55**	9
10-14 ans	28	2	**30**	7
15-19 ans	23	0	**23**	0
20-24 ans	27	1	**28**	4
25-29 ans	24	2	**26**	8
30-34 ans	11	1	**12**	8
35-39 ans	6	2	**8**	25
40 + ans	30	2	**32**	6

Total	234	17	**251**	7

Sex	**Outcome**		**Total number of cases**	**Case-fatality rate**
	Alive	**Dead**		
Féminin	114	8	**122**	7
Masculin	120	9	**129**	7
Total	234	17	**251**	7

APPENDIX 3B. How to manually plot a graph

Plotting a graph	
1.	Decide what information you want to show on the graph.
2.	Give a title by time, place, and person describing what the graph will contain (e.g., *monthly distribution of cases and deaths due to malaria with severe anaemia in patients hospitalized at UHC, year 2019).*
3.	Decide on the range of the numbers to show on the vertical axis: • take 0 as the lowest number; • write numbers, in the increasing order until you reach a number higher than the number of cases; • choose an interval if the numbers you will show on the vertical axis are large.
4.	Label the vertical axis, explaining what the numbers represent.
5.	Label the horizontal axis and mark the time units on it. The horizontal axis is divided into equal units of time. We usually start with the beginning of an outbreak or a calendar period (week, month or year).
6.	Draw bars of the same width
7.	Indicate the number of cases on the linear curve or histogram. For each unit of time on the horizontal axis, match the number of cases marked on the vertical axis. Fill in one square per case or for a number of cases, in the column corresponding to the day the patient was seen in consultation. If you choose to draw a curve, rather than bars or squares, draw a cross or dot where the horizontal and vertical lines intersect. Connect the crosses or dots together to represent the upward or downward trend over time. Indicate deaths by using a different type of line or a different colour.

Section 4: Investigating suspected outbreaks and other public health events

This section describes how to:

- decide when to investigate an outbreak or other public health event following their notification;
- record notifications of outbreaks, health events and rumours;
- check the notified information;
- prepare for the investigation;
- confirm the outbreak or the public health event;
- take immediate response measures;
- analyse investigation results to determine the origin of the health problem.

Introduction

This section presents the steps that should be taken when conducting an outbreak investigation. An **outbreak** is a sudden or unusual increase in the number of cases of a disease or a health event above what is normally expected within a population in a given geographical area over a particular period. When a suspected outbreak or any public health event is detected and notified, several steps need to be followed when conducting the investigation (see ***Figure13***.). Although these actions are listed in order of priority, they are not always conducted consecutively. It is important to be aware of these actions in order to conduct a proper investigation, using common sense and logic to determine when, how often, and to what extent these different actions should be conducted as part of a real investigation. These actions can also be carried out to investigate other public health events at the district level such as outbreaks of chronic or non-communicable diseases.

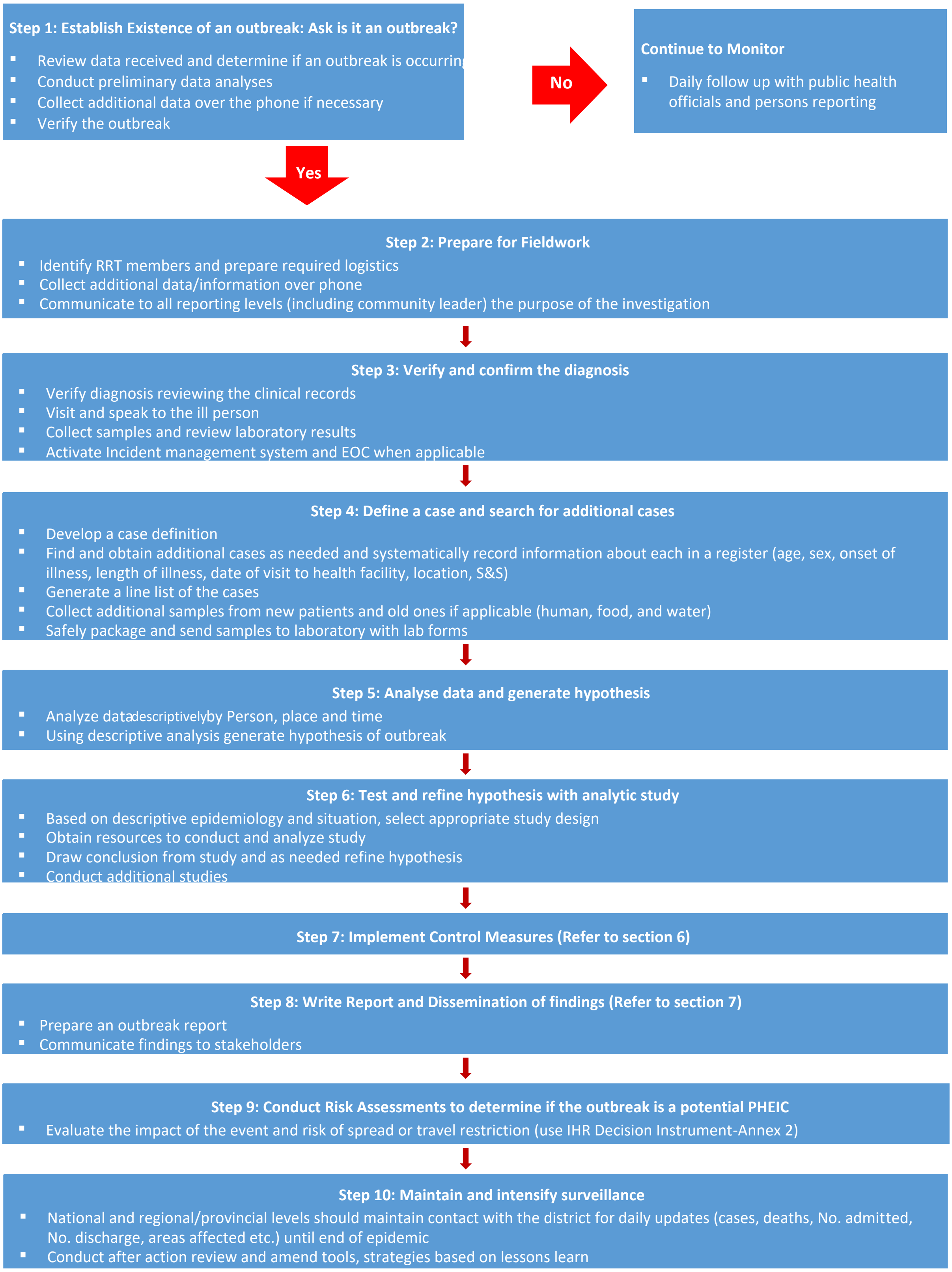
The results of an investigation during an outbreak or other public health event are the identification and evaluation of persons exposed to an infectious disease or affected by an unusual public health event.

4.1. Carrying out an investigation and confirming suspected outbreaks or other public health events

The results of an investigation of an outbreak or other public health event are the identification and evaluation of people exposed to an infectious disease or affected by an unusual public health event.

Its purpose is to:

- verify the existence of the outbreak or public health event and the risks;
- identify and treat additional cases that have not been reported or recognized;
- collect data and specimens for the laboratories to confirm the diagnosis;
- identify the source of the condition or the cause of the outbreak;
- describe the outbreak by time, place and person (personal characteristics)
- describe how the disease is transmitted and the populations at risk;
- identify appropriate response activities to address the outbreak or public health event;
- strengthen control activities to prevent a new outbreak from occurring.



Step 1: Establish Existence of an outbreak: Ask is it an outbreak?

- Review data received and determine if an outbreak is occurrin
- Conduct preliminary data analyses
- Collect additional data over the phone if necessary
- Verify the outbreak

Continue to Monitor

- Daily follow up with public health officials and persons reporting

Step 2: Prepare for Fieldwork

- Identify RRT members and prepare required logistics
- Collect additional data/information over phone
- Communicate to all reporting levels (including community leader) the purpose of the investigation

Step 3: Verify and confirm the diagnosis

- Verify diagnosis reviewing the clinical records
- Visit and speak to the ill person
- Collect samples and review laboratory results
- Activate Incident management system and EOC when applicable

Step 4: Define a case and search for additional cases

- Develop a case definition
- Find and obtain additional cases as needed and systematically record information about each in a register (age, sex, onset of illness, length of illness, date of visit to health facility, location, S&S)
- Generate a line list of the cases
- Collect additional samples from new patients and old ones if applicable (human, food, and water)
- Safely package and send samples to laboratory with lab forms

Step 5: Analyse data and generate hypothesis

- Analyze data descriptively by Person, place and time
- Using descriptive analysis generate hypothesis of outbreak

Step 6: Test and refine hypothesis with analytic study

- Based on descriptive epidemiology and situation, select appropriate study design
- Obtain resources to conduct and analyze study
- Draw conclusion from study and as needed refine hypothesis
- Conduct additional studies

Step 7: Implement Control Measures (Refer to section 6)

Step 8: Write Report and Dissemination of findings (Refer to section 7)

- Prepare an outbreak report
- Communicate findings to stakeholders

Step 9: Conduct Risk Assessments to determine if the outbreak is a potential PHEIC

- Evaluate the impact of the event and risk of spread or travel restriction (use IHR Decision Instrument-Annex 2)

Step 10: Maintain and intensify surveillance

- National and regional/provincial levels should maintain contact with the district for daily updates (cases, deaths, No. admitted, No. discharge, areas affected etc.) until end of epidemic
- Conduct after action review and amend tools, strategies based on lessons learn

Figure 12. The steps of an investigation

4.2. Deciding to investigate a suspected outbreak or a reported public health event

Health Districts (HDs) through their Rapid Response Teams (RRTs) are primarily responsible for investigating suspected outbreaks. Therefore, these guidelines are specifically intended for the HD responsible for organizing the investigation with support from the regional and national level. However, Health Facilities (HFs) and the community (Community health workers (CHWs), dialogue structures) are involved in this investigation.

The decision to conduct an investigation can be made under several conditions:

- the HD receives notification of a **single** suspected case of an immediately notifiable disease (some diseases are particularly dangerous because they can spread rapidly or result in high case fatality rates if not treated promptly; thus, a single suspected case may warrant notification to the higher level, investigation, and initiation of the response);
- there is an unusual increase in the number of cases or deaths during the systematic data analysis;
- alert or outbreak thresholds have been reached for given priority diseases;
- an emerging or re-emerging disease has been identified by the laboratory;
- information about an unexpected or unusual event is reported regardless of the source (administrative, political, religious, traditional authority, media, social networks etc.);
- communities or social media report rumours of deaths or large numbers of cases outside of care facilities;
- a clustered series of deaths with unexplained or unusual causes (e.g., an adult die from bloody diarrhoea, a series of illnesses among health care workers, a clustered series of deaths in domestic or wild animals; e.g., widespread deaths of birds from avian influenza, deaths among livestock from contact with anthrax, unusual abortion among livestock).

When any of the above conditions are met, health care personnel should promptly investigate and treat those affected. They will also prepare to implement broader public health responses. Alert and action thresholds are described in Section 11. HDs should conduct investigations of suspected events no later than 48 hours after notification.

4.3. Recording reports on outbreaks, public health events and rumours

Use forms to report outbreaks, public health events and rumours to the district and ensure that these reports are followed up with action and resolution. Recording these data will facilitate the collection of information to assess the timeliness and completeness of the outbreak investigation and response process.

A model outbreak notification form is included in ***APPENDIX 4A*** of this section. The data from this form should be added to the priority disease trend chart developed by the HDs.

Outbreak alerts should be, as far as possible, recorded and processed using electronic public health event management systems (see ***section 9***).

4.4. Checking the reported information

Since the investigation of outbreaks and public health events requires significant human, logistic and financial resources, it is important to promptly verify that the information is accurate and corresponds to an outbreak, in order to ensure proper resource management.

To verify the information, consider the following factors:

- the source of the information (e.g., is the source of the rumour reliable? is the report from a HF?);
- the severity of the disease or public health event and the use of standard case definitions when reporting;
- number of reported cases and deaths;
- age and gender of reported cases or deaths;
- mode of transmission of suspected pathogen and risk of spread;
- political or geographical considerations;
- the importance of maintaining good relationships with partners and the community;
- available resources;
- the distinction between an event of national or international scope.

Considering the above factors, it may appear that the situation requires a more urgent response than originally expected. Thus, notification of suspected cases of viral haemorrhagic fever is treated more urgently than notification of less virulent diseases because of the high mortality rate and the risk of rapid transmission and spread.

Regardless of all these factors, all suspected outbreaks or public health events (especially those that are immediately reportable) should be reported by HFs within 24 to 48 hours.

4.5. Mobilizing the Rapid Response Team (RRT) during public health emergencies

The RRT is a technical, multidisciplinary team that intervenes in the mobilization and rapid dispatch to lead the field response to a suspected or confirmed public health outbreak or event. The aim is to provide the RRT with appropriate information and data on the event or suspected disease in order to determine the direction of the investigation and the precautions to be taken. If the disease is known, the team should pay particular attention to symptoms, case definition, modes of transmission, diagnostic tests, and disease control measures.

This team must be trained and ready to be dispatched to any public health emergency. The HD surveillance focal person and any other identified and trained health personnel can be part of this team.

Note: **verify and periodically update the immunization status of personnel involved in infectious disease investigation and response activities.**

It is desirable to set up an emergency roster which is a database of Rapid Investigation and Intervention Teams (RIIT). These are qualified health personnel who could be quickly mobilized to perform the following functions:

- coordination;
- surveillance;
- laboratory confirmation;
- clinical case management;
- infection prevention and control (IPC);
- environmental health and sanitation;
- social mobilization and risk communication;
- logistics.

If resources are limited, experts who can perform more than one function within the RRTs could be called upon.

Note: In this text, job titles refer to the functions and not to the persons who perform these functions.

The RRT should be composed ideally of:

- an epidemiologist;
- a clinician;
- a biologist (laboratory technician);
- a psychologist/mobilizer;
- a hygienist;
- a driver.

Other team members will be selected based on the specific characteristics of the outbreak or public health event.

For example:

- an expert in chemicals or nuclear technology or even foodstuffs and pharmaceuticals in cases of suspected heavy metal poisoning;
- a water sector expert in the case of a cholera outbreak;
- an expert in animal or environmental health;
- an anthropologist/social mobilization expert;
- a data manager;
- an expert in infection prevention and control;
- scientific expertise required depending on the type of event.

The next step is to:

- define with the RRT the objectives of the investigation, to collect information essential to the development of appropriate and relevant response measures;
- discuss the actors and stakeholders who will be involved at the community level in the investigation and response to the outbreak or public health emergency within the framework of the national policy;
- use the guidelines and standard operating procedures (SOPs) for the priority disease or condition under investigation (e.g., type of specimen for laboratory analysis, case management guidelines, case investigation forms, line lists).

Note: The HD should have a mobilization plan for the RRTs (names, addresses, phone numbers, location and rotation schedule).

4.5.1. Clarify the respective tasks and roles of the RRT members

Each member of the team should be informed of the tasks and duties he or she is expected to perform during the investigation. Thus:

- the epidemiologist is responsible for the team coordination;
- the clinician is responsible for the case assessment and management;
- the biologist (laboratory technician) ensures the laboratory diagnosis;
- the psychologist/mobilizer provides psycho-social support to the cases, family and community as well as the risk communication;
- the hygienist is in charge of Infection Prevention and Control;
- the driver provides transportation for the team and equipment

However, in conditions where resources do not allow us to mobilize all of these people, one person can fulfil several functions.

4.5.2. Define reporting lines and communication strategy

A communication plan for the teams should be developed. This can be done by preparing a flowchart showing the different levels of authority and how information should flow both within the investigation team and between the district and other levels, including the community level. This involves deciding, for example, who will communicate with the Ministry of Health, the media and the community. Communication methods and frequency of releases to authorities will be defined throughout the outbreak. Consideration will be given to provide daily updates via radiophone, mobile phone, fax, e-mail, or audio conferencing. The levels of responsibility and the role of each team member will be indicated on the diagram. For non-health personnel, their role and how they will be supervised must also be defined.

It is essential to establish communication procedures with the community and key partners to share important information regarding the identification of risks associated with the outbreak or public health event and the response measures.

4.5.3. Decide on the area where the investigation will take place

After reviewing the available information on the suspected disease, including its mode of transmission and risk factors, the geographic boundaries and target population for the investigation will be defined. The search will begin in the most affected area. It is important to differentiate between the district of residence of the case and the district of notification, which may be different and require that actions be taken in both districts. This applies to regions.

Neighbouring HFs should be contacted to see if they have had similar cases or an increase in the number of cases with the same symptomatology. Involve the community and local HFstaff in planning and conducting the investigation. Listen and seek information about the customs, culture and habits of the local population that could undermine the success of the investigation.

4.5.4. Obtain the necessary authorizations

It is important to obtain the necessary authorizations and permissions to conduct the investigation and to respect ethical standards at all levels. It is necessary to obtain not only official authorizations, but also the agreement of influential local personalities in the community.

4.5.5. Prepare forms and methods for data collection and specimens' collection (if required)

Variables must be selected for the identification, recording and analysis of data related to the disease under investigation. A selection of survey forms with the key variables mentioned is appended to this section. Cameroon also has an investigation toolkit that contains several investigation forms for predefined priority diseases. Depending on the responsibilities of the staff, it is important to know how to:

- develop a line list mask that will be used later to synthesize the variables to be used for data analysis based on temporal, spatial and personal characteristics;
- develop appropriate laboratory forms, prepare labelling of specimens taking care to use a unique identification number for each case;
- prepare (and update if necessary) an outbreak curve;
- prepare a tool for geolocation of cases and deaths;
- prepare analysis tables by risk factor, age group, gender, vaccination status, etc.

4.5.6. Organize transportation and other logistical aspects

Transportation to and from the investigation site, as well as travel for the investigation, must be arranged. Also, verify that transportation of the collected specimens to the appropriate laboratories has been arranged prior to the team's departure. Other logistics such as medical supplies, vaccines and personal protective equipment should also be provided.

4.5.7. Gather the necessary equipment for laboratory tests.

Districts may have a response kit containing tools and equipment for use by investigators (including laboratory equipment).

If this kit is not available or the event is unusual, consult disease-specific program guidelines and contact laboratory operations specialists to determine laboratory input standards for collection, packaging, and transportation of required specimens (refer to ***APPENDIX 4B***).

The use of Personal Protective Equipment (PPE) and disinfection equipment is strongly recommended (refer to ***APPENDIX 4C***).

For laboratory equipment, reference should be made to the guidelines for the various diseases in ***Section 11.***

4.6. Verifying and confirming the outbreak or public health event

4.6.1. Take clinical and epidemiological histories.

Examine the patient(s) for signs and symptoms that meet the case definition. Ask the questions to the patient or a family member who can answer on his or her behalf. For example:

- identification (name, age, gender, occupation)
- where do you live?
- when did the symptoms occur?
- are there other sick people at home, at school, at work, in the village, in the neighbourhood?
- have you travelled recently? If yes, where?
- where did you stay during the period before the onset of the illness (week/day) (place of residence at the time of infection)? Has anyone visited you recently?
- who took care of you when you started feeling sick? Have you recently been in contact with sick or dead poultry, birds or animals (for zoonotic diseases)?
- have you been in contact with a sick person or a corpse?
- has anyone recently died in the community where you live?
- did you attend funerals? (Which role did you play)?
- what vaccines have you recently received (for post-vaccination adverse events)?

4.6.2. Collect specimens and obtain laboratory results to confirm the diagnosis

If the disease can be confirmed by laboratory tests, refer to the guidance in ***section 11*** to determine the diagnostic test to be perform and specimen to be collected. The specific laboratory requirements also indicate how to collect, store, and transport the collected specimens and the number of specimens to be collected to confirm the outbreak. See ***APPENDIX 4H*** for how to package the specimens using the triple packing technique. Note that for some conditions, additional sampling of food and environmental specimens may be required to establish the diagnosis and to ensure that these specimens are collected. Example: water sampling for cholera outbreaks and food sampling for food-borne illness outbreaks.

Review diagnostic test results with the investigation team, clinicians and laboratory technicians. Are test results consistent with clinical test results? To clarify certain points concerning the laboratory test results, further assistance can be sought from program managers or technical experts at the national level.

4.6.3. Define the case and search for additional cases and their contacts

Once the first cases have been clinically or biologically confirmed and treatment has begun, it is important to define the case, the contact and to actively find additional cases.

4.6.3.1. Define the case

Once an outbreak is confirmed and the diagnosis is known, it is essential to define what constitutes a case. In section 11, a list of standard case definitions for most priority diseases is available. However, other details can be added in the case definitions of specific outbreaks, namely: geographic area, participation in an event, or travel to a particular place.

In some circumstances, a new (emerging or re-emerging) condition not listed in Section 11 may emerge; in this case, an operational case definition will need to be developed. Common elements of a case definition include information about symptoms, date of symptom onset, laboratory test results, and personal characteristics of the person, place, and time.

4.6.3.2. Define the contact

It is also important to define contacts, especially during an outbreak related to a contagious disease (that can be transmitted from one person to another). This definition will be very useful to list all the contacts that will be monitored afterwards.

4.6.3.3. Search for additional cases

Once the first cases have been clinically confirmed and treatment has begun, additional cases should be actively sought in the community and elsewhere.

4.6.3.3.1. Search for suspected cases and deaths in HFregistries

In HFs that have reported cases, records will be searched for other cases and deaths, trying to identify patients who may have had the same or similar signs and symptoms to the disease or condition under investigation. Ask health personnel to look for similar cases in nearby HFs. The investigation team will actively search for cases in nearby HFs. ***Appendix 4D*** at the end of this section provides guidance for reviewing the registries.

4.6.3.3.2. Search for contacts of suspected cases and deaths in the community

The team will identify risk areas where patients have lived, worked or travelled, including ceremonies, tourism, zoo visits, hunting or animal husbandry areas, she will also discuss with sources of information in the community, notably pharmacists, school teachers, environmentalists, veterinarians (to learn about the animal health situation), farmers, community leaders (ministers of religion, traditional chiefs) and administrative officials.

The choice of search areas depends on the disease, its mode of transmission, and risk factors related to temporal, spatial, and personal characteristics, as well as the results of the registry review. These sites will be visited and people who have had contact or may have had contact with the case will be interviewed. You will also need to:

- ask whether they or anyone they know has ever had a disease or condition similar to the one being investigated;
- check whether another person living in an area close to where the case resides has been ill and has shown signs or symptoms that meet the case definition;
- ask whether there have been any recent deaths. If yes, ask about the signs and symptoms of the deceased;
- find out who cared for the deceased when they were ill and for their remains before and during burial;
- collect information that will help describe the magnitude and geographic extent of the outbreak or public health event;

- transfer newly identified cases to the nearest HFfor treatment.

  See ***Appendices 4E and 4F*** of this section for examples of forms used to record and trace contacts for additional case finding.

4.6.4. Isolate and treat cases as necessary

Based on the case definition, decide on isolation. This is an important measure to limit the spread of the disease. This will require:

- make HFs and other identified sites accessible for isolation and case management with dedicated staff. Depending on the suspected illness, immediate isolation may be required to protect staff, patients and community members;
- ensure that cases in isolation have access to clean water and sanitation facilities according to the case management guidelines;
- strengthen Infection Prevention and Control (by isolating the patient if necessary) and improve case management at home or in a health facility, by providing advice, assistance, materials and algorithms to HFs;
- take appropriate precautions with all patients in the health care facility and in the community, especially in the event of an outbreak of a disease that is transmissible through contact with contaminated objects and biological fluids.

4.7. Making a line list and recording information about additional cases

4.7.1. Make a line list

For each new case recorded in the facility registry or found during active community search that meets the case definition of the disease under surveillance, the information collected should be recorded either on an individual notification form or on a linear list or other specified form. Include geographic coordinates if possible. The registry will maintain primary data useful for cases already known or cases that will be identified over time (see ***APPENDIX 4E*** for a registry model). Record all contacts on the contact listing form and the contact tracing form. Ensure daily monitoring of signs and symptoms of illness during the indicated period ***(see Appendices 4F and 4G)***

Record information for all cases on a "Report Card" ***(see template in Appendix 2A).*** For each case, record at least the following information:

- name and address of the case, village/neighbourhood where he/she lives and his/her complete contact information If he/she does not have a specific address, record contact information in case additional information is needed or to communicate the results of the laboratory analysis and investigation;
- case status: deceased or alive? If deceased, record the date of death;
- age and sex of the case: this information will be used to describe the characteristics of the population affected by the disease;
- date of onset of symptoms and date the case was first seen at the health facility;
- relevant risk factor information such as immunization status if the disease being investigated is a vaccine-preventable disease; or occupation if you suspect that the outbreak targets a particular occupation;
- name and function of the person who transmitted the information: some diseases have more detailed individual investigation forms, which require specific information. Templates for these forms for the investigation of specific diseases are included in the Appendices to ***section 11***;
- case investigation form to be completed for all new cases ***(see Appendix 2A)*** and details recorded on the individual form (***Appendix 4E).***

Note: If the case is deceased, perform a verbal autopsy. Verbal autopsy is the method of collecting information about the symptoms and circumstances of the case's death to determine the cause of death. It is also a technique for collecting and using information about the deceased to reconstitute the events that preceded his or her death and deduce the causes.

The line list may be used as a referral slip to the laboratory when there are more than five cases to be collected. The specimens will then be shipped together.

4.7.2. Record information on contact cases

Contact tracing definition:

It is a process of identifying and tracing any person who has been exposed to a patient or other source of contamination for a defined period of time in order to identify the first signs

Contact tracing is defined as the identification and tracing of individuals who meet the contact definition for a defined period of time*. This is an important component, essential in the control of an outbreak because it strongly contributes to the early detection of cases.

Contact tracing is done from the last day of contact with the patient and ends according to the maximum incubation period (e.g., 21 days for Ebola and 14 days for Covid-19).

Contact Listing

A contact is any person who has been exposed to a confirmed or probable case of a communicable infectious disease or other source of contamination.

Good contact tracing is based on identifying anyone who has been in contact with the source case during the investigation.

The investigation team should establish a preliminary list of contacts based on the interview with the suspected case and with the people who are in contact with him/her on a daily basis. The investigation team should map the itinerary of the case for a period of time prior to the onset of symptoms; this period varies by disease (taking into account infectiousness), until the time of quarantine or formal isolation of the case.

Contact registration

At each step of the trajectory, the case's contacts should be listed with their home addresses, phone numbers, and the type of contact they had with the case. Contacts to be identified in advance include family members, relatives, friends, people who cared for the case or were geographically close to the case, including health workers, neighbours and travel companions.

The contacts are saved on the contact list and the contact tracing list. An itinerary map (case mapping) of the contacts should be drawn up.

4.8. Analysing outbreak data

The methods for analysing outbreak data are the same as those recommended for routine data analysis in ***section 3*****.** Epidemiological data are analysed several times during the outbreak.

During the initial analysis, summarize the situation regarding the outbreak or public health event and look for clues about where the outbreak or public health event is occurring and progressing, its source (e.g., a well or a funeral), and who is at risk (e.g., young children, refugees, people living in rural areas, etc.). Present the data with respect to time, place, and personal characteristics (see ***section 3***) as follows:

- draw the outbreak curve representing the evolution of the disease;
- represent the cases on a detailed map;
- draw up tables containing the main characteristics of the cases (comparing, for example, age groups and vaccination status, distribution by sex, incidence of cases by type of occupation, etc.);
- calculate case fatality rates (refer to the procedures described in ***section 3***);

- also calculate attack rates (refer to procedures described in ***section 3***), in addition to calculating case fatality rates in outbreak situations.

4.8.1. Interpreting analysis results

While identifying potential risk factors related to the outbreak, review the results of the analysis by seeking, for example, an answer to the following questions:

- what is the causal agent of the outbreak?
- what is the source of infection?
- what is the transmission pattern?
- what measures have been taken to contain the outbreak and what is the result?

4.8.1.1. Interpret the time analysis results

Look at the histogram. The shape of the outbreak curve reveals when exposure to the disease-causing agent occurred, the source of the infection, and the incubation period.

- If the curve bends sharply upward and then bends back down just as quickly, the exposure to the causative agent probably lasted a short time. The cases are likely to have a common source of infection.
- If exposure to the common source has lasted longer, the outbreak curve is more likely to level off, rather than peak.
- If the disease is the result of human-to-human transmission, the curve shows a series of increasingly high peaks, separated by periods corresponding to the duration of incubation.
- Below are some examples of outbreak curves and their possible interpretation.

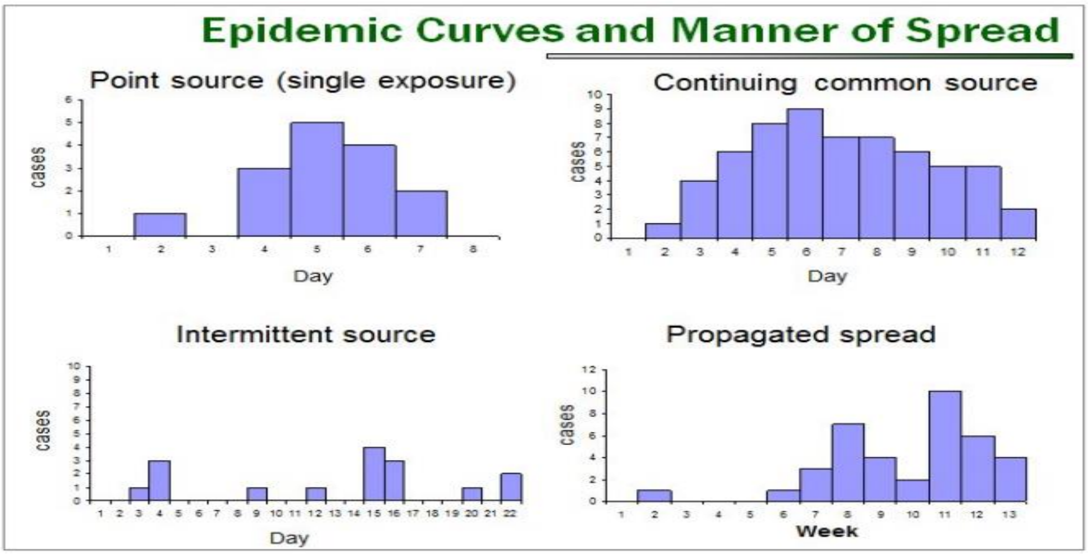


Figure 13: Types of outbreak curves and the manner of spread

4.8.1.2. Interpret analysis results by location

Use the map to:

- describe the geographic scope of the problem and identify areas at risk;
- identify and describe clusters of cases or transmission or exposure patterns. Depending on the agent that contributed to the extension of this outbreak, specify the proximity of the cases to the probable sources of the infection.

4.8.1.3. Interpret analysis results by personal characteristics

Information from the analysis of individual cases is essential for planning the response to the outbreak because it more accurately describes the group(s) at risk for the disease or condition concerned. For example, if cases of yellow fever have occured in children under 15 years of age, the vaccination response should target children of this age group.

Below is an analysis of data by personal characteristics (age), which shows how the results could be used to plan interventions. The table below shows the highest rates of disease among people aged 15 years and older.

Table IX: **Cholera attack rate by age group, Mankhowkwe Camp, Malawo, March-May 1988**

Age range	Number of cases	public	Attack rate (%)
<5	131	5303	2.5%
5-14	261	12351	2.1%
>15	391	12091	3.2%
Total	784	29745	2.6%

Source: Reprinted with the publisher's consent, Moren et al., 1991

4.8.2. Analyse data and formulate a hypothesis

(a) Conduct a descriptive analysis of the data (time, place and person).

- From the observations collected in the descriptive analysis, a hypothesis can be constructed about the causes of the observed trends and the factors that increase the risk for a given outbreak. For example, in ***Table 8*** above, one might make the hypothesis that the older the patient, the more likely they are to become ill. Therefore, it could be determined whether age is associated with the disease.
- Based on a descriptive analysis, formulate a hypothesis to determine the magnitude of an outbreak.

(b) Test a hypothesis and answer the questions of how and why the population was affected.

4.8.3. Verify and confirm the hypothesis with an analytical study

(a) select the appropriate type of study based on the descriptive epidemiology and the situation.
(b) obtain the necessary resources to conduct the analysis and study.
(c) draw conclusions from the study and, if necessary, confirm the hypothesis.

Various study designs can be used to conduct analytical studies. These include case-control studies, cohort studies and experimental studies. ***Appendix 4I*** provides an example of analytical study (case-control study) to test the hypothesis. Refer to the bibliography for more guidance on how to construct analytical study designs.

4.9. Reporting and communicating investigation results

All reports (preliminary, interim, and final) should always be disseminated, even if no critical risk factors have been identified for a given outbreak. Also, prepare situation reports (SitReps) of the outbreak concerned and distribute to relevant stakeholders. ***Section 7*** describes the different channels of communicating during an outbreak.

If risk factors are already known, draw conclusions and make recommendations regarding the outbreak:

(a) confirmed situation: it is an outbreak or a public health event;
(b) affected and at-risk population;
(c) possible causes of the outbreak or public health event, laboratory results, source of infection, transmission pattern, attack rate, case fatality rate, and possible risk factors;
(d) measures already implemented to contain the outbreak;
(e) recommendations: further investigations or studies may be recommended to bring the situation under control.

The district RRT should immediately prepare an outbreak investigation report. This detailed investigation report should be distributed immediately to the HFwhere the outbreak occurred, as well as to the district, region, national level and WHO.

See ***Appendix 7A*** in ***Section 7*** for a sample plan for writing the investigation report.

To understand the spread of the disease, transmission patterns must be described, starting with the probable sources of infection. In addition, transmission patterns provide insight into the contribution of different environments to the spread of the outbreak in a given geographic area, and are therefore essential for regulating the transmission of the infection and adopting measures to contain the outbreak. Reconstitution of a transmission chain or transmission patterns are possible if information is drawn from a priority list and the chronology of dates of infection or contact with other cases, field investigations, and rapid risk assessment are examined. Transmission profiles are very important because they facilitate the investigation of transmission itinerary in a given geographical area, as well as the planning of interventions. Transmission profiles should be regularly updated and if a new cluster of cases appears in any part of the district, questions should be asked to see if there is any connection. See ***Appendix 4J*** for a sample description of the transmission profiles.

4.10. Implementing measures to prevent and control outbreaks

Once an outbreak is identified, prevention and control measures are needed to interrupt disease transmission and limit exposure to the source of infection. If a suspected pathogen or other source of the outbreak is identified, these actions should be targeted at specific agents, sources, or reservoirs of infection. ***Section 11*** provides a description of some of the control measures for each priority disease and the bibliography for all useful purposes.

The measures implemented for the prevention and control of outbreaks aim to:

- identify the source of the infection;
- interrupt the secondary transmission;
- contain future outbreaks.

Note:

- Outbreak prevention and control measures should be implemented first at the beginning of the investigation and should run concurrently with other stages of the investigation. Often, non-specific measures can be implemented regardless of the type of disease or source.

- Multi-sector engagement is essential throughout the response, especially at the community level and with other non-health stakeholders who may play a critical role in managing particular outbreaks. For example, in order to enforce the regulations, it may be necessary to seek the assistance of the Ministry of Territorial Administration and of the General Delegation for National Security.
- At some point during the outbreak, the public health response could include testing of potential new countermeasures, including vaccines and therapeutic substances. Thus, biomedical research can be seen as an important but distinct aspect of the response. Public health efforts must always be at the forefront of the overall response to the outbreak. Research must be scientifically and ethically sound in order to reach conclusions about effectiveness and safety as quickly as possible. In such situations, it is the responsibility of the authority at national level to consult with local ethics committees to provide guidance for similar principles in the event of an outbreak.

4.11. Conducting an assessment to determine if the event is a potential Public Health Emergency of International Concern (PHEIC)

Risk assessment should be initiated as soon as possible by the designated investigation team to address the following questions:

- is the public health impact of the event serious?
- is the event unusual or unexpected?
- is there a significant risk of international scope?
- is there a significant risk of international travel or trade restrictions?

The national authority may be called upon to participate in the risk assessment to determine whether the event constitutes a potential Public Health Emergency of International Concern (PHEIC), which would warrant notification (refer to the IHR (2005) Decision Instrument at https://www.who.int/ihr/revised_APPENDIX2_guidance.pdf

4.12. Maintaining and intensifying surveillance

national and regional authorities should remain in contact with the district to receive daily updates (cases, deaths, number of people admitted to HFs, number of people discharged, affected areas, etc.) until the end of the outbreak. It will be necessary to:

- ensure that the same IDSR mechanism is used to improve outbreak surveillance and that the system is flexible enough to allow the adaptation of other variables to be collected using the existing system. This will prevent parallel reports that may create confusion about the evolution of the outbreak;
- report periodically on the status of the response and prepare daily situation reports that can be used to evaluate the response;
- update line lists, perform data analysis based on time, place and person;
- monitor the effectiveness of the outbreak response activity.

It is important to intensify surveillance during the investigation with neighbouring districts to ensure that the outbreak does not spread to another district, to share information and to plan joint surveillance and response activities. Neighbouring districts may also consider establishing cross-border disease surveillance and response committees to share surveillance data, epidemiological information, and other related information during the outbreak.

4.13. Conducting regular risk assessment after the outbreak confirmation

As soon as the outbreak is confirmed, regular assessment at each stage of the outbreak must be conducted to orient and focus interventions. The risk assessment should include:

- assess the susceptibility of the population to diseases and the potential for the spread of the outbreak both in the affected areas and in the surrounding areas;
- assess the risk of future transmission, morbidity and mortality.

The factors to be considered for this purpose are:

- demographic characteristics such as size, density, movement and diversity
- under-five mortality rate;
- the period of the year (taking into account the risk of seasonal outbreaks) and plans for any festivities or other social events that may increase the possibility of spread;
- access to health services, etc.

Risk assessment must be repeated as new information becomes available. It can also be done on a regular schedule. For some public health events, several risk assessment teams could collaborate to collect information to get a clear picture of the risk (e.g., clinical severity, transmission dynamics, and measures to contain the outbreak). At the end of the outbreak, all risk assessments should be reviewed. Systematic analysis of well-documented risk assessments identifies areas where improvements could be made in the management of acute public health events.

4.14: APPENDICES TO SECTION 4

Appendix 4A	District Register of Suspected and Rumoured Outbreaks
Appendix 4B	Checklist of laboratory supplies needed for investigation
Appendix 4C	List of recommended personal protective equipment
Appendix 4D	How to conduct the register review
Appendix 4E	Sample line lists
Appendix 4F	Contact Listing Forms
Appendix 4G	Contact Location Form (Tracing)
Appendix 4H	Types of triple packaging of specimens during an outbreak
Appendix 4I	Example of an analytical study to verify the hypothesis
Appendix 4J	Example of a transmission profile

APPENDIX 4A. Health District Register of Suspected and Rumoured Outbreaks

Record information provided verbally or written about suspected outbreaks, rumours or unexplained events from HF, the community or social media. Record actions taken and response activities implemented.

Disease or event	Rumor source (Media, phones, social networks, community)	Number of notified cases	Number of notified deaths	Place	Date of notification to the District	Date of investigation by the HD	Result of the investigation carried out by the HD	Date of outbreak onset	Index case notification date	Outbreak threshold crossing date	Consultation date of the index case at the health facility	Date of index case management	Treatment received	Date of outbreak notification	Date of response	Comments and observations	Name & signature

APPENDIX 4B. Checklist of laboratory supplies needed for investigation

For using standard safety precautions when collecting and handling all specimens:

_____ Pieces of bar soap for hand-washing

_____ Bleach for decontamination

_____ Supply of personal protection equipment (gloves, masks, gowns, etc.)

_____ Triple package and refrigerant for sample transportation,

_____ Safety boxes for collecting and disposing of contaminated supplies

_____ Equipment (Biosafety cabinet)

For collecting laboratory specimens :

Blood

_____ Sterile needles, different sizes

_____ Sterile syringes

_____ Vacutainers

_____ Test tube for serum

_____ Antiseptic skin disinfectant

_____ Tourniquets

_____ Transport tubes with screw-on tops

_____ Transport media (Cary-Blair, Trans-Isolate, Viral transport medium (VTM))

Blood films (malaria)

Sterile or disposable lancet

Glass slides and cover slips

Slide box

Respiratory specimens

_____ Swabs

_____ Viral transport medium

Cerebrospinal fluid (CSF)

_____ Local anaesthetic

_____ Needle and syringe for anaesthetic

_____ Antiseptic skin disinfectant

_____ Dry tube

_____ Slide box for microscope

_____ Trans-isolate transport medium

_____ Latex kit

_____ Gram stain

_____ May Grunewald Giemsa Kit

Stool

_____ Stool containers

_____ Rectal swabs

_____ Cary-Blair transport medium

Plague

_____ Gram stain kit

_____ Rapid diagnostic test (dipsticks AgF1)

_____ Cary-Blair transport medium

If HFhas a centrifuge:

_____ Sterile pipette and bulb

____ Sterile glass or plastic tube, or bottle with a screw-on top

For packaging and transporting samples:

_____ Cold box with frozen ice packs or vacuum flask

_____ Cotton wool for cushioning sample to avoid breakage

_____ Labels for addressing items to lab

_____ Labels for marking “store in a refrigerator” on outside of the shipping box

_____ Case forms and line lists to act as specimen transmittal form

_____ Marking pen to mark tubes with patient’s name and ID number (if assigned by the district)

Reagents and supplies for testing

_____ Reagents

_____ Media (MacConkey, Blood agar)

_____ Others

Appropriate personal protection equipment (PPE) (for all EPR diseases such as VHF, suspected avian influenza, etc.)

In some events which present with fever, it might be important to carry rapid diagnostic kits for malaria (mRDT)

APPENDIX 4C. List of Recommended Personal Protective Equipment (PPE)

The following protective equipment shall be made available to all personnel investigating a suspected case of highly contagious disease; for example, viral haemorrhagic fever, bird flu, etc. (See bibliography for guidelines and PPE selection at the end of the section). This equipment should be provided by the regional or HD health authority in case PPE kits are inappropriate. PPE should be available in regions or districts identified as a high-risk area for the occurrence of these specific diseases. See ***Appendix 5A*** for other equipment that may be needed in the response to a suspected outbreak.

Composition of the PPE	WHO hygiene kit
1 surgical gown	100 surgical gowns
1 suit	100 suits
1 hat	100 hats
2 pairs of safety glasses	50 pairs of safety glasses
1 pair of rubber gloves	100 pairs of rubber gloves
1 N95 face mask	200 N95 face masks
1 overboots*	0
1 box of 50 pairs of test gloves	800 boxes of 50 pairs of test gloves
1 re-usable plastic apron	20 re-usable plastic aprons
1 pair of rubber boots	20 pairs of rubber boots
1 hand-held sprayer	2 hand-held sprayers of 1.5 litres each
1 back pack sprayer	1 back pack sprayer of 10-12 litres
Specimen holder	
Roll of scotch tape	3 rolls
Anti-fog for goggles	3 bottles
Bleach	

Note: ***Depending on the type of disease or epidemic-prone condition (Ebola, COVID-19, bird flu etc.), it is important to make PPEs available in regions and HDs***

-Bleach can be locally purchased; Ziploc bags should also be bought for PPE waste management.

*** Not essential**

APPENDIX 4D. How to conduct registers review

1. Background

Register review is gathering information on cases admitted to the HFduring a given period. The information obtained will be used to determine the origin of the outbreak, to search for missed cases or for an increase in the number of cases. We will examine:

- reception and emergency registers;
- consultation registers;
- registers of hospitalization services;
- registers of major reference hospitals or university hospitals with paediatric services, which receive patients referred by other HFs;
- registers from neighbouring HFs serving isolated areas and high-risk populations (e.g., nomadic groups, refugees) or areas without regular health services.

2. Explain the purpose of the register review to HF staff.

Explain the purpose of this record review to the HF managers. The information obtained will help the HD and the HF determine the most appropriate action to limit the risk of the outbreak spreading and prevent new cases from occurring. Emphasize that this activity is not intended to evaluate the performance of health workers, but to collect information.

3. Organize the register review process

Agree on a period of time when the staff in charge of records will be available to provide assistance or answer questions.

4. Identify information sources

Depending on the disease, condition, or public health event being investigated, check the hospital registers for the paediatric and outpatient departments during the visit. The hospital registry for the paediatric ward is an excellent source of information, as it lists all children admitted to that ward. Annual summary reports are not always accurate, and outpatient department registers often list only presumptive diagnoses.

Review the system and procedures used by health workers to record diagnostic information. Verify that the data needed to investigate a suspected case are available. The register should at least include the following:

- patient's name, age, sex and address;

- signs and symptoms;
- date of symptom onset and outcome (e.g., date of death, if applicable); - immunization status, if relevant to the disease.

If the HF has not kept minimum information, discuss with managers how to improve data recording.

5. Perform register review on the scheduled day and time

Visit the selected departments at the agreed time. During the visit, search the registers for cases and deaths that meet the case definition of the disease under investigation. These cases and deaths should meet the standardized definition of suspected cases. Then determine whether the suspected cases have been investigated and reported according to national guidelines.

6. Report identified suspected cases on a line list

Record information about suspected cases. This information will be used in case of investigation activities.

7. Provide feedback to staff in the HF

Discuss the results of the record review with the HF manager. Simultaneously, review all aspects of the management of patients affected by the disease with the HF staff. Emphasize the importance of immediate notification and investigation of cases as a preventive tool for priority diseases and conditions. Take the opportunity to emphasize the importance of infection prevention and control and the use of minimal PPE.

8. Report all suspected cases to the next level

Report suspected cases according to local procedures. Further investigate cases to determine disease risk factors. Develop an appropriate case-based response.

Appendix 4E. Sample Line list

Note: - The line list is based on the major signs and symptoms of the suspected disease

- This line list may be revised depending on the suspected disease or condition (COVID-19, Cholera, Monkeypox, EVD, Meningitis, VPD, etc...)

SE	N°	Name	HD	HA	Village or neighbourhood	Age (in years)	Sex (M/F)	Profession	Date of symptom onset	Date of consultation at the health facility	Signs/ Symptoms 1	Signs/ Symptoms 2	Signs/ Symptoms 3	Sampling done (Y/N)	Laboratory result	Hospitalisation (Y/N)	Place of hospitalisation	Treatment received	Outcome (death, discharged, escaped)	Risk factors	HF that have notified
1																					
2																					
3																					
4																					
5																					
6																					

APPENDIX 4F: Contact Listing Form

Contact Registration Form 1______________________________

Patient's Name______________________ Number of cases______________________

Village/neighbourhood ______________________ Community head or leader ______________________

District/town ______________________ Region ______________________

Date of symptoms onset ______________________ Hospitalisation/Notification in the community ______________________

If hospitalised, hospital address ______________________ Hospitalisation date: ______________________

Name	First name	Relationship with the patient	Health personnel (Yes/No), If yes, indicate health facility?	Age (years)	Sex (M/F)	Telephone number	Head of the household	Village/ neighbourhood	Community head	District/town	Type of contact (1, 2 or 3, make a list)	Date of last contact	Date of last monitoring	first visit	Result

[1] **Any of the following are considered contacts:**

1. having slept in the same house with the case;
2. who have had direct physical contact with the case (living or deceased);
3. who has touched his linen or body fluids;
4. who has eaten or touched a sick or dead animal?

APPENDIX 4G: Contact tracing

Contact tracing form – by the village team ………………………………………. Volunteer name ....

Village Head …………………………………………………………. or Community Head..

District/Town.. Region ....

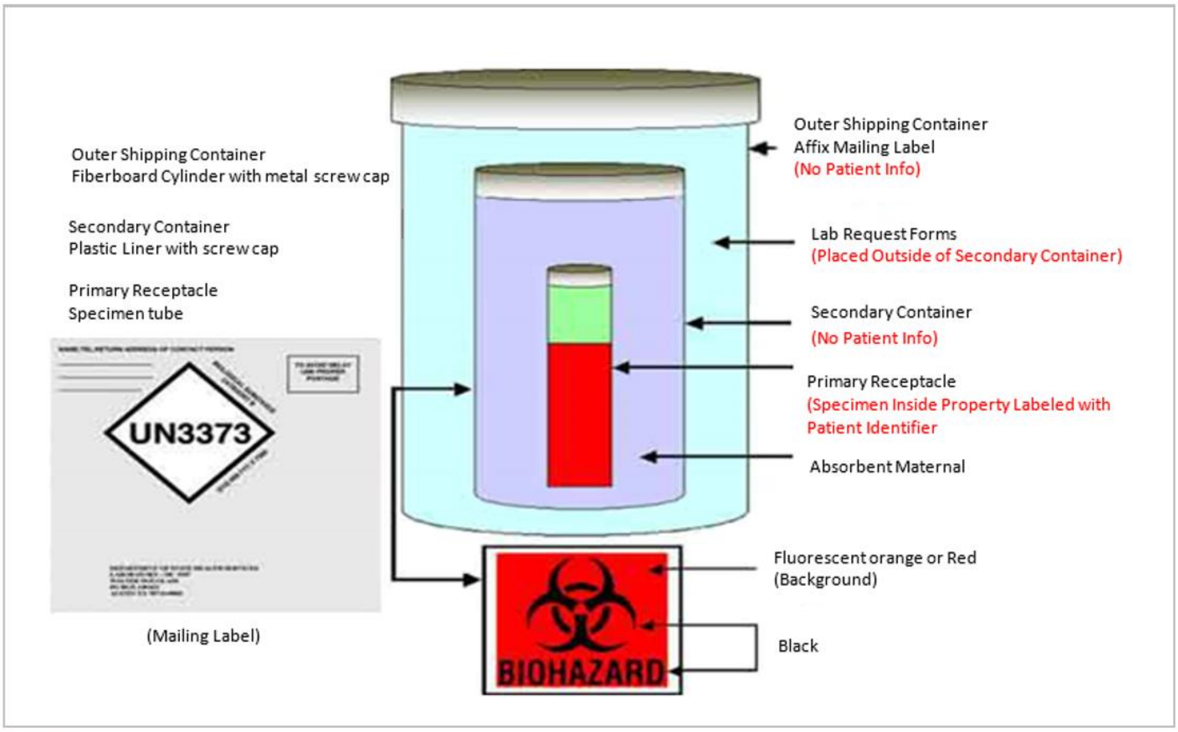
Case ID	Full name of the contact to monitor	Age	Sex	Telephone Number of the contact to monitor	Date of last contact	Symptoms to monitor	Date of first monitoring day: ____/____/____________ Date of last monitoring day: ____/____/_____________													
							1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**
						Symptom 1														
						Symptom 2														
						Symptom 3														
						Symptom 4														
						Symptom 5														
						Symptom 6														
						Symptom 7														
						Symptom 8														
						Symptom 9														
						Symptom 10														
						Others														
						Sampling 'yes/no)														

Note: - The number of days of contact monitoring depends on the disease and the date of last contact with the case

- This contact tracing form can be revised according to the disease or condition (EVD, COVID-19, VPD, etc.)

Code**:** **X=presence of signs** **0=absence of signs** **- = not seen**

APPENDIX 4H: Types of triple packaging of specimens during an outbreak


Outer Shipping Container
Fiberboard Cylinder with metal screw cap
Secondary Container
Plastic Liner with screw cap
Primary Receptacle
Specimen tube
UN3373
(Mailing Label)
Outer Shipping Container
Affix Mailing Label
(No Patient Info)
Lab Request Forms
(Placed Outside of Secondary Container)
Secondary Container
(No Patient Info)
Primary Receptacle
(Specimen Inside Property Labeled with
Patient Identifier
Absorbent Maternal
Fluorescent orange or Red
(Background)
Black
BIOHAZARD

Source: https://medicine-science-and-more.com

APPENDIX 4I: Example of an analytical study to verify the hypothesis

In a case study aimed at determining potential cholera exposures in the Central African Republic, unadjusted comparative analysis indicated that individuals who ate cold cassava leaves (one of the staple foods in the region (odds ratio (OR) = 3.07 with 95% confidence interval = 1.155; 8.163]; P = .020) were more likely to have cholera. This hypothesis was highly plausible at P <0.05.

Risk factors	**Odds ratio**	**Confidence interval at 95 percent**	**P value**
Drinking water from the Ubangi River	1.16	[0.415 ; 3.239]	0.983
Drinking water sold on the street	0.25	[0.027 ; 2.421]	0.422
Eating cold cassava leaves	3.07	[1.155 ; 8.163]	0.020
Eating hot cassava leaves	0.57	[0.090 ; 3.669]	0.900
Attending funerals as from September 2011	0.56	[0.192 ; 1.643]	0.627
Wash your hands after using the toilet	0.85	[0.295 ; 2.493]	0.395
Not eating at home	0.66	[0.259 ; 1.713]	0.206
Eating dried meat	0.45	[0.184 ; 1.208]	0.062
Eating raw meat	0.41	[0.143 ; 1.228]	0.060
Eating hot smoked fish	0.83	[0.328 ; 2.111]	0.354
Eating cold smoked fish	0.89	[0.360 ; 2.235]	0.410
Wash your hands before eating	1.05	[0.318 ; 3.512]	0.466

Excerpt from the site consulted

https://www.cdcfoundation.org/sites/default/files/upload/pdf/2011CholeraOutbreakReport.pdf

APPENDIX 4J: An example of a transmission profile

Consider the following scenario, which describes an outbreak of respiratory disease, for which the investigation team had information on 13 cases.

(a) The first case was a 25-year-old university student, who presented the first symptoms on March 21, 2012. He was admitted to Zarqa public hospital on April 4, 2012 after a week of cough, fever, and shortness of breath. The patient was diagnosed with pneumonia and pericarditis, and he was quickly transferred to the coronary care unit. As his condition worsened, he was transferred to Prince Hamzah Hospital for further treatment; the next day, he was intubated in the ICU and died on April 25, 2012. The investigation team learned that during his illness, the patient was in close contact with his mother (who had not reported illness) and two health workers (cases 2 and 3). His illness was subsequently confirmed in the laboratory as the new coronavirus (3).

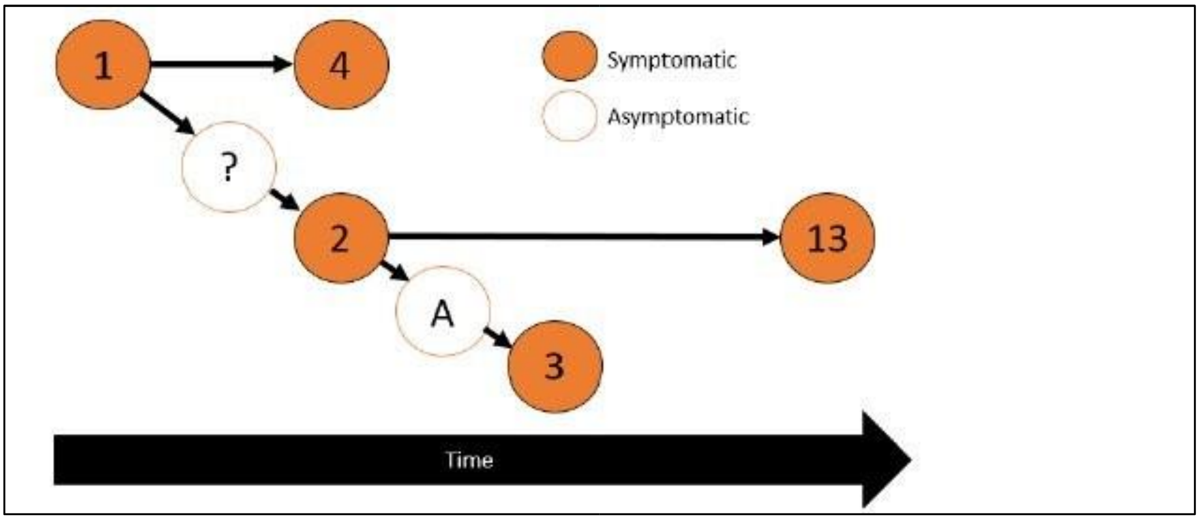
(b) The second case was a 30-year-old nurse from the intensive care unit of Zarqa hospital. His symptoms started around March 29, 2012. He had not travelled or been in contact with animals in the 10 days prior to his illness, although he had frequent contact with the first case that occurred in the ICU. On April 8, case 2 was admitted to Zarqa ICU with shortness of breath and pneumonia. He was later discharged without any sequelae from the Islamic hospital on April 23. The patient was in close contact with two family members, including his mother (case 13) and a man who was not ill (who was also the brother of case 3)

(c) Case 3 was a 40-year-old nurse from Zarqa hospital whose illness was confirmed by a laboratory after her death. Her symptoms started on April 2, 2012, and she was admitted to the intensive care unit of Zarqa Hospital after developing pneumonia 7 days later. She was then transferred to the ICU of the Islamic Hospital where she died on April 19. During her illness, she was in direct contact with 4 family members, including another brother who fell sick 10 days after exposure (case 9) and three others who were not affected. A month before her illness, her sister had travelled from Saudi Arabia (3).

(d) Case 4 was a 65-year-old male physician whose symptoms of fever and fatigue began on April 2, 2012, and progressed to pneumonia. The physician chose to stay home during his illness and quickly recovered. He had not travelled or been in contact with animals in the 10 days prior to his illness. His family members did not report any illness (3).

(e) Cases 5 to 13 occurred during the second phase of the outbreak, with symptom onset between April 11 and 26, 2012. All except case 13, who was the mother of case 2, had direct contact with one or two of the laboratory-confirmed cases. None of the health care workers travelled or had contact with animals. They reported that they only used gloves to care for patients to avoid stigmatising them.

Based on this information and a list, a transmission profile can be designed as follows:



Note: Excerpt from "Public health case study scenarios used for training public health professionals. Case studies developed under the CDC/AFENET agreement." Transforming Public Health Surveillance. (pages: In press). Location: Elsevier

Section 5: Preparing to respond to outbreaks and other Public Health Events

This section describes how to:

- create the Public Health Emergency Management Committee (PHEMC)
- establish a Public Health Emergency Operations Center (PHEOC);
- set up a Rapid Response Team (RRT);
- prepare an outbreak and health emergency preparedness and response plan;
- build up reserve stocks of drugs, vaccines, reagents and supplies;
- map the risks of outbreaks and other health events.

Introduction

A public health emergency, outbreak or health event, requires an immediate response. The HD, regional and national authorities must therefore be ready to detect and respond to such a situation; this is one of their essential missions. This involves identifying key members of the Public Health Emergency Management Committee (PHEMC), mapping risk areas and available resources, and estimating and meeting needs. If this is done in advance, the health care system will be able to respond promptly and effectively to minimize the risk of death and disability caused by emergencies.

This section describes the steps for organizing preparedness activities that need to be in place within the health system and that can be guided by an emergency preparedness and response plan. This section defines the roles and responsibilities of PHEMCs and RRTs at all levels of the health pyramid.

5.1. Establishing a Public Health Emergency Management Committee

PHEMCs should be established at all levels of the health pyramid. These coordinating committees, composed of technical and non-technical members from the health and other sectors, should operate at their respective levels to plan and monitor the implementation of public health emergency plans. The role of the PHEMC is to develop and supervise the implementation of emergency preparedness strategies, action plans and procedures.

This committee provides guidelines for the implementation and operation of PHEOCs, ensures the fund raising and supervision of its operations. It is the strategic structure for managing public health emergencies.

5.1.1. Identify the function of the Public Health Emergency Management Committee

The PHEMC[3] should meet to define the overall strategy for public health emergency preparedness and response at all levels of the health pyramid. Once this plan is established, the committee should review it regularly and update it as publi health events change. The primary functions of the PHEMC are to:

- ensure coordination of the overall management of response activities;

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[3] The health emergency management committees shall correspond as appropriate to the district or departmental epidemic control committees when the district overlaps with one or more districts

- coordinate the development of a comprehensive preparedness and response plan;
- monitor and assess the capacity to respond to public health emergencies;
- develop the communication and community engagement strategy;
- mobilize resources for emergency prevention and control, including the provision of necessary response and communication equipment;
- establish a mechanism to monitor the use of resources before, during and after the emergency;
- organize regular simulation exercises on emergency management at all levels of the health pyramid, including the community;
- coordinate after action review and provide for communication of results to affected populations.

5.1.2. Identifying the members of the PHEMC

The PHEMC must be made up of representatives from the public and private sectors, technical and financial partners, civil society and non-governmental organizations (NGOs).

Members from the public sector at the HD level

- administrative authorities, Divisional officer/Sub-Divisional officer (President);
- a departmental / district manager from the Ministry of Defence (gendarmerie, fire department);
- a divisional/sub-divisional law enforcement officer;
- municipal authority (mayor);
- the district health manager;
- the head of the referral HF;
- the manager of the laboratory of the referral HF;
- the Head of the Partnership Office;
- a sanitary engineering technician;
- a divisional/sub-divisional manager from the sectorial ministries of health, etc.
- members from non-governmental organizations involved in public health activities (NGOs, HCOs, CBOs);
- a traditional practitioner representative;
- a representative of discussion groups (HC/DHC).

Members from the private sector:

- a representative of the private HFs;
- a representative of the dispensing pharmacists;
- a representative of research institutions;
- a representative of the Community of Economic Operators.

The Chairman of the Committee may call upon any other expertise deemed necessary for the management of emergencies. Within the framework of its missions, the PHEMC is assisted by a Technical Committee.

5.1.3. Meet regularly before and during public health events

In the absence of an outbreak, the PHEMC shall:

- meet to review disease trends and updates on adopted preparedness steps;
- review the level of preparedness at the beginning of each outbreak season (e.g., before the period of increased meningitis cases);
- share the findings and recommendations of these meetings with authorities at all levels of the health pyramid;
- organize simulation exercises to test action plans.

If the PHEOC is already established, it will serve as the focal point for these activities.

When responding to an outbreak or emergency, the PHEMC shall:

- meet as soon as the outbreak or event is identified;
- conduct a situation analysis and determine the level of the event;
- activate the Incident Management System and appoint the Incident Manager;
- assess needs and, if necessary, request assistance from PHEMCs and the RRTs;
- meet daily at the beginning of the outbreak, then weekly for the duration of the response, or as necessary;
- regularly analyse the results of the response, including the stock situation, and take the necessary measures to improve the response to the outbreak;
- document response actions and inform higher level.

5.2. Making the Public Health Emergency Operations Center (PHEOC) operational

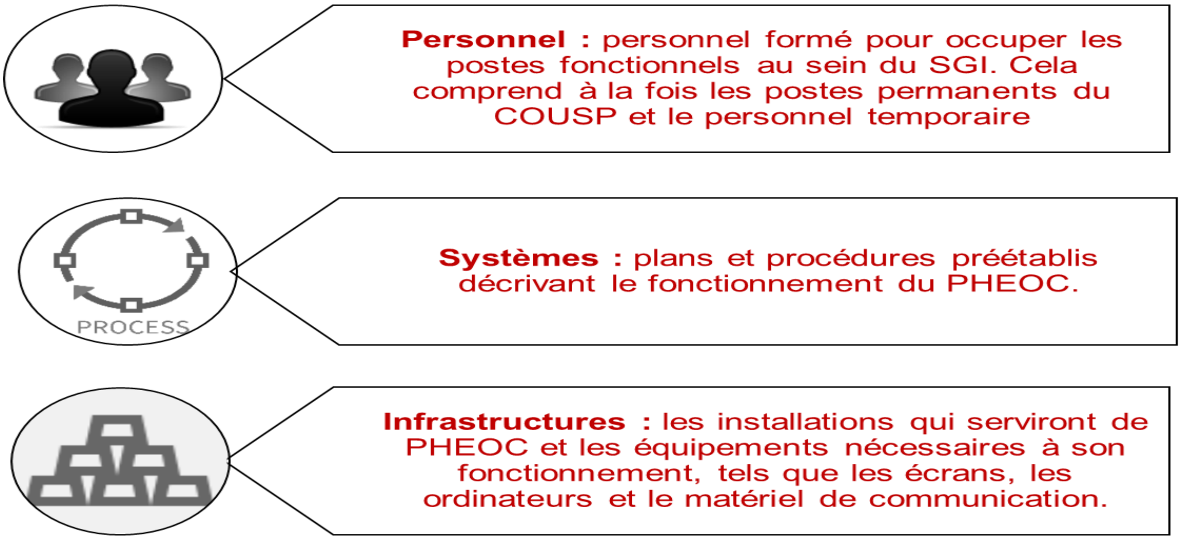
5.2.1. What is a PHEOC?

A PHEOC is a physical or virtual framework established for the coordination of information and resources to support public health incident management activities. This structure is an indispensable part of the Public Health Emergency Management system.

The WHO framework for a PHEOC (https://apps.who.int/iris/bitstream/handle/10665/258804/9789242565133-fre.pdf?sequence=1) has been developed to assist various states in this process. In Cameroon, this body is called the Public Health Emergency Operations Coordination Center. This structure was established by the decree No.51/PM of May 12, 2020 establishing the Public Health Emergency Operations Coordination Center.

To be fully functional in its support of emergency preparedness and response, the PHEOC will need to develop the following essential elements:

- plans and procedures;
- the information system for informed decision making, technology and telecommunication infrastructure necessary for timely communication;
- the information system for informed decision making, technology and telecommunication infrastructure necessary for timely communication;



Personnel : personnel formé pour occuper les postes fonctionnels au sein du SGI. Cela comprend à la fois les postes permanents du COUSP et le personnel temporaire

Systèmes : plans et procédures préétablis décrivant le fonctionnement du PHEOC.

Infrastructures : les installations qui serviront de PHEOC et les équipements nécessaires à son fonctionnement, tels que les écrans, les ordinateurs et le matériel de communication.

Figure 14: Essential elements of a PHEOC:

The PHEOC shall:

- ensure the implementation of response plans for public health emergencies;
- develop, implement and update the simulation exercise program for the preparation of the response to public health emergencies;
- prepare the alert, health watch and response system;
- ensure the immediate availability of resources for public health emergency response;
- ensure capacity building of actors involved in the public health emergency management;
- develop, communicate and implement standard operating procedures for public health emergency management;
- ensure the emergency supply of medical and sanitary equipment;
- ensure liaison with civil protection in case of public health emergency;
- serve as a framework for consultation and coordination between national and international actors involved in the management of public health emergencies.

If resources are available, regions and HDs will need to have PHEOCs with basic equipment.

It is important to develop/finalize the emergency operations plan that will define how the PHEOC will work and its various interactions with the actors involved in public health emergencies. There is also a need for a PHEOC strategic plan that will outline the country's vision for the operationalization of PHEOCs.

For optimal management of emergencies, the Ministry of Health has adopted the Incident Management System (IMS), which is the gold standard in public health emergency management.

5.2.2. What is an Incident Management System (IMS)?

The IMS is a standardized approach to managing and coordinating the response to public health events by the stakeholders involved in the response. This fully scalable and adaptable approach allows us to offer public health assistance. In addition, the IMS describes the specific roles and responsibilities of stakeholders in a public health event, and provides a multi-sector framework for working together in a transparent manner.

In the IMS, each person is assigned a specific role and follows a specific command structure. The system can be provided with additional teams of experts, analysts, logisticians, and support staff as the situation requires. This system is modular according to the requirements of the situation (***see WHO Framework for a Public Health Emergency Operations Center***).

The essential duties in the IMS are:

- management for the overall coordination of the response;
- operations for the implementation of tactical activities related to the response;
- planning for planning, monitoring and evaluation of activities, data management and training;
- logistics for resource acquisition and deployment;
- administration and finance for monitoring expenses, payments, and administrative services.

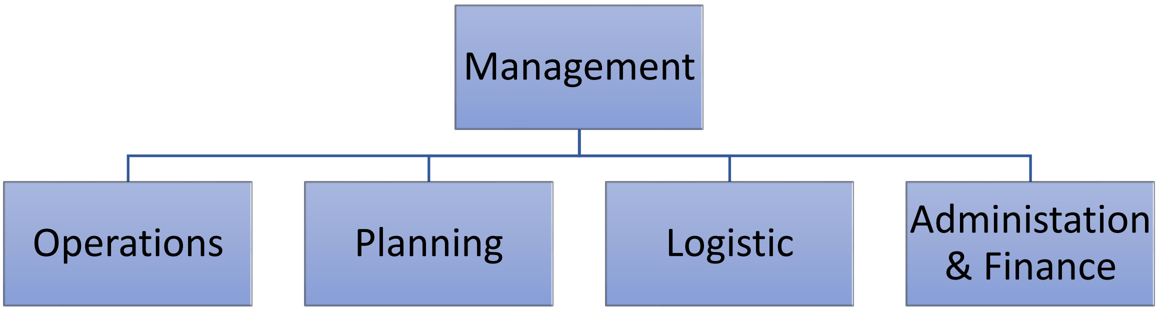


Figure 15: IMS fonctions

The management duty is carried out by an Incident Manager and includes liaison, health, safety, and public information. The other duties are carried out by sections in which there are many units. During the management of an incident, all duties must be taken into account, but it is possible to do without certain sections.

5.2.3. How does a PHEOC work?

A PHEOC has three modes of functioning:

- **Standby mode**: Mode during which a PHEOC operates with a minimum staff to carry out routine activities such as event-based surveillance. In fact, the standby staff selects the information to identify the triggers (Critical Information Requirement) that can lead to an activation.
- **Alert mode:** mode during which the PHEOC provides intensive monitoring of an event, with a view to possible activation of the IMS. The RRT can be deployed to the field for a thorough assessment of the situation. Very often, the alert mode involves an increase in

personnel and activities. This mode usually results in the activation of the IMS after the preliminary assessment meeting.

- **Response mode**: a mode during which the PHEOC directs all of its response activities to a specific public health incident or event. The response mode requires a significant increase in staff and activities; staff are mobilized from different jurisdictions to occupy IMS functional positions. It is in this mode that the Incident Manager is appointed and an incident action plan is developed.

It is important to note that the PHEOC remains permanently in standby mode.

5.3. Setting up a Rapid Response Team (RRT)

The RRT is a multidisciplinary technical team always available and ready to be mobilized/deployed quickly in case of emergency.

5.3.1. Define the roles and responsibilities of the Rapid Response Team (RRT)

The roles and responsibilities of the RRTs are:

- investigate rumours, reported outbreaks and other public health emergencies;
- propose appropriate strategies and control measures to the PHEMC, including risk communication activities;
- implement interventions proposed in the plan;
- prepare investigation reports;
- supervise, monitor and assess activities;
- participate in the Health Emergency Management Committee;
- participate in the final assessment of the response;
- participate in simulation exercises.

5.3.2. Identify the members of the Rapid Response TeamRRT

The RRT shall consist of:

- an epidemiologist, or public health officer or data manager;
- laboratory specialist;
- a clinician depending on the event;
- a specialist in environmental health or hygiene and sanitation;

- a veterinarian or wildlife specialist;
- a social mobilization expert;
- and any other resource person, depending on the availability of technical staff and the type of event constituting an emergency (infectiologist, administrator, chemical or industrial risk expert, investigator...).

5.4. Developing an outbreak and health emergency preparedness and response plan

The objective of this plan is to strengthen the capacities of the PHEOC, at the different levels of the health pyramid, to promptly react to the detection of an outbreak or other public health event.

This plan shall:

- rely on comprehensive risk assessments and identify available resources dedicated to outbreak preparedness and response;
- consider diseases with outbreak potential at all levels;
- provide population at risk estimates for diseases with outbreak potential and other health emergencies;
- clearly indicate the reference laboratory in charge of the confirmation for each suspected outbreak;
- estimate the need for drugs, vaccines and other items for each disease with outbreak potential and health emergencies that may occur;
- plan simulation exercises prior to implementation;
- include capacity building standard operating procedures (SOPs);
- integrating operational research on the determinants of health.

The preparedness plan should include the following:

- risk and vulnerability mapping;
- a communication plan: define the steps of communication around the risks, in particular through social mobilization;
- intervention measures according to the type and stage of the public health event;
- laboratory capabilities: sampling, packaging, transportation, analysis and containment of specimens, delivery of results;

- guidelines for psychological, pre-hospital and hospital management of patients (antimicrobial or other treatment as needed);
- IPC guidelines: identification of needs and pre-positioning of materials (pre- and post-exposure prophylactic drugs, decontaminants, disinfectants or others as needed);
- vaccination strategies where relevant;
- rapid control measures and additional measures in case of failure;
- capacity building strategies for field actors (training, awareness meetings and simulation exercises);
- indicators for monitoring/evaluating the response to the outbreak or public health event;
- The resource mobilization mechanism and procedures for reallocating resources to support the response.

5.5. Building up reserves of drugs, vaccines, inputs and other supplies

Outbreaks and other public health emergencies require quick mobilization of resources (drugs, vaccines, inputs and other supplies). It is therefore advisable to establish and pre-position stocks of materials before an emergency occurs.

Following the health risk assessment, HDs should stockpile drugs, vaccines, reagents, and supplies to enable rapid management of the first cases without waiting for assistance from higher levels. These stocks must be regularly and carefully controlled to avoid waste, stock-outs and expiration. The Appendices at the end of this section provide examples of tools for stock management.

The content of the reserve stocks depends on the type of diseases with outbreak potential in the HD and the identified public health emergencies. The risk assessment will therefore provide a list of items that need to be stored at the HD. A proposed list is provided in ***APPENDIX 5A.***

5.6. Managing stocks intended for the response to public health events

Stock management for response should follow the steps below:

- maintain a reliable supply of goods and equipment to face any public health event;
- use a check-list such as the one proposed in ***APPENDIX 5b*** to list stocks of inputs and other equipment that can be used in the response;

- ensure that items are well stored and ready for periodic use, e.g, every 3 months;
- estimate the necessary items for the response, identify those that are available and acquire those that are lacking, according to the level;
- list what is needed for surveillance activities, laboratory analysis and response to priority diseases and public health events:
    - forms;
    - supplies, laboratory reagents and other inputs;
    - patient management and field intervention equipment;
    - personal protective equipment.
- make an inventory and note the quantity of each item available;
- ill out and regularly update an inventory status form for each item;
- verify expiration dates and pack, ship, store and dispose of supplies and materials according to good logistical practices;
- for each item, define the minimum or essential quantity needed for investigation or response activities. Consider logistical capacity and risk mapping to define these minimum quantities;
- check stock levels compared to established minimum quantities;
- regularly report on the status of available supplies for public health events. An example of an IDSR movement and stock status form is shown in ***Appendix 5C***

5.7. Mapping the risks of public health events

Preparedness activities must be ongoing and regularly updated. They include the assessment of risks that may affect the health of the population (in the geographical area served by the HD) and may thus relate to sources of drinking water, seasonal variations, population movements, hygiene and sanitation conditions, gas hazards, flooding, domestic livestock, landslides, poor food preservation methods, etc. Once every two to three years, for example, the risk map and vulnerability analysis should be updated. This is useful information regarding supplies, transportation and other resources needed for the response.

Risk mapping should cover all public health risks specified by the IHR (2005), including chemical, zoonotic, radiological and nuclear hazards.

5.8. Appendices to Section 5

APPENDIX 5A	Main supplies to be stocked for public health events response
APPENDIX 5B	Stock situation report
APPENDIX 5C	IDSR stock movements and situation Form
APPENDIX 5D	Mission of the Outbreak Preparedness and Response Plan Committee

APPENDIX 5A. Main supplies to be stocked for outbreak response

Main supplies to be stocked for outbreak response				
Medicines	**Disinfectants, insecticides and rodenticides**	**Supplies**	**Vaccines**	**Equipment**
Ceftriaxone	**Disinfectants**	Auto-disable syringes	Meningitis AC, ACW135/ A, C, Y, W135, meningococcal conjugate vaccine (ACVM), meningococcal conjugated vaccines	PPE
			Cholera Vaccines	Body bags
Ciprofloxacine	2% Chlorine		Anti-tetanus serum	Buckets
Diazepam	Bleach	Mosquito nets	Anti amaryl vaccine	Camping kit
Doxycycline	Calcium hypochlorite	Personal protective equipment (see Appendix 4C)	Anti-rabies immunoglobulin	Candles
Comfort medicines	Cresol	Laboratory supplies (see Appendix 4B)	Other vaccines, for example, influenza vaccine	Computers
Erythromycine	Sodium hypochlorite			Containers
	Pesticides	Nasogastric tubes diameter 2.7 mm, length 38 cm		Kitchen utensils
	Cypermethrine	Nasogastric tubes diameter 5.3 mm, length 50 cm		Diesel fuel
Oral rehydration salts	Malathion	Needles and syringes		Headlamp
Paracetamol	Permethrine	intravenous injection kit (various sizes)		GPS
Penicillin V	Rodenticides	Spoons		Kerosene lamp
Rehydration fluids:	Brodifacom	Sprayers (pump and misting)		Lab: see appendix 4b
Ribavirin	Bromadione			Lamps
Ringer lactate				Cards
Oseltamivir				Kerosene
				Telephones
				Laminated paper
				Generator
				Radio
				Sprayer

***Note**: A detailed list is provided in Appendix 4B*

APPENDIX 5B. Stock Situation Report

Surveillance, Preparedness and Response to Public Health Events: Stock form

Year:

Evaluation date (day/month/year):

Reporting period:

Evaluation site:

Health Area:

District:

Region:

Country:

Items	Opening stock at the beginning of the inventory	Quantity received	Total stock	Quantity issued	Balance (available stock balance)	Observations, decisions and recommendations

Title, name and function of the person in charge:

APPENDIX 5C. IDSR stock movements and situation Form

IDSR Stock Item Transaction and Balance Sheet

Laboratory or Warehouse Name	Item Description Name (	Presentation (Unit of purchase)	Expiry date	Manufacturer	Batch number	Location in store	Airway bill	Allotment number	Shipment & operations cost (USD)	Transaction Date Day/Month/Year	Quantity received	Donor or Supplier	Quantity issued	Destination or Beneficiary	Stock Balance	Signature (Name and function)	Observations/ Remarks
															0		Inventory

se one sheet by stock item, and update the sheet every time any transaction takes place

APPENDIX 5D. Mission of the Outbreak Preparedness and Response Plan Committee

Tasks	Designated Committee Members
Designated coordinating structures, including committees	
Organisational framework of key actors and partners supporting health activities (human, animal, environmental, etc.), as well as roles and responsibilities	
Epidemiology and surveillance activities, including health information management	
Define the roles and responsibilities of members during an outbreak	
Make risk mapping	
Set up the supply and distribution chain in an emergency situation	
Steps for carrying out a risk communication strategy, including social mobilization	
Operational actions according to the expected outbreak phases	
Collection, handling, shipping, processing and management of laboratory specimens' information	
Case management, including treatment (antiviral, antimicrobial, decontamination, disinfection or other, as indicated), infection control, isolation facilities, management of a mass casualty event	
Prophylactic treatment before and after exposure to infection	
Immunization strategies	
Rapid isolation activities and additional methods for rapid isolation failure	
Psychosocial support for all those affected, including community members and stakeholders	
Communication on risks and social mobilization	
Capacity building, including required training, awareness meetings and simulation	
Logistic, including supply lists	
Environment, water and sanitation	
Patient and environmental decontamination, including management of corpses.	
Outbreak or public health event surveillance	
Resource mobilization and procedures for reallocation or mobilization of resources to support the response	

Section 6: Responding to outbreaks and other public health events

This section describes how to:

- convene the Health District Emergency Management Committee and choose the response mode;
- mobilize Rapid Response Teams;
- carry out response activities;
- regularly forward Public Health events reports;
- document the response.

Introduction

Integrated surveillance and disease response is about using data for appropriate public health interventions or actions. This section describes the different steps in the response interventions to notify an outbreak and activate response structures, conduct a public health response, and provide general guidance for immediate actions in response to the main causes of illness, death and disability. The WHO guidelines for responding to chemical, biological, radiological and nuclear public health events are attached at the end of this section.

When a disease outbreak, event or serious public health situation is detected, an investigation should be conducted to determine the causes, as described in ***Section 4***. The results of the investigation will guide the response. Within the framework of disease prevention and control programs, most of the interventions carried out have been successful: mass vaccination campaigns for vaccine-preventable diseases, increased food aid and nutritional support for malnourished children, administration of anti-malaria medicines, antibiotics or antivirals, depending on the disease. To be effective, these actions must:

- be conducted with the participation of the community;
- include an educational component;
- aim to change behaviours in the community.

It is also important to ensure effective coordination of response activities, as many actors or stakeholders will be involved in these activities. It is equally important to identify in advance all these actors or stakeholders, including their areas of activity, roles and responsibilities, for a coordinated response to an outbreak or other public health event. This is the role of the Public Health Emergency Management Committee (defined in ***Section 5***).

Whatever the specific response recommended, the national, intermediate (Regional Delegation of Public Health) and operational (HD) levels must play a crucial role in the choice and implementation of the recommended response in order to safeguard the health and well-being of communities.

According to the International Health Regulations (IHR, 2005), HDs should be involved in the response to public health events of infectious, zoonotic, chemical, radiological, nuclear or food safety origin, and other events of unknown origin if they are detected.

6.1. Notifying an outbreak and activating the response structures

6.1.1. Declaring an outbreak and notifying WHO

Once an outbreak threshold is reached at the HD level, the head of the HD must inform the region and then the central level. At the national level, the IHR National Focal Point (NFP) will determine, based on the Decision Instrument in ***APPENDIX 2*** of the IHR, whether it is a potential Public Health Emergency of International Concern (PHEIC). The IHR NFP will inform the WHO Regional Office. They will then alert neighbouring HDs (Regions, if applicable) of the outbreak to ensure coordination of response efforts. Once the laboratory diagnosis is confirmed, the Minister in charge of Public Health can declare the outbreak.

6.1.2. Activating response structures

See ***section 5***

6.1.3. Holding the Public Health Emergency Management Committee meeting at the HD.

When an outbreak or other public health event is confirmed, the chair of the Public Health Emergency Management Committee (local administrative authority) should convene a meeting to assess the risks, implement the response, and decide whether to activate the Incident Management System (IMS). (See **section 5)** Next, the following steps should be taken:

a) raise funds for the response to the outbreak or public health event;

b) alerting neighbouring HDs inside and outside the country about the outbreak. If they report a similar outbreak, coordinate response efforts with them. If a cross-border surveillance and response framework already exists with a neighbouring country, inform that country's neighbouring HD. If not, the IHR NFP should contact the neighbouring NFP to inform them of the Public Health event. This will facilitate the coordination of the response and limit the spread of the disease beyond the affected area;

c) Assign clear responsibilities for each of the specific response activities to lead the technical committee. (See **section 5**)

d) Provide guidance or training to the HD response team and relevant HF (HFs) staff and provide them with the necessary supplies;

e) Review available resources as outlined in the preparedness plan and determine if additional resources are needed. Consider for example:

    i. the human resources that can be mobilized to manage the outbreak;
    ii. available funding to support response activities;
    iii. material resources; for example, vehicles and fuel, telephones;

f) request emergency stocks consisting of: Personal Protective Equipment (PPE), disinfection equipment, medicines and other necessary medical supplies, such as specimen transport kits...;
g) provide a laboratory for the diagnosis and confirmation of pathogens responsible for outbreaks. If the HD does not have the capacity to safely collect, package, and transport specimens, the reference laboratory should be contacted for assistance. If it is difficult to get the sample to the reference laboratory, consider the use of rapid diagnostic kits or other point-of-care diagnostic systems, if available;
h) mobilize logistical resources (RRT travel, accommodation, communication, other essential equipment) for the HDs and the community;
i) if the supply is not locally possible:
    i. contact the regional level and possibly the central level to request support;
    ii. collaborate with other services, non-governmental organizations or private pharmacies/labs in the region;
    iii. identify practical and inexpensive substitutes.

6.1.4. Implement the health sector response: Activate the Incident Management System (IMS)

In 2014, WHO recommended that member countries establish Public Health Emergency Operations Centers. Cameroon has established the PHEOC and adopted the IMS model for the management of public health emergencies at the national level. It is also planned to set up PHEOC branches at the intermediate (RDPH) and operational (HD) levels for the implementation of response activities.

When an outbreak or any other public health event is suspected in a HD, the IMS is put on alert and once confirmed, a preliminary assessment of the situation will bring out the Critical Information Requirements (CIR) in order to assess the situation. At the end of this evaluation, the activation of the IMS can be decided by the competent authority. The level of IMS activation is selected based on well-defined criteria (see SOP on PHEOC activation). Depending on the magnitude of the situation, the regional or central IMS could be activated.

Once the IMS is activated:

- an Incident Manager is appointed based on his or her expertise in the type of event and his or her ability to mobilize various resources for the event management;
- the office managers, the section heads, the unit heads and the members of the various teams are identified and formally appointed;
- the Incident Manager defines the objectives of the response and the operational period; Under the coordination of the Planning Section Chief, the Incident Action Plan (IAP) is developed and includes the following:
    - the response objectives and operational period;
    - the list of response team leaders and the organigram;
    - planning hypothesis;
    - the strategic axes, actions and priority activities of the operational period;
    - the CIRs;
    - the response factors that may trigger a change;
    - frequency of response coordination meetings;
    - stakeholders and partners in the response;
    - the frequency with which status reports are prepared;
    - environmental health and safety considerations.

When the response objectives are met, the Incident Manager decides to deactivate the IMS.

6.2. Mobilizing RRTs for a prompt action during public health emergency situations

The RRTs are multidisciplinary teams ready to intervene 24 hours a day, 7 days a week. They are members of the operations section and are the working hand of the response. Therefore, the RRTs should have been identified and trained during the response preparedness phase to ensure mastery of basic response concepts. The composition of an RRT depends on the type of condition or event, its size, and the resources available for deployment. Refer to Section 5 of this guide for recommendations on the composition of the RRTs, their roles and responsibilities, and the SOPs for their deployment.

6.3. Identifying and conducting appropriate response activities to outbreaks and other public health events

Analysis of the results of the investigation and interpretation of the data provided by the RRT will allow the selection of appropriate response activities to contain the outbreak or other confirmed public health events. Regardless of the specific causes of the outbreak or public health event, the success of the response depends on the effectiveness of the health information management system and the implementation of response strategies within the IMS. Response strategies are selected by the Incident Manager based on the results of the risk analysis conducted and the priorities of the PHEMC. These strategies vary according to the type of public health event and are reflected in the Incident Action Plan (IAP). Globally, the response strategic axes are:

(a) general coordination;
(b) case management, Infection Prevention and Control (IPC);
(c) logistics and supply chain management;
(d) laboratory surveillance or diagnosis and epidemiology;
(e) social mobilization and risk communication;
(f) reactive immunization;
(g) water, sanitation, and hygiene (WASH);
(h) disease vector control.

Refer to ***section 11*** and disease-specific national guidelines to select response activities, including:

- tested measures to prevent deaths or avoidable disabilities due to specific causes;
- a set of activities to control the problem in the short term and reduce the risk of continued transmission in the long term through prevention activities;
- community, health facility, and HD staff involvement;
- involvement of other key stakeholders from private organizations, commercial entities, traditional practitioners, food vendor associations and others who can influence response activities.

6.3.1. Strengthen case management and Infections Prevention and Control (IPC)

This involves taking steps to help improve clinical practices in the HD. See recommendations in ***Appendix 6A*** and **Section 11** for the treatment of various illness cases during an outbreak and other public health events. This includes:

(a) training and equipping health personnel at the HD level to apply these measures;
(b) ensuring that clinicians receive the laboratory confirmation results, if necessary;

(c) ensuring that health personnel record all patients in a standard, easily identifiable register and on a line list;

(d) ensure the implementation of the mass influx management plans previously developed in the pre-outbreak phase by the managers of each HF;

(e) provide IPC Standard Operating Procedures in accordance with the guidelines;

(f) implement IPC measures to reduce risk, including:
    - implement a triage and isolation service for communicable diseases (Ebola, cholera, SARS, etc.). See ***APPENDIX 6H*** for a cholera treatment center plan;
    - ensure that health care workers receive safety and personal protection measures against infectious diseases;
    - ensure safe practices and protection of non-health care personnel (support staff, security guards, maintenance staff, administrative staff);
    - assess and ensure compliance with WASH standards in HF;
    - supervise the disposal of PPE and other soiled materials;
    - ensure compliance with biosafety and biosecurity standards for animal handling (farms, markets, etc.).

(g) ensure that necessary medicines and management inputs are available;

(h) ensure that appropriate treatment protocols are available.

6.3.2. Update the SOPs for patient referral system. Build the capacities of the response personnel

Response staff should be given the opportunity to learn and update their knowledge of disease case definitions and other public health events, case management procedures, the reporting process and data to be submitted. It is essential that RRT members have Personal Protective Equipment and are familiar with IPC practices. In the event of an outbreak of a disease requiring immunization, ensure that RRT members are up-to-date on their vaccinations.

To build the capacity of health personnel and of the RRT:

(a) provide clear and concise guidance to staff involved in response actions;

(b) select topics to be covered during training sessions. Focus on case management and IPC in accordance with disease-specific recommendations; for example:
    - case management algorithms;
    - enhance standard precautions (use of clean water, disposal of sharp objects);
    - barrier measures (use of PPE, etc.);

    - isolation measures;
    - safe and dignified burial;
    - safe disposal of animal remains;
    - other topics deemed necessary such as counselling skills, etc.

(c) guide and train:
    - direct or redirect training for all actors in the response chain based on priorities and on the event;
    - in emergency situations, there is often not enough time for conventional training. Therefore, on-the-job training will be provided as needed, ensuring that facilitators can follow learners during practice;
    - assess learner performance and verify skills, if necessary.

6.3.3. Strengthen surveillance during response activities

During the response to an outbreak, health personnel in all HF should remain vigilant in monitoring disease or public health events, keeping in touch with the Community Health Worker or other designated community focal point. For example, RRT members and health personnel assigned to HF should:

(a) search for other people who have contracted the disease and refer them to the approved HFs, treatment centers or, if necessary, quarantine the household and take care of the patient at home, ensuring that he/she has access to food, water, basic necessities (soap, chlorine, firewood, medicines, sanitary towels, etc.);
(b) share laboratory information with the team in a timely manner;
(c) update linear lists, analyse data by time-place-person;
(d) monitor the effectiveness of the outbreak response activities;
(e) produce a daily report at the beginning of the outbreak; as the outbreak progresses, the HD management team may decide to change the frequency;
(f) actively seek out contacts and monitor them as necessary (see ***Section 4*** on case definition and tracing).

6.3.4. Strengthen surveillance with neighbouring HDs

During response activities, it is important to work closely with neighbouring HDs to limit the spread of the outbreak. Information should be shared and surveillance and response activities should be planned jointly.

It is equally important to establish cross-border disease surveillance and response committees to serve as a platform for sharing surveillance data, epidemiological information and other related information during the outbreak. The committee should be composed of members from the two neighbouring HDs, namely:

(a) the surveillance focal point;
(b) the head of laboratory services;
(c) the person in charge of hygiene and sanitation;
(d) the person in charge of environmental health;
(e) the person in charge of HFs;
(f) the animal health officer;
(g) the emigration officers of the zone;
(h) the person in charge of the Security Forces (Forces of Law and Order);

Depending on the type of event, other members may also join the committee.

The cross-border committee will meet as soon as a public health emergency is identified (then weekly or every two weeks) and will continue to hold routine quarterly meetings during the inter-outbreak period. This is to review disease trends, other early warning systems and the level of HDs' preparedness.

6.3.5. Involve the community in response activities

Community Health Workers (see definition in the ***Introductory section***) can be the first to intervene and take action to make the situation as safe as possible for the community. Some of the actions to be taken are as follows:

(a) involving and informing community leaders about the situation and actions that can be taken to mitigate the situation;
(b) provide first aid and inform health care personnel;
(c) keep people away from a "risk" area (potentially contaminated water source);
(d) respectfully isolate anyone with symptoms of a potentially infectious disease, paying special attention to cultural sensitivities;
(e) quarantine animals, close markets, etc ;
(f) educate the community on specific actions they can take to protect themselves;
(g) engage in IPC and hygiene promotion by coordinating all efforts to make IPC and hygiene materials/infrastructure available;
(h) identify effective local ways to get information to the community;

(i) organize door-to-door campaigns with trusted individuals to reach every household in the affected area to stop the spread of the disease and to encourage those who have been exposed or who are suspected of being exposed to the event to come forward to the hospital for appropriate treatment.

(j) involve community members not only as problem-solvers, but also as beneficiaries.

6.3.6. Inform and educate the community

Effective risk communication is one of the essential elements of public health event management. It is a transversal activity that can have an impact on other technical aspects of the response, such as WASH, immunization, community-based surveillance, etc. It is also essential to build confidence between first responders and the community. When an actual or potential risk threatens the health of the population, treatment options may be limited, the response may take time to organize, and resources may be insufficient. Therefore, sharing guidelines is an effective way to manage this public health risk.

The public must be kept informed to allay their fears and encourage their collaboration in response activities. In addition, educational messages should be developed for the community to inform them of the symptoms to recognize the disease, prevent transmission, and know when to seek care for treatment. These community communication activities should begin as soon as the outbreak or public health event is identified. Community groups, local NGOs, or outreach teams that can help gather information and relay messages should also be identified. Ensure consistency of message content among all messengers (community leaders, health personnel, religious leaders, etc.).

The following elements should be considered for effective risk communication:

(a) decide what to communicate by referring to the disease-specific recommendations in ***section 11***. Ensure that the following information is included:
    - signs and symptoms of the disease;
    - means of treating the disease at home, if home treatment is recommended, and how disinfectants are prepared;
    - easy-to-apply preventive behaviours that can prevent the transmission of the disease;
    - appropriate times to go to the HFfor assessment or treatment;
    - recommendations for vaccination, if appropriate.

At the same time, continue to collect the information needed to confirm and manage rumours that are circulating.

(b) determine how to formulate the message. Ensure that messages:
   - use local terminology;
   - are culturally appropriate;
   - are clear and concise;
   - are based on local traditions;
   - take into account beliefs about the disease.

Note: Messages from similar sources should be checked for veracity before dissemination. Examples of community education messages can be found at the end of this section (***Appendix 6F***).

(c) Select appropriate and available means of communication in the HD. For example:
   - mass media (radio, television, newspapers);
   - meetings with health personnel, community, religious, opinion and political leaders;
   - educational and communication materials (posters, brochures);
   - multimedia presentations (films, videos, or narrated slides) in markets, health centers, schools, women's groups, other community associations, service providers, and religious facilities;
   - social media (Facebook, Twitter, WhatsApp, etc.);
   - drama groups/community drama groups;
   - public address systems;
   - corporate/institutional websites;
   - e-mail/SMS subscriptions.

(d) provide health education messages to community groups and service provider organizations by asking them to disseminate the messages at their meetings;

(e) forward health education messages to trusted and respected community leaders with a request to disseminate them in the community:
   - someone designated by the Ministry of Health should serve as the spokesperson. The name of the spokesperson should be communicated to the media and it should be made clear that all information about the outbreak will be conveyed through this person;
   - information should be released to the media only through the spokesperson to ensure that the community receives clear and consistent information.

(f) DMOs and Regional Public Health Delegates should meet regularly with local officials to communicate about:
- frequent and updated information on the outbreak and the response;
- clear and simple health messages to the media;
- clear instructions from PHEOC to the media on health education messages and information.

6.3.7. Conduct a mass immunization campaign

Collaborate with the Expanded Program on Immunization (EPI) manager to conduct a mass immunization campaign, if necessary. A microplan for the mass immunization campaign should be developed or updated as soon as possible. Promptness is essential for emergency immunizations, given the time required for vaccine procurement and distribution.

Determine the target population to be vaccinated based on the epidemiological profile drawn up at the end of the epidemiological survey (consult the EPI norms and standards concerning specific recommendations for vaccine administration).

Two work sheets entitled "Planning a mass immunization campaign" and "Assessment of vaccine stocks for immunization activities" can be found at the end of this section, in ***Appendices 6C*** and ***6D*** respectively. ***Appendix 6E*** describes recommended immunization practices for vaccination campaigns.

6.3.8. Improve access to safe water

Drinking water containers can be vectors for spreading outbreaks, including cholera, typhoid, shigella diseases, and hepatitis A and E. There is a need to ensure that the community has an adequate supply of potable water for drinking and other uses. The table below shows the daily water requirements per person in non-outbreak situations. During outbreaks, especially diarrheal disease outbreaks, the water requirements are much higher.

Table X: Daily water needs per person

Daily water needs per person*		
	Non-outbreak situation	During a diarrheal disease outbreak
At home	20 litres per day	50 litres
In a care facility	40 to 60 litres per day	50 litres in hospitalisation wards, 100 litres in surgery, 10 litres in kitchen

**Refugees Health. An approach to emergency situations. Doctors Without Borders, 1997. MacMillan.*

Drinking water includes:

(a) piped chlorinated water;
(b) safe drinking water obtained by chlorination at the point of use;
(c) water from protected sources (such as wells closed with a lid, rainwater collected in a clean container);
(d) boiled water from any source.

If during an emergency no local drinking water source is available, it is necessary to bring in water from outside. To ensure that families have safe drinking water at home (even if the source is safe), do the following:

(a) educate the community on how to keep drinking water at home. Refer to Appendix 6F for sample community messages and guidelines for storing and purifying drinking water at home;
(b) provide containers that are designed to prevent water contamination. For example, jars with narrow openings into which users cannot contaminate the water by inserting their hands;
(c) ensure that waste disposal points, including faecal matter, are located at least 30 meters from water sources.

6.3.9. Ensure disposal of potentially infectious waste in the community

Safe disposal of human excrement must be ensured to avoid secondary infections from contact with contaminated substances:

(a) assign teams to inspect local areas for human and animal waste disposal. Safe practices include disposing of faeces in a latrine or burying it in the ground more than 10 meters away from the water source;
(b) inform the community if practices such as open defecation are found to be unsafe, and have latrines built in cooperation with the community to suit local conditions;
(c) conduct effective community education on hygiene and sanitation practices.

6.3.10. Enhance food handling practices

Ensure that food is handled safely in homes, restaurants, food stores and factories. Refer to established national standards and procedures for food handling and processing.

To ensure good food hygiene, the following measures should be taken:

(a) conduct community education sessions on food hygiene practices for the general public and food industry stakeholders;
(b) inspect restaurants, food vendors, food processing plants and other sites to assess food handling practices, with particular attention to hygiene such as hand washing, cleanliness and compliance with national standards;
(c) close restaurants, retail outlets, or factories if inspection reveals unsafe food handling practices;
(d) strengthen national food safety controls where necessary.

6.3.11. Reduce exposure to infectious or environmental hazards

Measures should be taken to reduce exposure to risks or factors contributing to the outbreak or public health event. These may be chemical, physical, or biological agents. Technical requirements for reducing exposure will be determined based on national policy and in collaboration with those with experience in these areas. For example, exposure to heavy metals (e.g., lead) associated with occupation or industrial pollution will require coordination among several departments and partners. Education and habit-change efforts can help the community adopt behaviours that limit exposure to chemicals and other risks.

For vector-borne diseases, work with experts, such as an entomologist, to identify appropriate interventions to reduce exposure to vectors (e.g., the Anopheles mosquito). Also, work with the malaria control program at the HD level to:

(a) promote indoor residual spraying;
(b) raise community awareness of the proper use of mosquito nets and the prevention of mosquito bites from dusk to dawn;
(c) promote the use of insecticide-treated mosquito nets and other locally available insecticide-treated products (mosquito nets, blankets, clothing, sheets, curtains, etc.);
(d) encourage environmental cleanliness (e.g., draining standing water, cleaning bushes, etc.).

Promote prevention of rodent-borne diseases (which can be infested with plague-carrying fleas) by helping HD residents reduce their exposure to these animals. Work with the HD vector control officer to encourage the community to:

(a) avoid contact with rodent urine, faeces and other secretions;
(b) cover food and water stored at home to avoid rodent contamination;
(c) keep the house and kitchen clean and tidy to prevent rodent nesting;

(d) use chemicals (insecticides, rodenticides, larvicides, etc.) and traps appropriately, depending on environmental and entomological conditions;
(e) educate the community on personal protection to reduce exposure.

6.3.12. Ensure safe and dignified handling and burial of mortal remains

The management of corpses is critical in controlling the spread of infectious diseases. Viral haemorrhagic fevers, cholera, COVID-19, and unexplained deaths under suspicious circumstances require careful handling of remains. To this end, it is important to ensure that the remains are handled in a safe and dignified manner by qualified personnel. In addition, the systematic disinfection or decontamination of the premises where the victims have been staying should be carried out in accordance with national guidelines for disinfection and decontamination.

Safe and dignified burials may be conducted in dedicated locations or at the discretion of families. In the event of an infectious disease outbreak, the Public Health Emergency Management Committee (PHEMC) may develop specific guidelines for safe and dignified burial. These guidelines can be adapted as the outbreak evolves.

6.3.13. Ensure appropriate and adequate logistics and supplies

A team in charge of logistics only is essential for outbreak response activities. During the outbreak, it is necessary to:

- monitor the efficiency of the logistics and essential supplies system;
- conduct logistics planning to ensure that transportation is used in the most efficient manner;
- monitor the reliability of communication among teams during the outbreak and check for the need for additional equipment (e.g., communication packages);
- making available the necessary equipment that will enable teams to successfully carry out the response.

Monitoring and assessment of the outbreak or public health event management is crucial to adapt the response to its evolution. The results of the evaluation will be included in a report submitted to higher authorities and community leaders.

For example, ongoing monitoring may include:

(a) disease trends so that the effectiveness of the response, the spread of the outbreak, and risk factors can be assessed;

(b) the effectiveness of the response (case fatality rate, incidence);

(c) the implementation of the response (program coverage, outbreak management committee meetings, etc.)

(d) the availability and use of necessary resources, supplies and equipment;

(e) the acceptability of the response efforts by the community;

(f) regular reporting on the stock of supplies issued in public health emergencies.

Situation reports play an important role in assessing the response and preparing the final report. A suggested reporting template is provided in ***Appendix 7A***, ***Section 7***. ***Section 8*** describes the steps for supervising and assessing the response.

6.4. Documenting the response

During and at the end of an outbreak, the HD Incident Command System (ICS) coordination team should:

(a) collect all documents, including meeting reports, activity or procedure reports; investigation reports; assessment reports, and other relevant documents;

(b) prepare a cover page listing all of the above documents;

(c) document lessons learned and recommended improvements and, as a result, update the country's emergency preparedness and response plan, event/disease-specific plan, and other relevant Standard Operating Procedures (SOPs) and tools, including the After-Action Review (AAR).

This file will be a key source of data for assessing the response. Section 8 describes how to monitor and evaluate the activities of the IDSR.

6.5. Appendices to Section 6

APPENDIX 6A	Treating patients during the outbreak;
APPENDIX 6B	Preparing disinfectant solutions from common household products;
APPENDIX 6C	Planning an emergency vaccination campaign:
APPENDIX 6D	Assessing vaccine supplies for immunization activities:
APPENDIX 6E	Recommended immunization practices;
APPENDIX 6F	Sample educational messages for the community;
APPENDIX 6G	Communication during an outbreak;
APPENDIX 6H	Key Infection Prevention and Control (IPC) measures;
APPENDIX 6I	response to chemimcal and radio nuclear events

APPENDIX 6A. Treating Patients during the outbreak

Use appropriate medicines and treatments to treat patients during an outbreak. Below are recommended treatments for outbreaks of:

(a) cholera;
(b) dysentery;
(c) measles;
(d) bacterial meningitis,
(e) Viral Hemorrhagic Fever

Note: For detailed treatment guidelines for these and other priority diseases, please refer to the specific guidelines.

Cholera treatment in outbreak situations

In: *WHO guidelines for management of the patient with cholera, WHO/CDD/SER/91.15* and *The New Emergency Health Kit 98, WHO/DAP/98.10*

(a) Assess the patient's level of dehydration. See assessment guide below.
(b) Give fluids according to the appropriate treatment plan (see next page).
(c) Collect stool samples from the first five suspected cholera cases.
(d) Give an oral antibiotic to patients with severe dehydration.

Assess patient's level of dehydration	
• Assess the patient's general condition. Is he or she lethargic or unconscious? Agitated and irritable? • Does he or she have sunken? • Offer the patient something to drink. Is he or she: unable to drink, or drinking with difficulty, drinking greedily, is he or she thirsty? • Pinch the skin of the patient's abdomen. Does it come back into place very slowly (more than 2 seconds)? Or slowly?	
Determine whether or not the patient is severely or moderately dehydrated and give fluids according to the treatment plan	
If two of the following signs are present: • Lethargy or unconsciousness • Sunken eyes • Inability or difficulty to drink • Very slow return of the skin after pinching	**SEVERE DEHYDRATION*** Give rehydration fluids (Plan C)
* In adults and children over 5 years of age, the absence of a radial pulse and hypotension are additional signs of severe dehydration.	
If two of the following signs are present: • Agitation, irritability • Sunken eyes • Drinking greedily, intense thirst • Slow return of skin after pinching	**MODERATE DEHYDRATION** (Give rehydration fluids) (Plan B)

NO DEHYDRATION
Give food and drink to treat diarrhoea at home.
(Plan A)

If there are not enough signs to classify the patient as

in the moderate or severe dehydration category

Plan C: Intravenous therapy for severe dehydration

(a) severe dehydration is a medical emergency and patients should be treated urgently. Seconds can make a difference;

(b) patients with severe dehydration should begin to receive intravenous (IV) fluids immediately;

(c) as soon as the patient can drink, also give ORS solution 5ml/kg/hour simultaneously;

(d) ringer's lactate is the first choice among all intravenous fluids. If Ringer's lactate is not available, other sterile solutions may be an option:

- normal saline solution;
- 5% glucose in normal saline;
- cholera physiological saline (containing Na +, 133; K +, 20; Cl-, 98; acetate, 48 mmol / L).

(e) a simple 5% glucose solution (dextrose) is not recommended;

(f) give a total of 100 ml / kg of ringer's lactate solution divided into two periods, as shown below:

Age	First period	Second period	Total
< 1 year	30 ml/kg in 1 hour	70ml/kg in 5 hours	100 ml/kg in 6 hours
≥ 1 year and adults	30 ml/kg in 30 min	70 ml/kg in 2 ½ hours	100 ml/kg in 3 hours

(a) more than one intravenous (IV) may be needed to administer the first bolus treatment;

(b) when IV rehydration is not possible and the patient cannot drink, ORS may be administered via nasogastric tube. Do not use a nasogastric tube in patients who are unconscious or vomiting;

(c) if possible, measure fluid flow and add volumes equivalent to the amount described for initial treatment;

(d) monitor the patient closely and reassess frequently (every 15 to 30 minutes);

(e) if hydration does not improve, administer the IV infusion more rapidly. 200 ml/kg or more may be required during the first 24 hours of treatment;

(f) after 6 hours (infants) or 3 hours (older patients), perform a complete reassessment. Switch to ORS if hydration is improved and the patient can drink.

Complications - Pulmonary oedema may occur if too much intravenous fluid is given; renal failure if too little intravenous solution is given; and hypoglycaemia and hypokalaemia in malnourished children rehydrated with Ringer's lactate only. Rehydration should be closely monitored by medical personnel.

Antibiotic treatment

(g) the laboratory should be asked about the resistance patterns of the strain at the beginning and during the outbreak and adjust treatment accordingly;

(h) antibiotics should only be given in severe cases to reduce the duration of symptoms and carriage of the pathogen

(i) antibiotics are given as soon as the patient is able to take oral medication (once vomiting has stopped):

- doxycycline: a single dose (300 mg for adults; 2-4 mg / kg for a child aged 1-14 years) is the antibiotic of choice for all patients, including pregnant women;
- in case of resistance to doxycycline, use azithromycin (1 g orally as a single dose for adults and 20 mg / kg (max 1 g)) orally as a single dose for children under 12 years.

	First line	Alternative line
Adults (including pregnant women)	Doxycycline 300 mg single dose	Azithromycin PO 1g single dose
Children < 12 years	Doxycycline 2-4 mg/kg single dose	Azithromycin PO 20mg/kg single dose

Zinc supplementation for children

Zinc supplementation in the management of children aged 6 months to 5 years with watery diarrhoea reduces the frequency and severity of the episode and the frequency of subsequent diarrhoeas. If possible, supplementation (20 mg zinc per day) should be initiated immediately.

Plan B: Oral rehydration in patients with dehydration

(a) patients with signs of dehydration should be admitted to the CTC /CTU;

(b) initial treatment, give ORS according to patient's weight (75 ml / kg over the first 4 hours);

(c) cholera patients with some signs of dehydration do not require intravenous replacement fluids, but should be closely monitored for the first 4 hours:

- if at any time signs of severe dehydration appear, **proceed immediately** to Treatment Plan C;
- if signs of dehydration remain after the first 4 hours, **repeat Treatment Plan B** for 4 hours and reassess;

- if there are no signs of dehydration after the first 4 hours of treatment, patients may be discharged home with the same instructions as described above under Treatment Plan A.

(d) if the patient vomits while taking a liquid, wait for 10 minutes Then allow the patient to resume eating, but more slowly;

(e) continue to monitor the patient and replace the fluid until the diarrhoea stops;

(f) when the patient is ready to leave the facility, provide advice on treating diarrhoea at home;

(g) refer to the Integrated Management of Childhood Illness (IMCI) guidelines on treatment of children under 5 years of age and national guidelines for more information on treatment of acute watery diarrhoea and confirmed cholera.

Plan A. Oral rehydration in patients with no signs of dehydration

(a) patients with no signs of dehydration should be treated with oral rehydration solution (ORS).

(b) patients with no signs of dehydration do not need to be admitted to the CTC/CTU. They can be treated with ORS at home, in emergency treatment centres, or in the outpatient area of the health care facility;

(c) if the patient is seen in the health care facility, keep them under observation for 2-4 hours to ensure they are tolerating ORS;

(d) during observation and before returning home, give clear instructions for care. Advise patients or caregivers to continue to administer ORS after each wet bowel movement and to return immediately if the condition deteriorates (repeated vomiting, increased number of bowel movements, or if the patient is drinking or eating poorly);

(e) ORS should be prepared with drink water (boiled or treated with a chlorine-based product or household bleach). They should not be kept for more than 24 hours (refresh daily);

(f) ORS should be given regularly, in small quantities. If a patient vomits ORS, slow down the administration of ORS, then slowly increase it again when vomiting stops;

(g) patients should receive ORS after each loose stool to stay hydrated until the diarrhoea stops;

(h) patients should receive the following quantities of ORS after each loose stool:

Age	Quantity of ORS
<2 years	50-100 ml
2-9 years	100-200 ml
≥10 years	As much as desired

Hospital discharge

(a) consider discharge if the patient:
- shows no signs of dehydration;
- is able to take ORS without vomiting;
- has not had a loose stool for 4 hours;
- is able to walk without assistance;
- passes urine;
- has been informed of the date of return to the hospital/CTC.

(b) prior to discharge, provide patients and their caregivers with ORS and instructions on how to prepare it;

(c) inform the patient, family members, and caregivers of household precautions and instructions:
- for children, continue breastfeeding infants and young children;
- drink and use safe water;
- wash hands at critical times, especially after using the toilet (including after helping a child) and before preparing and eating food. When caring for a patient, always wash your hands after providing care and after handing over soiled items such as clothing or linens;
- cook food thoroughly and eat it while still hot;
- remove and wash any bedding or clothing that may have come into contact with diarrhoeal stool with the appropriate chlorine solution (0.02%). If chlorine is not available, patient' bedding and clothing can be disinfected by stirring for 5 minutes in boiling water, drying in direct sunlight, or washing with soap and drying thoroughly in direct sunlight;
- use a flush toilet or approved septic tank; double soiled bag materials when disposing of in the trash;
- use any household disinfectant or a 1/10 solution of bleach (one measure of bleach for every nine measures of water) to clean any area that may come in contact with faeces as soon as possible after it has been soiled;
- if a family member develops acute watery diarrhoea, administer an oral rehydration solution and seek health care immediately;
- when treating people with cholera, do not give food or drink to anyone outside the household;

- visitors may be allowed if the sick person wants company; visitors should also follow hand hygiene recommendations;
- inform patients about home care before they leave the health care facility, pointing out danger signs and when it is appropriate to return to the facility. Patients should resume treatment if they develop any of the following symptoms:
  - increasing number of loose stools;
  - difficulty eating and drinking
  - severe thirst;
  - repeated vomiting;
  - fever;
  - blood in the stool.

(a) Give an appropriate oral antibiotic for outbreaks of bloody diarrhoea due to *Shigella dysentariae type 1*

	NALIDIXIC ACID **# Give four times daily for 5 days**	**CIPROFLOXACIN** **# Give four times daily for 5 days**	**COTRIMOXAZOLE** **(trimethoprim + sulphamethoxazole)** **# Give four times daily for 5 days**		
WEIGHT	**TABLET** 250 mg	**TABLET** 250 mg	**ADULT TABLET** 80 mg trimethoprim + 400 mg sulphamethoxazole	**PAEDIATRIC TABLET** 20 mg trimethoprim + 100 mg sulphamethoxazole	**SYRUP** 40 mg trimethoprim + 200 mg sulphamethoxazole per 5 ml
Children's dose					
3–5 kg	¼	¼	¼	2	5 ml
6–9 kg	½	½	½	2	5 ml
10–14 kg	1	1	1	3	7.5 ml
15–19 kg	1	1	1	3	7.5 ml
20–29 kg	2	2	1	6	15 ml
Adult dose	**TABLET** 250 mg	**TABLET** 250 mg	**TABLET** 160 mg TMP +800 mg SMX		
	4 tablets	4 tablets	2 tablets		

Source: *WHO Guidelines for the control of outbreaks due to S. dysentariae type 1. WHO Geneva, 1995*

Give vitamin A to children with measles

(a) give the first dose in the HFor clinic;

(b) give the mother one dose to give at home the next day.

AGE	Vitamin A Capsules		
	200 000 IU	100 000 IU	50 000 IU
Up to 6 months		½ capsule	1 capsule
6 months up to 12 months	½ capsule	1 capsule	2 capsules
12 months up to 5 years	1 capsule	2 capsules	4 capsules

(b) Give appropriate antibiotic for bacterial meningitis cases during and outside an outbreak

Managing meningitis outbreaks in Africa: A quick reference guide for health authorities and health-care workers Revised 2015, WHO/HSE/GAR/ERI/2010.4. Rev. Meningitis outbreak response in sub-Saharan Africa. WHO guideline, WHO/HSE/PED/CED/14.5

Weekly Epidemiological Record No 51/52, 577-588, 19 December 2014 (http://www.who.int/wer)

Standard Operating Procedures for Surveillance of Meningitis, Preparedness and Response to Outbreaks in Africa, WHO document. WHO/AFRO/FRH October 2018, Brazzaville

(a) aqdmit patient to a HFfor diagnosis and treatment;
(b) following lumbar puncture, treat every new patient who is suspected of having meningitis with antibiotics as soon as possible; Ceftriaxone is the first line treatment for bacterial meningitis (Treatment protocols in the table below);
(c) ensure any child under 2 years of age or any patient with severe symptoms is admitted to the health centre for inpatient treatment and adjust the treatment as necessary;
(d) patient isolation is not necessary. Provide good supportive care and simplify case management.

Age	Treatment protocols for bacterial meningitis during outbreaks in Africa (without laboratory confirmation)
In children aged 0–2 months	Ceftriaxone 100mg/kg/day IM or IV once a day for 7 days
In children aged over 2 months	Ceftriaxone 100mg/kg/day once a day (maximum 2g) IM or IV for 5 days
In children aged >14 years and adults	Ceftriaxone 2g/day once a day IM or IV for 5 days
Note: Outside outbreaks, treatment duration should be 7–10 days for all ages	

Prophylaxis for household contacts

Antibiotics are recommended as a prophylactic measure for household contacts of all ages in non-outbreak periods, but not during outbreaks. Ciprofloxacin is the preferred prophylactic agent, with ceftriaxone as an alternative when ciprofloxacin is contraindicated.

Appendix 6B. Preparing disinfectant solutions from ordinary household products

During a response to an outbreak of any disease transmitted through direct contact with infectious body fluids (blood, urine, stool, semen, and sputum for example), an inexpensive system can be set up using ordinary household bleach.

The following table describes how to make 1:10 and 1:100 chlorine solutions from household bleach and other chlorine products.

Table XI: Preparation of chlorine solutions

Use this chlorine product	**To make a 1:10 solution for disinfecting:** **Excreta** **Cadavers** **Spills of infectious body fluids**	**To make a 1:100 solution for disinfecting:** **Gloved hands** **Bare hands and skin** **Floors** **Clothing** **Equipment** **Bedding**
Household bleach 5% active chlorine	1 litre bleach per 10 litres of water	100 ml per 10 litres of water, or 1 litre of 1:10 bleach solution per 9 litres of water
Calcium hypochlorite powder or granules 70% (HTH)	7 grams or ½ tablespoon per 1 litre of water	7 grams or ½ tablespoon per10 litres of water
Household bleach 30% active chlorine	16 grams or 1 tablespoon per 1 litre of water	16 grams or 1 tablespoon per10 litres of water

To disinfect clothing: we must:

- Rapidly and thoroughly disinfect the patient's personal belongings and immediate environment with one of the following disinfectants chlorinated lime powder;
  - 1% chlorine solution;
  - 1% to 2% phenol solution.
- Disinfect the patient's clothing quickly and thoroughly: boil or soak in disinfectant solution;
  - wash clothes with soap and water;
  - sun dry;
  - wash utensils with boiling water or disinfectant solution;

- do not wash contaminated articles in rivers or ponds that might be sources of drinking water, or near wells.

APPENDIX 6C. Planning for an emergency response

1. Review with health workers the need to plan vaccination campaigns and specify the target population for the immunization activity.
2. Estimate the necessary amounts of vaccine, diluent and immunization supplies such as sterile syringes and sterile needles, cold boxes, vaccine carriers and safety boxes.
   - (a) Coordinate with EPI, WHO country office and UNICEF offices to arrange for provision of necessary vaccines and supplies.
   - (b) A list of pre-qualified WHO vaccines is available at: http://www.who.int/immunization_standards/vaccine_quality/PQ_vaccine_list_en/en/. If a country already has an Immunization Inter agency Coordination Committee, ensure that there is discussion and agreement on the type of vaccine to be given, who to give and the methodology to be used.
   - (c) Contact the national level to request vaccines. If a national reserve stock is not available, the national IVD program manager will request an emergency supply from WHO.
3. Choose the immunization sites and inform the community.
   - (a) Coordinate with the EPI/IVD or disease control programme in your district to identify sites for conducting the immunization activity.
   - (b) Identify the facilities that can participate in the activity.
   - (c) Identify a mobile immunization team, if needed.
   - (d) Determine if there are any hard-to-reach areas; e.g., a transient workers' camp and identify a mobile immunization team to reach these areas.
   - (e) Contact the facilities and schedule the immunization sites.
   - (f) Make sure there is enough capacity to store extra amounts of the vaccine during storage and transportation to the immunization site.
4. Conduct a comprehensive microplanning for the campaign. A microplan is the operational plan for a campaign at the county or lower level. Ensure the plan has at least the following:
   - (a) estimate of the number of vaccination teams required and their composition including roles and responsibilities of team members, as well as number of supervisors and monitors;
   - (b) list of supervisors and their contact numbers;
   - (c) travel plan for teams and supervisors including transportation requirements;
   - (d) mapping of the coordination with other partners and regions/districts local partners like NGOs, religious and civic organizations, etc;
   - (e) maps of the targeted area;
   - (f) cold chain requirements and maintenance;

   (g) plan for distribution of logistics;
   (h) plans for disposal of waste from campaign;
   (i) social mobilization plan with community leaders mapped and engaged;
   (j) training schedule;
   (k) budget estimates for the various campaign components including training and planning prior to implementation and waste disposal following implementation.
5. Select immunization teams. For every 100 to 150 people expected at the immunization site, the following staff is required:
   (a) one to two vaccinators to give immunizations;
   (b) a person in charge of the registration on the immunization cards;
   (c) community health workers if already available or an identified community volunteer to verify age and immunization status.
6. Work with your EPI/IVD team to conduct refresher training for vaccinators on recommended immunization practices. Ensure instructions are given for the use of safe injection techniques.
7. Mobilize the community. Inform the public about the emergency immunization. Activity while ensuring that there is:
   (a) a clear communication plan that includes easy-to-understand information on the need for the campaign;
   (b) a clearly-defined target group for the campaign;
   (c) a clear understanding of the dates of the campaign;
   (d) a mechanism in the communication plan for rapidly identifying and addressing rumours that may arise during the campaign;
   (e) a single point of contact that is well versed in risk communication and the local culture;
   (f) a clear plan for monitoring any adverse effects.
8. Arrange staff transportation to the immunization site.
   (a) Plan their transportation to and from the site.
   (b) Schedule vehicles and plan for fuel and other costs.
   (c) Estimate per diem costs and make the necessary arrangements for lodging if the site is far from the health worker’s usual station.
9. Monitor the overall campaign process and the number of doses of vaccine given.
   (a) Collect daily factsheets from teams.
   (b) Calculate the number of remaining stocks and supplies necessary for the next day.
   (c) Ensure that the estimated number of individuals vaccinated is monitored daily and tracked against target population.

(d) Follow-up visit plans should be made for missed individuals based on tally/summary sheet information.

(e) Document any missing houses/individuals who should be followed up on subsequent days.

(f) Review the team available on site and, if necessary, reallocate/deploy the teams to other sites based on the workload.

(g) Conduct brief feedback sessions at the end of each day with vaccination teams and make the necessary mid-course corrections.

Note: A rapid guide to common Supplementory Immunization Activity problems and potential quick fixes is available at:
http://www.polioeradication.org/Portals/0/Document/Resources/PolioEradicators/1c.QuickFixesforSIA20100914.pdf

10. Give instructions for use of safe injection techniques. Review with health workers the need to plan vaccination campaigns.

Appendix 6D. Estimation of vaccine supplies

Outbreak: ____________________________ Confirmation date _______________________

Target population:

______children aged 0 to 5 years
______children aged 9 months to 14 years;
_______children and adults aged 0 to 30 years;
_______Women of childbearing age 15 to 45 years;
_______entire population, children and adults.

1. Size of the target population to be vaccinated. If the activity only targets a proportion of the general population, estimate the size of the target population. Multiply the general population times the percentage of children or adults in the target population. If you do not know the exact age distribution rates in your area, use recommended estimates such as the following:

•children aged 0 to 5 years	20 %
•children aged 9 months to 14 years	45 %
•children and adults aged 1 to 30 years	70 %
•women of childbearing aged 15 to 45 years	20 %

Note: these estimates are variable depending on regions

2. Find out how many doses each person should receive. Record the number below as "number of doses recommended."
3. Allow for wastage. Use a wastage factor of 20%. Multiply the size of the target population (see step 1) times the number of doses times 1.20.

_______________	**X**	_________________	**x 1.20 =**		_________________________
Size of the target		*Number of recommended doses*	*Wastage.*		*Number of doses to order including wastage*

Note: It is recommended that the wastage factor of 20% should be used only at the national level to estimate vaccine requirement during an outbreak. Use a wastage factor of 15% at the subnational and district levels and 10% at the HFlevel.

4. Allow for a contingency stock. Use a reserve factor of 25%. Multiply the estimated number of doses including wastage times 1.25 to obtain the total estimated number of doses.

___________________	**x 1.25**	**=**	________________________________
Number of doses	*Contingency factor*		*Total number of estimated doses including wastage*

Note: It is recommended that the contingency stock be kept only at the national level. However, if a subnational level has adequate capacity for vaccine storage, then it can also keep a contingency stock.

5. To obtain the total number of vials of vaccine to order, divide the total number of estimated doses by the number of doses contained in the vial. (This is usually printed on the label and may be one, two, five, ten or twenty doses)

____________________	÷	____________________	=	__________________________________
Total number of estimated doses		*Doses per vial*		*Number of doses required including wastage*

6. If the vaccine requires a diluent, multiply the number of millilitres of diluent per vial times the total number of vials required.

____________________	X	____________________	=	__________________________________
Diluent required		*Total number of vial*		*Total diluent to order per vial*

7. Estimate the number of sterile needles and syringes that will be needed to carry out the activity. If single- use needle and syringes are used, order the same amount as for the estimated number of doses in Step 4.
8. In addition, estimate the number of dilution syringes necessary for preparing the vaccine. Source: *Field Guide for Supplementary Activities Aimed at Achieving Polio Eradication, World Health Organization, Geneva 1997. District guidelines for yellow fever surveillance, Division of Emerging and other communicable disease surveillance and control, World Health Organization, Geneva 1998.*
9. Estimate the number of safety boxes required

Appendix 6E: Recommended immunization practices

Work with your EPI team to give refresher training to the vaccinator teams that will conduct the emergency immunization activity. As a minimum, make sure vaccinator teams know how to:

1. Reconstitute the vaccine correctly:
   (a) determine the appropriate quantity of diluent to reconstitute the freeze-dried vaccine;
   (b) use a sterile syringe and sterile needle for each dose;
   (c) using the dilution syringe, draw up and expel the diluent several times in the vial that contains the vaccine to mix the reconstituted vaccine well.
2. Wrap the vial in silver foil or cover it with a dark cloth. This will protect the vial from sunlight.
3. In a field situation, protect the vaccine and diluent from contamination. Cover the open top of the vial with foil to keep out dirt and flies.
4. Follow multidose vial policy as applicable; e.g., for measles (destroy the reconstituted vial after 6h) and polio (destroy the vial after 28 days).
5. Record the dose on an immunization card for each person immunized, if it is national policy to require immunized persons to have a card.
6. Collect data for monitoring the activity. For example, record the number of doses given on a tally sheet so that coverage from the campaign can be calculated.
7. Remind health workers about the risk of getting blood-borne diseases from an accidental needle stick. Review safe practices for handling and disposing of sharp instruments and needles.
8. Arrange for safe disposal of used injection materials at the end of the activity. They can be burned or buried in a pit according to medical waste disposal guidelines.

Appendix 6F: Sample messages for community education

Improve handwashing

Handwashing with water and soap may be the most effective way to prevent transmission of some organisms causing infectious diseases. For that reason, promote handwashing in every family. Handwashing is particularly important after defecation, cleaning a child who has defecated, disposing of a child's stool, before preparing or handling food and before eating.

Handwashing is practised more frequently where water is plentiful and within easy reach. If possible, water for washing should be stored separately from drinking water. During an outbreak, soap should be provided to those without it. If soap is not available, wood ash or ground can be used to scrub the hands. Do not dry washed hands with dirty cloths. Air-dry wet hands.

Message:

ARE YOU PROTECTED FROM DYSENTERY (bloody diarrhoea)?

Washing your hands protects you and others from disease.

Always wash your hands:

- After defecation
- After cleaning a child who has defecated
- After disposing of a child's stool
- Before and after eating
- Before preparing or handling food.

Message:

DO YOU HAVE WHAT IS NEEDED TO WASH YOUR HANDS?

- Clean water and soap (or if you do not have soap, use ash or ground to scrub your hands)
- Clean cloth for drying
- Air-dry your hands.

Safe Handling of food

Encourage the following food safety practices:

- thoroughly wash hands with soap before preparing food;
- thoroughly wash fruit and green vegetables with clean water before eating them;
- cook food until it is hot throughout;
- eat food while it is hot or reheat it thoroughly before eating;
- wash all cooking and serving utensils after use;

- keep cooked food and clean utensils separate from uncooked foods and potentially contaminated utensils;
- cover your food appropriately.

Message:

DO YOU PREPARE FOOD SAFELY?

Cooking kills germs

- *Thoroughly cook all meats, fish and vegetables.*
- *Eat cooked meats, fish and vegetables while they are hot.*

Washing protects from disease

- *Wash your hands before preparing or serving food.*
- *Wash your dishes and utensils with soap and water.*
- *Wash your chopping board especially well with soap.*

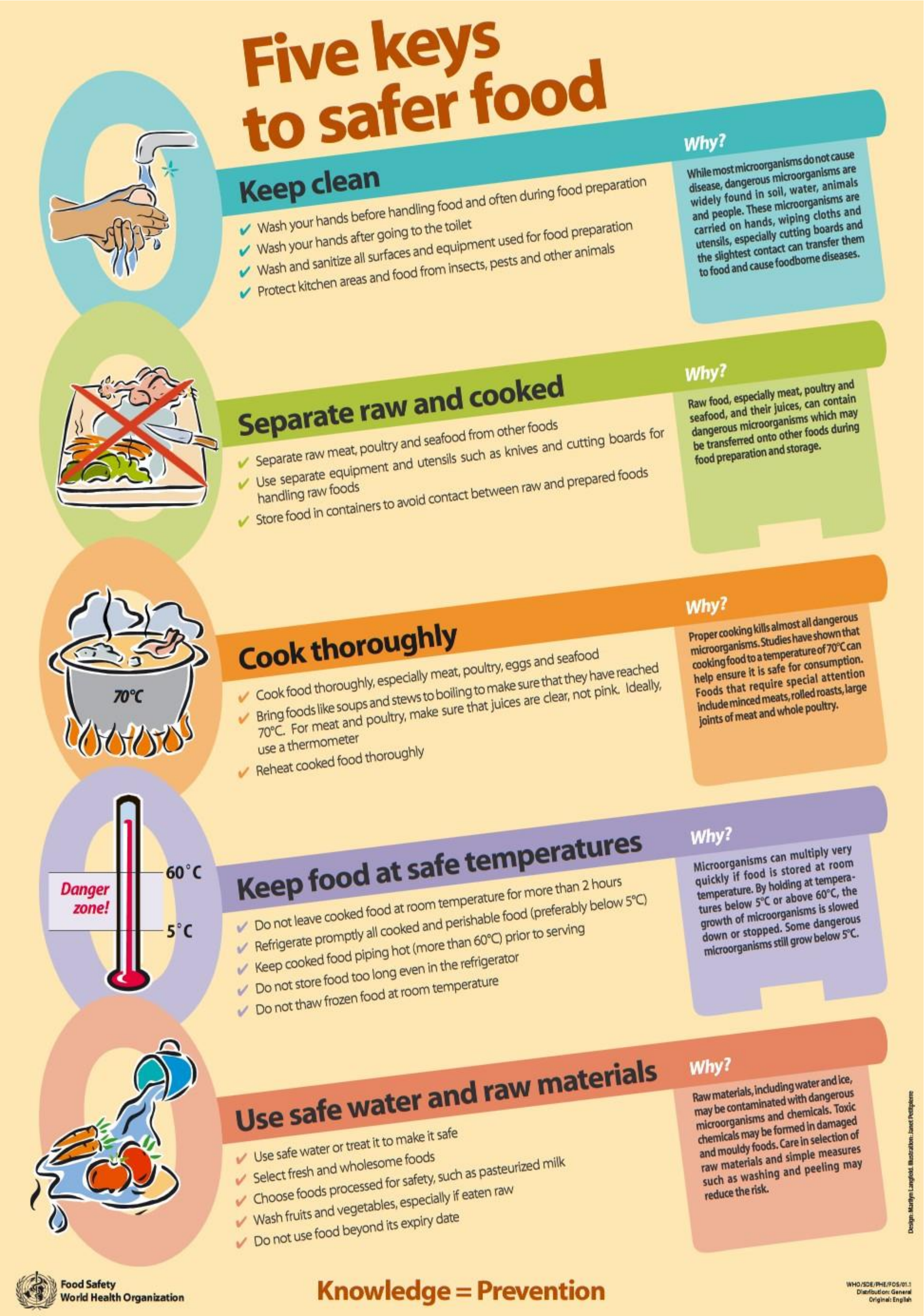
Peeling protects from disease

- *Only eat fruits that have been freshly peeled (such as bananas and oranges).*

KEEP IT CLEAN: COOK IT, PEEL IT, OR LEAVE IT.

Five Keys to Safer Food:

- keep clean;
- separate raw and cooked;
- cook thoroughly;
- keep food at safe temperature;
- use safe water and raw materials.



Five keys to safer food

Keep clean

- ✓ Wash your hands before handling food and often during food preparation
- ✓ Wash your hands after going to the toilet
- ✓ Wash and sanitize all surfaces and equipment used for food preparation
- ✓ Protect kitchen areas and food from insects, pests and other animals

Why?

While most microorganisms do not cause disease, dangerous microorganisms are widely found in soil, water, animals and people. These microorganisms are carried on hands, wiping cloths and utensils, especially cutting boards and the slightest contact can transfer them to food and cause foodborne diseases.

Separate raw and cooked

- ✓ Separate raw meat, poultry and seafood from other foods
- ✓ Use separate equipment and utensils such as knives and cutting boards for handling raw foods
- ✓ Store food in containers to avoid contact between raw and prepared foods

Why?

Raw food, especially meat, poultry and seafood, and their juices, can contain dangerous microorganisms which may be transferred onto other foods during food preparation and storage.

Cook thoroughly

- ✓ Cook food thoroughly, especially meat, poultry, eggs and seafood
- ✓ Bring foods like soups and stews to boiling to make sure that they have reached 70°C. For meat and poultry, make sure that juices are clear, not pink. Ideally, use a thermometer
- ✓ Reheat cooked food thoroughly

Why?

Proper cooking kills almost all dangerous microorganisms. Studies have shown that cooking food to a temperature of 70°C can help ensure it is safe for consumption. Foods that require special attention include minced meats, rolled roasts, large joints of meat and whole poultry.

Keep food at safe temperatures

- ✓ Do not leave cooked food at room temperature for more than 2 hours
- ✓ Refrigerate promptly all cooked and perishable food (preferably below 5°C)
- ✓ Keep cooked food piping hot (more than 60°C) prior to serving
- ✓ Do not store food too long even in the refrigerator
- ✓ Do not thaw frozen food at room temperature

Why?

Microorganisms can multiply very quickly if food is stored at room temperature. By holding at temperatures below 5°C or above 60°C, the growth of microorganisms is slowed down or stopped. Some dangerous microorganisms still grow below 5°C.

Use safe water and raw materials

- ✓ Use safe water or treat it to make it safe
- ✓ Select fresh and wholesome foods
- ✓ Choose foods processed for safety, such as pasteurized milk
- ✓ Wash fruits and vegetables, especially if eaten raw
- ✓ Do not use food beyond its expiry date

Why?

Raw materials, including water and ice, may be contaminated with dangerous microorganisms and chemicals. Toxic chemicals may be formed in damaged and mouldy foods. Care in selection of raw materials and simple measures such as washing and peeling may reduce the risk.

Design: Marilyn Langfold. Illustration: Janet Petitpierre

Food Safety
World Health Organization

Knowledge = Prevention

WHO/SDE/PHE/FOS/01.1
Distribution: General
Original: English

Safe disposal of human waste

High priority should be given to ensuring the safe disposal of human waste at all times, and especially during outbreaks of diarrhoea. Sanitary systems appropriate for local conditions should be constructed with the cooperation of the community.

Community messages should emphasize the following:

- everyone, including children, should use latrines properly;
- transfer children's excreta with a scoop or shovel to the latrine or bury in a hole;
- avoid defecating on the ground, or in or near the water supply.

When large groups of people congregate, as for fairs, funerals, or religious festivals, ensure the safe disposal of human waste. If there is no latrine, designate areas for defecation and provide a shovel to bury the excreta.

Message:

ARE YOU PROTECTED FROM DYSENTERY (bloody diarrhoea)?

DO YOU USE A TOILET OR LATRINE?

Germs that cause dysentery live in faeces. Even a person who is healthy might have dysentery germs.

- *Always use a toilet or latrine. If you do not have one – build one!*
- *Keep the toilet or latrine clean.*
- *Wash your hands with soap (or ash) and clean water after using the toilet or latrine.*

KEEP IT CLEAN: USE A TOILET OR LATRINE

Clean drinking water and storage

Community drinking water supply and storage

1. ***Piped water:*** To maintain safety, properly chlorinate piped water. To prevent entry of contaminated groundwater into pipes, repair leaking joints and maintain constant pressure in the system.
2. ***Closed wells:*** Equip with a well-head drainage apron, and with a pulley, windlass or pump.
3. ***Trucked in:*** If locally available water is likely to be contaminated, drinking water should be supplied by tankers or transported in drums, if it is adequately chlorinated and a regular supply can be ensured. The trucking of water, however, is expensive and difficult to sustain; it is usually considered a short-term measure until a local supply can be established.

Home drinking water storage and treatment

When the safety of the drinking water is uncertain, it should be chlorinated in the home or boiled.

To prevent contamination of drinking water, families should store drinking water using one of the following types of containers:

1. ***Covered containers*** that are cleaned daily and kept away from children and animals. Water should be removed from the containers using a long-handled dipper, kept especially for this purpose.
2. ***Narrow-mouthed containers*** with an opening too small to allow the insertion of a hand. Water should be removed by pouring from the opening or spout.

Water used for bathing, washing and other purposes other than drinking need not be treated and should be stored separately from drinking water.

Safe disposal of bodies

The body fluids of persons who die from diarrhoea or a viral haemorrhagic fever are still infectious. Use extreme caution when preparing the bodies of suspected cholera or viral haemorrhagic fever patients. Encourage safe funeral and burial practices.

Reducing Exposure to mosquitoes

Mosquito control is the main intervention for reducing malaria transmission. It can reduce malaria transmission from very high levels to close to zero. In high transmission areas, mosquito control can significantly reduce child and maternal deaths. Personal protection against mosquito bites is the first line of defence for malaria prevention.

Message:

ARE YOU PROTECTED FROM MOSQUITOE BITES?

Whenever possible:

- Avoid going out between dusk and dawn when mosquitoes commonly bite.
- Wear long-sleeved clothing and long trousers when going out at night, and avoid dark colours, which attract mosquitoes.
- Apply insect repellent to exposed skin (if the repellent is available).
- Use screens over doors and windows.
- Use an insecticide-treated mosquito net over the bed.
- Use anti-mosquito sprays or insecticide dispenser (if available).

Malaria transmission can rapidly be reduced through indoor residual spraying (IRS) with insecticides. IRS works for 3 to12 months, depending on the insecticide used and the type of surface on which it is sprayed.

Appendix 6G: Communication during an outbreak

Introduction

Following confirmation and verification of the event, the primary health and the district level authorities should liaise with the national level authorities to communicate and receive guidance on common positions to be delivered to the media.

From first announcement throughout the outbreak, communication from the district level should follow the directions and the key messages developed at national level in consultation with the field team, in order to ensure consistency and speaking with one voice.

Even though communication should be centrally coordinated by the national level, the media should approach local and district public health response level to obtain first-hand information from direct sources.

In addition, the director of the district level hospital should support the communication and provide scientific expertise as evidence for intervention.

Actions at the district level

- identify spokesperson(s) at district level (political and technical);
- liaise regularly with national authorities to provide them with first-hand information (received at the community local level, the media, local stakeholders);
- be in contact regularly with national authorities to receive common messages including guidelines and answers for frequently asked questions to feed the local media;
- be available for interviews by local media upon request to provide accurate, transparent and updated information following directions from the national level, using simple clear key messages;
- organize press briefings to provide regular information to local media, following directions from the national level;
- develop good relationships with local media for the delivery of accurate, transparent, timely messages to the population;
- use information materials developed at the national level with clear consistent messages to provide guidance to the population;
- identify local powerful channels for the delivery of information to the population;
- meet regularly with local stakeholders to disseminate a correct message of prevention and surveillance to the population;
- organize preventive door-to-door campaigns to reach remote rural areas and promote prevention and surveillance, following directions from national level.

Appendix 6H: Guide to infection prevention and control measures

Handwashing

Purpose: To protect the patient, staff and caregivers from cross infection

Responsibility: Clinicians, environmental health practitioner, caregiver

Steps in handwashing

- The hands are washed thoroughly for a minimum of 10-15 seconds with soap (plain or antimicrobial) and running water (tap or draining method).
- Remove jewelleries (rings, bracelets) and watches before washing hands, ensure that the nails are clipped short (do not wear artificial nails), roll the sleeves up to the elbow.
- Wet the hands and wrists, keeping hands and wrists lower than the elbows (to ensure that the water flows to the fingertips, avoiding arm contamination).
- Apply soap (plain or antimicrobial or ash) and lather thoroughly.
- Use firm, circular motions to wash the hands and arms up to the wrists, covering all areas including palms, back of the hands, fingers, between fingers and lateral side of the fifth finger, knuckles and wrists.
- Rub for a minimum of 10-15 seconds.
- Repeat the process if the hands are very soiled. Clean under the fingernails.
- Rinse hands thoroughly, keeping the hands lower than the forearms. If running water is not available, use a bucket and pitcher.
- Do not dip your hands into a bowl to rinse, as this re-contaminates them. Collect used water in a basin and discard in a sink, drain or toilet.
- Dry hands thoroughly with disposable paper towel or napkins, a clean dry towel or air-dry them. Discard the used towel in an appropriate container without touching the lid. Use a paper towel, clean towel or your elbow/foot to turn off the faucet to prevent re-contamination.

Different types of antiseptic disinfection:

Using antiseptics, hand rubs gels or alcohol swabs for hand antisepsis:

- apply the product to the palm of one hand. The volume needed for one application varies per product;
- rub hands together, covering all surfaces of hands and fingers, until hands are dry;
- do not rinse.

Note:

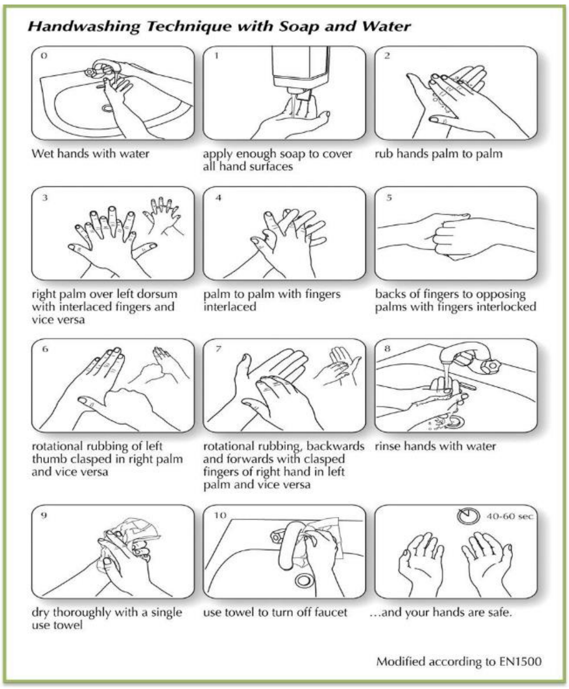
- When there is visible soiling of hands, they should first be washed with soap and water before using waterless hand rubs gels or alcohol swabs.
- In situations where soap is not available, ash can be used for washing hands.

Hand Hygiene Techniques

This is a process, which mechanically removes soil and debris from skin and reduces the number of transient microorganisms. Handwashing with plain soap and clean water is as effective in cleaning hands and removing transient microorganisms as washing with antimicrobial soaps and causes less skin irritation.

Steps:

- thoroughly wet hands;
- apply a hand-washing agent (liquid soap); an antiseptic agent is not necessary;
- vigorously rub all areas of hands and fingers for 10–15 seconds (tip: 10 average breaths), paying close attention to fingernails and between fingers;
- rinse hands thoroughly with clean running water from a tap or bucket;
- dry hands with paper towel or a clean, dry towel or air-dry them;
- use a paper towel or clean, dry towel when turning off water if there is no foot control or automatic shut off.


Handwashing Technique with Soap and Water
0
Wet hands with water
1
apply enough soap to cover all hand surfaces
2
rub hands palm to palm
3
right palm over left dorsum with interlaced fingers and vice versa
4
palm to palm with fingers interlaced
5
backs of fingers to opposing palms with fingers interlocked
6
rotational rubbing of left thumb clasped in right palm and vice versa
7
rotational rubbing, backwards and forwards with clasped fingers of right hand in left palm and vice versa
8
rinse hands with water
9
dry thoroughly with a single use towel
10
use towel to turn off faucet
40-60 sec
…and your hands are safe.
Modified according to EN1500

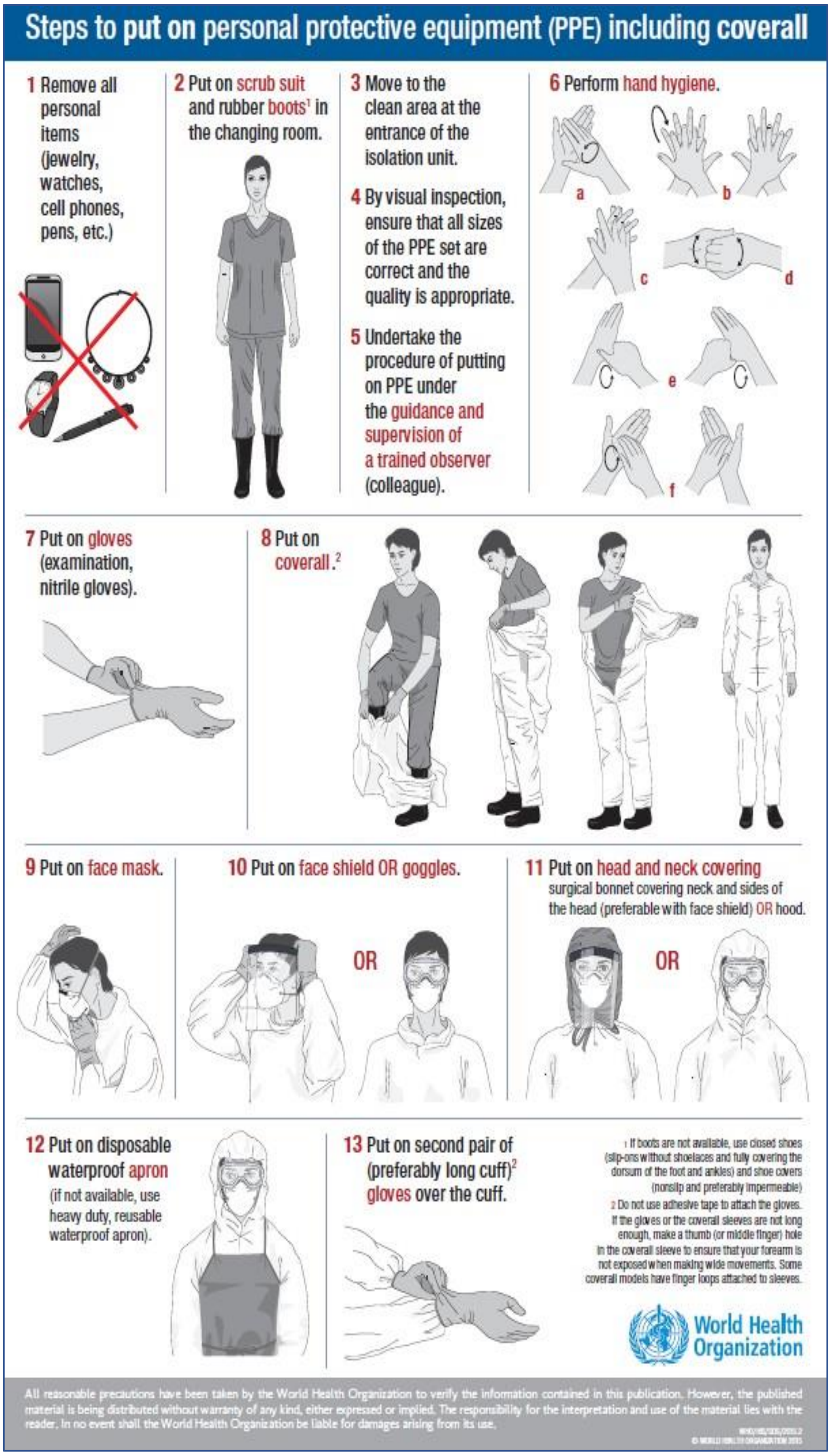
Note:

- If bar soap is used, provide small bars and soap racks that drain.
- Use running water and avoid dipping hands into a basin containing standing water; even with the addition of an antiseptic agent, microorganisms can survive and multiply in these solutions.
- Do not add soap to a partially empty liquid soap dispenser. This practice of "topping off" dispensers may lead to bacterial contamination of the soap.
- When soap dispensers are reused, they should be thoroughly cleaned before filling.
- When no running water is available, use a bucket with a tap that can be turned off to lather hands and turned on again for rinsing, or use a bucket and pitcher.
- Used water should be collected in a basin and discarded in a latrine if a drain is not available.

Tips on putting on and taking off PPE

Steps to put on PPE including gown

1. remove all personal items (jewellery, watches, cell phones, pens, etc.);
2. put on the scrub suit and rubber boots* in the changing room;
3. move to the clean area at the entrance of the isolation unit;
4. gather PPE beforehand. Select the right size coverall;
5. put on PPE under the guidance and supervision of a colleague;
6. perform hand hygiene.
7. Put on inner gloves (examination, nitrile);
8. put on coverall;
9. thumb (or middle finger) hole in the coverall sleeve or thumb loop;
10. put on face mask;
11. put on face protection (either face shield or goggles);
12. put on head covering: Surgical bonnet or hood.
13. Put on disposable waterproof apron;
14. put on outer gloves (examination, nitrile) over cuff;
15. self-check in mirror;
16. check friend and write name/ occupation/ time of entry.



Steps to put on personal protective equipment (PPE) including coverall

1 Remove all personal items (jewelry, watches, cell phones, pens, etc.)

2 Put on scrub suit and rubber boots[1] in the changing room.

3 Move to the clean area at the entrance of the isolation unit.

4 By visual inspection, ensure that all sizes of the PPE set are correct and the quality is appropriate.

5 Undertake the procedure of putting on PPE under the guidance and supervision of a trained observer (colleague).

6 Perform hand hygiene.

7 Put on gloves (examination, nitrile gloves).

8 Put on coverall.[2]

9 Put on face mask.

10 Put on face shield OR goggles.

11 Put on head and neck covering surgical bonnet covering neck and sides of the head (preferable with face shield) OR hood.

12 Put on disposable waterproof apron (if not available, use heavy duty, reusable waterproof apron).

13 Put on second pair of (preferably long cuff)[2] gloves over the cuff.

[1] If boots are not available, use closed shoes (slip-ons without shoelaces and fully covering the dorsum of the foot and ankles) and shoe covers (nonslip and preferably impermeable)

[2] Do not use adhesive tape to attach the gloves. If the gloves or the coverall sleeves are not long enough, make a thumb (or middle finger) hole in the coverall sleeve to ensure that your forearm is not exposed when making wide movements. Some coverall models have finger loops attached to sleeves.

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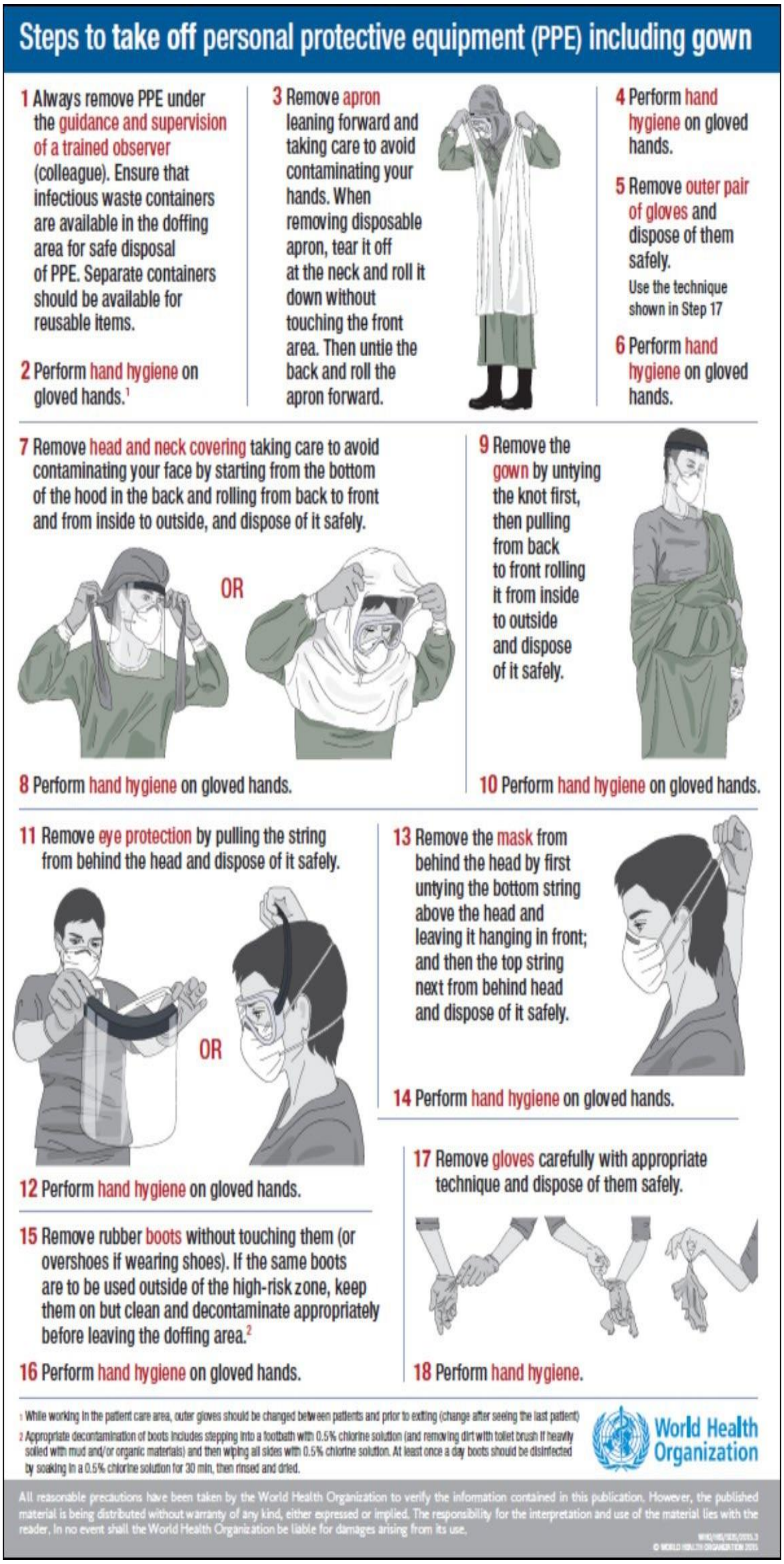
Steps to take off PPE including gown

1. Always remove PPE under the guidance and supervision of a trained observer (colleague).
2. Enter decontamination area by walking through chlorine tray.
3. Perform hand hygiene on gloved hands (0.5% chlorine).
4. Remove apron taking care to avoid contaminating your hands by peeling it off.
5. Perform hand hygiene on gloved hands (0.5% chlorine).
6. Remove hood or bonnet taking care to avoid contaminating your face.
7. Perform hand hygiene on gloved hands (0.5% chlorine).
8. Remove coverall and outer pair of gloves.
9. Tilt head back to reach zipper, unzip completely without touching any skin or scrubs, remove coverall from top to bottom.
10. After freeing shoulders, remove the outer gloves while pulling the arms out of the sleeves.
11. With inner gloves, roll the coverall, from the waist down and from the inside of the coverall, down to the top of the boots.
12. Use one boot to remove coverall from other boot and vice versa, and then step away from the coverall and dispose of it safely.



Steps to take off personal protective equipment (PPE) including gown

1 Always remove PPE under the guidance and supervision of a trained observer (colleague). Ensure that infectious waste containers are available in the doffing area for safe disposal of PPE. Separate containers should be available for reusable items.

2 Perform hand hygiene on gloved hands.[1]

3 Remove apron leaning forward and taking care to avoid contaminating your hands. When removing disposable apron, tear it off at the neck and roll it down without touching the front area. Then untie the back and roll the apron forward.

4 Perform hand hygiene on gloved hands.

5 Remove outer pair of gloves and dispose of them safely.
Use the technique shown in Step 17

6 Perform hand hygiene on gloved hands.

7 Remove head and neck covering taking care to avoid contaminating your face by starting from the bottom of the hood in the back and rolling from back to front and from inside to outside, and dispose of it safely.

8 Perform hand hygiene on gloved hands.

9 Remove the gown by untying the knot first, then pulling from back to front rolling it from inside to outside and dispose of it safely.

10 Perform hand hygiene on gloved hands.

11 Remove eye protection by pulling the string from behind the head and dispose of it safely.

12 Perform hand hygiene on gloved hands.

15 Remove rubber boots without touching them (or overshoes if wearing shoes). If the same boots are to be used outside of the high-risk zone, keep them on but clean and decontaminate appropriately before leaving the doffing area.[2]

16 Perform hand hygiene on gloved hands.

13 Remove the mask from behind the head by first untying the bottom string above the head and leaving it hanging in front; and then the top string next from behind head and dispose of it safely.

14 Perform hand hygiene on gloved hands.

17 Remove gloves carefully with appropriate technique and dispose of them safely.

18 Perform hand hygiene.

[1] While working in the patient care area, outer gloves should be changed between patients and prior to exiting (change after seeing the last patient)

[2] Appropriate decontamination of boots includes stepping into a footbath with 0.5% chlorine solution (and removing dirt with toilet brush if heavily soiled with mud and/or organic materials) and then wiping all sides with 0.5% chlorine solution. At least once a day boots should be disinfected by soaking in a 0.5% chlorine solution for 30 min, then rinsed and dried.

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13. Perform hand hygiene on gloved hands (0.5% chlorine)
14. Remove the goggles or face shield from behind the head (keep eyes closed).
15. Perform hand hygiene on gloved hands (0.5% chlorine).
16. Remove mask from behind the head (keep eyes closed).
17. Perform hand hygiene on gloved hands (0.5% chlorine).
18. Remove inner gloves with appropriate technique and dispose of safely.
19. Decontaminate boots appropriately and move to lower risk area one foot at a time and perform hand hygiene (0.05% chlorine).

Setting up a cholera isolation camp/unit/cholera treatment centre (CTC)

Site management

There are different recommendations for different situations/circumstances.

In urban settings and refugee camps:

Establish CTC + several oral rehydration points (ORPs)

Ideally, the CTC should be located inside the existing hospital premises but clearly separated and isolated from the other departments to avoid spread of infection to non-cholera patients. If the hospital premises are not suitable, another site must be found. In urban/camp settings, it is preferable to have one single CTC and several ORPs rather than setting up multiple CTCs, thereby increasing potential sources of infection. When affected areas are too far from the CTC, access can become a problem. Ambulances can be provided for referral, or a CTU may be established as an intermediate structure. Use of taxis/buses should be discouraged given the high contamination risk during the journey.

In rural settings:

Establish cholera treatment units (CTU)

The CTU should be located inside the health facility, or close to it. If this is not possible, other existing structures may be used. CTUs may paralyse routine health services as adequate case management is labour-intensive and other health services may suffer from staff shortage. In areas that are far from any treatment facility, it may be possible to decentralize the CTU to the level of the affected villages.

Oral rehydration points (ORPs)

ORPs points have two objectives: to treat patients, and to screen off and refer severely dehydrated patients to CTC/CTU(s). They reduce pressure on overburdened CTCs or CTUs. They can be decentralized to the community level. The community health worker should receive quick training and regular supplies, to be able to achieve given objectives.

Design of a CTC

Selection criteria

When establishing a cholera treatment centre, the following should be considered during site selection:

- proximity to the affected area;

- easy access for patients and supplies;
- protection against winds (there should be wind breaks);
- adequate space;
- compatibility with adjacent existing structures and activities;
- availability of adequate potable/safe water supply within a minimum distance to avoid contamination;
- good drainage from the site;
- provision of waste management facilities (clinical and general waste);
- availability of sanitary facilities (temporary);
- provision for extension of CTC (based on estimates from epidemiologist)

Setting up a temporary cholera treatment camp

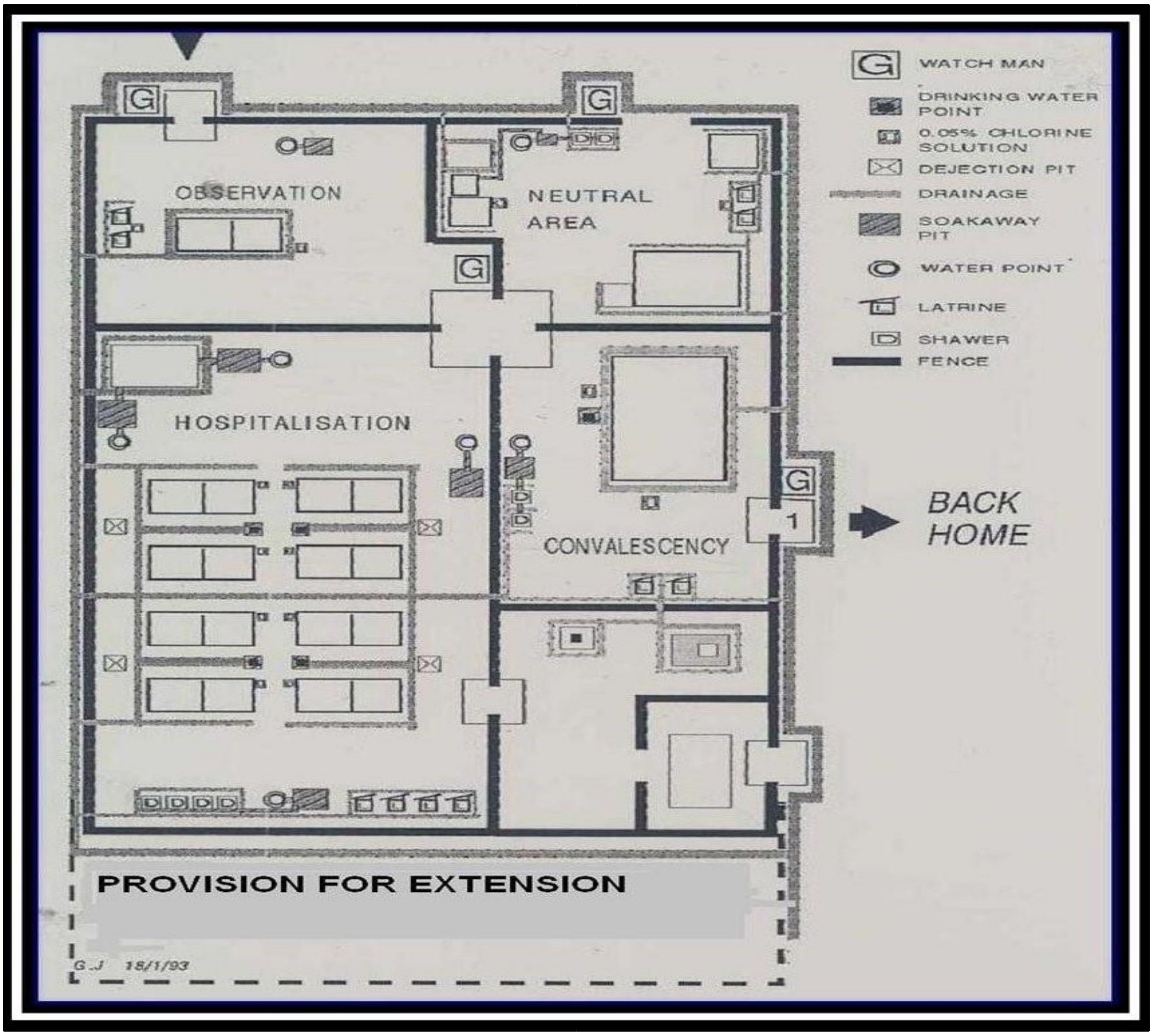
- In setting up a cholera camp, you can use an existing building or set up tents.
- It is important to consider the safety of patients and ventilation as high temperatures contributes to dehydration of the patients.
- The cholera camp should operate 24 hours a day independently of the other health facilities and therefore the necessary staff has to be recruited.
- It should be supplied with the necessary medical material specifically for the centre.
- An enclosure or other form of acceptable screen should be provided around the cholera camp.
- The various workstations should be clearly labelled and directions provided.
- The CTC must be a “closed system” where contamination is introduced through patients, and must be destroyed inside the structure. Under no circumstances should any contamination come out (through patients, water, material, solid and liquid waste etc.).

General rules for a good design:

- strict necessary movement for staff and patient;
- each zone is a “closed box”;
- systematic disinfection between zones;
- discipline and mutual control for the patient, attendant and staff on hygiene.

Good infection control means anything coming out is free of contamination

The diagram below shows the layout of a cholera treatment centre.


G WATCH MAN
DRINKING WATER POINT
0.05% CHLORINE SOLUTION
DEJECTION PIT
DRAINAGE
SOAKAWAY PIT
WATER POINT
LATRINE
SHAWER
FENCE
OBSERVATION
NEUTRAL AREA
HOSPITALISATION
CONVALESCENCY
1
BACK HOME
PROVISION FOR EXTENSION
G.J 18/1/93

1. **Triage and orientation**
   - (a) Patients are examined by a doctor for screening. If cholera, admit; otherwise send to normal HF.
   - (b) Patients are admitted with one attendant (caregiver) if necessary.
   - (c) Patients who are admitted are registered in the cholera line list.
   - (d) A footbath should be provided at the entrance.
   - (e) Toilets and water should be easily accessible for patients.
   - (f) Shower facilities should be provided for the patients.
   - (g) A disinfection area should be provided for the transporting vehicles and contaminated articles used on the patients.
   - (h) Tables, chairs, water containers fitted with taps, refuse receptacles should be provided in these areas.
   - (i) Provide safe water.

(j) Establish an ORP corner.

2. **Admissions area**

(a) Patients with severe dehydration and/or uncontrollable vomiting must be hospitalized for immediate rehydration.

(b) Each patient lies on a cholera bed with 1 bucket for stool collection underneath the hole in the bed and 1 bucket for vomit besides the bed. The following should be put in place or provided in the admissions area:

- separate rooms/tents for males and females, where possible;
- separate rooms for children, the old and pregnant women as risk of abortion increases with cholera;
- a foot bath and hand-washing facilities (with disinfectant) at the entrance;
- Provision for disinfection of soiled linen and clothing;
- access to toilets and washing facilities (with disinfectant) or showers where possible for the patients;
- cholera beds with receiving buckets, buckets for those who vomit and water containers for patients;
- tables and chairs for staff;
- refuse receptacles

(c) Patients should be screened by medical staff and categorized according to their status.

3. **Convalescence/recovery area**

(a) The convalescence or recovery area is meant for oral rehydration after hospitalization when less surveillance is required. Patients can lie on mats or sit on benches, as in the observation area.

(b) Patients who are no longer vomiting or having diarrhoea and who therefore require less medical attention can be put in this ward.

(c) Separate rooms/tents should be provided for males and females.

Appendix 6I. Response to chemical and radio nuclear events

Response to radiological events

If an accident is suspected:

- prevent inadvertent ingestion of contamination (e.g., wear gloves, do not smoke or eat);
- perform life saving measures and provide first aid for serious injuries immediately, before conducting radiological monitoring;
- keep people away from any potential source of exposure (at least 10 m from the public);
- arrange to transport seriously injured people to local medical facility;
- wrap them in a blanket to control the spread of contamination and tell those transporting victims and the receiving medical facility that the person may be contaminated and that the risk to those treating such a patient is negligible, but care should be taken to prevent inadvertent ingestion of contamination;
- identify and register potentially exposed/contaminated individuals; gather information that could be useful in reconstructing their dose, including medical symptoms and description of events;
- report to the competent officials and obtain instructions. In case of less serious injury, remain in the area until monitored.

Respond to action threshold

If an accident is confirmed:

- reassess and review medium to long-term protective actions such as food-chain restrictions with relevant departments and agencies;
- provide the population with useful, timely, truthful, consistent and appropriate information as to the likely health effects of the emergency by reference to existing knowledge;
- arrange for detailed clinical and radiological review of affected persons;
- promptly provide the public with the results of any medical examinations;
- establish and maintain an appropriate disease surveillance program;
- establish a registry of persons to be tracked and to receive long-term follow-up;
- base inclusion in the registry on objective criteria that indicate potential for an increase in the incidence of radiation-induced cancer;
- begin surveillance of any identified groups at risk, e.g., screening for thyroid disease in children in an area affected by radioactive iodine release;
- assist government authorities in planning a return to normal life for the affected population.

External contamination

Use instrumental contamination monitoring. Use cotton swabs for skin, nostrils, ear canals, wounds or any contaminated object. Each swab should be placed in a labelled test tube for counting.

Internal contamination

Use instrumental detection methods such as whole body counting, gamma camera, thyroid counting. Radionuclides may be in the blood or excreted in the faeces or urine. Excreta should be placed in appropriate containers and blood samples in test tubes for counting.

Decontamination procedures

- Materials: Lukewarm water, soap or ordinary detergent, soft brush, sponges, plastic sheets, tape, towels, sheets, iodine tablets or solution.
- Procedural priority: Remove all clothing and place in plastic bags. Carry out life saving measures first. Identify contaminated areas, mark clearly and cover until decontamination takes place. Start with decontamination of wounds when present and move on to the most contaminated area of the body.

Local contamination

- Cover areas still uncontaminated with plastic sheet and tape edges. Soak the contaminated area, gently scrub with soap and rinse thoroughly. Repeat the cycle and observe changes in activity. One cycle should not last longer than about 2–3 min. Avoid vigorous scrubbing. A stable isotope solution may facilitate the process.
- For wounds, irrigate with normal saline solution repeatedly. Surgical debridement might be considered in some instances. Eyes and ears may be irrigated gently with isotonic saline solution.

***Extensive contamination**:*

- Shower those not seriously injured. Bathing may be done on the operating table or stretcher for the seriously injured.
- Soak– scrub–rinse cycle should also be observed.

Inhalation: Irrigate nasopharynx and mouth.

Ingestion: Administer cathartics for insoluble materials. Administer diuretics by forcing fluids for soluble contaminants.

Prophylactic measures

- Cover areas still uncontaminated with plastic sheet and tape edges.
- Repeat washing after allowing the skin to rest.

Treatment

- Erythema and dry desquamation can be treated symptomatically. Lotions or sprays containing hydrocortisone can be used to relieve the symptoms associated with severe erythema accompanied by oedema. To treat moist desquamation, daily dressings and bathing of the affected skin in antiseptic solutions is helpful. Antibiotic creams can also be used.
- For ulceration, isolation of the limb in a sterile environment or daily dressing and bathing of the ulcer in antiseptic solutions is recommended. Analgesics or stronger opioids may be necessary. In the event of suspected or verified secondary infection, topical or systemic antibiotic therapy should be considered.
- For necrosis, only surgical treatment is effective. Surgical toilet is indicated. Excision of deep necrosis followed by skin grafts or other kinds of grafting may be conducted when indicated.
- Indications for amputation include very severe lesions with destruction of underlying tissues, including vascular damage, intractable pain and lack of infection control.

Expected outcome

Radionuclide activity is no longer detectable or is decreasing.

Response to chemical event/attack

Components of rescue and medical services:

- search and rescue teams;
- emergency medical teams used for day-to-day emergencies (medical officers, nurses, first aiders, ambulance);
- field medical services (field medical teams and posts);
- medical emergency response plans and procedures;
- personnel and equipment to reinforce the resources available for day-to-day emergencies;
- a transport service for medical evacuations;
- hospitals with casualty and surgical units.

At the emergency site:

- operate as close as possible (but within safe distance) to the emergency site;

- collaborate closely with different rescue teams (engineering, fire fighters, decontamination and human rescue groups);
- ensure all rescue workers don appropriate PPEs;
- assess the situation to determine that there is no eminent danger;
- rescue teams should locate casualties and remove them from danger;
- rescue team should do primary medical assessment to identify and manage life-threatening conditions. Assess:
  - airway;
  - breathing;
  - blood circulation.
- Rescue team should provide first aid and record details of first aid provided before forwarding casualties to field medical teams;
- field medical services post/s: establish field medical post/s;
- assign triage category to casualties based on the medical assessment;
- initiate appropriate treatment;
- prepare casualties for evacuation to hospital according to triage category;
- continue documentation of casualties;
- provide surveillance of casualties awaiting evacuation;
- liaise with casualty transport service;
- evacuate casualties to appropriate medical facilities according to priorities;
- ensure continuity of medical care for casualties along the whole length of the chain from the emergency site to the hospital;
- provide information to receiving medical facilities as necessary; along the whole length of the chain from the emergency site to the hospital;
- treat minor injuries not requiring hospitalization.

Hospital services:

- prepare for casualty reception;
- do a medical assessment to identify and manage life-threatening conditions;
- assign triage category based on assessments;
- provide appropriate treatment according to triage priorities and available hospital resources;
- continue medical documentation of casualties;
- undertake surgical procedures where necessary;
- provide postoperative care and release casualties.

Recognizing and diagnosing health effects of chemicals in chemical events

Agent Type	Agent name	Any unique characteristics	Initial effects
Nerve	Cylohexyl sarin Sarin (GB) Soman (GD) Tabun (GA) VX	Miosis (pinpoint pupils) Copious secretions Muscle Twitching/fasciculation	Miosis (pinpoint pupils) Blurred/dim vision Headache Nausea, vomiting Diarrhoea Copious secretions Sweating Muscle twitching/ fasciculation Breathing difficulty Epileptic seizures
Asyphxiant/Blood Arsine	Cyanogen chloride Hydrogen cyanide	Possible cherry red skin Possible cyanosis	Possible frostbite Confusion Nausea Patient may gasp for air similar to asphyxiation but more abrupt onset. Epileptic seizures before death
Choking/Pulmonary damaging	Chlorine Hydrogen chloride Nitrogen oxide Phosgene	Chlorine is greenish yellow with a pungent smell Phosgene gas smells like freshly cut hay or grass Possible frostbite	Eye and skin irritation Airway irritation Dyspnoea, cough Sore throat Chest tightness
Blistering/Vesicant	Mustard/Sulfur Mustard (HD, H) Mustard (gas) Nitrogen mustard Lewisite (L)	Immediately decontaminate skin; flush eyes with water or normal saline for 10-15 minutes; if breathing difficulty, give oxygen and any supportive care	Possible pulmonary oedema Mustard has an asymptomatic latent period There is no antidote or treatment for mustard Lewisite has immediate burning pain, blisters later Specific antidote British Anti Lewisite (BAL) may decrease systemic effects of Lewisite

Incapacitating/behaviour-altering	Agent 15/BZ	May appear as mass drug intoxication with ecstatic behaviour, distinct hallucinations and confusion Hyperthermia Mydriasis (dilated pupils)	May cause death Dry mouth and skin Initial tachycardia Altered consciousness, delusions, denial of illness, belligerence Hyperthermia Ataxia (lack of coordination) Hallucinations Mydriasis (dilated pupils)

Decontamination and treatment

Agent Type	Decontamination	First Aid Access	Other patient consideration
Nerve	Remove clothing immediately Gently wash skin with soap and water Do not abrade skin For eyes, flush with plenty of water or normal saline	Atropine before other measures Pralidoxime (2PAM) chloride	Onset of symptoms from dermal contact with liquid forms may be delayed Repeated antidote administration may be necessary
Asyphxiant/Blood Arsine	Remove clothing immediately – if no frostbite Gently wash skin with soap and water	Rapid treatment with oxygen For cyanide, use antidotes (sodium nitrite and then sodium thiosulfate)	Arsine and cyanogen chloride may cause delayed pulmonary oedema
Choking/Pulmonary damaging	Remove clothing immediately Gently wash skin with soap and water Do not abrade skin For eyes, flush with plenty of water or normal saline	Fresh air Forced rest Semi upright If signs of respiratory distress are present, oxygen with or without positive airway pressure may be needed Other supportive therapy as needed	May cause delayed pulmonary oedema, even following a symptom free period that varies in duration with the amount
Blistering/Vesicant	Immediate decontamination is essential to minimize damage Gently wash skin with soap and water Do not abrade skin For eyes, flush with plenty of water or normal saline	Immediatly decontaminate skin, flush eyes with water or normal saline for 10–15 minutes	Possible pulmonary oedema Mustard has an asymptomatic latent period, there is no antidote for mustard Lewisite has immediate burning pain, blisters later Specific antidote British Anti Lewisite (BAL) may decrease systemic effects of Lewisite Phosgene oxine causes immediate pain

Incapacitating/behaviour-altering	Remove clothing immediately Gently wash skin with soap and water Do not abrade skin	Remove heavy clothing Evaluate mental status Use restraints as needed Monitor essential temperature carefully Supportive care	Hyperthermia and self-injury are target risks Hard to detect because it is an odourless and non-irritating substance Possible serious arrhythmias Specific antidote (physostigime) may be available

Antidote recommendations following exposure to cyanide

Patient	Mild (Conscious)	Severe (unconscious)	Other treatment
Child	Antidotes may not be necessary	Sodium nitrite: 120.33ml/kg, not to exceed 10ml of 3% solution. Slow intravenous, no less than 5 minutes, or slower if hypotension develops Sodium thiosulfate: 1.65ml/kg of 25% solution intravenous over 10-20 minutes	For sodium nitrite-induced orthostatic hypotension, normal saline infusion and supine position are recommended. If still apnoeic after antidote administration, consider sodium bicarbonate for severe acidosis.
Adult	Antidotes may not be necessary	Sodium nitrite: 10-20 ml of 3% solution through slow intravenous, no less than 5 minutes, or slower if hypotension develops Sodium thiosulfate: 50 ml of 25% solution through intravenous over 10-20 minutes	

Note:

1. Victims whose clothing or skin is contaminated with hydrogen cyanide liquid or solution can secondarily contaminate response personnel by direct contact or through off-gassing vapours.
2. Avoid dermal contact with cyanide-contaminated victims or with gastric contents of victims who may have ingested cyanide-containing materials.
3. Victims exposed only to hydrogen cyanide gas do not pose contamination risks to rescuers. If the patient is a victim of recent smoke inhalation (may have high carboxyhemoglobin levels), administer only sodium thiosulfate.
4. If sodium nitrite is unavailable, administer amyl nitrite by inhalation from crushable ampoules.
5. Available in Pasadena Cyanide Antidote Kit, formerly Lilly Cyanide Kit.

Section 7: Risk communication

This section describes how:

- how to communicate within the IDSR;
- key principles for effective communication;
- how to create an enabling environment for effective communication to at-risk populations;
- how to communicate before, during and after the outbreak

Introduction

Risk communication is an essential element of disaster and emergency preparedness and response and is one of the essential capacities in the International Health Regulations 2005 (IHR 2005). Risk communication is a two-way exchange of information, perceptions and advice among risk assessors, risk managers, and various groups of people in the society about the likelihood and consequences of harm from the event (WHO, 2005). Its ultimate purpose is to ensure that everyone at risk is able to take informed decisions to mitigate the effects of the threat (hazard) such as a disease outbreak and take protective and preventive action. Risk communication uses a mix of communication and engagement strategies and tactics, including but not limited to, media communication, social media, mass awareness campaigns, health promotion, stakeholder engagement, social mobilization and community engagement. The current 21st century has been marked by an exponential growth in travel, trade, migration, as well as a communication technology revolution that has widened access to a variety of means of communication and information. The public and communities have been exposed to a variety of dynamic, fast-changing, formal and informal media, social media and complex social networks that influence how risk is communicated, perceived and acted on. The latest evidence shows that the practice of risk communication is a complex task that is an essential public health intervention in any response to disease outbreaks, pandemics and other public health emergencies. It is therefore important for risk communication to be conducted effectively, to promote the primary public health goal of rapid outbreak containment and prevention of avoidable death and disease with the least possible disruption to economies and society. During outbreaks and pandemics as well as humanitarian crises and natural disasters, effective risk communication enables people who are most at-risk to understand and adopt protective behaviours. It enables the authorities and experts to heed and address people’s concerns and needs, and to offer advice that is relevant, trusted and acceptable. This **section** describes how to conduct risk communication before, during and after an outbreak. Effective communication equips those at risk with the knowledge they need to make informed decisions for protecting themselves from disease, priority conditions or public health events. It also provides decision-makers with summary information especially regarding outbreak response, so that they review how resources were applied to contain the event.

7.1. Risk communication in the context of IDSR

The Integrated Disease Surveillance and Response (IDSR) strategy is an approach for improving public health surveillance and response for priority diseases, conditions and public health events at the community, health facility, district and national levels. Since IDSR has the potential to ensure the reliable supply of information at the national level with a view to fulfilling IHR requirements, risk communication should be included in all IDSR core functions and activities, particularly detection, sample collection, reporting, analysis and interpretation, feedback, response and preparedness. IDSR core functions and activities for each level of the health system are well illustrated in the **Introduction section** of this guideline. Effective risk communication is therefore needed to achieve IDSR objectives. If risk communication is well planned and integrated into IDSR, it can improve decision-making and the adoption of recommended behaviours by communities and also contribute to the prevention, control and response to epidemic-prone diseases, priority affections and other public health events. Such communication needs to be carefully planned, implemented and properly integrated with emergency management activities and operations at the community, district, region and national levels to support all relevant core IDSR functions and related activities.

7.1.1. Benefits of risk communication

Risk communication improves decision-making, compliance with treatment and the required behaviours for preventive actions. It also promotes transparency and delegation of tasks and builds trust with individuals, community leaders, health workers and policy-makers. When risk communication is properly conducted, it promotes the primary public health goal of rapid outbreak containment, thereby preventing avoidable death and disease with the least possible disruption to economies and society. During outbreaks and pandemics as well as humanitarian crises and natural disasters, effective risk communication enables people who are most at-risk to understand and adopt protective behaviours. It enables the authorities and experts to heed and address people’s concerns and needs, and to offer advice that is relevant, trusted and acceptable.

It is critical that risk communication is not only targeted at outbreak response; risk communication plans should include activities conducted before, during and after the outbreak. When the public is at risk of a real or potential health threat, direct interventions may take time to organize and resources may be limited. Hence, communicating advice and guidance is often the first and most important public health tool in managing a risk. Proactive risk communication encourages the public and service providers to adopt protective behaviours when they are linked to functioning systems and services. It facilitates heightened disease surveillance, reduces confusion, and minimizes miscommunication and falsehoods (rumours) related to the cause and transmission of a disease as well as proven effective protective actions. It allows for better use of resources, which is crucial to effective response (WHO, 2008).

7.1.2. Target audiences for risk communication

Target audiences include:

- community: All people at risk of acquiring a disease or in need of health services within the context of the public health event;
- health-care providers and first responders (rescuers);
- HF and healthcare staff;
- surveillance officers;
- laboratory staff;
- points of entry;
- airlines staff;
- immigration officers;
- travellers;
- stakeholders (policy-makers, Ministry of Public Health and partner ministries, maternal and child health organizations, partners, community organizations, etc.)
- media as a channel to reach these audiences;
- schools and workplaces;
- traditional and religious leaders.

7.1.3. Community engagement and its relevance to public health emergency preparedness and response

Community engagement is crucial to risk communication. It is the process of working collaboratively with, and through people affiliated by geographical proximity, special interest, or similar situations to address issues affecting their well-being and is often used as an active method of implementing change. During risk communication, the emphasis is on building and upholding trust-based relationships.

The steps for community engagement involve:

- setting the goals of the plan;
- determining who to engage;
- developing engagement strategies;
- prioritizing of activities;
- designing an implementation plan; and
- monitoring and evaluating progress.

Effective community engagement helps you to:

- know the community (problems and needs);
- understand existing health beliefs, attitudes and practices;
- listen to the community carefully;
- analyse community dynamics; and
- involve the community in all aspects of the response beginning from planning stages.

7.1.4. Risk communication approaches

The components needed for effective emergency risk communication include:

- health education;
- social mobilization;
- community engagement;
- media and social media;
- outbreak communication;
- crisis communication;
- messaging (Information, Education and Communication (IEC) and Behaviour Change Communication (BCC));
- rumour monitoring and management; and
- advocacy.

7.1.5. Integrated risk communication model

Since risk communication is a complex activity involving different audiences, it is crucial to adopt an integrated approach. The key components for integrated emergency risk communication are presented in **Figure 16**. This model allows for the successful design and implementation of an effective communication strategy. It highlights the need for a collaborative approach between different target audiences across the board.

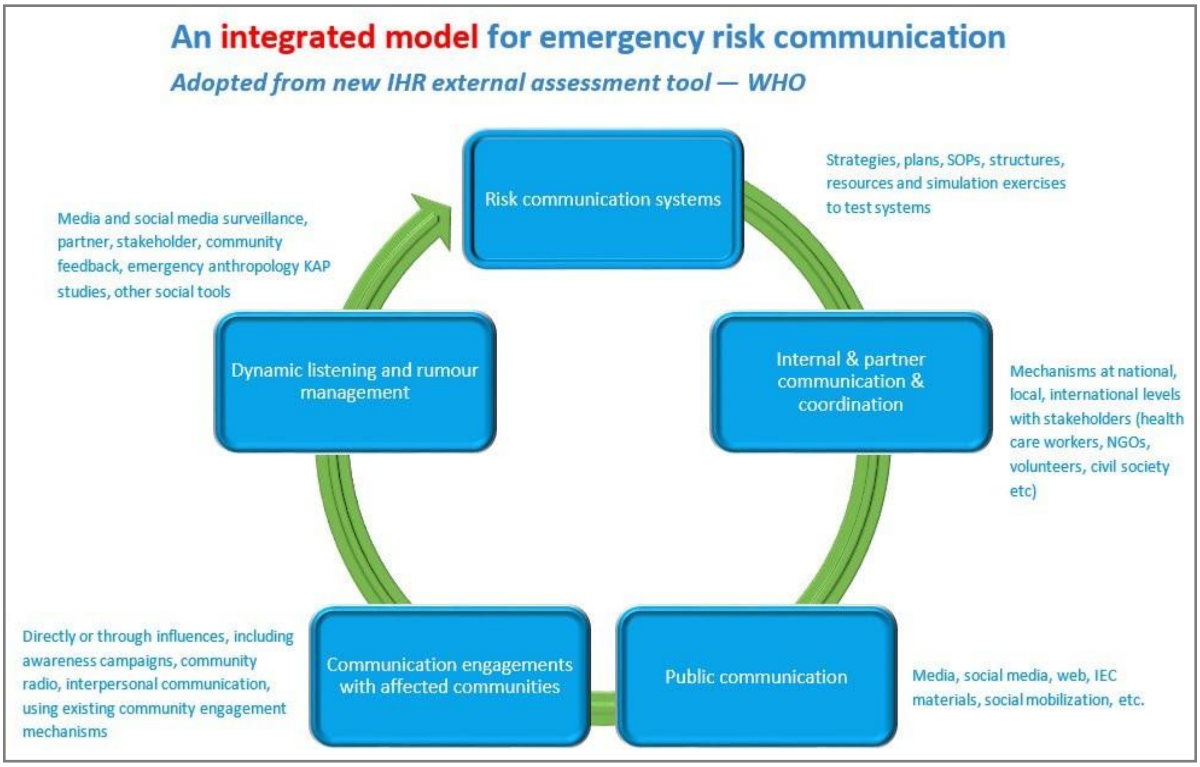

An integrated model for emergency risk communication
Adopted from new IHR external assessment tool — WHO
Risk communication systems
Strategies, plans, SOPs, structures, resources and simulation exercises to test systems
Internal & partner communication & coordination
Mechanisms at national, local, international levels with stakeholders (health care workers, NGOs, volunteers, civil society etc)
Public communication
Media, social media, web, IEC materials, social mobilization, etc.
Communication engagements with affected communities
Directly or through influences, including awareness campaigns, community radio, interpersonal communication, using existing community engagement mechanisms
Dynamic listening and rumour management
Media and social media surveillance, partner, stakeholder, community feedback, emergency anthropology KAP studies, other social tools

Figure 16: An integrated model for emergency risk communication

7.2. Key principles for effective communication

There are five key principles for effective communication as outlined below:

7.2.1. Building and maintaining trust

Building and maintaining trust is, arguably, the most important function in effective communication during an outbreak or a public health event and should include:

- timely, transparent information regarding the nature of the threat;
- the response to the event; and
- actionable advice on protective actions people can take that, together with functioning services, increase self-efficacy.

This creates trust in the response between communities and the response teams and increases the likelihood that they will follow the advice given.

Trust is now considered the most important requirement for effective risk communication.

According to the latest evidence, risk communication in health emergencies should include ensuring genuine participation of the population, taking into account three key elements, namely:

(a) understanding of the specific context, concerns, beliefs, practices and traditions of the population concerned in order to develop scientific and logistical information and explanations that address community concerns (social science intelligence);

(b) provision of understandable and trusted advice that they are likely to follow to save lives and curb the outbreak within the shortest possible time; such advice is provided in their own languages, adapted to their educational levels and preferences (i.e., oral or visual) and disseminated through their preferred channels and messengers (interpersonal communication); and

(c) meaningful community engagement and the participation of the community trusted interlocutors/messengers.

Risk communication should include timely, transparent, understandable information relayed to the affected and at-risk population on:

1. ***the nature of health risk they face;***
2. ***the response that is being organized; and***
3. ***what they can do to protect themselves and their loved ones***.

Trust is therefore the currency for all public health interventions, and has, in the current era of information overload, emerged as the critical element for effective risk communication (i.e., ensuring that expert advice is acted on by key stakeholders, affected and at-risk populations). Risk communication should therefore be aimed at building, maintaining or restoring public trust in those tasked with risk management.

The latest evidence from 21st century outbreaks reveal that, in order to build trust, risk communication activities should:

- be integrated with functioning and accessible services;
- be transparent and timely;
- be easy to understand for target populations (i.e., in their preferred oral or visual formats; in their own languages or dialects; and tailored to their educational levels and cultural references);
- acknowledge and communicate uncertainty (neither over-reassure nor speculate; rather, communicate frequently so that the evolution of the event and public understanding are transparent and not destroy trust);
- link to self-efficacy (can people really do what you ask them? Do they have the ability, equipment, services, education they need to adopt our advice?);
- be disseminated using multiple channels, platforms and methods;
- identify, involve and collaborate with people that the community trusts when making decisions (and not just in information dissemination).

This ensures that interventions and any communication on them are contextually appropriate and community-owned.

7.2.2. Timely announcements and transparency

In most cases, community response to any outbreak or public health event depends on the way the first announcements are made. This means that a public health event or threat must be announced as soon as it emerges, even when the information is incomplete or changing fast. This implies that communicating uncertainty is a cornerstone of risk communication.

Communication by authorities, response managers or front-line personnel must include:

- information about the uncertainties associated with the risk, event and interventions;
- information indicating what is known and unknown at each given moment in time;
- a commitment and follow-up to keep people frequently informed on the changing, uncertain situation;
- all platforms, mechanisms and trusted messengers to ensure that stakeholders and the population reach consistent and coordinated information.

7.2.3. Listening to, understanding and respecting community concerns

Understanding community perceptions, concerns, fears and expectations is as critical to risk communication as understanding the risky practices and behaviours that affect risk. The understanding of communities must start before and continue during an emergency. There are many ways to improve awareness of community concerns and understand the contexts that determine whether the advice given to them on corrective or preventive practices will actually be accepted and acted upon.

These include Knowledge, Attitude and Practice (KAP) surveys or mini-surveys, home visits, group discussions, key informant interviews, feedback from stakeholders, social media and media monitoring, etc. Using these methods and other social science approaches enables the execution of health interventions and offer health advices, based on evidence gathered.

7.2.4. Advance planning of risk communication

Risk communication is most effective when it is integrated with public health emergency preparedness, risk analysis and response (risk management). This means that a risk communication plan must be prepared during the preparedness stage. Emergency risk communication planning must occur in advance and be a continuous process focused on preparedness, prevention and response. Planning should be participatory, responsive to the context of affected groups and should include feedback from such groups. The International Health Regulations (IHR, 2005) require all governments to build national capacity for detection signal and response to public health emergencies. One of the core capacities is risk communication. Accordingly, risk communication planning should include the systems required (strategies, plans, SOPS and mechanisms at the national, regional and district levels); the coordination of partners, sectors and stakeholders; the capacity for fast, effective public communication in the preferred languages and channels of the population; the ability to track and quickly manage concerns, perceptions, rumours and misinformation; and communication engagement with affected and at-risk communities.

7.2.5. Ensuring equity in risk communication

All citizens have a right to appropriate information about health risks, including what needs to be done in response to threats to their health. Unfortunately, large segments of society are excluded from routine communication about threats to health.

Risk communication must therefore ensure equitable sharing of information to the public and avoid exclusion of marginalized members of society from health action. This means paying attention to the reach of communication, using trusted channels and messengers; avoiding jargon or technical language; using the people's own languages and dialects; adapting messages to people's levels of understanding, education or people in any possible situation of disability or destitution; and ensuring that the actions promoted are those that people can realistically change. Special attention should be paid to analysing power dynamics in communities and taking special measures to reach those hardest to reach (women, minorities, the very old and young, people with disabilities, the poor, migrants and refugees, etc.).

7.3. Creating an enabling environment for effective communication to at-risk populations

Creating an enabling environment for effective risk communication requires:

(a) Establish risk communication systems and structures at the district, regional and national levels

- if unavailable, establish multisector communication committees/structures across all levels; i.e., national, regional and district levels (See ***Appendix 5E*** for examples of members of the communication subcommittee and their roles). Term of reference can be expanded depending on the pre-outbreak, outbreak and post outbreak phase in line with each function (See ***Appendix 7F*** for an expanded list of possible stakeholders);

  - review the existing risk communication structures and mechanisms.

(b) Ensure that the communication system has a link to the community leadership structure since they wield great influence within the community. A quick assessment can be made to evaluate the framework for risk communication and this can include:
  - conducting an assessment to identify risk communication needs based on risk profile;
  - preparing a mapping and developing a database of risk communication stakeholders at all levels; and
  - preparing a resource mapping for risk communication.

(c) map languages and dialects, religions, preferred and trusted means/channels and interlocutors (sources) for communication, as well as traditional practices relevant to the top priority health risks and use all this information to shape risk communication strategies and plans;

(d) if no interlocutor is available at the district and regional levels, identify a government spokesperson and ensure that he/she is trained in public communication procedures;

(e) in addition to risk communication personnel, all frontline personnel should receive basic training in risk communication (surveillance, contact tracing, case management, social mobilization, community engagement, burial teams, health personnel, volunteers, etc.);

(f) develop a risk communication plan for Public Health Emergencies at district, regional and national levels and ensure that all stakeholders are given some orientation on risk communication procedures;

(g) develop a coordination platform as well as internal and partner communication mechanisms for engaging any stakeholders, including media outlets and community radio networks and a definition of roles and responsibilities;

(h) have detailed budgets and advocate strongly for resources mobilization, and multisector collaboration to implement public health emergency and risk communication activities at all levels;

(i) create a system for dynamic listening and rumour management.

(See ***Appendix 7E*** for checklist on risk communication monitoring).

7.4. Communicating before, during and after the outbreak

7.4.1. Pre-outbreak/Routine risk communication

A large proportion of communication activities should be implemented in the pre-outbreak/pre-emergency phase to ensure better preparedness. Those managing communication activities should take advantage of the absence of an emergency to build the national communication capacity and develop communication plans and tools that will bring the country to a high level of communication preparedness. The pre-emergency phase should also be used to develop the necessary communication messages and materials and promote the practice of risk-prevention behaviours.

Before an outbreak or other public health event, the following measures should take place:

- Ensure that the Public Health Emergency Management Subcommittee for Risk Communication meets at least once monthly or quarterly to:
  - review the risk communication plan and required risk communication materials/logistics;
  - develop, pre-test, produce and disseminate appropriate communication materials based on identified public health risks; and
  - Organize the training of risk communication human resources.
- ensure that the communication coordination mechanism is in place with clear term of references and well- defined roles and responsibilities for each entity;
- organize periodic interactions with stakeholders who will be involved in risk communication for prevention and preparedness or in response should an event or emergency occur. These include district, regional or national media; community radios, civil society, and stakeholders from other sectors, like the animal and environmental health;
- review past emergency communication interventions to draw lessons learnt, build on successful practices and avoid negative ones;
- collect and analyse epidemiological and social data about periodic disasters and outbreaks: seasons of common diseases; at-risk communities/populations; as well as accessible and credible channels of communication;
- build capacity for outbreak communication, identify and train spokespersons;
- alert all relevant entities and notify them on their role(s) in case the outbreak occurs;
- ensure that messages and materials have been developed, pre-tested and are ready for production and dissemination;
- ensure that all required training modules, guidelines and monitoring checklists are developed and updated;
- develop and share Standard Operating Procedures (SOPs) for social mobilization and community engagement;
- ensure the integration of risk communication in the overall preparedness and public health emergency response plan;
- identify and prepare the database of stakeholders and partners, such as youth/women groups or organizations, schools, religious institutions, Civil Socialety of Organization, theatre

groups, and other community groups that can disseminate messages at the grassroots level and involve them in preparedness activities;

- identify all the channels of communication available to spread the message and assess the reach and credibility of these channels
- produce a “Response Kit” which includes frequently asked questions, media briefs, training manual, micro-planning tools, monitoring checklists/tools, communication plan templates and key messages/materials for rapid distribution. This kit is intended for the use of communication stakeholders at all levels;
- establish communication lines with the media (journalists and radio/TV stations, written press, bloggers) train and regularly update them;
- organize activities with theatre groups, musicians and traditional community entertainers.
- identify and train Community Health Workers, community leaders, religious leaders, influential people, women’s groups, youth groups and other social mobilizers in Communication for Development (C4D) and risk communication;
- identify mechanisms for communicating with hard-to-reach and vulnerable populations (the aged, persons with disabilities, children, the nomadic, people living in insecurity) and with isolated communities to ensure that they have access to health protection information and assistance;
- define communication channels that can be used to reach vulnerable groups;
- disseminate messages that describe the actions that the government is taking to protect the community and health care professionals, promote awareness of the imminent health threats and preventive behaviours and actions that they can take to reduce the risk. This can be done through the mass media, such as local community radios, government communications, community drama groups, television, print media and social media (Facebook, twitter, etc.).
- conduct community engagement activities and build trusted relationships between those in authority and communities through training, dialogue, consultations and capacity- building. It is important to note that effective community engagement is based on trusted relationships between those in authority and communities. It is important, therefore, to use every opportunity to strengthen these relationships during non-public health emergency periods.
- use ongoing health education, health promotion and other means to create, test and build trust in the systems. Interlocutors can be used for risk communication during emergencies;
- make arrangements for a functioning hotline facility, which can be started immediately when the emergency occurs;
- establish a media monitoring team to monitor the news and social media, as well as practises and behaviours related to the emergency;

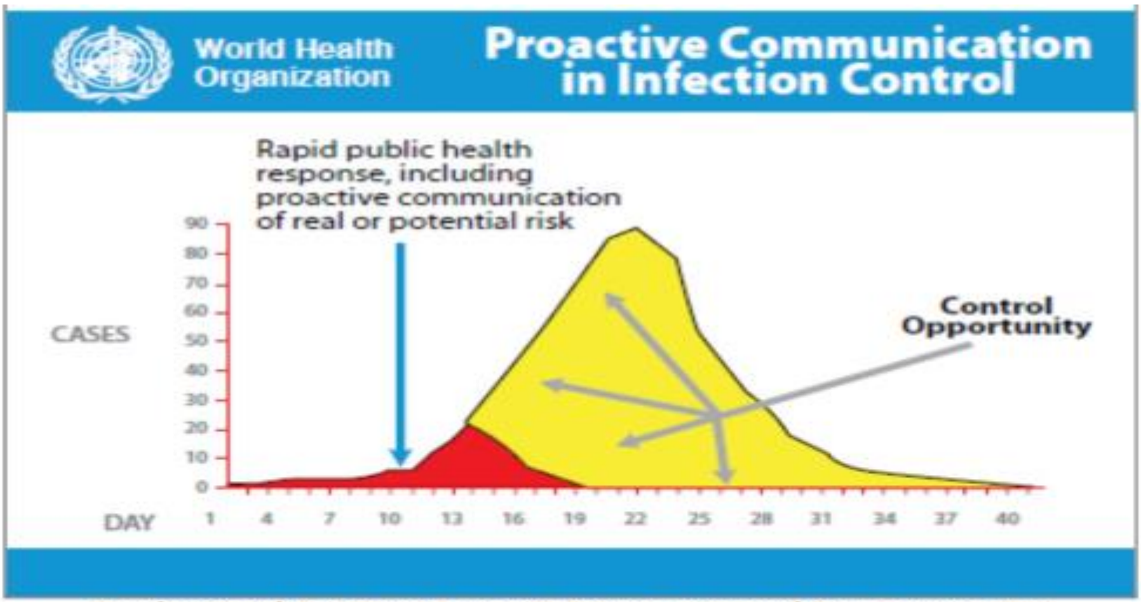
- maintain and update a list of press organs.
- develop plans for routine monitoring of rumours and misinformation.

7.4.2. During outbreak response

During an outbreak response, and when there is a real or potential risk on health, treatment options may be limited, direct interventions may take time to organize and resources may be few. Providing advice and guidance, therefore, often stands as the most important public health tool in managing a risk. The focus of risk communication is to control outbreak and mitigate disruption to society by communicating with the community in ways that build, maintain or restore trust. Proactive communication encourages the population to adopt protective behaviours, facilitates heightened disease surveillance, reduces confusion and fear and allows for a better use of resources, all of which are necessary for an effective response. It also shows that health authorities are in control of the situation and care about the public. Hence, it builds trust between such authorities and the community at large.

People have a fundamental right to information and to participation. In addition to the public health objectives, the populations have a right to information on protective actions and also to participate in and shape interventions that are acceptable to them.

The figure below illustrates an outbreak curve showing the importance of proactive communication.



D'après : adapté de la figure 2, page XII, Rapport sur la santé dans le monde 2007

Figure 17: Outbreak curve showing the importance of proactive communication

The yellow area represents the number of cases that could be avoided through actions of rapid response to the threat.

The blue arrow indicates the point at which proactive communication plays a crucial role in supporting such a rapid response. By alerting a population and partners to an infectious disease risk, surveillance of potential cases increases, protective behaviours are adopted, confusion is limited and communication resources are more likely to be focused. Effective communication can help limit the spread of a disease and ultimately save lives. It also minimizes damage to societies and economies and can help communities recover faster from a health event or public health emergency.

7.4.2.1. Identify and coordinate partners and other stakeholders during an outbreak

Outbreaks usually create fear in the community. The involvement of different stakeholders sometimes leads to lack of coordination and the duplication of efforts. Provision of timely and accurate information through a well-coordinated mechanism is important.

Internal coordination of communication among national stakeholders is crucial during an emergency. The risk communication and social mobilization Subcommittee described in Section 5 ensures that an internal communication system is established among national stakeholders to ensure the timely sharing of information to various government sectors.

Partner coordination is another key essential element during outbreak and event response. It is aimed at fostering ownership, effective participation of key players and efficient use of resources (See Appendix 7F for the potential partners and stakeholders who can be involved). It establishes routine communication structures among health professionals, community and partners. It helps ensure that this vital link is available and functional during an emergency. A district, regional or national communication plan should consider the partner communication.

Coordination helps ensure that messages reaching the population are consistent and not contradictory or confusing, thereby promoting trust and the likelihood that expert advice will be followed. The Public Health Emergency Management Committee (PHEMC) through the Public Health Emergency Operation Coordination Centre (PHEOC) ensures that communications are consistent and reflect the data that has been analysed, and that the focus of communication activities is transparent and accurate, and considers community experiences and expectations regarding the outbreak.

Distinction should be made between communication with experts and other stakeholders of the response who require a more down to earth description and explanation. The latter, including the media, the civil society and the general population will require targeted and adapted products and messages. This means that carefully segmenting and targeting audiences, as well as adapting materials, messages and mechanism to suit each of them is essential.

7.4.2.2. Communicate with the affected community and stakeholders

Communication with affected communities and stakeholders, including the media is essential during outbreak and public health event response. Thus, establishing routine communication structures and processes between the health and community partners helps to make this vital link available and functional during a public health emergency. Various options for communicating include press releases and conferences, television and radio messages, meetings (health personnel, community, religious, opinion and political leaders), educational and communication materials (posters, fliers), to multimedia presentations (films, video or narrated slide presentations) at the markets, health centres, schools, women's and other community groups and service organizations, religious centres, local community media, Social media (Facebook, Twitter, WhatsApp, etc.), SMSs, telephone, hand-carried messages, community drama groups/play groups; site visits, faxes, email updates and exchanges of communication materials through more formal decision-making committees.

Whatever the mechanism, the focus should be on transparent and trustworthy communication that considers community experiences.

Consider the following points when preparing messages:

- **make sure messages are clear and understandable to the audience:** What is happening? Why and how is it happening? What threats to health do exist or are likely to occur? What should the community do? Where can populations get services or information? What assurances can be given? Are the messages written in an understandable language and tailored to the audience's level of understanding?

Research shows that risk should not be explained in technical language.

- **Consider these factors when providing messages**: who is your target audience? What do you want your target audience to do after hearing the message? Do the target audience have an enabling environment to do as advised? Are there functioning and accessible services that enable the target audience to follow the advice?

- **Promote dialogue**: ensure that there is two-way communication/exchange; listen to the target audience's concerns and respond appropriately rather than just informing.
- **Demonstrate empathy and be caring**: are you showing empathy? Are you insensitive? Are you respectful?
- **Provide harmonized and consistent messages**: Ensure that consistent messages reach the community, notwithstanding the variety of partners involved in the dissemination of information. Use memory aid and other tools to keep the same frame and logic for the messaging as this would enable partners to adapt to the context of more segmented target audiences. Are messages consistent regardless of who is issuing them? Inconsistent or conflicting messages create confusion and destroy population trust in the response and with authorities;
- **Establish a mechanism for continuous collation of factual data** about the public health event.
- **Update messages intended for the public and share them** with stakeholders involved in information dissemination.
- **Ensure the relevance** of information: communicate data/information that best illustrate the situation, factoring in community concerns. Use examples that relate to the audience.

Note: Consider pre-testing messages from similar settings before dissemination.

In case of rumours, quickly address them and any inaccuracies, and especially within the specific community where they occur. Set up a rumour management system. Widespread rumours should be counteracted through public statements or press conferences or by providing comprehensive information.

We should build, maintain and restore trust as we communicate and be as courteous as possible, share health information with trusted and respected community leaders and ask them to transmit to the community. Only credible and authorized persons should communicate during crisis period.

On a regular basis, district and regional medical officers should meet with the local leaders to provide:

- regular and up-to-date information on the outbreak and response;
- clear and simple health messages for the media;
- clear instructions to communicate to the media only the information and health education messages provided by the Public Health Emergency Management Committee.

7.4.2.3. Distribute communication tools and develop fact sheets or aide-mémoire

Fact sheets or memory aids are brief summaries of 1 to 2 pages. They are prepared by health staff for consumption by the general public and deal with a single topic or message. For example, a fact sheet on a bloody diarrhoea (Shigella) outbreak in a district may contain the following information: the cause of bloody diarrhoea, how it is transmitted, steps for prevention and updates on the number of cases and deaths. The fact sheets could be posted on a bulletin board or distributed to community groups in paper or electronic form during awareness campaigns.

Where possible, transform the fact sheets into audio files (short audio recordings on a phone), scripts or visual files (posters). These can be used depending on the target audience (oral, visual, written or illustrated communication). See attached example in the ***APPENDIX 7A***. Also distribute any other developed communication materials, ensure that they have been pre-tested for a good understanding.

7.4.2.4. Develop and distribute outbreak or public health event reports

The national or regional level publishes a situation report (SITREP) describing outbreak and its trends. These situation reports have a wider audience than just the health staff in a concerned district or health facility. They are usually brief (2 to 8 pages) and are submitted by administrative authorities, legislators and other decision-makers. They are thus valuable advocacy tool for reaching technical, and financial partners, and donor

The reports contain at least:

- a summary table showing the number of updated cases and deaths due to EPD, priority affections and other public health events;
- a commentary or message on a disease or a given public health event;
- relevant social science data on risky practices, behaviours and other factors.

Upon receipt of the public health situation report at regional or district level, it is important to display it where everyone can see it, make copies and distribute to HFstaff, submit a copy of the report during supervision to show health workers how useful are data transmitted during outbreak management. (See **Appendix 7C** on a template for drafting a situation report).

7.4.2.5. Communicating to the media

The media is a major influence and should be seen as a partner in risk communication. However, the media is often associated with political parties or private interests and can therefore have biases of their own. They are also able to research and report on community concerns, sensationalize stories and may not always rely on facts and evidence. Therefore, it is essential to meet regularly with the media, brief and educate them on priority hazards and response systems, and also provide them with appropriate information to cultivate a respectful and trusted relationship with them.

The media ensure wider dissemination of messages on radio, TV or other appropriate channels.

As part of the risk communication plan, determine how to announce news of the outbreak and then keep the media regularly informed. Often, regular press releases and media briefings are appropriate tools for communicating with the media.

If the emergency is complex, organizing a workshop with targeted media is helpful to ensure correct information is disseminated, as most journalists are not trained in medicine or public health. In addition, it is good to develop media kits including fact sheets or memory aid and community messages about the priority diseases and public health events. Prior to the outbreak, it is necessary to reach out to the media and identify the key outlets to work with during an outbreak.

It is also necessary to identify, prior to a public health emergency, the clearance process for media products and appreciate the following:

- ensure prompt and frequent access to experts, officials and spokespersons who will speak authoritatively and credibly on the issue at hand;
- provide media training to spokespersons;
- ensure spokespersons can speak in a nonprofessional language, clearly explain scientific ideas and terms, avoid using scientific jargon, and illustrate the information provided with easy-to-understand stories or examples. Talking points having the latest information could be used, with the messages kept as simple as possible;
- ensure that the identified spokespersons are able to clearly communicate the uncertainty in an evolving event and to admit it when they do not have any answer. Community case definitions and communication aids will help the spokesperson to deliver correct messages.
- promptly answer journalists` calls to show your consideration for them;
- provide them with accurate and well-explained information;

- provide information and give exclusive interviews to present a different point of view or perspective;
- propose reports;
- Provide them with clear easy-to-use documents (written, audio, visual or audio-visual).

Note: release information to the media only through the spokesperson to make sure that the community receives clear and consistent information, and monitor the media daily to see how the outbreak is being reported.

Also, include social media in your monitoring strategy. If you feel that the wrong messages are being disseminated, devise a strategy for correcting this misinformation.

7.4.2.6. Communicate to health professionals

During the response, it is important to:

- communicate regularly with health professionals by providing correct information pertaining to the outbreak. It is important to communicate with health staff at the various levels of the health pyramid about the data reported (including missing data), analysis results for such data and the measures being taken to respond to the potential public health event that they have reported. Communication can also include providing participating health care professionals with any outbreak or public health event response reports for future reference;
- make sure that health professionals provide correct information on number of cases and any deaths that have occurred, on case management or on any other response intervention;
- ncourage health professionals to always use updated information, in real-time during an event or a public health emergency using reliable sources such as the Ministry of Public Health, WHO platforms (www.OpenWHO.org) on common, re-emerging and emerging epidemic-prone diseases and on risk communication.

Increasingly, during emergency response to disease outbreaks, WHO provides real-time online, off-line or face-to-face training to update health care professionals and response teams. These provide them an opportunity to update or acquire knowledge and skills.

7.4.3. Post-outbreak response

7.4.3.1. Prepare an outbreak or public health event response report

After any outbreak or public health event response has taken place, district staff who led the investigation should prepare a report, the purpose of which is to document how the problem was identified, investigated, responded to, what the outcome was, which decisions were taken and what recommendations were made. It is important to make sure that the HFthat reported the initial cases receives a copy of the report. (See ***Appendices 7B and 7D*** for examples of recommended formats).

7.4.3.2. Evaluate lessons learnt in order to strengthen the response to possible similar public health emergencies

In order to strengthen the response to public health emergencies in the future, it is worth to assess:

- the effectiveness of the communications team in each phase and area;
- the effectiveness of meetings;
- the effectiveness of the internal flow of communications;
- the media monitoring;
- the response of various media;
- the outputs and outcomes of risk communication and community engagement.

7.4.3.3. Periodic assessment of the risk communication plan

To assess the risk communication plan, it is necessary to:

- carry out simulations to test the risk communication plan in order to detect possible weaknesses or gaps that need to be corrected before an emergency;
- revise the plan based on lessons learnt from the simulation exercise, AAR or other assessment done.

WHO provides ready-made desktop and other simulation exercises on www.OpenWHO.org

7.5. Appendices to Section 7

APPENDIX 7A	Sample fact sheet or memory aid
APPENDIX 7 B	Template for drafting district outbreak report
APPENDIX 7 C	Template for preparing public health event situation report
APPENDIX 7 D	Template for outbreak investigation report
APPENDIX 7 E	IHR core capacity monitoring questionnaire for risk communication
APPENDIX 7 F	List of stakeholders and partners for risk communication

APPENDIX 7A: Sample fact sheet or memory aid

Influenza A virus

General information about influenza A virus infections in humans
(*Reference* *http://www.who.int/influenza/human_animal_interface/faq_H7N9/en/*)
Influenza A H7 viruses are a group of influenza viruses that normally circulate among birds. The influenza A (H7N9) virus is a subtype among the larger group of H7 viruses. Although some H7 viruses (H7N2, H7N3 and H7N7) have occasionally been found to infect humans, no human infections with H7N9 virus has been reported until recent reports from China.

What are the main symptoms of human infection with influenza A virus (H7N9)?

Thus far, most patients with this infection have had severe pneumonia. Symptoms include fever, cough and shortness of breath. However, information is still limited about the full spectrum of diseases caused by the influenza A (H7N9) virus infection.

Why is this virus infecting humans now?

The answer to this question is still unknown, because the source of exposure for these human infections is not yet identified. However, analysis of the genes of these viruses suggests that although they have evolved from avian (bird) viruses, they show signs of adaptation to growth in mammalian species. These adaptations include an ability to bind to mammalian cells, and to grow at temperatures close to the normal body temperature of mammals (which is lower than that of birds).

What is known about previous human infections with H7 influenza viruses globally?

From 1996 to 2012, human infections with H7 influenza viruses (H7N2, H7N3, and H7N7) were reported in the Netherlands, Italy, Canada, United States of America, Mexico and the United Kingdom. Most of these infections occurred in association with poultry outbreaks. The infections mainly resulted in conjunctivitis and mild upper respiratory symptoms, with the exception of one death, which occurred in the Netherlands. Until now, no human infections with H7 influenza viruses have been reported in China.

Is the influenza A (H7N9) virus different from influenza A (H1N1) and A (H5N1) viruses?

Yes. All three viruses are influenza A virus but they are distinct from each other. H7N9 and H5N1 are considered animal influenza viruses that sometimes infect people. H1N1 viruses can be divided into those that normally infect people and those that normally infect animals.

How are people becoming infected with the influenza A (H7N9) virus?

Some of the confirmed cases result from contact with animals or with an animal environment. The virus was detected in a pigeon in a market in Shanghai. It is not yet known how humans became infected. The possibility of animal-to-human transmission is being investigated, as well as the possibility of person-to-person transmission.

How can infection with influenza A (H7N9) virus be prevented ?

Although both the source of infection and the mode of transmission are uncertain, it is prudent to follow basic hygienic practices to prevent infection. They include hand and respiratory hygiene and food safety measures.

Hand hygiene: Wash your hands before, during and after you prepare food; before you eat; after you use the toilet; after handling animals or animal waste; when your hands are dirty; and when providing care to someone in your home who is sick. Hand hygiene will also prevent the transmission of infections to yourself (from touching contaminated surfaces) and in hospitals to patients, health care workers and others. Wash your hands with soap and running water when visibly dirty; if they are not visibly dirty, wash them with soap and water or use an alcohol-based hand cleanser.

Respiratory hygiene: Cover your mouth and nose with a medical mask, tissue or a sleeve or flexed elbow when coughing or sneezing; throw the used tissue into a closed bin immediately after use; perform hand hygiene after contact with respiratory secretions.

Is it safe to eat meat i.e., poultry and pork products?

Influenza viruses are not transmitted through consuming well-cooked food. Because influenza viruses are inactivated by normal temperatures used for cooking (so that food reaches 70°C in all parts – “piping” hot, no “pink” parts), it is safe to eat properly prepared and cooked meat, including from poultry and game birds. Sick animals and those that died of diseases should not be eaten. In areas experiencing outbreaks, meat products can be safely consumed if they are properly cooked and properly handled during food preparation. The consumption of raw meat and rare dishes is a high-risk practice and should be discouraged.

Is it safe to visit live poultry markets and farms in areas where human cases have been recorded?

When visiting live markets, avoid direct contact with live animals and surfaces in contact with animals. If you live on a farm and raise animals for food, such as pigs and poultry, keep children away from sick and dead animals; keep animal species separated as much as possible; and report immediately to local authorities any cases of sick and dead animals. Sick or dead animals should not be butchered and prepared for food.

Is there a vaccine for the influenza A (H7N9) virus?

No vaccine for the prevention of influenza A (H7N9) infections is currently available. However, viruses have already been isolated and characterized from the initial cases. The first step in developing a vaccine is the selection of candidate viruses that could go into a vaccine. WHO, in collaboration with partners, will continue to characterize available influenza A (H7N9) viruses to identify the best candidate viruses? These candidate vaccine viruses can then be used to manufacture vaccines if this step becomes necessary.

Does treatment exist for influenza A (H7N9) infection?

Laboratory testing conducted in China has shown that the influenza A (H7N9) viruses are sensitive to the anti-influenza drugs known as neuraminidase inhibitors (oseltamivir and zanamivir). When these drugs are administered at the early stage of the disease, they have proven to be effective against seasonal influenza virus and influenza A (H5N1) virus infection. However, until date, there is no experience on the use of these drugs for the treatment of H7N9 infection.

Is the general population at risk from the influenza A (H7N9) virus?

Information about these infections is not enough to determine whether there is a significant risk of community spread. This possibility is already subject to epidemiological investigations.

Are health workers at risk from the influenza A (H7N9) virus?

Health care workers are often in contact with patients with infectious diseases. Therefore, WHO recommends that appropriate infection prevention and control measures be consistently applied in health care settings, and that the health status of health workers be closely monitored. Together with standard precautions, health workers caring for those suspected or confirmed to have influenza A (H7N9) infection should use additional precautions.

Does this influenza virus pose a pandemic threat?

Any animal influenza virus that develops the ability to infect people is theoretically a risk to cause a pandemic. However, whether the influenza A (H7N9) virus could actually cause a pandemic is unknown. Other animal influenza viruses that have been found to occasionally infect people have not gone on to cause a pandemic.

Preventing human infection with avian influenza A viruses?

The best way to prevent infection with avian influenza A virus is to avoid sources of exposure. Most human infections with avian influenza A viruses occurred following direct or close contact with infected poultry.

Seasonal influenza vaccination will not prevent infection with avian influenza A virus, but can reduce the risk of coinfection with human and avian influenza A viruses.

Because rare episodes of limited, non-sustained human-to-human transmission of HPAI H5N1 virus have been reported, people should avoid sick patients who have suspected or confirmed HPAI H5N1 virus infections. Health personnel managing patients with suspected or confirmed HPAI H5N1 virus infection should wear recommended personal protective equipment and follow recommended infection control measures (standard, droplet, contact and airborne precautions).

APPENDIX 7B: Template for preparing district outbreak report

Title/Description (include disease/condition investigated):_______________________________

Period _____________ location (village, neighbourhood, district, region):___________________

Executive summary: ____

____

____

____

I. Introduction:

- Background;
- reasons for investigation (public health significance, threshold met, etc.);
- investigation and outbreak preparedness.

II. Methods:

- dates of investigation;
- site(s) of investigation (health care facilities, villages, other);
- case finding (indicate what was done regarding case finding, e.g., register review, contact investigation, alerting other HF, others);
- laboratory specimen collection;
- description of response and intervention (include dates);
- data processing.

III. Results:

- date and location of the first detected (or index) case;
- date and address of HFwhere first case was seen by the health system;
- results of additional case finding;
- laboratory analysis and results;
- key features description of results analysis in time, place and person;
- detailed results in time (epi curve), location (map), and individual characteristics (tables) and line lists;
- results of response and evidence of its impact.

IV. Self-evaluation of the timeliness and quality of preparedness, outbreak detection, investigation and response

Outbreak preparedness			
Indicator		Yes	No
Were drugs and medical supplies available at the onset of the outbreak?			
Were treatment protocols available to health workers?			
Does the district public health emergency preparedness and response committee regularly meet as part of the outbreak response?			
Outbreak detection			
Indicator	Date 1	Date 2	Interval
Interval between onset of index case (or occurrence of an unusual cluster at the community level) [date 1] and arrival of first outbreak case at the HF [date 2] (Target: <3 days) (Target: <3 days)			
Interval between initial outbreak case seen at the HF (or date of outbreak threshold crossing at the health facility) [date 1] and reporting to the district health team [date 2] (Target: within 24 hours)			
Cumulative interval between the onset of index case (or occurrence of an unusual cluster at the community) [date 1] and notification to the district [date 2] (Target: <7 days)			
Outbreak investigation			
Indicator		Yes	No
Were case forms and line lists completed?			
Were laboratory specimens taken (if required)?			
Indicator	Date 1	Date 2	Interval
Interval between notification of district [date 1] and field investigation [date 2] (Target: within 48 hours)			
Interval between sending specimens to the lab [date 1] and receipt of results by the district [date 2] (Target: 3-7 days, depending on type of test)			
Outbreak response:			
Indicator	Date 1	Date 2	Interval
Interval between notification of outbreak to district [date 1] and concrete response by the district [date 2] (Target: within 48 hours of notification)			
Evaluation and Feedback:			
Indicator	Date 1	Date 2	Interval

Interval between end of the outbreak [date 1] and finalization of outbreak report with case forms/line list sent to national level [date 2] (Target: 2 weeks)			
Indicator		Yes	No
Did the outbreak management committee meet to review investigation results?			
Was feedback given to HF and population?			

V. Evaluation of other aspects of the response:

VI. Interpretations, discussion, and conclusions:

VII. Recommended public health actions:

Comment on following levels: community, health facility, district, partners, region and national level.

District Health Emergency Management Committee Chairperson:

Name____ Signature_______________________

District Medical Officer:

Name____ Signature_______________________

Reporting date: ____________________________________

APPENDIX 7C: Template for preparing public health event situation report

District __________ Epidemiological week _______Week end (date) ___________

I. Epidemiological situation: Week (insert week number and date here) ______________________

Table XII. *Epidemiological situation: Week_*

Disease	Cases	Deaths	Case fatality rate (%)	Districts in alert	Districts affected by the epidemic	Week notified	Timeliness	Completeness
D1								
D2								
Dn								
Total								

Comments:

Contact us:

II. Summary of the epidemiological situation (insert reported weeks here)

Table XIII. *Epidemiological situation: Weeks____*

Districts	Cases	Deaths	Case fatality rate (%)	Districts in alert	Districts affected by the epidemic	Week notified	Timeliness	Completeness
D1								
D2								
Dn								
Total								

Comments:

Contact us:

III. Graphs (This section provides a graphic representation of data)

IV. Outbreak trends

APPENDIX 7D: Template for outbreak investigation report

INVESTIGATION OF ANTHRAX OUTBREAK IN KILIMANJARO REGION, DECEMBER 2015–JANUARY 2016

1. INTRODUCTION

Anthrax is an acute disease caused by *Bacillus anthracis*, a spore-forming Gram-positive bacterium. The disease commonly affects wild and domestic herbivores, and humans and carnivores are accidental hosts. There are three outbreak forms in humans: cutaneous, respiratory and gastrointestinal. The disease can be transmitted through the intestine (ingestion), respiratory tract (inhalation) and skin (cutaneous) from infected animal tissues and from infected persons.

The worldwide estimate of the disease burden is not well known. However, occasional outbreaks do occur (WHO, 2005).

An outbreak of anthrax in Marangu, Moshi DC in Kilimanjaro region was reported to the Ministry of Health and Social Welfare (MOHSW) by the Kilimanjaro Regional Medical Officer. Hence, there was a need to carry out an investigation to ascertain what was happening in the affected district

Objectives of the outbreak investigation

- confirm and determine the magnitude of the outbreak by actively searching for cases;
- characterize the outbreak in terms of time, place and persons;
- identify the source of infection through collection of both animal and clinical samples;
- generate and test the hypothesis of the outbreaks; and
- come up with recommendations and assist the district teams in the response and control of the outbreak.

Hypothesis for the anthrax outbreak at Moshi Rural District

- The slaughtering of a dead cow killed by anthrax is associated with contracting of the disease in humans.
- The handling/eating of meat from a dead cow killed by anthrax is associated with contracting of the disease in humans.

2. METHODOLOGY

This was a cross sectional study that involved the patients presenting with signs and symptoms for anthrax who were either admitted or at home. Active case finding in the community was done to identify the cases. Clinical samples were collected and transported to the laboratory for confirmation

Study area: The suspected anthrax outbreak occurred in Rauya Village in Marangu Mashariki district and Mae Juu Village in Siha District; both districts are located in Kilimanjaro region in Tanzania

Places visited:

We identified the homes of the cases that sought treatment from the Marangu and Siha HF, with the assistance of the Kilimanjaro Regional Health Officer, District Health Officers (Moshi DC and Siha), District Veterinary Officers (Siha and Moshi DC) and Community leaders. We visited two most affected villages i.e., Mae Juu and Rauya.

Study period: The outbreak investigation was carried out from 8-14 January 2015

Case definition used:

Case definition: The standard case definitions used to identify the cases were:

Suspected anthrax case	**IF THERE IS AN ANTHRAX OUTBREAK:** Any resident of Marangu presenting with a clinical illness and who is epidemiologically linked to a confirmed or suspected animal case or contaminated animal product from 12 December 2015
Probable anthrax case definition	Any resident of Marangu presenting with cutaneous ulcers developing within two weeks of contact with a sick or dead animal, confirmed to have anthrax or dying of unknown diseases since 12 December 2015
Confirmed anthrax case definition	A suspected case or a probable case in whom laboratory testing confirms Bacillus anthracis by Gram Staining or culture or PCR

Data collection methods:

Semi structured questionnaires were used to collect information from patients and their families. Where the patient was not available or could not speak, well informed agent was used.

3. FINDINGS

Rauya Village, Moshi DC, Kilimanjaro region

The index case reported at the Rauya RC Dispensary on 6/12/2015. The index case (Erasto Kingstone) was a male aged 39 years living in Rauya Village in Marangu Mashariki. The index case is the son of the owner of the dead cow. (This family had one cow and one goat). The outbreak is said to have occurred after the slaughter of the dead cow. It was noted that the cow died on 3/12/2015. The family did not notify the veterinary officer for inspection. The cow was slaughtered at home by Erasto with the help of a neighbour called Elibariki who also distributed the meat to relatives and neighbours. The cow skin was given to the dogs. The next day (4/12/2015) the index case opened the head of the dead cow when he accidentally pricked one of his fingers with a cow bone. On 6/12/2015, the index case had a swollen finger (the pricked finger). On 7/12/2015, the goat died and was slaughtered by the index case. The goat meat was eaten by only the family members. On 8/12/2015 the index case realized that the pricked hand was swollen and he was rushed to Marangu RC Hospital. Suspecting the patient to have anthrax, the doctor administered amoxicilin to the patient for five days as well as painkillers. After a few days without recovery, the patient returned to the hospital on 12/12/2015 and presented with swelling on the right upper limb and chest and chest with eschar on the wound. The hospital in charge referred him to Marangu Lutheran Hospital where he was also refereed to KCMC Hospital.

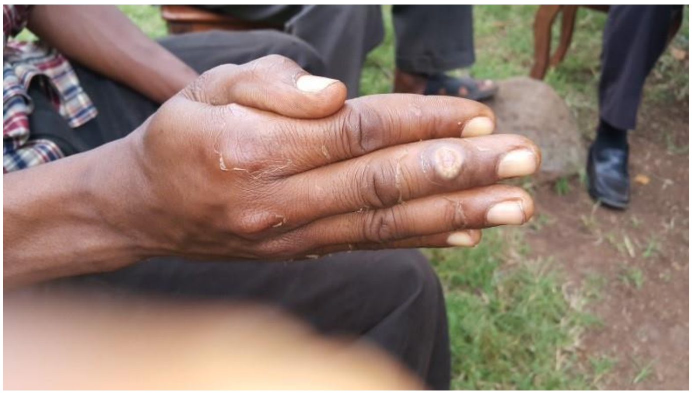


Photo 1: A photo showing a lesion on the hand of the index case identified in Rauya Village, Moshi, Kilimanjaro Region

Mae Juu Village, Siha District, Kilimanjaro Region

The first case in Siha District was reported on 17/12/2015. This was a male, 25 years of age, living in Mae Juu Village in Siha District. This case was brought unconscious to the Siha health facility, with blisters and lesions on his arms. This case was involved in selling the meat of a dead cow that was slaughtered.

The community was notified of the presence of an anthrax outbreak on 22/12/2015. They were also informed that those who might have had contact with or eaten the infected meat should go to the hospital. From 23/12/2015 to 28/12/2015, 760 people presented at the hospital and were all treated with doxycycline (7 days) or amoxicilin (5 days).

In Siha District there were four deaths of cows that were identified. The team was able to trace three cows that had died, were slaughtered and the meat sold at one of the butcheries. It also found that the meat was not inspected by the veterinary officer. One household had had its cow inspected, but we found that the person who inspected the cow was an artificial inseminator who had no knowledge on how to inspect cows, and not a veterinary officer.

A total of 904 contacts linked to the infected meat were obtained (68 were linked to the dead cow in Rauya Village, Moshi DC and a total of 836 contacts were obtained in Siha District). Of the total 904 contacts, 23 subjects met the standard case definition and there were no deaths reported. The median age for the cases was 36 years with the youngest being 1 year old and the oldest being 98 years old. The 10-19 years age group constituted 29.1% of the cases. Other social demographic characteristics of the contacts and cases are as shown in Tables XIV and XV below.

***Table XIV*:** General characteristics of the study subjects of the anthrax outbreak in Kilimanjaro Region

Variable		Number	Percentage	95% CI
Gender				
	Male	427	47.2	44.0, 50.6
	Female	477	52.8	49.4, 56.0
Occupation				
	Peasant	128	14.1	11.9, 16.5
	Formal employment	138	15.3	13.0, 17.8
	Unemployed	85	9.4	7.6, 11.6
	Student	314	34.8	31.7, 38.0
	Cattle keeper	113	12.5	10.5, 14.9
	Others	126	14.0	11.8, 16.4
Educational level				
	Uneducated	91	10	8.1, 12.2
	Primary education	480	53.2	49.8, 56.4
	Secondary education	237	26.2	23.4, 29.3
	Tertiary education	96	10.6	8.7, 12.9
Age group				
	0 to 9 years	157	17.5	15.1, 20.2
	10 to 19 years	233	26.1	23.2, 29.1
	20 to 29 years	108	12.1	10.1, 14.4
	30 to 39 years	80	9.0	7.2, 11.1
	40 to 49 years	111	12.4	10.4, 14.8

	50 to 59 years	83	9.3	7.5, 11.4
	60 and above	122	13.6	11.5, 16.1
Place of residence				
	Mae Juu Village, Siha District	836	92.5	90.5, 94.1
	Rauya Village, Moshi DC	68	7.5	5.9, 9.5

Others includes those who were not available physically for the interview of the under five.

Table XV: Social demographic distribution of anthrax cases in the Kilimanjaro Region, January 2016

Variable		**Number**	**Percentage**	**95% CI**
Gender				
	Male	13	60.9	34.0, 90.6
	Female	8	39.1	9.9, 65.1
Occupation				
	Peasant	4	17.4	1.9, 36.5
	Unemployed	6	26	7.6, 41.6
	Student	5	21.7	6.7, 48.0
	Cattle keeper	8	34.9	10.5, 54.9
Educational level				
	Uneducated	2	8.7	1.1, 22.2
	Primary education	11	47.9	29.8, 66.4
	Secondary education	4	17.4	3.4, 39.3
	Tertiary education	6	26.1	8.7, 42.9
Age group				
	0 to 9 years	2	8.7	2.1, 20.2
	10 to 19 years	5	21.7	6.2, 49.1
	20 to 29 years	6	26.1	10.1, 44.4
	50 to 59 years	4	26.1	7.5, 41.4
	60 and above	6	17.4	3.5, 36.1
Place of residence				
	Mae Juu Village, Siha District	19	82.6	65.5, 99.1
	Rauya Village, Moshi DC	4	17.4	1.9, 39.5

Table XVI: Case classification, anthrax outbreak, Kilimanjaro Region, January 2016

	Cases classification	Number	Percentage
1	Suspected	10	43.5
2	Probable	11	47.8
3	Confirmed	2	8.7

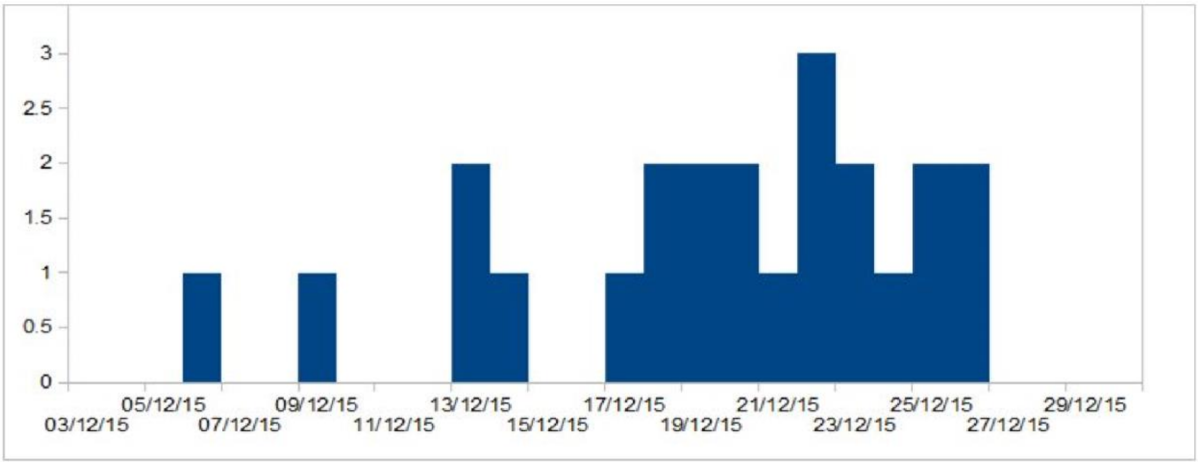


Figure 18: Outbreak curve, anthrax outbreak, Kilimanjaro Region, January 2016

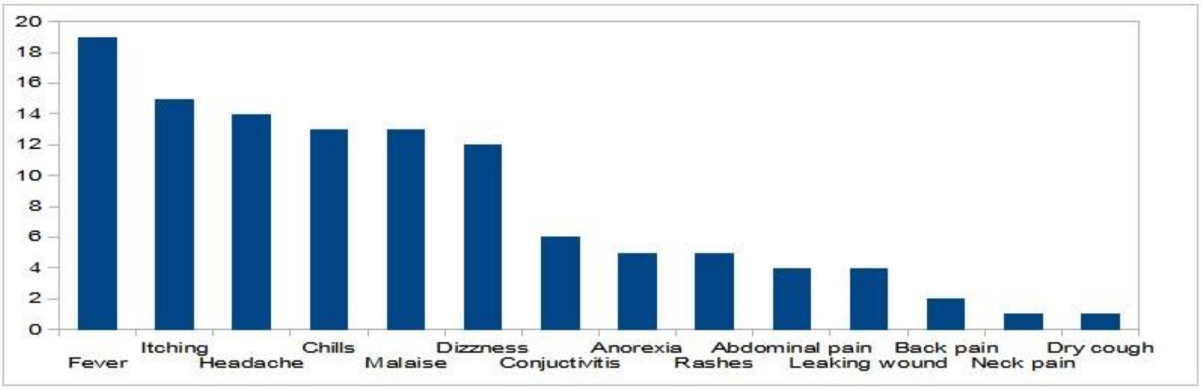
The outbreak curve shows that the first cow died in Rauya Village on 3/12/2015 and the first human case that was involved in slaughtering the dead cow developed symptoms on 6/12/2015. The peak of the outbreak was on 22/12/2015. The last case was reported on 28/12/2015. Of all the potential exposures indicated in Table 4 below, slaughtering and meat handling were significantly associated with an increased risk of contracting the disease.

Table XVII: Potential risk exposure for acquiring anthrax disease in Kilimanjaro Region, January 2016

		Prevalence Odds Ratio	95% CI	P value
1	Involved in slaughtering the cow	5	1.2, 23.5	0.01
2	Involved in handling the meat	3.2	0.9, 11.3	0.02
3	Contact with a dead cow, goat (burial)	2	0.7, 10.5	0.2
4	Eating the meat	7.6	2.2, 18.9	0.07
5	Contact with other animal products (skin, milk)	1.7	0.3,15.6	0.6

Disease characterization

The period from the exposure of the dead cow to the onset of symptoms ranged from 2 to 15 days with an average of 7.6 days. Among the 23 patients of anthrax, 3 (13%) were hospitalized. Infectivity rate was 23/904 (2.5%). The most presenting symptoms was fever as shown in Figure 2 below.



Other findings

(a) The national team worked hand in hand with veterinary officers who claimed that there were many unvaccinated cattle. Hence, the vaccination coverage is quite low; few cows within the village were presented for vaccination. The community indicated that it could not afford the vaccination fee of 2000 Kenyan shillings per cow and 1000 shillings per goat.

(b) The community was not well informed on anthrax and how to handle sick and dying animals.

Measures taken

- Quarantine of meat and meat products imposed in the affected areas.
- Vaccination of cows against anthrax and focused on the affected neigbourhoods.
- Provision of education to the community through village meetings, churches, mosques and schools. Community leaders were present to address the population.
- All cases presented to the hospital with cutaneous lesions were treated and prophylaxis was given to all identified contacts: Doxycycline (7 days) or Amoxicilin (5 days).
- Officers from the DVO, DHO and the Ministry of Health and Social Welfare held a joint meeting as more discussions arose on different interventions.
- Disinfection of cow sheds and slaughter sites using limestone powder.

4. CHALLENGES

- Immunization coverage was quite low, few cows within the villages were vaccinated due to shortage of vaccines.
- The community was not well informed on anthrax and how to handle dying animals.
- There are no close links between the DMO and the District Veterinary Officer (DVO) except when zoonotic outbreaks emerge.

5. RECOMMENDATIONS

- The veterinary sector should intensify immunization programs by increasing coverage and providing regular and timely vaccines to livestock especially against epidemic-prone zoonotic diseases.
- The community should be well informed on anthrax and other zoonotic diseases and how to handle sick and dying animals.
- There should be a close link between the RMO/DMO and the DVO in dealing with zoonotic diseases. This will promote the “one-health” approach.
- Preliminary diagnostic tests for animals should be the starting-point before anything else. Such tests will provide the clue for identifying the event.

6. CONCLUSION

The outbreak of Anthrax in the Kilimanjaro Region was established through the laboratory confirmation of human and animal cases. All the cases were linked to the dead cows, making them the main cause for the outbreak. Slaughtering of the dead cows and handling of their meat were the major risk factors for transmission of the disease to humans. The outbreak was declared over on 14 January 2016.

Appendix:7E. IHR core capacities for monitoring risk communication

Questionnaire on the monitoring of IHR core Capacities: Risk Communication

1. Have risk communication partners and stakeholders been identified?
2. Has a risk communication plan been developed?
3. Has the risk communication plan been implemented or tested through actual emergency or simulation exercise and updated in the last 12 months?
4. Are policies, SOPs or guidelines developed on the clearance[B] and release of information during a public health emergency?
5. Are regularly updated information sources accessible to media and the public for information dissemination?
6. Are there accessible and relevant IEC (Information, Education and Communications) materials tailored to the needs of the population?
7. In the last three national or international public health emergencies, have populations and partners been informed of a real or potential risk within 24 hours following confirmation?
8. Has an evaluation of the public health communication been conducted after emergencies, for timeliness, transparency and appropriateness of communications, been carried out?
9. Have results of evaluations of risk communications efforts during a public health emergency been shared with the global community?

Notes:

A. Plan includes inventory of communication partners, focal persons, stakeholders and their capacities in the country

B. Procedures are in place for scientific, technical and communication staff to give advice before information is released at public health events.

C. This may include website/webpage (national level), community meetings, radio broadcasts nationally as appropriate etc.

D. The views and perceptions of individuals, partners and communities affected by public health emergencies should be systematically taken into account; this includes vulnerable, minority, disadvantaged or other at-risk populations.

E. Transparency here implies openness, communication and accountability, i.e., all information about public health risk is open and freely available.

Appendix 7F: List of stakeholders and partners for risk communication

- Ministry of Public Health
- Ministry of Basic Education
- Ministry of Secondary Education
- Ministry of Higher Education
- Ministry of Agriculture and Rural Development
- Ministry of Decentralisation and Local Development
- Ministry of Water Resources and Energy
- Ministry of Communication
- Ministry of Transport
- Ministry of Forestry and Wildlife
- Ministry of the Environment, Nature Protection and Sustainable Development
- Ministry of Tourism and Leisure
- Departement of Civil Aviation
- Departement of Civil Protection
- Institute of Agricultural Research for Development
- Ministry of Labour and Social Security
- National Fire Brigade
- Hospitals
- Laboratories
- Police Department
- Drug manufacturers
- Drug suppliers (CENAME)
- United Nations : WHO/UNICEF, UNAIDS
- African Centers for Disease Control and Prevention
- US Centers for Disease Control and Prevention
- Doctors Without Border
- Red Cross
- Non-governmental organisations
- Medical Associations

Section 8: Supervising, monitoring, evaluating and providing feedback to improve surveillance and response

This section describes how to:

- Identify IDSR objectives and indicators;
- Monitor the quality of surveillance activities at the HD level;
- Supervise surveillance and response stakeholders;
- Evaluate the surveillance and response system;
- Perform measures to improve surveillance and response

Introduction

Monitoring of surveillance and response system refers to the routine and continuous tracking of planned surveillance activities (prompt delivery of reports, for example). Evaluation is done periodically (weekly, monthly, quarterly or annually) and determines the extent to which surveillance and response objectives have been achieved. Both monitoring and evaluation help to understand if the system has been working effectively. Evaluating information regularly, for example at the end of a given year, helps to determine whether surveillance and response objectives have been achieved and whether outcomes obtained are of expected quality.

As regards supervision, this enables supervisors and actors of surveillance to work together to review progress made, identify problems, determine problem causes and develop feasible solutions. Sustainable supervision and feedback have been shown to contribute to improve performance of the national disease surveillance system.

8.1. Monitoring, evaluating and improving surveillance and response

Supervision of surveillance actors consists of:

- systematically and continuously track surveillance activities (e.g., checking that reports are received in good time);
- identify problems and assist actors in resolving problems;
- periodic evaluation (e.g., annual) help to assess whether surveillance and response objectives have been achieved. Supervision and evaluation are used to improve surveillance and response.

Constant monitoring of the IDSR includes:

(a) tracking progress of implementation of planned activities and ensuring that planned targets are achieved in good time;
(b) tracking progress of improvements in targeted indicators of the quality and attributes of the system, such as timeliness and completeness of reporting;
(c) identifying problems encountered and promptly institute realistic and feasible solutions;
(d) ensuring that all stakeholders are held responsible and accountable for their assigned activities;
(e) ensuring that stakeholders can receive feedback on performance of the surveillance and response activities.

The evaluation of the surveillance system helps to:

(a) ensuring that the surveillance system meets the objectives for which it was formulated;
(b) documenting surveillance system status (performance level) and change in performance;
(c) providing evidence, based on which surveillance objectives, implementation strategy and planned activities can be modified;
(d) enabling planning of resource allocation;
(e) providing explanations for achievements and failures in the system; and
(f) providing specific recommendations for improving the system.

Providing feedback after supervision is helpful for:

(a) reinforcing health staff efforts to participate in the surveillance system;
(b) motivating those who provide data, hence scaling up compliance for reporting;
(c) improving quality of data provided by data collectors;
(d) enhancing planned public health action;
(e) complementing planning of appropriate actions; and
(f) strengthening communication and spirit of teamwork.

Section 3 of these guidelines describes how, each month, health staff responsible for surveillance at HFand district levels review and analyse data that is reported during the month. Conclusions are drawn about the following:

(a) timeliness and completeness of reporting;

(b) quality of preventive interventions and control activities taking place, so that when problems are detected, districts respond with appropriate action.

The same information can also be used during supervision to routinely monitor, and annually evaluate:

(a) timeliness in reporting immediately notifiable diseases, conditions or public health events;
(b) outbreak investigations and responses; and
(c) reporting of summary data on a routine basis.

When problems are detected in the surveillance and response system, action can be taken to strengthen it. By providing feedback to health workers for implementing identified corrections, it is more likely that results of desired outcomes will be evident. For example, one may use the monthly monitoring data to do an evaluation at the end of the year, and questions to help carry out an evaluation may include:

- Are surveillance objectives for existing activities being met?
- Were surveillance data used for taking public health action?
- Did surveillance, laboratory and response activities have an impact on the outcome of health events in the district?

8.2. Identifying IDSR objectives and indicators

Using indicators is helpful in measuring the extent of achievement for a particular program or activity. Outcomes obtained are compared to overall recommended standard quality practices and that method may be used to evaluate the progress made in achieving the general objective of a program. For example, a district may have a goal of reaching 100% completeness of reporting by a certain period. An indicator can be developed to measure the proportion or percentage of facilities that are reporting. This proportion is then compared with the desired goal or target, and can be used to evaluate progress and, therefore, quality of service or activity.

Use indicators according to national goals or specific plans to improve integrated disease surveillance and response activities in a district. Also, select indicators that are most relevant to the district's strategy for improving surveillance in the current year, and that will provide information the district can use.

Below are given some examples of indicators:

- **Indicators used to measure the quality of the overall surveillance system**: To evaluate timeliness and completeness, select as an indicator the percentage of HF that have regularly reported in time.
- **Indicators used to measure the quality of the surveillance of particular diseases and public health events**: For instance, to track the meningitis outbreak response, select as an indicator the percentage of districts in which meningitis outbreak clusters have been detected, i.e., where the outbreak threshold has been reached with laboratory case confirmation.

Sometimes, adding other indicators is necessary to measure the impact of public health interventions.

Appendix 8A provides examples of indicators and ***Appendix 8E*** a table that are helpful for monitoring key indicators at HF. Key indicators are provided in ***Appendices 8B and 8F*** for a district level, in ***Appendix 8E***, for regional level, and in ***Appendix 8F,*** for national level.

Table XVIII: Performance indicators for the core functions of the IDSR

1.	**Proportion of HF that submit weekly (or monthly) surveillance reports on time at the district level.**
2.	Proportion of districts that submit weekly (or monthly) surveillance reports on time at the regional level.
3.	Proportion of notifications submitted to districts using case-based forms or line of diseases targeted for elimination, eradication or under case-based surveillance.
4.	Proportion of suspected outbreaks of epidemic-prone diseases notified to the regional level within 2 days of crossing the outbreak threshold.
5.	Proportion of districts with updated trend analysis (curve or histogram) for selected priority diseases.
6.	Proportion of outbreak investigation reports including a case-based data analysis
7.	Proportion of investigated outbreaks with laboratory results.
8.	Proportion of confirmed outbreaks dealt with by recommended national response actions
9.	Case-fatality rate for each epidemic-prone disease notified.
10.	Attack rate for each priority disease outbreak
11.	Number of outbreaks detected at the national level but not identified at the district level over the previous year
12.	Proportion of districts that report laboratory data for diseases under surveillance
13.	Proportion of districts laboratories supervised at least once and the supervision of which resulted in a written report at the regional or national level over the previous year
14.	Proportion of regions that report a monthly analysis of laboratory data to the national reference laboratory

Table XIX: Performance indicators for the essential functions of the IHR (2005)

1.	**Proportion of HF that comply with the prevention and control requirements**
2.	Proportion of districts with health risk and resource mapping
3.	Proportion of districts that report information from event surveillance
4.	Proportion of districts provided with necessary regulations or instruments from their national governments for the implementation of the IHR
5.	Proportion of districts with coordination mechanisms of sectors concerned by the application of IHR

8.2.1. Select data for measuring indicators

After determining appropriate indicators, specify the numerator and the denominator to be used. For example, if a district would like all HF to draw trend lines for a number of priority diseases, the numerator and denominator are defined as follows:

***Indicator**: The proportion of HF in the district having trend lines for priority diseases*

Numerator: The number of HF having trend lines for priority diseases.

Denominator: The total number of HF in the district.

8.2.2. Ensure availability of data sources

Each level should make sure that the level it supervises has the following sources of data available. Thus, the national level will have data available from operational and intermediate levels to conduct required monitoring activities.

Table XX: Types of sources of data at various levels

Data source	**Level**			
	Health facility	District	Regional	National
Indicator monitoring curve (*Sample charts are in this Section APPENDIX*)	X	X	X	X
Outpatient register	X			
Inpatient register	X			
HFreporting forms	X			
Case-based and/or line listing reporting forms	X	X	X	X
Monthly Activity Report (MAR)	X	X	X	
Outbreak investigation report	X	X	X	X
Log of suspected outbreaks and rumours	X	X	X	X
Supervisory reports from district or region	X	X	X	X
Laboratory reports received	X	X	X	X
SFE documents and tools	X	X	X	X

8.3. Monitoring core functions for IDSR at district level

Indicators for core functions measure processes and outputs from the surveillance system. In the introductory section, core surveillance functions have been described and so one may refer to the table of core surveillance functions. This subsection describes key indicators at district level, in relation to core functions. Here, core functions are succinctly described, but further detail of said functions is provided in ***Appendix 8A-D***, where indicators for each core function are available.

The core functions are:

(a) *Identifying cases and public health events*

- Case identification may be done through the formal health structures (public, private and religious) or community structures. Case definitions and a functioning signal-verification system are vital for case and outbreak detection. Once a case has been identified, it has to be recorded in a register (outpatient or inpatient register, clinical cases register etc.). Health care workers use any of these registers to extract the IDSR priority diseases.
- Indicators should be established to monitor this core surveillance function. These indicators could be:
  - Proportion of HF that have standardized registers for recording diseases. Further assessment could also be done to examine the validity and quality of information recorded as well as factors that affect registration.
  - Proportion of HF using standard case definitions to identify IDSR priority diseases of cases.

(b) Report cases and events

- Reporting refers to the process by which surveillance data move through the surveillance system from the point of generation to the next level.
- It also refers to the process of reporting any suspected and confirmed outbreaks and any public health emergency of international concern (PHEIC) under the IHR, using the decision instrument mentioned in Section 2.
- There may be different reporting systems, depending on the type of data and information being reported, purpose and urgency of submitting data/information, and where the latter is being reported.

- Timely submission of data is critical for prompt outbreak detection and response to prevent widespread outbreaks. HF should therefore strive to submit reports on time, as prescribed in national guidelines.
- Examples of indicators for this core surveillance function could be:
  - Proportion of complete surveillance reports submitted on time to the district.
  - Proportion of cases of diseases targeted for elimination, eradication and any other disease selected for case-based surveillance reported with case-based forms or line lists.

(c) Analyse and interpret data

- Analysing data is the systematic process of examining data to extract relevant information for decision-maker.
- Surveillance data should be analysed routinely and the information extracted should be interpreted for use promptly in public health interventions.
- Capacity for routine data analysis and interpretation should be established and maintained for epidemiological and laboratory data.
- Examples of indicators which can be used to review analysis are:
  - proportion of priority diseases for which a current line graph is available.
  - proportion of districts that report laboratory data for diseases under surveillance.

(d) Investigate and confirm suspected case or outbreak

- Case or outbreak confirmation depends on the capacity for surveillance system and laboratory for confirmation.
- Capacity for case confirmation is enhanced through improved referral systems, networking and partnerships. This implies having the capacity for appropriate specimen collection, packaging and transportation.
- Internal and external quality-control mechanisms are important elements for case confirmation; they help to ensure the validity and reliability of test results.

Examples of indicators for monitoring this core function may be:

- Proportion of suspected outbreaks of epidemic-prone disease notified to the district level within 24 hours of crossing the outbreak threshold.
- Proportion of investigated outbreaks with laboratory results.

(e) Prepare

- Outbreak preparedness refers to the level of preparedness at any given time, to deal with a possible outbreak. This implies the following: developing preparedness and response plans, emergency stockpiling, designation of isolation facilities, and setting aside resources for outbreak response.
- Examples of indicators which can be used to monitor preparedness include:
  - Proportion of HF with stock of key items (PPE, specimen collection kits, case-investigation forms, treatment kits) for possible response.
  - Proportion of districts with emergency preparedness and response plans.

(f) Respond

- Public health surveillance systems are particularly useful because they provide data for appropriate public health intervention and response. For an early warning system, the capacity to respond to detected outbreaks and emerging public health threats needs to be assessed. This can be done, following a major outbreak response and control, to document the quality and impact of public health response and control.
- Some examples of indicators for monitoring response include:
  - Proportion of districts with functional multisector emergency public health preparedness and response committees.
  - Proportion of districts with functional RRT.
  - Case-fatality rate for the epidemic-prone disease notified.

(g) Feedback

- Feedback is a process in which the effect or output of an action is returned (fed back) to modify the next action. It is an important function of all surveillance systems. Item ***8.5.2*** of this section provides a thorough description of types of feedback which may be used to improve performance of IDSR.
- Some examples of indicators for feedback are:
  - Proportion of districts producing regular epidemiological bulletins.
  - Proportion of feedback bulletins or reports received from national levels (when evaluating feedback from national to sub-national levels).
  - Proportion of HF with at least one IDSR technical support supervision visit in the previous quarter.

NOTE: while all indicators for the IDSR core functions are important, the WHO/AFRO will measure overall performance of core functions of IDSR in the countries, using 14 key performance indicators described in ***Appendix 8J***.

8.4. Monitoring the quality of ISDR activities at the HD level

The quality of the surveillance system is defined by attributes such as:

(a) completeness;
(b) timeliness;
(c) usefulness;
(d) sensitivity;
(e) positive predictive value (PPV);
(f) specificity;
(g) representativeness;
(h) simplicity;
(i) flexibility;
(j) acceptability;
(k) reliability;

The quality of the surveillance system should be assessed, based on these indicators.

Surveillance attributes can be evaluated using quantitative and qualitative methods. Some tools that may be used to comprehensively evaluate surveillance systems include:

- the updated version of morbidity mortality weekly bulletin;
- updated guidelines for evaluating public health surveillance systems, produced by the United States Centres for Disease Control and Prevention
- the framework for evaluating public health surveillance systems for early detection of outbreaks (CDC, 2001).

Field Epidemiology Training Program (FETP) should be helpful in evaluating the surveillance and response systems of IDSR and other disease surveillance systems.

8.4.1. Monitor timeliness and completeness of monthly reporting;

Key indicators of a good-quality reporting system are timeliness and completeness rates of reporting at each level. If reports are sent and received on time, detecting problems and conducting prompt response is easier. Completeness indicator determines whether all units that send reports have submitted data as it should. If reports are late or not submitted at all, aggregated information for a given district will not be accurate. In such an event, outbreaks can go undetected, and any opportunity to respond to health problems will be missed.

8.4.1.1. Timeliness

The single most important measure of timeliness is whether data are submitted in good time to begin investigations and implement control measures. Timeliness of reporting should be measured against standards developed by each country, in accordance with timelines set by the WHO/AFRO. Important aspects of timelines of reporting in a communicable disease surveillance system include:

- timeliness of immediate notification, i.e., within 24 hours;
- timeliness of weekly reporting; data should be entered in DHIS 2, by the latest on the monday of the following week; and
- timeliness of monthly reporting, data should be entered in DHIS 2, by the latest on the 05th of the following month.

(a) Monitor and evaluate detection and notification of immediately reportable diseases or public health events

Monitor how well the system is able to detect immediately notifiable diseases or public health events and evaluate the interval between the onset of the disease in an identified index case and when the case was seen in the health facility. If this interval is too long, it will seriously affect the health outcome of patients and will alter the spread of outbreak.

Other intervals to monitor for detection of immediately reportable diseases include, monitoring reporting from community to HF (target: within 24 hours of onset of illness); from HFto district (target: within 24 hours); and from the time threshold is reached to the time of concrete response (target: within 48 hours).

(b) Timeliness monitoring of weekly and monthly reporting

To evaluate reporting timeliness, the reception of notification should be routinely monitored. In doing so, a monitoring tool, such as a record of received reports or a computer database, may be used. A sample form for verifying timeliness of reporting is provided in ***APPENDIX 8G*** at the end of this Section.

If dates on which reports are received are routinely recorded and reviewed, system effectiveness can easily be assessed each week or month, in the course of analysing routine and case-based data. For example, use the record of reports received to:

- determine the number of reporting units that submitted reports for a given epidemiological week or month;
- identify which reporting units have reported; and
- measure how many monthly reports were timely, during a period of time in accordance with national guidelines.

Ensure deadlines are given for each level to enable effective monitoring.

8.4.1.2. Completeness

Completeness in surveillance can have varying dimensions and may include the following:

(a) Completeness of reporting sites submitting surveillance reports:

Completeness of reporting sites refers to the proportion of reporting sites that submitted surveillance reports, irrespective of the time that report was submitted. Measuring completeness of reporting sites for each of the surveillance reports can:

- provide a trend analysis on completeness of reporting for each of the surveillance reports over a period of time; and assist in identifying how each site is performing;
- trigger further investigation for reasons of poor performance, and possibly help to identify solutions to correct such performance.

(b) Completeness of case reporting

Completeness of case reporting refers to the match between the number of cases reported and the actual number of cases. This can be obtained by comparing the number of cases reported to the higher level (over a given period), with the number of cases recorded in the patient register, over the same period.

(c) Completeness of surveillance data

Completeness of surveillance data is the match between the expected data and what is received. The following questions are useful in determining completeness of surveillance data and its implications on public health actions:

- Are all data on each of the required variables in a surveillance form collected, registered, validated and compiled?
- If not, which variables are not routinely collected, and what problems are encountered in their collection?
- What is the implication of missing data on the quality of surveillance data?
- How can this problem be resolved?

8.4.2. Identify problems and take action accordingly

If monitoring information shows that a HFor any other reporting site has not provided a report, or if the report is not on time, the surveillance focal person at the HF or site should be contacted. Work with the appropriate staff to identify what has caused the problem, and develop corrective actions. For example, find out if health care staff has adequate supply of reporting forms or other means of communication (SMS, phone or internet). Newly recruited staff in HFmay not be familiar with reporting procedure, or the staff may not receive any feedback following the submission of notification, or may not have resources for conducting appropriate actions following feedback.

It is worth working with the reporting site to develop strategies aimed at improving this situation. Explain that the district can more efficiently help healthcare staff plan and implement responses if the information is complete. For example, if lack of supplies is a problem, the district can use the information from reporting to increase awareness at the higher levels of health pyramid.

8.4.3. Submit to higher levels information for timeliness and completeness notifications

When routine reports of the number of cases are being sent to the regional or national level, include also necessary data for timeliness and completeness to help the other levels of the health system to understand the situation much more clearly, and to evaluate quality of the data that is being sent. For example, if the report to the national level states that two cases of measles were detected during the month, it should also include information about the number of HF that have reported during the same period. It will make a difference to the other levels, when they evaluate the information, if the two cases occurred with only 20%, rather than 100% of the reporting sites.

8.4.4. Monitor other criteria for assessing quality of the surveillance system

Other relevant attributes are summarized in the table below and can be used to assess quality of surveillance systems during periodic evaluation assessments ***(Table 8.2)***. Refer to the updated framework for evaluating surveillance systems for a complete list of attributes.

Table XXI: Summary of other criteria for assessing quality of the surveillance system

Criteria	Definition	Examples of questions that can be asked while evaluating
Usefulness	Describes if the surveillance system has been able to contribute to the prevention and control initiatives or has been useful in measuring performance. Example: Usefulness of surveillance data in an early warning system	Is the system e.g., the early warning system able to detect outbreaks early? Example: A useful system, over time must, demonstrates that a certain intervention has been instituted and has worked effectively. In a malaria control program, data collected over time might show if long lasting impregnated net has been useful in reducing incidences of malaria among children under five years.
Simplicity	Simplicity refers to the structure of the system and ease of its implementation from the end-user to those at higher levels.	Is the system simple? E.g., is the standard-case definition simple? Does it have multiple reporting structures? Example: Should a health worker have to report to the district, as well as to another vertical programme on the same disease?
Acceptability	Acceptability of a system consists in the willingness of the surveillance staff to implement the system, and of the end-users to accept and use data generated through the system	How is the participation rate of surveillance sites? How is the completeness rate of reports? Example: number of HF submitting reports on time
Representativeness	Representativeness refers to the degree to which reported cases reflect incidence and distribution of all cases in the population.	Is the system covering all geographical areas to ensure accurate capture of cases? Note: A good system should be able to cover all population, including marginalized and special population.

Data quality	Data quality reflects completeness and validity of data recorded in the surveillance system.	For completeness one can examine the percentage of "unknown" or "blank" filled in on surveillance tools Note: Validity depends on data quality. Error-prone systems and data prone to inaccurate measurement can negatively affect detection of unusual trends.

For further information on the other attributes mentioned above but not cover here, please refer to Centres for Disease Control and Prevention (CDC) (2001). Updated guideline for evaluating public health surveillance systems. MMWR: 50 (RR-13) 1–35.

8.5. Monitoring the quality of community-based surveillance activities

8.5.1. Monitor events from community-based surveillance

It is important to monitor the implementation of the EBS as well as the IBS in districts and regions. CHWs involved in the implementation of the EBS must understand the benefits of the system and know that their input is of value and helps to improve and strengthen early detection and rapid response system to better serve their community.

Feedback from CHWs and other community members on the implementation of the EBS is a key element in contextualising this surveillance, understanding and improving the quality of the data collected. A system should be developed at the beginning of surveillance to collect information and feedback from the community and CHWs, and one or more of the following approaches can be adopted:

- open and regular community meetings where all issues are noted and resolved;
- organizing focus group discussions with CHWs and community leaders;
- provision of complaint or suggestions boxes for community members;
- appointment of a community representative(s) to gather feedback and complaints;
- establishment of the communication platform via mobile phones, which can be used by CHWs to provide feedback.

There are tools for collecting (defining signals) and transmitting (dashboards, signal register, EBS database) the signals received in the community. It is advisable to encourage the CHW to understand the data by analyzing it and identifying outbreaks and events that have occurred, those that have not been alerted to, and the corrective measures to be implemented.

The table below shows examples of indicators applied to the EBS.

Table XXII : Examples of indicators for community-based surveillance

Number of signals detected	A signal is unofficial information about a disease, condition or event of public health importance which may be true or invented	Number of signals detected from each CHW or FPS HF	Community surveillance reports/dashboards
Proportion of signals responded to within 24hr–28hr	**Numerator:** number of alerts responded to on time. Denominator: Total number of alerts detected from CBS focal person Note: responding to alerts is defined as visit by the nearby HFfor case investigation, case management, health promotion, community sensitization and distribution of materials (must be defined according to response plan)	Number of alerts responded to within 24hr–48hr divided by total number of alerts reported	Community surveillance reports and response reports
Proportion of signals which turned into events	Number of true events detected	Total number of true events detected divided by total number of alerts reported	Community surveillance reports and response reports

8.6. Supervising surveillance and response actors

Supervision aims at improving performance. Supervision is not an inspection. Rather, supervision aims at sustaining good-quality services, and not seeking to punish. For the system to work correctly, supervisors and health professionals must work together to review progress, identify problems, determine the cause of difficulties and propose feasible solutions.

8.6.1. Prepare job descriptions for surveillance staff

Job descriptions and standards operational procedures are the basis for conducting supervision and assessing performance. It is important to review those relating to health staff with a role in the surveillance and response system to ensure that the following information is included

- the surveillance tasks to perform;
- the authority responsible for the personnel involved in surveillance and response;
- a defined scope of work, as well as standard operational procedures that are adhered to in practice.

8.6.2. Prepare a supervision plan

Include surveillance and response targets in the overall plan for supervision. For example:

- to determine how often to monitor the performance of health workers, a HD should organise supervision according to national guidelines (each HFvisited at least once a month);
- ask supervisors to plan their annual supervision in HF and in all community sites that submit notification reports to them;
- make sure that transport is available for supervision and for surveillance activities that require transportation means. For example, coordinate travel or logistics for surveillance supervision with visits made by other programs or activities.
- include in the overall plan, other reporting sites in supervision of district surveillance activities, such as integrated health centres (IHC), medicalised health centres (MHC), district hospitals (DH), regional hospitals (RH), central hospital (CH) and others, general hospitals including private HF and community reporting sites.

8.6.3. Prepare a supervisory checklist (supervision grid)

Each HFhas unique problems and priorities that require specific problem solving and corrections. To maintain positive motivation of HFstaff for their efforts at ensuring improvement, consider developing a graduated checklist to guide the supervision. (See ***Appendix 8I*** *an example of checklist)*. The items on the list are selected based on what has already been achieved in the health facility. Once this has been achieved (for example consistent use of standard case definitions), ensure that staff to move on to the next performance monitoring item or indicator (e.g., use of action levels). The checklist should be modified accordingly. This will enable health workers, in subsequent visits, to monitor their activities and assess their progress in improving the system.

When supervising, also use a checklist to see if the health worker is correctly applying the recommended surveillance functions. For example, the supervisor at the HD level should check the following functions during the site supervision visit:

Identifying and registering cases	Check the HFregister to see if the case diagnoses (**signs and symptoms**) correspond to the recommended case definition. Check that the register is well drawn and that all the columns are filled out correctly.

Case confirmation	Compare the number of cases of priority diseases recorded in the laboratory registers with the number of cases consulted at the HFduring the same period. For example, compare the number of positive malaria smears with the reported number of hospitalised malaria cases.
Reporting	Ask for a copy of the latest reports covering the most recent period. Compare the number of notified cases of priority diseases with the figure in the register. Check the date on which the case report was sent against the date recommended for sending the report. Check reports to make sure the reports are complete, accurate and correct.
Reviewing and analysing data	Check whether trend lines are set for priority diseases. Ask to see the health facility's data analysis plan. Check whether the trends for the selected diseases are up to date
Preparedness	Look at the stocks of medicines, supplies and personal protective equipment (PPE) to ensure that they are of sufficient quality and quantity

Note - A sample supervision checklist is provided in **Appendix 8I** at the end of this section. The questions to be asked during the supervisory visit may be adapted or modified to fit particular issues and progress in the implementation of the integrated surveillance system at the HFand community reporting sites.

8.6.4. Conduct supervision

During the periodically scheduled supervision visits in the HD, check that:

- the necessary materials (forms, guidebook, etc.), guidelines and standardized case definitions are available;
- health workers know how to identify suspected cases of priority diseases seen in their facility using standard case definitions;
- priority diseases are recorded in the case register according to the case definition;
- the data are analysed in the HFor in the HD in order to identify signal and/or intervention thresholds for both systematically notified priority diseases (diseases of public health

importance) and for diseases subject to case-based surveillance (diseases with outbreak potential or subject to eradication or elimination measures);

- notifications of diseases for which a single case constitutes a suspected outbreak rapidly trigger an investigation;
- response measures are taken after confirmation of the outbreak or identification of health problems following routine notification;
- response actions are monitored and action is taken by the HFto improve surveillance and readiness activities for outbreak response.

During the visit, do not forget to:

- debrief facility managers and health workers. They need to be told what is working well and what is not, they need to be given feedback on how previously reported data has been used to detect outbreaks, and they need to take action to reduce morbidity, mortality and disability in the HD. If improvements are needed, solutions should be discussed with staff;
- provide on-site training (at the workstation), if necessary, if a problem is identified. For example, during the review of the registers, the supervisor notices that during the data synthesis, the case fatality rates were not calculated correctly. He will meet with the staff responsible for these calculations and show them the process to follow;
- rto requests for assistance, including equipment or other materials for emergency response;
- if the solution to a particular problem was identified during a previous visit, check that the solution was correctly applied, if problems persist, find another solution if necessary;
- ensure that the supervisor(s) and supervisee(s) sign the supervision record and indicate the dates on which the supervision took place;
- develop with staff, a plan for resolving identified problems (especially those not resolved during supervision) as illustrated below:

Problems	**causes**	**Corrective activities**	**Monitoring of indicators**		**Officials**	**Completion time**

8.6.5. Write and share a supervisory report

The report should include strengths, points for improvement, good practices, progress made during the visit and recommendations. Planned interventions with health personnel and possible requests for additional resources should also be specified, as well as any special problems.

The supervision report should be shared with all stakeholders involved in the supervision as well as with the next level of management. The report must also be archived.

8.6.6. Use supervisory visits to improve surveillance activities in the HD

Regional or national supervision visits are an opportunity to discuss and improve disease control in the HD. For example, when supervising a regional or national malaria programme manager, he or she may discuss with the HD managers the reasons why malaria-specific mortality has not decreased in the HF. Subsequently, discussions should be held to identify additional resources or other measures that the malaria program could provide to address the problems identified.

8.7. Provide feedback

In most cases, HF and HDs report surveillance data to the next level of management as required. When managers at the HD, regional or national level receive these data, they should respond to the HF that submit it to them. The purpose of feedback is to reinforce health workers' efforts at participating in the surveillance system. Another purpose is to raise awareness about certain diseases and any achievements made by disease-control and prevention projects in the area. Feedback is classified as "supportive" when it reinforces and acknowledges good performance, and "corrective", when a change in behaviour and improvement is required. It also strengthens the communication and spirit of team work Feedback should be both vertical and horizontal targeting different audiences as provided by different levels in the health system. Effective feedback should be:

(a) specific to ensure that recipients understand the subject of the feedback;
(b) based on the report submitted or the actual events and activities observed in the field;
(c) given as soon as feasible, after receiving the report or field visit, so that recipients will remember activities that should be sustained or corrected.

If the facility does not receive information from the next level on how data were used or what data meant, health staff may think that their reporting is not important. As a result, subsequent reporting may not be reliable, since health staff will not know whether the information, they sent to other levels was important or necessary. These health workers may have a good understanding of the health situation at their own level, but lack the information needed to understand the situation at the HD, regional or national level. At community level, communication includes building relationships, communicating and coordinating with other community key informants, resource persons and existing formal and informal networks for information dissemination and reporting.

Feedback can be written, e.g., in the form of a letter/newsletter, email, WhatsApp messages, SMS, or official periodic information media such as publications; it can be oral, through phone calls or periodic meetings. Although this section focuses on feedback at the HD level, the analysis is also valid at the national and HFlevel. Feedback can also be given as part of formative supervision, from the HD to the HF, or from the region to the HDs, or from the national level to the HDs and regions. Supervision can focus on the implementation of health programs and feedback can be given during these supervision visits.

(a) Developing and disseminating routine epidemiological bulletins

Feedback should also be given periodically of IDSR reportable diseases, and this can be done through weekly, monthly or quarterly epidemiological bulletins. Bulletins provide information on disease patterns and achievement of program objectives in the country. They are usually brief and are important to inform policy-makers, legislators, development partners, program technical staff and stakeholders. As a minimum, they contain:

- a summary table with the number of reported cases and deaths, to date, for each priority disease;
- a commentary or message on a given disease or topic;
- any relevant social, economic or cultural information or data on the context that can provide real insight into an event.

Appendix 8I shows examples of an epidemiological bulletin.

(b) Developing information summary sheets

An information summary sheet is a report that presents data and its interpretation in a table or other graphic format. For example:

- at a staff meeting, or during a supervision, give a verbal report or comment about data that were reported by the HFduring a given period;
- display data in a simple table. Sit with health staff and show them the data. Talk together about the likely conclusions that may be drawn from said data. Consider conclusions not only for the health facility, but also for the district as a whole;

- prepare a single sheet with a simple table that shows how data reported for a given period are different from data reported for some other period or target population. For instance, show the number of cases of diarrhoea with dehydration in children aged less than 5 years, from the same period last year, and compare them with a corresponding period in the current year, after a safe water project was implemented in a high-risk area, for example; use summary sheets to support requests made to higher levels for additional funds, supplies and resources.

(c) Drafting of HD newsletters

The purpose of a district newsletter is to provide shorter updates than those provided in a more detailed feedback bulletin. The district newsletter is useful for informing and motivating health staff. The target audience for a newsletter could be health staff in the HD. The newsletter can be 2 to 4 pages long and can be a simple text entered on a computer or typewriter.

Examples of articles that could be carried in a newsletter are:

- summary of national or district data for a given priority disease;
- report of progress towards a specific public health target;
- report of specific achievements towards public health by an individual health worker or a group of health workers;
- description of special events or activities (for example, a change in market day).

8.8. Evaluating the performance of the surveillance and response system

The aim is to assess the effectiveness of the surveillance and response system in terms of data completeness, data timeliness, preparedness, case management, overall performance and use of indicators to identify problems or areas for improvement. A comprehensive evaluation should thus include the surveillance system and, if already available, the IDSR Implementation Plan. Evaluation of the surveillance system should:

(a) determine the extent to which the desired outcomes and results are achieved;
(b) provide explanations for achievements, disparities and failures;
(c) describe the quality of the system and show any changes in its implementation;
(d) demonstrate the extent to which overall surveillance objectives are achieved.

Indicators corresponding to the HD's annual priorities and objectives will be selected according to the stage of development of the surveillance.

If there is already an IDSR implementation plan, with clearly defined objectives, then it is appropriate to conduct mid-term and end-of-term evaluations. Otherwise, surveillance systems should be evaluated every 2, 3 or 5 years.

8.8.1. Key steps in evaluation include:

The objectives defined must be specific, measurable, achievable and time-bound.

8.8.1.1. Develop evaluation indicators

Indicators should be identified for each of the evaluation objectives, and should be harmonized, as much as possible, with monitoring indicators.

8.8.1.2. Develop evaluation methods and tools

In the light of these indicators, an evaluation protocol should be developed, describing the evaluation process, methods, target group, data sources, data collection methods and the plan for analysis and use of the data.

8.8.1.3. Identify people to conduct evaluation

During planning, evaluators should be identified according to the type of evaluation, its scope, purpose and available resources. The type of evaluation is determined by the evaluators selected:

- a self-evaluation: the evaluators are members of the system being evaluated (HD staff, etc.);
- external evaluation: the evaluators are external to the system being evaluated (staff from other structures, partners/donors, etc.);
- joint evaluation: the evaluators are both extern and members of the system being evaluated.

To ensure objectivity and transparency during the evaluation process, all three types of evaluation should be conducted on a regular basis.

8.8.1.4. Conduct the evaluation

(a) Compiling and organising monitoring data

The HD should compile the surveillance data received from all the Health Units located in the HD and submit the report to the regional level. The transmission of the report should not be delayed due to late receipt of reports from some HF. All reports received must be submitted on time. Late reports should be submitted as they arrive. It is advisable to follow up on Health Units that did not send a report or that regularly send it late.

It is important to help the HF to solve problems that might delay the transmission of their summary notifications. HF should be regularly informed about the results of the indicator. Feedback is a positive tool to encourage health workers to provide information in a timely manner and to participate in the surveillance system.

The Regional Delegation of Public Health should compile the surveillance data received from all the HDs in the region and submit the report to the national level. The transmission of the report should not be delayed while waiting for late reports. Reports received within the deadline must be compiled and submitted immediately. Reports received late may be sent separately.

The national level should compile the surveillance data received from all regions and investigate whether any outbreaks have not been detected at the HD level. Follow up with areas where reporting continues to be unreliable or does not happen at all. It will provide support to the region to assist the HDs in assessing their performance indicators and in implementing corrective measures. Finally, it should provide feedback at all levels.

Each level will use a chart (model presented below) to monitor the performance of its own indicators and will inform staff of the results by reporting successes (strengths) in order to help health workers, continue to progress. If a problem arises, identify with them the cause and the means to solve it. Seek assistance of the next level, as needed, for obtaining additional help or resources.

Gather information from different sources:

- the current year's targets in the HD's annual plan to improve surveillance and response;
- weekly and monthly summary notifications of cases and deaths to the district, detailed maps and other analytical results obtained by the HD;
- all results of specific investigations or studies carried out in the district during the past year;

- individual case investigation forms and reports of response activities in the HD;
- summary information from the community and from health staff.

(b) Analysing data

When evaluating the summary data for the past year, determine the following:

- Were the reports complete, on time and accurate?
- What have been the significant changes in disease trends or health events in the past year? If there has been an upsurge, has the problem been identified?
- If additional cases continue to be recorded, what is their cause? Where do they occur?
- Have timely and appropriate actions been taken in response to the monitoring data?
- Did the supervision visits take place as planned?
- Does the community feel that the response activities have been successful?
- Have steps been taken to respond to requests or suggestions from health personnel regarding services or monitoring?
- Have appropriate measures been taken to prevent the occurrence of similar events?

(c) Identifying problems and their causes

If, due to certain problems, the district has not achieved one of the objectives set or if one of the indicators shows that it has not achieved the desired level of performance, investigate the cause of the gap between the expected result and the result achieved. If a problem is identified, talk with the district team and HFstaff to find out possible causes of the problem.

8.9. Update plans for improving the IDSR system

Activities that work well and should be continued should be included in the district program. The practical solutions planned based on the results of the evaluation for the past year should also be indicated. Plan to implement the solution. For example:

- state the new activity and its objectives;
- identify the staff to carry out the activity;
- develop a timetable for this activity and set the logical sequence of interventions;
- deetermine the logistics required for the new activity (equipment, personnel, transport, resource allocation);
- estimate the cost of the activity

8.10. Provide feedback to HF and share a copy of the evaluation report with them

A report and feedback on the evaluation should be provided to the HF and other interested parties in the district. State in the evaluation report:

- the objectives set;
- the results achieved;
- the likely reasons for any discrepancies between the projections and the results;
- recommended solutions and prioritised activities to improve surveillance and response in the HD.

Appendices to Section 8

APPENDIX 8A	Follow-up indicators of core functions of the IDSR at the HFlevel
APPENDIX 8B	Follow-up indicators of core functions of the IDSR core Indicators at the District level
APPENDIX 8C	Follow-up indicators of core functions of the IDSR core Indicators at the Regional level
APPENDIX 8D	Follow-up indicators of core functions of the IDSR core Indicators at the National level
APPENDIX 8E	Monitoring chart for IDSR performance of indicators at the HFlevel
APPENDIX 8F	Monitoring chart for IDSR performance of indicators at the district,
APPENDIX 8G	regional levels
APPENDIX 8H	Sample form for recording timeliness and completeness of monthly reporting from HFto the district level
APPENDIX 8I	Checklist for monitoring IDSR activities in health facility
APPENDIX 8J	Checklist for supervising surveillance and response activities at the health facility
APPENDIX 8K	Sample weekly and monthly epidemiologic bulletin
APPENDIX 8L	Indicators for monitoring the implementation of IDSR core functions

APPENDIX 8A. Follow-up indicators of core functions of the IDSR at the HFlevel

IDSR core functions	Indicator	Purpose	Numerator	Denominator	Source of information	Target	When to be done
Identifying	Availability of Standard-case definition (SCD) and IDSR forms/registers	Correctly identifying and filling cases/events	NA	NA	Check-list for the officer in charge of the HF	NA	Quarterly
	Existence of a mechanism to capture unusual or public health events from non-formal sources	Measure the ability of the system to capture unusual events	NA	NA	- Interviews HFlog of suspected outbreaks and signals in the HF Electronic event platforms	NA	Monthly
Reporting	Proportion of complete surveillance reports and submitted on time to the district	Practice in HF regarding the timely submission of surveillance reports to the higher level	The proportion of complete monitoring reports sent to the district on time	Number of expected surveillance reports from the health facility	Monitoring chart for timely submission of report Electronic reporting platforms	80 %	Monthly
	Proportion of cases of diseases targeted for elimination, eradication and any diseases selected for case-based surveillance reported with case-based forms or line lists.	Measure the reporting of monitoring data with detailed information to be used for further analysis	The number of cases of surveillance diseases reported through forms or line lists	Total number of case diseases covered by surveillance occurring at the health facility	Summary of reports and line lists	80 %	Monthly
Analysing and interpreting	Proportion of priority diseases for which a current line graph is available.	Measure practice and capacity to analyse surveillance data	Proportion of priority diseases for which a current line graph is available.	Total number of priority diseases	- Check-list for the officer in charge of the HF HFsummary reports on the IDSR	80 %	Quarterly

Analysing and interpreting	Proportion of priority diseases for which a distribution map is available	Measure practice and capacity to analyse surveillance data	Number of priority diseases for which a distribution map is available.	Total number of priority diseases	- Check-list for the officer in charge at the HF HFsummary reports on the IDSR	80 %	Quarterly
	The proportion of priority diseases for which there is laboratory data analysis (if the HFhas a laboratory)	Evidence of systematic analysis and interpretation of laboratory data	Number of priority diseases for which a current lab data analysis is available.	Total number of priority diseases	Laboratory register	80 %	Quarterly
Investigating and confirming suspected outbreaks	Proportion of suspected outbreaks of epidemic-prone diseases notified to the district within 24 hours or surpassing the outbreak threshold	Measure early detection and timely reporting of outbreaks	Number of suspected outbreaks of epidemic-prone diseases notified to the district within 24 hours or surpassing the outbreak threshold	Total number of suspected outbreaks of epidemic-prone diseases	HF logbook of suspected outbreaks and signals in the HF	80 %	Monthly
	Proportion of specimens from suspected cases within 24 hours of collection****	Measure capacity to refer samples on time	Number of suspected cases for which samples were sent within 24hrs	The total number of suspected cases	Laboratory register	80 %	Weekly
	Proportion of samples of suspect cases whose laboratory test results are provided within an acceptable processing time	Measure the functioning of the sample transmission network and the functioning of the reference laboratory	Number of samples of suspected cases whose lab test results have returned on time	Total number of samples of suspected cases submitted	Transmission slip Laboratory registers;	80 %	Weekly

Preparing	Availability of supplies for emergency response (see kit)***	Measure the health facility’s preparedness	NA	NA	HFstock register or stock sheet	NA	Monthly
	Availability of an emergency preparedness and response plan	Measure the health facility’s preparedness	NA	NA	Annual work plans		Yearly
Responding	Presence of a functioning public health emergency management committee (based on HFstaff and national context)	Measure response capacity at the HFlevel	NA	NA	Committee meeting reports and attendance sheets		Quarterly
	Case-fatality rate for each epidemic-prone disease reported	Measure response activities (early request of management and quality of health care services)	Number of deaths from each of the epidemic-prone diseases	Number of cases from the same epidemic-prone diseases.	Routine reports and outbreak investigation	Depends on disease	Monthly
	The attack rate for each declared disease with epidemic-prone disease	Measure response activities	Number of new cases detected	Population at risk	Routine reports and outbreak investigation	Depends on disease	Monthly
	Existence of infection prevention and control measures in all HF, including a triage area	Measure the capacity to prevent nosocomial infections	NA	NA	Observation (demarcation of triage area, hand washing posters).		Monthly
	Availability of an isolation facility in all hospitals	Measure ability to effectively manage highly infectious patients	NA	NA	Observation		Quarterly
	Proportion of health workers trained in infection prevention and control within the last 12 months in the health facility	Measure the capacity to prevent nosocomial infections	Number of health workers trained in IPC in last 12 months in a given care facility	Total number expected to be trained	Training reports	80 %	Quarterly

Providing feedback	Community feedback sessions at least once quarterly	Measure continuous community mobilization	NA	NA	Community feedback reports		Quarterly
	Proportion of feedback bulletins or reports received from the higher level	Presence of a feedback mechanism	The number of bulletins or feedback reports received from the higher level	number of feedback bulletins or reports expected from the higher level	Observation		Quarterly

*** Emergency kit: e.g. gloves, intravenous solutions, masks, aprons, boots, disinfectants, sampling kits,

APPENDIX 8B. Follow-up indicators of core functions of the **IDSR at the district level**

IDSR core functions	Indicator	Purpose	Numerator	Denominator	Source of information	Target	When to be done
Identifying	Proportion of HF with standard-case definition	Correctly identifying and filling cases/events	Number of HFwith standard case definition	Total number of HF	Check-list for the officer in charge of the HF	100%	Quarterly
	Proportion of HF reporting information using EBS	Measure the ability of the system to capture unusual events	Number of HF reporting information from EBS	Total number of HF	Summary reports and supervisory reports	80%	Monthly
Reporting	Proportion of HF including hospitals (referral/zonal/specialised) with standardised registers and IDSR forms	Measure the availability of registers and IDSR forms	Number of HF with registers and IDSR forms	Total number of HF	Check-list for the officer in charge of the HF	100%	Quarterly
	Proportion of HF including hospitals (referral/zonal/specialised) submitting IDSR reports on time to the district	Measure the timeliness of submission of surveillance reports	Number of HF that submitted surveillance reports on time to the district	Total number of HF in the district	Monitoring chart for timely submission of report	80%	Monthly
	Proportion of cases of diseases targeted for elimination, eradication and any diseases selected for case based surveillance reported with case based forms or line lists	Measure the reporting of monitoring data with detailed information to be used for further analysis	Number of cases of diseases targeted for elimination, eradication and any diseases selected for case based	Total number of suspected cases selected for case based surveillance	Summary of reports and line lists for diseases targeted for elimination and eradication and for any other disease targeted for surveillance	80%	Quarterly

			surveillance reported with case based forms or line lists	which occurred in the district			
	Proportion of hospitals submitting their IDSR reports on time	Measure reporting rates by HF	Number of hospitals that submitted report on time	Total number of HF	Monitoring chart for timely submission of report	100%	Monthly
Analysing and interpreting	Proportion of HF that have current trend analysis	Measure the practice and capacity of HFteam to detect trends of suspected possible outbreaks	Number of HF that have current trend analysis for selected priority diseases	Total number of HF in the district	-Supervisory report HFdata analysis tools	80%	Quarterly
	Proportion of HF that have current laboratory analysis data for priority disease analysis (if applicable)	Evidence of systematic analysis and interpretation of laboratory data	Number of HF that have laboratory data analysis for selected priority diseases.	Total number of HF in the district	Laboratory register	80%	Quarterly
	Proportion of priority diseases for which a current line graph is available.	Measure the practice and capacity of district health management team to analyse surveillance data	Number of selected diseases (at least malaria and bacterial meningitis in districts at high risk for Meningitis) for which a line graph is available and updated.	Total number of selected diseases with a line graph (at least malaria and bacterial meningitis if district is at high risk for meningitis)	Monitoring of indicators dashboard	80%	Quarterly
Investigating and confirming suspected outbreaks	Proportion of suspected outbreaks of EPDs notified to the district within 24 hours or surpassing the outbreak threshold	Measure use of data and thresholds for early detection of outbreaks and timely reporting	Number of suspected outbreaks of EPDs notified to the region within 24 hours of surpassing the outbreak threshold	Total number of suspected outbreaks of EPDs in the district	- Register of suspected outbreaks and rumours -District analysis booklet or other systematic analysis tool	80%	Quarterly
	Proportion of outbreak investigation reports that include analysed case based data	Measure availability of additional variables for further analysis	Number of outbreak investigation reports that include case-based data	Total number of outbreak investigation reports drafted in the district	-Investigation report - Map Epidemiological curve -Analysis table of persons ; Line lists or case reporting forms	80%	Quarterly
	Proportion of investigated outbreaks for which laboratory results are available within 7 days	Measure capacity of laboratories to confirm diagnosis and their involvement in surveillance activities	Number of investigated outbreaks for which laboratory results are	Total number of outbreaks Investigated within a given time period	- Register of suspected outbreaks and rumours; Laboratory reports	80%	Quarterly

			available within a defined time frame		- Total number of investigated outbreaks		
	Proportion of confirmed outbreaks with a nationally recommended public health response	Measure the capacity of Districts to respond to outbreaks	Number of confirmed outbreaks with a nationally recommended response	Number of confirmed outbreaks in in the district	- Register of suspected outbreaks and signals; outbreak investigation reports ; - Supervisory reports	80%	Quarterly
	Proportion of samples from suspected outbreak transported within 24 hours	Measure capacity to refer samples on time	Number of suspected outbreaks of which samples were sent on time (within 24 hours)	Number of samples collected from suspected outbreaks	Laboratory register	80%	Quarterly
Preparedness	Presence of effective public health emergency management committee in the district	Measures district's preparedness	NA	NA	-Meeting reports - Annual work plans		Yearly
	Proportion of HsF with emergency preparedness and response (EPR) plans	Measure the HFs preparedness	Number of HFs with emergency preparedness and response plans	Total number of HF	Annual work plans		Yearly
	Availability of a district emergency preparedness and response plan	Measures district's preparedness	NA	NA	Annual work plans		Yearly
	Existence of funds for emergency response (or budget line for emergency funds)	Measure HFs preparedness	NA	NA	Annual work plans		Yearly
	Proportion of HFs that experienced shortage of drugs and supplies for the most recent outbreak (define the time frame e.g. 3, 6, 12 months)	Measure HFs preparedness	Number of HFs that experienced shortage	Total number of HF	Stock register and stock sheet of the health facility		Quarterly
	Proportion of HFs that have contingency stocks for 3–6 months	Measure HFs preparedness	Number of HF with contingency stocks	Total number of HF	Stock register and stock sheet of the health facility		Quarterly
	Proportion of HFs with availability of laboratory diagnostic reagents	Measure capacity of preparedness of HFs	Number of HFs with available laboratory reagents	Total number of HF	Stock register and stock sheet of the health facility		Quarterly

	Proportion of HFs with supplies for sample collection and transportation	Measures preparedness capacity of HFs	Number of HFs with available sample collection and transportation	Total number of HF	Stock register and stock sheet of the health facility		Quarterly
	Proportion of laboratories with performance reports of quality assurance	Measures preparedness capacity of HF	Number of laboratories that systematically carry out a performance evaluation on quality assurance	Total number of laboratories	Quality assessment reports		Quarterly
Responding	Presence of a functional PHEMC	Measures ability to respond at district level	NA	NA	Meeting reports		Quarterly
	Proportion of HFs with functional PHEMC	Measure response capacity at the HFlevel	Number of HFwith functional committee	Total number of HF	Reports from HFmeetings		Quarterly
	Availability of public health emergency RRT	Measure response capacity at the HFlevel	NA	NA	Service note designating RRTs		Quarterly
	The case fatality rate for each reported disease with EPD	Measure quality of case management	Number of deaths from each of the EPDs	Number of cases from the same EPDs	Summary of reports and outbreak investigation report	Depends on disease	Monthly
	Attack rate for each outbreak of a priority disease	To identify the population at risk and measure the effectiveness of the intervention	Number of new cases with EPD that occurred during an outbreak	The number of people at risk during the outbreak	- Demographic data of the district District demographics; outbreak investigation report based on descriptive sheets or case forms	Depends on disease	
	Proportion of outbreaks or any public health event responded to in the previous 12 months	Measure early detection and timely reporting of outbreaks	Number of suspected outbreaks of EPDs responded	Total number of suspected outbreaks of EPDs/events	HFregister of suspected outbreaks and signals in the FOSA	80%	Quarterly
	Proportion of hospitals with infection prevention and control (IPC) requirements established including isolation ward/unit	Measure the practice and capacity of hospitals to apply infection control requirements	Number of hospitals that reported having put in place infection prevention and control (IPC) requirements	Total number of HF in the district	-Summary of reports - Supervisory reports; Review of infection prevention and control practices		Yearly
	Availability of	Presence of a feedback mechanism	NA	NA	Observation		Quarterly

Providing feedback	feedback reports/letters/ bulletins						
	Proportion of feedback bulletins or reports sent to lower level	Presence of a feedback mechanism	Number of reports or bulletins or any documentation actually sent to lower level and received	Total number of reports/bulletins or any form of feedback document expected to be sent to lower levels	Observation		Quarterly

APPENDIX 8C. Follow-up indicators of core functions of the **IDSR at the regional level**

IDSR Core functions	**Indicator**	**Purpose**	**Numerator**	**Denominator**	**Source of information**	**Target**	**When to be done**
Identifying	Proportions of districts with IDSR guidelines to guide identification of cases	Correctly identifying and filling cases/events	Number of Districts with Guidelines	Total number of districts	Discharge slip	100%	Quarterly
	Proportion of districts reporting information using EBS mechanism	Measure the ability of the system to capture unusual events	Number of districts reporting information using event-based surveillance methods	Total number of districts	Summary of reports - Supervisory reports	80%	Monthly
	Number of events recorded in the register rumour	Measure ability of the region to capture unusual events from unofficial report sources	NA	NA	Regional rumour register		Quarterly
	Proportion of districts with routine data validation mechanism	Measure routine validation of data	Number of districts having routine data validation system	Total number of districts	Journal reports		
Reporting	Proportion of monthly surveillance reports submitted from the district to the region on time in the last 3 months the last 3 months	Measure the practice of timely submission of surveillance data	Number of districts that submitted IDSR reports on time to the region	Total number of districts that report to the regional level	Monitoring chart Summary of reports	80%	Quarterly
	Proportion of cases of diseases targeted for elimination, eradication and any disease selected for case based surveillance reported with case based forms or line lists	Measure the reporting of monitoring data with detailed information to be used for further analysis	Number of cases of diseases targeted for elimination eradication and any diseases selected for case based surveillance reported with case based forms or line lists	Total number of diseases targeted for elimination, eradication and any other disease selected for case based surveillance	Summary of Reports and line Lists	80%	Quarterly
Analysing and interpreting	Proportion of districts in which a current line graph is available for selected priority diseases	Measure practice and capacity to analyse surveillance data	Number of priority diseases for which a current line graph is available in districts.	Total number of districts	- Supervisory reports; - District analysis booklet	80%	Quarterly
	Proportion of districts in which an updated spot map of cases is available	Measure practice and capacity to analyse surveillance data	Number of priority diseases for which an updated spot map is available in districts.	Total number of districts	- Supervisory reports; - District analysis booklet	80%	Quarterly

	for selected priority diseases						
	Proportion of districts that report laboratory data for diseases under surveillance	Measure if districts are collecting and reporting lab data to higher level	Number of district labs that submitted monthly data to higher level	Total number of district laboratories	- Supervisory reports; - District analysis booklet		
Investigating and confirming reported cases and outbreaks	Proportion of suspected outbreaks of epidemic-prone diseases notified to the national level within 24 hours of surpassing the outbreak threshold	Measure early detection and timely reporting of outbreaks	Number of suspected outbreaks of epidemic-prone diseases notified to the national level within 24 hours of surpassing the outbreak threshold	Total number of suspected outbreaks of epidemic-prone diseases	- Log of suspected outbreaks and alerts Summary of reports	80%	Quarterly
	Proportion of reports of investigated outbreaks that includes analysed case-based data	Measures availability of additional variables for further analysis including possible risk factors involved	Number of outbreak investigation reports that include outbreak curve, mapping, personal tables and case based forms or descriptive sheets	Total number of outbreak investigation reports	-Investigation reports Summary of reports	80%	Quarterly
	Proportion of investigated outbreaks with laboratory results	Measure capacity of laboratories to confirm diagnosis and their involvement in surveillance activities	Number of investigated outbreaks with laboratory results	Total number of investigated outbreaks	- Total number of investigated outbreaks Laboratory reports Summary of reports - Register of outbreaks and rumours	80%	Quarterly
	Proportion of confirmed outbreaks with a nationally recommended public health response	Measure capacity of the region or province to respond to outbreaks	Number of confirmed outbreaks with a nationally recommended public health response	Total number of confirmed outbreaks	- Register of suspected outbreaks and alerts - Total number of investigated outbreaks - Supervisory reports	80%	
	Proportion of laboratories performing routine testing and reporting of antimicrobial resistance	Measure capacity in readiness	Number of laboratories reporting antimicrobial resistance results	Total number of laboratories	- National Laboratory Policy Document -Laboratory register		Quarterly

Preparing	Presence of an effective coordination of the Public Health Emergency Management Committee (Emergency Operations Centre) at regional level	Measure the Regional readiness	NA	NA	-Reports of the meetings; - Annual work plans		Six months
	Proportion of districts with established functional Public Health Emergency Management Committee	Measure the Regional readiness	Number of districts with the functional Public Health Emergency Management Committee	Total number of districts	Supervisory reports		Quarterly
	Proportion of districts with emergency preparedness and response plans	Measure preparedness of districts	Number of districts with emergency preparedness and response plan	Total number of districts	Supervisory reports		Yearly
	Proportion of districts with public health risk and resource mapping	Measure the practice and capacity of the district to conduct mapping of available resources and risks	Number of districts that reported having conducted public health risks and resources mapping	Total number of districts	Risk assessment, mapping reports and resource mapping reports. Resource mapping	80%	Yearly
	Proportion of districts with funds for emergency preparedness and response	Measure district's preparedness	Number of districts with budgets/ budget line	Total number of districts	Annual work plans		Yearly
	Proportion of districts that have contingency stocks including laboratory supplies for 3–6 months	Measure district's preparedness	Number of districts with contingency stocks	Total number of districts	Stock register or district/regional stock register	80%	Quarterly
	Proportion of districts laboratories with performance reports on quality assurance	Measure capacity preparedness	The number of laboratory districts that have carried out a performance evaluation on quality assurance	Total number of districts	A performance evaluation on quality assurance reports		Quarterly
Responding	Proportion of districts with established functional Public PHEMC	Measure ability to respond at district/regional level	Number of districts with functional committee	Number of districts/regions	- Supervisory reports; Minutes of the PHEMC meetings		Quarterly
	Proportion of districts with effective rapid response teams (RRTs) in public health emergencies	Measure ability of region and district's preparedness towards emergencies	Number of districts with functional RRTs	Total number of districts	- Supervisory reports; Minutes of RRT intervention meetings		Quarterly

	Attack rate for each priority disease outbreak	Identify the population at risk and measure the effectiveness of the intervention	The number of new cases of a disease with outbreak potential occurring during an outbreak	The number of people at risk during the outbreak	- Demographic data of the district -Outbreak investigation report based on line lists or case forms	Depends on disease	
	Case-fatality rate for each epidemic-prone disease reported	Measure quality of case management and response to outbreak	Number of deaths from each of the epidemic-prone disease during an outbreak	Total number of cases from the same epidemic-prone disease during an outbreak	Summary of reports and outbreak investigation report	Depends on disease	
	Proportion of outbreaks or any public health event responded to within the previous 12 months	Measure early detection and timely reporting of outbreaks	The number of suspected outbreaks with epidemic-prone diseases that resulted in an intervention within the last 12 months	The total number of suspected outbreaks with epidemic-prone diseases or events that led to an intervention in the last 12 months	Suspected outbreaks and signal register in FOSAs	80%	
	Proportion of hospitals with infection prevention and control (IPC) requirements	Measure the practice and capacity of hospitals to apply infection control measures	Number of HF that reported having put in place infection prevention and control (IPC) requirements	Total number of HF in the region	Routine summary reports and supervisory reports	80%	Yearly
Providing feedback	Proportion of districts with epidemiological bulletin/newsletters/briefs summaries	Presence of a feedback mechanism in the region and districts	Number of districts with epidemiological bulletin	Total number of districts	Supervisory reports		

APPENDIX 8D. Follow-up indicators of core functions of the IDSR at the national level

IDSR Core functions	Indicator	Purpose	Numerator	Denominator	Source of information	Target	When to be done
Identifying	Proportion of districts with IDSR guidelines to identify cases	Correctly identify and fill cases/events	Number of districts with Guidelines	Total number of all the HF	List of districts	100%	Yearly
	Proportion of districts reporting information using EBS	Measure the ability of the system to capture unusual events	Number of districts reporting information using event-based surveillance methods	Total number of all the districts	Summary reports and supervisory reports	80%	Six months
	The number of events recorded in the signal logbook	Measure ability of national level to capture unusual events from unofficial reported sources	NA	NA	National signal logbook		Six months
	Proportion of districts with routine data validation system	Measure routine validation of data	Number of districts having routine data validation system	Total number of all the districts	National reports		Six months
Reporting	Number of HFs that submit IDSR reports to the district on time	Measure practice of timely submission of surveillance data from HF to district	Number of HF submitting reports on time to the district	Number of districts	Reporting forms and summary report	80%	Monthly
	Proportion of monthly surveillance reports submitted on time by the region during the last 3 months	Measure the practice concerning timely submission of surveillance data	Number of regions that submitted IDSR reports on time	Total number of regions that report to the national level	-Follow-up table ; summary of reports	80%	Six months
	Proportion of cases of diseases targeted for elimination, eradication and any disease selected for case-based surveillance reported with case based forms or line lists	Measure the reporting of monitoring data with detailed information to be used for further analysis	Number of cases of diseases targeted for elimination, eradication and any diseases selected for case-based surveillance reported with case based forms or line lists	Number of diseases targeted for elimination, eradication and any other disease selected for case based surveillance	Summary of systematic reports and case line lists	80%	Quarterly
Analysing and interpreting	Proportion of districts with an updated line graph for selected priority diseases	Measure practice and capacity to analyse data from the surveillance	Number of priority diseases for which a line	Number of districts	Supervisory reports; District analysis booklet	80%	Six months

IDSR Core functions	Indicator	Purpose	Numerator	Denominator	Source of information	Target	When to be done
			graph is available in districts.				
	Proportion of regional laboratories reporting laboratory analysis data to the national laboratory	Measure the data analysis quality of district laboratories at the regional level	Number of regional laboratories that report their analyses to the national public health laboratory on a monthly basis	Total number of regional laboratories	National public health laboratory		Monthly
	Proportion of districts that report laboratory data for diseases under surveillance	Measure if districts are collecting and reporting laboratory data to the higher level	Number of district laboratories that submitted monthly data to the higher level	Total number of laboratories in the districts	National register of received reports		Monthly
Investigating and confirming suspected outbreaks	Proportion of suspected outbreaks of epidemic-prone diseases notified to the national level within 2 days of surpassing the alert threshold	Measure early detection and timely reporting of outbreaks	Number of suspected outbreaks of epidemic-prone diseases notified to the national level within 2 days of surpassing the alert threshold	Total number of suspected outbreaks of epidemic-prone diseases	Register of suspected outbreaks and signals; Summary of reports	80%	Monthly
	Proportion of reports of investigated outbreaks that includes analysed case data	Measure availability of additional variables for further analysis including possible associated risk factors	Number of outbreak investigation reports that include an epidemiological curve, mapping, individual tables and forms or line lists cases	Number of outbreak investigation reports	Investigation reports; Summary of reports	80%	Monthly
	Proportion of investigated outbreaks with laboratory results	Measure capacity of laboratories to confirm diagnosis and involvement of laboratories in surveillance activities	Number of investigated outbreaks with laboratory results	Number of investigated outbreaks	Investigated outbreaks Reports; laboratory reports Summary of reports Register of outbreaks and rumours	80%	Monthly

IDSR Core functions	Indicator	Purpose	Numerator	Denominator	Source of information	Target	When to be done
	Proportion of confirmed outbreaks with a nationally recommended public health response	Measure capacity of the region to respond to outbreaks	Number of confirmed outbreaks with a nationally recommended public health response	Number of confirmed outbreaks	Register of suspected outbreaks and signals; Investigated outbreak reports; Supervisory reports	80%	
	Number of outbreaks detected at the national level, but not detected at the district level	Check the capacity of the entire health system to detect outbreaks, and show that the national level is checking whether districts are observing trends	Number of outbreaks detected by the regional or national level from analysing district-specific data	Total number of outbreaks reported by districts	District forms Summary reports and supervisory reports; District analysis booklet; Standard surveillance reports	Zero	Six months
Preparing	Presence of a functional coordination public health emergency management committee at the national level	Measure national level preparedness	NA	NA	Minutes of meetings; annual work plan		Yearly
	Proportion of regions with functional coordination mechanism/committee for outbreaks/public health emergencies	Measure national level preparedness	Number of regions with the functional mechanism or body for coordination of public health emergencies	Total number of all regions	Supervisory reports		Quarterly
	Proportion of regions with emergency preparedness and response (EPR) plans	Measure preparedness of regions	Number of regions with emergency preparedness response plan	Total number of all regions	Supervisory reports		Yearly
	Proportion of regions with public health risk and resource mapping	Measure practice and capacity of the regions to conduct mapping of available resources and risks	Number of districts that reported having conducted public health risks and resources mapping	Total number of regions targeted for public health risks and resources	Risk assessment and mapping reports	80%	Yearly
	Proportion of regions with funds for emergency preparedness and response	Measure preparedness of regions	Number of regions with budgets/budget line	Number of all regions	Annual work plan		Yearly

IDSR Core functions	Indicator	Purpose	Numerator	Denominator	Source of information	Target	When to be done
	Proportion of regions that have contingency stocks, including lab supplies for 3–6 months	Measure preparedness of regions	Number of regions with contingency stocks	Number of all regions	District/regional stock register	80%	Quarterly
	Proportion of regional laboratories with performance reports on quality assurance	Measure capacity preparedness	Number of regional laboratories that perform a quality assurance review	Number of regional laboratories	Quality assurance reports		Quarterly
Responding	Proportion of regions with established functional Public Health Emergency Management Committee	Measure ability to respond at the regional level	Number of regions with functional committee	Number of regions	Yearly		Yearly
	Proportion on regions with a functional rapid response intervention team	Measure ability to respond at the regional level	Number of regions with a functional rapid response intervention team	Total number of regions	Yearly		Yearly
	The attack rate for each outbreak of a priority disease	Identify the population at risk and measure the effectiveness of the intervention	The number of new cases of a disease with outbreak potential occurring during an outbreak	The number of people at risk during the outbreak	Demographic data of the district; Outbreak investigation report based on line lists or case forms	Depends on disease	
	Case-fatality rate for each epidemic-prone disease reported	Measure quality of case management	Number of deaths from each of the epidemic-prone diseases	Number of cases from the same epidemic-prone diseases	Outbreak systematic and investigation reports	Depends on disease	Monthly
	Proportion of outbreaks or any public health event responded to within the last 12 months	Measure early detection and timely reporting of outbreaks	Number of suspected outbreaks of epidemic-prone diseases responded	Total number of suspected outbreaks of epidemic-prone diseases/events	HFregister of suspected outbreaks and signals in HFs	80%	Yearly
	Proportion of HFs with infection prevention and control (IPC) requirements	Measure the practice and capacity of HFs to apply infection control measures	The number of hospitals that reported having infection prevention and control measures in place recorded	Total number of HF in the country	Summary of reports Supervisory reports	80%	Yearly

IDSR Core functions	Indicator	Purpose	Numerator	Denominator	Source of information	Target	When to be done
Providing feedback	Proportion of regions with epidemiological bulletins/letters/summary notes	Presence of a feedback mechanism	Number of regions with epidemiological bulletin	Total number of regions	Supervisory reports		Yearly

APPENDIX 8E: Monitoring Chart for IDSR performance indicators at the HFlevel

Instructions: Please use this table to monitor HF results regarding those indicators that are related to the institution's IDSR implementation

Summarise and compile HF summary data for priority diseases on a monthly basis. Report summary data to the district level in the time period expected. Note the results of the indicators in this table. Submit this table to the district supervisor during a visit to the health facility, or present it at the meeting

Indicator	Jan	Feb	Mar	Apr	May	June	July	August	Sep	Oct	Nov	Dec
Availability of Standard-case definition (SCD) and IDSR forms/registers												
Existence of a mechanism to capture unusual or public health events from non-formal sources												
The proportion of complete monitoring reports sent to the HD on time												
The proportion of cases of diseases targeted by the monitoring team that were reported to the HD through sheets or case report forms												
The proportion of priority diseases for which there exist a current trend chart												
The proportion of priority diseases for which there is a current analysis of laboratory data												
Availability of an emergency preparedness and response plan												
Availability of equipment for sample collection and transportation												
Availability of emergency supplies												
Proportion of suspected outbreaks of epidemic-prone diseases reported at district level within 24 hours, following the crossing of the outbreak threshold												
Proportion of samples collected and transported on time for laboratory testing in a situation of a suspected outbreak												
Availability of a functional outbreak preparedness committee												
Case-fatality rate for each epidemic-prone disease reported												
Proportion of suspected outbreaks of epidemic-prone diseases reported at district level within 24 hours, following the crossing of the outbreak threshold												
Availability of an isolation facility												
The declared attack rate of each epidemic-prone disease												
Availability of feedback reports to the community												

The number of bulletins or feedback reports received from the higher level												
Answer YES or NO for each item in the check list												
Were the Surveillance reports sent on time?												
Are the trend charts up-to-date?												
If YES, have you noticed a change in trends?												
If YES, has the threshold been crossed?												
If YES, have you taken measures to alert the district?												

APPENDIX 8F: Monitoring Chart of IDSR performance indicators at the district and regional levels

District: ____________________ Region: ____________________ Year: ____________________

Indicators	Jan	Feb	Mar	Apr	May	June	July	August	Sep	Oct	Nov	Dec
Proportion of HF with standard-case definitions												
Proportion of districts or regions sending reports through surveillance event system												
Number of HF in the district with standardised registers and IDSR forms												
Number of events recorded in the district rumour register												
Number of HF that submit IDSR reports to the district on time												
Number of cases of diseases targeted for elimination, eradication and any disease selected for case-based surveillance reported with case based forms or line lists												
Proportion of hospitals submitting their IDSR reports on time												
Proportion of priority diseases for which a line graph is updated												
Proportion of HF with a current trend analysis												
Proportion of HF with current laboratory data analysis from priority disease analysis												
Proportion of suspected outbreaks of epidemic-prone diseases reported at the regional or provincial level within 24 hours, following the crossing of the outbreak threshold												
Proportion of outbreak surveillance and investigation reports that include data from analysed cases												
The proportion of investigated outbreaks for which laboratory results are available												
Proportion of confirmed outbreaks for which a public health intervention is recommended at the national level												
Proportion of samples from suspected outbreak cases transported on time (within 24 hours) to laboratories												
Existence of a functional public health emergency management committee												
Proportion of HF with emergency preparedness and response plans												
Availability of a district emergency preparedness and response plan												
Proportion of HF with an effective outbreak preparedness committee												
Availability of public health emergency rapid response team												
Case-fatality rate for each epidemic-prone disease reported												
Attack rate for each outbreak of a priority disease												

Proportion of outbreaks or public health events that resulted in a timely response in the last 12 months												
Proportion of hospitals with isolation facilities												
Availability of feedback reports/letters/bulletins												
Proportion of feedback bulletins/reports sent to the lower level												

Note: Please calculate the actual proportion for each unit.

APPENDIX 8G: Sample form for recording timeliness and completeness of weekly report from HF to the district level

Country: ____________________ Region: ____________________ District: ____________________

HF____ Year: ____________________

Legend

T = arrived on time

L = arrived late

W=report not received

Name of the HF	**W1**	**W2**	**W3**	**W4**	**W5**	**W6**	**W7**	**Wn**
HF1								
HF2								
HF3								
HF4								
HF5								
HF6								
Total number of reports expected (N)								
Total number of reports sent on time (T)								
Total number of reports sent late (L)								
Total number of reports not received (W)								
Timeliness of reports =100 * T / N								
Completeness of reports =100 * (N-W) / N								

* Completeness and timeliness are reported as a percentage (%). When the surveillance system is effective, the promptness and completeness rates should be close to 100%. This table enables to follow the progress of these two indicators in the district so that measures can be taken to improve the timeliness of each of the HF in the district

APPENDIX 8H: Checklist for supervising surveillance and response activities at the health facility

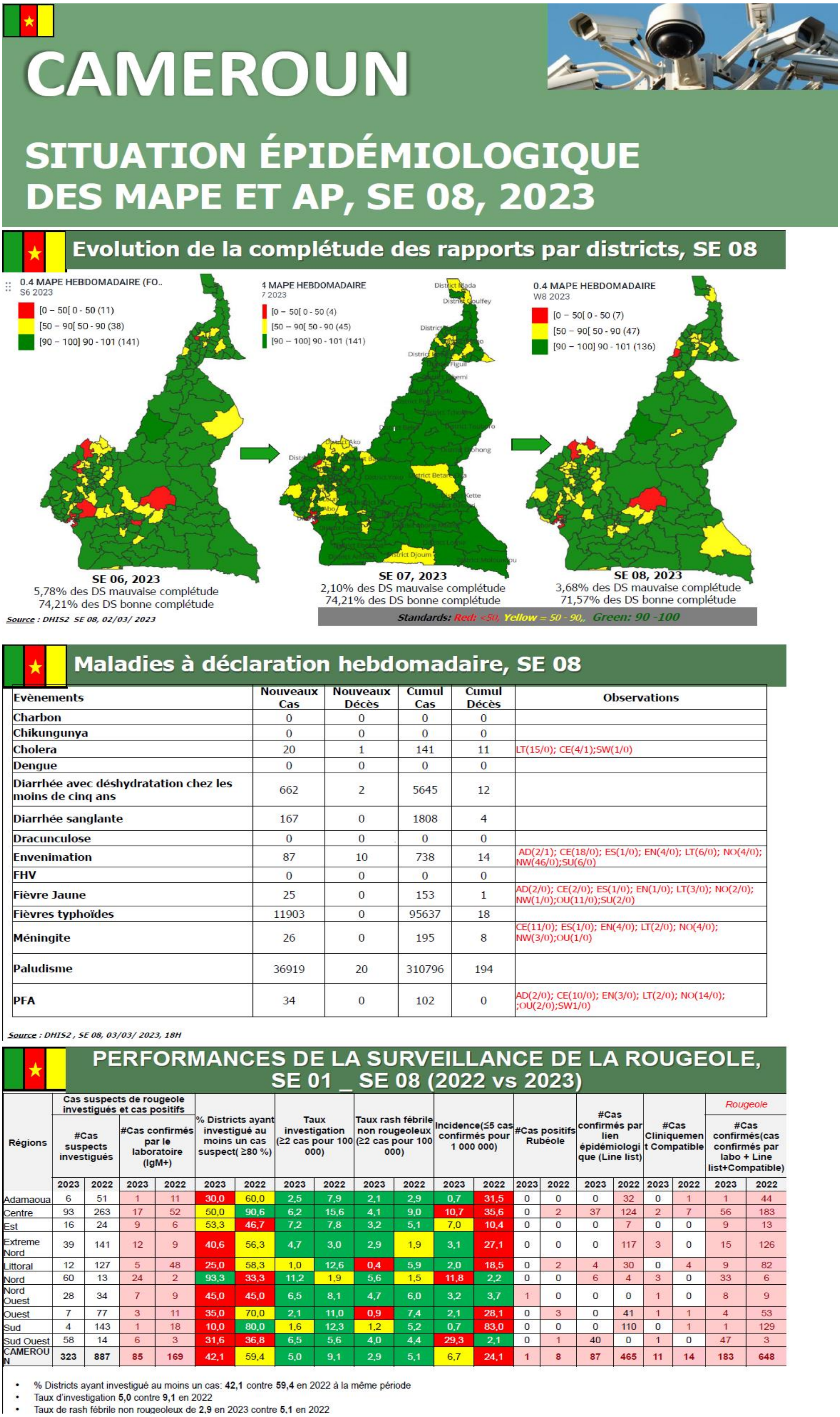
Health Facility: _________________________________Date of Supervision: _____________________________

ACTIVITY	SUPERVISORY QUESTION	ANSWER	COMMENT (What Caused Problem)
Data collection to identify suspected cases within the health facility	1. How often do you collect information from the community about reports of suspected cases or deaths due to a priority disease or condition?	____________________	
Registering cases	1. Are the diagnosis for priority diseases recorded in the clinic register according to the standard-case definition?	Yes [ ] No [ ]	
Reporting	1. Does the health staff use a standard-case definition to report the suspected cases and outbreaks? 2. Do you record information about immediately notifiable diseases on a case form or a line list?	Yes [ ] No [ ] Yes [ ] No [ ]	
Analysing and interpreting	1. Do you plot the number of cases and deaths for each priority disease on a graph? (Ask to see the health facility's data analysis plan. Check whether trend lines are up-to-date) 2 Do you plot distribution of cases on a map?	Yes [ ] No [ ] Yes [ ] No [ ]	
Investigating and confirming reported cases and outbreaks	1. If an epidemic-prone disease was suspected, was it reported immediately to the district office? 2. For cases of priority diseases needing laboratory tests seen since the last supervisory visit, how many had laboratory results? 3.	Yes [ ] No [ ] Number of results obtained: ____________ Number of expected cases seen: ____________ Yes No	

	Are appropriate supplies available or set aside for collecting laboratory specimens during an urgent situation? May I see the supplies?	
Riposte	1. Les fournitures appropriées sont-elles disponibles pour riposter à un cas ou une épidémie confirmée (par exemple, le matériel de vaccination et les vaccins, les SRO, les antibiotiques, etc.) ? 2. Veuillez me montrer les fournitures qui serviraient à exécuter une intervention recommandée. 3. Qui est le coordonnateur des situations d’épidémie dans la formation sanitaire? 4. À quelle fréquence communiquez-vous des informations et dispensez-vous des formations en riposte aux épidémies et aux urgences de santé publique ?	Désignation : Formation suivie :
Rétro information	1. À quelle fréquence communiquez-vous des informations à la communauté ? 2. Avez-vous reçu le dernier bulletin de santé publique du niveau supérieur	L’indiquer Oui [ ] Non [ ]
Évaluation amélioration système	1. Les 3 derniers rapports mensuels ont-ils été envoyés au district ? Oui [ ]Non [ ] 2. Les 3 derniers rapports mensuels systématiques ont-ils été Oui [ ] Non [ ]	

		envoyés à temps	
Préparation à l’épidémie	1.	Quelles précautions prend systématiquement le personnel de santé (y compris le personnel de laboratoire) avec tous les malades quel que soit leur statut infectieux ?	Niveau minimum de précautions standard :
	2.	Comment estimez-vous la quantité de fournitures à mettre en réserve pour les situations d’urgence ?	Comment les fournitures sont estimées :

APPENDIX 8I. Sample weekly epidemiological situation report



CAMEROUN

SITUATION ÉPIDÉMIOLOGIQUE DES MAPE ET AP, SE 08, 2023

Evolution de la complétude des rapports par districts, SE 08

0.4 MAPE HEBDOMADAIRE (FO..
S6 2023
[0 – 50[0 - 50 (11)
[50 – 90[50 - 90 (38)
[90 – 100] 90 - 101 (141)
4 MAPE HEBDOMADAIRE
7 2023
[0 – 50[0 - 50 (4)
[50 – 90[50 - 90 (45)
[90 – 100] 90 - 101 (141)
0.4 MAPE HEBDOMADAIRE
W8 2023
[0 – 50[0 - 50 (7)
[50 – 90[50 - 90 (47)
[90 – 100] 90 - 101 (136)

SE 06, 2023
5,78% des DS mauvaise complétude
74,21% des DS bonne complétude

SE 07, 2023
2,10% des DS mauvaise complétude
74,21% des DS bonne complétude

SE 08, 2023
3,68% des DS mauvaise complétude
71,57% des DS bonne complétude

Source : DHIS2 SE 08, 02/03/ 2023

Standards: Red: <50, Yellow = 50 - 90,, Green: 90 -100

Maladies à déclaration hebdomadaire, SE 08

Evènements	Nouveaux Cas	Nouveaux Décès	Cumul Cas	Cumul Décès	Observations
Charbon	0	0	0	0	
Chikungunya	0	0	0	0	
Cholera	20	1	141	11	LT(15/0); CE(4/1);SW(1/0)
Dengue	0	0	0	0	
Diarrhée avec déshydratation chez les moins de cinq ans	662	2	5645	12	
Diarrhée sanglante	167	0	1808	4	
Dracunculose	0	0	0	0	
Envenimation	87	10	738	14	AD(2/1); CE(18/0); ES(1/0); EN(4/0); LT(6/0); NO(4/0); NW(46/0);SU(6/0)
FHV	0	0	0	0	
Fièvre Jaune	25	0	153	1	AD(2/0); CE(2/0); ES(1/0); EN(1/0); LT(3/0); NO(2/0); NW(1/0);OU(11/0);SU(2/0)
Fièvres typhoïdes	11903	0	95637	18	
Méningite	26	0	195	8	CE(11/0); ES(1/0); EN(4/0); LT(2/0); NO(4/0); NW(3/0);OU(1/0)
Paludisme	36919	20	310796	194	
PFA	34	0	102	0	AD(2/0); CE(10/0); EN(3/0); LT(2/0); NO(14/0); ;OU(2/0);SW1/0)

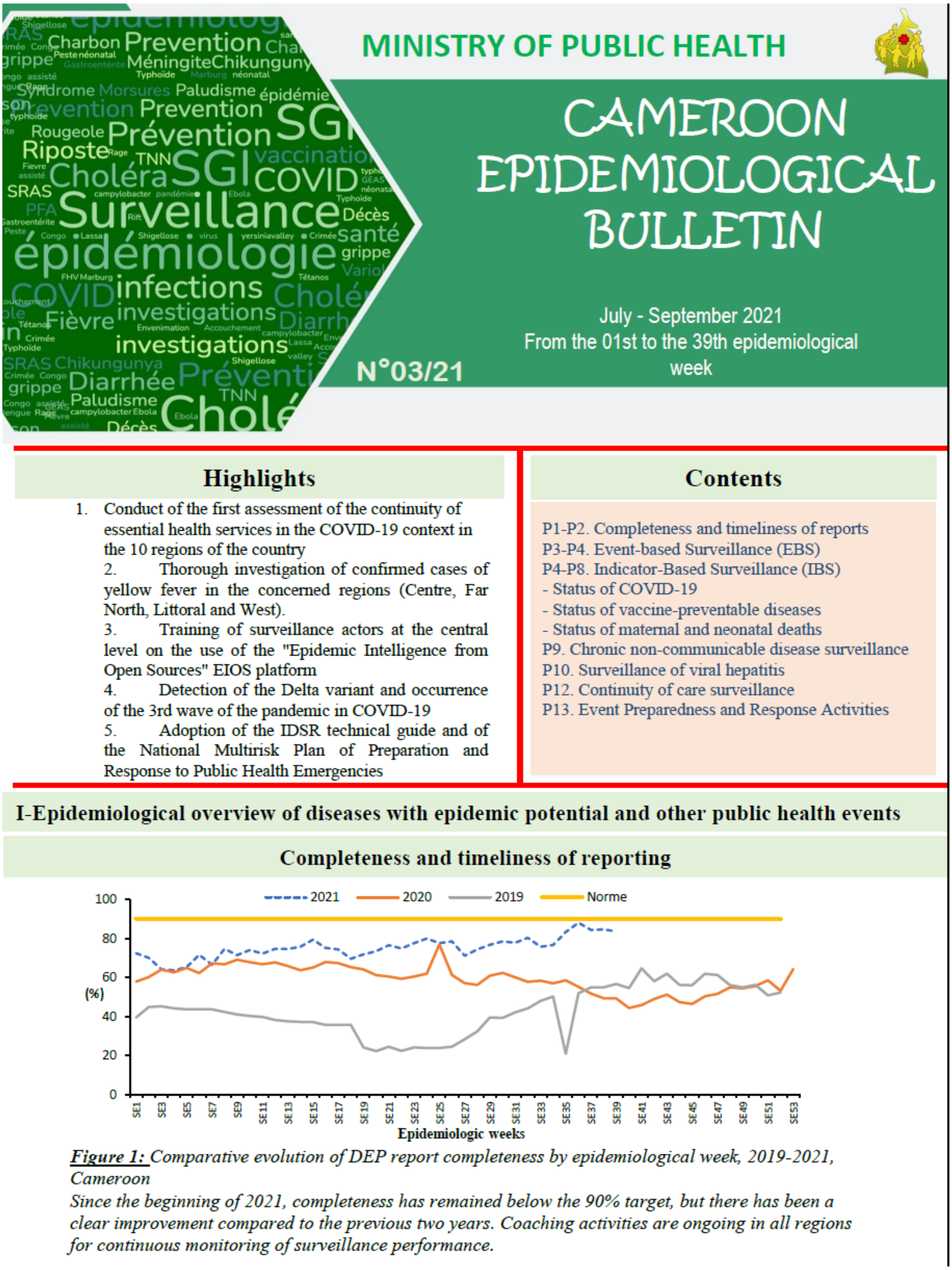
Source : DHIS2 , SE 08, 03/03/ 2023, 18H

PERFORMANCES DE LA SURVEILLANCE DE LA ROUGEOLE, SE 01 _ SE 08 (2022 vs 2023)

Régions	Cas suspects de rougeole investigués et cas positifs				% Districts ayant investigué au moins un cas suspect(≥80 %)		Taux investigation (≥2 cas pour 100 000)		Taux rash fébrile non rougeoleux (≥2 cas pour 100 000)		Incidence(≤5 cas confirmés pour 1 000 000)		#Cas positifs Rubéole		#Cas confirmés par lien épidémiologique (Line list)		#Cas Cliniquement Compatible		*Rougeole*	
	#Cas suspects investigués		#Cas confirmés par le laboratoire (IgM+)																#Cas confirmés(cas confirmés par labo + Line list+Compatible)	
	2023	2022	2023	2022	2023	2022	2023	2022	2023	2022	2023	2022	2023	2022	2023	2022	2023	2022	2023	2022
Adamaoua	6	51	1	11	30,0	60,0	2,5	7,9	2,1	2,9	0,7	31,5	0	0	0	32	0	1	1	44
Centre	93	263	17	52	50,0	90,6	6,2	15,6	4,1	9,0	10,7	35,6	0	2	37	124	2	7	56	183
Est	16	24	9	6	53,3	46,7	7,2	7,8	3,2	5,1	7,0	10,4	0	0	0	7	0	0	9	13
Extreme Nord	39	141	12	9	40,6	56,3	4,7	3,0	2,9	1,9	3,1	27,1	0	0	0	117	3	0	15	126
Littoral	12	127	5	48	25,0	58,3	1,0	12,6	0,4	5,9	2,0	18,5	0	2	4	30	0	4	9	82
Nord	60	13	24	2	93,3	33,3	11,2	1,9	5,6	1,5	11,8	2,2	0	0	6	4	3	0	33	6
Nord Ouest	28	34	7	9	45,0	45,0	6,5	8,1	4,7	6,0	3,2	3,7	1	0	0	0	1	0	8	9
Ouest	7	77	3	11	35,0	70,0	2,1	11,0	0,9	7,4	2,1	28,1	0	3	0	41	1	1	4	53
Sud	4	143	1	18	10,0	80,0	1,6	12,3	1,2	5,2	0,7	83,0	0	0	0	110	0	1	1	129
Sud Ouest	58	14	6	3	31,6	36,8	6,5	5,6	4,0	4,4	29,3	2,1	0	1	40	0	1	0	47	3
CAMEROUN	**323**	**887**	**85**	**169**	**42,1**	**59,4**	**5,0**	**9,1**	**2,9**	**5,1**	**6,7**	**24,1**	**1**	**8**	**87**	**465**	**11**	**14**	**183**	**648**

- % Districts ayant investigué au moins un cas: **42,1** contre **59,4** en 2022 à la même période
- Taux d'investigation **5,0** contre **9,1** en 2022
- Taux de rash fébrile non rougeoleux de **2,9** en 2023 contre **5,1** en 2022

APPENDIX 8J. Sample quarterly and annual epidemiological bulletins


Charbon
Prevention
Méningite
Chikungunya
Typhoïde
Syndrome
Morsures
Paludisme
épidémie
Rougeole
Prévention
SGI
Riposte
TNN
vaccination
Choléra
COVID
SRAS
PFA
Surveillance
Décès
épidémiologie
Santé
grippe
infections
Fièvre
investigations
Diarrhée
Tétanos
Crimée
Congo
Lassa
Ebola
Shigellose
campylobacter

MINISTRY OF PUBLIC HEALTH

CAMEROON EPIDEMIOLOGICAL BULLETIN

July - September 2021
From the 01st to the 39th epidemiological week

N°03/21

Highlights

1. Conduct of the first assessment of the continuity of essential health services in the COVID-19 context in the 10 regions of the country
2. Thorough investigation of confirmed cases of yellow fever in the concerned regions (Centre, Far North, Littoral and West).
3. Training of surveillance actors at the central level on the use of the "Epidemic Intelligence from Open Sources" EIOS platform
4. Detection of the Delta variant and occurrence of the 3rd wave of the pandemic in COVID-19
5. Adoption of the IDSR technical guide and of the National Multirisk Plan of Preparation and Response to Public Health Emergencies

Contents

P1-P2. Completeness and timeliness of reports
P3-P4. Event-based Surveillance (EBS)
P4-P8. Indicator-Based Surveillance (IBS)
- Status of COVID-19
- Status of vaccine-preventable diseases
- Status of maternal and neonatal deaths
P9. Chronic non-communicable disease surveillance
P10. Surveillance of viral hepatitis
P12. Continuity of care surveillance
P13. Event Preparedness and Response Activities

I-Epidemiological overview of diseases with epidemic potential and other public health events

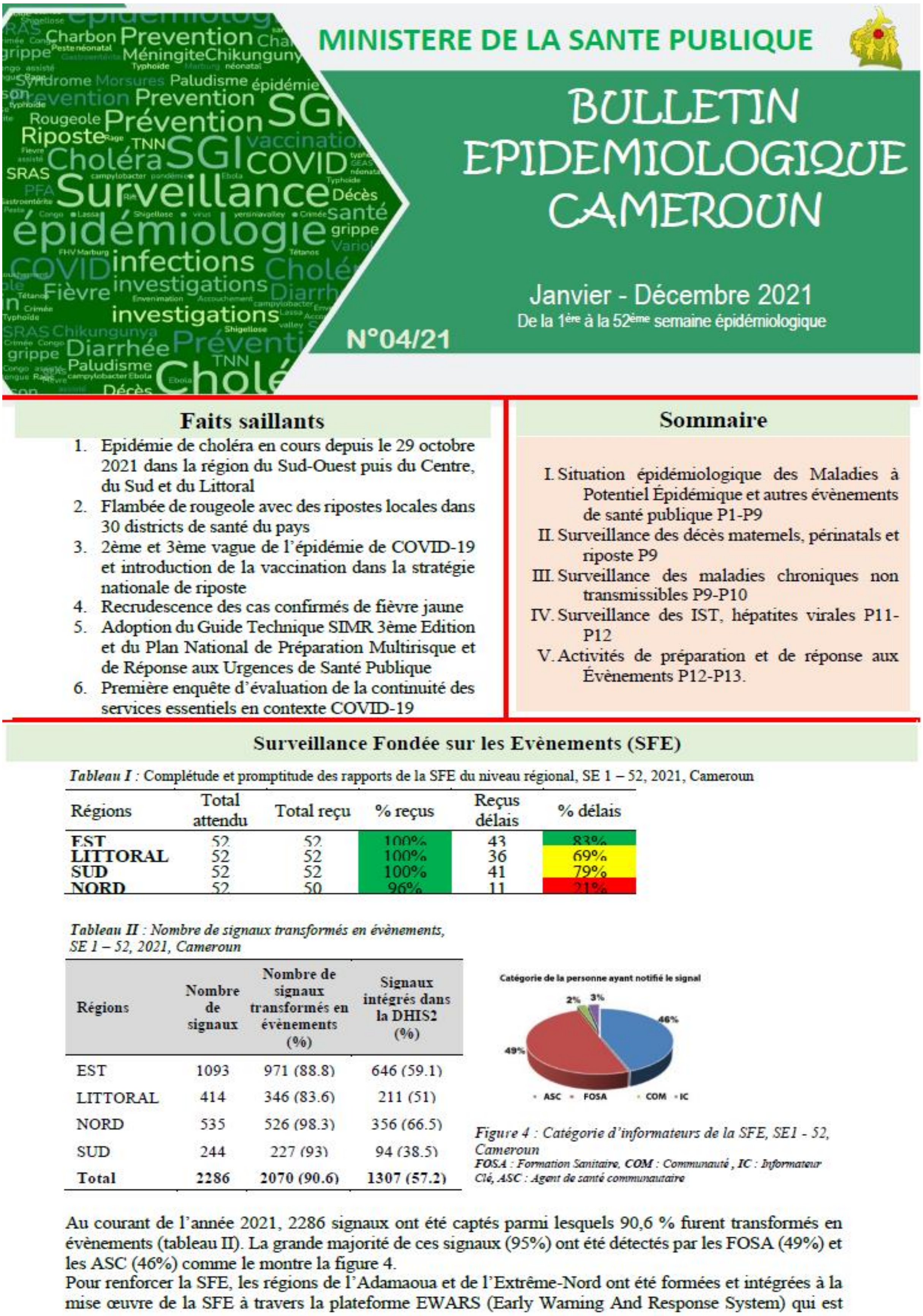
Completeness and timeliness of reporting

2021
2020
2019
Norme
100
80
60
40
20
0
(%)
SE1
SE3
SE5
SE7
SE9
SE11
SE13
SE15
SE17
SE19
SE21
SE23
SE25
SE27
SE29
SE31
SE33
SE35
SE37
SE39
SE41
SE43
SE45
SE47
SE49
SE51
SE53
Epidemiologic weeks

__Figure 1:__ Comparative evolution of DEP report completeness by epidemiological week, 2019-2021, Cameroon

Since the beginning of 2021, completeness has remained below the 90% target, but there has been a clear improvement compared to the previous two years. Coaching activities are ongoing in all regions for continuous monitoring of surveillance performance.

Sample quarterly epidemiological bulletin



MINISTERE DE LA SANTE PUBLIQUE

BULLETIN EPIDEMIOLOGIQUE CAMEROUN

Janvier - Décembre 2021

De la 1ère à la 52ème semaine épidémiologique

N°04/21

Faits saillants

1. Epidémie de choléra en cours depuis le 29 octobre 2021 dans la région du Sud-Ouest puis du Centre, du Sud et du Littoral
2. Flambée de rougeole avec des ripostes locales dans 30 districts de santé du pays
3. 2ème et 3ème vague de l'épidémie de COVID-19 et introduction de la vaccination dans la stratégie nationale de riposte
4. Recrudescence des cas confirmés de fièvre jaune
5. Adoption du Guide Technique SIMR 3ème Edition et du Plan National de Préparation Multirisque et de Réponse aux Urgences de Santé Publique
6. Première enquête d'évaluation de la continuité des services essentiels en contexte COVID-19

Sommaire

I. Situation épidémiologique des Maladies à Potentiel Épidémique et autres évènements de santé publique P1-P9
II. Surveillance des décès maternels, périnatals et riposte P9
III. Surveillance des maladies chroniques non transmissibles P9-P10
IV. Surveillance des IST, hépatites virales P11-P12
V. Activités de préparation et de réponse aux Évènements P12-P13.

Surveillance Fondée sur les Evènements (SFE)

Tableau I : Complétude et promptitude des rapports de la SFE du niveau régional, SE 1 – 52, 2021, Cameroun

Régions	Total attendu	Total reçu	% reçus	Reçus délais	% délais
EST	52	52	100%	43	83%
LITTORAL	52	52	100%	36	69%
SUD	52	52	100%	41	79%
NORD	52	50	96%	11	21%

Tableau II : Nombre de signaux transformés en évènements, SE 1 – 52, 2021, Cameroun

Régions	Nombre de signaux	Nombre de signaux transformés en évènements (%)	Signaux intégrés dans la DHIS2 (%)
EST	1093	971 (88.8)	646 (59.1)
LITTORAL	414	346 (83.6)	211 (51)
NORD	535	526 (98.3)	356 (66.5)
SUD	244	227 (93)	94 (38.5)
Total	**2286**	**2070 (90.6)**	**1307 (57.2)**

Figure 4 : Catégorie d'informateurs de la SFE, SE1 - 52, Cameroun

***FOSA** : Formation Sanitaire, **COM** : Communauté , **IC** : Informateur Clé, **ASC** : Agent de santé communautaire*

Au courant de l'année 2021, 2286 signaux ont été captés parmi lesquels 90,6 % furent transformés en évènements (tableau II). La grande majorité de ces signaux (95%) ont été détectés par les FOSA (49%) et les ASC (46%) comme le montre la figure 4.

Pour renforcer la SFE, les régions de l'Adamaoua et de l'Extrême-Nord ont été formées et intégrées à la mise œuvre de la SFE à travers la plateforme EWARS (Early Warning And Response System) qui est

APPENDIX 8K. Indicators for monitoring the implementation of IDSR core functions

1. Percentage of HFs that sent their weekly (or monthly) surveillance reports on IDSR to the district on time.
2. Percentage of districts that sent their weekly (or monthly) surveillance reports to the next higher level on time.
3. Percentage of cases of diseases targeted for elimination and eradication and any other disease targeted for surveillance, notified to the district using case-based forms or line lists.
4. Percentage of suspected outbreaks of epidemic-prone diseases reported to the next level within 24 hours following the crossing of the epidemic threshold.
5. Percentage of care structures having a current annalysis of trends (line graph or histogramme) for some priority diseases.
6. Percentage of districts having a current annalysis of trends (line graph or histogramme) for some priority diseases.
7. Percentage of investigation reports on outbreaks having analysed data on cases.
8. Percentage of outbreaks investigated with available laboratory results within seven days.
9. Percentage of confirmed outbreaks for which public health intervention is recommended at the national level between 24 to 48 hours following reporting **(target >80%).**
10. Lethality rate of each epidemic-prone disease notified.
11. Attack rate of each outbreak of a priority disease.
12. Number of outbreaks detected at the national level, but which went unnoticed to the district the previous year.
13. Percentage of selected laboratories producing monthly reports on laboratory data on priority diseases targeted for surveillance.
14. Percentage of district laboratories which received at least one supervision and feedback a written from the regional or national level in the preceding year.

- **Intervention standard between 24 to 48 hours**
  1. Conduct a first evaluation or quick analysis of the situation.
  2. Inform WHO about the outbreak or the public health event.
  3. Activate national response structures to emergencies and assign essential functions.
  4. Launch response activities using a pillar-by-pillar approach.
  5. Convene the first emergency intersector coordination meeting.
  6. Develop the first response strategy, objectives and an action plan.
  7. Publish the first internal situation report (Sitreps).

Section 9: Electronic integrated disease surveillance and response (electronic idsr (e-IDSR)

This section dwells on how to:

- Integrate electronic IDSR into NHIS;
- Develop and implement an electronic IDSR;
- Grasp the roles and functions of e-IDRS at all the levels of the health-care pyramid.

Introduction

Electronic IDSR (eIDSR) is the application of electronic tools to principles of IDSR to facilitate prevention, prediction, detection, reporting and response to priority diseases and public health events. It is based on:

(a) standardised interoperable and interconnected information systems administered within a national context;
(b) the rapid collection, reporting, analysis, and use of disease/public health events data in real time situation for appropriate public health action.

Since the adoption of IDSR in 1998, using paper-based tools for its implementation has been an instrumental strategy for strengthening public health surveillance in the African Region. With the adoption of the IHR in 2005, which requires countries to strengthen capacity for disease surveillance and response, application of electronic tools to enhance real-time surveillance can improve timeliness of outbreak detection and other public health events.

In recent years, technological and analytical innovations have emerged as an approach which can be used to facilitate rapid transmission of public health surveillance information, thus aiding timely detection of and response to outbreaks and other public health events. Application of e-tools in the health sector has the potential to provide real-time validated data for public health surveillance, investigation and prompt outbreak response and other public health events. eIDSR provides new opportunities for accelerating achievement of the IHR (2005) core capacities in the country.

9.1. eIDSR as part of the National Health Information System (NHIS)

The National Health Information System (NHIS) facilitates routine collection of data to support planning, management and decision-making in health service provision. These systems routinely collect data on diseases, public health events and conditions as well as other administrative and service-provision data. The primary source of these data is the HF outpatient department (OPD) or inpatient department (IPD) register. In Cameroon, the most widely used electronic platform of NHIS are the Early Warning Alert and Response System (EWARS), WHONET, and *District Health Information System version 2* (DHIS2). The latter is used in more than 30 countries in Africa. It is used in low- and middle-income countries.

The eIDSR is a part of NHIS. Source data is derived from HF outpatient (OPD) or inpatient (IPD) registers, laboratory logbooks, medical records and community. In the eIDSR platform, there is an active and timely means of collecting data on IDSR priority diseases, conditions and public health events. Data is reported immediately, weekly or monthly.

In the circular letter **No. D36-59/L/MINSANTE/SG/CIS of 22 November 2017** addressed to all health structures from the central services to the peripheries, Cameroon, through the Minister of Public Health, has set new requirements in terms of health information management and has chosen DHIS2 (www.dhis-minsante-cm.org), as the national electronic platform for the collection and transmission of health data. All the health services, projects and programs should therefore comply with this new tool by integrating their data into this system that serves as a national health data warehouse. In parallel, the EWARS platform allows for the collection of signals at community level, which are reported directly by the CHWs on their phones.

9.2. eIDSR in the context of e-Health

In 2013, the African Region adopted an e-Health resolution (AFR/RC60/R3) to address the use of Information and Communication Technologies (ICT) in health and health-related fields, including disease surveillance. The actions recommended from this resolution include the development of national policies, strategies, standards and appropriate governance mechanisms, long-term strategic plans and frameworks for strengthening e-Health in countries. e-Health encompasses a range of services and systems, including:

- health and medical informatics;
- telemedicine, which is the transmission of health-related services or information through telecommunication means;
- e-learning whereby technologies are used in education outside of the traditional classroom setting; and
- m-health, which is a general term for the use of mobile phones and other wireless technologies in the medical field.

Member States echoed the same sentiment in the recent Seventy-first World Health Assembly, and unanimously agreed that digital health solutions should complement and enhance existing health service delivery models. Subsequently, they adopted the new resolution on e-Health, which underscores the importance of nationally supported digital health strategies (including policy, standards, capacity, interoperability, privacy and security, and much more), and transitioning to sustainability and government ownership.

e-Health provides cost-effective and secure use of ICTs in health and health-related fields. e-health, as defined by the ***Broadband Commission for Sustainable Development***, is an umbrella term that encompasses all concepts and activities at the intersection of health and information and communications technologies (ICTs). This includes delivery of health information using ICTs to improve public health services and using health information systems to capture, store, manage or transmit information on patient health or HFactivities. ICTs are defined as tools that facilitate communication, processing and transmission of information by electronic means and these encompass a full range of tools like radio, television, telephones (fixed and mobile), computers, and the internet.

eIDSR, which is part of e-Health, is one of the essential innovations in the implementation of recommendations of the WHO Regional Committee for Africa on the use of information technology, which is essential in achieving IHR (2005) requirements by countries. Standardisation of electronic tools and sustained infrastructure across the region will promote easy generation and sharing of country and regional profiles of priority diseases, conditions and public health events.

9.3. Rationale of eIDSR

Limitations of the current approaches to IDSR data collection and transmission are attributed to the fact that many countries still use manual procedures and paper-based methods to collect and transmit data. Submitting and transmitting data on time is a challenge as health workers have to travel long distances on difficult terrains to submit their files. This leads to delays in getting information on time for action, especially in the event of a suspected outbreak.

The eIDSR system aims to facilitate the work of every staff member in a health system by improving disease surveillance using electronic tools. This will strengthen the capacity of the surveillance and response team and in the long term reduce morbidity and mortality caused by epidemic-prone diseases and other public health events.

eIDSR is thus likely to improve the following:

- timeliness and completeness of reports;
- early detection, investigation and response to outbreak or public health events;
- manual data entry that is prone to errors;
- systematic information sharing across levels and sectors;
- combining data streams;
- data archiving;

- data use, data analysis and analytics.

There have recently been various supporting initiatives and resolutions at the regional and global levels, which have acknowledged the potential of digital technologies to promote the Sustainable Development Goals (SDGs), and particularly to support health systems in health promotion and disease prevention in all countries. Hence, the aids is developed to reflect the following recently adopted overarching frameworks:

- integrated Disease Surveillance and Response ((AFR/RC/48.8);
- IHR (2005) (WHA58.3);
- regional Strategy for Health Security and emergencies strategy (AFRO/RC66/6);
- e-Health resolution and decision (WHA58.28);
- e-Health (WHA71.7).

9.4. Benefits of eIDSR

The eIDSR provides real-time information for immediate action. Potential benefits of eIDSR include:

9.4.1. Early alert and detection;

With eIDSR, the speed of outbreak detection can be improved on as information may be more rapidly captured, and in some cases, the time and place of an outbreak can be predicted with varying degrees of accuracy. Thus, enabling opportunities for prevention and control (Refer to a study done by CDC). 2008b. *Potential effects of electronic laboratory reporting on improving timeliness of infectious disease notification—Florida, 2002– 2006. Morbidity and Mortality Weekly Report 57(49):1325–1328.* http://www.cdc.gov/mmwr/preview/mmwrhtml/mm5749a2.htm).

9.4.2. Timely reporting

eIDSR tools allow rapid and timely transmission of data from lower primary reporting units to subsequent higher levels to enable appropriate public health action.

9.4.3. Standardisation of data

Standardisation of tools in the eIDSR system enables data collection to be more consistent and complete for ease of data exchange and comparison across HF.

9.4.4. Better data transmission and management including storage

- A major challenge of paper-based data is the need to compile reports from various sources and provide final reports to higher offices and to different administrative levels at regular intervals. Moreover, data storage and transport can be difficult and there is a risk of data damage and loss.
- With eIDSR, there is faster data transmission. Moreover, data is also organised into a format that is more accessible for use and interpretation.

9.4.5. Interoperability and sharing of data

eIDSR provides an opportunity for exchange and use of information across entities, especially if standards and workflow have been well developed for the eIDSR system to allow interoperability with other information systems.

9.4.6. Automated transmission, analysis and improved data quality

- Paper-based reporting runs the risk of omitting valuable information when reporting to higher administrative levels;
- eIDSR reduces the number of data entry errors and facilitates automated data analysis, thus saving considerable effort for health staff.

9.4.7. Ultimate contribution towards good response, better monitoring and evaluation

eIDSR provides a platform for data storage and automatic analysis across HF for better monitoring and evaluation of various public health interventions. For example, studies carried out by Valenciano et al (2003); Pinto el al (2005) demonstrated that the simple advance of a computerised reporting system using country-identified thresholds for priority diseases, showed improvements in monitoring trends and in predicting outbreaks in a matter of weeks post-implementation.

9.4.8. Cost reduction

eIDSR makes for early detection of outbreaks which in effect, can contribute to overall reduction of high costs associated with management of these outbreaks and other public health events.

9.5. Key guiding principles in establishing eIDSR

The following are key guiding principles in establishing eIDSR:

(a) **Use of existing infrastructure**: as far as possible, eIDSR should be built on existing frameworks and systems such as paper-based IDSR, NHIS, DHIS2, etc. This enables easy adaptability for users and promotes smooth transitions. In a situation where the infrastructure already exists, the introduction of eIDSR may possibly not require major customisation that could be costly.

(b) **Standardisation:** standardisation of data and electronic tools will promote uniformity in data collection and aggregation. Standardisation promotes comparison between various levels of the health systems and between countries.

(c) **Integration**: IDSR builds on the premise of integration and so eIDSR should be implemented in the spirit of integration. This could entail integration of various data sources and information systems from other health programs (e.g., malaria, EPI, cancer registry, non-communicable diseases, HIV/AIDS etc.) into a common platform or data warehouse.

(d) **Interoperability:** it is the ability of different health information systems to work together within and across organisational boundaries to exchange data and use information that has been exchanged. It is important that eIDSR developers make sure standards and workflow are easily interoperable with other information systems.

(e) **Multisector collaboration**: it is essential to collaborate with stakeholders such as telecommunication companies. Such collaboration could be in the form of waivers, corporate social responsibility, and partial tax holidays etc. Effective collaboration could accelerate eIDSR rollout and coverage. Collaboration with other sectors like animal and environment is also key as this will facilitate efficient utilisation of scarce resources, effective and prompt leveraging of various sector capabilities for better disease prevention and control.

(f) **Near real-time approach**: efforts must be made to ensure provision of near real-time transfer of information about events incorporated into the design and implementation of eIDSR. This is a long-term goal and countries will have to start progressively with the planning phase and next the development phase.

(g) **One Health**: the “One Health” approach offers an innovative approach where various disciplines work together to address health at the human-animal-environment interface. Owing to the fact that majority of emerging and re-emerging infections are often zoonotic and are responsible for large number of outbreaks in recent times, maintaining a focus on diseases that affect both humans and animals is a worthwhile investment.

(h) **Data Security**: protection of health information is core in every HIS. Security of data will ensure that information is only accessible to authorised personnel who need to take action. This also promotes ethical handling of data. Caution should be taken to ensure there are processes for patient privacy.

(i) **User-friendly system**: the system should ideally be simple enough to be used by staff at all levels. It should be easy to log in, enter data or receive information. The system should also be flexible to adapt to change of disease and public health event profiles over time.

9.6. eIDSR Development and implementation process

Developing an eIDSR system should be planned carefully with all relevant stakeholders getting involved. The system should fit capabilities and needs of the country and a plan for securing resources should be developed prior to initiating eIDSR. The most important considerations for the process of developing and implementing eIDSR are shown below. Depending on the country, more considerations may need to be made. Countries also have to ensure that they are ready to embark on eIDSR by weighing the costs and benefits and by assessing eIDSR feasibility options in their country.

9.6.1. Process of establishing eIDSR

(a) Engage stakeholders and establish technical working group

- The success of eIDSR requires an effective involvement of all the relevant stakeholders.
- When initiating an eIDSR system, the surveillance group should work in collaboration with the department that handles the HIS. These two departments should bring together all relevant stakeholders in the country to develop a technical working group required to coordinate eIDSR implementation.
- eIDSR may require the mobilisation of ICT expertise from other administrations, especially those in charge of ICT. The Ministry of Public Health should collaborate with the Ministry of Post and Telecommunications, and agencies in charge of ICT to ensure sufficient coverage and proper management of digital healthcare.

In Cameroon, this technical working group was set up by Order **No. 1899/MINSANTE of 21 August 2020** on the creation, organisation and functioning of the interministerial committee in charge of monitoring the implementation of Cameroon's national strategic plan for e-Health 2020-2024.

The members of this technical working group are:

- the Ministry of Public Health;
- the Ministry of Posts and Telecommunications;
- the Ministry of Higher Education;
- the Ministry of Economy, Planning and Regional Development;
- the Ministry of Finance;
- the Ministry of Communication;
- the General Delegation for National Security;
- Cameroonian society of medical informatics;
- national AIDS Control Committee (NACC);
- national Civil Status Registration Office (BUNEC);
- the National Agency for Information and Communication Technologies (ANTIC);
- telecommunications Regulatory Board.

This Committee is tasked with:

- defining the main guidelines for the implementation of the above plan;
- ensuring the mobilisation of national and international resources to finance the provision of e-Health services;
- validating annual action plans;
- monitoring and evaluating the implementation of e-Health in compliance with the annual action plan;
- validating the scale-up plans;
- proposing strategies and mechanisms for the sustainability of e-Health;
- implementing any other mission entrusted to it by the Minister of Public Health.

(b) Assessing the implementation level of the IDRS

The IDRS has to be assessed at all levels, including the political commitment to put ICTs at the heart of any development and social transformation process. The Ministry of Public Health, through the CDC, is tasked with monitoring the implementation of disease surveillance activities program and should make electronic disease surveillance system a priority. The designation of National Focal Persons for monitoring at the HD level is an essential factor for the successful implementation of e-IDRS.

(c) Determine country capabilities and needs

- A crucial step in the development of an electronic system is to assess capabilities and infrastructural needs of the country. For an efficient implementation of e-Health in Cameroon, the eIDSR technical working group should carefully consider the capabilities, infrastructure, resources and the country's needs. WHO has developed a standardised eSurveillance assessment tool which may also be used as a resource tool.
- Network coverage

Assessing network capability of a country is a critical step in determining the type of system that can be developed. Internet and mobile network coverage is a key component to consider.

- Mobile network: number of providers; cost of text messaging; cost of phone calls; provider coverage in all areas of the country; distribution of providers by customers; common operating systems (android/iOS);
- Internet: number of providers; cost of subscription; internet speed; internet coverage in all areas of the country; internet connectivity at the national, regional and district levels.
- Equally explore alternative sophisticated models to extend connectivity such as TV white space, and balloon-, drone-, or low-orbiting satellite-based internet connectivity to extend coverage in remote or hard-to-reach areas. Some of these options can be done through public–private partnership.

- Power supply option
    - Availability of power supply is key to a successful eIDSR. Reliable power supply to suit the needs of the system must be available throughout the implementation phase.
    - Countries should determine available and potential power supply options for each level of the health system in the entire territory. For example, connection to the power grid; consistent generator with fuel provision; consistent generator but no fuel provision; inconsistent generator or inconsistent fuel; power banks or solar power.
    - In addition, countries may consider seeking alternative models to ensure reliable power supply. Solar panels, for example, could be outfitted to key government ministries, prioritising those responsible for managing critical data sets in emergencies and to district HF.
- Equipment – data capture, data management, data analysis
    - Equipment is an essential component of eIDSR. It is important to assess the equipment available for eIDSR in a given country at each level of the health system. If equipment is not available, the feasibility of using options for each type of equipment should be considered.

  - You should also consider the life cycle of all of your hardware and ensure you develop a plan for replacing/renewing hardware as needed.

- Hardware
  - Consider how to address housing of data on servers; will servers be cloud-based? (Easier maintenance but monthly/yearly payment) or will it be housed within physical structures? (Which requires cool room with consistent power and with a high-speed internet connection). Potential costs, including initial/set up and operation cost must be considered;
  - Consider types of computers and quantities required. It is worth noting that desktop computers are cheaper but must have power. Laptops are portable, but expensive and tablets are portable and convenient;
  - Consider the minimum characteristics required for mobile devices especially for smart phones;
  - Assess mobile broadband penetration and internet connectivity;
  - Consider availability of power and how you will ensure uninterrupted power supply.
- Software
  - Develop a mapping of software used for monitoring activities;
  - Assess the use of open-source software or commercial software adapted to the needs;
  - Create platforms for collaboration with software developers for the development of flexible and easily adaptable software;
  - Set up a good data backup system.
- Human resources – technical capacity
  - The country will need a pool of software development staff available to be able to support open-source systems. These would ideally be government staff.
  - Computer literacy of staff is key for those who will be using the electronic system that will be put in place;
  - It might be necessary to train and retrain, as technology evolves (continuous training).

(d) Partnerships

- Partnership frameworks for public–private partnerships with telecommunication operators to support eIDSR systems should be explored jointly with the Ministry of Post and Telecommunications, and agencies in charge of ICT;
- Explore if such partnerships are available for the implemention of eIDSR

(e) Determining appropriate scope of eIDSR implementation, including One Health approach

- Based on the above assessments, the country should determine the scope of implementing eIDSR (alert notification, case-based reporting, routine weekly reporting, routine monthly reporting, and outbreak/ public health emergency management). The country may start with any approach that fits its needs and capacities and later include other aspects;
- The country should obtain estimates for initial investments, operating costs and current costs and should determine potential investors.

(f) Rolling out eIDSR plan

- Develop a strategic plan for the implementation of the IDRS that incorporates eIDSR;
- Develop annual operational plan (time lines, costs, duties);
- Consider a step-wise incremental process in implementing plan and training;
- Incorporate routine monitoring and regular assessment of its implementation.

Cameroon has developed a national strategic plan for e-Health for the period 2020-2024.

9.6.2. Important considerations for a successful eIDSR

The following are deemed important for the successful implementation of eIDSR in a country.

(a) Laboratory integration

- the system should integrate laboratory data or be able to do so in the future;

(b) Data privacy and use of a unique identifier (ID number)

- individual patient data should be stored on a secure server;
- access to data should be controlled through user-access rights.

(c) Data security and user-agreement policies

- there should be clear guidelines on how to access data;
- there should be scheduled data backups (local and remote);
- physical data storage devices should be secured.

(d) IT System Maintenance

- software upgrades, hardware upkeep or replacement and server maintenance should be taken into consideration if it is in-house system.

(e) Sustainability

- in order to ensure sustained support of the eIDSR programme, a sustained financial basis should be established to account for one-time and operation costs such as hardware system maintenance, training of personnel, connectivity costs and production of end-user support materials;
- the software and hardware should be maintained;

- there should be adequate resources to support operational infrastructure;
- resources should be made available to bear all the operational costs and to purchase equipment such as mobile devices and computers;
- resources should be mobilised for continued capacity-building, training, re-training, etc.;
- eIDSR should be anchored within national e-Health policy and strategy;
- all the stakeholders (including private companies and telecommunication companies) should be involved from the conception to the implementation of the system.
- an appeal has to be made for the allocation of financial resources as well as innovative financial solutions including leveraging resources from the private sector such as telecommunication service providers.

(f) Interoperability

- Data and information may be ideally shared across systems. For example, the animal health sector and other relevant sectors

9.6.3. Potential available tools for eIDSR

There are several data collection and data aggregation systems in Cameroon. The functionality of these systems is presented in the table below. When considering the use of commercial software, countries should ensure that there is a budget allocated for licensing costs. Negotiations should be done to ensure that suppliers provide enhancements or adaptations. It is important to note that open source does not mean “free” as there are always implementation and customisation costs to fit each country’s specific context and needs.

Table XXIII:Functionalities of existing systems

FUNCTIONALITY OF EXISTING SYSTEMS			
Application	**Summary**	**Actual Status**	**SOP Status**
Community-based systems and HF			
1. Open MRS-EMRs (electronic medical records)	Open MRS-based electronic medical record system called Bahmni, has been locally designed for patient-level data in the Cameroon health system	Planning for 10 more sites	None
2. DAMA (Patient Data Management System)	Electronic version of the GRC health registers built in the Microsoft C # .NET programming language that generates reports for MOH and PEPFAR	Functional in the PEPFAR area, 154 sites	None
3. BLIS (Basic Laboratory Information System)	Laboratory software for electronic laboratory results and quality assurance	Functional in 10 high-volume laboratories	Ongoing documentation with C4G
4. N-SAMBA/ CommCare	Dimagi / Commcare is used for beneficiary registration and case management; data collection on smartphones/tablets to a server that is not materialised	Pilot functional in 8 centres	SOPs have been installed for data collection and analysis and not for their usage
5. VINDATA	Java / Linux based software requiring a LAN connection for HIV drug and stock management	Functional in the PEPFAR area, 29 large sites	Training guide but no SOPs
National / sub-national information systems			
6. PSO-AIDS	Web reports at the regional level on the number of pharmacies / supplies in institutions	Set up in all 10 regions	None
7. DHIS 2-CIS	Web-based software that facilitates the compilation and analysis of health indicators at	Installed in all districts, that is	Data collection guide in force.

	district, regional and national levels.	about 6,000 installations	
1.**DHIS 2 - NACC**	NACC web-based software to compile and analyse HIV indicators at district, regional and national levels	Installed in all districts, that is about 6,000 installations	A user guide exists but is not widely used
2. **EWARS**	Web-based software that allows the collection, compilation and processing of signals from the community to the central level.	Scaling up in the 3 regions	Training guide

9.7. Using eIDSR in essential surveillance functions

There are many components that will ensure successful implementation of eIDSR in the public health sector. These components include understanding the scope and operational environment, using the right tools, and building capabilities within the local context. The One Health approach also provides an opportunity to create interoperable, interconnected electronic reporting systems between human and animal surveillance systems.

The use of e-tools to conduct Data Quality Assessment/Assurance (DQA) is also part of a monitoring and evaluation strategy of IDSR functions that may be used for continuing improvement of data quality. Such tools can identify errors, inconsistencies and other data anomalies which can affect reliable, accurate, precise and complete data.

Establishing an electronic platform can facilitate the implementation of IDSR activities as described in previous sections of the document: They include:

(a) real-time reporting (indicator and event-based surveillance). (Refer ***to introduction section***);
(b) alert notification (community and HFreporting); refer to ***Section 2;***
(c) case-based reporting; refer to ***Section 2***
(d) routine reporting (weekly aggregates) and routine monthly reporting; refer to ***Section 2. Figure 9.1*** below illustrates how information flows in eIDSR system;
(e) outbreak/emergency management; refer to Sections 4 and 6;
(f) case investigation, see ***Section 6;***
(g) contact tracing; refer to ***Section 6***;
(h) logistics and supply chain management; refer to ***section 6***;

(i) real-time outbreak line-listing; refer to ***Sections 4 and 6***;
(j) event management (hazard description, characterisation, risk assessment and outcomes), refer to ***Section 6***;
(k) information products i.e., situation reports (SitReps), epidemiological bulletin etc. refer to ***Section 7***;
(l) supportive supervision; refer to ***Section 8***;
(m) monitoring and the Evaluation and Data Quality Assessment (DQA); refer to ***Section 8***.

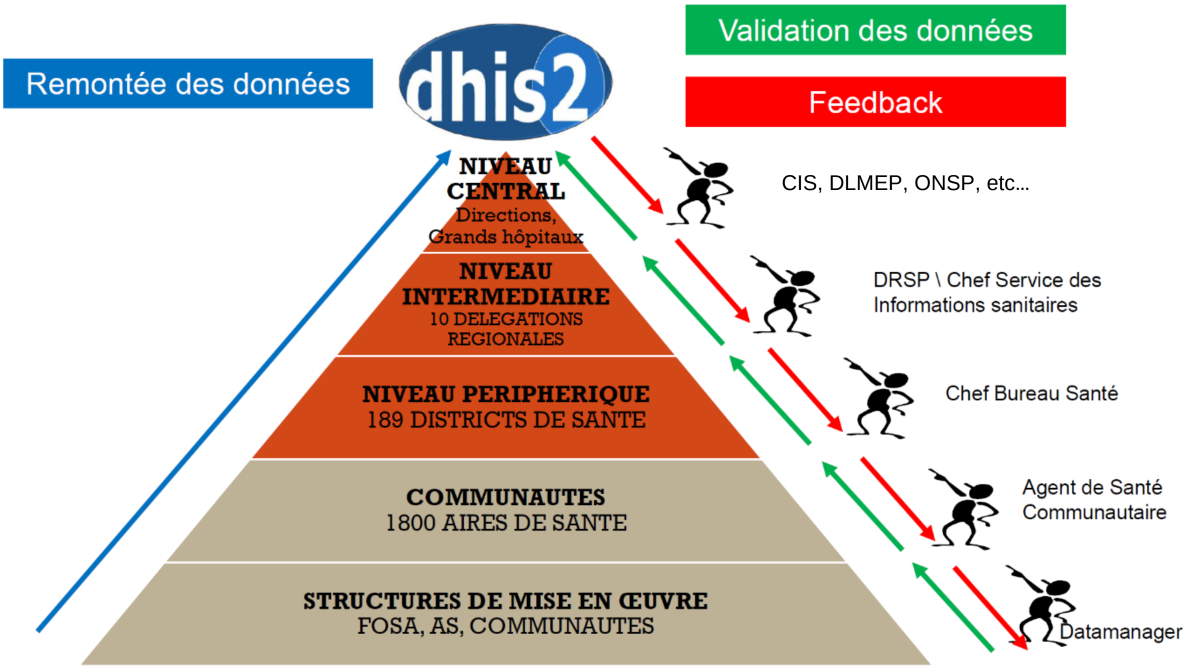

Remontée des données
dhis2
Validation des données
Feedback
NIVEAU CENTRAL
Directions, Grands hôpitaux
NIVEAU INTERMEDIAIRE
10 DELEGATIONS REGIONALES
NIVEAU PERIPHERIQUE
189 DISTRICTS DE SANTE
COMMUNAUTES
1800 AIRES DE SANTE
STRUCTURES DE MISE EN ŒUVRE
FOSA, AS, COMMUNAUTES
CIS, DLMEP, ONSP, etc…
DRSP \ Chef Service des Informations sanitaires
Chef Bureau Santé
Agent de Santé Communautaire
Datamanager

Figure 19: Information flow for eIDSR

9.8. Roles and responsibilities at different levels in the context of real-time reporting and outbreak/emergency management

The following are some roles and responsibilities with regard to eIDSR at various levels.

(a) Community level

- detecting and reporting signals/alerts on events e.g., through toll-free helplines;
- acting on alert message sent from health authorities.

(b) HF level

- detecting and immediately reporting priority diseases and public health events requiring immediate action;
- submitting weekly and monthly IDSR reports;
- following up on events that are reported by community;
- investigating on priority diseases, public health events and responding as recommended for their area of jurisdiction;
- ensuring compatibility of their handset with eIDSR;
- ensuring maintenance and ownership of handset and other tools.

(c) District level

- training staff on the eIDSR;

- providing staff access to the eIDSR;
- verifying and approving onward transmission of reported events from upper HF;
- issuing alerts to other facilities and authorities regarding events within the district;
- providing feedback to reports on events from HF;
- updating HF and authorities on progress made regarding response;
- training, mentorship and supervision of health staff;
- mobilising resources to support effective implementation of eIDSR;
- ensuring availability and compatibility of ICT equipment with eIDSR;

(d) Regional level

- Training and supervision;
- Collaborating with national level to develop and update electronic tools;
- Providing feedback and issuing alerts to districts;

(e) Central/ national level

- maintaining the server;
- developing and updating electronic tools;
- managing the eIDSR system, including troubleshooting;
- maintaining system administration (registration of health staff using server);
- training and supervision;
- providing feedback;
- coordination of partners and stakeholders;
- ensuring connection with other platforms to facilitate interoperability;
- monitoring alerts;
- doing advocacy with policy-makers, and resource mobilisation to sustain the system;
- ensuring data security;
- overseeing development and implementation of national e-Health/digital health strategy;
- aligning eIDSR investments and working with national e-Health/digital health strategy;
- country e-Health/digital health architecture with consideration for re-usable components;
- system governance

(f) WHO and other regional bodies (AU, CEMAC etc.

- facilitating creation of formal platform for sharing of information and data across countries;
- technical assistance to Member States;

- sharing best practices and facilitating exchange of expertise.

9.9. Supervision, monitoring and evaluation

eIDSR development and implementation requires constant monitoring. system functionality can be evaluated by looking at issues such as:

(a) acceptability or willingness to participate. i.e., number of people who are accessing and using the system correctly;
(b) accessibility – Is the system accessible from the place where the reporting site is situated? In some areas where mobile telephone is used for eIDSR, accessibility is an important aspect, and this can hamper prompt report of outbreaks;
(c) data quality and completeness – Check for any data errors;
(d) timeliness of data submission;
(e) system flexibility, transferability and stability;
(f) cost.

To improve data use at the service level, users should be encouraged to use the system with regular feedback of information to lower levels. Information flow should not be one-way.

Other system performance indicators include essential surveillance indicators for monitoring IDSR (***refer to section 8***). The IDSR support supervision checklist should be used during supervision, while considering integrated needs from other teams in terms of joint supervision. The supportive supervision checklist has to be updated to incorporate eIDSR and uploaded as part of the eIDSR platform. Overall evaluation of the eIDSR system, and its interoperability with the NHIS and e-Health system should be done periodically, using a blend of internal and external experts.

Section 10: Tailoring idsr to emergency or fragile health system contexts

This section dwells on:

- the steps to implement IDRS in an emergency situation;
- the different stakeholders to be involved;
- structure and tools to be used.

Introduction

Humanitarian emergencies have major implications for the populations where they occur and for their health service surveillance systems (WHO, 2012). Emergencies typically result in population displacement to congested settings where access to basic needs like water, food, shelter and other social services are constrained. These conditions increase the risk of death from common outbreak and endemic diseases.

Consequently, effective public health surveillance and outbreak response is a priority during public health emergencies in affected populations. Due to the disruption of health and othersocial services during the emergencies, the routine IDSR system must be enhanced to meet the public health surveillance and outbreak response needs in humanitarian contexts.

It should be based on the IDSR strategy, structures, tools, guidelines and resources, but should ensure the flexibility required in addressing the surveillance and response needs of affected populations in emergency situations. This should be done within the existing national IDSR system.

This section introduces key principles of implementing IDSR in complex humanitarian emergencies. This will involve enhancing IDSR core functions to ensure early detection, assessment and response to acute public health events. For a more detailed description, please refer to the WHO document on early detection, assessment and response to acute public health events- ‘‘implementation of early warning and response with focus on event-based surveillance’’ (WHO, 2014).

10.1. Health Information System in Emergency Contexts

Acute and protracted crises, conflicts and disasters have major immediate and long-term effects on population health and health systems and can create disruptions in the overall functionality of the health system. In such situations, the routine IDSR system may be under performing or may be disrupted. IDSR must therefore be tailored to adequately meet the surveillance information needs of a humanitarian emergency. Examples of such humanitarian emergencies include armed conflict, famine, natural disasters and other major emergencies.

Key definitions in Emergency contexts

a) Disaster

A serious disruption of the functioning of a community or a society causing widespread human, material, economic or environmental losses which exceed the ability of the affected community or society to cope using its own resources (*International Strategy for Disaster Reduction [ISDR], 2009).* A disaster is also defined as a situation or event which overwhelms local capacity, necessitating additional national or international assistance (*Centre forResearch on Environmental Decisions [CRED, ReliefWeb, 2008*).

b) Humanitarian emergency

A situation where the basic human needs of a population are threatened and therefore requires extraordinary measures and urgent action (ReliefWeb, 2008).

c) Complex emergency

A humanitarian crisis in a country, region or society where there is total or considerable breakdown of authority resulting from internal or external conflicts and which requires an international response that goes beyond the mandate or capacity of any single and/or ongoing UN country programme (ReliefWeb, 2008).

10.2. Early Warning and Response

Early warning is an organized mechanism to detect, as early as possible, any abnormal occurrence or any divergence from the usual or normally observed frequency of diseases, conditions and public health events. It relies on a network of people from functional static or mobile health facilities/clinics whose responsibility it is to collect, investigate, report, analyse and disseminate information from the field to the central level for appropriate action. ‘‘*implementation of early warning and response including event-based monitoring*’’ *(WHO, 2014).*

10.2.1. Why is it needed?

The enhanced surveillance needs during humanitarian emergencies demand that surveillance systems are in place for systematic collection, collation, analysis, and interpretation of data, and for dissemination of information to facilitate public health response to prevent excess morbidity, mortality and disability (WHO, 2009). Consequently, during the acute phase of a humanitarian emergency, IDSR should be modified as soon as possible to focus on priority health problems during the emergency phase. The tailored IDSR should focus on diseases, conditions or public health events for a given emergency context and should be flexible enough to respond to other emerging public health priorities (WHO, 2009).

During emergencies, populations are more vulnerable to morbidity, mortality and disability resulting from endemic and epidemic-prone diseases. Thus, IDSR should be enhanced within 3–10 days of grading the public health emergency to facilitate rapid detection and response to disease outbreaks and public health events (WHO, 2009). Ultimately, this will contribute to the overall goal of reducing avoidable mortality, morbidity, and disability during humanitarian crises (WHO, 2009b).

10.2.2. What are the objectives of tailoring IDSR to emergency context?

The main objective is to rapidly detect and control acute public health events of any origin, with particular attention to prioritized health risks. The aim is to increase sensitivity of detection, quality of risk assessment, and timeliness and effectiveness of the response to acute public health risks in order to minimize the negative health consequences to the affected population.

The specific objectives are to:

a) detect acute public health events and health risks early;
b) ensure immediate communication of information from local and intermediate levels to national levels as well as from any source identified at the national level;
c) verify the initial information (i.e. the signal);
d) document the nature of the event through investigation, characterization, and etiological confirmation;
e) perform risk assessment to determine the level of risk posed by the detected event;
f) ensure immediate alert mechanisms from national and/or intermediate levels to local levels;
g) ensure prompt investigation as necessary and implement an adequate response through mitigation and control measures, as required by the continuous risk assessment;
h) alert and maintain communication/coordination with national/international stakeholders.

10.2.3. Critical components

During humanitarian crises, all functional static and mobile HF/clinics, including those in refugee camps for internally displaced persons which provide curative, disease prevention and health promotion interventions should be included in the IDSR network to enhance the sensitivity of the system (WHO, 2009).

Depending on the extent of the crisis, the surveillance network may include government and/or partner-supported clinics. (WHO, 2012)

To ensure efficiency, the data collection and analysis processes need to be systematized and formalized. Outbreak intelligence should be based on the two main IDSR event detection systems, namely: indicator-based surveillance (immediate and weekly reporting of data aggregated by HF) and event-based surveillance, which is the organized collection, monitoring, assessment and interpretation of mainly unstructured ad hoc information regarding health events or risks. These complementary systems increase the sensitivity of IDSR to ensure timely detection and verification of outbreaks and effective monitoring of morbidity patterns (WHO, 2012).

10.3. Implementation of IDSR in humanitarian emergencies

10.3.1. Rapid assessment of the situation

During the acute phase of the emergency, it is helpful to undertake a systematic assessment of the risk of acute public health events. This involves gauging both the likelihood of a disease occurring and its eventual impact. The assessment can identify theepidemic-prone diseases that have the potential to cause the greatest amount of morbidity and mortality in the affected population, and determine the geographical scope of surveillance. A prioritization of these epidemic-prone diseases will be made and the signals drawn up based on these priority diseases.

An assessment of the status of key surveillance infrastructure is also done, including existing surveillance capacity, identification of resource needs for IDSR implementation, including staff with relevant skills, communication and IT equipment laboratory support and transport. The assessment should be based on consensus-building, analysis of existing data, establishment of working groups and conducting of in-depth interviews, as required. It should be based on an all-hazards approach and be repeated as the emergency evolves, to account for changes.

10.3.2. Gap analysis

The gap analysis should be performed to complement the situation analysis and the assessment of the surveillance system. It aims to assess the specific needs and environment and review the strengths, weaknesses, threats and opportunities around the existing national surveillance system in order to identify available resources to reinforce IDSR. Gap analysis does not require a new or additional formal evaluation to be carried out. The results of previous evaluations of the surveillance system can be reused. In cases where not all of theinformation needed is available, focus groups or in-depth interviews with stakeholders at all levels of the surveillance system could be considered.

10.3.3. Prioritization

In order to ensure the most efficient use of resources, the strategy should be based on a prioritization exercise, the results of the gap analysis and the list of priority events for surveillance. For each selected disease, condition or public health event, surveillance objectives need to be specified based on the country’s context. The objectives will depend on the characteristics of the disease, condition or public health event (e.g. attack rate, morbidity and mortality), the mode of transmission (e.g. person to person, point source outbreaks, exposure to toxic substances), and the nature of the public health interventions required to control spread.

10.3.4. Development of a plan of action for the implementation of IDSR

Once the prioritization exercise has been completed and all potential sources of information listed, a plan of action should be developed and implemented at the national, intermediate and local levels. The plan of action should be well integrated with the national IDSR system, including monitoring and evaluation.

10.3.5. Designate a coordination mechanism

A coordination structure should be established at the national and intermediate levels to ensure a single-entry point for reporting, analysis and triaging of information, verifying signals, assessing risks, and monitoring and responding to acute public health events.

10.4. Various actors in enhancing IDSR to improve early warning and response

During acute or complex emergencies where the capacity of the national and sub-national IDSR system in the Ministry of Public Health is greatly constrained, the roles and responsibilities of various actors may have to be reinforced.

10.4.1. National/Central level (DLMEP)

The overall coordination of data collection, entry, analysis and dissemination during humanitarian crises should be undertaken by the DLMEP within the Ministry of Public Health (MOH) with support from partners, and forwarded as necessary to the Public Health Emergency

Operations Center (PHEOC). PHEOC coordinates and provides response to the affected regions or districts. However, during acute crises or complex emergencies where the capacity of the national surveillance coordination unit is greatly constrained, PHEOC comes in and a coordinator (typically an epidemiologist or public health expert with experience in disease surveillance and disease control in emergencies) can be recruited to provide support during the acute phase of the crisis. The functions of the coordinator will be guided by the initial rapid assessment and should include, but not be limited to:

a) providing dedicated technical oversight;
b) coordinating the supervision of surveillance and outbreak response activities in crisis affected areas;
c) guiding coordination of health workers and partners for effective disease surveillance and outbreak/public health response in crisis-affected populations;
d) supporting districts or regions to investigate and respond to outbreaks or public health events including reorientation of staff in IDSR;
e) conducting regular analysis of epidemiological trends and production of regular surveillance bulletins and situation reports;
f) providing aids for reporting and notifying priority diseases, conditions and public health events;
g) supporting evaluation.

10.4.2. District level

The surveillance focal person at the district level should coordinate surveillance and response activities in crisis-affected populations. However, during acute crises or complex emergencies where the capacities of district surveillance focal person are constrained, the Regional Delegation for Health (RDPH) working in close collaboration with the health cluster, should activate the Regional Emergency Operations Center (REOC) and designate an incident manager to:

a) coordinate disease surveillance and outbreak response in crisis-affected populations;
b) ensure timely reporting of priority diseases, conditions and public health events related to the crisis;
c) conduct trend analysis and provide feedback to HF and clinics;
d) conduct initial investigation of disease outbreaks and public health events;
e) respond to disease outbreaks and public health emergencies in collaboration with the Ministry of Public Health, partners, and local HF.

10.4.3. Health Facilities

All identified focal persons working in HF or mobile clinics offering curative, preventive and health promotion services should implement the following:

a) detect, collect and report priority diseases, conditions and public health events;
b) support the verification and investigation of outbreaks and public health events;
c) undertake public health and outbreak response measures, with support from the community focal persons, district and national surveillance focal points.

10.4.4. Community level

Community members and community health workers assist in detecting and reporting signals to the next level using appropriate reporting tools including phones with the EWARS application. They equally participate in community and key informant outreach activities and surveys.

10.5. Key structures and tools to be put in place during an acute humanitarian crisis

10.5.1. List of diseases/conditions events

During the acute phase of a humanitarian crisis, a rapid risk assessment should be done to identify diseases, conditions, and public health events that pose a threat to the population. These should be prioritized in addition to the national IDSR priority list. In identifying the list of additional priority diseases, conditions and public health events, criteria for inclusion should take into account WHO guidelines for inclusion of an event under a surveillance system (WHO, 2012), namely:

a) outbreak;
b) vaccine preventable diseases due to disruption of immunization in most of the emergencies;
c) ability to cause severe morbidity or death;
d) international surveillance requirements (IHR, 2005);
e) availability of prevention and control measures;
f) availability of reliable and meaningful case definitions and simple laboratory tests, where appropriate.

It is critical that clinicians register the most important diagnosis per patient and that only new case visits are counted and not the follow-up visits.

The sources of data on new cases include:

a) outpatient clinics;
b) inpatient clinics;
c) laboratory;
d) mobile clinics in the community or from some focal persons identified in the community;
e) IDP/refugee camp clinics;
f) other sources of event-based information.

10.5.2. Case definitions

The existing case definitions should be used for diseases, conditions and public health events already included on the IDSR priority disease list. Sensitive case definitions that increase the chances of detecting new outbreaks should be developed for the additional diseases, conditions, events and syndromes identified as part of the risk assessment. These case definitions should be simple, standardized and harmonized with the national IDSR case definitions.

10.5.3. Laboratory support

Quality assured WHO approved Point of care Rapid Diagnostic Test (RDT) kits for diseases like malaria, cholera, meningitis, and hepatitis A and E are essential for timely treatment and outbreak response decisions.

Laboratory confirmation is more critical for suspected outbreaks in crisis-affected populations and the following should be in place to facilitate timely investigation of new outbreaks:

a) adequate stocks of outbreak and sample collection kits/SOPs (standard operating procedures) at the local level;
b) cold chain and shipping arrangements that are linked to the national specimen transportation network;
c) field or mobile laboratories that are set up to address the routine and outbreak laboratory testing needs of crisis-affected populations;
d) existing laboratories at national and sub-national levels that have been strengthened to address the extra demands of crisis-affected populations;
e) referral laboratories (national and international) should be identified to facilitate laboratory confirmation, antibiotic susceptibility testing and quality control;

f) the existing IDSR laboratory and case investigation forms should be used for routine collection of laboratory aggregate and case-based data;
g) harmonization of laboratory reporting between surveillance and laboratory systems for timely dissemination of results.

10.5.4. Methods of data collection

Data should be collected from reportable signals that are generated from data sources such as HF community health workers and other sources of disease surveillance data. The data will be recorded in the electronic platforms DHIS2 or EWARS according to the nature of the case. Health workers should observe the following standards:

a) strict adherence to the case definitions while collecting data on the disease, conditions or public health event;
b) each patient should be assigned one main diagnosis and should be counted once.

The data collection will entail the following:

a) the CSA logbook for recording signals detected;
b) posters with descriptive lists of signals;
c) the specific forms found in the EWARS electronic platform;
d) phones with the EWARS application installed.

10.5.5. Methods of reporting and transmitting data

Humanitarian crises tend to disrupt existing national disease surveillance platforms for transmitting data. In the same way, crisis-affected populations may have additional public health needs beyond the ones established through the routine IDSR. Flexibility should be exercised to update the existing IDSR/NHIS reporting tools to capture diseases, conditions and public health events unique to crisis-affected populations. Consequently, the existing IDSR/NHISpaper-based tools and/or electronic reporting platforms should be updated to capture such additional diseases, condition and public health events

The reporting platforms should provide for the following reporting time lines:

a) immediate reporting of epidemic-prone disease signal;
b) daily reporting of aggregated and/or case-based data on priority diseases, conditions, public health events during the acute phase of the crisis and after a new outbreak is confirmed;
c) weekly reporting of aggregated data on priority diseases;

d) weekly mortality line listing should be updated with community and HF deaths.

10.5.6. Data analysis and interpretation

The principles of data analysis utilised as part of the routine IDSR should be used in crisis-affected populations. Analysis on aggregated data is therefore conducted to document and describe disease trends and crossing of thresholds. The data is also used to calculate ratios and rates.

Before embarking on any analysis, data validation and cleaning should be undertaken for missing entries, outliers, and duplicates. The basic analysis entails case/death descriptive analysis by time, person, and place.

Morbidity indicators in crisis-affected populations include:

a) absolute counts of cases and deaths by priority disease;
b) incidence of disease (new cases by week divided by the total population) with a graph to show trends recorded in recent weeks. This can be disaggregated by location and the individual's characteristics;
c) proportional morbidity (new cases of disease in a week divided by the new consultations in the week);
d) case fatality ratio– the proportion of cases that die from a specific disease;
e) attack rate during outbreaks as the cumulative incidence of outbreak disease in a population over a period of time.

The mortality indicators in crisis-affected populations:

It is critical that mortality rates (CMR and U5MR) are monitored for crisis-affected populations to ensure that rates exceeding the established emergency threshold are detected and responded to promptly.

a) Crude Mortality Rate is the mortality of a defined population, over a specified period of time in a given territory. Crude Mortality Rate (CMR), as deaths per 10 000 per day, is calculated as the number of deaths divided by the population present during the period and the total number of days over which the deaths were reported.
b) Under-five Mortality Rate (U5MR), as deaths per 10 000 per day is calculated as the number of deaths in under-fives divided by the population of under-fives present during the period and the total number of days over which the deaths were reported.

The existing electronic platforms offer the advantage of automated analysis for both routine and case-based outbreak data thus saving time and ensuring analysed data is available in real time to inform disease surveillance and outbreak response decisions at all levels.

10.5.7. Feedback and dissemination

Feedback is critical for ensuring full engagement of the stakeholders. Feedback helps to provide information on disease control efforts. Information providers must be included in the feedback section. Weekly surveillance summaries, bulletins and presentations should be presented and reviewed during:

a) weekly IDSR coordination meetings within the Department for the Control of Disease, Epidemics and Pandemics (DLMEP);
b) health and Water, Sanitation and Hygiene (WASH) (and other relevant) cluster meetings;
c) other relevant disease control meetings;
d) extension of all reporting units and stakeholders to inform public health response decisions;
e) during outbreaks, regular situation reports should be shared with all the stakeholders involved.

The existing electronic platforms have the advantage of producing automated disease surveillance and outbreak bulletins or situation reports to inform disease surveillance and outbreak response decisions at all levels.

10.5.8. Support functions for surveillance in crisis-affected populations

To optimise the functioning of disease surveillance and outbreak response in crisis-affected populations, it is critical that IDSR guidelines be adapted and used to improve access to the following:

a) surveillance and outbreak response guidelines at all levels;
b) training of health workers, surveillance focal persons and rapid response teams on surveillance functions including outbreak preparedness, investigation and response;
c) support to communication (computers, phones, internet connectivity) based on local context and surveillance needs;
d) regular supervision and support to enhance surveillance functions at all levels;
e) periodic evaluation to improve the performance of the surveillance system (refer to framework for evaluating surveillance systems).

10.5.9. Outbreak preparedness

Outbreak preparedness is paramount given the heightened risk of disease outbreaks in crisis-affected populations. Preparedness efforts should, as much as possible, is integrated in the existing national IDSR framework at national and sub-national levels with the Ministry of Public Health leading the efforts and supported by WHO and partners. However, during acute or complex emergencies where the capacities of the Ministry of Health are greatly compromised or diminished, WHO, working in collaboration with the health cluster partners, should take lead to enhance outbreak preparedness (WHO, 2012). The key preparedness efforts in crisis-affected populations should entail the following:

a) strengthening existing or forming new multisector outbreak control teams at national and sub national levels, with roles and responsibilities designated for each team member;
b) updating existing or developing new outbreak prevention and response plans that incorporate risks unique to crisis-affected populations;
c) development or updating (if necessary) of standard line-list forms for data collection during an outbreak;
d) development and distribution of standard treatment protocols for key diseases, with strategies for staff training;
e) calculation of potential attack rates for epidemic-prone diseases, where possible;
f) pre-positioning stocks of essential treatment supplies to initiate outbreak control (e.g. oral rehydration salts, intravenous fluids, vaccination material, personal protective equipment, transport media for samples, water purification supplies, disinfectants, spray pumps and information leaflets on preventive measures for health staff or the community;
g) procurement of laboratory sample collection for the priority diseases, and identification of a competent laboratory for confirmation of cases;
h) identifying potential sites for isolation and adequate treatment of patients, or for extra capacity, in the event of a surge in cases (e.g. a cholera treatment center, COVID-19 crisis management center, and a center for measles management);
i) implementing relevant prevention measures based on the risk assessment of diseases (e.g. measles and cholera vaccination, indoor-residual spraying of dwellings and distribution of long-lasting insecticide-treated nets to prevent outbreaks of measles, cholera, and malaria);

j) scaling up preparedness and response efforts/ activating rapid response teams at the points of entry (PoE).

10.5.9.1. Alert and outbreak thresholds

The following thresholds are used in crisis-affected populations:

a) Assess the severity of the humanitarian crisis based on the crude mortality rate (CMR) and under five mortality rate (U5MR).
   - The crude mortality rate threshold should be less than 1 death per 10 000 people per day.
   - The under-five mortality rate threshold should be less than 2 deaths per 10 000 people per day.

b) Signal system for detecting possible outbreaks based on doubling of weekly incidence compared to the weekly average recorded 2 or 3 weeks earlier.

c) Detection of a case of potentially severe epidemic-prone disease like measles, polio, cholera, viral haemorrhagic fevers (VHF) or meningitis based on the IDSR signal and action thresholds specific to crisis-affected populations.

Once the thresholds are exceeded, verifications, investigations and response should be instituted promptly to prevent further morbidity and mortality.

10.5.9.2. Alert verification

To minimize morbidity and mortality, signal verification should start immediately once the signal is received by sub-national and national surveillance focal persons (WHO, 2012). The verification can be done by telephone or site visit and can include the collection of information on:

a) cases based on standard case definitions;
b) symptoms and signs (consider differential diagnosis);
c) date of onset of symptoms of the first and the most recently detected cases;
d) place and date seen or admitted at the health facility;
e) age, sex and vaccination status of patients, where relevant;
f) place of residence at onset of illness;
g) where cases are occurring (community-level data);
h) geographical, personal and time relationships between cases;
i) prompt laboratory investigation of samples from suspected cases;

j) outcomes including, for example, deaths, case management details and the health-care staff affected.

10.5.10. Outbreak investigation

Outbreak investigation involves determining the cause of an outbreak and who is at risk so that control measures can be implemented. The main objective of an outbreak investigation is to control the outbreak and thus reduce morbidity and mortality. The investigation should begin as soon as a signal is detected and has been verified.

In acute and complex emergencies, dedicated and trained teams will be identified to undertake the investigations, which should follow existing IDSR outbreak investigation guidelines that have been adapted to address the unique needs of crisis-affected populations.

10.5.11. Outbreak response

Outbreak response should follow the existing national IDSR framework at national and sub national levels. However, during acute or complex emergencies where the capacities of the Ministry of Public Health are greatly compromised or diminished, WHO, working with the health cluster partners should take lead in coordinating and implementing outbreak response activities.

The additional risks in crisis-affected populations will demand strengthening existing or the putting in place of new multisector outbreak control teams at national and sub national levels, with roles and responsibilities designated for each team member as set out in the IDSR outbreak response guidelines. Health, Water, Sanitation and Hygiene (WASH) and other relevant cluster partners should support outbreak response activities in crisis-affected populations.

10.6. Exit Strategy

During the recovery phase of the crisis, the Ministry of Public Health should work with WHO and partners to re-establish all the surveillance structures and focal points in the crisis-affected populations. Countries should aim at conducting an evaluation to assess what happened, why it happened and draw lessons learnt and gaps identified in order to make recommendations for the prevention of future occurrences.

Section 11: Summary guidelines for specific priority diseases

This section provides summary guidelines for each of the priority diseases and events. These include:

- Identification of surveillance goals and objectives,
- Surveillance data analysis and interpretation,
- Measures to respond to alerts when alerts have crossed action thresholds,
- Proper usage of laboratory tools and proper analysis of lab results

Introduction

This section provides summary guidelines for each of the priority diseases, events and conditions targeted for surveillance by WHO/AFRO. This section provides the following summary guidelines for each of the priority diseases and events:

- Take action to respond to alerts and action thresholds,;
- Identify surveillance goals and objectives;
- Conduct surveillance data analysis and interpretation,
- Prepare to use the district analysis workbook or database
- Propose standardised case definitions for reporting priority diseases, conditions and public health events.

In adapting these guidelines, each country should establish a list of priority diseases, conditions and public health events based on the local epidemiological situation. The list of priority diseases may vary from country to country, depending on national policies and resources.

This section provides a quick referral system. For more information, refer to references listed under Summary. The table below shows how information is organised in this section.

Priority Disease or Event for IDSR in Cameroon

Background

In this sub-section, you will find general information about :

- The disease or event, the causative agents, geographic areas affected and other epidemiological information.
- Transmission routes such as person-to-person, unprotected contact with infectious body fluids or contaminated materials, vector-borne, and so on.
- Why the disease/event is a priority for surveillance. For instance, because the disease/event is responsible for high rates of mortality, morbidity and disability in Cameroon.
- General and specific risk factors in the country.
- Any additional background information that might serve the district health management team.

Surveillance Goal

This sub-section states how the surveillance data is used for taking appropriate response measures.

Standard case definition

Suspected case: A definition is provided for identifying suspected cases or outbreak of this disease or event.

Probable case: A definition is provided for a suspected case with epidemiological link to a confirmed case or an outbreak

Confirmed case: A definition is provided for classifying a case as confirmed through laboratory diagnostic testing.

Responding to alert threshold

Some diseases, conditions or events have programme specific thresholds for alerting the HFor district to a potential problem.

For *epidemic-prone diseases, diseases targeted for elimination or eradication, or public health events of international concern*, a single case is a suspected outbreak and requires immediate reporting followed by patient treatment, collection of specimens for case confirmation, and investigation of the case to determine the risk factors and potential interventions.

For other priority diseases of public health importance, an outbreak or event is suspected when there is any unusual cluster, morbidity, or increase in the number of cases when compared with previous periods. This should prompt a response such as investigating what might have caused the unusual events. If laboratory confirmation is indicated, specimens should be collected for laboratory confirmation.

Responding to outbreak or action threshold

For epidemic-prone diseases, diseases targeted for elimination or eradication, or public health events of international concern, a confirmed case should trigger a response such as conducting an emergency immunization activity, enhancing access to safe drinking water, community education campaigns, and improving case management.

For other priority diseases of public health importance, a confirmed outbreak should prompt an appropriate response such as improving coverage for specified immunizations, strengthening case management, providing information, education and communication about preventing and controlling the disease, and so on.

Analysing and Interpreting data

This sub-section contains generic information about the minimum data elements to collect, analyse and interpret. The key points to consider for interpreting the data and specific elements for analysis are also stated (time, place, and personal information).

Laboratory confirmation

In this sub-section, guidelines on laboratory confirmation are provided including: relevant diagnostic tests, how to collect, store and transport the specimens needed for lab confirmation, and information on the results of laboratory work.

References

For more information, relevant references are provided for each disease Most are available on WHO website.

Acute Viral Haemorrhagic Fever Syndrome

Background

Acute haemorrhagic fever syndromes can be attributable to Ebola and Marburg viral diseases (*filoviridae)*; Lassa fever (*arenaviridae*), Rift Valley fever (RVF) and Crimean-Congo Haemorrhagic fever (CCHF) (*bunyaviridae*); dengue haemorrhagic fever (DHF)) and yellow fever (*flaviviridae*) and other epidemic-prone viral, bacterial or rickettsial diseases.

All cases of acute haemorrhagic fever syndrome whether single or in clusters should be immediately notified without waiting for the causal agent to be identified

Surveillance Goal

Early detection of acute viral haemorrhagic fever syndrome cases and outbreaks, rapid investigation, and early laboratory verification of the cause of all suspected cases. Investigation of all suspected cases with contact tracing. During outbreaks, most infected patients do not show haemorrhagic symptoms and a specific case definition according to the suspected or confirmed disease should be used (e.g. case definitions for Ebola-Marburg, Crimean-Congo, Lassa, Dengue haemorrhagic fever and Yellow Fever).

Standard case definition

Cas communautaire : Tout décès inexpliqué ou soudain survenant dans les zones frontalières à un pays en épidémie OU toute personne présentant des saignements des orifices (nez, bouche, anus…) ou muqueuses (yeux…) ou des vomissements avec du sang, Diarrhée ou urine avec du sang OU toute personne avec un début brusque de fièvre ne répondant pas au traitement habituel.

Cas Suspect : Toute personne, vivante ou décédée, résidant ou de passage dans des zones en épidémies

- présentant ou ayant présenté une fièvre aigue (> 38 ° C), et (ayant été en contact avec une personne présentant des saignements, une fatigue et des céphalées ou ayant une notion de voyage en dans une zone en épidémie)
- OU : toute personne présentant une fièvre aigue élevée à début brutal et au moins trois des symptômes suivants : maux de tête, vomissement, anorexie/perte d'appetit, diarrhée, fatigue intense, douleurs abdominales, douleurs musculaires ou articulaires, difficulté à avaler, difficulté à respirer, hoquet, eruption cutanée
- OU : toute personne présentant des saignements inexpliqués
- OU : toute personne morte subitement et dont le décès ext inexpliqué

Cas probable : Tout cas suspect évalué par un clinicien ;

> OU tout cas suspect décédé (et pour lequel il n'a pas été possible d'obtenir des échantillons biologiques pour confirmation au laboratoire) ayant un lien épidémiologique avec un cas confirmé.

Cas confirmé : tout cas suspect ou probable avec un résultat de laboratoire positif. Les cas confirmés au laboratoire doivent être positifs soit pour l'antigène du virus, soit pour l'ARN viral détecté par transcription inverse suivie de la réaction en chaîne par polymérase (RT- PCR), soit pour les anticorps IgM dirigés contre Marburg ou Ebola.

Cas alerte : toute personne présentant une fièvre élevée à début brutal qui ne répond à aucun traitement des causes habituelles de fièvre dans la region; OU toute personne ayant présenté une hémorragie ou une diarrhée sanglante ou une hématurie ; OU Toute personne morte subitement.

***Note**: During an outbreak, case definitions may be changed to correspond to the local event. It is important to note that during outbreaks, most cases might not show haemorrhagic manifestation; a proper history taking is crucial*

Responding to alert threshold

If a single case is suspected:

- Report case-based information immediately to the appropriate levels.
- Suspected cases should be isolated from other patients/people and strict infection prevention procedures should be implemented. Standard precautions should be enhanced throughout the healthcare setting and in communities.
- Treat and manage the patient with supportive care.
- Collect the appropriate specimen while observing strict infection prevention and control procedures to confirm the case.
- Complete a laboratory request form, use triple packaging of the specimens (see detailed SOPs for triple packaging) and properly label the containers to warn of a potential laboratory biosafety risk.
- Ensure case-contact tracing and follow-up and conduct active case finding (See detailed SOPs for contact tracing and follow up).
- Begin or enhance deaths reporting and surveillance; as well as screening procedures for fever and VHF related symptoms

Respond to epidemic or action threshold

If a single case is confirmed:

- Maintain strict VHF infection prevention and control (IPC) practices* throughout the outbreak.
- Mobilize the community for early detection and management; conduct community education about how the disease is transmitted and how to implement IPC in the home care setting and during funerals and burials. Consider social distancing strategies.
- Conduct case-contact follow-up and active search for additional cases that may not come to health services.
- Request additional help from other levels as needed.
- Establish an isolation ward or treatment centre to handle additional cases that may come to the health centre and ensure strict IPC measures to avoid transmission in health care settings.
- Isolate suspected cases and treat for more common conditions with similar symptoms, which might include malaria, typhoid, louse borne typhus, relapsing fever or leptospirosis. Ensure a barrier is instituted between suspected and confirmed cases.
- Provide psychosocial support for the family, community and healthcare workers.
- Consider quarantine for high-risk contacts with home support during the incubation period and ensure daily follow up of their movements.
- There are promising vaccine candidates under development for some VHDs that might be useful in the event of outbreak in a ring vaccination approach and for health care workers.
- Treat conservatively the symptoms which might be presented; severe cases require intensive additional support and care; if dehydrated, ensure fluid replacement with fluids that contain electrolytes.
- A range of potential treatment options including blood products, immune therapies, and drug therapies are currently being evaluated

Analysing and Interpreting data

Time: Graph cases and deaths daily/weekly. Draw up an outbreak curve during the outbreaks.
Place: Map locations of cases' homes and work sites. If you have a GPS, this will add to exactly locate the cases; and identify contacts.
Personal information: Implement immediate case-based reporting of cases and deaths. Analyse age and sex distribution. Assess risk factors and plan outbreak response interventions accordingly.

Laboratory confirmation	
Diagnostic tests	Laboratory confirmed cases must test positive for the virus antigen, either by detection of virus RNA by reverse transcriptase-polymerase chain reaction (RT- PCR), or by detection of IgM antibodies directed against Ebola/Marburg, CCHF, Lassa or West Nile Fever
Specimens	For ELISA: Whole blood, serum or plasma For RT-PCR: Whole blood or blood clot, serum/plasma or tissue For immunoshisto-chemistry: Skin or tissue specimens from fatal cases **Note**: RDTs theoretically can be performed in any HF and without additional equipment. However, use of an RDT may result in both false positive and false negative test results. A nucleic acid based (e.g., PCR) diagnostic assay, such as GeneXpert, must be used to confirm the RDT result. Recent guidance from WHO recommends that antigen detection RDTs for VHDs have no role in the routine management of VHDs in settings where PCR testing is available. However, they may have utility in settings without laboratory infrastructure and where rapid sample transport to a diagnostic laboratory is not possible, if their benefits and limitations are understood.
When to collect the specimen	Collect specimens from all suspected patients. All cases must be investigated, with contact tracing. Blood samples and appropriate clinical specimens must be collected to confirm a diagnosis as rapidly as possible
How to prepare, store, and transport the specimen	HANDLE AND TRANSPORT SPECIMENS FROM SUSPECTED VHF PATIENTS WITH EXTREME CAUTION. WEAR PROTECTIVE CLOTHING AND USE FULL PERSONAL PROTECTIVE EQUIPMENT (PPE) *For ELISA or PCR:* ▪ Refrigerate serum or clot ▪ Freeze (-20C or colder) tissue specimens for virus isolation *For Immunohistochemistry:* ▪ Fix skin snip specimen in formalin. Specimens can be stored up to 6 weeks. Once it is in formalin, the specimen is not infectious. ▪ Store at room temperature. Formalin-fixed specimens may be transported at room temperature.
Results	Diagnostic services for VHF are not routinely available. Advance arrangements are usually required for VHF diagnostic services. Contact the national competent authority or WHO.
References	

- Interim Infection Control Recommendations for Care of Patients with Suspected or Confirmed Filovirus (Ebola, Marburg) Haemorrhagic Fever. BDP/EPR/WHO. Geneva, March 2008.
- World Health Organisation (1998) *Contrôle de l'infection en cas de fièvre hémorragique virale en milieu hospitalier africain*. OMS/EMC
- WO Recommended Surveillance Standards WHO/CDS/CSR/ISR/99.2
- WHO Guidelines for preparedness qnd response to outbreaks: Ebola Haemorrhagic Virus (FHE). WHO/EMC/DIS/97.7.
- *Contrôle de l'infection en cas de fièvre hémorragique virale en milieu hospitalier africain* WHO/EMC/ESR/98.2
- Viral Infections of Humans; Epidemiology and Control. 1989. Evans, A.S. (ed). Plenum Medical Book Company, New York

Acute and Chronic Viral Hepatitis

Background

Viral hepatitis A and viral hepatitis E

- Enterically transmitted HAV and HEV are a worldwide public health problem.
- Common sources of outbreaks have been related to contaminated water and to contamination via infected food handlers.
- In general, both HAV and HEV are self-limiting viral infections and case fatality is low (0.1 – 0.3%). Women in the third trimester of pregnancy are especially susceptible to fulminant HEV disease.
- Both HAV and HEV are transmitted via the faecal-oral route.
- Prevention and control measures for hepatitis A and hepatitis E include adequate supplies of safe-drinking water and improvement of sanitary and hygienic practices to eliminate faecal contamination of food and water.

Viral hepatitis B and viral hepatitis C:

- Estimates indicate that worldwide, there are 257 million carriers of hepatitis B virus and 71 million carriers of hepatitis C virus.
- Acute B and C Hepatitis may be anicteric thus not recognised, but acute outbreaks are rare.
- Chronic infection and severe sequelae occur with hepatitis B – an estimated 15% to 25% of chronically infected persons will die prematurely of either cirrhosis or hepatocellular carcinoma. Chronic infection is common in hepatitis C and 5% to 20% of those infected with HCV may develop cirrhosis. There seems to be a connection between HCV infection and hepatocellular carcinoma.
- Hepatitis B is transmitted by percutaneous or per mucosal exposure to blood or other infectious body fluids. In most countries where HBV is highly endemic, most infections occur during infancy and early childhood or by perinatal transmission from mother to infant. Major modes of transmission include nosocomial exposure (transfusions, unsafe injection practices), shared needles or syringes among injecting drug users, shared personal items (e.g., communally used razors and toothbrushes) and sexual contact with an infected person.
- Hepatitis C is transmitted by parenteral exposure to blood and plasma derivatives. It is found in highest concentrations in blood. The major causes of HCV infection worldwide are use of unscreened blood transfusions and re-use of needles and syringes that have not been adequately sterilised.
- Prevention and control measures for hepatitis B and C include transfusion safety, safe and appropriate use of injections and vaccination (hepatitis B). Screening and early treatment are effective secondary prevention modes
- To address the increasing burden of viral hepatitis, in 2016, African member states adopted the document entitled *Prevention, Care and Treatment of viral hepatitis in the African Region: Framework for action 2016-2020*

There is no specific treatment for acute viral hepatitis.

Surveillance Goal

Acute viral Hepatitis

- Detect hepatitis outbreaks.
- Identify areas/populations at high risk to target prevention and control measures.
- Estimate the burden of disease.
- If countrywide surveillance is not possible, surveillance in sentinel areas or hospitals may provide useful information on potential sources of infection.

Chronic viral hepatitis

- Estimate the burden of chronic viral hepatitis B and C
- Measure the impact of control measures/treatment on mortality reduction. To this effect, data is captured on persons diagnosed with hepatocellular carcinoma or cirrhosis

Standard case definition

I) Acute viral Hepatitis

Suspected case: Any person with discrete onset of an acute illness with signs/symptoms of;

(in) Acute infectious illness (e.g. fever, malaise, fatigue) and (ii) Liver damage (e.g. anorexia, nausea, jaundice, dark coloured urine, right upper quadrant tenderness of body), **AND/OR**

(iii) Raised alanine aminotransferase (ALT) levels more than ten times the upper limit of normal values

Confirmed case: A suspected case that is laboratory confirmed by virus specific biomarkers

- **Acute Hepatitis A**: IgM anti-HAV positive or positive RNA HAV
- **Acute Hepatitis B**: detection of the surface antigen of the VHB virus (HBsAg) AND presence of the essential antigen of the HBV virus (anti-HBc) class IgM, detection of DNA HBV.
- **Acute Hepatitis C:** presence of RNA HCV (viral load), presence of the essential antigen of the HCV (if available) and detection of anti-HCV IgM. Absence of acute hepatitis A markers (IgM anti-HAV) and Hepatits E (IgM anti-HEV).
- **Acute Hepatitis D:** detection of the antigen HBsAg (or presence of IgM anti-HBc) and presence of anti-HDV antibodies (usually IgM), and detection of RNA HDV (infection with HDV occurs ONLY as infection or superinfection of hepatitis B).
- **Acute Hepatitis E**: IgM anti-HEV positive

II) Chronic Viral Hepatitis Case definition (HBV and HCV):

Chronic Hepatitis B:

- Persistence of HBsAg for at least 6 months following acute infection indicates chronic infection by hepatitis B
- Presence of HBsAg and anti-HBc antibodies (usually IgG) in asymptomatic persons or patients suffering from chronic hepatic disease and/or hepatic tumour is an indication of chronic infection with hepatitis B virus.

Chronic Hepatitis C:·

- Hepatitis C virus RNA present in a person with antibodies against hepatitis C (usually IgG)
- HCV RNA positive OR HCV essential antigen positive

Note: Antibody detection (i.e HVC Ab positive) cannot help differentiate between acute, chronic infection and past infection.

Surveillance for the detection of chronic hepatitis B and C

- Conduct seroprevalence tests of HBsAg antigens and anti-HCV antibodies within the general population and in all patients with chronic liver disease;
- These may involve:
  - Approaches to screen the general population using the possibilities or community-based screening programmes or in existing HFs, namely antenatal services and in HIV/TB management structures.
  - Periodic surveys on seroprevalence within the population using serological markers of viral hepatitis B and C.
  - Patients presenting at the HF with a chronic disease or tumour in the liver.

Responding to alert threshold

If hepatitis cases are suspected:

- Report case-based information to the appropriate authorities (the Case-based reporting form is available in the appendices).
- Collect specimens and send to laboratory to identify the aetiology of the disease
- Treat and manage acute viral hepatitis patient(s) with supportive care as required.

Respond to the outbreak and action threshold

If hepatitis cases are confirmed

- Determine the mode of transmission
- Identify the population exposed to the risk of infection
- Eliminate common source(s) of infection
- Implement appropriate prevention and control interventions
- Patients with chronic viral hepatitis should be referred to tertiary specialist level centres or designated treatment centres for treatment, care and follow-up

Analysing and Interpreting data

Time: Analyse suspected and confirmed cases per week and month. Graph cases and deaths weekly and monthly. Draw an outbreak curve during outbreaks.

Place: Plot location of case households.

Personal information: Analyse by age and gender. Assess risk factors to plan and monitor prevention and control measures. Calculate the Incidence Rate for Acute Viral Hepatitis cases and Prevalence Rate for Chronic Viral Hepatitis B and C cases and Case Fatality Rate

Laboratory confirmation

Diagnostic tests	**Hepatitis A**: IgM anti-HAV positive **Hepatitis B**: positive for Hepatitis B surface antigen (HbsAg) or IgM anti-HBc positive **Hepatitis C**: Anti-HCV positive **Hepatitis D**: HBsAg positive or IgM anti-HBc positive plus anti-HDV positive (only as co-infection or super-infection of hepatitis B) **Hepatitis E**: IgM anti-HEV positive and/or IgG anti-HEV positive
Specimen	Serum, whole blood or stool (for hepatitis A and E viruses)

When to collect the specimen	Specimens should be collected from the suspected patient. IgM anti-HAV becomes detectable 5-10 days after exposure. HBsAg can be detected in serum from several weeks before onset of symptoms to days, weeks or months after onset. It persists in chronic infections. IgM anti-HBc positive usually disappears within 6 months.
How to prepare, store, and transport the specimen	Apply universal precautions to minimize exposure to sharps and any body fluid. ▪ Collect 5-10 ml of venous blood. ▪ Let clot retract for 30 to 60 minutes at room temperature or centrifuge to separate serum from red blood cells. ▪ Aseptically pour off serum into sterile, screw capped tubes. ▪ Store serum at 4°C. ▪ For storage >5 days, samples are frozen at -20°C Transport serum samples using appropriate packaging to prevent breakage or leakage
Results	Results are usually available within one to 3 days from arrival in the laboratory
Laboratory Tests for Chronic Viral Hepatitis	

I) Chronic Viral Hepatitis B (HBV)

Basic initial laboratory investigations:

The following laboratory tests should be requested after thorough history and physical examination in HBsAg positive individuals:

a. Establish chronicity: persistence of the antigen HbsAg for more than six months following an acute infection or presence of a chronic disease or a tumour in the liver.
b. Determine the status of the antigen HBe and antibody anti-HBe: antigen HBe and antibody anti-HBe.
c. Establish inflammatory activity: Liver Function Tests (LFT),
d. Determine the level of viraemia – viral load: HBV DNA
e. Screen for chronic liver disease or complications using a clinical test of stigmata of a chronic liver disease, abdominal ultrasound, coagulation profile or Full blood count
f. Screen for other co-infections: HCV Ab, HIV, HDV if available
g. Supportive investigation: determine blood urea and creatinine
h. Consider liver biopsy or fibro-scan if indicated

II) Chronic Viral Hepatitis C (HCV)

Initial investigations for HCV Patients:

a. The screening test for HCV is HCV Ab test. Unlike HBV testing, a positive HCV screening test (anti-HCV Ab) does not equate to active infection. Also, the HCV testing often provides several false positive results.
b. The following steps are recommended to establish active infection;
   - Confirm HCV Ab testing using ELISA
   - Confirm active infection using RNA testing; detectable RNA confirms active infection; if RNA is undetectable, no further testing is indicated. This result indicates a past infection or false positive serological test.
   - Further testing for RNA positive cases include LFT, abdominal ultrasound, genotyping, FBC, alpha fetoprotein, BUE and Cr;
   - Screen for co-infections (HIV, HBV)
   - Assess degree of inflammation and fibrosis by conducting the following test:
     - Aspartate aminotransferase-to-platelet ratio index (APRI) Score
     - Fibrosis-4 (FIB4) score (the score uses a combination of age, number of platelets, aspartate aminotransférase (AST) and alanine aminotransférase (ALT) to obtain the score).
     - Fibroscan

Liver biopsy is the gold standard.

References

- *WHO Recommended Strategies for Prevention and Control of Communicable Diseases*; WHO/CDS/CPE/SMT/2001.13
- *WHO Recommended Surveillance Standards* WHO/CDS/CSR/ISR/99.2
- WHO Fact Sheet No 328, Hepatitis A, revised in May 2008
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- WHO Fact Sheet No 204, Hepatitis B, revised in August
- WHO Fact Sheet No 2008 164, Hepatitis C.

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- US Centers for Disease Control and Prevention http://www.cdc.gov/hepatitis/
- Control of Communicable Diseases Manual, 18th edition
- World Health Organisation... *WHO Guidelines for the prevention, care and treatment of persons with chronic hepatitis B infection*; March 2015. Geneva, March 2015
- World Health Organisation. *WHO Guidelines for the prevention, care and treatment of persons with chronic hepatitis B infection*; March 2015. Geneva, April 2016
- WHO Global Report on Hepatitis, 2017
- WHO Guidelines on hepatitis B and C testing, February 2017

Adverse Events Following Immunization (AEFI)

Background

AEFI surveillance serves as a quality assurance mechanism for national immunization programmes and, in most countries, it must be continuously strengthened. There are five possible causes of AEFIs: 1) Reaction to a vaccine product: 2) Reaction to a defect in vaccine quality; 3) Reaction to an immunization error; 4) a stress-related reaction to immunization; and 5) a coincidental health issue. It is important to identify and treat patients with AEFIs. Serious AEFIs should equally be investigated thoroughly to determine their cause.

Surveillance Goal

To monitor the safety of vaccines and immunization after homologation and provide answers to safety concerns.

Standard case definition

An adverse event following immunization is an untoward medical incident occurring after immunization and which does not necessarily have a direct causal link with the administration of the vaccine. The adverse event may equally be a negative or unintentional sign, any abnormal lab result, a symptom or a disease.

Reaction to mild AEFIs

If a case is identified

- Treat the patient
- Inform the parents and the community that an AEFI may occur after the administration of a vaccine.
- Reply to rumours or requests for information by the public
- Fill in the Case-based investigation form.

Reaction to serious AEFIs

An AEFI is said to be serious if it causes death, endangers life, requires hospitalization or extending a current hospitalization time, causes a disability or persistent or significant incapacity, causes an anomaly or congenital defect, or requires surgery to prevent a deficiency or permanent damages. Before any AEFI case:

- Treat the patient
- Inform the parents and the community that an AEFI may occur after the administration of a vaccine.
- Reply to rumours or requests for information by the public
- Fill in the Case Reporting Form and send it immediately to initiate an investigation.
- If a problem related to a product or an immunization error is identified, take corrective measures to avoid the occurrence of another AEFI for the same reason.

Analysing and Interpreting data

Determine the cause of the event. Beware of very psychological illness if a number of school-aged or older individuals are involved at the same time.

References

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- *Global Manual on Surveillance of Adverse Events Following Immunization* *http://www.who.int/vaccine_safety/publications/Global_Manual_revised_12102015.pdf?ua=1*
- *Definition and application of terms for vaccine pharmacovigilance. Report of CIOMS/WHO Working Group on Vaccine Pharmacovigilance. Geneva, Council for International Organizations of Medical Sciences, 2012*

Anthrax

Background

- Anthrax is a widespread zoonotic disease caused by the spore-forming bacterium *Bacillus anthracis*, a Gram-positive rod-shaped bacterium. It is transmitted from infected domestic livestock (cattle, sheep, goats, buffaloes, pigs and others) or wild game animals to humans by direct contact or indirect contact with animals or their products.
- The incubation period typically ranges from 1 to 7 days, but may be longer (up to two to three weeks for cutaneous anthrax and up to 42 days for inhalation anthrax). Persons exposed to occupational hazards include those handling infected carcasses and those employed in the processing of bones, hides, wool and other animal products. People may also become infected by handling or consuming meat from animals that are sick with or have died of the disease. Biting flies have been reported to transmit the disease from infected animals to humans however how readily or often this occurs is unknown.
- Human anthrax is a serious problem in several countries and can cause explosive outbreaks (especially the gastrointestinal form that is contracted from eating infected meat).While pulmonary (inhalation) anthrax is mainly occupational, the threat of biological warfare attacks should not be forgotten. Anthrax has a serious impact on the trade of animal products.
- The control of anthrax is based on its prevention in livestock. Programmes based only on prevention in humans are costly and likely to be ineffective except for those industrially exposed.
- There is an effective vaccine for those persons considered at risk for occupational exposure, and successful vaccines are used for livestock, particularly for herds with ongoing exposure to contaminated soil or vegetation.

Surveillance Goal

- To detect outbreaks.
- To monitor control and prevention programmes

Standard case definition

Suspected Case:

Any person with acute onset characterized by several clinical forms which are:

(e) ***Cutaneous form****:* Any person with skin lesion evolving over 1 to 6 days from a papular through a vesicular stage, to a depressed black eschar invariably accompanied by oedema that may be mild to extensive

(f) ***Gastro-intestinal*****:** Any person with abdominal distress characterized by nausea, vomiting, anorexia and followed by fever

(g) ***Pulmonary (inhalation)*****:** any person with brief prodrome resembling acute viral respiratory illness, followed by rapid onset of hypoxia, dyspnoea and high temperature, with X-ray evidence of mediastinal widening

(h) ***Meningeal*****:** Any person with acute onset of high fever possibly with convulsions, loss of consciousness, meningeal signs and symptoms; commonly noted in all systemic infections, but may present without any other clinical symptoms of anthrax.

AND has an epidemiological link to confirmed or suspected animal cases or contaminated animal products

Confirmed case

A confirmed case of anthrax in a human can be defined as a clinically compatible case of cutaneous, inhalational or gastrointestinal illness that is laboratory-confirmed by:

(c) isolation of *B. anthracis* from an affected tissue or site; or

(d) Other laboratory evidence of *B. anthracis* infection based on at least two supportive laboratory tests

Note: it is not always possible to reveal the presence of *B. anthracis* din clinical specimens when the patient had been treated with antimicrobial agents.

Responding to alert threshold

If a single case is suspected:

- Report case-based information immediately to the appropriate levels (public health sector and animal health sector)
- Use standard barrier precautions for all forms. Use protective equipment and clothing (gloves, gowns, face shields), and respiratory protection if there is a risk of aerosols. Disinfect and dress any cuts and abrasion before putting on the PPE.
- Perform environmental cleaning (disinfection) with hypochlorite
- Treat and manage the patient with supportive care and using antibiotics such as Penicillin V, procaine penicillin (uncomplicated cases), or penicillin G (severe cases)
- Collect specimens safely to confirm the case.
- Conduct joint (public health and animal health sectors) investigation of cases/deaths
- Vaccinate exported/imported animals
- In humans, selective preventive vaccination may be considered in case of occupational exposure. It is important to take thorough history to determine if there is occupational exposure, as unnecessary administration of antibiotics might led to antimicrobial resistance (AMR)

Responding to outbreak or action threshold

If a single case is confirmed:

- Standard infection control precautions are sufficient and should be used when managing the patients
- Particular attention should be paid to body fluid spills that should be managed by the usual methods for cleaning and decontamination of any body fluid spills. This should be done promptly and thoroughly because organisms which remain on surfaces may form spores which are infectious
- As is usual practice, personal protective equipment should be used in situations where there is potential for splashes and inoculation injuries. Any incidents should be reported immediately
- Mobilize the community for early detection and care.
- Bury or cremate (if practiced) dead bodies (humans and animals) properly
- Conduct community education about the confirmed case, how the disease is transmitted, and how to use infection control measures for home care.
- Conduct active searches for additional cases that may not come to the health units (old people or small children, for example) and provide information about prevention in the home and when to seek care.

Request additional help from other levels as needed.

Analysing and Interpreting data

Time: Graphs of number of suspected / probable / confirmed cases by date.
Place: Map out suspected and confirmed human and animal cases by geographical area (district)
personal information: Table showing the number of suspected / probable / confirmed cases by date, age and sex

Laboratory confirmation

Diagnostic tests	Isolation of *Bacillus anthracis* from a clinical specimen (e.g. blood, lesions, discharges). Demonstration of *B.anthracis* in a clinical specimen by microscopic examination of stained smears (vesicular fluid, blood, cerebrospinal fluid, pleural fluid, stools). Positive serology (ELISA, Western blot, toxin detection, chromatographic assay, fluorescent antibody test) Detection of nucleic acid by genetic amplification
Specimen	**Cutaneous** 1. For vesicular lesions, two swabs of vesicular fluid from an unopened vesicle using sterile cotton swabs 2. For eschars, the edge should be lifted and two swab samples rotated underneath 3. For ulcers, collect the specimen at the base of the lesion with two saline moistened swabs Blood specimens for bacteriological culture should be collected before antimicrobial therapy, if the patient has evidence of systemic symptoms. 4. A full thickness punch biopsy of a papule or vesicle including adjacent skin should be obtained from all patients with a lesion being evaluated for cutaneous anthrax, to be soaked in 10 percent formalin for histopathology. 6. In patients not on antibiotic therapy or on therapy for <24 hours, collect a second biopsy specimen 7. collect acute and convalescent serum samples for serologic testing. **Gastro-intestinal form** 1. Blood specimen obtained for culture prior to antimicrobial therapy. 2. Ascites fluid for culture and PCR. 3. Stool or rectal swab for culture and PCR. 4. Oropharyngeal lesion, if present, for culture and PCR. 5. Acute and convalescent serum samples for serologic testing. 6. Autopsy tissues from fatal cases for histopathology. **Pulmonary (Inhalation)** Blood cultures obtained prior to antimicrobial therapy. Pleural fluid, if present, for culture and PCR. CSF, in patients with meningeal signs, for culture and PCR. Pleural and/or bronchial biopsies for IHC. Acute and convalescent serum samples for serology Autopsy tissues from fatal cases for histopathology.

When to collect the specimen	Specimens should be collected from any patient being evaluated for cutaneous *Bacillus anthracis* infection. Note: it may not be possible to demonstrate B. anthracis in clinical specimens if the patient has been treated with antimicrobial agents. Microorganism is best demonstrated in specimen taken at the vesicular stage Specimens for culture should be obtained prior to initiation of antimicrobial therapy If available at reference laboratories specimens may be submitted for PCR Caution: *B.anthracis* is highly infectious
How to prepare, store, and transport the specimen	***Vesicular stage***: collect fluid from intact vesicles on sterile swabs. ***Eschar stage***: without removing eschar, insert swab beneath the edge of eschar, rotate and collect lesion material. Do not store specimen for more than 24 hrs and transport to the laboratory at room temperature within 2hrs maximum. **Stool**: collect 5-10 g in a clean sterile leak-proof container. Store for ≤24hrs at 4°C. Transport within 1hr at room temperature. **Blood**: collect per institution's procedure for routine blood culture. Collect 10 ml of blood in EDTA for PCR. Transport within 2hrs at room temperature. **Sputum**: collect expectorated specimen into a sterile leak-proof container. Store for ≤24 hrs at 4°C. Transport within 2hrs at room temperature.
Results	*Diagnostic services for Anthrax are not routinely available. Advance arrangements are usually required for VHF diagnostic services. Contact the national competent authority or WHO.*

References

- *World Health Organisation, Anthrax in humans and animals. Geneva. 2008 (Available onhttp://www.who.int/csr/resources/publications/ AnthraxGuidelines2008/en/index.html*
- *WHO Recommended Surveillance Standards WHO/CDS/CSR/ISR/99.2*
- *WHO Recommended Strategies for Prevention and Control of Communicable Diseases; WHO/CDS/CPE/SMT/2001.13*
- *2003 WHO Manual for Laboratory Diagnosis of Anthrax (http://www.searo.who.int/en/Section10/Section17/ Section58/Section909.htm)*
- *Anthrax Information for Health Care Providers, CDC (http://emergency.cdc.gov/agent/anthrax/anthrax-hcp-factsheet.asp)*
- *Recommended Specimens for Microbiology and Pathology for Diagnosis: Inhalation, Cutaneous, and Gastrointestinal Anthrax, CDC*

  (http://emergency.cdc.gov/agent/anthrax/lab-testing/recommended specimens.asp)

Bacterial Meningitis

Background

- *Neisseria meningitidis, Haemophilus influenzae* type b (Hib), and *Streptococcus pneumoniae* are responsible for most cases of bacterial meningitis and 90% of bacterial meningitis in children.
- Meningococcal Meningitis is the main bacterial meningitis form causing outbreaks. It remains a key public health challenge across the 'meningitis belt' that stretches from Senegal to Ethiopia In these areas during the dry season (November to May); populations are at high risk of outbreaks of this disease. Meningitis outbreaks still occur in the "meningitis belt" at a lower magnitude throughout the year;
- Outbreaks in the "meningitis belt' were generally associated with serogroup A of *Neisseria meningitidis* before the introduction of the conjugate vaccine against Meningitis (MenAfriVac®) in countries of the meningitis belt since 2010. The conjugate vaccine against Meningococcus A is immunogenic in infants and adults and gives long-term protection. The vaccine has considerably reduced the circulation of Meningococcus A and eliminated meningitis outbreaks caused by this meningoccus.
- Outbreaks related to other serotypes continue to occur: since 2013, major outbreaks caused by serogroup C meningococcus broke out in Nigeria and Niger. Between 2016 to 2018, mixed major outbreaks of *Neisseria meningitidis serogroup W* and *Streptococcus pneumoniae* were reported in Ghana. In 2016 and in 2017, Togo reported outbreaks related to meningococcus serogroup W. in addition, in 2006, Burkina and Niger reported an outbreak caused by meningococcus serogroup X.
- The disease is airborne from one person to the other carried by big droplets of rhinopharyngeal secretions from infected individuals.
- The incubation period lasts two to ten days.
- The highest attack rates were noted in children aged below 15 years. The case fatality rate generally stands between 8 and 15% of patients treated, and above 70% in the absence of treatment. Many survivors will have long-term sequelae, namely mental retardation, loss of hearing and the use of limbs.
- Ceftriaxone is the preferred medicine for treating meningitis during outbreaks for its effectiveness against predominant pathogenic agents of the disease. Moreover, antimicrobial resistance to ceftriaxone has not yet been reported in Africa.
- During outbreaks in the 'meningitis belt' antibiotherapy is not recommended.
- The current response to meningitis outbreaks consists in organising mass immunization reactive campaigns with bivalent (A and C) or trivalent/quadrivalent (A, C, W/A, C, Y W) polysaccharide vaccines, as early possible, following the oubreak. Polysaccharide vaccines do not protect toddlers (below two years of age) and their effect lasts only for three years.

Surveillance Goal

- Quickly detect meningitis outbreaks and confirm their etiology
- Use surveillance data to organise the supply in vaccines and medicines, and take additional prevention and control measures.
- Assess and monitor the spread of the outbreak and the effectiveness of control measures
- Monitor the situation throughout the year, namely changes in serotypes
- Follow up antibiotic sensitivity

Standard case definition of bacterial Meningitis

Suspected Case:

Any person with sudden onset of fever (>38.5 °C rectal or 38.0 °C axillary) and one of the following signs: neck stiffness, altered consciousness or other meningeal signs.

Any infant with sudden onset of fever (rectal temperature ≥ 38.5°C or axillary temperature ≥ 38.0°C) with any of the following signs: neck stiffness or limp neck, bulging fontanel, gaze ceiling, convulsion or any other meningeal sign

Probable case

Any suspected case with macroscopic aspect of CSF turbid, cloudy or purulent; or with a CSF leukocyte count >10 cells/mm^3; or with bacteria identified by Gram stain in CSF; or antigen detection (for instance, by latex agglutination) in CSF.

In infants: CSF leucocyte count >100 cells/mm3; or CSF leucocyte count 10–100 cells/mm^3 AND either an elevated protein (>100 mg/dl) or decreased glucose (<40 mg/dl) level

Confirmed case

Any suspected or probable case that is laboratory confirmed by culturing or identifying (i.e. by polymerase chain reaction,) a bacterial agent of *Neisseria meningitidis, Streptococcus pneumoniae or Haemophilus influenzae* type b in the CSF or blood

Responding to alert threshold

Alert threshold

For a population between 30 000–100 000 people, an attack rate of 3 suspected cases / 100 000 inhabitants / week (minimum of 2 cases in one week)

- Under 30 000 inhabitants, 2 suspected cases in one week or an increased incidence compared to previous non-outbreak years

Responding to alert threshold

- Inform the higher level of the health system
- Record cases on a line list
- Investigate cases and request laboratory confirmation
- Treat suspected cases with appropriate antibiotics in line with the national protocol
- Enhance surveillance to find other cases in the region
- Prepare for a possible response
- Be ready to launch a mass vaccination campaign

Responding to outbreak or action threshold

Outbreak threshold

- For 30 000 - 100 000 inhabitants : 10 suspected cases / 100 000 inhabitants / week
- Under 30 000: 5 suspected cases in one week** or doubling of the number of cases in a three-week period

*For districts with more than 100 000 inhabitants, it is recommended to calculate the attack rate per sub-districts between 30 000 and 100 000 inhabitants.

**In special situations such as such as mass gatherings of refugees, displaced persons or closed institutions, two confirmed cases in a week should prompt mass vaccination.

Respond to outbreak threshold

- Proceed with mass vaccination within four weeks after the outbreak threshold is crossed***
- Mobilise the community to enable the quick detection and treatment of cases and improve the vaccination coverage during mass vaccination campaigns.
- Proceed with data collection, transmission and analysis.
- Keep with the regular collection of specimens weekly, 5 to 10 specimen of CSF, throughout the outbreak season, in all affected districts to detect any possible change in serogroup. Distribute drugs to health centres.
- Treat all suspected cases with appropriate antibiotics as per the national protocol.

***Include any neighbouring region of a population targeted for vaccination considered at risk (cases recorded at the beginning of the dry season, absence of recent relevant vaccination campaign, high population density) in the vaccination programme.

Analysing and Interpreting data

- **Time**: during the outbreak season, in countries of the African Meningitis belt, graph weekly cases and deaths. Or, plot the monthly trends of cases and deaths. Draw the outbreak curve of outbreak-related cases.
- **Place**: during outbreaks (not in endemic situations), map out the locations of cases and estimate the distance from the nearest HF.
- **Personal information**: sum up sporadic cases and outbreak-related cases. Analyse their age group distribution. Goal set for case fatality rate: < 10 %

Laboratory confirmation

Diagnostic tests	Microscopic examination of CSF for the presence of Gram negative diplococcus Culture and isolation of *N. meningitidis Streptococcus pneumoniae and Haemophilus influenzae, type b* from CSF or blood RT-PCR (national reference laboratory)
Specimen	Cerebrospinal Fluid **Note:** CSF is the best specimen for culture and microscopic examination of *N. meningitidis*. If CSF cannot be collected, collect blood (10ml in adults, 1-5ml in children) for culture
When to collect the specimen	Collect specimen from 5 to 10 cases, once the alert or outbreak threshold is reached (see “Meningitis”, Section 8.0

How to prepare, store, and transport the specimen	▪ Prepare the patient and collect CSF in an aseptic manner in cap assay tubes. ▪ Fill in the dry tube (culture) and a cryotube (PCR) ▪ If the dry tube cannot reach the laboratory within two hours, inoculate 1ml of CSF in a pre-heated trans-isolate. ▪ Incubate between 36°C and 37°C (body temperature). ▪ Never refrigerate specimen that will be cultured. Store CSF for microscopic analysis and biological analysis in the sampling syringe (re-cap). Refrigerate the syringe and send it to the lab as soon as possible.
Results	*Neisseria meningitidis, Streptococcus pneumoniae, and Haemophilus influenzae b. Neisseria meningitidis* are sensible microorganisms, which are costly and difficult to isolate. They require excellent sampling techniques and treatment of specimen, and expensive transport media and anti serum. Carry out serotyping and antibiogramme of *N. meningitidis or Neisseria meningitidis* isolates of first specimens from sporadic cases or related to an outbreak, to determine the best treatment. Trans-Isolate is stable. It may be stored at 4°C for two years after preparation. At 4°C, the liquid phase becomes gelatinous, but becomes liquid again at room temperature. Unused Trans-Isolates should be tightly closed. Do not use if the liquid colour changes (yellow or opaque) or if the agar medium dries up or shrinks.

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Buruli Ulcer (*Mycobacterium Ulcerans Disease*)

Background

- Skin infection caused by *Mycobacterium ulcerans* (an AFB)
- Occurring mainly as skin lesions (nodules, plaques and ulcers) than can be complicated by bone and joint involvement. Involvement of other organs like the eyes is rare
- Spreading in inter-tropical areas, in swampy soils or water body surroundings, forestry or surface mining zones
- Patients are classified into three categories:
  - **Category I**: patient with a single lesion which size is less than 5 cm of diameter (early lesion)
  - **Category II**: patient with single lesion which size is between 5 and 15 cm of diameter
  - **Category III**: patient single lesion which size is over 15 cm of diameter or with multiple lesions or lesion located in critical site (face, head & neck, breast, perineum, genitalia, lesion spanning over joints)
- BU case management has improved greatly through use of WHO recommended antibiotics (rifampicin and streptomycin) in 2004. Since 2017, full oral combined antibiotics (rifampicin and clarithromycin) are now recommended for treatment of ulcers. Surgery is still needed for late cases (category III). Cumulative number of cases in the WHO African Region that is the most affected (95% of global cases) is around 90,000 in 2017.
- Mode of transmission is still unknown. *M. ulcerans* could penetrate the skin through insect bite (water bugs); micro trauma or small wounds
- Confirmation of diagnosis is done by PCR, AFB search with Zielh-Nielsen (ZN) staining, culture or histology. Specimens of lesions are taken by swab in ulcer, fine needle aspiration (FNA) or biopsy in case of surgery. New diagnostic tests based of the presence of mycolactone, a toxin released by *M. ulcerans* in lesions, are under development.

Surveillance Goal

- Geographical distribution of the disease to locate endemic areas and districts and focus on early case finding, proper management with WHO recommended antibiotics and prevention of disabilities

Standard case definition

Suspected case: A person presenting a painless skin nodule, plaque or ulcer, living or having visited a BU endemic area

Confirmed case: A suspected case confirmed by at least one laboratory test (ZN for detection of AFB, PCR, culture or histology). Confirmation of presence of mycolactone in skin lesions

Responding to alert threshold

If a single case is suspected:

- Report the suspected case to the appropriate level of the health system

At HF level:

- Take a specimen for laboratory confirmation (Swab or FNA)
- Begin wound dressing and antibiotic treatment with Rifampicin 10 mg/kg daily oral intake for 8 weeks (56 days).
- Clarithromycin 7.5 mg/kg twice daily oral intake for 8 weeks (56 days)
- Refer category III patients to reference hospital/centre
- Fill in Case report form (BU 1 or BU 2) with origin village GPS data and report to HD, Regional and National levels

Search other cases in the origin village of a confirmed case of BU

Responding to outbreak or action threshold

Not applicable to BU

Analysing and Interpreting data

Time: Graph of cases by year of diagnosis, graph of cumulative number of cases.

Place: Plot cases by location of households and colour shade endemic districts

Personal information: Count newly detected cases monthly by category of patients (Cat I, II or III). Analyse age and disability distribution and treatment outcomes (cases cured without limitation of movement or amputation, cases cured with disability, relapse after recommended antibiotic treatment).

Laboratory confirmation	
Diagnostic tests	*Mycobacterium ulcerans*: Smears and biopsy specimens can be sent to the laboratory for confirmation by: Ziehl-Neelsen stain for acid-fast bacilli Culture PCR Histopathology Mycolactone detection in lesion (new)
Specimen	Smears Fine needle aspirations (FNAs) Biopsy
When to collect the specimen	Specimens should be collected from suspected patient with clinical symptoms (nodule, plaque, ulcer, osteomielitis ...) Specimen should be collected before any antibiotic is given. Another specimen should be collected at the end of the treatment (in case the treatment is not efficacious or surgery is indicated)

How to prepare, store, and transport the specimen	It is important to avoid cross contamination between the collection of samples Materials: Dry swabs and recipients Types of specimens: No ulcerative forms, Ulcerative forms, Bone Store at 4°C
Results	Buruli ulcer is usually diagnosed clinically and by finding acid fast bacilli (AFB) in smears from infected ulcers and tissue biopsies. It can also be identified using PCR. *M ulcerans* can be cultured in a reference lab using the same culture media used to grow *M.tuberculosis*. The organism grows very slowly, usually requiring several weeks to provide visible colonies. Diagnostic services are not routinely available. Contact the appropriate national competent authority or WHO.

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Chikungunya

Background

- Chikungunya fever is a viral illness that is spread by the bite of infected mosquitoes. The disease resembles dengue fever. It is characterized by severe, sometimes persistent, joint pain (arthritis), as well as fever and rash. It is rarely life threatening. Nevertheless widespread occurrence of diseases causes substantial morbidity and economic loss
- The word "Chikungunya" is a Swahili word for "that which bends up," in reference to the stooped posture of patients afflicted with the severe joint pain associated with the disease. Outbreaks of fever, rash and arthritis, resembling Chikungunya fever were recorded as early as 1779. However, the virus was first isolated between 1952-1953 from both man and mosquitoes during an outbreak, in Tanzania.
- Chikungunya fever historically displayed interesting epidemiological profiles in that: major outbreaks appeared and disappeared cyclically, usually with an inter-outbreak period of 7-8 years and sometimes as long as 20 years. After a long period of absence, outbreaks appeared in Indonesia in 1999 and have been virtually ongoing since 2004.

Surveillance Goal

- Detect Chikungunya sporadic cases and outbreaks promptly, and seek laboratory verification
- Identify high-risk areas in order to improve prevention of outbreaks by taking steps to avoid mosquito bites and elimination of breeding sites.

Standard case definition

i. **Acute clinical case**
   a. Clinical criteria: Sudden onset of fever >38.5 °C (101.3 °F) and pain in the joints[a] (usually incapacitating[b]) **AND**
   b. Epidemiological criteria : person residing in an area of local transmission of Chikungunya or who had been in that area within the previous 15 days (suspected case for epidemiology surveillance) **OR**
   c. Laboratory criteria: laboratory confirmation: PCR, serology or viral culture (confirmed case for epidemiological surveillance).

ii. **Atypical case:**

Chikungunya clinical case confirmed in the lab along with other manifestations: neurological, cardiological, dermatological, ophtalmological, hepatic, renal, respiratory, haematological, etc.

iii. **Severe acute case**

Chikungunya clinical case confirmed in the lab presenting a malfunctioning in at least one organ or system putting life at risk and requiring hospitalization.

iv. **Suspected chronic case:** a person who received a clinical diagnosis of Chikungunya 12 weeks after onset of symptoms, and presenting at least one of the manifestations in joints below: persistent or recurrent pain, stiffness or edema.

Confirmed chronic Case: each chronic case confirmed by a positive result in the laboratory.

a) Generally accompanied by rash, myalgia, back pain, headaches and, sometimes vomiting and diarrhoea (pediatric age group).
b) In children below three years of age, pain in the joints is expressed through unsoothable crying, irritability, and refusal to move or walk.

Responding to alert threshold

If Chikungunya cases are suspected:

- Report case-based information immediately to the next level.
- Collect specimens for confirming the cases
- Conduct an investigation to determine the risk factors for transmission
- Manage and treat the cases using anti-inflammatory agents

Responding to outbreak or action threshold

If Chikungunya cases are confirmed

- Symptomatic treatment for mitigating pain and fever using anti-inflammatory drugs along with rest usually suffices. Persistent joint pain may require painkillers and long-term anti-inflammatory therapy.
- Prevention is entirely dependent upon taking steps to avoid mosquito bites and elimination of mosquito breeding sites.

To avoid mosquito bites:

- Wear full sleeve clothes and long dresses to cover the limbs.
- Use mosquito coils and repellents.
- Use mosquito nets – to protect babies, old people and others, who may rest during the day. The effectiveness of such nets can be improved by treating them with permethrin (pyrethroid insecticide). Curtains (cloth or bamboo) can also be treated with insecticide and hung at windows or doorways, to repel or kill mosquitoes.
- Mosquitoes become infected when they bite people who are sick with Chikungunya. Mosquito nets and mosquito coils will help prevent mosquitoes from biting sick people

Analysing and Interpreting data

Time: Graph cases and deaths weekly. Construct an outbreak curve during outbreaks.
Place: Precisely map out the location of case houses.

Personal information: immediately report case-based information for cases and deaths. Report summary totals monthly.

During outbreak, count cases and deaths weekly. Analyse by age. Assess risk factors to improve prevention of outbreaks.

Laboratory confirmation

Diagnostic tests	Serological tests show a rise in antibody titer to Chikungunya virus. The virus may be isolated from the blood of acutely ill patients by inoculating newborn mice, on mosquitoes cells or stem cell culture. Detection using IFA or Reverse Transcription Polymerase Chain Reaction (RT-PCR)
Specimen	Serum
When to collect the specimen	Collect specimen from the first suspected case (s). Suspected CHIK cases generally occur in clusters. Collect representative specimens from suspected cases. If outbreak is confirmed, collect more specimens from cases and mosquitoes from the affected homes for testing. Type of Specimen - Acute-phase blood (0-10 days after onset) - Convalescent-phase blood (7 - 21 days after onset Time of collection: When patient presents at the HF; collect second sample during convalescence. Between days 7 and 21 after onset.

How to prepare, store, and transport the specimen	Transport of specimens should comply with the WHO guidelines for the safe transport of infectious substances and diagnostic specimens (WHO, 1997). For ELISA: ▪ Refrigerate at 2º to 8ºC serum for testing within 24 hour. If kept for longer, store at -80ºC. For virus Isolation or RT_PCR ▪ Store at -80ºC for a short period or at -70ºC, or transport in fully charged dry shipper. Mosquitoes for testing should be transported in fully charged dry shipper. Focus on *Aedes* species
Results	Diagnostic services for Chikungunya are not routinely available. C Contact the appropriate national competent authority or WHO. Samples should be sent to WHO reference labs namely KEMRI laboratories. ▪ Preliminary results are ready within 24 hours after samples arrive in the laboratory. Confirmatory results are ready within a week from sample reception.

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Cholera

Background

- Acute illness with profuse watery diarrhoea caused by *Vibrio cholerae* serotypes O1 or O139. The disease is transmitted mainly through the faecal-oral route; that is through eating or drinking contaminated food or water.
- Cholera causes over 100 000 deaths per year. It may produce rapidly progressive outbreaks or worldwide pandemics. In endemic areas, sporadic cases (less than 5% of all non-outbreak related diarrhoea cases) and small outbreaks may occur.
- Incubation period is from a few hours to 5 days, usually in the range of from 2 to 3 days.
- There has been a resurgence of cholera in Africa since the mid-1980s, where over 80% of the world's cases occurred in 1999. The majority of cases occurred from January through April. In 2016, globally, 38 countries reported a total of 132 121 cases. Of cases reported globally, 54% were from Africa, 13% from Asia and 32% from Hispaniola. Imported cases were reported in 9 countries.
- Cholera may cause severe dehydration in only a few hours. In untreated patients with severe dehydration, the case fatality rate (CFR) may exceed 50%. If patients present at the HFand correct treatment is received, the CFR is usually less than 1%. At least 90% of the cases are mild, and they remain undiagnosed.
- Risk factors: eating or drinking contaminated foods such as uncooked seafood or shellfish from estuarine waters, lack of continuous access to safe water and food supplies, attending large gatherings of people including ceremonies such as weddings or funerals, contact with persons who died of cholera.
- Other enteric diarrhoea may cause watery diarrhoea, especially in children less than 5 years of age. Please see Diarrhoea with dehydration summary guidelines.

Surveillance Goal

- Detect and respond promptly and appropriately to cases and outbreaks of watery diarrhoea. To confirm an outbreak, collect and transport stool specimens transported in Cary-Blair medium.
- Do immediate case-based reporting of cases and deaths when an outbreak is suspected.

Standard case definition

Suspected case: In areas where a cholera outbreak has not been declared. Any patient aged two years and older presenting with acute watery diarrhoea and severe dehydration or dying from acute watery diarrhoea.

In areas where a cholera outbreak is declared: any person presenting with or dying from acute watery diarrhoea.

Confirmed cholera case: A suspected case with *Vibrio cholerae* O1 or O139 confirmed by culture or PCR and, in countries where cholera is not present or has been eliminated, the *Vibrio cholerae* O1 or O139 strain is demonstrated to be toxigenic.

Responding to alert threshold

If a single case is suspected:

- Report case-based information immediately.
- Manage and treat the case according to national guidelines.
- Enhance strict hand-washing and isolation procedures.
- Conduct case-based investigation to identify similar cases not previously reported.
- Obtain stool specimen from 5 patients within 5 days of onset of acute watery diarrhoea, and before antibiotic treatment is started. See laboratory guidelines for information on how to prepare, store and transport the specimens.

Responding to outbreak or action threshold

If a suspected case is confirmed:

- Establish treatment centre in locality where cases occur. Treat cases onsite rather than asking patients to go to standing treatment centres elsewhere.
- Initiate a line listing of suspected and confirmed cases and ensure laboratory results are linked with cases
- Strengthen case management including treatment.
- Mobilize community early to enable rapid case detection and treatment. Survey the availability of clean drinking water.
- Work with community leaders to limit large gatherings for ceremonies, funerals or other reasons, especially during an outbreak. If seen mandatory, establish bylaws
- Reduce sporadic and outbreak-related cases through continuous access to safe water. Promote safe preparation of food (especially seafood, fruits, and vegetables)
- Promote safe disposal of human waste.
- Ensure adequate collaboration with various sectors including water and sanitation to ensure appropriate interventions are addressed
- Cholera vaccine is available but its utilization must be accompanied with strategies to improve water and sanitation

Analysing and Interpreting data

Time: Graph weekly cases and deaths and construct an outbreak curve during outbreaks. Report case-based information immediately. Send a summary information monthly for routine surveillance.

Place: Plot the location of case houses.

Personal information: Count weekly total cases and deaths for sporadic cases and during outbreaks. Analyse distribution of cases by age and according to sources of drinking water. Assess risk factors to improve control of sporadic cases and outbreaks.

Laboratory confirmation

Diagnostic tests: Isolate *V. cholerae* from stool culture and determine O1 serotype using polyvalent antisera for *V. cholerae* O1. If desired, confirm identification with Inaba and Ogawa antisera.

If specimen is not serotypable, consider *V. cholerae* O139 (see note in Results column)

Specimen: liquid stool or rectal swab

When to collect the specimen

For each new area affected by the outbreak, a laboratory confirmation should done. Collect stool sample from the first suspected cholera case. If more than one suspected case, collect until specimens have been collected from 5 to 10 cases. Collect stool from patients fitting the case definition and onset within the last 5 days, and before antibiotics treatment has started

Do not delay treatment of dehydrated patients. Specimens may be collected after rehydration (ORS or IV therapy) has begun.

If possible, specimens should be collected from 5 – 10 suspected cases every 1 – 2 weeks to monitor cessation of the outbreak, changes in serotypes, and antibiotic sensitivity patterns of *V.cholerae*

How to prepare, store, and transport the specimen	▪ Place specimen (stool or rectal swab) in a clean, leak proof container and transport to lab within 2 hours. ▪ If more than 2- hour delay is expected, place stool-soaked swab into Cary-Blair transport medium. If Cary-Blair transport medium is not available and specimen will not reach the lab within 2 hours: Store at 4°C to 8°C ▪ Do not allow specimen to dry. Add small amount of 0.85% NaCl if necessary ▪ To transport, transport in well marked, leak proof container ▪ Transport container in cold box at 4ºC to 8ºC
Results: cholera diagnostic tests in the lab	▪ Cholera tests may not be routinely performed in all laboratories. ▪ Culture results usually take 2 to 4 days after specimen arrives at the laboratory. ▪ Cary-Blair transport medium is stable and usually good for at least one year after preparation. It does not require refrigeration if kept sterile and in properly sealed container. If colour changes (medium turns yellow) or shrinks (depressed meniscus), do not use the medium. ▪ The O139 serotype has not been reported in Africa and only in a few places in southwest Asia. Serological determination of Ogawa or Inaba is not clinically required. It is also not required if polyvalent antisera results are clearly positive.

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Dengue Fever

Including Dengue haemorrhagic fever (DHF) and Dengue shock syndrome (DSS)

Background

- Dengue fever is an arbovirus transmitted by *Aedes* mosquitoes (both *Ae. aegypti* and *Ae. albopiticus*). Dengue is caused by four serologically distinct, but closely related viruses: dengue virus (DENV) 1, 2, 3, and 4 of the Flaviviridae family.
- Dengue fever is an emerging pandemic that has spread globally during the past 30 years as a result of changes in human ecology. Dengue is found in tropical and sub-tropical regions around the world, predominately in urban and semi-urban areas. During dengue outbreaks, infection rates among those who have not been previously exposed to the virus are often 40% to 50%, but can reach 80% to 90%
- Dengue fever is a severe, influenza-like illness that affects infants, young children and adults, but seldom causes death. Dengue haemorrhagic fever (DHF) is a potentially deadly complication that has become a leading cause of hospitalization and death among children in Asia. There is good evidence that sequential infection with the different serotypes of dengue virus increases the risk of more severe disease that can result in shock syndrome (DSS) and death.
- Dengue outbreaks in Africa have mostly been classical dengue fever caused by DENV-1 and DENV-2 without associated mortality. The first major outbreak of DENV-3 in Africa was documented in Mozambique in 1984-1985. During this outbreak, most patients experienced secondary infections and 2 deaths were attributed to DHF and shock. In 2008, yellow fever and DENV-3 were found to be co-circulating in Abidjan, Cote d'Ivoire, however, no severe dengue cases or deaths attributable to dengue were identified.
- There is no specific treatment for dengue, but appropriate medical care frequently saves the lives of patients with dengue haemorrhagic fever.
- Infected humans are the main carriers and multipliers of the virus, serving a source of the virus for uninfected *Aedes aegypti* mosquitoes which maintain the urban dengue transmission cycle. The virus circulates in the blood of infected human for 2-7 days, at approximately the same time that they have a fever. A sylvatic transmission cycle has been documented in West Africa where DENV-2 has been found in monkeys. There is no evidence of person-to-person transmission.
- At present, the only method of controlling or preventing dengue virus transmission is to combat the vector mosquitoes using environmental management and chemical methods.

Surveillance Goal

- Surveillance for suspected cases and investigation of clusters of suspected cases in areas with *Ae. aegypti* and Ae. *albopiticus mosquitoes*

Standard case definition

Dengue Fever Suspected case: Any person with acute febrile illness of 2-7 days duration with 2 or more of the following: headache, retro-orbital pain, myalgia, arthralgia, rash, haemorrhagic manifestations, leukopenia.

Dengue Fever Confirmed case: A suspected case with laboratory confirmation (positive IgM antibody, rise in IgG antibody titres, positive PCR or viral isolation).

Dengue Haemorrhagic Fever: A probable or confirmed case of dengue with bleeding tendencies as evidenced by one or more of the following: positive tourniquet test; petechieae, ecchymoses or purpura; bleeding of mucosa, gastrointestinal tract, injection sites or others; hematemesis or melena; and thrombocytopenia (100 000 cells or less per mm^3) and evidence of plasma leakage due to increased vascular permeability, manifested by one or more of the following: 20% rise in average haematocrit for age and sex, 20% drop in haematocrit following volume replacement therapy compared to baseline, signs of plasma leakage (pleural effusion, ascites, hypo-proteinaemia).

Dengue Shock Syndrome: All the above criteria, plus evidence of circulatory failure manifested by rapid and weak pulse, and narrow pulse pressure (≤ 20 mm Hg) or hypotension for age, cold, clammy skin and altered mental status.

Responding to alert threshold

If a single case is suspected:

- Report case-based information immediately to the next level.
- Conduct active search for additional cases
- Collect specimens for confirming the cases

Responding to outbreak or action threshold

If a single case is confirmed:

- Report case-based information immediately to the next level and initiate a line list/register of suspected cases
- Conduct active search for additional cases
- Collect specimens for confirming the cases and ensure results are linked with cases
- Survey the community to determine the abundance of vector mosquitoes, identify the most productive larval habitats, promote and implement plans for their elimination, management or treatment with appropriate larvicides.
- Educate the public and promote behaviours to remove, destroy or manage mosquito vector larval habitats.
- Manage and provide supportive treatment to dengue fever cases. Implement standard infection control precautions. Prevent access of mosquitoes to patients by using mosquito bed nets.

Refer suspected DHF/DSS cases to more advanced facilities.

Analysing and Interpreting data

Time: Graph cases and deaths weekly/monthly. Construct an outbreak curve during outbreaks.

Place: Plot location of case houses and work sites using precise mapping.

Personal information: calculate the case-fatality rate. Analyse age and sex distribution. Percentage of DHF / DSS cases and of hospitalisations.

Laboratory confirmation

Diagnostic tests	Rapid diagnostic testing (RDT) for simultaneous detection of IgA and IgG antibodies Demonstration of IgM and IgG by Antibody Assays. Detection of viral genomic sequences by PCR. Isolation of the dengue virus using cell culture. Antigen detection assays for acute phase samples when PCR or isolation is negative. Demonstration of dengue virus antigen in autopsy tissue by immunohistochemistry or immunofluorescence or in serum samples by EIA. *Note: there are several diagnostic techniques available to document an infection by the dengue virus. The IgM ELISA is the basic test for serologic diagnosis.*

Specimen	ELISA: Whole blood, serum or plasma from acute (0-5 days) and convalescent 6 or more days) depending on each case. PCR: Whole blood or blood clot, serum/ plasma or tissue preferably from acute specimens (0-5 days) The samples should be collected for diagnosing a suspected dengue fatality: a blood sample to attempt PCR, virus isolation and serology. If an autopsy is performed, blood from the heart should be collected.
When to collect the specimen	Collect specimen from the first suspected case. If more than one suspected case, collect until specimens have been collected from 5 to 10 cases. Type of Specimen ▪ Acute-phase blood (0-5 days after onset of symptoms) ▪ Convalescent-phase blood (≥ 6 days after onset) Time of collection ▪ Collect 2nd sample during convalescence. Between days 6 and 21 after onset. Lab diagnosis of fatal cases is indispensable for understanding the risk factors for severe cases
Laboratory confirmation	
How to prepare, store, and transport the specimen	Transport of specimens should comply with the WHO guidelines for the safe transport of infectious substances and diagnostic specimens. *For ELISA or PCR:* ▪ Refrigerate serum or clot. For long term storage, freeze at -20°C ▪ Freeze (-20°C or colder) tissue specimens for virus isolation If an autopsy has been performed and no fresh tissues are available, tissues fixed in formalin should be submitted for immunohistochemical studies.
Results	**Diagnostic services for Dengue fever and Dengue haemorrhagic fever are not routinely available.** Advance arrangements are usually required. Contact the appropriate national competent authority or WHO.
References	
▪ World Health Organisation, WHO Recommended Surveillance Standards WHO/CDS/CSR/ISR/99.2 Geneva. ▪ *Dengue:* Clinical and Public Health Aspects, CDC	

Diabetes

Background

- *Diabetes mellitus* (DM) is a widespread chronic disease that occurs when the pancreas does not produce enough insulin, or when the body cannot effectively use the insulin it produces. Diabetes can cause serious health complications including heart disease, blindness, kidney failure, and lower-extremity amputations.
- The most common form is Type 2 diabetes that represents more than 85% of the cases. Other forms are less common such as Type 1 (10% of cases), specific diabetes 'secondary) and gestational diabetes (5% of cases).
- The risk factors that affect the onset of diabetes are well known. They comprise non-modifiable factors like old age (over 45 years of age), family history, and insulin sensitivity in pregnancy. Modifiable risk factors for diabetes are obesity, physical inactivity and excessive alcohol consumption.
- The global prevalence in 2000 was estimated at 2.8%, with projections of 4.8% by 2030. The total number of persons affected will rise from 171 million in 2000 to 366 million in 2030 if no action is taken. Annual mortality linked to diabetes worldwide is estimated at more than one million.
- Diabetes is no longer considered rare in Africa. Recent estimates based on the WHO STEP-wise approach for monitoring the risk factors of non-communicable diseases indicate prevalence of diabetes between 1% and 20%. In some countries such as Mauritius, it reaches 20%.
- The rate of limb amputations due to diabetes varies from 1.4% to 6.7% of diabetic foot cases. In some African countries, the mortality rate is higher than 40 per 10,000 inhabitants.
- In the African Region, efforts made to create an environment that enhances the fight against diabetes include adoption of resolutions on non-communicable diseases in 2000, cardiovascular diseases strategy in 2005, and diabetes mellitus strategy in 2007. The World Health Organisation and the International Diabetes Federation (IDF) have also jointly carried out actions to contribute to promoting diabetes awareness in Africa.

Surveillance Goal

- Estimate the magnitude of the disease
- Monitor trends and risk factors
- Identify populations at highest risk (e.g.; age groups, urban vs. rural)
- Monitor prevention and control programme activities

Standard case definition

Suspected new case :

Any person presenting with the following symptoms:

- Increased thirst (polydipsia)
- Increased hunger (polyphagia)
- Frequent urination

Confirmed new case:

Any person with a fasting 6.1 mmol/L (110 mg/dl) or venous plasma glucose measurement of ≥ 7 mmol/L (126 mg/dl) or capillary glucose ≥

Any person with a non-fasting glucose ≥ 11,1 mmol/L or venous plasma glucose measurement of ≥ 11.1mmol/L (200 mg/dl) or capillary glucose (mg/dl)

*Report only the first lab-confirmed

Recommended public health action

For people with diabetes:

- Treat confirmed cases according to the standardized case management guidelines (WHOPEN).

District-level Prevention:

- Implement an integrated prevention and control programme for non-communicable diseases focusing on diabetes through community awareness and education activities conducted in accordance with national prevention and control programmes for non-communicable diseases. These activities would include multisector strategies and plans of action on diet, weight reduction and physical activity.
- Implement clinical preventive measures and treatment interventions using evidence-based guidelines (screening high-risk patients, for example).

Analysing and Interpreting data

Time: Graph case numbers quarterly to analyse trends.
Place: Compare district trends with national and regional trends.
Personal information: Analyse the distribution of cases by age and other demographic factors.
**Data for non-communicable diseases is analysed for long term trends*

Laboratory confirmation

Diagnostic tests	Measuring glucose in capillary blood using a reagent strip test and reference meter Measuring glucose in plasma using a glucose-oxidase colorimetric test method Lab case definition (see section 8.0)
Specimen	Plasma Capillary blood
When to collect the specimen	Blood glucose measurements must be carried out on the day and at the time requested. Fasting specimen: for adult the fasting time is usually 10 to 16 hours. For children the fasting time is 6 hours. Post-prandial specimen: 2hrs after a meal.
How to prepare, store, and transport the specimen	Specimen should be examined as soon as possible (before 2 hours) at HFwhere the specimen is taken.
Results	Results are ready within few hours

References

- Non communicable Diseases: A strategy for the African Region, AFR/RC50/10
- Cardiovascular Diseases in the African Region: Current situation and perspectives, AFR/RC55/12
- Diabetes prevention and control: a strategy for the African Region, AFR/RC57/7
- Steps manual: http://www.who.int/chp/steps/en/
- Gojka R et al, Global prevalence of diabetes, Diabetes Care 27(5): 1047–1053, 2004
- IDF, Diabetes Atlas, 2nd Edition, Brussels, International Diabetes Federation, 2003
- WHO, Preventing chronic diseases: A vital investment, Geneva, 2005
- World Health Organisation, *The burden of mortality attributable to diabetes*. Geneva. 2004
- WHO-PEN: Protocols for health promotion, prevention and management of NCDs at primary care level http://www.afro.who.int/en/divisions-a-programmes/ddc/division/2257-who-pen-protocols.html
- District Laboratory Practice in Tropical Countries, Cambridge

Diarrhoea with Blood (Shigella dysenteria)

Background

- Shigella dysenteriae type 1 (SD1) is the most common cause of enteric infections and is transmitted from person-to-person through faecal-oral spread
- Large scale outbreaks may be caused by Shigella dysenteriae type 1 (SD1) with up to 30% of populations infected. The case fatality rate may approach 20% among young children and elderly persons with severe dehydration.
- The incubation period is from 1 to 4 days.
- Clinical illness is characterized by acute fever and bloody diarrhoea, It can also present with systemic symptoms and signs as well as dehydration especially in young children
- Risk factor: overcrowded areas with unsafe water and poor sanitation (for example, refugee and famine populations).
- SdD1 is frequently resistant to multiple antibiotics including trimethoprim-sulfamethoxazole.
- Enterohaemorrhagic and enteroinvasive *E. coli* and other bacteria or parasites such as *Entamoeba histolytica* may also cause bloody diarrhoea.

Surveillance goal

- Detect and respond to dysentery outbreaks promptly.
- Improve percentage of laboratory-confirmed cases and evaluate proportion verified as type 1 (SD1).
- Determine antibiotic sensitivity pattern of the agents isolated (especially SD1) both for routine surveillance and during outbreaks.

Standard case definition

Suspected case:
A person with abdominal pain and diarrhoea with visible blood in stool.

Confirmed case:
Suspected case with stool culture positive for *Shigella dysenteriae* type1.

Responding to alert threshold

If you observe that the number of cases or deaths is increasing over a period of time:

- Report the increase to the next level of the health system.
- Treat the suspected cases with oral rehydration and antibiotics based on recent susceptibility results, if available.
- Obtain stool or rectal swab specimen for confirming the SD1 outbreak.
- Investigate the case to determine the risk factors contributing to transmission.

Responding to outbreak or intervention threshold

If a suspected outbreak is confirmed:

- Search for additional cases in locality of confirmed cases. Initiate a descriptive list or keep a register of suspected cases
- Strengthen case management and treatment.
- Mobilize community to enable rapid case detection and treatment.
- Identify high-risk populations using, time, place, and personal data.
- Reduce sporadic and outbreak-related cases by promoting handwashing with soap or ash and water after defecating and before handling food.
- Ensure access to safe water (supply and storage), and use of latrines and safe disposal of human waste.
- Ensure adequate collaboration with various sectors including water and sanitation to ensure appropriate interventions are addressed

Analysing and Interpreting data

Time: Graph monthly trends in cases and deaths. Construct an outbreak curve for outbreak cases.
Place: Plot location of case houses.
Personal information: Count cases and deaths each month. During an outbreak, count outbreak-related cases by week. Routinely analyse age distribution of cases and deaths. Assess risk factors to improve prevention and control of sporadic disease and outbreaks.

Laboratory confirmation of Bloody diarrhoea (Shigella dysenteria).

Diagnostic tests	Isolate *Shigella dysenteria* type 1 (SD1) in stool culture to confirm shigella outbreak. If SD1 is confirmed, perform antibiotic sensitivity tests with appropriate drugs.
Specimen	Stool or rectal swab.
When to collect the specimen	For each new geographical area affected by a cholera outbreak, a laboratory confirmation should done. Collect sample when an outbreak is suspected. Collect stool from 5-10 patients who have bloody diarrhoea ▪ within four days of the onset of symptoms; and ▪ before starting antibiotic treatment. Preferably, collect stool in a clean, dry container. Do not contaminate sample with urine Using a swab, remove portions of stool with blood or mucus. If stool cannot be collected, obtain a rectal swab sample with a clean cotton swab.
How to prepare, store, and transport the specimen	Place the stool or rectal swab sample in Cary-Blair transport medium. Transport the refrigerated specimen to the laboratory. In the absence of Cary-Blair transport medium, send the stool specimen to the laboratory within two hours of collection in a clean, dry, tightly closed container. The quantity of Shigella decreases sharply after 24 hours in specimens that are not preserved in Cary-Blair medium. If storage is required, keep specimens at 4°C to 8°C, and do not freeze.

Results	Stool culture results are usually available two to four days after receipt by the laboratory. Sd1 isolates should be characterized by their sensitivity to antibiotics. After confirmation of initial 5-10 cases in an outbreak, sample only a small number of cases until the outbreak ends to monitor cessation of the outbreak, and antibiotic sensitivity patterns, which will guide the final treatment. Refer to disease specific guidelines in Section 11.0, for additional information about the outbreak potential of Sd 1.

Reference

- Guidelines for the control of outbreaks due to Shigella dysenteriae type 1. WHO/CDR/95.4
- Safe Water Systems for the Developing World: A Handbook for Implementing Household-based Water Treatment and Safe Storage Projects. Department of Health & Human Services. Centers for Disease Control and Prevention. Atlanta. 2000.
- Laboratory Methods for the Diagnosis of Outbreak Dysentery and Cholera. CDC/WHO, 1999 CDC, Atlanta, GA, USA

Diarrhoea with dehydration in children below five years old

Background

- Diarrhoea with dehydration in children of less than five years old is due to infections of the gastrointestinal tract caused by viruses (especially Rotavirus), bacteria (*E. Coli, Salmonellae, shigellae, Campylobacter, Yersinia*, and others), and by parasites (*Giardia, Entamoeba, cryptosporidia*, and *cyclospora*). These diseases are transmitted by the fecal-oral route, notably through the consumption of contaminated water or food.
- In many African countries, diarrhoeal diseases are the second leading cause of death among children under five years old, with more than three million deaths per year.
- Different epidemiological patterns (for example, seasonality) are observed for different pathogens.
- WHO and UNICEF advocate that each district team use the Integrated Management of Childhood Illnesses (IMCI) strategy to reduce morbidity and mortality of childhood diarrhoea.

Surveillance goal

- Promptly detect diarrhoea outbreaks. Laboratory confirmation helps to confirm a pathogen-specific outbreak, but is not required for routine surveillance of diarrhoea with dehydration.
- Monitor antibiotic resistance patterns during bacterial outbreaks.

Standard case definition

Suspected case:
Passage of at least three loose or watery stools in the past 24 hours, with or without dehydration and :

Moderate dehydration - at least two of the following signs: restlessness, irritability, sunken eyes, thirst, skin that relaxes slowly after pinching, OR
Severe dehydration - two or more of the following signs: lethargy or loss of consciousness, sunken eyes, inability or difficulty drinking, skin that relaxes very slowly after pinching

Confirmed case:
Suspect case confirmed by stool culture revealing the presence of a known enteric pathogen. ***Note*** Laboratory confirmation of the specific pathogen causing the outbreak is not routinely recommended for surveillance purposes.

Responding to alert threshold

If you observe that the number of cases or deaths is increasing over a period of time:

- Report the increase to the next level.
- Investigate the cause for the increase and identify the problem.
- Make sure that cases are managed according to IMCI guidelines.
- Encourage home-based therapy with oral rehydration.

Responding to outbreak or intervention threshold

If the number of cases or deaths doubles compared to the same period in previous years

- Assess the application of IMCI guidelines by health workers for case management, and improve performance in classifying cases of diarrhoea with dehydration in children below the age of five.
- Train mothers in home-based treatment with oral rehydration.
- Educate the community on the importance of boiling and chlorinating water, drinking water storage processes and food preparation.

Analysing and Interpreting data

Time: Graph cases and deaths to compare with same period in previous years. Graph non-hospitalized cases of diarrhoea with mild to severe dehydration. Construct an outbreak curve when outbreaks are detected.
Place: Plot location of case houses.
Personal information: report monthly totals of cases of diarrhoea with mild dehydration and of cases of diarrhoea with severe dehydration not hospitalized. Also, report the monthly total of inpatient cases and deaths due to diarrhoea with severe dehydration.

Laboratory confirmation

Laboratory culture of stools may be used to confirm possible outbreaks of specific agents, but is not necessary for case definition.

Reference

- *Management of childhood illness: Clinical skills training course for first level HF. The World Health Organisation. WHO/CDR/95.14*
- *Integrated Management of Childhood Illness: a WHO/UNICEF Initiative Bulletin of the World Health Organisation. Vol. 75, 1997, Supplement 1, 1997. ISBN 92 4 068750 5*

Dracunculiasis (Guinea worm disease)

Background

- Dracunculiasis is commonly known as Guinea worm disease. It is caused by a large nematode, a disabling parasite that emerges through the skin of the infected person.
- Known since antiquity, this disease inflicts excruciating pain on those affected and usually results in temporary disability, leaving many patients with unfortunate socio-economic consequences. It is transmitted by ingestion of water containing a crustacean (cyclops) infested with an immature form of the nematode (larva). The cyclops lives in stagnant water (marshes, traditional shallow wells), in rural areas. The female releases her eggs through the skin of the host on contact with water. The incubation period usually last from 10 to 14 months. There is neither treatment nor vaccine against the disease.
- Thanks to effective disease control strategies by endemic countries and an international coalition of partners, Guinea worm disease is on the verge of being eradicated. In 2017, only 30 cases of Guinea worm were reported to WHO, worldwide, compared to 892,000 in 1989, a 99.99% reduction.
- In 1989, the disease was endemic in 24 countries: Benin, Burkina Faso, Cameroon, Central African Republic, Chad, Ghana, Ethiopia, India, Pakistan, Kenya, Mali, Mauritania, Niger, Nigeria, Sudan, Senegal, Togo, Uganda and Yemen.
- Africa remains the only continent affected, with five countries reporting infection due to indigenous transmission of the parasite in humans or animals in 2018: Angola, Chad, Ethiopia, Mali, and South Sudan.
- Since 2012, new worms from animals, mainly dogs and, in a few cases, cats and baboons, have been reported in some of the remaining endemic countries and confirmed by the WHO Collaborating Center at CDC for the eradication of dracunculiasis, *Dracunculus medinensis*, in the laboratory. Therefore, eradication of dracunculiasis, which previously relied on interruption of transmission in humans, will now include interruption of transmission in human and animal hosts[2].

Surveillance goal

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[4] Since 2011, when Sudan was divided into two parts, the number of countries has increased to 21. Then with the reporting and confirmation of an indigenous case in Angola in 2018, the number of countries is now 22, including 18 in the WHO African Region.

[2] At its February 2018 meeting, the International Independent Commission for the Certification of Dracunculiasis Eradication (ICCED) revised the operational definitions of dracunculiasis elimination and eradication as follows:

- ***Elimination****: Confirmed absence of clinical disease (interruption of transmission of Dracunculus medinensis* ***in humans and animals)*** *for three years or more* ***from a country*** *with such a low risk of reintroduction of the parasite that preventive measures could be reduced to a minimum.*
- ***Eradication****: Confirmed absence of clinical manifestations (interruption of transmission of Dracunculus medinensis* ***in humans and animals****) for three years or more* ***worldwide.***

- Active detection and containment of cases at the community level, and immediate notification to the health center and relevant authorities at the district, regional and national levels. Thereafter, weekly and monthly case reports should be submitted to the next level.
- In areas where local transmission of Guinea worm disease has been interrupted, actively pursue case finding in high-risk areas and promptly follow up and investigate all rumours of Guinea worm disease (within 24 hours of notification) reported through the national surveillance system and/or directly by community members.
- Report all imported cases to the countries or regions of origin for follow-up investigation to trace the source of infection so that further action can be taken.
- Integrate dracunculiasis surveillance into national surveillance systems and continue to report cases immediately on a weekly or monthly basis, or as prescribed by the national reporting system.
- Use opportunities provided by other community health activities (e.g., polio and other immunization campaigns, neglected tropical disease mapping, mass drug administration, distribution of insecticide-treated nets and other health products) to actively search for cases of guinea worm disease and document the results.
- Continue to advertise rewards for reporting dracunculiasis cases.
- Systematically document and properly store Guinea worm surveillance information and data so that it can serve as evidence for future certification and beyond, until global eradication is declared

Standard case definition for Dracunculiasis

Rumour

- ***Information*** about the occurrence of Dracunculiasis from any source.

Suspected case:

- Anyone with an itchy or blistering skin lesion living in a Guinea worm endemic region or areas, and Guinea worm emergence.

Confirmed case:

- Someone with a skin lesion with guinea worm emergence, and when the laboratory confirms that the worm is D. *medinensis*. The person is considered a case only once during the calendar year, when the first worm emerges. Worm specimens should be collected from each patient for laboratory confirmation and sent to the U.S. Centers for Disease Control and Prevention. All cases should be monitored at least twice a month for the remainder of the calendar year for early detection of the possible emergence of other guinea worms.

Respond to alert threshold

As a disease to be eradicated, any rumour or suspected case of dracunculiasis is an emergency.

- Monitor and investigate any rumours of dracunculiasis (within 24 hours of notification), using national program guidelines and WHO-recommended forms, to determine whether or not there is a suspected case requiring follow-up, surveillance, and additional sampling for laboratory analysis.

If a single case is suspected

- Report the case according to national programme guidelines for eradication of Dracunculiasis.
- Treat the wound (if there is any) to reduce the disability associated with painful leg injuries.
- Collect and store specimens of any emerged worms in a 70% alcohol solution according to WHO national guidelines for specimen handling and send them to the WHO country office for onward transmission to the WHO Collaborating Centre at CDC for laboratory analysis.
- Conduct case investigation to confirm risk factors and assess the source and burden of the disease.
- Improve access to safe water in accordance with national guidelines.

Analysing and Interpreting data

Time: Graph cases monthly.
Place: Map the location of communities where cases have been reported.
Personal information: count the number of cases per month and analyse their age distribution. Use data to forecast interventions. Report monthly to next levels

Laboratory confirmation

The diagnosis of dracunculiasis is usually made when the anterior end of the female worm is seen emerging from a skin lesion (when the blister ruptures). Current programme standards require that the emerging worm be sent to the laboratory for confirmation that it is D. *medinensis.* Several other worms emerging from the skin may mimic Guinea worms, including those responsible for onchocerciasis and sparganosis, and should be differentiated from *D. medinensis* by laboratory confirmation. Collect and store specimens of any emerged worms in accordance with WHO national guidelines for specimen handling and send them to the WHO country office for onward transmission to the WHO Collaborating Centre at CDC for laboratory analysis (mandatory).

Reference

- World Health Organisation, Dracunculiasis or Guinea worm, Geneva, WHO/CDS/CEE/ DRA/99.2, 1999 and WHO/WER N°37 September 2003
- Control of Communicable Diseases Manual, 18th Edition
- District Laboratory Practice in Tropical Countries, Cambridge
- Dracunculiasis Eradication: (https://www.who.int/dracunculiasis/surveillance-control/fr/)
- Weekly epidemiological Records, 2018, 93, 33–44(http://www.who.int)
- Reports of meetings of International Task Force for Disease Eradication (ITFDE)https://www.cartercenter.org/news/publications/health/itfde_reports.html)
- Report of the International Independent Commission for the Certification of Eradication of Guinea Worm Disease (ICDEW) meeting, February 2018.

Ebola or Marburg Virus Disease

Background

The Ebola and Marburg viruses are both filoviruses.

Almost 3,000 cases of Ebola with over 1,900 deaths have been documented since the Ebola virus was discovered in 1976. Major Ebola outbreaks have occurred in Sudan, DRC, Cote d'Ivoire, Gabon, Uganda and Congo.

More than 500 cases of Marburg fever with over 400 deaths were reported during outbreaks of Marburg virus that occurred in DRC (1998-2000), Angola (2004-2005) and Uganda (three cases in 2007). Between 2018 and 2023, 3 countries in Africa were affected: DRC, Equatorial Guinea and Uganda

These two viruses are transmitted by direct contact with the blood, secretions, organs or other body fluids of infected persons. Transmission of Ebola virus to humans has also been reported when handling infected chimpanzees, gorillas, and wood antelopes (live or dead).

Ecological studies are ongoing to identify natural reservoirs of Ebola and Marburg viruses. From the observations, it appears that bats are involved.

Outbreaks can take on dramatic proportions in health care facilities if precautions or barrier measures for infection control are not properly implemented.

The incubation period is two to 21 days for Ebola and Marburg viruses.

Between 20% and 80% of patients have haemorrhagic manifestations, depending on the Ebola or Marburg virus strain. As the disease progresses, patients become more contagious.

High fatality rates have been reported during Ebola (25% to 90%) and Marburg (25% to 88%) fever outbreaks

There is no specific treatment for these diseases. Severe cases require intensive care, as patients who are often dehydrated require rehydration by intravenous infusion or oral rehydration with solutions containing electrolytes.

Close contact with a severely ill patient, during home or hospital care, and certain funeral practices are common sources of infection. Transmission through contaminated injection equipment or accidental needle-stick injuries is associated with more severe forms of the disease. Infection may also be spread via contact with soiled clothing or bed linens from an infected patient.

Surveillance goal

Early detection of cases and outbreaks, rapid investigation and immediate laboratory verification of the aetiology of all suspected cases.

Investigation of all suspected cases with contact tracing.

During outbreaks, most infected patients do not show haemorrhagic symptoms and a specific case definition according to the suspected or confirmed disease should be used.

Support of prevention efforts such as social distancing and vaccination

Monitoring case fatality, assess spread of illness (infection chains), and death.

Guiding the support and care of survivors

Standard case definition Ebola or Marburg virus diseases

Routine Surveillance:

Suspected case: Anyone with a high fever that does not respond to any treatment of the usual causes of fever in the area, and who has at least one of the following signs: bloody diarrhoea, gum bleeding, skin bleeding (purpura), conjunctival injection and blood in the urine.

Confirmed case: Laboratory-confirmed suspect case (positive IgM serology, positive polymerase chain reaction (PCR), or virus isolation, or with an epidemiological link to confirmed cases or an outbreak

Community-based surveillance:

Community Alert Case:

a. disease with onset of fever and no response to treatment of the usual causes of fever in the area;
   OR
b. At least one of the following signs: bleeding, bloody diarrhoea, bleeding into urine
   OR
c. Any sudden death

Actions to be taken: If an alert case (living or dead) is identified, report the case to a surveillance team or to the closest health centre

This definition of "alert cases" for Ebola or Marburg virus disease has been developed for use by the community or community-based volunteers. It may be used for community-based surveillance during the pre-outbreak phase and during the outbreak

Note: During an outbreak, case definitions may be adapted to the new clinical picture(s) or to the different modes of transmission related to the local event.

In outbreak setting, the following standard case definitions may guide appropriate detection of cases:

Suspected case: Any person, alive or dead, suffering or having suffered from a sudden onset of high fever and having had contact with: a suspected, probable or confirmed Ebola or Marburg case; a dead or sick animal (for Ebola); a mine (for Marburg) **OR**

Any person with sudden onset of high fever and at least three of the following symptoms: headaches - lethargy - anorexia / loss of appetite - aching muscles or joints - stomach pain - difficulty swallowing - vomiting - difficulty breathing - diarrhoea - hiccups; **OR**

Any person with inexplicable bleeding; **OR**

Any sudden, inexplicable death; **OR**

A person (alive or dead) suffering or having suffered from a sudden onset of high fever and having had contact with a dead or sick animal (for Ebola); a mine (for Marburg)

Standard case definition Ebola or Marburg virus diseases

Note During outbreaks, most infected patients do not show haemorrhagic symptoms, therefore, the case definition for suspected or confirmed cases does not include it.

Probable case

Any suspected case evaluated by a clinician;

OR

Any deceased suspected case (where it was not possible to collect specimens for laboratory confirmation) with an epidemiological link to a confirmed case.

Note: In the event of timely collection of laboratory specimens during the course of the illness, the preceding categories are reclassified as "laboratory confirmed" and "non-case".

Laboratory confirmed case: Any suspected or probable case with a positive laboratory result for the presence of the virus. Laboratory-confirmed cases should test positive for the virus, either by detection of virus ribonucleic acid (RNA) by reverse transcriptase polymerase chain reaction (RT-PCR) or by detection of IgM antibodies to Marburg or Ebola virus.

• **Non-case:** Any suspected or probable case with negative laboratory results. "Non-case" showed no specific antibodies or detectable specific RNA or antigen.

Respond to alert threshold: Ebola or Marburg virus diseases

If a single case is suspected:

- Immediately report (by phone or SMS with information from a generic case investigation form) any information related to a case to the appropriate authorities;
- Collect specimen to confirm the case(s). Carefully complete specimen request form and mark containers to warn laboratory of risk.
- Suspected cases should be isolated from other patients and strict barrier techniques should be applied. Eliminate exposure to body fluids and wear personal protective equipment for viral haemorrhagic fever;
- Standard precautions should be enhanced throughout the healthcare setting.
- Conduct monitoring of case contacts (using the survey form) and actively search for additional cases. Start contact tracing (see contact tracing forms).

Begin or enhance death reporting and surveillance

Respond to outbreak or intervention threshold of Ebola or Marburg virus diseases

If a single case is confirmed:

- Notify next level and WHO
- Maintain strict haemorrhagic fever viral infection control measures for the duration of the outbreak (see separate infection prevention and control guidelines);
- Mobilize the community for early detection and care of cases and conduct community education about how the disease is transmitted and how to implement infection control in the home care setting and during funerals.
- Conduct case contact tracing and active searches for additional cases that may not come to the health care setting.
- Ensure psychosocial support for family, community, and staff.
- Begin screening procedures for fever and VHF-like symptoms at the entrances to health care facilities with hand washing
- Request additional help from other levels as needed.
- Establish isolation ward to handle additional cases that may come to the health centre. Ensure there is a barrier between suspected cases and confirmed cases in an isolation unit.
- Quarantine high-risk contacts with home support during the incubation period. Low risk contacts under daily monitoring should be encouraged to limit their movements;
- Begin surveillance and screening of corpses, including: any person five years of age or older, who dies within 14 days of the onset of symptoms from an undetermined cause, OR stillbirths);
- treat similar symptoms that accompany the disease, especially malaria, typhoid, fever, louse-borne typhus, recurrent fever or leptospirosis ;
- Implement infection prevention and control measures and avoid nosocomial transmission through strict implementation of barrier measures. In the absence of barrier measures, avoid invasive procedures (e.g., blood collection, injections, placement of infusion lines or nasogastric tubes) and put on at least one layer of gloves for direct patient contact; double gloving is recommended during invasive procedures (e.g., surgery) that present an increased risk of blood exposure;
- There is no specific treatment for either disease. Severe cases require intensive supportive care, as patients are frequently dehydrated and in need of intravenous fluids or oral rehydration with solutions containing electrolytes.
- For Ebola Virus Disease, a range of potential treatments, including blood products, immunotherapies and drug therapies are being evaluated.

Analysing and Interpreting data: Ebola or Marburg virus diseases

Time: Graph cases and deaths daily or weekly. Construct an outbreak curve during the outbreak.
Place: Map locations of case houses.
Personal information: Immediately transmit case and death information on a case-based basis. Analyse age and sex distribution. Assess risk factors and plan disease control interventions accordingly.

Laboratory confirmation Ebola or Marburg virus diseases	
Diagnostic tests	Laboratory-confirmed cases must test positive for Ebola or Marburg virus antigen, either by detection of virus RNA by reverse transcriptase polymerase chain reaction (RT-PCR) or by detection of IgM antibodies to Ebola or Marburg virus.
Specimens	For ELISA: Whole blood, serum or plasma For RT-PCR: Whole blood or blood clot, serum or plasma or tissue For immunohistochemistry: skin or tissue specimens from deceased cases. **Note: Rapid diagnostic tests can theoretically be performed in any health care facility and without additional equipment, but the use of a rapid diagnostic test can produce both false positive and false negative test results. A nucleic acid-based diagnostic test (e.g. PCR), such as GeneXpert, should be used to confirm the result of the rapid diagnostic test. Recent WHO guidance recommends that rapid diagnostic antigen tests for viral haemorrhagic diseases (VHD) play no role in the routine management of these diseases in settings where PCR testing is available. However, they can be useful in environments without laboratory infrastructure and where samples cannot be quickly transported to a diagnostic laboratory, if their advantages and limitations are well understood.**
When to collect the specimens	Collect specimen from the first suspected case. If more than one suspected case, collect until specimens have been collected from 5 to10 suspected cases.
How to prepare, store, and transport the specimens	HANDLE AND TRANSPORT SPECIMENS FROM SUSPECTED VHF PATIENTS WITH EXTREME CAUTION. WEAR PROTECTIVE CLOTHING AND USE BARRIER PRECAUTIONS. *For ELISA or PCR:* ▪ Refrigerate serum or clot ▪ Freeze (at -20°C or lower) tissue specimens for virus isolation *For Immunohistochemistry:* ▪ Fix skin snip specimen in formalin. They can be stored for up to six weeks at room temperature. The specimen is no more infectious once it is fixed in formalin. ▪ Formalin-fixed specimens may be transported at room temperature.

Laboratory confirmation

Results	Diagnostic services for Viral Haemorrhagic Fever (VHF) are not routinely available. Arrangements are usually required. Contact the appropriate National authority or WHO.

References

- WHO Interim Guidelines -Case Definitions Recommendations for Ebola and Marburg Virus diseases. August 9 2014
- Interim Infection Control Recommendations for Care of Patients with Suspected or Confirmed Filovirus (Ebola, Marburg) Haemorrhagic Fever. BDP/EPR/WHO, Geneva March 2008.
- Infection control for VHF in the African health care setting, WHO, 1998. WHO/EMC
- WHO Recommended Surveillance Standards WHO/CDS/CSR/ISR/99.2
- WHO Fact Sheet No 103, Ebola haemorrhagic fever, revised in December 2008
- WHO Fact Sheet, Marburg haemorrhagic fever, revised in July 2008
- Interim Infection Control Recommendations for Care of Patients with Suspected or Confirmed Filovirus (Ebola, Marburg) Haemorrhagic Fever. BDP/EPR/WHO, Geneva March 2008.
- WHO recommended Guidelines for Outbreak Preparedness and Response: Ebola Haemorrhagic Fever (EHF). WO/EMC/DIS/97.7
- Dengue Haemorrhagic Fever: diagnosis, treatment, prevention and control. 2nd Edition, Geneva, World Health Organisation 1997.

Epilepsy

Background

- Epilepsy is defined as the recurrence of at least two epileptic seizures with sudden occurrence of abnormal signs which could be: motor, tonic, sensitive, sensorial, neuro-vegetative, or psycho-behavioural. These symptoms could or could not be associated to a loss of conscience. It can appear at any age.
- Epilepsy is the most common result of brain cells disturbance that lead to excessive nerve-cell discharges. Depending on the disturbance on some or many groups of cells, seizures could be partial or generalized
- Seizures with tonic-clonic muscle movements are named convulsion or attack. Convulsions can appear at any age. All convulsions are not systematically epilepsy seizure.
- Epilepsy is frequent in Africa where its prevalence rate ranges from 2.2 to 58 per 1000. Studies in five sub-Saharan African countries have shown an incidence of between 64 and 156 per 100,000 persons/year.
- This high incidence could be the consequence of predisposing factors, such as inadequate perinatal care, head trauma, consanguinity.
- Many etiological factors are also related to communicable diseases (malaria, tuberculosis, meningitis, neurocysticercosis, and HIV), non-communicable diseases (hypertension, diabetes, alcoholism, and drug abuse), poor medical infrastructure, poor health in general, and low standard of living. Misunderstanding due to religious beliefs, stigmatization and exclusion of people with epilepsy do not facilitate access to appropriate care.
- The risk of mortality increases significantly in people with epilepsy, especially when the disease is detected late because of the lack of qualified health care personnel capable of diagnosing and treating neurological disorders.
- In epileptics, the causes of death and injury are primarily related to *status epilepticus* (especially in cases of abrupt cessation of treatment), burns and drowning.

It has been estimated that in developing countries, up to 80% of people with epilepsy are not receiving treatment, or are often not even identified. Although the etiological diagnosis of epilepsy is more difficult in developing countries due to lack of investigative resources, it is still possible to diagnose many epileptics based on simple medical and epidemiological knowledge.

Standard case definition

Suspected case: Any person with one epileptic seizure
New suspected case: Report only the first diagnostic of the case to the health facility.
Confirmed case:
Any person with recurrence of, at least, two epileptic seizures. Any positive response to treatment with an antiepileptic drug reinforces the hypothesis of a confirmed case. Seizures can last from 30 seconds to three minutes. When they are intricate without a pause, they can lead to *status epilepticus*.

Responding to alert threshold

Suspected case:

- All health care personnel should look for the warning signs of epilepsy. Diagnosis should include good interviews (describing as precisely as possible the seizure type) and clinical examination.
- Once diagnosed, search for underlying and associated causes. Check for abnormal increases on number of cases and propose appropriate environmental measures if needed.

Confirmed case:

- Treatment should be given immediately, starting with low doses of antiepileptic drugs and gradually increasing until a stable condition is achieved. If the seizure is difficult to control, increase the dose or try another antiepileptic drug, refer the patient to a higher level of care.
- Referral to higher level health structure should be done if seizures continue regardless of pharmacological treatment or if first seizure occurs in an adult aged 30 and above.

Responding to outbreak or intervention threshold

All cases: Conduct information and education activities on epilepsy and risk factors at community level

Analysing and Interpreting data

Time: Graph quarterly cases
Place: Plot the distribution by area of residence
Personal information: Analyse the distribution of cases by gender and age (by age group beyond six years).

Laboratory confirmation

Diagnostic test	▪ Random blood glucose (capillary and venous blood), electrolyte assay to exclude other conditions such as diabetes and renal pathology. ▪ Perform appropriate medical tests to exclude other conditions such as cerebral malaria, meningitis, toxoplasmosis, cerebral calcifications following tuberculosis (cerebral tuberculoma), parasitic diseases and others
Specimens	Blood and cerebrospinal fluid
When to collect the specimens	Glucose - Sampling during emergency patient admission (random blood glucose) Subsequent sampling for confirmation (fasting blood glucose)
How to prepare, store, and transport the specimens	Apply universal precautions to minimize the risk of exposure to blunt instruments and body fluids.
Results	Results are usually available one to three hours after the specimens arrive at the laboratory.

References :

- WHO, Epilepsy in the WHO African Region: Bridging the Gap, WHO Regional Office for Africa, Congo, 2004.
- WHO, Epilepsy: a manual for medical and clinical officers in Africa, WHO, Geneva 2002.

Foodborne Illnesses

Background

- Foodborne illnesses are caused by a variety of bacterial, viral, parasitic and fungal pathogens or their toxins that enter the body through the ingestion of contaminated food or water. In addition to diseases such as cholera and shigellosis (discussed in this guide), foodborne disease surveillance may reveal other causes, such as salmonella, hepatitis A virus or chemical contamination.
- A foodborne illness occurs when two or more people have consumed the same food or drink and then develop similar symptoms within the same time period.
- Most people with a foodborne illness do not seek medical care, so cases and outbreaks of foodborne illness usually are neither recognized nor reported.
- The first symptoms often occur in the gastrointestinal tract with nausea, vomiting, abdominal cramps and diarrhoea.
- Foodborne illness outbreaks can be localized, affecting only two people who have consumed the same meal or product. They can also be significant and geographically widespread, when a food is contaminated before distribution and is widely consumed by many people in many areas.
- Foodborne disease surveillance monitors food safety and targets health promotion efforts among food handlers for better food practices and personal hygiene.

Surveillance goal

- Promptly identify any unusual clusters of potentially foodborne illness that may require a public health investigation or response.
- Assess the extent of foodborne illnesses.
- Identify high-risk foods or food practices.
- Monitor risk factors to target public health interventions and health promotion actions to specific foods and dietary practices.

Standard case definition

A foodborne illness is suspected when at least two people who have consumed the same food or drink show similar symptoms.

The case definition of a foodborne disease depends on the specific agent that causes it (e.g. cholera, hepatitis A, salmonellosis, shigellosis).

A foodborne illness is confirmed when the laboratory confirms the presence of a particular agent in a common food or drink source.

Responding to alert threshold

If at least two people are ill and they have consumed food from the same source:

- immediately report suspected cases to the highest level of the health system;
- Collect specimens from patients and that of suspected food and beverage products for laboratory confirmation;
- Treat suspected cases

Respond to outbreak or intervention threshold

If an outbreak of a foodborne illness is confirmed:

- Search for additional cases in the locality of confirmed cases;
- Strengthen case management and treatment;
- raise community awareness on early detection and prompt treatment of cases;
- Identify high risk groups
- Remove from the restaurant menu or the supermarkets shelves, food items from which evidence of unsafe food may be obtained
- initiate a thorough investigation of food chains that may be associated with the outbreak;

- reduce sporadic cases and outbreaks of foodborne illness by promoting hand washing with soap and water after using the toilet and before handling food and eating; by increasing access to safe drinking water, use of latrines, and safe human waste disposal systems;
- Intensify food safety health promotion activities using the WHO manual, Five Keys to Safer Food (see reference below), and the Hazard Analysis and Critical Control Point (HACCP) system;
- Scale-up food inspection activities

Analysing and Interpreting data

- **Time:** Graph monthly trends in cases and deaths; construct an outbreak curve during a foodborne illness outbreak.
- **Place:** Plot location of houses for cases and deaths
- **Personal information**: Count cases and deaths each month. During an outbreak, count outbreak-related cases each week.
- Regularly review clinical data and laboratory results of case and food testing to identify clustered cases in time, space or personal data. Investigate any suspected foodborne illness outbreaks based on the data.
- Investigate all suspected foodborne illness outbreaks.

Reference

WHO Foodborne disease outbreaks: Guidelines for investigation and control
http://www.who.int/foodsafety/publications/foodborne_disease/outbreak_guidelines.pdf

High blood pressure

Background

- Hypertension or high blood pressure (HBP) is a chronic condition in which blood pressure is high. It is classified as primary hypertension (the majority) and secondary hypertension. “Primary” Hypertension is elevated blood pressure where no medical cause is found. “Secondary” Hypertension is caused by other conditions that affect the arteries, heart, endocrine system or kidneys
- Hypertension is an important risk factor for cardiovascular disease (heart attack and stroke). According to the World Health Report 2001, deaths from cardiovascular disease are increasing in the African Region, where they represent 9.2% of total deaths in 2000. Prevalence rates range from 25% to 35% among adults aged 25 to 64 years.
- Hypertension affects approximately one billion people worldwide and it is estimated that over 20 million people are affected in Africa.
- Age, lack of physical activity, obesity and a diet high in salt and fat are the main risk factors for hypertension. Alcohol and tobacco use are also risk factors.
- Lifestyle changes can reduce blood pressure: weight loss in overweight or obese individuals, decreasing salt and fat intake in the diet, increasing consumption of fresh fruits and vegetables, increasing physical activity, and reducing tobacco and alcohol consumption.

Surveillance goal

- Provide secondary prevention through early detection of hypertension and standardized treatment.
- Estimate the burden of disease and reduce identified risk factors.
- Monitor hypertension prevention and control activities.

Standard case definition

New suspected case at first visit:

Anyone with a resting blood pressure (measured over an average of three readings, five minutes apart) ≥140 mm Hg for systolic pressure, or ≥90 mm Hg for diastolic pressure.

Confirmed case

Anyone who, on at least two occasions, has a resting blood pressure (three readings) ≥140 mm Hg for systolic pressure, or ≥90 mm Hg for diastolic pressure.

Report only the first diagnosis of the case in the health facility

Recommended public health action

- Promote health for non-communicable diseases, particularly hypertension, including education to the community about behavioural changes and healthier lifestyles.
- Promote secondary prevention and treatment interventions at HF according to national guidelines.

Analysing and Interpreting data

Time: Graph cases quarterly to analyse trends.

Place: Compare district trends with regional and national trends.

Personal information: Analyse the distribution of cases by age and other demographic factors.

**** Data for non-communicable diseases is often analysed for long term trends***

Laboratory confirmation

Diagnostic is clinical.

Reference

- *WHO, Atlas of heart disease and stroke, Geneva, World Health Organisation, 2004*
- *Non communicable Diseases: A strategy for the African Region, AFR/RC50/10*
- *Cardiovascular Diseases in the African Region: Current situation and perspectives, AFR/ RC55/12 http://www.who.int/chp/steps/en/ http://www.afro.who.int/dnc/databases/afro infobase/index.html*
- *WHO CVD-risk management package for low-and medium resource settings.*

- *« The Seventh Report of the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure » U.S. Department of health and Human Services, National Institutes of Health, National Heart, Lung, and Blood Institute, NIH Publication No. 03-5233, December 2003*
- *Handbook of Hypertension, Vol 20. Editor; C.J. Bulpitt, 2000 http://www.cdc.gov/bloodpressure/*

Influenza caused by a new sub-type

Background

- An influenza pandemic occurs when a new influenza A virus emerges and spreads efficiently and sustainably from person to person in populations with low immunity to the virus. Influenza pandemics occurred in 1918, 1957 and 1968. It is estimated that the one that occurred in 1918 killed between 40 million and 50 million people. A pandemic of the same magnitude is expected to kill 62 million people, 96% of them in developing countries.
- Influenza caused by a new sub-type has been reported in: i) human infections caused by A virus (H5N1) HPAI virus in 1997; ii) the A influenza pandemic (H1N1) in 2009; and iii) human infections caused by the A virus (H7N9) in 2013. Other avian influenza viruses that have caused sporadic human infections include A (H7N7) and A (H9N2). Some countries have also reported sporadic human infections with swine influenza viruses, particularly A (H1) and A (H3) subtypes.
- Successful mitigation or control of pandemic influenza is determined by the early detection of sustained human-to-human transmission of a new influenza A virus. Countries have been encouraged, as part of pandemic preparedness planning, to enhance surveillance to: i) detect the emergence of a new disease; ii) characterize the disease (epidemiology, clinical manifestations, severity); and iii) monitor its progress and take control measures.
- **Under the International Health Regulations (IHR) (2005), a State Party must immediately notify WHO of any laboratory-confirmed case of recent human infection with influenza A virus of pandemic potential. No evidence of disease is required for this report.**

Surveillance goal

- Detect and investigate any first observation of sustained human-to-human transmission of an influenza virus with pandemic potential.
- Study the first cases of an influenza pandemic occurring in a country, in order to characterize the new disease, including its clinical characteristics, risk factors and epidemiological and virological features.
- Monitor the evolution of the pandemic within the country, but also regionally and globally.

Standard case definition

1- For infections with other non-seasonal influenza viruses, case definitions must be adapted to the situation. The following case definitions are proposed for future adaptation:

- ***Suspected case***: Fever (temperature >38°C) **and** [cough **or** shortness of breath or difficulty breathing] with onset of symptoms within the past 10 days in a person who has had one or more of the following epidemiologic exposures during the two weeks prior to symptom onset in [Area X] since/during [date Y/date Y to Z[b]] The different types of exposures are ;
    - Close contact (within one meter) with a person who is a suspected, probable or confirmed case;
    - exposure to animals or their remains or environments contaminated by their excrement in an area where cases of non-seasonal influenza infections in animals or humans have been suspected or confirmed in the past month;
    - consumption of raw or undercooked animal products in an area where there have been suspected or confirmed cases of influenza infection in animals or humans in the past month;
    - Close contact with a confirmed influenza-infected animal;
    - handling specimens suspected of containing non-seasonal influenza virus in a laboratory or other environment
- **Probable case**

  Any suspected case with **either:**
    - positive laboratory confirmation of influenza A virus infection, but insufficient laboratory evidence for the subtype; or
    - a person dying from an unexplained acute respiratory illness that is considered to be epidemiologically linked to a probable or confirmed human case of non-seasonal influenza.

- **Confirmed case:** laboratory confirmation of recent infection[d] with non-seasonal influenza virus in a person.

- **Rejected case:** a suspected or probable case with a negative test for non-seasonal influenza virus.

[b] *When a case has been confirmed, set the onset date at least 28 days (two maximum incubation periods) before the start of the first confirmed case.*
[c] *Whose non-seasonal influenza virus test results are accepted by WHO for confirmation.*
[d] *An infection is considered recent if it has been confirmed by positive polymerase chain reaction (PCR) results, virus isolation, or paired acute and convalescent serologic tests. A single serum antibody titer is often not sufficient to confirm a recent infection and should be evaluated according to valid WHO case definitions for human infections with specific influenza A subtypes.*

2- For some subtypes of zoonotic influenza, **there are specific case definitions such as for H5N1 and H7N9.**

- Link to WHO H5N1 case definitions: ***http://www.who.int/influenza/resources/documents/case_definition2006_08_29/en/***
- Link to WHO H7N9 case definitions: ***:*** ***https://www.who.int/influenza/human_animal_interface/influenza_h7n9/en/***

3- IHR case definition of human influenza caused by a new subtype

- An influenza A virus is considered to have the potential to cause a pandemic if the virus has demonstrated the ability to infect a human being and if the hemagglutinin gene (or protein) is not a variant or mutated form of these, i.e., A/H1 or A/H3, that is widely circulating among the human population. An infection is considered recent if it has been confirmed by positive polymerase chain reaction (PCR) results, virus isolation, or paired acute and convalescent serologic tests. A single serum antibody titer is often not sufficient to confirm a recent infection and should be evaluated according to valid WHO case definitions for human infections with specific influenza A subtypes.

Respond to alert threshold: Influenza caused by a new subtype

Responding to a suspected case of human influenza caused by a new subtype or an unusual event of severe acute respiratory infection:

Survey triggers

Examples of triggers include:

- respiratory disease in humans that is associated with recent exposure to animals;
- groups[1] of acute respiratory infection[2] (SARI) or pneumonia in families, workplaces or social networks;
- severe respiratory tract infection occurring in a health care worker caring for patients with respiratory diseases;
- severe respiratory tract infection or pneumonia in travellers from countries or regions affected by new acute respiratory infections;
- severe respiratory tract infection occurring in a laboratory worker or researcher handling new influenza and other emerging respiratory pathogens;
- higher than expected number of hospitalizations or deaths due to respiratory diseases;
- Laboratory detection of human infection with a non-seasonal influenza virus or a new respiratory pathogen;
- Unexplained sudden changes in trends in respiratory disease occurrence or clinical outcomes observed in routine surveillance activities; and
- Unusually high levels of sales of pharmaceutical drugs used to treat respiratory diseases that cannot be explained by known or expected disease patterns.

[1] *A "group" is defined as a collection of two or more people who show symptoms in the same 14-day period and who are associated with a particular setting, such as a classroom, workplace, household, extended family, hospital, other residential facility, military barracks, or recreational camp.*
[2] *A severe respiratory tract infection (SARI) is an acute respiratory infection with a history of fever or measured fever of ≥38°C with cough, where the onset of symptoms has occurred within the past 10 days and requires hospitalization.*

Key steps in a survey: Influenza caused by a new subtype

- Preparing for the survey:
    - build a multidisciplinary survey team;
    - inform the competent authorities;
    - collect information and materials.
- Investigate the first reported cases.
- Protecting surveyors.
- Develop case definitions.
- Find other cases:
    - identify and trace case contacts;
    - search for active cases.
- Strengthen surveillance.
- Collect specimens.
- Undertake animal health and environmental surveys.
- Manage and analyse data (time, location, personal data).
- To address certain public health issues that may require further study.
- Implement response and control measures:
    - manage patients ;
    - prevent further transmission;
    - prevent and control infections;
    - communicate risk;
    - monitor the event and the response.
- Report and inform:
    - report the results of the survey;
    - Inform local, sub-national and national public health authorities.

Respond to outbreak or intervention threshold

If a single case of human influenza caused by a new subtype is confirmed or if another acute respiratory illness with outbreak or pandemic potential is confirmed:

- manage patients ;
- prevent any other transmission;
- prevent and control infections;
- communicate on risk;
- monitor the event and the response: an event is considered contained if active surveillance in the risk population has not resulted in new cases for twice the presumed incubation period for that disease.

For more details, see the *WHO Protocol for the Study of Non-seasonal Influenza and Other Emerging Acute Respiratory Diseases*, 2018. https://www.who.int/influenza/resources/publications/outbreak_investigation_protocol/en/

Analysing and Interpreting data

1-Manage data

- use a line list; and
- Establish procedures for recording and validating data.

2-Analyze data

Time: Draw an outbreak curve, with the number of cases per week on the y-axis, and the date or time of onset of illness on the x-axis. Construct secondary outbreak curves by case classification status (suspected, probable and confirmed cases), death status, exposure types, etc.

These curves can provide information on the magnitude of the event, patterns of spread and exposure, temporal trend of the event, incubation period of the disease, type of exposure, outliers, impact of implemented interventions.

Place: Map out cases by geographic location, e.g. by village, by home, or by location in a health facility. The maps can be local, regional or national, depending on the geographical distribution of the event.

Visual interpretation of the maps can provide important etiological clues, identify clusters, and provide details on the geographic extent of disease spread. Two types of cards should be used:

- dot map - use dot maps to assess the likely pattern of spread (clustered cases, dispersion of cases, etc.);
- area map - use area maps to take into account the underlying population at that location (allowing direct comparison of incidence rates between sites, regions, etc.).

Personal information:

To identify the clinical spectrum and dynamics of the disease, it is necessary to analyse :

- epidemiological and clinical parameters of the cases;
- attack rates by age, gender, occupation and exposure history; and
- for clinical parameters, the level of severity of illness, including the proportion of cases with pneumonia, cases requiring hospitalization, hospitalizations in the intensive care unit and the proportion of cases who died.

For more details, see the *WHO Protocol for the Study of Non-Seasonal Influenza and Other Emerging Acute Respiratory Diseases*, 2018 https://www.who.int/influenza/resources/publications/outbreak_investigation_protocol/en/

Laboratory detection and confirmation

1-Collection and handling of specimens.

The list of specimens to be collected to test for respiratory disease pathogens includes: sputum, bronchoalveolar lavage, tracheal aspirate, nasopharyngeal aspirate, nasal lavage, nasal and pharyngeal smears, nasopharyngeal swab, biopsy or autopsy tissues including lung, serum, whole blood and urine. All these types of specimens should be stored at 4°C and sent to the national reference laboratory for influenza. If testing for influenza is to be performed within 48 hours, specimens should be stored at 4°C, and stored at -70°C if testing is to be performed within more than 48 hours. When the etiology of the event is unknown, it is useful to take a variety of specimens when possible to maximize the possibility of detection and characterization.

2-Testing of specimens

Various laboratory techniques can be used to identify human influenza virus infections:

1) detection of influenza-specific RNA by reverse transcriptase polymerase chain reaction (RT-PCR);
2) isolation in cell culture ;
3) direct detection of the antigen (low sensitivity).

If influenza is suspected as the causative agent, a specific protocol provides an algorithm for laboratory testing with RT-PCR (see references).

Laboratory detection and confirmation

Reference

WHO protocol for the study of non-seasonal influenza and other emerging acute respiratory diseases, 2018

WHO Fact Sheet on Avian Influenza and Other Zoonotic Diseases, 2018

WHO Guidance for Surveillance during an Influenza Pandemic, Updated 2017

WHO summary of key practical information for countries affected by outbreaks of influenza A(H5N1) and other avian influenza subtypes, 2016

Prevention and control of outbreak and pandemic prone acute respiratory infections in the WHO Health Care Guidelines, 2014

WHO manual for laboratory diagnosis and virological surveillance of influenza, 2011

WHO Operational Guidelines on Sharing Influenza Viruses with Human Pandemic Potential (IVPP) in Pandemic Influenza Preparedness (PIP)

WHO operational guidelines on sharing seasonal influenza viruses with WHO Collaborating Centres (CCs) as part of the Global Influenza Surveillance and Response System (GISRS)

WHO Standard guidance for the clinical management of influenza infections, to be published in 2019.

WHO Collaborating Centres for Influenza Contacts can be found on the WHO website, link *: https://www.who.int/influenza/gisrs*

WHO reference laboratories for contact diagnosis of influenza A/H5 infection: *https://www.who.int/influenza/gisrs*

Page dedicated to the WHO health section on influenza: (http://www.who.int/influenza/en/)

Influenza-like illness

Background

- Respiratory infections are a significant cause of infectious disease morbidity and mortality in the world. The mortality rates are particularly high among infants, children and the elderly. However, the burden of disease is not well characterized in Africa.
- The most common pathogens causing respiratory infections are *Streptococcus pneumoniae*, *Haemophilus influenzae type b (Hib)*, *Staphylococcus aureus* and other bacterial species, Respiratory Syncytial Virus (RSV), measles virus, human parainfluenza viruses type 1, 2, and 3 (PIV-1, PIV-2 and PIV-3), influenza virus and varicella virus.
- An better understanding of the epidemiology and seasonality of respiratory infections in Africa is essential for optimizing public health strategies for their prevention and control (e.g., vaccines and antivirals for prophylaxis and treatment, infection control).
- The threat of respiratory infections due to new organisms that have outbreak or pandemic potential warrants special precautions and preparedness.
- Surveillance of respiratory infections, primarily viral infections is based on the case definition of influenza-like illness.

Surveillance goal

- Describe the seasonality of influenza.
- Report the beginning and end of the influenza season.
- Establish baseline or average rates of influenza and severe influenza-related illness.
- Describe the viruses in circulation.
- Identify locally circulating virus types and subtypes and their relationship to global and regional trends.
- Monitor antiviral susceptibility
- Identify and monitor high-risk groups for serious illness and infection-related complications.
- Help understand the relationship between viral strains and disease severity.
- Produce influenza data that can be used to estimate the burden of influenza.
- Detect unusual and unexpected events.

Standard case definition

An acute respiratory infection with:

- fever measured at a temperature ≥ 38 ºC;
- cough;
- onset of symptoms within the last 10 days;
- not requiring any admission.

Responding to alert threshold

Please refer to the *WHO Protocol for the Study of Non-Seasonal Influenza and Other Emerging Acute Respiratory Diseases*, 2018, if there is an unusual respiratory infection event (e.g., acute respiratory infection or atypical respiratory infection clusters, death cluster).

Responding to outbreak or intervention threshold

Please refer to the *WHO Protocol for the Investigation of Non-Seasonal Influenza and Other Emerging Acute Respiratory Diseases*, 2018, if a single case of acute respiratory disease with pandemic tendencies is suspected.

Analysing and Interpreting data

Time: Reports Frequency: Epidemiological and virological data collected at sentinel sites should be analysed **weekly.** Graph the cases each week. Draw an outbreak curve for the entire year and describe transmission trends and changes in the level of respiratory activity compared to the previous week(s), year(s)

Place: Map cases by geographic location e.g. by village, by home, or by site within a health care facility.

Personal information: For patients with influenza-like illness who have been tested for influenza viruses, **the minimum data to be collected and analysed for each patient**, especially if a specimen is collected are: unique identifier (to link laboratory and epidemiologic data), sex, age, history of fever and body temperature at presentation, date of symptom onset, date of specimen collection, use of antivirals for current illness at the time of specimen collection, pregnancy status, presence of pre-existing chronic disease(s) (chronic respiratory disease, asthma, diabetes, chronic heart disease, chronic neurologic or neuromuscular disease, hematologic disorders, HIV). Data on influenza-like illness can be grouped by age group to facilitate analysis and reporting. The main **age groups** recommended **for reporting** are 0 to <2 years, 2 to <5 years, 5 to <15 years, 15 to <50 years, 50 to <65 years, and ≥ 65 years.

For laboratory data, it is recommended that, as a minimum, the following data be collected:

- number of specimens tested for influenza within the week;
- proportion of positive specimens for influenza-like illness;
- virus types and subtypes detected during the week;
- Antiviral resistance test results (if applicable).

Finally, the following **indicators or aggregate data should be collected and reported from each sentinel site**:

1. the number of new influenza-like illness cases from which specimens were collected during the week, grouped by standard age group, and the proportion of each that tested positive for influenza;
2. the total number of new ILI cases reported during the week, grouped by standard age group (including cases that were not tested or for which detailed data were not collected);
3. the total number of new out-patient visits during the week in the out-patient clinics where ILI surveillance is conducted or the population served by the sentinel site, ideally grouped according to the recommended age groups;
4. the proportion of cases with each of the pre-existing chronic conditions for ILI cases testing positive for influenza.

More technical information on the role of the laboratory can be found at the following website:

- WHO global epidemiological surveillance standards for influenza, 2014. *https://www.who.int/influenza/resources/documents/influenza_surveillance_manual/en/*

Laboratory test

1- For the influenza virus:

- Specimens may be positive seven or more days after illness onset, but the ability to detect the virus decreases significantly after five to seven days, depending on the test used;
- Reverse transcriptase polymerase chain reaction (RT-PCR) is the most sensitive method for detecting influenza virus and is the recommended influenza surveillance test for most laboratories;
- viral culture is also required on at least a subset of specimens to allow detailed antigenic and genetic characterization of the virus;
- Antiviral resistance testing should be considered for high-risk patients if the laboratory has the capacity to do so in addition to sampling non-high-risk patients.

More technical information on the role of the laboratory can be found at the following websites:

- WHO global epidemiological surveillance standards for influenza, 2014. *https://www.who.int/influenza/resources/documents/influenza_surveillance_manual/en/*
- WHO manual for laboratory diagnosis and virological surveillance of influenza, 2011. https://www.who.int/influenza/gisrs_laboratory/manual_diagnosis_surveillance_influenza/en/

***2- For other viruses and pathogens, please consult the specific WHO guidelines on pathogens*.**

References

- *WHO Global Standards for Influenza Epidemiological Surveillance, 2014.*
- *AFR generic protocol for influenza sentinel surveillance,* 2015https://afro.who.int/publications/protocol-national-influenza-sentinel-surveillance
- Protocol for the Investigation of Outbreaks of Acute Respiratory Illness of Unknown Aetiology *https://afro.who.int/publications/protocol-investigation-acute-respiratory-illness-outbreaks-unknown-etiology*
- *WHO Fact Sheet on Seasonal Influenza, 2018*

- *WHO protocol for the study of non-seasonal influenza and other emerging acute respiratory diseases, 2018*
- *Prevention and control of outbreak and pandemic prone acute respiratory infections in the WHO Health Care Guidelines, 2014*
- *WHO manual for laboratory diagnosis and virological surveillance of influenza, 2011*
- *WHO operational guidelines on sharing seasonal influenza viruses with WHO Collaborating Centres (CCs) as part of the Global Influenza Surveillance and Response System (GISRS)*
- *WHO Operational Guidelines on Sharing Influenza Viruses with Human Pandemic Potential (IVPP) in Pandemic Influenza Preparedness (PIP)*
- *WHO Standard guidance for the clinical management of influenza infections, to be published in 2019.*
- *Page dedicated to the WHO health section on influenza: (http://www.who.int/influenza/en/)*

Trauma (due to road traffic accidents)

Background

- Trauma is physical damage caused to a person when his or her body has been subjected to sudden or brief levels of energy beyond its physiological tolerance. It can also be a functional deficiency, resulting from a deprivation of one or more vital elements (water, air, heat). The energy causing the trauma can be mechanical, electrical, thermal, ionizing radiation or chemical. Trauma is divided into two groups: intentional and unintentional.
- Trauma accounts for 10% of deaths worldwide. 5.8 million people die each year as a result of trauma. Of all the systems that people deal with on a daily basis, road transport is the most complex and dangerous.
- Road traffic accidents cause unintentional trauma.
- A road traffic accident (motor vehicle collision, car accident) occurs when a vehicle collides with another vehicle, a pedestrian, an animal, debris on the road, or any other geographical or architectural obstacle. Road traffic accidents can cause trauma, property damage and death.
- Worldwide, the number of people killed in road traffic accidents each year is estimated at 1.2 million, while the number of injured could be as high as 50 million.
- Road traffic accidents are a major but neglected global public health problem requiring concerted efforts to prevent them sustainably and effectively.
- Road traffic crashes remain one of the leading causes of death for individuals aged 5 to 44 years and the leading cause of death for those aged 15-29 years. The majority of these deaths currently involve "vulnerable road users" - pedestrians, cyclists and motorcyclists.
- Without increased efforts and new initiatives, the total number of road traffic deaths worldwide and injuries is forecast to rise by some 67% by 2020, and in low income and middle-income countries deaths are expected to increase by as much as 83%
- The African Region has the highest mortality rate attributed to traffic accidents, with 32 per 100,000 inhabitants.

Traffic injuries and significant reductions in injuries and fatalities can be prevented by implementing measures to address risk factors (excessive and inappropriate speed, drunk driving, failure to use seat belts and child restraints, failure to use helmets for cyclists)

Surveillance goal

- To estimate and monitor the incidence of traffic injuries and related outcomes
- Identify risk factors and locations to guide prevention policies and programmes.
- Evaluate programmes to prevent injuries from road traffic accidents.
- Establish alert thresholds for fatalities to allow HFpersonnel review care and services provided to injured persons.
- Establish incidence alert thresholds and monitor trends to enable district health personnel inform relevant stakeholders

Standard case definition

Road traffic accident injury: any person suffering from a traffic accident injury, seen in consultation for the first time.

Traffic accident fatality: Any person killed immediately or dying within 30 days because of an injury crash.

Responding to alert threshold

- Promote primary prevention by supporting measures to address risk factors.
- Review and monitor the services and care provided to the injured
- Review arrangements for handling large numbers of victims.

Responding to outbreak or intervention threshold

- Intensify implementation of measures to address risk factors.
- Activate the mass casualty management system.

Analysing and Interpreting data
Personal information: Analyse the distribution of cases by gender, age and other demographic factors **Time:** Graph the monthly injury and fatality figures and plot them over the year to describe trends. **Place:** Map the location of cases and identify high-risk areas.
Laboratory confirmation
Radiological examination of the injured person - when necessary.
References
▪ *World Health Report*, 2004, WHO ▪ *WHO- 2010 Status report on Road Safety in Africa*, 2010, WHO ▪ Peden, M. et. al (eds), *World Report on Road Traffic Injury Prevention,* 2004, WHO ▪ Holder Y., Peden M., Krug E. et. al (eds), *Injury Surveillance Guidelines*, 2001, Geneva, WHO ▪ Harvey A, (Ed)., *Data systems*, Geneva, World Health Organisation, 2010

Lassa and Crimean-Congo Haemorrhagic Fevers

Background

- Crimean-Congo haemorrhagic fever (CCHF) is caused by a virus belonging to the *Bunyaviridae* family and Lassa fever by a virus of the *Arenaviridae* family.
- CCHF is endemic in parts of Africa and outbreaks have been reported in Uganda, Mauritania and South Africa. Each year, Mauritania reports a few cases. South Africa reported 165 laboratory-confirmed cases between 1981 and March 2006.
- Lassa fever is known to be endemic in Guinea, Liberia, Nigeria, and Sierra Leone, but is probably also present in other West African countries. According to some studies, there are between 300,000 and 500,000 cases per year in West Africa, of which 5,000 are fatal.
- CCHF is transmitted to humans either by tick bite or by direct contact with contaminated animal tissue immediately after slaughter.
- The animal reservoir of Lassa virus is a rodent of the genus Mastomys. The infection does not make him sick, but he can excrete the virus (urine and faeces). Humans usually become infected through aerosols or direct contact with the excreta of infected rodents. The virus can also be transmitted from person to person through direct contact with blood, nasopharyngeal secretions, urine, faeces or other biological fluids of an infected person.
- Human-to-human transmission of Crimean-Congo and Lassa Fevers has occurred in HF after exposure to blood or secretions of infected patients.
- The incubation period for CCHF following a tick bite is usually one to three days (maximum nine days). It is five to six days (maximum 13 days) after contact with blood or tissue. The incubation period for Lassa fever ranges from 6 to 21 days.
- The onset of CCHF symptoms is brutal: fever, myalgia and other signs and symptoms. The case fatality rate is between 3% and 30%.
- About 80% of human Lassa virus infections are mild or asymptomatic. The remaining 20% have severe multi-organ damage. In symptomatic patients, the disease progressively manifests itself through fever, generalized weakness and a feeling of malaise. Lassa fever is difficult to distinguish from many other febrile illnesses, such as malaria, shigellosis, typhoid fevers, yellow fever and other viral haemorrhagic fevers. The overall case fatality rate is between 1% and 15% in hospitalized patients.
- Management of CCHF cases is primarily based on supportive systemic therapy. Intensive monitoring for volume and blood component replacement is essential. The antiviral drug ribavirin has been used to treat established cases of CCHF. It appears that both oral and intravenous formulations are equally effective. Ribavirin is also effective in the treatment of Lassa fever, if given early in the clinical course of the disease.

Surveillance goal

- Detect cases and outbreaks of haemorrhagic fevers as early as possible, investigate promptly, and verify the aetiology of all suspected cases in the laboratory immediately.
- Investigate all suspected cases, including contact tracing.
- Assess and monitor the spread and evolution of outbreaks and the effectiveness of control measures.

Standard case definition

***Suspected CCHF case*:** Anyone with sudden onset of fever, malaise, generalized weakness, irritability, headache, severe pain in the limbs and lumbar region, and marked anorexia. Early congestion of the face and chest, injected conjunctiva, haemorrhagic enanthema of the soft palate, uvula, and pharynx, and very often a rash of fine petechiae extending from the chest and abdomen to the rest of the body, sometimes with large purpuric areas.
Confirmed CCHF case: Laboratory-confirmed suspect case (positive IgM serology, positive PCR, virus isolation or IgG seroconversion by ELISA or IFA) or with an epidemiological link to confirmed cases or an outbreak

Suspected case of Lassa fever: Any person with a progressive illness with at least one of the following: malaise, fever, headache, sore throat, cough, nausea, vomiting, diarrhoea, myalgia, chest pain, hearing loss; and a history of contact with rodent excreta or with a confirmed case of Lassa fever

Confirmed case of Lassa fever: Laboratory-confirmed suspect case (positive IgM serology, positive PCR, virus isolation) or epidemiologically linked to a laboratory-confirmed case

Responding to alert threshold

If there is only one presumed case:

- immediately report the case information to the appropriate levels;
- Isolate suspected cases from other patients and strictly apply isolation care techniques;
- Strengthen standard infection control precautions throughout the medical environment;
- treat and manage the patient by providing supportive care;
- Collect specimens for case confirmation;
- Monitor the case contacts and actively seek out other cases.

Responding to outbreak or intervention threshold

If a single case is confirmed:

- maintain strict infection control measures* for the duration of the outbreak;
- mobilize the community for rapid case detection and management, and conduct community-based education on disease transmission and infection control in home-based care. For CCHF, inform the public about the mode of transmission by ticks. For Lassa fever, intensify rodent control measures;
- actively seek out other cases;
- Request additional help from other levels as needed.
- create an isolation unit to handle additional cases that may come to the HF.

Analysing and Interpreting data

- **Personal information:** Immediately communicate case and death information on a case-based basis. Analyse the distribution according to age and gender. Assess risk factors and organize disease control actions accordingly.
- **Time:** Graph cases and deaths daily or weekly. Draw an outbreak curve during outbreaks.
- **Place:** Map locations of cases' houses.

Laboratory confirmation

Diagnostic test	Presence of IgM antibodies against CCHF or Lassa fever
Specimens	*For ELISA*: Whole blood, serum or plasma *For PR*: Whole blood or blood clot, serum or plasma or tissue *For Immunohistochemistry:* Skin or tissue specimens from deceased cases
When to collect the specimens	Collect specimen from the first suspected case. If there is more than one suspected case, collect specimens from 5 to 10 suspected cases.
How to prepare, store, and transport the specimens	HANDLE AND TRANSPORT SPECIMENS FROM PATIENTS WITH PRESUMED VIRAL HEMORRAGIC FEVER WITH EXTREME CAUTION. WEAR PPE AND USE BARRIER PRECAUTIONS. *For ELISA or PCR:* ▪ refrigerate the serum or clot; ▪ Freeze (at -20°C or lower) tissue specimens for virus isolation. *For Immunohistochemistry:* ▪ Fix skin snip specimen in formalin. Specimens can be stored for up to six weeks. Once fixed in formalin, specimens are no longer infectious; ▪ Formalin-fixed specimens can be stored and transported at room temperature.
Results	Diagnostic services for viral haemorrhagic fevers are not always available. Early arrangements should usually be made. Contact the appropriate National competent authority or WHO.

References : Lassa and Crimean-Congo haemorrhagic fevers

- *Interim Infection Control Recommendations for Care of Patients with Suspected or Confirmed Filovirus (Ebola, Marburg) Haemorrhagic Fever. BDP/EPR/WHO, 2008.*
- *Infection control for viral haemorrhagic fever in African hospitals, WHO, 1998. WHO/EMC*
- *Ergonul O. Crimean-Congo haemorrhagic fever. Lancet Infect Dis 2006, 6:203-14.*
- *WHO Recommended Surveillance Standards WHO/CDS/CSR/ISR/99.2*
- *WHO Fact Sheet No. 208: Crimean-Congo haemorrhagic fever, revised November 2001*
- *WHO Fact Sheet No. 179: Lassa fever, revised April 2005*

Leprosy

Presentation

- Leprosy is a chronic mycobaterial disease that affects the skin, peripheral nerves and mucous membranes of the upper airways. It is mainly transmitted by air from the nasal secretions of patients infected with Hansen's bacillus (Mycobacterium leprae), but also by inoculation into a skin lesion. Leprosy is endemic in several tropical regions around the world, especially in Africa.
- Depending on the presence of cutaneous and nervous signs, two groups of patients are distinguished:
  - multibacillary (MB) patients with more than five skin spots and several nerve thickenings;
  - paucibacillary (PB) patients with one to five skin spots and a single nerve thickening.
- The fight against leprosy has improved considerably thanks to multidrug therapy (MDT) recommended by the WHO. MDT combining two or three drugs (rifampicin, clofaximin and dapsone) is indeed very effective in the treatment of leprosy. At the end of 1999, the prevalence of the disease in African countries was 1.6 cases per 10,000 inhabitants, with nearly 70,000 cases recorded. Seventeen years later, at the end of 2016, this prevalence rate has been reduced to 0.25 cases per 10,000 inhabitants and less than 25,000 cases have been recorded.
- The incubation period lasts from six months to 20 years or more. Infection is probably common, but clinical illness is rare, even among the closest contacts of patients. Multibacillary patients are highly contagious. However, the infectious nature decreases rapidly, from the start of MDT. The disease sometimes presents complications (neuritis and leprosy reactions) leading to deficiencies and infirmities in the hands, feet and eyes.
- Historically, leprosy was synonymous with social exclusion and had significant psychosocial consequences. This social stigma still persists in some African countries.
- Certain skin diseases, such as pytiriasis versicolor, fungal infections, vitiligo, scleroderma, psoriasis, systemic lupus erythematosus and Von Recklinghausen's disease can be confused with leprosy.

Purpose of surveillance

- Monitor national trends towards the leprosy elimination goal, defined as reducing the prevalence rate to less than one new case with grade two disabilities per 1,000,000 population.
- Constantly monitor the resistance of Hansen's bacillus to the drugs used for MDT.
- As leprosy is on the verge of elimination, complement routine surveillance with community surveillance activities, including active case finding among household contacts of leprosy patients, especially during administration mass drugs or vaccination campaigns.

Standard case definition

Suspicious case:

An individual showing at least one of the three essential signs of leprosy: hypo-pigmented or reddish skin lesion, loss or reduction of skin sensitivity, thickening of the peripheral nerve.

***Confirmed case*:**An individual with at least two essential signs of leprosy who has not yet completed multidrug therapy (MDT).

Responding to the alert threshold: Leprosy

If only one case is suspicious:

- report the suspected case to the appropriate level of the health system;
- investigate the case for risk factors;
- begin appropriate care:

-- *multibacillary (MB) patients: treatment for 12 months with a MDT combining three drugs (12 strips of MB to be taken for 18 months).*

-- *sickpaucibacillary (PB): treatment for six months with a MDT combining two drugs (six packs of PB to be taken for nine months)*

Respond to the epidemic or intervention threshold

If a suspected case is confirmed:

- examine patients for skin and nerve signs at each medical consultation to diagnose and treat reactions and deficiencies caused by leprosy;
- determine the risk factors for treatment interruption (eg insufficient supplies of MDT in the health facility, difficult access to the villages where the patients reside, etc.). Give patients unable to go to a health center every month a quantity of platelets sufficient for the complete treatment;
- identify any rapid increase or decrease in the number of new cases over a period of time. Assess the quality of surveillance in areas where underreporting or overreporting is suspected. Control the distribution of MDT drugs.

Analyze and interpret data

Time : Make a chart of cases, showing dates of diagnosis and start of treatment.
Place : Map cases according to where they live and disease classification (MB or PB)
Individual characteristics: Make a monthly count of new cases detected, according to the type of leprosy (multibacillary disease or paucibacillary disease). Analyze their distribution according to age, degree of disability and treatment results (recovery, failure, relapse).

Laboratory confirmation

Leprosy surveillance does not require routine laboratory confirmation.

References

- *Global Leprosy Strategy 2016-2020 (SEA-GLP-2016.2)*
- *WHO recommended surveillance standards WHO/CDS/CSR/ISR/99.2*

Lymphatic filariasis

Presentation

- Lymphatic filariasis is the second leading cause of permanent or prolonged disability worldwide. Of the 120 million people affected in 80 countries, more than 40 million are severely disabled by the disease. The risk of infection threatens 20% of the world's population. About a third of those infected live in India, another third in Africa, and the rest in the Americas, Asia and the Pacific. In 1997, Resolution WHA50.29 called for the elimination of lymphatic filariasis as a global public health problem. The adopted strategy is based on:
  - reducing transmission below a threshold at which new infections cease to appear;
  - taking into account the problems associated with preventing and combating disabilities.
- Causative agents: in Africa, only the filariae Wuchereria bancrofti, Brugia malayi and Brugia timori are involved.
- Modes of transmission: the disease is transmitted by different species of mosquitoes. Filiform parasitic worms take up residence in the human lymphatic system, where they produce millions of immature microfilariae that circulate in the blood. Microfilariae appear in the peripheral blood three to six months after infection for Brugia malayi, six to 12 months later for Wuchereria bancrofti, very often with a nocturnal periodicity. When a mosquito bites an infected person, it ingests microfilariae which develop in its body to reach the infective stage after about two weeks.
- Clinical picture:
  - the infection may have no external clinical manifestations (even if the laboratory confirms lymphatic and renal damage). It can also be manifested by one or more acute symptoms (fever, localized swelling, tropical eosinophilic lung, lymphangitis).
- Chronic complications:
  - lymphedema or elephantiasis of the limbs;
  - genital damage (including hydrocele in men);
  - kidney damage (including chyluria) and damage to the lymphatic system.

Purpose of SURVEILLNCE

There are currently three monitoring options depending on the local situation:

1. regular monthly reporting of summary data on suspected and confirmed cases from peripheral level to intermediate level and central authority;
2. surveys conducted in sentinel populations (standardized and periodic);
3. active case finding in surveys of selected population groups or in mass surveys. At the international level: annual notification from the central authority to the WHO (for a limited number of countries).

Standard case definition

Suspicious case:
An individual with clinical signs of hydrocele or lymphedema, and residing in an endemic area, after excluding any other cause.
Confirmed case:
An individual with a positive laboratory diagnosis: microfilaraemia on a blood smear, filaria antigens or positive ultrasound.

Respond to alert threshold: Lymphatic filariasis

- confirm the prevalence of infection in the community through surveys.

Meet the epidemic or intervention threshold

Case management
Hygiene measures of the affected body parts (and when necessary, antibiotic and antifungal treatments) can reduce the risk of adenolymphangitis:

- wash affected parts with soap and water twice a day;
- elevate affected limb at night;

- exercise the limbs to stimulate lymphatic circulation;
- keep fingernails short and clean;
- wear comfortable shoes;
- use antiseptic or antibiotic creams to treat small wounds or abrasions, or in severe cases, use systemic antibiotics.

For the treatment of individuals with microfilariae, the protocol recommended by the country should be followed:

- in regions where there is neither onchocerciasis nor loiasis: a single dose of 6 micrograms per kilogram of diethyl carbamazine citrate (DEC);
- in regions where onchocerciasis is excluded, but not loiasis: individual clinical decision.

The current filariasis control strategy relies mainly on antiparasitic measures. Interrupting transmission requires treatment of the entire population at risk. To do this, an annual dose of medication will be given as follows:

Areas endemic for onchocerciasis:

- 400 micrograms (mg) albendazole + 150 micrograms ivermectin per kilogram (kg) body weight, once a year, for four to six years.

Areas without onchocerciasis:

- 6 mg of diethylcarbamazine per kg of body weight + 400 mg of albendazole, once a year; Or
- cooking salt fortified with diethylcarbamazine for daily use, for at least six to 12 months.

NOTE: in regions also affected by loiasis (rain forest in sub-Saharan Africa), systematic mass interventions cannot be envisaged (unless onchocerciasis represents a serious public health problem), because of the risk of severe side effects in patients heavily infected with loa loa (approximately one in 10,000 treatments).

It is essential to educate the population on the importance of good adherence to treatment during mass chemotherapy. With regard to lymphatic filariasis, it is not necessary to promote specific vector control efforts. These efforts should be part of existing vector control programs, such as malaria vector control operations.

Analyze and interpret data	
- Map the distribution of lymphatic filariasis cases and identify implementation units that will require mass drug administration. - Analyze the drug coverage of the implementation units. - Assess the decrease in parasitological indices of microfilaremia, before starting MDA and after at least four rounds of MDA, until the criteria of 1% microfilaremia in the population is reached and less than 0.1% antigenemia in school-aged children.	
Diagnostic tests	Blood smear from an overnight collection Filariasis antigen test
Specimens	Blood smear Blood
When to take samples	At night between 10:00 p.m. and 2:00 a.m. At any time of the day
How to prepare, store and transport samples	Place three drops of blood on a glass slide and spread them out to make three lines. After heat fixation, stain Giemsa and observe under the microscope. The antigen is tested either by an immunochromatographic test (ICT) on a rapid card or by an ELISA test in the laboratory.
Results	Positive result in case of microscopic observation of microfilariae of W. bancrofti. Positive result in case of detection of microfilariae antigens.

References

- *WHO. Epidemiological monitoring and evaluation of the lymphatic filariasis elimination program at the level of the implementing units WHO/CDS/CPE/CEE/2005.50*
- *WHO. Lymphatic filariasis. WHO/CDS/CPE/SMT/2001.7http://whqlibdoc.who.int/hq/2001/WHO_CDS_CPE_SMT_2001.7_eng.pdf*
- *WHO. Training module on lymphatic filariasis for drug distributors (in countries where onchocerciasis is not co-endemic). WHO/CDS/CPE/CEE/2000.10 (parts 1 and 2)*
- *WHO. The lymphatic filariasis elimination program - essential elements for medical personnel (in countries where onchocerciasis is co-endemic). WHO/CDS/CPE/CEE/2000.11 (parts 1 and 2)*
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- *WHO. The lymphatic filariasis elimination program - essential elements for medical personnel (in countries where onchocerciasis is co-endemic). WHO/CDS/CPE/CEE/2000.15*
- *WHO. The lymphatic filariasis elimination program - essential elements for medical personnel (in countries where onchocerciasis is co-endemic). WHO//CDS/CPE/CEE/2000.16 Webpage:www.who.int/lymphatic_filariasis*

Malaria

Presentation

Malaria is a tropical and endemic disease with fever as the main symptom. It follows the bite of an infected female Anopheles mosquito which transmits the parasite. Five species of parasites cause malaria in humans, namely: Plasmodium falciparum (the most common), P. ovale, P. vivax, P. malariae and P. knowlesi. Serious malaria infections are usually due to P. falciparum, which can lead to severe illness.

Malaria is one of the leading causes of morbidity and mortality in many African countries. In most parts of Africa, malaria transmission is highly seasonal. In areas of high transmission, malaria is primarily a disease of children under five and pregnant women. However, some countries have seen a dramatic reduction in malaria transmission and in these countries it has become a disease that affects all age groups and epidemics are likely to occur.

The incubation period, from mosquito bite to onset of symptoms, ranges from seven to 30 days. It may be longer, especially with species other than P. falciparum.

Purpose of monitoring

- Rapidly detect malaria epidemics in areas of high transmission and rapidly detect epidemics in epidemic-prone areas or in areas with a high population at risk.

Standard case definition

Uncomplicated malaria:THEmalariasimple to P. falciparum is highly variable and has symptoms similar to those of many other diseases. Although fever is common, it is often intermittent and may even be absent in some cases. The fever is usually irregular at first and usually associated with chills. Solemn chills are unusual in acute falciparum malaria. The patient often complains of fever, headache, aches and pains elsewhere in the body, and sometimes abdominal pain and diarrhea. In a young child, irritability, refusal to eat and vomiting can be observed. On physical examination, fever may be the only sign. In some patients, the liver and spleen are palpable. This clinical picture is usually clinically indistinguishable from that of influenza and a variety of other common causes of fever. If the disease is not diagnosed and treated quickly, a patient with falciparum malaria can see their condition deteriorate rapidly. As a result,anyone living in a malaria risk area with fever or a history of fever in the previous 24 hours, without signs of severe malaria, who has a positive rapid diagnostic test or microscopic examination result, should be considered a case of uncomplicated malaria (Note: WHO currently recommends that all cases of malaria be confirmed by rapid diagnostic test or microscopic examination).

Severe malaria:THEmalariaSevere is defined by clinical or laboratory signs of dysfunction of vital organs. Almost all deaths from severe malaria result from P. falciparum infections. Strict definitions of severe malaria have been published for epidemiological and research purposes, but in practice there should be a low threshold for starting parenteral therapy in any patient about whom a health worker is concerned. Even if some of the laboratory measures are not immediately available, this should not delay the start of intensive treatment. A general overview of the characteristics of severe malaria includes:

1. the alteration of theconsciousness (including a lethargic coma);
2. prostration, which is generalized weakness that prevents the patient from sitting, standing, or walking without help;
3. multiple seizures: more than two episodes in 24 hours;
4. deep breathing and respiratory distress (acidotic breathing);
5. acute pulmonary edema and acute respiratory distress syndrome;
6. circulatory collapse or shock, systolic blood pressure: < 80 mm Hg in adults and < 50 mm Hg in children;
7. acute kidney injury;
8. clinical jaundice and signs of other vital organ dysfunction; and abnormal bleeding.

Note : These manifestations may occur singly or, more commonly, in combination in the same patient.

Hospitalized patient with laboratory-confirmed P. falciparum parasitaemia (asexual forms) with signs and symptoms of severe disease (laboratory-diagnosed vital organ dysfunction).

Respond to alert threshold

If there is an unusual increase in the number of malaria cases or deaths compared to the same period, in previous non-epidemic years:

- report suspected outbreaks to higher level;
- treat cases with appropriate antimalarial drugs according to national treatment guidelines;
- investigate the cause of the increase in the number of cases;
- ensure that cases in children aged two months to five years are managed according to the guidelines of the Program for the Management of Childhood Illnesses (IMCI);
- conduct community education activities to quickly detect cases and ensure access to health facilities.

Meet the epidemic or intervention threshold

If the number of new cases exceeds the upper limit of the number of cases observed in the same period, in previous non-epidemic years:

- assess and improve, if necessary, prevention strategies, such as the use of insecticide-treated mosquito nets (ITNs) and indoor residual spraying (IRS) for all those exposed malaria;
- ensuring appropriate case management;
- ensure adequate availability of supplies and drugs in health facilities.

Analyze and interpret data

Time : Graph the number of cases by month or week. Draw an epidemic curve during epidemics.
Place : Map where new cases and deaths live.
Individual characteristics: Count the number of new malaria cases and deaths per month, and analyze their distribution by age and period of onset of illness.

Laboratory Confirmation: Malaria	
Diagnostic tests	▪ Microscopic examination: presence of malaria parasites in blood smears of suspected cases. ▪ Malaria rapid diagnostic test: presence of malaria antigen.
Specimens	Blood In general, collect with vaccinopens at any age or with other methods approved for blood collection in very young children.
When to take samples	*For the blood smear:*prepare a smear for all suspected cases hospitalized, or as per national malaria case management guidelines.
How to prepare, store and transport specimens	*Blood smear :* ▪ collect blood directly onto clean, properly labeled microscope slides; prepare thin smears and thick smears; ▪ let the smears dry well; ▪ use the appropriate staining technique; ▪ store the stained and perfectly dried slides at room temperature, taking care not to expose them to direct light. *For the rapid diagnostic test:* ▪ *take the samples and perform the test following the manufacturer's instructions.*
Results	Thick film and thin film results can be obtained the same day of collection. Microscopic examination of slides for malaria may also reveal the presence of other blood parasites. The result of the rapid diagnostic test is immediate. **Noticed**: In a hospital setting, do a hemoglobin test to confirm severe anemia in children aged two months to five years.
References	
▪ *Malaria epidemics: Detection and control, forecasting and prevention. Geneva. World Health Organization. WHO/MAL/98.1084.* ▪ *"Basic Laboratory Methods in Medical Parasitology" WHO, Geneva, 1991.*	

Malaria: continued…

Note: Determine an endemic threshold:

In areas where malaria is endemic, the National Malaria Control Program can help DSs and FOSAs to determine appropriate thresholds allowing them to detect possible epidemics. In the absence of a threshold defined by the national programme, the following method can be used to determine the threshold for detecting a malaria epidemic. The threshold is determined using the average and the 3rd quartile of a given period (for example, monthly or weekly data over five years from a FOSA or a DS):

1. record the number of monthly or weekly cases of malaria in a specific healthcare facility or DS, over the past five years;
2. determine the average for each month or each week (for example, each January for the last five years). Rank monthly or weekly data for each of the five years in ascending order. Identify the number in the middle of each monthly or weekly series over the past five years. This number corresponds to the median. Repeat this process for each month or each week, for the five years;
3. determine the 3rd quartile of each monthly or weekly series, by identifying the fourth highest number in each series, starting from the bottom (as long as the data is ranked in ascending order). The 3rd quartile represents the upper limit of the normal number of expected malaria cases;
4. plot on a diagram the 3rd quartile of each monthly or weekly data series, over these five years, and connect the dots. The line obtained represents the upper limit of the normal number of cases expected;
5. plot on a diagram the average of each monthly or weekly data series, over the five years, and connect the dots. The resulting line represents the lower bound of the expected number of cases;
6. the area between the two lines (between the median and the 3rd quartile) represents the “normal”. If the number of currently observed malaria cases falls between the two lines, the number of new cases for that month or week is considered “normal”. If the number is above the 3rd quartile (upper limit), this is an indication of a possible malaria epidemic.

   Note: To ensure early detection and control of malaria epidemics, it is best to use weekly surveillance data in epidemic-prone geographic areas.

   In areas in the pre-elimination or elimination phase of malaria, a single case of locally transmitted malaria should prompt proactive interventions, including active case finding in the locality of origin of the case.

Source: WHO/AFRO Regional Malaria Control Program

Malnutrition

Presentation

- Globally, undernourishment of women and children remains the underlying cause of 3.5 million deaths. In particular, it accounts for 35% of the disease burden in children under five. There are 23 African countries among the 40 countries in which the prevalence of underweight is 40% or more.
- Severe malnutrition can be a direct cause of death, but also an indirect cause, as it dramatically increases the mortality rate in children suffering from common childhood illnesses, such as diarrhea or pneumonia.
- Despite these facts, the burden of child mortality due to severe malnutrition is rarely on the international health agenda, and few countries, even in areas of high prevalence, have national policies to address this issue comprehensively.
- Children under five, pregnant and breastfeeding women are the most vulnerable. The poor nutritional status and poor diet of pregnant women can contribute to the birth of low birth weight babies (weight measured immediately after birth). A newborn weighing less than 2500 grams (2.5 kilograms or 5.5 pounds) is considered a low birth weight newborn. This low birth weight is a major determinant of infant and child mortality, morbidity and disability. It also has an impact on adult health.
- The main causes of malnutrition include socio-economic conditions, lack of access to safe drinking water and sanitation systems, poor nutritional education of mothers regarding infant and young child feeding, and infections repeated
- Programs to eradicate malnutrition are based on food security, drinking water supply and sanitation, promotion of infant and young child feeding practices, micronutrient supplementation programs, management of serious cases of malnutrition within the communities and in the health structures, the management of infections, in particular diarrheal diseases.
- Many sporadic studies are organized, but to date, nutrition surveillance is very little applied and does not allow taking measures for the prevention and management of malnutrition.

Purpose of surveillance

- Early warning and problem identification.
- Decision making and action planning.
- Program management and evaluation.
- Verification of the effectiveness of Public Health measures aimed at remedying the causes of low birth weight and malnutrition in children and pregnant women.

Standard case definition: Malnutrition

Low birth weight:

Newborn with a birth weight of less than 2500 grams (or 5.5 pounds)

Malnutrition in children:

- underweight children under five (height-weight index:ZScore weight for age<-2);
- children aged 6 to 59 months with an arm circumference <11.5 centimeters (high risk of mortality);
- bilateral edema of the extremities.

Malnutrition in pregnant women:

Pregnant women giving birth to babies with low birth weight < 2.5 kg (poor nutritional status and poor health status of mothers predict which population groups are likely to benefit from improved antenatal care for women and neonatal care for infants).

Respond to alert threshold

If more than 20% of children are underweight:

The program should emphasize:

- breastfeeding support;
- nutrition education;
- food supplements for children and mothers;
- prevention and treatment of diarrhoea;
- prevention and treatment of severe malnutrition;
- socioeconomic support.

Upon detection of a case with MUAC < 11.5 centimeters (cm) or upon identification of bilateral oedema:

Alert, carry out additional examinations and refer the child to a therapeutic feeding program.

If 15% or more of low birth weight infants weigh less than 2.5 kilograms:

Target interventions to improve prenatal care for women and neonatal care for infants, including nutritional care (anti-tobacco and anti-alcohol campaigns, nutritional care for women before and during the prenatal period and during breastfeeding, malaria prophylaxis, newborn care facilities, etc.) for women who are at risk of complications during pregnancy and childbirth, and treat newborns, in order to prevent morbidity and mortality.

Analyze and interpret data

Time : Graph the number of monthly cases to analyze trends, and the number
of weekly cases in emergencies.

Place : Map homes and communities with cases.

Individual characteristics: Count monthly or weekly cases and analyze their distribution by age and gender.

Laboratory confirmation

Surveillance does not require routine laboratory confirmation.

References

- *BlackRE et al. Maternal and child undernutrition: global and regional exposures and health consequences. Lancet 2008; 371: 243–260.*
- *Gross R, Webb P, Wasting time for wasted children: severe child undernutrition must be resolved in nonemergency settings. Lancet 2006; 367: 1209-1211.*
- *Physical status: the use and interpretation of anthropometry. Report of a WHO Expert Committee. WHO Technical Report Series, 1995, No 854: 81, 128-130, 198-208.*
- *WHO growth standards and identification of severe acute malnutrition in children. Joint statement by the World Health Organization and the United Nations Children's Fund*

Maternal deaths

Presentation

- Maternal deaths include maternal deaths that occur during or late in pregnancy, during childbirth, and within six weeks (42 days) after childbirth or termination of pregnancy. (Note: Deaths due to accidental or fortuitous causes are not considered maternal deaths)
- Typically, about 80% of maternal deaths are caused by severe bleeding (most often after childbirth), infections (mostly immediately after childbirth), hypertensive disorders during pregnancy (eclampsia), and obstructed labor . Complications after unsafe abortion account for 13% of maternal deaths.
- In developing countries, maternal mortality remains unacceptably high, with more than 500,000 women dying each year from complications during pregnancy and childbirth. About half of these deaths occur in sub-Saharan Africa, where the lifetime risk of maternal death is one in 22 women, compared to one in 8,000 women in industrialized countries.
- Haemorrhages are the leading cause of maternal mortality in sub-Saharan Africa, with a particularly high risk for unassisted deliveries, especially in rural areas where transport to health facilities is a problem.
- Reporting on the Sustainable Development Goals (SDGs) in 2030 requires active monitoring and counting of maternal deaths. The report is no longer proportionate as it was in the Millennium Development Goals (MDGs) (Reduce maternal mortality by 75%), but countries will instead report a real number – in the sense that no country should have a maternal mortality ratio (MMR) >70 deaths per 100,000 live births.
- Review of progress towards the fifth millennium development goal indicates that most African countries were not able to achieve this MDG in 2015. Intensive actions and increased investments are needed to improve coverage and quality of maternal health care services and addressing the issues and factors that contribute to these deaths is essential if we are to achieve the MDGs.

Purpose of surveillance

1. Active surveillance for improved and accurate identification and reporting of maternal deaths at community and health facility levels.
2. Estimate and monitor maternal mortality rates.
3. Identify risk factors and geographical areas at high risk of maternal mortality, in order to guide program orientations.
4. Evaluate programs aimed at reducing maternal mortality.

Standard case definition

Death of a woman occurring during pregnancy or within 42 days after its term, regardless of the duration or location of the pregnancy, for any cause related to or aggravated by the pregnancy or its management, but that is neither accidental nor fortuitous.

Respond to alert threshold

- After determining that a woman's death occurred during pregnancy or within 42 days of her term, initial notification of presumed death should be made immediately (within 24 hours), by the quickest means possible .
- Each maternal death is significant, placing the alert threshold at ONE (1).
- The facility should contact the DS authority and provide information on the Integrated Disease Surveillance and Response (IDSR) case alert form. Additionally, the HF's Maternal Death Review Committee is required to review the case within seven days.
- The initial notification should be followed by a written report using a maternal death review form, which should be shared with the maternal death review coordinator at the district or regional levels.
- Maternal death review should be anonymous and unrelated; and reports should not be used for disciplinary purposes in the event of a dispute.
- Initial notification should be followed by a written report using a maternal death review form or case investigation form.

Recommended Public Health Action

- In matters of reproductive health, any death of a woman should be investigated in order to rule out pregnancy and thus establish whether or not it is a maternal death. .
- Maternal death surveillance should not only be done in delivery rooms, but also in the community and in all service areas where women are seen or die.
- Monitor trends and respond to any maternal deaths based on recommendations arising from the maternal death review.
- Increase the availability and use of antenatal care and skilled attendance at birth.
- Implement essential high-impact, evidence-based interventions for maternal health.
- Educate and engage communities in emergency preparedness and preparedness for complications, including evidence-based nutrition and dietary interventions for safe pregnancy and childbirth.
- Address socio-cultural norms and practices that negatively impact maternal health.
- Ensure >80% emergency obstetric care coverage with recommended signaling functions by level of care.

Analyze and interpret data

Time : Graph cases and draw an epidemic curve over the year to identify trends.
Place : Map the location of cases and analyze their geographical distribution.
Individual characteristics: Analyze the distribution of cases by age and other demographic factors.

Laboratory confirmation

Surveillance does not require routine laboratory confirmation.

References

- *WHO. Maternal mortality :http://www.who.int/makingpregnancy safer/topics/maternal*
- *mortality/en/index.html; ICD MM;*
- *http://apps.who.int/iris/bitstream/handle/10665/70929/9789241548458_eng.pdf;jsessionid=862B3C6054CED65E30EDE6605FFAEDF4?sequence=1*
- *WHO Technical Guide for Maternal Death Surveillance and Response; MEBC guide*
- *UNICEF: http://www.unicef.org/index.php*

Measles

Presentation

- Measles is a febrile rash disease, caused by a virus of the Paramyxoviridae family (Morbillivirus) which is transmitted by air, from one person to another, by respiratory droplets suspended in the air. In many African countries, measles is the fourth leading cause of death among children under 5 years old.
- The incubation period lasts from 7 to 18 days, between exposure and the onset of fever.
- In children suffering from malnutrition and vitamin A deficiency, infection with the measles virus can cause serious illness due to the virus itself and bacterial superinfections, in particular pneumonia; severe cases are a minority.
- Measles is one of the most contagious human diseases. Major epidemics occur each year in areas where vaccination coverage is low and where there is a high concentration of people who have never been infected or vaccinated. The actual incidence of measles far exceeds the reported cases.
- Risk factors include the low vaccination coverage (<85 to 90%) which allows the concentration of a large number of people at high risk of contracting measles. Measles outbreaks can be explosive in areas with high population density.
- Other viral diseases, such as rubella, can contribute to or cause outbreaks similar to measles.

Purpose of surveillance

- Detect outbreaks of febrile rash illness early:

in the WHO African Region, in line with the regional measles elimination goal: immediate notification, on a case-by-case basis, of suspected cases of febrile rash illness and of deaths attributable to febrile rash illness; laboratory confirmation (usually IgM serology) of all suspected cases.

Standard case definition

- ***Suspicious case***: anyone with fever, generalized maculopapular (non-vesicular) rash and cough or a cold or conjunctivitis (red eyes), or anyone in whom a doctor suspects measles.
- ***Confirmed case***: Suspected case confirmed by the laboratory (positive serology of IgM) or having an epidemiological link with confirmed cases or an epidemic outbreak.

Respond to alert threshold

If an outbreak is suspected:

- notify the suspected case to the higher level;
- take blood samples to confirm the outbreak;
- treat cases with oral rehydration salts, vitamin A, and antibiotics to prevent secondary bacterial infections. If possible, take aerial precautions;
- Investigate the case or outbreak to identify the causes.

Meet the epidemic or intervention threshold

If there is confirmation of an outbreak:

- improve routine immunization coverage as part of the EPI, and organize supplementary immunization activities in areas with low immunization coverage;
- mobilize the community as soon as possible to enable the rapid detection and treatment of cases;
- give vitamin A:
    - dose 1: immediately, dose 2: the next day
    - age: 0-6 months = 50,000 IU, 7-11 months = 100,000 IU; 12 months and over =200,000 IU

Analyze and interpret data

Time : chart weekly cases and deaths. Plot the epidemic curve of cases linked to the outbreak.
Place : map where the cases live.
Individual characteristics:count the total number of cases and analyze their distribution according to age group and vaccination status.

Laboratory confirmation
Diagnostic tests: research of IgM directed against the measles virus.

Specimens :serum
Whole blood, gum fluid, throat swab

When to take samples

- Collect between the 3rd and the 28th day after the onset of the eruption.
- Take blood samples from the first 5 suspected measles cases, when the number of cases exceeds the measles epidemic threshold (usually more than 5 cases per month, in a district).
- In countries that have set a measles elimination goal:
- Take a swab for each suspected measles case.
- Take a serum sample for antibody testing, at the first opportunity or at the first consultation in FOSA.

How to prepare, store and transport specimens

- In children, collect 1 to 5 ml of venous blood, depending on the size of the child. Take the sample in a test tube, capillary tube or microvessel.
- Separate serum from red blood cells. Allow the blood clot to retract at room temperature for 30 to 60 minutes. Centrifuge at 2000 rpm for 1020 minutes and pour the serum into a clean glass tube.
- If no centrifuge is available, refrigerate sample overnight (4-6 hours) until clot retracts. The next morning, pour the serum into a clean tube.
- In the absence of a centrifuge and refrigerator, tilt the sample for at least 60 minutes (without shaking or transporting in a vehicle). Pour the serum into a clean tube.
- Store sera at 4°C.
- Transport serum samples in appropriate packaging to avoid any risk of breakage or leakage during transport.

Results :samples should be delivered to the laboratory within 3 days of collection. Results are usually available after 7 days.

If 2 out of 5 suspected measles cases are laboratory confirmed, the outbreak is confirmed.

Avoid shaking blood samples before the serum has been separated.

To avoid bacterial growth, pour the serum into a clean glass tube. The tube does not have to be sterile, it just needs to be clean.

Transport serum specimens in a refrigerated EPI vaccine carrier at 4°C to 8°C to avoid bacterial growth. If there is no possibility of refrigeration, the serum can be stored for a minimum of 3 days, in a clean tube.

References

- *"Response to meal outbreaks in meal mortality reduction settings" http://apps.who.int/iris/bitstream/10665/70047/1/WHO_IVB_09.03_eng.pdf*
- *WHO–recommended standards for surveillance of selected vaccine-preventable diseases (2018). http://www.who.int/immunization/monitoring_surveillance/burden/vpd/WHO_SurveillanceVaccinePreventable_11_Measles_R1.pdf?ua=1*
- *World Health Organization. Regional Office for Africa. African Regional guidelines for measles and rubella surveillance - Revised, April 2015. http://www.afro.who.int/index.php?option=com_docman&task=doc_download&gid=10814&Itemid=2593*

Middle East Respiratory Syndrome (MERS)

Presentation

- Middle East Respiratory Syndrome (MERS) is a viral respiratory disease caused by a novel coronavirus (Middle East Respiratory Syndrome Coronavirus -MERS-CoV) first detected in 2012 in Saudi Arabia.
- Coronaviruses are a large family of viruses that can cause a variety of illnesses, ranging from the common cold to severe acute respiratory syndrome (SARS).
- Common symptoms of MERS include fever, cough, and difficulty breathing. The presence of pneumonia is frequent, but not systematic. Gastrointestinal symptoms, including diarrhea, have also been reported. Some laboratory-confirmed cases of MERS-CoV infection are reported as asymptomatic, which means they do not show clinical symptoms while testing positive for MERS in a laboratory test. Most of these asymptomatic cases were detected following very active contact tracing of a lab-confirmed.
- Approximately 35% of reported cases of MERS-CoV infection result in patient death.
- Camels are the main reservoir host for the virus and humans are infected through direct or indirect unprotected contact with infected camels. However, the precise role that these animals play in the transmission of the virus and the exact mode(s) of transmission are not known.
- The virus does not appear to spread easily from person to person unless there is close contact with an infected person, such as when providing unprotected care to a patient. Healthcare-related outbreaks have occurred in several countries, with the largest seen in Saudi Arabia, the United Arab Emirates and the Republic of Korea.
- Approximately half of human cases of MERS are attributed to human-to-human infections.

Purpose of surveillance

- Detect early cases of MERS-CoV infection and any signs of prolonged human-to-human transmission
- Determine geographic areas at risk of virus infection

Standard case definition

http://www.who.int/csr/disease/coronavirus_infections/technical-guidance-surveillance/en/

The following people should be investigated and tested for MERS-CoV (extract fromSurveillance of human infection with Middle East respiratory syndrome coronavirus (MERS-CoV))

1. Anyone with acute respiratory infection, with a history of fever and cough, and evidence of lung parenchymal disease (eg, pneumonia or acute respiratory distress syndrome), based on clinical or radiological evidence, who requires hospitalization, with no other etiology fully explaining the clinical picture[5](clinicians should also consider the possibility of an atypical presentation in immunocompromised patients);

   AND meet one of the following criteria:

   a. *the patient resides in the Middle East[6], particularly where human infections have been reported, and in countries where MERS-CoV is known to circulate in camels;*

   b. *the patient is part of a household[7]cases of acute respiratory illness that has occurred within the previous 14 days, regardless of place of residence or travel history;*

   c. *the patient is a healthcare worker who has worked in an environment where patients with severe acute respiratory infection have been cared for, regardless of place of residence or travel history;*

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[5] Testing should follow local guidelines for the management of community acquired pneumonia. Other etiologies may be Streptococcus pneumoniae, Haemophilus influenzae type b, Legionella pneumophila, other recognized primary bacterial pneumonias, influenza, and respiratory syncytial virus.

[6] For a map of the Middle East, see: http://www.un.org/Depts/Cartographic/map/profile/mideastr.pdf

[7] A “household” is defined as a group of two or more people in whom symptoms appeared during the same 14-day period and who are associated with a specific environment (classroom, workplace, household, extended family , hospital, other residential establishment, military barracks or holiday camps).

d. *the person has an unusual or unexpected clinical course, including a sudden worsening of their condition despite appropriate treatment, regardless of place of residence or travel history, even if another etiology has been identified that fully explains the clinical picture*.

2. Person with acute respiratory infection, with a history of fever and cough, and indications of lung parenchymal disease (eg, pneumonia or acute respiratory distress syndrome), based on clinical or radiological evidence, and who has traveled within 14 days of illness onset in the Middle East2 or in countries where MERS-CoV is known to circulate in camels or where human infections have recently been reported.
3. Person presenting with an acute respiratory illness, whatever its severity, who, in the 14 days preceding the onset of the illness, found himself in one of the following exposure situations (note: see the section "Recommendations screening in homes associated with health facilities”):

a. *close physical contact[8]with a confirmed or probable case of MERS-CoV infection at the time of illness;*

b. *healthcare facility in a country with reported hospital-associated MERS-CoV infections;*

c. *direct contact with a camel or consumption or exposure to camel products (raw meat, unpasteurized milk, urine) in countries where MERS-CoV is known to circulate in camel populations or where human infections have occurred. produced as a result of suspected zoonotic transmission.*

4. Countries in the Middle East2 are also strongly encouraged to consider adding MERS-CoV testing to current testing algorithms, as part of routine sentinel respiratory disease surveillance, and expert diagnostic panels for the pneumonia.

Case definitions for MERS-CoV can be viewed at https://www.who.int/csr/disease/coronavirus_infections/case_definition/en/

Confirmed case definition

Anyone with a laboratory-confirmed MERS-CoV infection regardless of clinical signs and symptoms.

Probable case definition

- Anyone suffering from an acute febrile respiratory disease, presenting clinical, radiological or histopathological signs of pulmonary parenchyma (pneumonia or acute respiratory distress syndrome, for example) ANDhaving a direct epidemiological link2 with a confirmed case of MERS-CoV ANDfor whom the test test for MERS-CoV is not available, is negative on an inappropriate single sample3 or is found to be inconclusive4.
- Anyone suffering from an acute febrile respiratory disease, presenting clinical, radiological or histopathological signs of pulmonary parenchyma (pneumonia or acute respiratory distress syndrome, for example) which cannot be fully explained by any other etiology AND residing or having stayed in the Middle East, or in countries where MERS-CoV is known to circulate in camels or where human infections have recently occurred AND for which testing for MERS-CoV is inconclusive4.
- Anyone suffering from an acute febrile respiratory illness, whatever its severity ANDhaving a direct epidemiological link2 with a confirmed case of MERS-CoV ANDfor which the screening test for MERS-CoV proves to be inconclusive4.

Ratings

1. A case can be laboratory confirmed by viral nucleic acid detection or serological testing. The presence of the viral nucleic acid is confirmed if the result of gene amplification after reverse transcription (RT-PCR) is positive for at least two specific genomic targets or for a single target with sequencing of a second target. A case is confirmed by

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[8] A “close contact” is defined as:
- healthcare-related exposure, including providing direct care to patients infected with MERS-CoV, working with healthcare workers infected with the virus, visiting people infected with the virus, or being in in their immediate environment;
- working in close proximity or sharing the same classroom with patients infected with the MERS-CoV virus;
- traveling with a patient infected with MERS-CoV by any means of transport;
- living under the same roof as a patient infected with MERS-CoV.

The epidemiological link may have occurred within a period of 14 days before or after the onset of the disease in the case in question.

serological testing if this reveals seroconversion in two samples, ideally taken at least 14 days apart, using an ELISA or IFA test and a neutralization test.

Nevertheless, the Interim Recommendations for Laboratory Testing for MERS-CoV should be consulted for the most up-to-date criteria for laboratory confirmation (http://www.who.int/csr/disease/coronavirus_infections/en/).

2. The existence of a direct epidemiological link with a confirmed case of MERS-CoV can assume:

- exposure associated with medical care, including when providing direct care to patients infected with MERS-CoV, when working with health care workers infected with the virus, when visiting people infected with the virus or when we are in their immediate environment;
- working in close proximity or sharing the same classroom with people infected with the MERS-CoV virus;
- traveling with people infected with the MERS-CoV virus by any means of transport;
- living under the same roof as a person infected with the MERS-CoV virus;
- the epidemiological link may have occurred within a period of 14 days before or after the onset of the disease in the case in question.

3. An unsuitable sample may be a nasopharyngeal swab not accompanied by another sample from the lower respiratory tract, a sample that has been mishandled, a sample deemed poor quality by the testing laboratory, or a sample taken too late in the course of the illness.

4. An inconclusive test may be:

- a positive screening test performed by gene amplification after reverse transcription for a single target without further confirmation;
- demonstration of seroreactivity in a single serum sample from a convalescent patient, ideally taken at least 14 days after exposure, by means of an ELISA or IFA test and a neutralization test, in the absence of molecular confirmation on samples taken from the respiratory tract.
- **Inconclusive test:**patients with an inconclusive initial screening test should undergo additional virological and serological testing to determine if they can be classified as confirmed cases of MERS-CoV. It is strongly advised to collect and analyze multiple samples from the patient's lower respiratory tract such as sputum, endotracheal aspirate, or bronchoalveolar lavage fluids. If these patients have no signs or symptoms of lower respiratory tract infection and specimens from these tracts are not available or collection of such specimens is not clinically indicated, specimens should be collected. nasopharyngeal and nasopharyngeal swab specimens. If the initial nasopharyngeal specimen test is negative in a patient with strong suspicion of MERS-CoV infection, retesting of a specimen from their lower respiratory tract or a second nasopharyngeal specimen and an additional oropharyngeal sample if samples from the lower respiratory tract are not possible, as well as on paired acute phase and convalescent phase sera. Other types of clinical specimens could also be considered for molecular testing if needed, including blood/serum, urine, and stool. These generally have lower virus titers than samples taken from the respiratory tract, but were used to confirm cases when other specimens were inadequate or unobtainable. Laboratories that obtain discordant PCR test results and have limited experience in detecting MERS-CoV should consider referring their samples to more experienced laboratories for confirmation.

Consult the WHO case notification form (available inEnglishand inFrench)

Respond to an alert

If a single case/outbreak or outbreak is suspected:

- All HCWs collecting specimens from patients with suspected or confirmed MERS-CoV infection should wear appropriate personal protective equipment, and
    - standard and droplet precautions are sufficient when collecting biological samples from suspected patients;
    - additional precautions are necessary when aerosol-generating procedures are applied to a patient.
- Every individuals involved in specimen collection and transport should be trained in safe handling practices and spill decontamination procedures.
- WHO requests that probable and confirmed cases be reported within 24 hours of classification, through the International Health Regulations regional contact point, to the relevant WHO regional office.

- Bury or cremate (if applicable) dead (human) bodies in a proper and respectful manner
- Provide community education about the confirmed case, mode of disease transmission and home infection control (seehttp://www.who.int/csr/disease/coronavirus_infections/technical-guidance-infection/en/)
- Measures to identify additional cases beyond close contacts are essential for infection prevention and control, and determining the full extent of transmission within the community. Active case finding in the area under investigation should focus on:
    - patients admitted to FOSAs in the community where the confirmed case of MERS-CoV was discovered. Consider screening for MERS-CoV in all patients in hospital presenting with unexplained SARI.
    - health care providers in the community; health workers should be asked about recent cases of unexplained pneumonia and instructed to immediately report any patient with signs and symptoms that meet the case definition for investigation presented in section 3.4 .1. Patients meeting the case definition should be tested for MERS-CoV infection;
    - patients who recently died of unexplained illness meeting the established investigational case definition should be tested for MERS-CoV infection if appropriate clinical specimens are available.
- Close contacts of confirmed or probable cases should be identified and monitored for the onset of respiratory symptoms for 14 days after the last exposure to the confirmed or suspected case, during which the case was symptomatic. Any contact who becomes ill during this time should be tested for MERS-CoV. If possible, all contacts, especially healthcare workers and other hospital contacts, regardless of onset of symptoms, should be tested for MERS-CoV.
- Seek additional support at the national level, if needed.

Analyze and interpret data

- **Time :**graph of the number of suspected/probable/confirmed cases by date (epidemic curve).
- **Place :**map of probable and confirmed human and animal cases by geographical area (district)
- **Individual characteristics:**table showing the number of suspected/probable/confirmed cases by date, age and sex

Laboratory confirmation

This section contains guidelines on laboratory confirmation, including: relevant diagnostic tests, methods of collecting, storing and transporting specimens needed for laboratory confirmation, and information on laboratory test results. laboratory.

Guidelines for Laboratory Testing for MERS-CoVhttp://www.who.int/entity/csr/disease/coronavirus_infections/technical-guidance-laboratory/en/index.html

Recommendations for sample collection

Samples taken from the lower respiratory tract have a higher diagnostic value than those taken from the upper respiratory tract in detecting MERS-CoV infection. Upper respiratory tract specimens have produced negative results in some symptomatic close contacts of confirmed cases, who later developed pneumonia and tested positive on respiratory tract specimens lower.

Whenever possible, WHO strongly advises collecting and testing specimens from the lower respiratory tract, such as sputum, endotracheal aspirate, or bronchoalveolar lavage fluids. If patients have no signs or symptoms of lower airway disease and collection of lower airway specimens is not possible or not clinically indicated, upper airway specimens should be collected such as than nasopharyngeal aspirate or combined nasopharyngeal and oropharyngeal swabs.

If the initial test is negative in a patient strongly suspected of being infected with MERS-CoV, repeat swabs should be taken from multiple airway sites in that patient. Matched acute and convalescent sera should also be collected to detect antibodies. The presence of the virus in biological fluids such as blood, urine and faeces has also been demonstrated, but generally at lower titers than those encountered in the respiratory tract. This type of sample can be taken in the absence of good quality respiratory tract samples or to monitor the presence of the virus in different compartments of the body.

Specimens to be collected from symptomatic patients and asymptomatic contacts

Patient	**Test**	**Sample Type**	**When the sample should be taken**	**Storage and transport**	**Remarks**
Symptomatic	Nucleic Acid Amplification Test (NAAT)	**Lower respiratory tract** - sputum - aspiration - washing **Upper respiratory tract** - nasopharyngeal swabs - and oropharyngeal - nasopharyngeal washout/nasopharyngeal aspirate	When the patient arrives To confirm virus clearance, sample collection should be repeated until results are negative for 2 consecutive samples	If the sample can be shipped to the laboratory in less than 72 hours, store and transport it to 4°C. If shipping the sample to the laboratory takes more than 72 hours, store it at -80°C and transport it in dry ice or liquid nitrogen.	Follow national regulations for shipments within the country and WHO guidelines for the international transport of specimens, including the use of triple packaging systems
Symptomatic	Serology	Serum for a serological test. Only if NAAT is not available	Paired specimens are required to confirm test results of the initial specimen taken during the first week of illness and the second specimen ideally taken 2-3 weeks later. If only one sample can be	As stated above, storage and shipping at -20°C is sufficient.	As stated above.

			collected, the sample should be taken at least 3 to 4 weeks after the onset of symptoms to diagnose a probable case.		
Asymptomatic contact (particularly in outbreaks linked to a health-care setting or other outbreak situations where contact is intense) It is not recommended to test people not associated with outbreaks.	NAAT	Nasopharyngeal and oropharyngeal swabs; samples taken from the lower respiratory tract if possible.	Within 14 days of the last recorded contact.	As indicated above for NAAT.	As stated above.
	Serology	Serum	Initial serum collected as soon as possible within 14 days of contact and convalescent serum collected 2-3 weeks after contact. If only one sample can be collected, collect at least 3-4 weeks after last recorded contact.	As indicated above for serology.	As stated above.

See the reference for the detailed algorithm.

Reference

Surveillance(link (english))*)*

WHO guideline.

- *Case definitions for reporting confirmed MERS-CoV cases to WHO (link)*
- *Surveillance of human infection with MERS-CoV (link)*
- *Investigation of cases of human infection with MERS-CoV (link)*
- *Questionnaire for initial interview with MERS-CoV cases (link)*
- *Summary form for rapid notification of probable or confirmed cases of MERS-CoV infection to WHO (link)*
- *The latest WHO MERS-CoV global summary and risk assessment and archives (link)*

Investigative tools

- *Cross-sectional seroprevalence study of MERS-CoV infection in presumed high-risk populations (link)*
- *Case-control study to assess potential risk factors related to human illness caused by MERS-CoV (link)*
- *Assessment of potential risk factors of infection of MERS-CoV among health care personnel in a health care setting (link)*
- *Seroepidemiological investigation of contacts of MERS-CoV patients (link)*
- *Update on animal-to-human transmission of MERS-CoV and interim recommendations for at-risk groups*

Laboratory

- *WHO guideline; Laboratory testing for Middle East respiratory syndrome coronavirus (MERS-CoV) (link)*

Case management and infection prevention and control

- *WHO guideline. Clinical management of severe acute respiratory infections when MERS-CoV infection is suspected (link)*
- *Home care for patients with MERS-CoV infection presenting with mild symptoms and management of contacts Infection prevention and control during health care for probable or confirmed cases of MERS-CoV infection (link)*
- *Management of asymptomatic subjects with a positive RT-PCR test result for MERS-CoV (link)*

Other Resources

- *Infection prevention and control of epidemic-and pandemic prone acute respiratory infections in health care*
- *Natural ventilation for infection control in health-care settings (link)*
- *Preventive measures (posters and videos) from the WHO Regional Office for the Eastern Mediterranean for the general public, health workers and candidates for hajj and umrah pilgrimages*

Travels

- *WHO guideline: Travel advice on MERS-CoV for pilgrimages*and Considerations for mass gathering events and MERS-CoV (*link)*

Simian orthopox virus (Monkey pox)

Presentation

- Simian orthopoxvirus, or monkeypox, is a rare viral zoonosis whose symptoms are less severe than those observed in the past in human subjects suffering from smallpox (now eradicated). It is generally a self-limiting disease whose fatality rate can still reach 10%, especially in children.
- Monkeypox occurs mainly in the tropical forests of West and Central Africa. Its main animal reservoir is unknown, but it has been detected in various mammals, including rodents and monkeys. The animal species in which the presence of signs of simian orthopoxvirus has been detected include Cricetomys gambianus (the savannah cricetoma or Gambian rat), various species of squirrel of the genera Funisciurus and Heliosciurus, Graphiurus kelleni (African dormouse) and various species non-human primates.
- Communities living in the tropical rainforest regions of West and Central Africa need to be educated to avoid direct contact with animals, especially wildlife. Transmission prevention efforts in endemic areas should emphasize the proper cooking of all animal products (blood, meat) before consumption.
- Human-to-human transmission is limited (no evidence that this mode of transmission alone can sustain monkeypox in human populations) and occurs through prolonged contact with respiratory droplets and through contact with lesions or biological fluids containing the virus. Household members and health workers are most at risk of an outbreak.
- Monkeypox is an emerging disease that has become the most common form of orthopoxvirus since the global eradication of smallpox declared by the World Health Assembly in 1980. This is partly because the smallpox vaccine that offered cross-protection against other orthopoxviruses had been halted at the time, meaning that younger people no longer have the immunity conferred by the vaccine.
- Human monkeypox was first identified in humans in 1970 in the Democratic Republic of the Congo, which remains the country that consistently reports the highest number of cases (over 1,000) per year since 2005. La Sierra Leone, Liberia, Ivory Coast, Nigeria, Cameroon, Gabon, Republic of Congo, Central African Republic and Sudan (in an area that is now South Sudan) have also reported human cases. , since 1070. Since the end of 2016, more and more cases of monkeypox have been reported in countries that had not seen one for 40 years.
- Clinical recognition, especially differential diagnosis from other rash and fever-causing diseases, such as chickenpox, as well as laboratory diagnosis and prevention remain major challenges in endemic areas. Two distinct clades or subtypes have been identified. Infection with a West African strain of simian orthopoxvirus is thought to cause less severe infection, fewer deaths, and a lower rate of human-to-human transmission compared to outbreaks involving Central African strains.
- The incubation period for monkeypox is 6 to 16 days (range 5 to 21 days). The infection can be divided into two periods: 1) the invasion period (0 to 5 days) characterized by fever, severe headache, lymphadenopathy (swollen lymph nodes), back pain, myalgia (muscle pain) and intense asthenia (lack of energy); and 2) the rash period (1-3 days after fever onset), during which different phases of the rash appear, often starting on the face and then spreading to other parts of the body. body.
- Lymphadenopathy is the most distinctive symptom of monkeypox. The face (in 95% of cases), as well as the palms of the hands and the soles of the feet (75%) are the most affected by the rash. Rashes progress from maculo-papules (flat-based lesions) to vesicles (small fluid-filled blisters) and pustules, followed by scabs, occurring within about 10 days. It may take three weeks for the scabs to completely disappear.
- Chickenpox, which is often confused with orthopoxvirus, is distinguished from the latter and from smallpox by its much more superficial lesions, their predominant presence on the trunk rather than on the face and extremities and their development in several outbreaks in the same area. In varicella, fever and rash occur simultaneously and progress more rapidly, with death being a rare complication. Co-infection with varicella viruses and simian orthopoxvirus has been reported. However, the frequency of this phenomenon, the relationship and the impact between the pathogenesis and epidemiology of viruses are not clear.

Purpose of surveillance

- Detect suspected cases of orthopox and respond immediately.

Standard case definition

- **Suspicious case**: anyone with a fever > 38.5°C and a deep rash in the form of macules or uniform papules (same stage) generalized or localized (face, palms of the hands, soles of the feet, ano-genital parts, etc.) AND one or more of the following symptoms: asthenia, lymphadenopathy, headache, back pain or myalgia.
- **Probable case**: any suspected case having had proven contact with an animal with skin lesions OR an epidemiological link with a confirmed case.
- **Confirmed case**: any suspected or probable case in which the PCR test for monkeypox virus is positive.

Differential diagnosis :other causes of clinical symptoms to consider include other rash diseases, such as smallpox, chickenpox, measles, bacterial skin infections, scabies, syphilis, and drug allergies.

Respond to alert threshold

If there is only one suspected case:

- report case information immediately to appropriate levels;
- ensure that the patient is isolated, take measures to prevent airborne transmission, and if possible, allow smallpox-vaccinated health care personnel to care for patients;
- treat and care for the patient by providing supportive care and symptomatic management;
- collect and transport swabs (preferably swab taken from rash area) under strict secure conditions to confirm case.
- risk communication, community mobilization, contact tracing and management;
- conduct case surveillance to identify additional cases;
- send a notification to WHO.

Meet the epidemic or intervention threshold

If only one case is confirmed:

- maintain strict infection control measures throughout the outbreak;
- mobilize the community for early disease detection and prompt treatment;
- to carry out an educational action within the community on the confirmed case of smallpox, the mode of transmission of the disease and how to fight against the infection in the places of residence and during funerals;
- actively search for other cases;
- seek further assistance at national and international levels.

Analyze and interpret data

Time: graph daily/weekly/monthly cases and deaths. Draw an epidemic curve.
location: map where the cases live.
Individual characteristics: transmit immediately, on a case-by-case basis, information relating to cases and deaths. During the outbreak, count and report the number of cases and deaths (including suspected and confirmed cases). Analyze their distribution according to age and sex. Immediately assess risk factors (contact with wild animals or another confirmed case).

Laboratory Confirmation: Monkeypox

Diagnostic tests	Identification of orthopoxvirus DNA by polymerase chain reaction (PCR) from a clinical specimen – preferred technique Or Note: Only in level C or D labs.
Specimens	Optimal specimens: vesicular swabs of lesion exudate or scab which may be as follows: 1) biopsies* 2) scabs* 3) vesicular fluid* 4) skin lesion (top of lesion) 5) pustules* Blood/serum samples – mainly for serological testing, as viremia is short-lived. Require dates and detailed case and disease information for proper interpretation Note: It may be difficult to draw blood from people with a severe, very dense rash because the skin may peel off. It may be necessary to use a central line if it proves too difficult to draw peripheral blood. * Specimens preferred for diagnosis of acute disease during the eruption phase
When to take samples	A suspected case of orthopoxvirus represents a medical and health emergency. Take samples from any Suspected Case, as soon as possible, to obtain the different types of recommended samples.

Laboratory confirmation	
How to prepare, store and transport specimens	Conventional sampling practices are also suitable for obtaining specimens of lesions caused by orthopoxviruses. Personal protective equipment, including gloves, should be worn and the skin at the sampling site should be disinfected beforehand. If alcohol is used, it is important to let it dry well before taking the sample. **Biopsies:** Aseptically place two to four portions of the sampled tissue in a sterile, dry, leak-proof container that can withstand freezing. Storage between -20°C and -70°C. For transport, samples can be stored at 4°C for approximately 6 hours. *Note: For shipping, pack formalin-unfixed lesion biopsies on dry ice; leave the biopsies fixed in formalin at room temperature. Do not freeze formalin-fixed biopsies.* **Crusts:** Aseptically place scabs/portions of harvested tissue in a sterile, dry, leak-proof, freezing-resistant container. No viral transport medium. Storage between -20°C and -70°C. For transport, samples can be stored at 4°C for approximately 6 hours. **Vesicular fluid:** Collect vesicular fluid from each lesion using different sterile cotton swabs. Check that the sample contains cells from the base of each of the vesicles. Storage between -20°C and -70°C. For transport, samples can be stored at 4°C for approximately 6 hours. No viral transport medium. **Blood :** Collect 10 ml of blood in a plastic tube with a marble top, or in a plastic tube with serum separation gel with a yellow top. *Note: Authorization must be obtained to be able to send clinical samples from a patient suspected of having orthopoxvirus disease to a reference laboratory.*
Results	Orthopoxvirus diagnostic services are currently available at the CPC, CREMER and CRESAR.
Reference	
WHO Fact Sheet, Monkeypox.[*http://www.who.int/mediacentre/factsheets/smallpox*](http://www.who.int/mediacentre/factsheets/smallpox)	

Neonatal tetanus

Presentation

- The disease is caused by a neurotropic toxin produced by Clostridium tetani, a spore-forming anaerobic bacterium living in the soil. The disease is contracted when spores enter the body through a wound (injections, section of the umbilical cord) or skin lesions.
- Although adults can be affected, tetanus mostly affects newborn babies. In countries where maternal tetanus immunization coverage has improved, the incidence of neonatal tetanus has declined significantly. The WHO African Region aims to eliminate maternal and neonatal tetanus, aiming to reduce incidence rates of this disease to less than 1 case per 1,000 live births.
- The incubation period lasts from 3 to 21 days, with an average of about 6 days.
- Risk factors: at the time of birth, section of the umbilical cord performed in poor hygienic conditions. Absence of protective antibodies in mothers who have not been fully vaccinated.

Purpose of surveillance

- Immediately detect neonatal tetanus cases to obtain confirmation and prevent additional cases, by vaccinating at least pregnant women residing in the vicinity of the confirmed case.
- Identify geographic areas at risk and organize tetanus vaccination campaigns there for women of childbearing age.

Standard case definition

Suspicious case:

Neonatal tetanus-- Any newborn able to cry and latch on normally for the first 2 days of life, and who, between the 3rd and 28th day, can no longer suckle normally, becomes stiff and/or has convulsions.

Non-neonatal tetanus-- Anyone over 28 days of age with acute onset of the following signs: contracture of jaw muscles, sustained spasms of facial muscles, or generalized muscle spasms.

Confirmed case:

Laboratory confirmation is not advised.

Respond to alert threshold

If there is only one suspected case:

- immediately notify the next level of information relating to the case;
- conduct investigations to determine the risk of transmission;
- treat and manage cases, according to national guidelines, usually providing supportive care, if possible, in an intensive care unit. Isolation measures are not necessary.

Meet the epidemic or intervention threshold

If a suspected case following investigations:

- vaccinate the mother of the case and other pregnant women living in the same locality, with at least 2 doses of tetanus toxoid;
- organize local supplementary immunization activities for all women of childbearing age;
- improve routine immunization coverage as part of EPI and maternal immunization program activities;
- educate birth attendants and women of childbearing age on the importance of hygienic conditions when cutting and caring for the umbilical cord. Multiply the number of skilled birth attendants.

Analyze and interpret data

Time :graph cases and deaths by month.
Place :map residential areas where cases have occurred and the location of midwives.
Individual characteristics:count monthly cases and deaths. Analyze each case of neonatal tetanus by district, maternal characteristics (age, parity), place of delivery, and umbilical cord care practices.

Laboratory confirmation: Neonatal tetanus

Laboratory confirmation is not required.

Reference

- *WHO–recommended standards for surveillance of selected vaccine-preventable diseases. WHO/V&B/03.01*[*http://apps.who.int/iris/bitstream/10665/68334/1/WHO_V-B_03.01_eng.pdf?ua=1*](http://apps.who.int/iris/bitstream/10665/68334/1/WHO_V-B_03.01_eng.pdf?ua=1)

New cases of HIV/AIDS

Presentation

- Acquired immunodeficiency syndrome (AIDS) is an infection that affects human lymphocytes (white blood cells) and other organs. It is caused by a retrovirus, the human immunodeficiency virus (HIV). HIV is transmitted from one person to another through unprotected sexual intercourse, contaminated needle sticks, blood transfusions, transplacental or transvaginal routes, breastfeeding, or occasionally direct contact with biological fluids of infected persons.
- AIDS appears in the later stages of HIV infection, following a weakening of the immune system, with reduced levels of T lymphocytes and a decline in their activity. The damage to vital organs by HIV and many opportunistic infections lead to death, if viral multiplication is not interrupted by drugs capable of killing the virus (antiretroviral therapy). When the infection progresses to disease, the symptoms are usually due to the inability of the immune system to resist other so-called opportunistic infections. These can be tuberculosis, bacterial pneumonia, sepsis, oropharyngeal candidiasis, chronic diarrhea, chronic skin infections, recurrent herpes, etc.
- Twenty-six million Africans, or nearly one in ten adults between the ages of 15 and 49, are living with HIV/AIDS. The impact of the epidemic is already measurable through the sharp increase in morbidity and mortality rates among adults and children. HIV/AIDS is today the leading cause of adult mortality in the African Region.
- The incubation period is approximately 1 to 3 months, from the time of infection to the time when antibodies to HIV can be detected by laboratory techniques. It generally takes 7 to 9 years between HIV infection and the onset of AIDS.
- Risk factors: the populations most at risk of contracting HIV are sex workers with or without other sexually transmitted infections (STIs). Some STIs could increase the risk of HIV transmission. Intravenous drug users, people receiving untested blood products, and newborns of HIV-positive mothers are also at-risk populations.
- Tuberculosis, visceral ionhmaniasis, trypanosomiasis and other subacute or chronic bacterial, parasitic and viral infections can cause similar syndromes.

Purpose of surveillance

- Monitor the impact of HIV/AIDS interventions, studying the changing incidence and prevalence of HIV infections, AIDS and STIs through sentinel sites, surveys and special studies (according to the guidelines for second generation surveillance of HIV/AIDS).
- Estimate the HIV/AIDS burden in the district, based on information collected from HIV sentinel populations, and thus count each new AIDS case.
- Monitor the local epidemiology of STIs as possible co-factors of HIV transmission.
- Monitor the local epidemiology of opportunistic infections, especially tuberculosis.
- Improve the percentage of suspected HIV/AIDS cases confirmed by a serological test.
- Improve HIV/AIDS screening.

Standard definition of new cases of HIV/AIDS

The WHO Regional Office for Africa recommends that countries use the Bangui or Abidjan HIV/AIDS case definitions. A positive ELISA test confirming HIV infection, coupled with a rapid test confirming the positive result, is sufficient to define an epidemiological case of HIV infection.

Public Health Measures

- Monitor for STIs and opportunistic infections, including tuberculosis, as possible co-factors of HIV infection.
- Improve the percentage of suspected HIV/AIDS cases confirmed by a serological test.
- Monitor condom use by sex workers.
- Provide voluntary counseling and testing services at local and district level.
- Treatment of individual cases with antiretrovirals is not yet widespread in most African countries. Prompt diagnosis and treatment of opportunistic infections associated with AIDS can prolong life expectancy, but this aspect has not been widely evaluated in developing countries.
- Promote condom use, especially among at-risk individuals.
- Treat STIs, especially syphilis, chancroid and other genital ulcers.
- Mobilize voluntary blood donors and promote appropriate use of blood.
- Promote good infection control practices in health facilities in the district.
- Educate patients and their sexual partners to refrain from donating blood, tissue, semen, or breast milk.

Analyze and interpret data

Time :count new cases of HIV/AIDS and report them monthly. Analyze the number of cases confirmed by serological test. At the end of the year, review the total number of cases and incorporate trends in HIV serosurveillance, STI surveillance data and the results of special studies (socio-behavioural studies, antimicrobial susceptibility, etc.)

Laboratory confirmation: new cases of HIV/AIDS

Diagnostic tests	**Adults and children 18 months or older:** Diagnosis of HIV infection is based on: -. the positive search for anti-HIV antibodies (rapid test or immunoenzymatic test). The result will be confirmed by a second test for anti-HIV antibodies (rapid test or immunoenzymatic test) based on different antigens or different operating techniques; AND OR - the positive result of the virological test for the detection of HIV or its constituents (RNA-HIV, DNA-HIV or ultrasensitive test for the detection of the p24 antigen of HIV), confirmed by a second virological test on a separate sample. **Children under 18 months:** The diagnosis of HIV infection is based on the positive result of the virological test for the detection of HIV or its constituents (HIV-RNA, HIV-DNA or ultrasensitive test for the detection of HIV p24 antigen), confirmed by a second virological test on a separate sample, taken more than 4 weeks after birth. HIV antibody testing is not recommended for definitive diagnosis or confirmatory diagnosis of HIV infection in children under 18 months of age.
Specimens	Serum
When to take samples	Take samples in accordance with the strategy of the national HIV/AIDS control program for clinical and epidemiological sampling.

How to prepare, store and transport specimens	Take universal precautions to minimize any risk of exposure to sharp or pointed objects or body fluids. ELISA: Collect 10 ml of venous blood. ▪ Allow blood clot to retract at room temperature for 30-60 minutes or centrifuge to separate serum from red blood cells. ▪ Aseptically pour the serum into sterile screw cap tubes. ▪ Store sera at 4°C. Transport serum samples in suitable packaging to prevent breakage or leakage.
Results	HIV testing is strictly regulated and the dissemination of related information is subject to strict controls. The results are generally available one week after the samples arrive at the laboratory.

Reference

- *Guiding principles for sexually transmitted infection surveillance. Geneva. UNAIDS and World Health Organization. WHO/CDS/CSR/EDC/99.3. UNAIDS/99.33E*
- *WHO Case definitions of HIV for surveillance and revised clinical staging and immunological classification of HIV-Related disease in adults and children*
- *WHO recommended standards for surveillance. WHO/CDS/CSR/ISR/99.2*
- *Guidelines for second-generation HIV surveillance: an update: knowing your epidemic, WHO and UNAIDS, WHO/CDS/CSR/ISR/2000.5*
- *Consultation on technical and operational recommendations for clinical laboratory testing harmonization and standardization, January 2008, WHO, CDC*

Noma

Presentation

- Noma (cancrum oris, stomatitis gangrenosa) is an opportunistic bacterial infection affecting children aged 1 to 4 years, characterized by rapidly spreading orofacial gangrene, which develops from gingival inflammation.
- Noma is the result of complex interactions between several risk factors: poor hygiene conditions, malnutrition, recurrent illnesses and weakened immunity. Measles, malaria, severe diarrhea and ulcerative-necrotizing gingivitis are among the diseases that frequently precede noma.
- Noma is found all over the world, but is more common in sub-Saharan Africa. According to WHO estimates, dating from 1998, 140,000 children contract noma each year and 79% of them die of the disease and associated complications.
- In Africa, it is in the countries bordering the Sahara Desert that the prevalence rate of noma is highest, with an annual incidence of 25,000, according to estimates in a recent report. However, noma is found wherever people live in conditions of extreme poverty.
- Early detection of noma cases and prompt treatment with antibiotics are essential to prevent severe disfiguring mutilation or death. In the acute phase of the disease, the administration of large doses of penicillin makes it possible to prevent death. However, only expensive surgery can treat the disfigurements.
- Prevention should focus on education and awareness of the disease, better nutrition, improved hygiene conditions, promotion of exclusive breastfeeding during the first 3 to 6 months of life, access to prenatal care, and vaccinations against common childhood illnesses.
- The clinical features of noma include irritation of the mouth, pronounced halitosis (bad breath), bad taste, increased tenderness of the lips or cheeks, cervical lymphadenopathy, foul-smelling and purulent discharge from the mouth, edema and bluish-black tint of the skin in the affected area.
- Health workers should know how to recognize the risk factors for noma:
  - significant growth retardation in the first 6 months of life
  - signs of malnutrition and poor eating habits
  - persistent diarrhea
  - mouth ulcers in children in high-risk areas
  - striking bad breath

Purpose of surveillance

- Detect and treat cases quickly.
- Identify communities and families at risk.
- Estimate disease incidence and identify risk factors.

Standard case definition

New suspected case:

Any child with a mouth ulcer and other warning signs, such as malnutrition, poor hygiene, recent illness (measles, persistent diarrhea, or malaria).

New confirmed case:

Anyone with a gangrenous condition beginning with an ulcer of the gingival mucosa that spreads rapidly from the mouth to the face, destroying soft and hard tissues.

Recommended Public Health Measures

When a suspected case is detected:

- treat the case with nationally recommended antibiotics;
- conduct health promotion activities in the community to:
  - raise awareness of noma within the community and households
  - improve environmental and personal hygiene conditions
  - keep livestock away from residential areas
  - practicing exclusive breastfeeding for the first 6 months of life
  - improve diet and food preparation techniques
- increase immunization coverage in the district
- improve access to drinking water sources in at-risk communities
- train Public Health personnel in the early identification of oral lesions likely to develop into noma.

Analyze and interpret data

Time : monitor the number of cases detected in time to receive standard treatment. Track cases over time to estimate disease burden and determine trends.

Place : map where the cases live and analyze their geographic distribution.

Individual characteristics:analyze the distribution of cases by age and other demographic factors.

Laboratory confirmation

Monitoring does not require routine laboratory confirmation.

Reference

- *Enwonwu, C. (2006). Noma--the ulcer of extreme poverty. New England Journal of Medicine, The 354(3): 221-224*
- *Enwonwu, C., W. Falkler, et al. (2006). Noma (cancrum oris). The Lancet 368(9530): 147-156.*
- *Fieger, A., K. Marck, et al. (2003). An estimation of the incidence of noma in north-west Nigeria. Tropical medicine & international health 8(5): 402-407.*
- *Enwonwu, CO (1995). Noma: a neglected scourge of children in sub-Saharan Africa. Bulletin of the World Health Organization 73(4): 541-545.*
- *Enwonwu, CO, WA Falkler, et al. (1999). Pathogenesis of cancrum oris (noma): confounding interactions of malnutrition with infection. The American journal of tropical medicine and hygiene 60(2): 223-232.*

Onchocerciasis

Presentation

- Filarial infection of the skin and eye caused by Onchocerca volvulus, transmitted by the bite of female black flies of the genus Simulium.
- Almost all of the 18 million people believed to be infected worldwide (250,000 of whom have lost their sight) reside in 26 African countries. Onchocerciasis is the second major infectious cause of blindness in the world. It also causes debilitating skin problems, resulting in severe productivity declines in endemic regions. Entire villages have resettled far from fertile land, close to rivers where blackflies breed.
- The incubation period lasts several years, even decades, since the disease manifests itself only as a result of repeated infections. The clinical form of the disease is therefore rare in children, even in endemic areas.
- Other heartworms (eg, Loa loa and Mansonella), as well as other chronic skin and eye diseases, can cause similar clinical manifestations.

Purpose of monitoring

- Detect infection early, in order to reduce resumption of parasite transmission in areas where it has been eradicated (areas covered by the Onchocerciasis Control Programme).
- Ensure periodic monitoring in sentinel villages: screening using diethylcarbamazine (DEC); in the event of a positive reaction to DEC, confirmation by microscopic examination of a skin biopsy of each suspected case.

Standard case definition

Suspicious case: In an endemic area, anyone with fibrous nodules in the subcutaneous tissue.

Confirmed case: Suspected case confirmed in the laboratory by the presence of one or more of the following elements: microfilariae in skin biopsies, adult worms in excised nodules, or characteristic ocular manifestations (observation under a slit lamp of microfilariae in the cornea, the anterior chamber of the eye or the vitreous humor).

Respond to alert threshold

If a suspected case is detected:

- report case according to national guidelines
- take samples to confirm the case
- undertake an investigation of the case to determine the cause
- treat the case according to national guidelines.

Meet the epidemic or intervention threshold

If a suspected case is confirmed:

- investigate population movements to identify the origins of infection and initiate control activities;
- carry out vector control activities, in accordance with the guidelines of the Onchocerciasis Control Programme;
- administer periodic mass treatments with ivermectin in areas where onchocerciasis has been endemic for the past ten years;
- actively search for cases through population surveys and skin swabs.

Noma

Background

- Noma (*cancrum oris*, *stomatitis gangrenosa*) is an opportunistic bacterial infection affecting children between 1–4 years. It is characterized by quickly spreading orofacial gangrene, evolving from a gingival inflammation.
- Noma results from complex interactions between risk factors such as poor sanitation, malnutrition, recurrent illnesses, and impaired immunity. Diseases that commonly precede Noma include measles, malaria, severe diarrhoea, and necrotizing ulcerative gingivitis.
- Noma occurs worldwide, but is most common in sub-Saharan Africa. In 1998, WHO estimated that worldwide 140 000 children contract Noma each year, and 79% of them die from the disease and related complications.
- In Africa, the highest prevalence of Noma occurs in countries bordering the Sahara desert, where a recent report estimates an annual incidence of 25,000. However, Noma can occur wherever there is extreme poverty.
- Early detection and treatment with antibiotics is key to preventing severe disfigurement or death. In the acute stage, death can be prevented with high doses of penicillin. However, disfigurement can only be treated with costly surgery.
- Prevention should focus on education and awareness of the disease, improved nutrition and household hygiene, promotion of exclusive breastfeeding in the first 3–6 months of life, access to prenatal care, and immunizations against common childhood diseases.
- Clinical features include soreness of the mouth, pronounced halitosis (bad smelling breath), fetid taste, increased tenderness of the lip or cheek, cervical lymphadenopathy, a foul-smelling purulent oral discharge, and a blue-black discoloration of the skin and swelling in the affected area.
- Health workers should recognize risk factors for Noma:
    - Severe stunting in the first 6 months of life
    - Evidence of malnutrition and poor dietary habits;
    - Persistent diarrhoea
    - Oral ulcers in children from high risk areas
    - Prominent bad smelling breath

Surveillance goal

- Early detection and treatment of cases
- Identification of high risk communities and families
- Estimation of disease incidence and identification of risk factors

Standard case definition

Suspected new case

Any child with a mouth ulcer and other warning signs such as malnutrition, poor hygiene, recent illness (measles, persistent diarrhoea, or malaria).

Confirmed new case

Any person with a gangrenous disease which starts as gingival ulceration and spreads rapidly through the tissues of the mouth and face, destroying the soft and hard tissues.

Recommended public health actions

When a suspected case is detected:

- treat the case with nationally recommended antibiotics;
- conduct health promotion activities in the community for:
  - awareness of Noma among the community and in the household
  - improve environmental sanitation and personal hygiene
  - Separate livestock from areas where humans live
  - Exclusive breastfeeding for the first 6 months of life
  - improve nutrition and food preparation techniques
- increase vaccination coverage in the district
- Improve sources of drinking water in at-risk communities
- Train public health personnel on early recognition of oral lesions that can lead to Noma.

Analysing ant interpreting data

Time: Monitor number of cases detected in time for treatment and use of standardized treatment. Monitor cases over time to estimate burden of disease and identify trends.

Place: Plot the location of case households and analyse their geographical distribution.

Personal information: Analyse the distribution of cases by age and other demographic factors.

Laboratory confirmation

Routine laboratory confirmation for surveillance is not required.

Reference

- *Enwonwu, C. (2006). Noma--the ulcer of extreme poverty. New England Journal of Medicine, The* ***354****(3): 221-224*
- *Enwonwu, C., W. Falkler, et al. (2006). Noma (cancrum oris). The Lancet* ***368****(9530): 147-156.*
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- *Enwonwu, C. O., W. A. Falkler, et al. (1999). Pathogenesis of cancrum oris (noma): confounding interactions of malnutrition with infection. The American journal of tropical medicine and hygiene* ***60****(2): 223-232.*

Onchocerciasis

Background

- Filarial infection of the skin and eye caused by *Onchocerca volvulus* transmitted by the bite of female *Simulium* black flies.
- Nearly the entire world's estimated 18 million infected persons (of whom more than 250 000 are blind) live in 26 African countries. Onchocerciasis is the second leading infectious cause of blindness worldwide. It causes debilitating skin problems, leading to significant decreases in productivity in areas where it is endemic. Entire villages have relocated away from the fertile lands near rivers where black flies breed.
- Incubation period lasts many years to decades since disease occurs only following repeated infections. Clinical illness is unusual in children even in endemic areas.
- Other filaria (for example, *Loa loa* and *Mansonella*) and other chronic skin and eye diseases can produce similar clinical findings.

Surveillance goal

- Early detection of the infection to reduce the recurrence of transmission of the parasite in areas where it has been eradicated (zones covered by the Onchocerciasis Control Programme).
- Conduct periodic surveillance in sentinel villages: screen using diethylcarbamzaine (DEC). In case of a positive reaction to DEC, confirm with a microscopic examination of a skin biopsy from each suspected case.

Standard case definition

Suspected case**:** In an endemic area, any person with fibrous nodules in subcutaneous tissues.

Confirmed case: A suspected case that is laboratory confirmed by presence of one or more of the following: microfilariae in skin snips, adult worms in excised nodules, or typical ocular manifestations (such as slit-lamp observations of microfilariae in the cornea, the anterior chamber of the eye, or the vitreous body).

Responding to alert threshold

If a suspected case is detected:

- report the case according to national guidelines
- collect specimen for confirming the case
- investigate the case to determine the cause
- treat the case according to national guidelines.

Respond to the outbreak or intervention threshold

If a suspected case is detected:

- conduct an investigation on population migration to identify the origins of the infection and initiate control activities.
- Carry out vector control activities according to Onchocerciasis Control Programme guidelines.
- Conduct periodic mass treatment with ivermectin in areas with endemic onchocerciasis during the last 10 years.
- Conduct active case finding via population-based surveys and skin snips.

Analysing ant interpreting data

Time: chart the cases quarterly
Place: Plot distribution of patients' household and workplaces.
Personal information: Count quarterly cases and analyse their distribution according to age.

Laboratory confirmation

Diagnostic tests	Microscopic test. Laboratory criteria for confirmation: One or more of the following: - presence of microfilaria in skin snips taken from the iliac crest - presence of adult worms in excised nodules - presence of typical ocular manifestations, such as slit-lamp observations of microfilariae in the cornea, the anterior chamber of the eye, or the vitreous body.
Specimen:	Bloodless skin snips from : - Nodule fluids - Iliac crests - scapula area
When to collect specimen	Collect bloodless skin snips and nodule fluids from suspected cases 1 hour after administration of Diethyl carbomazine
How to prepare, store and transport the specimen	Put the sample in a general container. Add a few drops of normal saline solution. Close it tightly before transporting it to the laboratory. Transport at ambient temperature.
Results	Results should be ready within the day.
Reference	
▪ *WHO Recommended Surveillance Standards.* Second edition. WHO/CDS/CSR/ISR/99.2	

Perinatal (stillbirths and neonatal) deaths

Background

The Global Strategy for Maternal, Child and Adolescents Health with its three objectives of Survive, Thrive and Transform sets targets for the coming 15 years, which Member States have agreed and committed to achieve. This includes reducing neonatal mortality to less than 12 deaths per 1,000 births and stillbirths to less than 12 per 1,000 total births, in line with the multi-stakeholders action plan. “Every Newborn: an action plan to end preventable deaths” (ENAP), which encompasses two goals: ending preventable newborn deaths and stillbirths.

Globally there are 2.7 million annual neonatal deaths, of these 1 million take place in the African Region. Three main causes of neonatal death represent about 80% of deaths: birth asphyxia, prematurity and neonatal infections. Equally, there are about 2.6 million annual stillbirths globally, of which 98 percent occur in developing countries. About half of all stillbirths occur in the intrapartum period, representing the greatest time of risk. Causes of stillbirths may be a consequence of maternal conditions and diseases like pre-eclampsia, obesity, diabetes, malaria, syphilis and HIV. There are however no available global estimates on causes of stillbirths.

The reduction of neonatal mortality reached 38% in the African Region during the MDG era. However, the reduction has been much slower than that of the under-5, which stood at 54%. Achieving the set SDG target for the reduction of both stillbirths and neonatal deaths will require up to a seven-fold reduction of the current neonatal and stillbirth mortality rates in the African Region. This will require addressing current challenges for the efficient delivery of high quality services for mothers and newborns, but also efforts of strengthening the health information systems to understand the real number of deaths and the causes of deaths.

One of the biggest challenge in addressing both stillbirths and neonatal deaths is lack of information on the correct numbers and causes of deaths. Yet, majority of the stillbirths and neonatal deaths can be prevented with clearly targeted quality interventions by appropriately skilled health workers. To fill these gaps, there is a need to invest in the registration of births and deaths through surveillance and response, alongside perinatal screening and taking action.

The primary goal is to eliminate preventable stillbirths and neonatal deaths by:

- counting every stillbirths and neonatal death through an active identification and reporting at community and HFlevel to enable an assessment of the true magnitude of stillbirths and neonatal mortality and the impact of actions to reduce them.
- Identifying underlying causes, contributing factors and high-risk areas for stillbirths and neonatal deaths to effectively guide immediate as well as longer term actions and to inform programme decisions to reduce these deaths.

Standard case definition

Perinatal death includes the death of a baby of at least 28 weeks of gestation and/or 1,000 g in weight and early neonatal death (the first seven days after birth).

A stillbirth is defined as any death of a baby before birth and with no signs of life at birth of at least 1,000g birth weight and/or at least 28 weeks’ gestation and 35 cm long.

Early neonatal death is defined as any death of a live newborn that occurs before the first 7 full days of life. Day 1 is clinically seen as the first day of life.

Late neonatal death is defined as any death of live newborn occurring between 8 through 28 days after birth.

Responding to alert threshold

After determining that a perinatal death has occurred, the initial notification should be done immediately (within 24 hours), by the fastest means possible.
The HFshould contact the district authority and provide information about the IDSR Case Alert form. Moreover, the HFor the district perinatal death review committee is required to review the case within seven (7) days.
The review of perinatal deaths should be anonymous.
It should be linked to the maternal condition where applicable.
The reports should not be used for disciplinary purposes.

Recommended public health actions

- In many low-income countries, it is not possible to review all perinatal deaths given the large numbers of deaths and the limited capacity in human resources and time. However, it is important to accurately identify and classify the causes of those deaths.
- Some perinatal deaths need to be investigated and explored to determine the cause.
- Surveillance for perinatal deaths should be conducted not just in the labour wards, but also in the community, and all service areas where they occur.
- Response to any perinatal death is based on recommendations from the perinatal death review.
- Findings from review of the selected perinatal death should lead to actions to prevent similar deaths by identifying gaps that should be addressed at the political level and in both HF and communities.
- Monthly, quarterly or semi-annual analysis of aggregated data at larger HF and at the district level can lead to a more comprehensive approach to address a particular problem across multiple facilities or communities or a problem in particular geographical areas where they are occurring in greater numbers. These analyses should be carried out alongside those on maternal deaths by the Maternal and Perinatal Death and Prevention Review Committee

Analysing and Interpreting data

Indicators of magnitude
Number of stillbirths
Number of early neonatal deaths
Causes of stillbirths
Causes of early neonatal deaths
Percentage of stillbirths and neonatal deaths due to preventable factors
Descriptive analysis by person, place and time:
Gestational age at time of death
Socio economic status of family, parents' level of education

Time and date of death, weekday or weekend. Graph the cases in order to draw a yearly chart to see the evolution of the situation.
Place where the family lived or where women died.
Analyse the distribution of the cases
Place of birth - HFor community

Laboratory confirmation

Routine laboratory confirmation for surveillance is not required.

Review committee

This should be the same committee as that of the Maternal Deaths Committee and can be renamed the Maternal and Perinatal Deaths and Prevention Review Committee.

Reference

- WHO Every Newborn Action Plan http://www.EveryNewborn.org;
- WHO Application of ICD-10 to deaths during the perinatal period: ICD- PM ; 2016 ; http://www.who.int/maternal_child_adolescent/en
- ICD 10 PM : http://apps.who.int/iris/bitstream/handle/10665/249515/9789241549752-eng.pdf?sequence=1

Bubonic Plague

Background

- Zoonotic systemic bacterial infection caused by *Yersinia pestis* (plague bacillus) usually transmitted to humans by rodents and their fleas.
- Main forms of the disease: bubonic, pulmonary and septicaemic. Major outbreaks can occur in both urban and rural areas. If left untreated, bubonic plague can develop into pneumonic or septicaemic plague.
- Inter-human transmission occurs only in the case of pneumonic plague, through infectious droplets.
- Incubation period is 1 to 6 days.
- In the absence of treatment, the case fatality rate can exceed 50-60% for bubonic plague and approach 100% for pneumonic or septicaemic plague. However, it is usually less than 1% with appropriate treatment.

Risk factor: exposure to infected populations of wild or domestic rodents and their fleas.

Surveillance goal

- Detect any case of plague promptly as, included in its bubonic form; a single can be at the origin of an outbreak.

Standard case definition

Suspected case of bubonic plague:
- very painful swelling of the lymph nodes - buboes
And
-fever (or history of fever) or at least three of the following signs: headache, chills or generalized or severe asthenia and
- consistent epidemiological features, such as exposure to infected animals or humans and/or evidence of fleabites and/or residence in or stay in a known endemic area within the past 10 days.

Confirmed case of bubonic plague:
Any suspected case confirmed by isolation of *Yersinia pestis* in blood or by bubonic aspiration, or by specific seroconversion or a rapid diagnostic test detecting F1 antigen in endemic areas.

Responding to alert threshold

If a single case is suspected:

- Report case-based information to the next level.
- Collect specimen (blood or puncture) to confirm the case;
- Isolate the patient if suspicion of pneumonic plague with precautions against airborne spread (the patient and the staff managing the patient must wear appropriate masks);
- Undertake an investigation of the case, including identifying all known contacts and determining exposure history;
- Start treatment as soon as the patient is suspected using gentamicin and fluoroquinolones -levofloxacin, ciprofloxacin, moxifloxacin), chloramphenicol, doxycycline
- The duration of treatment is 10 to 14 days, or up to 2 days after the end of the fever;
- It is important to treat patients quickly to prevent pneumonic or septicaemic plague, which leads to more deaths.
- Lower doses are recommended for children and pregnant women. All antibiotics recommended against plague have contraindications related to their use in children and pregnant women, but their use is justified in life-threatening situations.

Responding to outbreak or action threshold

If the suspected case is confirmed:

- Isolate the Bubonic Plague patient while taking precautions against airborne spread of the bacteria (patient and carers should wear appropriate masks) until at least 48 hours of appropriate antibiotic therapy. Comply with infection prevention and control standards;

- mobilise the community to enable rapid detection and treatment of cases;
- identify high-risk population groups through an analysis of people, place and time;
- Reducing the number of sporadic and outbreak-related cases by improving rodent control (removing rubbish, food scraps and destroying rat harbourage areas), protecting against flea bites by using repellents on the skin or clothing, and controlling fleas (especially in homes, ports and airports);
- distribute awareness and risk reduction communication materials

Analysing ant interpreting data

Time: Graph the monthly evolution of cases, deaths and treatment outcome. Plot the outbreak curve of outbreak-related cases.
Place: Map the location of the cases' living areas.
Personal information: immediately report cases and deaths on a case-based basis for routine surveillance. Conduct weekly counts of cases and deaths during outbreaks. Analyse their age distribution and identify risk factors to improve control of sporadic disease and outbreaks.

Laboratory confirmation

Diagnostic tests	Isolation of *Yersinia pestis* in blood or sputum. *Yersinia pestis* specific seroconversion for the detection of F1 antigen in serum. Culture (gold standard); or PCR AND rapid diagnostic test to detect F1 antigen (only properly trained clinicians can perform the test on presumptive cases according to the clinical diagnosis and must be able to obtain) an adequate specimen; or seroconversion or increase in IgG antibody titer from 4 to 15 d
Specimen:	Blood, sputum or autopsy specimens for culture and blood for serological testing
When to collect specimen	Take specimens from all suspected plague cases, if possible before antibiotics are administered. However, treatment should not be delayed and should be well monitored to control serious side effects. Serum should be collected within 5 days of the beginning of the disease and again after 2-3 weeks.
How to prepare, store and transport the specimen	▪ Specimens should be taken using aseptic techniques. Samples for culture should be sent to the laboratory in Cary Blair transport. Unpreserved samples must reach the laboratory on the same day. ▪ Liquid specimens (punctures) should be swabbed onto a sterile cotton swab and placed in Cary-Blair transport medium. Refrigerate. ▪ If transport time exceeds 24 hours and Cary Blair transport medium is not available, freeze the samples before shipping with ice packs.
Results	A rapid diagnostic test (RDT) can be performed on site by well-trained clinicians. Clinical specimens should only be sent to a laboratory with established plague diagnostic capacity or to a WHO Collaborating Centre for plague. The results of the culture will be obtained 5 working days after receipt of the samples by the laboratory. Antibiotic therapy should be started as soon as possible. In plague patients, antibodies to the F1 antigen of *Yersinia pestis* appear 7-10 days after the beginning of the disease.

References

- Plague Manual : Epidemiology, Distribution, Surveillance and Control/Manuel de la Peste : épidémiologie, répartition, surveillance et lutte. WHO/CDS/CSR/EDC/99.2
- Laboratory Manual of Plague Diagnostic tests. CDC/WHO publication, 2000, Atlanta, GA

https://www.cdc.gov/plague/resources/Recommended-antibiotics-for-plague-web-site-rev-Jan2018-P.pdf

Pneumonic plague

Background

- Zoonotic systemic bacterial infection caused by *Yersinia pestis* (plague bacillus) usually transmitted to humans by rodents and their fleas.
- Main forms of the disease: bubonic, pulmonary and septicaemic. Major outbreaks can occur in both urban and rural areas. If left untreated, bubonic plague can develop into pneumonic or septicaemic plague.
- Inter-human transmission occurs only in the case of pneumonic plague, through infectious droplets.
- Incubation period is 1 to 6 days.
- In the absence of treatment, the case fatality rate can exceed 50-60% for bubonic plague and approach 100% for pneumonic or septicaemic plague. However, it is usually less than 1% with appropriate treatment.

Risk factor: exposure to infected populations of wild or domestic rodents and their fleas.

Surveillance goal

- Detect any case of plague promptly as, included in its bubonic form; a single can be at the origin of an outbreak.

Standard case definition

Presumed case of bubonic plague:
- very painful swelling of the lymph nodes - buboes
And
-fever (or history of fever) or at least three of the following signs: headache, chills or generalized or severe asthenia and
- consistent epidemiological features, such as exposure to infected animals or humans and/or evidence of fleabites and/or residence in or stay in a known endemic area within the past 10 days.

Confirmed case of bubonic plague:

Any suspected case confirmed by isolation of *Yersinia pestis* in blood or by aspiration of buboes, or by specific seroconversion or a rapid diagnostic test detecting the F1 antigen in endemic areas.

Responding to alert threshold

If a single case is suspected:

- Report case-based information to the next level. Isolate the patient if pneumonic plague is suspected while taking precautions against the spread of the bacteria by air (the patient and his or her carers must wear an appropriate mask);
- Take specimens (sputum, blood) to confirm the case;
- Undertake an investigation of the case, including identifying all known contacts and determining exposure history;
- Start treatment as soon as the patient is suspected using gentamicin and fluoroquinolones -levofloxacin, ciprofloxacin, moxifloxacin), chloramphenicol, doxycycline
- The duration of treatment is 10 to 14 days, or up to 2 days after the end of the fever;
- It is important to treat patients quickly to prevent pneumonic or septicaemic plague, which leads to more deaths. Lower doses are recommended for children and pregnant women. All antibiotics recommended against plague have contraindications related to their use in children and pregnant women, but their use is justified in life-threatening situations.

Suspected case of pneumonic plague:

- Any person of any age with a cough less than 5 days old accompanied by any of the following signs

Bloody sputum or dyspnoea or chest pain

And fever (or history of fever) or at least three of the following signs: headache or chills or generalized or severe asthenia

And

Epidemiological context (contact with a suspected or confirmed case of pneumonic plague, etc.)

Deaths from suspected plague:

Any person who dies suddenly without apparent cause, but with an established epidemiological link to plague and without biological specimen.

Probable case of plague :
Any suspected live or dead plague case that has undergone a rapid diagnostic test (RDT) against the F1 antigen
Or
Positive PCR test only
Confirmed case of pneumonic plague:
Any suspected plague case from which *Yersinia pestis* has been isolated by culture
Or
Suspected plague case with a positive F1 rapid diagnostic test (RDT)
and positive PCR test
Or
Seroconversion or increase in IgG antibody titre from 4 to 15 d

IMPORTANT: A case with a positive rapid diagnostic test (RDT) against the F1 antigen alone is not a confirmed case. Culture and PCR tests should be done in an appropriate facility.

Responding to outbreak or action threshold

If the suspected case is confirmed:

- Isolate the patient with pneumonic plague while taking precautions against the spread of the bacteria by air (patient and carers should wear appropriate masks) until at least 48 hours of appropriate antibiotic therapy. Comply with infection prevention and control standards;
- mobilise the community to enable rapid detection and treatment of cases;
- identify high-risk population groups through an analysis of people, place and time;
- Reducing the number of sporadic and outbreak-related cases by improving rodent control (removing rubbish, food scraps and destroying rat harbourage areas), protecting against flea bites by using repellents on the skin or clothing, and controlling fleas (especially in homes, ports and airports);
- distribute awareness and risk reduction communication materials

Analysing ant interpreting data

Time: Graph the monthly evolution of cases, deaths and treatment outcome. Plot the outbreak curve of outbreak-related cases.
Place: Map the location of the cases' living areas.
Personal information: immediately report cases and deaths on a case-based basis for routine surveillance. Conduct weekly counts of cases and deaths during outbreaks. Analyse their age distribution and identify risk factors to improve control of sporadic disease and outbreaks.

Laboratory confirmation

Diagnostic tests	Isolation of Yersinia pestis in blood or sputum. *Yersinia pestis* specific seroconversion for the detection of F1 antigen in serum. Culture (gold standard); or PCR AND rapid diagnostic test to detect F1 antigen (only properly trained clinicians can perform the test on presumptive cases according to the clinical diagnosis and must be able to obtain) an adequate specimen; or seroconversion or increase in IgG antibody titer from 4 to 15 d
Specimen:	Blood, sputum or autopsy specimens for culture and blood for serological testing
When to collect specimen	Take specimens from all suspected plague cases, if possible before antibiotics are administered. However, treatment should not be delayed and should be well monitored to control serious side effects. Serum should be collected within 5 days of the beginning of the disease and again after 2-3 weeks.

How to prepare, store and transport the specimen	▪ Specimens should be taken using aseptic techniques. Samples for culture should be sent to the laboratory in Cary Blair transport. Unpreserved samples must reach the laboratory on the same day. ▪ If transport time exceeds 24 hours and Cary Blair transport medium is not available, freeze the samples before shipping with ice packs.
Results	Clinical specimens should only be sent to a laboratory with established plague diagnostic capacity or to a WHO Collaborating Centre for plague. The results of the culture will be obtained 5 working days after receipt of the samples by the laboratory. Antibiotic therapy should be started as soon as possible. In plague patients, antibodies to the F1 antigen of Yersinia pestis appear 7-10 days after the beginning of the disease.

References

- Plague Manual : Epidemiology, Distribution, Surveillance and Control/Manuel de la Peste : épidémiologie, répartition, surveillance et lutte. WHO/CDS/CSR/EDC/99.2
- Laboratory Manual of Plague Diagnostic tests. CDC/WHO publication, 2000, Atlanta, GA https://www.cdc.gov/plague/resources/Recommended-antibiotics-for-plague-web-site-rev-Jan2018-P.pdf

Poliomyelitis (Acute Flaccid Paralysis)

Background

- Poliovirus (genus Enterovirus) serotypes 1, 2, and 3 are transmitted from person-to-person via faecal-oral route.
- Incubation period is 7 to 14 days for paralytic cases and the range is approximately 3 to 35 days. The virus may be shed for several years by immuno-compromised persons.
- Infection is usually asymptomatic, but may cause a febrile syndrome with or without meningitis. In less than 5% of infections, paralysis results often in a single limb.
- The infection affects almost exclusively children. Infection may occur with any of 3 serotypes of Poliovirus. Immunity is serotype-specific and lifelong.
- Paralytic polio, though not fatal in most cases, has devastating social and economic consequences among affected individuals.
- The Polio Eradication Program has nearly halted ongoing wild-type polio transmission worldwide through use of oral poliovirus (OPV) vaccine.
- Areas with low vaccine coverage may allow ongoing wild-type transmission.
- Other neurological illnesses may cause AFP, for example, Guillain-Barré syndrome and transverse myelitis.

Surveillance goal

- Immediate case-based reporting of all poliomyelitis cases. Weekly summary reporting of cases for routine surveillance and outbreaks.
- Detect cases of acute flaccid paralysis (AFP) and obtain laboratory confirmation of the aetiology of all suspected AFP cases. Obtain at least two stool specimens, within 14 days of the onset of paralysis.
- Surveillance for AFP is used to capture all true cases of paralytic poliomyelitis. Target for surveillance performance to provide certification of polio eradication is 1 case of AFP per year per 100 000 population aged less than 15 years.
- The ultimate goal of AFP surveillance is eradication of poliovirus.

Standard case definition

Suspected case: Any child under 15 years of age with acute flaccid paralysis or any person with paralytic illness at any age in whom the clinician suspects poliomyelitis.

Confirmed case: A suspected case with virus isolation in stool.

Responding to alert threshold: poliomyelitis (Acute Flaccid Paralysis)

If a single case is suspected:

- immediately report the suspected case according to the National Polio Eradication Programme guidelines;
- investigate each case. Include a vaccination history for the patient.
- Collect two stool specimens. Collect the first one when the case is investigated. Collect the second one from the same patient 24 to 48 hours later See laboratory guidelines for information on how to prepare, store and transport the specimen.
- Obtain virological data from reference laboratory to confirm wild-type poliomyelitis or Vaccine Associated Paralytic Polio (VAPP).

Responding to the outbreak or intervention threshold

If a suspected case is detected:

- If wild poliovirus is isolated from stool specimen, refer to national polio eradication program guidelines for recommended response actions. The national level will decide what actions to take. They may include the following:
    - identifying the reasons why each unvaccinated case was not vaccinated, and then addressing the failures identified;
    - Immediately conduct “door-to-door” vaccination campaign around the vicinity of the case.
    - Conduct surveys to identify areas of low OPV coverage during routine Expanded Programme on Immunisation (EPI) activities, and improve routine vaccine coverage of OPV and other EPI antigens.
    - lead house-to-house vaccination in supplemental vaccination campaigns during National Immunization Days (NIDs) or local Immunization Days (LIDs). Focus supplemental vaccination activities in areas of low routine immunisation coverage.

Analysing ant interpreting data

Time: make a weekly graph of cases or a graph by date of onset of the disease. Evaluate the percentage of suspected cases reported within 48 hours and the percentage with adequate laboratory specimen collection.

Place: Plot location of case households. Investigate the circumstances of poliovirus transmission in each case thoroughly. Examine the possibility of other potential areas of transmission.

Personal information: Count cases. Analyse their age distribution and number of polio vaccine doses received. Assess risk factors for low vaccine coverage.

Laboratory confirmation

Diagnostic tests	Isolation of polio virus from stool
Specimen:	Stools. Note: If no specimen is collected, re-evaluate patient after 60 days to confirm clinical diagnosis of polio (AFP).
When to collect specimen	Collect stool sample from every suspected AFP case. Collect the first specimen when the case is investigated. Collect a second specimen from the same patient 24 to 48 hours later.
How to prepare, store and transport the specimen	▪ Place stool in clean, leak-proof container and label clearly. ▪ Immediately place in refrigerator or cold box not used for storing vaccines or other medicines. ▪ Transport specimens so they will arrive at designated polio laboratory within 72 hours of collection When there is a delay, and specimen will not be transported within 72 hours, freeze specimen at -20 C or colder. Send it in dry ice or with frozen ice packs at the same temperatures.

Results	Confirmed results are usually available within 21days after receipt of specimen by the laboratory. If poliovirus is detected, the national programme will decide on the appropriate response.

References

- Field Guide for Supplementary Activities Aimed at Achieving Polio Eradication. World Health Organisation.
- WHO global action plan for laboratory containment of wild polioviruses. WHO/V&B/99.32, Genève, 1999
- Manual for the virological investigation of polio, WHO/EPI/GEN/97.01, Genève, 2004
- Supplement to the Manual for the virological investigation of Polio. WHO/EPI 2007

Human rabies

Background

- Rabies is a zoonotic disease (a disease that is transmitted to humans from animals) caused by a virus. Rabies infects domestic and wild animals, and is spread to people through close contact with infected saliva (via bites or scratches).
- The rabies virus infects the central nervous system, causing disease in the brain and, eventually, death. In humans, the disease initially manifests itself as fever, headache, generalized malaise or weakness. As the disease progresses, symptoms include; insomnia, anxiety, confusion, slight or partial paralysis, excitation, hallucinations, increase in saliva, difficulty swallowing, and fear of water.
- In unvaccinated humans, rabies is almost always fatal if post-exposure prophylaxis is not administered before the onset of symptoms. Death usually occurs within days of the onset of neurological symptoms.
- Dogs are responsible for most (approximately 97%) of the human rabies deaths worldwide. They are the main vector of the virus in Africa.
- WHO estimates approximately 55,000 human deaths worldwide due to rabies each year. In Africa, the annual death toll is 24,000.
- People most at risk of rabies live in rural areas, and children are at highest risk of dog rabies. About 30% to 60% of the victims of dog bites (the primary mode of virus transmission) are children less than 15 years of age. Children often play with animals and are less likely to report bites or scratches.
- Canine rabies control and access to post-exposure prophylaxis can significantly reduce the burden of rabies in human populations.
- Rapid and accurate diagnostic tests for rabies in humans and animals are essential for the timely administration of post-exposure prophylaxis. Within a few hours, a diagnostic laboratory can determine whether an animal is rabid and inform the medical personnel in charge.

Surveillance goal

- Detect and respond promptly and appropriately to cases and outbreaks of rabies.
- Identify risk areas
- Estimate the disease burden of rabies.
- Report cases immediately and make monthly routine summary reports.

Standard case definition

Suspected case: A person with one or more of the following: headache, neck pain, nausea, fever, fear of water, anxiety, agitation, abnormal tingling sensations or pain at the wound site, when contact with a rabid animal is suspected.
Confirmed case: A suspected case that is laboratory confirmed

Recommended Public Health Actions

If a single case is suspected:

- Post exposure prophylaxis to prevent rabies;
- Isolate patient if rabies develops to prevent infection of others;
- Immunize contacts if patient develops rabies
- Vaccinate local dogs and cats to prevent outbreaks

General preventive measures:

- Promote public awareness of rabies;
- Target immunization campaign for domestic or wild animals in risk areas
- Maintain active surveillance of rabies in animals.

Analysing ant interpreting data

Time: graph the number of cases per month.
Place: Plot the location of case households and animal exposures.
Personal information: Analyse distribution of cases by age, exposing animal, and circumstances of infection. Assess risk factors to improve control of cases

Laboratory confirmation

Diagnostic tests	Detection of rabies virus antigens by direct immunofluorescence (IF) in clinical specimens, preferably brain tissue (collected post-mortem) ▪ Detection of viral antigens by IF on skin biopsies or corneal biopsies (taken intra-vitam). ▪ Positive IF after inoculation of brain tissue, saliva or CSF into cell cultures of mice or young mice. ▪ Neutralising rabies antibody titre detectable in the CSF of an unvaccinated person. ▪ Identification of viral antigens by PCR on fixed tissue taken post-mortem or in a clinical specimen (brain tissue, skin, cornea or saliva). ▪ Isolation of rabies virus from clinical specimens and confirmation of viral antigens by direct IF.
Specimens:	▪ Brain tissue (collected post mortem) ▪ Skin biopsy (usually from the neck) ▪ Cornea ▪ Saliva ▪ CSF ▪ Head of suspected rabid animal (dog)

hen to collect specimens	When a person is bitten by a pet that appears sick or by a wild animal, the biggest concern is the risk of rabies infection. No test can determine whether the rabies virus has been transmitted to the person immediately after the bite. So, the animal is evaluated to determine whether the person requires treatment. A wild animal that has bitten a person is killed if possible, so that its brain can be examined. If a person who was bitten by an animal becomes increasingly confused and agitated or paralysed, the diagnosis is probably rabies. At this point, tests can detect the rabies virus. Post mortem: within 4-6hrs after death of patient, as soon as the suspected animal dies or is killed
How to prepare, store and transport the specimen	Safety precautions in handling rabies virus should be taken to avoid infection. Remove the head of the suspected animal, wrap head completely such that no blood is oozing out. When possible, request a veterinarian to assist in the collection and preservation of the specimen. Specimens should be sent to the Reference Lab for Rabies virus.
Results	Post-exposure treatment should be administered without waiting for the results of the laboratory diagnosis, especially as the latter can be delayed for a variety of reasons. Results can be obtained from the reference lab within 1-2days.

Reference: rabies (human)

- *WHO Recommended Surveillance Standards. WHO/CDS/CSF/ISR/99.2*
- *Laboratory techniques in rabies, Fourth Edition, WHO, edited by F-X. Meslin et al.*
- World Health Organisation Rabies Fact Sheet. *https://www.who.int/fr/news-room/fact-sheets/detail/rabies*
- *Council of State and Position Statement. Territorial Epidemiologists (CSTE). National Surveillance for Human Rabies. CSTE 09-ID-70. Atlanta: CSTE; June 2009. Available from: http://www.cste.org*
- *Center for Disease Control and Prevention (CDC). Human Rabies Prevention — United States, 2008 :*
- *Recommendations of the Advisory Committee on Immunization Practices. MMWR 2008; 57(RR03):1–26, 28. Available from: http://www.cdc.gov/mmwr/*
- *Bleck TP, Rupprecht CE. Chapter 160 – Rhabdoviruses. In: Mandell GL, Bennett JE, Dolin R, editors. Principles and Practice of Infectious Diseases, 6th edition. Philadelphiea: Churchill Livingstone ; 2005*

Rift Valley Fever (RVF)

Background

- Rift Valley Fever (RVF) is a viral zoonosis disease that affects mainly animals and but can sometimes humans. The virus is a member of the *Phlebovirus* genus, one of the five genera in the family *Bunyaviridae*. The disease is frequently reported following heavy rainfall and floods. It was first identified in 1930 in the Rift Valley, a province of Kenya. The disease was reported in Kenya after the El Nino flooding of 1997/98 and more recently in 2006 to 2007. In 2007 and 2010, Tanzania and South Africa respectively were also affected. Other outbreaks have previously been reported in Somalia, Egypt, Saudi Arabia and Yemen.
- RVF is mainly transmitted from animals (sheep, cattle, goats, camels) to humans through close contact with infected animals (such as handling meat and body fluids and consumption of raw milk). During established RVF outbreaks in animals, humans can also be infected through bites of infected mosquitoes and other biting insects.
- The incubation period of RVF varies from 2 to 6 days. The clinical symptoms include an influenza like illness, with sudden onset of fever, headache, myalgia and backache. These symptoms usually last from 4 to 7 days. Most of the infected people recover on their own. However, a small proportion (about 1%) get complications such as vomiting blood, nose bleeding and passing bloody stool. Other severe types of the disease are eye disease and meningo-encephalitis. Because the symptoms of RVF are varied and non-specific, it is often difficult to make a clinical diagnosis, especially early in the disease. Rift Valley fever is difficult to distinguish from other viral haemorrhagic fevers and many other febrile diseases, including malaria, shigella, typhoid fever and yellow fever.
- In the absence of specific treatment for RVF, case management consists mainly of supportive care. Early detection and management of the disease is important. Control of human RVF is through control of the disease in animals through a sustainable vaccination programme and limiting human-animal contact. Use of insecticide treated nets and mosquito repellents can reduce infections in human. In addition to human suffering and death, RVF has far-reaching economic implications to the livestock industry. In outbreak settings, the disease manifestation includes non-haemorrhagic febrile syndromes, and laboratory testing should be considered among persons with milder symptoms suggestive of viral illness.
- Immediate Notification to WHO is formally required by IHR (Appendix 2)

Standard case definition of Rift Valley Fever (RVF)

Suspected case

Early stage of the disease :

- Acute febrile illness (axillary temperature >37.5 ºC or oral temperature of >38.0ºC) of more than 48 hours that does not respond to antibiotic or antimalarial therapy, and is associated with:
    - Direct contact with sick or dead animal or its products **AND** / **OR**:
    - Recent travel (during last week) to, or living in an area where, after heavy rains, livestock die or abort, and where RVF virus activity is suspected/confirmed **AND** / **OR**:
    - Abrupt onset of at least 1 of the following signs: exhaustion, backache, muscle pains, headache (often severe), discomfort when exposed to light, and nausea/vomiting **AND** / **OR**:
    - Nausea/vomiting, diarrhoea OR abdominal pain with 1 or more of the following:
      - pale skin (or Hb < 8 gm/dL)
      - Low platelets (thrombocytopenia) as evidence by presence of small skin and mucous membrane haemorrhages (petechiae) (or platelet count < 100x109 / dL)
      - evidence of kidney failure (oedema, reduced urine output) (or creatinine > 150 mol/L) **AND** / **OR**:
      - Evidence of bleeding into skin, bleeding from puncture wounds, from mucous membranes or nose, from gastrointestinal tract and unnatural bleeding from vagina **AND** / **OR**:
      Clinical jaundice (3-fold increase above normal of transaminases)

Late stages of diseases or complications (two to three weeks after onset of the first symptoms):

- Patients who have experienced, in the preceding month a flu-like illness, with clinical criteria, who additionally develop the following:
  - CNS disorders which resemble meningo-encephalitis **AND/OR**:
  - Unexplained visual loss **OR**
  - Unexplained death following sudden onset of acute flu-like illness with haemorrhage, meningoencephalitis, or visual loss during the preceding month.

Confirmed case: any patient who, after clinical screening, has a positive anti-RVF IgM ELISA (characteristic appearance of antibodies between 4 and 6 days after the onset of symptoms) or quantitative reverse transcriptase polymerase chain reaction (RT-PCR) tests.

Responding to alert threshold

If a single case is suspected:

- Report case-based information immediately to the appropriate levels;
- Enhance the usual standard precautions throughout the health care setting;
- treat and manage the patient with supportive care;
- collect specimen safely to confirm the case.

Respond to the outbreak or intervention threshold

If a single case is confirmed:

- Mobilise the community for early detection and rapid care;
- Develop a descriptive list/case registry;
- Conduct community education about the confirmed case, how the disease RVF is transmitted, and how to prevent contact with tissues of infected animals and avoid mosquito bites.
- Provide information about prevention in the home and when to seek care.
- Provide supportive treatment to all cases identified
- Request additional help from national levels as needed.
- Collaborate with the animal health specialists to search and document cases among animals as well.

Analysing ant interpreting data

Time: graph cases and deaths monthly. Construct an outbreak curve during the outbreak.

Place: Plot location of case households and work sites using precise mapping.

Personal information: Immediate case-based reporting of cases and deaths. During the outbreak, count and report cases and deaths. Analyse age and sex distribution Assess risk factors immediately and consider request for assistance to improve outbreak control.

Laboratory confirmation : Rift Valley Fever (RVF)		
Diagnostic tests	Acute RVF can be diagnosed using several different methods. Serological tests such as ELISA may confirm the presence of specific IgM antibodies to the virus the virus itself may be detected in blood during the early phase of illness or in post-mortem tissue using a variety of techniques including, antigen detection tests by ELISA, RT-PCR, virus propagation (in cell cultures), immunohistochemistry in formalin-fixed tissues. IgG detection by ELISA allows retrospective diagnosis. Same test can be used for animal diagnosis	
Specimens:	ELISA (serology) ▪ Whole blood ▪ Serum or plasma ▪ Whole blood or clot ▪ Tissues (fresh frozen)	RT-PCR – Virus isolation ▪ Blood ▪ Serum or plasma ▪ Liver biopsy from fatal cases ▪ Anatomopathology ▪ Tissue biopsy of deceased cases ▪ Identical specimen in animals
When to collect specimens	Collect specimen from the first suspected case. If more than one suspected case, collect until specimens have been collected from 5 to10 suspected cases.	
How to prepare, store and transport the specimen	Laboratory staff are at risk. Samples from suspected human cases of RVF for diagnostic purposes should be handled by trained personnel and analysed in properly equipped laboratories. ELISA/PCR/ISOLATION : ▪ Preparation and storage (freeze or refrigerate/as colder as possible) ▪ Shipping: frozen on dry ice or ice packs or both *Note: if dry ice or ice packs are not available, sample may be transported at room temperature and still provide valid results in most cases.* **Immunohistochemistry:** ▪ Preparation and storage: Fix in formalin (can be stored up to 6 wks.) ▪ Shipping: Room temperature (do not freeze). *Same shipping conditions for animal specimens*	
Results	Diagnostic services for RVF are not routinely available. Advance arrangements are usually required for RVF diagnostic services. Contact the appropriate National authority or WHO. Contact national Veterinary Services for animal diagnostic	
References		
▪ *WHO/EMC Infection control for VHF in the African health care setting, WHO, 1998.* ▪ *WHO Recommended Surveillance Standards.WHO/CDS/CSR/ISR/99.2* ▪ *Fact sheet Nº207, Revised September 2007.* ▪ *Infection Control for VHF in the African Health Care Setting /CDC (Appendices 11-12)*		

COVID-19 and other Severe Acute Respiratory Infections (SARI)

Background

- Severe acute respiratory infections (SARIs) are a significant cause of infectious disease morbidity and mortality in Africa. By the end of December 2017, an estimated 290,000 to 650,000 deaths are associated with seasonal influenza each year. The mortality rates are particularly high among infants, children and the elderly.
- An improved understanding of the epidemiology and seasonality of SARIs in Africa is essential for optimizing public health strategies for their prevention and control (e.g., vaccines and antivirals for prophylaxis and treatment, infection control).
- The threat of SARIs due to novel organisms that have outbreak or pandemic potential warrants special precautions and preparedness. Respiratory disease events that may constitute a public health emergency of international concern1 include severe acute respiratory syndrome (SARS); human influenza caused by a new subtype, including human episodes of avian influenza; pneumonic plague; and novel agents that can cause large-scale SARI outbreaks with high morbidity and mortality.

Surveillance goal

- To detect, in a timely manner, unusually severe morbidity and mortality caused by both known and unknown respiratory pathogens that have the potential for large scale outbreaks or pandemics.
- To determine and monitor trends in morbidity and mortality attributable to SRI.

Standard case definition

suspected case of COVID-19 is any person with :

-SUSPECT CASE A: any person who meets the clinical AND epidemiological criteria

Clinical criteria:

o Recent onset of AT LEAST TWO of the following signs or symptoms: fever, cough, general weakness/fatigue, headache, myalgia, sore throat, coryza, dyspnoea, anorexia/nausea/vomiting, diarrhoea, altered mental status.

Epidemiological criteria: o Stay in an area of high risk of virus transmission (e.g. closed residential facilities, humanitarian settings such as camps or IDP centres) at any time during the 14 days prior to symptom onset;

OR o Staying in a community transmission area at any time during the 14 days prior to symptom onset;

OR o Working in the health care sector, including health facilities or in the community, at any time during the 14 days prior to symptom onset. -

-SUSPECT CASE B: anyone with severe acute respiratory illness (severe acute respiratory difficulty with or without fever).

-SUSPECTC CASE C: any asymptomatic person meeting at least one epidemiological criterion with evidence of contact with a probable or confirmed case. -

A probable case of COVID-19 is any person with:

- PROBABLE CASE A: any person who meets the clinical criteria of a suspected case AND is a contact of a confirmed case, or who is linked to an outbreak of COVID-19.

- PROBABLE CASE B: any person with anosmia (loss of smell) or agueusia (loss of taste) in the absence of any other identified cause.

- PROBABLE CASE C: Unexplained death in an adult who was a contact of a probable or confirmed case or who was linked to an outbreak of COVID-19. -

A confirmed COVID-19 case is any person whose diagnostic test (RDT antigen) or PCR has confirmed infection with SARS-CoV2 COVID-19.

A contact case is a person who has been exposed to a probable or confirmed case in one of the following ways within 2 days before and 14 days after the onset of symptoms in the case

- Face-to-face contact with a probable or confirmed case, less than 1 metre apart and for at least 15 minutes;
- Direct physical contact with a probable or confirmed case;
- Direct care of a probable or confirmed case of COVID-19, without Operational Guidelines for COVID-19 Surveillance in Cameroon OR - Other situations as indicated by local risk assessments

Definition of death due to COVID-19

A COVID-19 death is defined, for surveillance purposes, as a death resulting from an illness clinically consistent with a probable or confirmed COVID-19 case, unless there is another obvious cause of death that cannot be linked to COVID-19 disease (e.g. trauma). There should be no period of full recovery between illness and death.

Community definition of COVID19

All of the situations listed in the table below are suggestive of community-acquired COVID-19. history of fever or fever measured at ≥ 38ºC.

Resurgence thresholds

Resurgence of COVID-19 is defined as an increase in the number of new cases of COVID-19 after a period of low or no transmission for a period of at least two consecutive weeks.

Standard case definition (SARI)

Acute respiratory infection with:

- history of fever or fever measured at ≥ 38ºC;
- and cough or sore throat;
- with appearance for less than 10 days;

and requiring hospitalization.

Responding to alert threshold

Please refer to the *WHO protocol to investigate non-seasonal influenza and other emerging acute respiratory diseases, 2018* in case of an unusual event (e.g. outbreak of respiratory infections or atypical respiratory infections, outbreak of death) of respiratory infection.

Respond to the outbreak or intervention threshold
Please refer to the WHO protocol to investigate non-seasonal influenza and other emerging acute respiratory diseases, 2018 if a single case of acute respiratory disease with pandemic potential is suspected.

Severe Acute Respiratory Syndrome (SARS)

Background

- Severe acute respiratory syndrome (SARS) was first recognized as a global threat in 2003 when international spread resulted in 8,098 SARS cases in 26 countries, with 774 deaths.
- Nosocomial transmission of SARS-CoV was a striking feature of the SARS outbreak.
- The majority of the cases were adults. The case fatality ratio of SARS is estimated to range from 0% to more than 50% depending on the age group affected and reporting centre, with a crude global CFR of approximately 9.6%.
- The incubation period lasts from 2 to 10 days, with an average duration of 5 days. Patients initially develop influenza like prodromal symptoms including fever, malaise, myalgia, headache and stiffness. Cough (initial dry), dyspnoea and diarrhoea may be present in the first week but more commonly reported in the second week of illness. Severe cases develop rapidly progressing respiratory distress. Up to 70% of the patients develop diarrhoea.
- Disease transmission occurs mainly during the second week of illness.
- The SARS coronavirus (SARS-CoV) which causes SARS is believed to be an animal virus that crossed the species barrier to humans recently.
- In the inter-outbreak period, all countries must remain vigilant for the recurrence of SARS and maintain their ability to detect and respond to the possible re-emergence of SARS
- Immediate Notification to WHO is formally required by IHR (Appendix 2, IHR).

Surveillance goal

- Early detection and investigation of individuals with clinically apparent SARS-CoV.

Standard case definition

Suspected case of SARS is an individual with:

1. a history of a febrile flare-up, or a documented fever ≥ 38°C **AND**
2. One or more symptoms of lower respiratory tract illness (cough, difficulty breathing, shortness of breath) **AND**
3. Radiographic evidence of lung infiltrates consistent with pneumonia or ARDS or autopsy findings consistent with the pathology of pneumonia or ARDS without an identifiable cause **AND**
4. No alternative diagnosis can fully explain the illness.

Confirmed case of SARS: An individual who tests positive for SARS-CoV infection by the WHO recommended testing procedures.

Respond to suspected case

- Report case-based information immediately to the appropriate levels.
- Practice infection control precautions for an acute respiratory disease with outbreak/pandemic potential immediately and enhance Standard Precautions throughout the health care setting.
- Treat and manage the patient according to national guidelines.
- Collect and transport laboratory specimens from case-patient and from symptomatic contacts and arrange for laboratory testing.
- Review clinical history and exposure history during 2-10 days before disease onset.
- Identify and follow-up close contacts of case-patient.
- Conduct active searches for additional cases.
- Expedite the diagnosis. *(WHO will assist in the investigation of SARS alerts as appropriate, including facilitating access to laboratory services)*

Responding to alert threshold

Response to SARS alert is same as response to suspected case (see above).

SARS ALERT:

An individual with clinical evidence of SARS AND with an epidemiological risk factor for SARS-CoV infection in the 10 days before the onset of symptoms **OR**

Two or more health-care workers with clinical evidence of SARS in the same healthcare unit and with onset of illness in the same 10-day period **OR**

Three or more persons (health-care workers and/or patients and/or visitors) with clinical evidence of SARS with onset of illness in the same 10-day period and epidemiologically linked to a health-care facility.

Analysing ant interpreting data

Time: Graph cases and deaths daily/weekly/monthly. Construct an outbreak curve during the outbreak.

Place: Plot locations of case households and work sites.

Personal information: immediate case-based reporting of cases and deaths. During the outbreak, count and report cases and deaths. Analyse age and sex distribution Assess risk factors immediately.

Laboratory confirmation

Diagnostic tests	Confirmed positive PCR for SARS virus: (Note: tests meet the criteria for laboratory diagnosis of SARS and almost always include two or more different tests or the same test on two or more occasions during the course of the illness or from different clinical sites) ▪ At least 2 different clinical specimens (e.g. nasopharyngeal and stool) OR ▪ The same clinical specimen collected on 2 or more days during the course of the illness (e.g. 2 or more nasopharyngeal aspirates) OR ▪ New extract from the original clinical specimen tested positive by 2 different assays or by repeating the PCR each time. ELISA or Immunofluorescence for seroconversion : ▪ negative antibody test on acute phase serum, followed by a positive test on convalescent phase serum OR ▪ at least 4-fold increase in antibody titer between acute and convalescent phase serum tested simultaneously. Virus isolation: isolation of SARS-CoV in cell culture from any sample; and confirmation by a PCR test using a validated method.
Specimens:	Nasopharyngeal lavage/aspiration: sample of choice for respiratory viruses. Nasopharyngeal or oropharyngeal swabs Stool Serum
When to collect specimens	The respiratory tract specimen can be collected at any time, but are best taken during the acute phase of illness. The time collection of paired blood samples is very important: ▪ Collect an acute illness sample at first contact with the patient at days 7, 14, 28 and 90 after onset where possible. ▪ Collect blood on discharge if collection of a convalescent sample is unlikely.

How to prepare, store and transport the specimen	▪ SARS specimens should be handled according to appropriate biosafety practices in order to avoid laboratory-related infections and spread of disease to close contacts. ▪ Clinical samples from patients should be collected by trained personnel. Nasopharyngeal wash/aspirate: have the patient sit with the head titled slightly backward. Instil 1.5 ml non-bacteriostatic sterile saline (Ph. 7.0) into one nostril. Flush a plastic catheter or tubing (e.g., mucus trap tubing) with 2-3 ml of saline Insert the tubing into the nostril parallel to the palate. Aspirate nasopharyngeal secretions. Repeat for the other nostril Collect aspirates in sterile vial or mucus trap. Remove tubings and discard in plastic bag. Nasopharyngeal or oropharyngeal swabs: use only sterile Dacron or rayon swab with plastic shafts. Place each swab immediately in a tube containing Virus Transport Media (VTM). Serum collection: Collect 5-10 ml of whole blood in a serum separator tube. Allow blood to clot Respiratory / stool / blood/serum specimens: Refrigerate immediately (4°C). If transport/ shipping will be international or will occur > 5 days after collection of last specimen, freeze the specimens at – 20 °C (serum), -20/-70 °C (respiratory specimens) for planned shipping with dry ice if available. Fixed tissues (formalin fixed) from all major organs. Store and ship fixed tissue at room temperature.
Results	Diagnostic services for SARS are not routinely available Advance arrangements are usually required for SARS diagnostic services. Contact the appropriate National authority or WHO. If there is a high level of suspicion, WHO will support countries to contact a reference laboratory if necessary.

Reference

- *WHO Guidelines for the Global Surveillance of SARS, Updated Recommendations, October 2004*
- *WHO Interim Guidelines, Infection Prevention and Control of Outbreak- and Pandemic-Prone Acute Respiratory Diseases in Health Care. June 2007. WHO/CDS/EPR/2007.6*
- *Use of laboratory methods for SARS diagnosis, WHO*
- *WHO Biosafety guidelines for handling of SARS specimens*
- *A practical Guide for SARS laboratories: from samples collection to shipment. WHO, 29 Dec 2003.*

Severe Pneumonia in Children Under 5 Years of Age

Background

- Infection of the lower airways caused by bacteria or viruses transmitted person-to-person via aerosolized respiratory droplet spread. The main bacterial causes of pneumonia among children are Streptococcus pneumonia (the pneumococcus) and Haemophilus influenza type b (Hib).
- Acute respiratory infections (ARIs) and pneumonia represent the number one cause of mortality among children less than 5 years of age.
- Incubation period is usually less than 7 days, depending on the aetiology.
- WHO and UNICEF recommend use of Integrated Management of Childhood Illness (IMCI) strategy to reduce morbidity and mortality attributable to childhood pneumonia. Early antimicrobial therapy has been shown to reduce mortality.
- Resistance of the pneumococcus and Hib to beta-lactams (for example, ampicillin), sulphonamides (for example, trimethoprim-sulfamethoxazole) and other antimicrobials is increasing.
- Viruses such as respiratory syncytial virus (RSV) may also cause ARI and pneumonia.

Surveillance goal

- Early identification of pneumonia cases and outbreaks using clinical definitions.
- Monitor antimicrobial resistance routinely and during outbreaks.
- Reducing the proportion of severe pneumonia cases compared to non-severe pneumonia cases to monitor quality of interventions.

Standard case definition

Clinical case definition (IMCI) for pneumonia:
A child presenting with cough or difficult breathing and:

- 50 or more breaths per minute for infant age 2 months up to 1 year
- 40 or more breaths per minute for young child 1 year up to 5 years.

Note: A young infant age 0 up to 2 months with cough and fast breathing is classified in IMCI as "serious bacterial infection" and is referred for further evaluation.

Clinical case definition (IMCI) for severe pneumonia: a child presenting with cough or difficult breathing and any general danger sign, or chest indrawing or stridor in a calm child. General danger signs for children 2 months to 5 years are unable to drink or breastfeed vomits everything, convulsions, lethargy, or unconsciousness.
Confirmed case: radiographic or laboratory confirmation of pneumonia will not be feasible in most districts.

Responding to alert threshold

If you observe that the number of cases or deaths is increasing over a period of time:
- Report the problem to the next level.
- Investigate the cause for the increase and identify the problem.
- Make sure that cases are managed according to IMCI guidelines.
- Treat cases appropriately with recommended antimicrobial drugs

Respond to the outbreak or intervention threshold

If the number of case or deaths increases to two times the number usually seen during a similar period in the past:

- Assess health worker practices of IMCI guidelines for assessing, classifying and treating children with pneumonia and severe pneumonia.
- Identify high-risk populations through analysis of person, place and time.
- Conduct community education about when to seek care for pneumonia.

Analysing ant interpreting data

Time: Conduct month-to-month analysis for unexpected or unusual increases. Graph cases and deaths by month. Plot the outbreak curve of outbreak-related cases. Plot month-to-month data and compare to previous periods.
Place: Plot location of case households.
Personal information: count monthly pneumonia and severe pneumonia cases. Count pneumonia deaths. Analyse age distribution.

Laboratory confirmation

Routine laboratory confirmation for surveillance is not required.

Reference

- Integrated Management of Childhood Illnesses. World Health Organisation. WHO/CDR/95.14.1

Sexually Transmitted Infections

Background

- Infections of the human genito-urinary and reproductive systems transmitted via human sexual contact (sexually transmitted disease, STIs). The most common causes of male urethral discharge are a) the gonococcus *Neisseria gonorrhoea* and b) *Chlamydia trachomatis*. The most common causes of male and female genital ulcer are a) syphilis (*Treponema pallidum*), herpes simplex virus (HSV1 or 2) and cancroid (*Haemophilus ducreyi*).
- STIs are endemic in most countries of the world, including countries in Africa. Multiple simultaneous STIs are common (for example, gonorrhoea plus Chlamydia). STIs may be most highly prevalent in areas where HIV occurs and may facilitate HIV transmission. STIs may be primary or from repeated attacks of urethral discharge.
- STIs are a leading cause of abortion and stillbirth, prematurity, and congenital infections. They may lead to pelvic inflammatory disease (PID), a major cause of decreased fertility.
- Incubation periods for gonorrhoea are 2 to 7 days; Chlamydia 7 to 14 days (or longer); syphilis, 10 days to 12 weeks (usually around 3 weeks), and cancroid, 3 to 14 days.
- STIs may be more commonly diagnosed in men, in whom clinical evidence of infection may be more readily apparent.

Surveillance goal

- Early detection and treatment of STI reduces transmission rates. Active efforts to diagnose latent syphilis may prevent significant disability.
- Improve early and effective treatment of STIs using simple algorithms based on syndromic diagnosis for index cases and partners.
- Carry out laboratory-based anti-microbial sensitivity monitoring and modify treatment guidelines accordingly at the national level.
- Compare surveillance data for both STIs and HIV/AIDS since STIs may reflect co-presence of HIV.

Standard case definition

Genital ulcer syndrome (non-vesicular):
Suspected case: Any male with an ulcer on the penis, scrotum, or rectum, with or without inguinal adenopathy, or any female with ulcer on labia, vagina, or rectum, with or without inguinal adenopathy.
Confirmed case: Any suspected case confirmed by a laboratory method.

Urethral discharge syndrome:
Suspected case: Any male with urethral discharge with or without dysuria.
Confirmed case: A suspected case confirmed by a laboratory method (for example Gram stain showing intracellular Gram-negative diplococci).

Public health action

- Conduct active case finding for specific target groups.
- Conduct primary prevention activities such as promotion of safer sexual behaviours and provision of condoms.
- Assess use of algorithms for detection and treatment of STIs, and improve health worker practice with algorithms.
- Include STI prevention and care services in maternal and child health, and family planning services.
- Target acceptable and effective STI prevention and care services to populations identified as vulnerable to STI transmission.
- Promote early recourse to health services in case of STI.

Analysing ant interpreting data

Time: Graph cases each quarter.
Place: No recommendation for analysis of place.
Personal information: Count quarterly cases and Analyse age distribution.

Laboratory confirmation

Routine laboratory confirmation for surveillance is not required.

Reference

- Guidelines for Sexually Transmitted Infections Surveillance. Geneva. UNAIDS and World Health Organisation. WHO/CDS/CSR/EDC/99.3. UNAIDS/99.33E

Smallpox

Note: Smallpox was eradicated worldwide in 1980 and there has been no disease in humans since 1977. Information in this section is provided to educate public health professionals to enable detection of re-emergence and to differentiate smallpox from similar diseases.

Background

- Smallpox is an acute contagious disease caused by *Variola virus*, a member of the *Orthopoxvirus*, of the *Proxviridae* family. Other members of the genus that can cause disease in humans include cowpox, camelpox, and monkeypox. Monkeypox virus has caused the most recent human poxvirus infections.
- The case fatality rate for smallpox was 30%. In 1967, when WHO launched an intensified programme to eradicate smallpox, annually there were 10-15 million cases and 2 million deaths globally.
- **The global eradication of smallpox was certified by a commission of eminent scientists in December 1979 and subsequently endorsed by the World Health Assembly in 1980.**
- The incubation period of smallpox is 12–14 days (range 7–17) during which there is no evidence of viral shedding i.e. the person is not infectious. During this period, the person looks and feels healthy and cannot infect others.
- Incubation is followed by the sudden onset of high fever and symptoms such as malaise, headache, backache, nausea and vomiting. Two to three days later, the temperature falls and the patient feels somewhat better, at which time the characteristic rash appears, first on the face, hands and forearms and then after a few days progressing to the trunk. Lesions also develop in the mucous membranes of the nose and mouth, and ulcerate very soon after their formation, releasing large amounts of virus into the mouth and throat. The centrifugal distribution of lesions, more prominent on the face and extremities than on the trunk, is a distinctive diagnostic feature of smallpox and gives the trained eye cause to suspect the disease. Lesions progress from macules to papules to vesicles to pustules. All lesions in a given area progress together through these stages. From eight to 14 days after the onset of symptoms, the pustules form scabs, which leave depressed depigmented scars upon healing.
- Smallpox had two main forms: variola major and variola minor (the latter was less common). The disease followed a milder course in variola minor, which had a case-fatality rate of less than 1 per cent. The fatality rate of variola major was around 30%. There are two rare forms of severe smallpox: haemorrhagic and malignant. In the first one, always fatal, the rash was accompanied by haemorrhage into the mucous membranes and the skin. Malignant smallpox was characterized by lesions that did not develop to the pustular stage but remained soft and flat. It was almost invariably fatal.
- Varicella (chickenpox) is often confused with smallpox can be distinguished from smallpox by its much more superficial lesions, their presence more on the trunk than on the face and extremities, and by the development of successive crops of lesions in the same area. Fever and rash occur simultaneously in chickenpox and develop more rapidly; with death being a rare complication.
- Prior to the eradication of smallpox, simian orthopoxvirus infections were first reported in humans in 1970 and may have been misdiagnosed as smallpox because of similarities in skin presentation and progression. Smallpox and human orthopoxvirosis have similar clinical features; however, patients with smallpox do not develop lymphadenopathy, a major clinical sign of orthopoxvirosis. The progression of the disease through the incubation period, pre-eruptive phase and eruptive phase is also similar for both diseases. Human orthopoxvirosis is less virulent with a lower case fatality rate (up to 10%) compared to smallpox (up to 30%).
- Smallpox transmission is human-to-human and occurs after the onset of fever, through aerosols and fine droplets in the air, during direct and fairly prolonged contact with an infected person, especially when coughing. The disease can also be transmitted through contaminated linen (clothes and sheets), but the risk of infection is much lower.

- The frequency of contamination is highest after direct contact with a patient, once fever has set in and during the first week of the rash, when the virus is released through the respiratory tract. The environments most at risk are residential areas and HF with active cases, but the spread within the community is low because sick people remain bedridden.
- In the absence of immunity induced by vaccination, humans appear to be universally susceptible to infection with the smallpox virus. Since vaccination with smallpox vaccine was discontinued globally after the eradication of smallpox in 1980, most of the world's population under 40 years of age are not immune and the older age groups have waning immunity.
- WHO maintains smallpox vaccine contingency stocks to be deployed in the event of a smallpox re-emergence in order to contain the outbreak. First responders are prioritized to receive the vaccine. Vaccine administered up to 4 days after exposure to the virus, and before the rash appears, provides protective immunity and can prevent infection or minimize the severity of the disease.
- The IHR (2005) formally requires immediate notification of smallpox cases to the WHO. The risk of emergence of smallpox is extremely low as the remaining global live variola virus stocks are held in two high security laboratory facilities in Russia and the US and the disease has no animal reservoir.

Surveillance goal

To detect and immediately respond to a potential re-emergence or any suspected case of smallpox.

Standard case definition of Smallpox

***Suspected case*:** An acute illness with sudden onset of high fever > 38.3 C (101 F) followed by a characteristic rash (macules, vesicles, pustules, crusts) with centrifugal distribution in the same stage of development without other apparent cause.

***Probable case*:** A case that meets the clinical case definition, is not laboratory confirmed, but has an epidemiological link to a confirmed or probable case.

***Confirmed case*:** A clinically compatible case that is laboratory confirmed.

Responding to alert threshold

If a single case is suspected:

- Report case-based information immediately to the appropriate levels;
- Ensure patient is isolated and personnel attending have been vaccinated with smallpox vaccine;
- Implement airborne infection control precautions;
- Treat and manage the patient with supportive care; (The antiviral agent used to treat smallpox, tecovirimat, was approved in July 2018)
- Collect and transfer specimen (prefer swab of rash) under strict safety conditions to confirm the case.
- Implement contact tracing and contact management.
- Conduct active surveillance to identify additional cases.
- Notify WHO

Respond to the outbreak or intervention threshold

If a single case is confirmed:

- Maintain strict infection control measures practices throughout the duration of the outbreak;
- Mobilise the community for early detection and rapid care;
- conduct community education on the confirmed case of smallpox, how the disease is transmitted and how to control infection in the home and at funerals;
- Conduct active search for additional cases.
- Request additional help from national and international levels.
- Establish isolation ward to handle additional cases that may be admitted to the health facility

Analysing ant interpreting data

- **Time**: Graph cases and deaths daily/weekly/monthly. Construct an outbreak curve during the outbreak.
- **Place:** Map location of case households.
- **Personal information**: Immediate case-based reporting of cases and deaths. During the outbreak, count and report cases and deaths. Analyse age and sex distribution Assess risk factors (contact with another active confirmed case) immediately.

Laboratory confirmation

Diagnostic tests	Isolation of smallpox (Variola) virus from a clinical specimen Or Polymerase chain reaction (PCR) assay identification of Variola DNA in a clinical specimen Note: Level C or D laboratories only.
Specimens:	Biopsy specimens*, Scabs*, Vesicular fluid*, skin lesion (roof)* Pustule material* Blood sample *Note: blood samples from person where severe, dense rash may be difficult to draw as the skin may slough off. A central line may be needed for access in cases where a peripheral blood draw is difficult.* *preferred specimens for diagnosis of acute illness during rash phase
When to collect specimens	A suspected case of smallpox is a public health and medical emergency. Collect samples from every suspected case at available times to obtain the various specimen types recommended.

How **To prepare, store, and transport**	Typical practices associated with collection of patient specimens are appropriate for collection of orthopoxvirus lesions as well. Personal protective equipment, including gloves, should be worn and the skin at the sampling site should be disinfected beforehand. If alcohol is used to prepare the lesion for collection, it is important to allow the lesion to dry before it is collected. **Biopsy**: Aseptically place two to four portions of tissue into a sterile, leakproof, freezable container. Storage -20 °C to - 70 °C. For transport, samples can be stored at 4 °C for about 6 hours. *Note: For shipping, pack biopsies from the lesion not fixed in formalin on dry ice; leave biopsies fixed in formalin at room temperature. Do not freeze formalin-fixed biopsy sample.* **Crusts**: aseptically place the crusts/portions of the removed tissue in a sterile, airtight, freezable container. Storage -20°C to - 70°C. For transport, samples can be stored at 4°C for about 6 hours. **Vesicular fluid**: collect the vesicular fluid from each lesion using different sterile cotton swabs. Check that the sample contains cells from the base of each vesicle. Storage -20°C to - 70°C. For transport, samples can be stored at 4°C for about 6 hours. **Blood:** Draw 10cc of blood into a plastic marble-topped tube, or a plastic yellow-topped serum separator tube. *Note: approval must be obtained prior to the shipment of potential smallpox patient clinical specimens to a Reference laboratory*
Results	Diagnostic services for smallpox are not routinely available Advance arrangements are usually required for smallpox diagnostic services. Contact the appropriate National authority or WHO.
Reference	
WHO Fact Sheet, Smallpox. http://www.who.int/mediacentre/factsheets/smallpox	

Trachoma

Background

- Trachoma is the leading cause of preventable blindness worldwide. It is caused by infection with *Chlamydia trachomatis* bacteria, and is both treatable and preventable.
- Infections often begin during infancy or childhood and can become chronic. If left untreated, the infection eventually causes the eyelid to turn inwards, which in turn causes the eyelashes to rub on the eyeball, resulting in intense pain and scarring of the front of the eye. This ultimately leads to irreversible blindness, typically between 30 and 40 years of age.
- Trachoma is spread through direct personal contact with the infected person's eye secretions, when sharing towels or clothing, and by flies that have landed on the eyes or nose of infected people.
- WHO estimates that approximately 6 million cases of blindness due to trachoma and 11 million cases of trichiasis occur worldwide each year. Prevalence of active disease in children varies from 10- 40% in some African countries.
- The infection primarily affects young children, with blindness occurring later in life. Females are three times more likely than males to suffer from trichiasis, the in-turning of the eyelashes that can lead to blindness. People are most at risk for trachoma infection in areas where there is poor sanitation, lack of latrines, poor sources of clean water, and the presence of flies.
- Primary interventions advocated for preventing trachoma infection include improved sanitation, reduction of fly breeding sites and increased facial cleanliness (with clean water) among children at risk of disease. Scarring of the cornea and progression to blindness can be reversed by a simple, in situ surgical procedure that reverses the lashes.

Surveillance goal

- Prevention of blindness by early detection.
- Identification of high-risk areas and epidemiologic trends.
- Estimation of disease burden
- Monitoring of disease control programmes.

Standard case definition

Suspected case:
Any patient with red sticky eyes who complains of pain and itchiness of the eyes.
***Confirmed case*:**
A suspected case in whom eye examination confirms one of the stages of Chlamydia trachomatis infection according to the **WHO Simplified Trachoma Grading System.** (See reference below).

Recommended public health action

To combat trachoma, the World Health Organisation has developed a strategy including a series of interventions to control trachoma known by the acronym SAFE: Surgery, Antibiotics, Facial cleanliness, and Environmental improvement.
Effective Trachoma control has four main components:

- Eyelid surgery for those at immediate risk of blindness.
- Antibiotics to treat individual cases and to reduce infection in a community.
- The promotion of facial cleanliness and hygiene to reduce transmission.
- Environmental improvements such as provision of water and household sanitation

Analysing ant interpreting data

Time: Monitor epidemiologic trends over time.
Place: Plot the location of case households and Analyse the distribution.
Personal information: Analyse the distribution of cases by age and other demographic factors.

Laboratory confirmation	
Routine laboratory confirmation for surveillance is not required.	
Diagnostic test	Detection of specific antigen. Genomic sequence research and tissue culture techniques. Occasionally, direct microscopic examination, in epithelial cells of Giemsa or iodine stained smears.
Specimens:	Collection of conjunctival scrapings
How to prepare, store and transport the specimen	After anaesthetising the conjunctiva with anaesthetic eye drops, remove the secretions with a filter paper and, with a blunt-tipped spatula, scrape the entire conjunctiva. Spread the specimen evenly on a slide. As soon as the preparation is air-dry, fix it with methanol for 2-3 minutes if the preparation is to be Giemsa stained.
Results	Outside of specialist laboratories, most ocular infection is diagnosed clinically (see appendix 8 on the recommended case definition for the confirmed case) or immunologically.

References : Trachoma

- *WHO Trachoma Page - Page dédiée au trachoma*http://www.who.int/topics/trachoma/en/
- *World Health Organisation. Trachoma control: A guide for program managers Geneva: World Health Organisation, 2006.* http://www.who.int/blindness/publications/tcm%20who_pbd_get_06_1.pdf
- *World Health Organisation. Achieving Community Support for Trachoma Control. Geneva: World Health Organisation, 1993.* http://www.who.int/blindness/achieving_en.pdf
- *World Health Organisation. Primary Health Care Level Management of trachoma. Geneva: World Health Organisation, 1993.* http://www.who.int/blindness/publications/trachoma_english.pdf
- World Health Organisation. Trachoma epidemiologic survey protocol. Geneva: World Health Organisation, 1993. http://www.who.int/blindness/prevalence_protocol_trachoma_english.pdf
- CDC Trachome http://www.cdc.gov/healthywater/hygiene/disease/trachoma.html
- The Carter Center http://www.cartercenter.org/health/trachoma/index.html

Trypanosomiasis

Background

- Trypanosomiasis is an infection of blood, lymphatics and central nervous system. African trypanosomiasis (also known as sleeping sickness) is caused by the protozoan *Trypanosoma burcei rhodesiense* and *T. b. gambiense*, which are transmitted by the bit of infected Glossina (tsetse) flies.
- Trypanosomiasis is endemic in over 30 African countries in West, Central and East Africa. It is highly outbreak in the Democratic Republic of Congo, Angola, and other areas of civil conflict, where 80% of some village populations may be infected. Cattle are the major reservoir of Trypanosoma brucei rhodesiense, and humans are the major reservoir for T. b. gambiense.
- Incubation period is usually days to weeks with T. b. rhodesiense, and months to years with T. b.gambiense infections. Without treatment, both forms are usually fatal.
- Trypanosomiasis control strategies include human and cattle population surveys to treat infected persons and diminish cattle reservoirs, and tsetse fly habitat control (for example, removal of bushes and tall grasses near villages, and use of residual insecticides).
- Tuberculosis, malaria, bacterial meningitis, HIV/AIDS, and other central nervous system or systemic infections can produce similar clinical findings.

Surveillance goal

- Increase percentage of cases confirmed by laboratory methods.
- Use population-based surveys and serologic screening for active case finding in endemic areas.
- Monitor trypanosomiasis-free areas by screening humans and livestock.

Standard case definition

Suspected case

Early age: A painful chancre originating as a papule and then evolving into a nodule at the primary fly bite site. There may be fever, intense headache, insomnia, painless lymphadenopathy, anaemia, local oedema and rash.

Late stage: Cachexia, somnolence, and central nervous system signs

Confirmed case:

A suspected case confirmed by card agglutination trypanosomal test (CATT) or by isolation of trypanosomes in blood lymph nodes or cerebrospinal fluid.

Responding to alert threshold

If you observe that the number of cases or deaths is increasing over a period of time:

- Report the problem according to national guidelines.
- Treat any individual suspected and confirmed cases with appropriate therapy in closely monitored setting.
- Collect specimen for laboratory confirmation.
- Investigate cause of increasing number of cases to identify problems with prevention activities.

Respond to outbreak or action threshold

If the number of cases or deaths increases to two times the number usually seen in a similar period in the past

- Assess prevention activities in the area around the cases and take action to improve them as indicated.
- Conduct active case finding activities if it is an endemic area.
- Conduct vector control activities specified by national guidelines.

Analysing and Interpreting data

Time: Graph quarterly cases. **Place:** Plot the distribution of case households. **personal information:** Count the number of monthly cases and analyse their age group distribution	
Laboratory confirmation	
Diagnostic test	Suspected Case: Serological: card agglutination trypanosomiasis test (CATT) Confirmed Case: Parasitological: detection (microscopy) of trypanosomes in blood, lymph nodes aspirates or cerebrospinal fluid (CSF)
Specimen	Whole blood Lymph nodes aspirates Cerebrospinal fluid
When to collect the specimen	Suspected cases from endemic places with fever Any patient with fever and may have been exposed to tsetse flies.
How to prepare, store, and transport the specimen	For slides: Put the slides in a slide box and close properly. Store at room temperature in a dust-free place. In case there is no slide box, the slides can be wrapped in soft tissue paper (filter papers, towels, toilet paper, etc.) For blood in anticoagulant bottles, refer to reference lab.
Results	Results should be available the same day.
References	
• *Control and Surveillance of African Trypanosomiasis. Report of a WHO Expert Committee, Geneva, World Health Organization, 1998 (WHO Technical Report Series, No. 881).* • *WHO Recommended Surveillance Standards WHO/CDS/CSR/ISR/99.2*	

Tuberculosis

Background

- Infection of the lungs and other organs usually caused by Mycobacterium tuberculosis transmitted person-to-person by droplet infection through coughing, sneezing or spitting. Clinically, the pulmonary form of the disease is more common than the extra-pulmonary forms. The main symptoms of pulmonary TB are chronic cough, weight loss, fever, loss of appetite and night sweats
- Tuberculosis (TB) is a leading cause of infectious illness and death worldwide with over 8 million new cases and 3 million deaths per year. In African countries, approximately 1.6 million of the new cases and over 600 000 cases occur each year. It is also estimated that between 30 and 50% of all new TB cases detected are infected with HIV and 40% of all AIDS deaths are due to TB. Those who are at highest risk of dying from TB include people with HIV/AIDS, malnutrition and other immuno-compromising conditions, the very young, and the very old.
- The global HIV pandemic has been a major cause of increasing TB cases, especially in African countries.
- Incubation period is approximately 1 to 3 months.
- WHO recommends the Directly Observed Therapy, Short-course (DOTS) strategy to maximize compliance and treatment efficacy and to reduce development of drug-resistant strains. The DOTS strategy has been implemented by at least 40 of 46 Member States in the African Region. Varying degrees of success have been achieved in controlling TB where resources and motivation for diagnosis, treatment, and patient follow up are adequate.
- Clinically, bacterial pneumonia, malaria, trypanosomiasis, HIV/AIDS and a variety of other bacterial, parasitic, and viral infections may cause similar syndromes of fever, cough, fatigue, and weight loss, or may themselves precipitate active TB in an already infected individual. Abdominal or other extra-pulmonary sites of infection may occur after ingestion of un-pasteurized cow's milk (M. bovis).

Surveillance goal

Early detection of persons with infectious lung disease to improve chances of clinical improvement and reduce transmission of TB.

Improve percentage of TB cases confirmed by microscope

Standard case definition Tuberculosis

Suspected case:

Any person with a cough of 3 weeks or more.

Confirmed case:

Smear-positive pulmonary TB: a) a suspected patient with at least 2 sputum specimens positive for acid-fast bacilli (AFB), or b) one sputum specimen positive for AFB by microscopy and radiographic abnormalities consistent with active PTB as determined by the treating medical officer, or c) one positive sputum smear by microscopy and one sputum specimen positive on culture for AFB

Smear negative PTB: a patient who fulfils all the following criteria: a) two sets taken at least 2 weeks apart of at least two sputum specimens negative for AFB on microscopy, radiographic abnormalities consistent with PTB and a lack of clinical response despite one week of a broad spectrum antibiotic, a decision by a physician to treat with a full course of anti-TB chemotherapy, or b) a patient who fulfils all the following criteria: severely ill, at least two sputum specimens negative for AFB by microscopy, radiographic abnormalities consistent with extensive pulmonary TB (interstitial and military), a decision by a physician to treat with a full course of anti-TB chemotherapy, or c) a patient whose initial sputum smears were negative, who had sputum sent for culture initially, and whose subsequent sputum culture result is positive.

Responding to alert threshold

If you observe that the number of cases or deaths is increasing over a period of time:

- Report problem to the next level, or according to national guidelines.
- Treat individual cases with direct observation (DOTS) including a treatment supporter.
- Where feasible, isolate persons using respiratory infection control practices, especially if multi-drug resistant TB is suspected.
- Investigate cause of increase, including performance of DOTS programme in your area

Respond to outbreak or action threshold

If the number of cases or deaths increases to two times the number usually seen in a similar period in the past

- Assess health worker performance with detection and treatment of smear-positive PTB and improve practices as needed.
- Assess DOTS programme and take action to make identified improvements.
- Conduct drug susceptibility tests to establish patterns of resistance.

Analysing and Interpreting data

Time: Graph cases and deaths monthly

Place: Plot distribution of case households and workplaces.

Personal information: Count monthly cases and deaths. Analyse age and sex distribution quarterly.

Laboratory confirmation

Diagnostic test	Microscopy: Presence of acid fast bacillus (AFB) in Ziehl Neelsen (ZN) stained smears Culture and identification Drug susceptibility test : Anti-tuberculosis drug resistance occurs when a strain of Mycobacterium tuberculosis isolate is resistant to one or more antimicrobial agents as evidenced by internationally recommended methods for susceptibility tests) MDR =Resistance to Isoniazid and Rifampicin; X-DR= Resistance to Isoniazid and Rifampicin (MDR); plus additional resistance to a fluoroquinolone and a second-line injectable agent

Specimen	Deep-chest sputum aspirates
When to collect the specimen	Collect sputum (not saliva) for direct smear microscopy and examine at least two stained specimens taken on different days.
How to prepare, store, and transport the specimen	Smear should be examined at HFwhere the specimen is taken. TB cultures should be packaged in leak proof containers, wrapped in cotton wool. Transport in waterproof container to reference lab.
Results	TB microscopy is read daily. Quantification of observed mycobacterium are reported using various reporting methods. Refer to the criteria used by the examining laboratory. Culture: after 6-8 weeks Anti-tuberculosis drug resistance: The national reference laboratory should be linked to an Supranational reference laboratory by strain exchange to ensure quality control
References	
• *Treatment of Tuberculosis: Guidelines for National Programmes. WHO/TB/97.230* • *Policy Statement of Prevention Therapy Against TB in People Living with HIV, WHO/TB/98.255* • *Laboratory Services in Tuberculosis Control, Parts I, II and III. WHO publications WHO/TB/98.258* • *Guidelines to surveillance of drug resistance in tuberculosis 4th ed. WHO/HTM/TB/2009.422*	

Typhoid Fever

Background

- Typhoid fever is a bacterial disease, caused by Salmonella typhi. Symptoms usually develop 1–3 weeks after exposure, and may be mild or severe. They include high fever, malaise, headache, constipation or diarrhoea, rose-coloured spots on the chest, and enlarged spleen and liver. Healthy carrier state may follow acute illness.
- Typhoid fever remains a serious public health problem throughout the world, with an estimated 16– 33 million cases and 500 000 to 600 000 deaths annually.
- In virtually all endemic areas, the incidence of typhoid fever is highest in children from 5–19 years old. The disease is almost exclusively transmitted by food and water contaminated by the faeces and urine of patients and carriers.
- Polluted water is the most common source of typhoid transmission. In addition, shellfish taken from sewage-contaminated beds, vegetables fertilized with night-soil and eaten raw, contaminated milk and milk products have been shown to be a source of infection.
- Typhoid fever has been virtually eliminated in most areas of the industrialized world with the advent of proper sanitary facilities. Most cases in developed countries are imported from endemic countries.
- People can transmit the disease as long as the bacteria remain in their body; most people are infectious prior to and during the first week of convalescence, but 10% of untreated patients will discharge bacteria for up to 3 months.
- Typhoid fever can be treated with antibiotics. However, resistance to common antimicrobials is widespread. Healthy carriers should be excluded from handling food.

Surveillance goal

Detect Typhoid Fever sporadic cases and outbreaks promptly, and seek laboratory verification
Identify areas/population at high risk in order to improve prevention of the disease by taking hygienic measures

Standard case definitions

Suspected case: Any person with gradual onset of steadily increasing and then persistently high fever, chills, malaise, headache, sore throat, cough, and, sometimes, abdominal pain and constipation or diarrhoea.

Confirmed Case: Suspected case confirmed by isolation of Salmonella typhi from blood, bone marrow, bowel fluid or stool.

Responding to alert threshold

If Typhoid fever cases are suspected:

- Arrange for laboratory testing of stool specimens or rectal swabs of suspected cases, especially in situations where food or water-borne transmission is suspected.
- Report and investigate all suspected outbreaks of typhoid. Search for case/carrier that is the source of infection and for the vehicle (water or food) through which infection is being transmitted.
- Treat typhoid fever patients with antibiotics. Severe cases should be provided supportive measures such as oral or intravenous hydration, the use of antipyretics, and appropriate nutrition.

Respond to outbreak or action threshold

If Typhoid Fever cases are confirmed

- Initiate a line list/register for cases
- Identify areas/populations at high risk to identify source(s) and mode(s) of transmission in order to prevent and control the disease.
- Conduct health education programmes on hygiene with simple messages on safe water, safe food handling practices, hygiene and handwashing.
- Work with water authorities to support provision of clean water and proper sanitation to affected population(s). Chlorinate suspected water supplies. All drinking water should be chlorinated or boiled before use.
- More than 90% of patients can be managed at home with oral antibiotics, reliable care and close medical follow-up for complications or failure to respond to therapy. Patients with persistent vomiting, severe diarrhoea and abdominal distension may require hospitalization and parenteral antibiotic therapy.

Analysing and Interpreting data

Time: Graph cases and deaths weekly. Draw up an outbreak curve during outbreaks.
Place: Plot location of case households with precise mapping.
Personal information: Report immediate case-based information for cases and deaths.

Report summary totals monthly.
During outbreak, count cases and deaths weekly.
Analyse by age.
Assess risk factors to improve prevention of outbreaks.

Laboratory confirmation

Diagnostic test	Culture: Isolation of salmonella spp. from stool or blood of a patient The WIDAL Test should not be used for diagnostic purpose
Specimen	Blood, Stool
When to collect the specimen	Collect samples preferably before antibiotics are administrated
How to prepare, store, and transport the specimen	5-10 ml of blood distributed in a blood culture bottle. Stool in stool container Store specimens at 4-8 C or ambient temperature away from heat and direct sunlight.
Results	Blood culture: 4 days to 2 weeks Stool culture: 3-4 days.

References

The Diagnosis, Treatment and Prevention of Typhoid Fever; WHO/V&B/03.07
Weekly Epidemiological Record; N° 1, 2005, 80, 1-8; http//www.who.int/wer
WHO Recommended Surveillance Standards WHO/CDS/CSR/ISR/99.2

Cluster of Public Health events and unexplained deaths

Background

- A large number of public health events in the history of mankind appeared locally in the form of outbreaks and have spread with displacements, their causes remaining unknown until an explanation was finally found. An effective public health system is characterized by their will to alert on uncertain and worrying events.
- The mere nature of these events does not even allow for their accurate description, but cases have been used to illustrate aspects likely to raise concerns. The International Health Regulations (IHR) involves a “decision tool” prescribing guidelines to WHO Member States (See Section 2 of these Guidelines). A positive answer to either two of the four questions below shall mean that an event may constitute a Public Health Emergency of International Concern notifiable to WHO by the Member State: Are the repercussions of the event on public health serious? Is the event unusual or unexpected? Is there a major risk of international spread? Is there an important risk on restrictions to travels and international exchanges?
- Reporting of a possible outbreak or unusual event may originate from various sources, namely:
    - Routine analysis of surveillance date (for instance, the systematic report presents an unexpected increase in the number of cases of a reportable disease)
    - A health professional (medical doctor, nurse or community health worker, environmental health worker) signalling a cluster of patients suffering from a particular disease in a health structure or in the community.
    - The community chief noticing an unusual event among his population reports it to the opportunities.
- For this type of events to continue to be notified by the concerned actors at local level, district or regional and national stakeholders must be ready to listen to them and trust their reports. The system’s capacity to respond to alert ensures continuous reporting and maintaining vigilance.
- An analysis highlighted the major hurdles below following the reporting of public health events of international concern:
- Ignorance of the reporting process by clinicians who do not know for instance the reportable diseases and the information to be communicated. There is generally a confusion about whether it is the hospital or the laboratory, which should report the event, and if laboratory confirmation should occur before reporting.
- A poor understanding of the use of reporting information and the feeling that reporting of diseases is a useful activity.
- Many surveyed people have revealed to deplore them the real or perceived negative consequences associated with reporting, like the additional workload, unwelcome information request, attention of the media, judgements, sanctions and criticisms.
- The strategies below seeking to enhance completeness of the reporting of reportable diseases and public health events as understood by the IHR:
        - Give front-line professionals clear information on the following points:
            - Why notify unusual health events?
            - What are the reportable events?
            - How to notify an unusual health event?
    - What happens after reporting?
    - Build capacities in asking questions and obtaining immediate answers between clinicians and other key partners to promote greater completeness of reporting, for instance by granting access of reporting platforms to public health professionals during emergency situations and by installing a toll free number for reporting.
    - More frequent field visits or telephone conferences may also be useful.
    - Feedback to clinicians and other stakeholders in the reporting chain showing that preventive actions are carried out following reporting enables to underscore the importance of timely and complete reporting. Provide feedback to those reporting to build confidence and transparency in information exchange on unusual health events, improve understanding on how information is used and highlight the consequences of the lack of understanding.
    - The surveillance facility relies on good personal relationships or the knowledge of people involved in the reporting process. Encourage the development of interpersonal relationships.
- How transmitted information are managed:
    - IHR country focal points contact their peers in WHO Regional Offices. Regional Offices enter epidemiological date and other information needed for risk analysis and management in the public health events management system in which information to be made available later on are stored Feedback to countries through their

national IHR focal points concludes the reporting process and, if a country needs help in responding to an outbreak, a request shall be sent to WHO to this end.

- This recent instruction by WHO/AFRO insists on public health events whose aetiology is unknown from the onset, that is public health events of undetermined cause. For this type of event, the guideline recommends the "One Health" approach requiring the ministry of Health to closely collaborate with other ministries and intersector partners to strengthen teamwork and improve the effectiveness of preparedness, response and monitoring and evaluation.

Surveillance goal

- The possible international scope of an event is determined at the national level in line with Appendix 2 of the IHR (2005) and not at the peripheral level.
- In this definition of an "event" or death, emphasis is laid on awareness raising to facilitate reporting and minimise lateness by insisting on the fact that a possible false alert shall not lead to negative consequences.
- Detect cases.
- Immediately send data on the cases. Weekly summary reporting of cases for the routine surveillance and outbreaks.

Standard case definitions

These events are not defined in details or standardised for now. Two types of events were selected in the 2005 IHR to provide surveillance guidelines and enable early detection and response.

- Unexplained deaths
- Clusters of diseases

Community alert triggers

Clusters of unknown health issues Any health issue of unknown origin, affecting many humans or animals in the same community.

For instance:

- An outbreak or a cluster: A group of sick (or deceased) individuals presenting similar symptoms in the same place (community, school or health structure) at the same time.
- An unusual death or cluster of deaths: two people or more who died from an unknown cause after experiencing the same symptoms in the same place (village, school, health structure) at the same time.
- A group of people who fell sick or have an unusual reaction after consuming the same food or drinking at the same water source.
- A sick person showing symptoms never seen before or for a long time (an emerging infectious disease is suspected, for instance)
- One or many members of the community is/are sick at the same time when animals are sick or dying in the village
- Sickness or death of animal from an unknown cause

HF

The proposed definition for events that should be reported by clinicians and **HF** is as follows: "Any outbreak of disease, OR any unusual disease potentially significant for public health, OR any infectious or pseudo-infectious syndrome deemed unusual by the clinician based on its frequency, circumstances surroundings its onset, clinical manifestations or seriousness".

Any infectious or pseudo-infectious syndrome deemed unusual by the clinician based on the elements below:

- Frequency - for example, sudden, unexplained and significant increase in the number of patients, especially out of the normal season.
- Circumstances surrounding its onset - for example, many patients coming from the same locality or taking part in similar activities.
- Clinical manifestations - for example, rapid worsening of a patient's health non-concurrent with the apparent symptoms or the diagnosis.

References

- *MacDonald et al: Detection of events of public health importance under the international health regulations: a toolkit to improve reporting of unusual events by frontline healthcare workers. BMC Public Health 2011. 11:713.*
- *International Health Regulations 2005 http://www.who.int/ihr/9789241596664/en/ 2nd edition. ISBN : 9789241580410*
- *Public health events of initially unknown aetiology: A framework for preparedness and response in the African Region. Bureau régional de l'OMS pour l'Afrique, 2014. ISBN : 978 929 023 2476 (classification NLM : WA 105)*

- Seriousness - for example, a certain number of patients unresponsive to treatments.
- A patient in contact with animals (wild or domestic) presenting unusual clinical manifestations.

Standard case definitions

The proposed definition for events that should be reported by **laboratories** is as follows:

- "Any situation deemed unusual in connexion with the specimens received (frequency, circumstances of the onset or clinical manifestation) OR lab results (unexpected number of the same species/sub-species, type/sub-type of strain or the antimicrobial resistance profile, or impossible/uncertain diagnosis)"

Responding to alert threshold

If there is a single death or cluster of deaths or presumed unexplained diseases:

- Report the suspected case immediately using the IDSR alert form
- Start active surveillance
- Investigate each case
- Report clusters of potentially worrying events by indicating the personal information, place or time.

Respond to outbreak or action threshold

If the case is validated at the HD level, the measures to take shall be determined at the regional or central level. These may involve the response measures below concerning systematic outbreaks up to the possible involvement of a rapid intervention team in public health emergencies. See **Section 6** of IDSR Guidelines.

- Implement anti-infectious control measures by applying standard precautions pertaining to cases and health workers.
- Ensure worthy and safe burials.
- If animals are also involved, communicate and ensure coordination with the local officer in charge of animal husbandry or an official of the ministry of Agriculture.

Analysing and Interpreting data

Time: Determine the time (date) of onset of the disease or symptoms and death.

Place: Plot the location of families and communities to which the cases belong. In each case, seriously investigate the circumstances and possible transmission means. Study the possibility for other areas to be affected. Seek for any environmental associations. Identify whether displacement is involved. Map out the cases and check for clusters or links between cases locations and the public health event investigated.

Personal information: Count the number of cases and monitor demographic factors. Analyse the age group distribution, links with the profession and recent exposures. Assess the risk factors.

Laboratory confirmation

The diagnosis of public health events of international concern, namely unexplained deaths and clusters of diseases is made based on their onset or following other most common options. No specific test can be done.

West Nile Fever

Background

- West Nile Fever is a febrile illness resulting from a mosquito-borne arbovirus in the Flavivirudae family. It is a zoonotic disease transmitted from birds to humans and other animals. Serological evidence suggests that the infection is present throughout practically the entire African continent. West Nile Fever most likely emerged in Africa and is now found worldwide. Outbreaks occur in humans, birds and horses.
- Most cases are mild and may not come to the attention of the health system. Patients seeking healthcare usually present with flu-like symptoms such as fever, headache and body aches. Occasionally patients present with a skin rash on the neck, trunk, arms or legs.
- People of all ages and conditions may be affected. However, those who are above age 50 years or who have had an organ transplant are at increased risk of severe illness
- Very severe cases include signs of encephalitis, meningoencephalitis or meningitis. Symptoms include high fever, headache, neck stiffness, stupor, tremors, convulsions, flaccid paralysis and coma.
- The case fatality rate in patients with neurological involvement ranges from 4% to 14% and as high as 29% in elderly patients
- West Nile Fever can be prevented by avoiding mosquito bites especially at dusk when mosquitoes are most active. Insect repellents, wearing long sleeves and trousers, staying indoors and draining breeding sites like pools of standing water can reduce exposure to mosquitoes.
- Confirmation of West Nile Fever in patients with clinical symptoms requires laboratory confirmation of specific IgM antibodies in cerebrospinal fluid and serum specimens.
- Because there is no specific treatment for West Nile Fever, patients with severe disease are usually hospitalized for supportive treatment and nursing care.

Surveillance goal

- Identify risk factors for infection and determine high-risk populations for targeted prevention activities
- Identify geographic areas for targeted prevention and control activities
- Identify most severe cases for referral to hospitalized care

Standard case definitions

Suspected Case:
A hospitalized case of encephalitis due to unknown cause
Confirmed Case:
Confirmation of West Nile Fever is through laboratory diagnostics to identify WNV-specific IgM

Responding to alert threshold

If a single case is suspected:

- Report case-based information immediately to the appropriate levels.
- Treat and manage the patient with supportive care.
- Collect specimen safely to confirm the case.

Respond to outbreak or action threshold

Treat and manage the patient with supportive care

- Treat and manage the patient with supportive care.
- Mobilise the community through education in order to promote adoption of behaviours that reduce disease risk such as protection against mosquito bites and reduction of mosquito breeding sites
- Conduct community education on how WNV is transmitted and on how to prevent being infected

Analysing and Interpreting data

Time: Draw up an outbreak curve during the outbreak.
Place: Plot distribution of case households and workplaces.
Personal information: Immediate case-based reporting of cases and deaths. During an outbreak, count and report cases and deaths. Analyse age and sex distribution. Assess risk factors immediately and consider request for assistance to improve outbreak control.

Laboratory confirmation West Nile Fever

Diagnostic test	Presence of IgM antibodies against West Nile Fever
Specimen	*For ELISA:* Whole blood, serum or plasma *For PCR:* Whole blood or blood clot, serum/plasma or tissue *For immunohistochemistry:* Skin or tissue specimens from fatal cases.
When to collect the specimen	Collect specimen from the first suspected case. If more than one suspected case, collect until specimens have been collected from 5 to10 suspected cases.
How to prepare, store, and transport the specimen	HANDLE AND TRANSPORT SPECIMENS FROM SUSPECTED VHF PATIENTS WITH EXTREME CAUTION. WEAR PERSONAL PROTECTIVE EQUIPMENT AND USE BARRIER MEASURES. *For ELISA or PCR:* Refrigerate serum or clot Freeze (-20C or colder) tissue specimens for virus isolation *For immunohistochemistry:* • Fix skin snip specimen in formalin. Specimen can be stored up to 6 weeks. The specimen is not infectious once it is in formalin. Store at room temperature. Formalin-fixed specimens may be transported at room temperature.
Results	Diagnostic services for VHF are not routinely available. Advance arrangements are usually required for VHF diagnostic services. Contact the appropriate National authority or WHO.

References: West Nile Fever
Global Alert and Response; West Nile Fever outbreak updates http://www.who.int/csr/don/archive/disease/west nile fever/en/ *Pedro N. A and Boris Szyfres. Zoonoses and Communicable Diseases Common to Man and Animals. Third edition, Volume II. Chlamydioses, Rickettsioses and Viroses, Part II: Viroses Pages 372-376. Pan American Health Organization, WHO* *Outbreak/Epizootic West Nile Virus in the United States: Guidelines for Surveillance, Prevention and Control. http://www.cdc.gov/ncidod/dvbid/westnile/resources/wnv-guidelines-aug-2003.pdf* *Infection Control for Viral Haemorrhagic Fevers in the African Health Care Setting WHO/EMC/ESR/98.2* *Evans, A.S. (ed). Viral Infections of Humans; Epidemiology and Control. 1989. Plenum Medical Book Company, New York* *Evans, A.S. (ed). Viral Infections of Humans; Epidemiology and Control. 1989. Plenum Medical Book Company, New York*

Yaws and Endemic Syphilis or Bejel

Background

- Endemic trepanometoses in the WHO African Region include two Neglected Tropical Diseases caused by two different sub species of Treponema pallidum (T.p.) : yaws, due to *T. p. pertenue* and bejel caused by *Treponema pallidum (T.p.)*
- Yaws initially presents as a papilloma teemed with bacteria (primary yaws). The papilloma is a typical presentation of yaws and clinical diagnosis is straightforward. Without treatment, the papilloma will ulcerate. Papilloma and ulcers are very infectious and in the absence of treatment can quickly spread to other persons. Other clinical forms of yaws exist but they are not very infectious. Apart of papilloma and ulcers, other lesions of yaws and bejel range from macules, papules, nodules, plaques to secondary yaws that occurs weeks to months after the primary infection. It typically presents with multiple raised yellow lesions or pain and swelling of long bones and fingers (dactylitis).
- Yaws spreads in inter-tropical areas, in humid and warm zones such as equatorial rain forests and their surroundings, while bejel is found in most dry and arid regions such as the Sahel trip
- Children from 2 to 14 years old are the most affected age-group, especially in school-age children where outbreaks of yaws or bejel could be observed
- Yaws treatment which was based on single injection of long lasting penicillin (benzathine benzyl penicillin) has improved greatly by the confirmation of the efficacy of a single dose of Azithromycin for curing yaws lesion in 2010. Further to this confirmation, the WHO has designed a yaws eradication strategy, titled “The Morges Strategy” from the name of a city near Geneva, where the Strategy was drafted in 2012. This eradication strategy consists mainly in mass administration of azithromycin (MAA) to at-risk communities and achieving at least 90% coverage of targeted populations
- The mode of transmission is through direct contact with skin lesions or items already contaminated by primary lesions (papilloma and ulcers)
- Confirmation of diagnosis is done by dual treponemal and non-treponemal rapid tests, a syphilis test which is not specific for yaws followed by a dual path platform (DPP) test specific for *T. p pertenue.* These rapid tests can be performed in the fields and are able to detect recent and past infections.

Surveillance goal

Yaws is targeted for eradication by 2020, eradication being defined as complete interruption of transmission (zero new case of yaws) globally. The surveillance goals are to:

- 1) ensure detection of any new case of yaws in a given area for implementing the eradication strategy
- and 2) after stopping transmission, maintain active case search for at least three years to certify yaws eradication

Standard case definitions

Suspected case: a person with a history of residence in an endemic area (past or present) who presents with clinically active (visible) yaws lesions

Confirmed case: a suspected case with a positive serological test (rapid treponemal test for syphilis confirmed by DPP test)

Imported case: a person who presents with clinically active yaws serologically confirmed in an area where yaws is not known to be endemic

Index case: first case of yaws which is detected in a community
Contact of a case: a person who has close, frequent contact with the infected person. A contact for the purpose of yaws eradication is the household, classmates or close playmates as identified by the contact

Responding to alert threshold

If a single case is suspected:

- Report the suspected case to the appropriate level of the health system (peripheral HFor HD) for serological confirmation and exclusion of imported case.

If the suspected case is not confirmed:

Maintain surveillance for three years during the post-elimination of transmission period

Respond to outbreak or action threshold

If a single case is confirmed and importation excluded:
The area is confirmed endemic and eradication strategy is implemented
If a single case is confirmed and is an imported case:
Treat the case and his contacts as identified by the case and re-start post-elimination of transmission surveillance for again a three-year period

Analysing and Interpreting data

Time: Graph of cases by year of diagnosis, graph of cumulative number of cases.
Place: Plot cases by location of households and colour shade endemic districts.
Personal information: Count newly detected cases which were treated and number of contacts identified and treated. Estimate the number of persons in endemic communities or districts and calculate treatment coverage of Mass Azithromycin Administration (at least 90 %.)

Laboratory confirmation

Diagnostic test	Positive rapid Syphilis test confirmed by positive dual path platform (DPP) test PCR Histopathology
Specimen	Blood from finger stick for serological tests Swab samples from papilloma and ulcerated lesions for PCR Biopsy of lesions for histopathology
When to collect the specimen	Specimens should be collected from suspected patient with clinical symptoms (papilloma and ulcers mainly)

How to prepare, store, and transport the specimen	During collection of specimen for PCR test, it is important to avoid cross contamination between the collection of samples Materials: Dry swabs and recipients. Types of specimens: swabs from papilloma and ulcers, stored at 4°C
Results	Positive Rapid Syphilis test and positive DPP test Positive PCR for Treponema pallidum pertenue for yaws or Treponema pallium pallidum for bejel Evidence of causative organisms in histopathological samples

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Yellow fever

Background

- Yellow fever virus is an RNA virus that belongs to the genus Flavivirus. It is related to West Nile, St. Louis encephalitis, and Japanese encephalitis viruses. Acute viral haemorrhagic disease caused by a flavivirus transmitted human-to-human via the domestic species of Aedes mosquitoes (Urban outbreaks) or to humans from primate reservoir via a forest mosquito species (Sylvatic cycle).
- Large scale outbreaks occur every 3 to 10 years in villages or cities in the absence of large scale immunisation. Sporadic cases can occur regularly in endemic areas. Resurgence of disease in Africa since mid-1980s. True incidence far exceeds reported cases. Real incidence is largely higher than declared cases.
- Incubation period 3 to 6 days after the bite from an infected mosquito. About 15% of infections progress to fever and jaundice. While only the minority of cases are severe, case fatality rate may be 25% to 50% among patients with syndrome of haemorrhage, jaundice, and renal disease.
- Risk factor: sporadic cases often linked to occupation or village location near woods or where monkeys are numerous.

  Also non-vaccinated persons.
- International reporting to WHO required within 24 hours.
- Viral haemorrhagic fevers (VHF) and other parasitic, viral, or bacterial diseases such as malaria, Dengue Chikungunya, leptospirosis, hepatitis A-E, Epstein-Barr virus, West Nile, Q fever, anthrax, rickettsial diseases, etc., and toxic exposures may mimic yellow fever.
- Infection and disease can be prevented by vaccination. With a vaccine efficacy > 95% and duration of immunity is lifetime

Surveillance goal

- Seek confirmation of yellow fever and rule out other possible aetiologies of fever with jaundice
- Provide information in order to adopt appropriate control measures
- Identify populations at risk of yellow fever
- Monitor the epidemiology of the disease and the impact of control measures
- Support operational research and innovation

Standard case definition Yellow fever

Suspected case:

Any person with acute onset of fever, with jaundice appearing within 14 days of onset of the first symptoms.

Probable case: A suspected case

AND

One of the following:

- Epidemiological link to a confirmed case or an outbreak
- Positive post-mortem liver histopathology

Confirmed case: A probable case

AND

One of the following:

- Detection of YF-specific* IgM
- Detection of four-fold increase in YF IgM and/or IgG antibody titres between acute and convalescent serum samples
- Detection of YFV-**specific*** neutralizing antibodies

**YF-specific means that antibody tests (such as IgM or neutralizing antibody) for other prevalent flavivirus are negative. This testing should include at least IgM for Dengue and West Nile and may include other flavivirus depending on local epidemiology*

OR

One of the following:

- Detection of YF virus genome in blood or other organs by PCR
- Detection of yellow fever antigen in blood, liver or other organs by immunoassays Isolation of the yellow fever virus

Laboratory confirmation Yellow fever	
Diagnostic test	ELISA for the presence of yellow fever Specific IgM and IgG antibodies. Exclusion of Dengue, West Nile virus and other locally prevalent flavivirus will be necessary for the confirmation of yellow fever. PCR, YF specific seroneutralization, virus isolation or histopathology
Specimen	Serum in the acute and convalescent phases of the illness; In the event of death, post-mortem liver specimen
When to collect the specimen	Within 14 days of onset of first symptoms Collect specimen from at least the first to 10th suspected cases of yellow fever. Collect specimen from last cases (based on outbreak curves) to decide on the end of the outbreak.

How to prepare, store, and transport the specimen	Collect 10 ml of venous blood from adults, 1-5 ml from children, in a capillary tube, microtainer, or if necessary in a standard glass test tube. Separate blood cells from serum: • Let clot retract for 30 to 60 minutes at room temperature. Centrifuge at 2000 rpm for 10-20 minutes and pour off serum into a clean glass tube. • If no centrifuge, put sample in refrigerator overnight (4 to 6 hours) until clot retracts. Collect serum the next morning. • If no centrifuge and no refrigerator, let blood sit at an angle for at least 60 minutes (without shaking or being driven in a vehicle. Using a pipette, pour off serum into a clean, labelled tube for shipment and storage. Store serum at 4°C. Transport serum samples using appropriate packaging to prevent breaking or leaks during transport. Avoid glass tubes for shipment and transport if possible. The specimen should arrive at the laboratory within 3 days of being collected. Avoid shaking of specimen before serum has been collected. To prevent bacterial overgrowth, ensure that the serum is poured into a clean glass test tube. The test tube does not need to be sterile – just clean. Transport the serum in an EPI hand vaccine carrier at 4°C-8°C to prevent bacterial overgrowth (up to 7 days). If not refrigerated, serum stored in a clean tube will be good for at least 3 days.
Results	Laboratory results should be received within 7 days of reception of the specimen in the laboratory.

References

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Zika Virus Infection

Background

- Zika virus is a flavivirus that is transmitted primarily through the bite of an infected mosquito, primarily Aedes Aegyptus, and also Aedes albopictus, the same mosquitoes that transmit dengue, chikungunya, and yellow fever.
- Zika virus can also be transmitted in-utero from mother to foetus, and through sexual contact, blood transfusion, and organ transplantation.
- Zika virus infections are usually asymptomatic. When symptoms occur, they tend to be mild and include mild fever, rash, conjunctivitis, and muscle and joint pain that last for 2 to 7 days. There is no specific treatment but symptoms can be treated with common fever medicines, rest and drinking fluids.
- Zika virus infection during pregnancy can result in preterm birth, foetal loss, stillbirth, and congenital malformations including microcephaly, limb contractures, eye abnormalities, brain calcifications, and other manifestations of Congenital Zika Syndrome.
- Zika virus is also associated with an increased risk of Guillain-Barré syndrome, requiring close medical management and possibly intensive care and mechanical ventilation.

History

Zika virus was first identified in 1947 in a rhesus monkey the Zika forest of Uganda, and was first identified in humans in 1952 in Uganda and the United Republic of Tanzania. Over the following decades, Zika virus caused rare, sporadic cases of disease in Africa and Asia, generally causing mild and self-limited illness of fever, rash, malaise, and other mild symptoms.

The first outbreaks were reported in Yap Island (Federated States of Micronesia) in 2007 and French Polynesia in 2013. The virus subsequently spread to other Pacific islands including New Caledonia, Cook Islands, Vanuatu and Easter Island (Chile), Fiji, Samoa, Solomon Islands, and Vanuatu. Zika virus was not known to cause severe disease until the 2013-2014 outbreak in French Polynesia, where increased incidence of Guillain-Barré Syndrome was first reported.

The Zika virus outbreak in the Region of the Americas began in Brazil in 2015; in July 2015, Brazil reported an association between Zika virus infection and Guillain-Barré syndrome (GBS) and few months later, in October 2015, an association between Zika virus infection and microcephaly.

Since 2015, outbreaks of Zika virus disease have now been recorded in Africa, the Americas, Asia and the Pacific; over 86 countries and territories have reported the presence of this mosquito transmitted disease.

Since 2017, Zika virus transmission in the Americas has waned, but transmission continues with intermittent areas of emergence and re-emergence.

In the African Region, only rare, sporadic Zika virus infection had been reported until 2015. Since 2015, outbreaks of Zika virus have been reported in Cabo Verde, Guinea-Bissau, and Angola.

There are two strains of Zika virus known as the African and Asian strains. The Asian strain was associated with the outbreaks in the Pacific and in the Americas. The Asian strain was also identified in the Cabo Verde outbreak and in Angola. To date, microcephaly has only been identified following infection with the Asian strain. Little information is available on the spectrum of disease and pregnancy risk associated with the African strain

Aedes mosquitoes that transmit Zika, dengue, yellow fever, and chikungunya primarily bite during daylight hours. Aedes sp. breed in small collections of water such as in trash, used tyres, flowerpots, and open water storage containers. Efforts to prevent transmission focus on elimination of these breeding sites around homes and near other areas of human-vector contact such as around schools and work sites. Other prevention strategies include use of personal protection measures such as use of protective clothing, insect repellent, and screens on windows and doors.

Surveillance goal

- The goal of surveillance is to develop, strengthen and implement integrated surveillance systems at all levels for Zika virus disease, its complications, and other arboviral diseases and their vectors, in order to provide up-to-date and accurate epidemiological and entomological information, to guide response.
- Existing surveillance systems should be enhanced for early detection and reporting of Zika virus and unusual clusters of neurological disorders or neonatal malformations.
- Timely notification of any event compatible with Zika virus is important, and in particular any associated with neurological disorders and neonatal malformations through established channels, including IHR.
- The establishment or strengthening of event-based or syndromic surveillance should be supported, potentially targeting specific groups for surveillance, such as pregnant women through antenatal and postnatal care, sentinel based surveillance systems for birth defects and Guillain-Barré syndrome, and existing lab-based disease specific surveillance systems (e.g. measles, polio) to facilitate detection of Zika virus infection and associated disorders. The similarity in vector and epidemiological transmission profiles of Dengue, Zika and Chikungunya calls for integrated surveillance of arboviroses.

Standard case definitions

Suspected Case:

A person presenting with rash and/or fever and at least one of the following signs or symptoms:

- Arthralgia; or
- Arthritis; or
- Conjunctivitis (non-purulent/hyperaemic).

Probable case:

A suspected case with presence of IgM antibody against Zika virus and an epidemiological link (with no evidence of infection with other flaviviruses).

Confirmed Case:

A person with laboratory confirmation of recent Zika virus infection:

- presence of Zika virus RNA or antigen in serum or other samples (e.g. saliva, urine, tissue, whole blood); or
- IgM antibody against Zika virus positive and PRNT90 for Zika virus with titre ≥20 and Zika virus PRNT90 titre ratio ≥ 4 compared to other flaviviruses; and exclusion of other flaviviruses.

These case definitions may change based on new knowledge.

Response to Zika virus disease

If Zika virus cases are suspected:

- Immediately report suspected cases to the next level using the case-based reporting form.
- Collect specimens for laboratory confirmation of cases
- Conduct active search for additional cases.
- Strengthen event-based surveillance to detect the emergence of clusters of cases presenting with rash and febrile syndrome of unknown aetiology.
- Conduct an investigation to determine risk factors for transmission.
- Manage and treat cases with supportive care

If Zika virus cases are confirmed:

Coordination and leadership

- Develop a national contingency plan for the prevention and control of Zika virus transmission and disease.
- Reinforce the Incident Management System to strengthen their coordination [including emergency operations centre (EOC)] to include the preparedness to respond to Zika, dengue, chikungunya and yellow fever.
- Actively engage other sectors (e.g., environment, agriculture, tourism) to respond to Zika virus through a multisector approach (One Health approach).

Surveillance, data management and laboratory

- Notify WHO through Ministry of Health using the IHR decision instrument.
- Enhance surveillance of Zika virus disease and of the conditions that may be associated with it, including microcephaly and Congenital Zika Syndrome and Guillain-Barré syndrome (GBS).
- Enhance surveillance at prenatal and postnatal clinics to monitor possible congenital infections and complications.
- Conduct active search for additional cases.
- Ensure the rapid and timely reporting and sharing of information of Zika virus disease using the IDSR/IHR tools.
- Ensure proper collection, transport, and storage of specimens for laboratory diagnostic testing.
- Conduct community-based assessments to determine the abundance of vector mosquitoes, identify the most productive larval habitats, promote and implement plans for appropriate vector control.
- Report any identified unusual increase in the incidence of congenital neurological malformations including microcephaly in neonates and adverse pregnancy outcomes not explained through alternate causes, to the relevant public health authorities using IDSR framework.

Vector control and personal protection

Zika

- Intensification of efforts to reduce mosquito populations including elimination of potential breeding sites (e.g., removal of trash and standing water sites around homes, covering home water storage containers, and use of larvicides) and adult mosquito control methods
- Promotion of personal protection measures such as use of light-coloured protective clothing (long sleeves and pants), insect repellent, and physical barriers such as screens, closed doors and windows, and sleeping under mosquito nets including during the day when Aedes mosquitoes are most active.
- All operators and other persons involved in vector control, such as larvicide application and indoor residual spraying, should be given protective measures including personal protective equipment.

Social mobilization, community engagement and communication

- Develop risk communication messages to address population concerns, enhance community engagement, improve reporting, and ensure application of vector control and personal protective measures targeting reduction of contact with the vector.
- Provide women of childbearing age and particularly pregnant women with the necessary information and materials on family planning and to reducing risk of exposure.
- Provide clinical and psychosocial support services for affected children and families.

Transmission prevention and case management

- Engage community health workers to inform them of the disease and risks and to build capacity
- Reinforce preventative measures for pregnant women through targeted interventions (including primary antenatal, postnatal and neonatal health care settings).
- Pregnant women who feel they may have been exposed to Zika virus may wish to consult with their health-care providers for laboratory testing for Zika virus infection, ultrasound assessment, and close monitoring throughout pregnancy, labour, delivery, and the post-natal period.
- After delivery, all infants should have head circumference measured and be examined for evidence of congenital malformations, including microcephaly, eye abnormalities, limb contractures, and other anomalies associated with congenital Zika syndrome. http://apps.who.int/iris/bitstream/10665/204475/1/WHO_ZIKV_MOC_16.3_eng.pdf?ua=1
- Zika can be transmitted through blood and blood products. Precautions already in place for ensuring safe blood donations, transfusions, and prevention of blood borne pathogens should be followed.
- Zika can be transmitted sexually. Men and women need to get counselling on safer sexual practices, and be offered condoms and full range of contraceptive methods.
- Ensure that pregnant women who have been exposed to Zika virus be counselled and followed for birth outcomes based on the best available information and national practice and policies.
- Refer most severe cases with complication to hospitalized care.

Operational research

- Assess the methods for practical and sustainable surveillance of Zika virus transmission, including integrated surveillance strategies of arboviruses.
- Conduct studies including case-control studies to investigate disease outcomes of infants exposed in-utero to Zika virus infection.
- Promote research in the areas of vaccines, drugs, diagnostics, vector biology and appropriate mosquito control methods.
- Entomological surveillance of Aedes mosquitoes is used for operational research purposes to determine changes in geographical distribution, for monitoring and evaluating control programmes, for obtaining relative measurements of the vector population over time, and for facilitating appropriate and timely decisions regarding interventions.
  Sampling of Aedes mosquitoes, pupae and oviposition should be conducted.
- As part of entomological surveillance, insecticide resistance monitoring in field populations of Aedes should be conducted to identify and select the appropriate insecticides.

Note: Application of strategic intervention in different country contexts:

- The described interventions will be packaged and applied in countries depending on the context. In countries where there is the spread of Zika virus as well as the associated complications, a full suite of strategies will be applied from enhanced surveillance, engaging communities, vector control and personal protective measures, care for people with complications and public health research to better understand risk and evaluate mitigation measures.
- For countries are already experiencing widespread Zika transmission or presence of Aedes vectors, enhanced surveillance should be put in place, communities engaged, and vector control and personal protective measures enhanced.

- For all other countries, risk communications for the public regarding trade and travel will be the main line of engagement. Table 1 below outlines the application of the strategies in the varying country context. Table 1 below describes the strategies to enforce in the various national contexts.

Table 1: Application of strategies to country context

Country Context	**Engage communities Communicate on risks**	**Monitor for Zika virus transmission and disease**	**Control transmission and prevent exposure**	**Manage complications associated with Zika virus**	**Investigate associated risks**
Aedes + Zika virus + Associated complications	☐	☐	☐	☐	☐
Aedes + Zika virus	☐	☐	☐		
Aedes	☐	☐	☐		
Other	☐				

Analysis and interpretation of data

Time: Graph cases of Zika virus infection, Guillain-Barré syndrome, and deaths weekly, by date of onset of symptoms. Graph cases of microcephaly and congenital Zika syndrome by date of birth. Construct an outbreak curve during the outbreak.
Place: Plot location of case households and worksites using precise mapping.
Personal information: Report case-based information for cases including Zika virus associated complications, hospitalizations, and deaths.
Analyse age and sex distributions and rates of associated complications. Assess risk factors to improve prevention of outbreaks and to better understand the rate of neurological complications among those infected with Zika virus.

Note: Entomological Analysis
In affected and high risk areas map infected and uninfected mosquito populations, breeding sites and case households

Laboratory confirmation

Diagnostic tests	• **Reverse transcriptase-polymerase chain reaction (RT-PCR) for viral RNA** • Serology for IgM detection • Plaque reduction neutralization test (PRNT)
Specimen	• RT-PCR: Serum, whole blood, or urine collected in a dry tube within 7 days of onset of symptoms • Serology (IgM): whole blood or serum collected in a dry tube >7 days after onset of symptoms. Whenever possible, a convalescent specimen should be collected at least 2-3 weeks after first specimen for IgG
How to prepare, store, and transport the specimen	Transport of specimens should comply with the WHO guidelines for the safe transport of infectious substances and diagnostic specimens. • Keep refrigerated (2-8°C) if specimen will tested within 48 hours of collection. • If testing will be done >48 hours, separate and freeze serum at -20°C and store for up to 7 days. • If storage >7 days, serum specimens should be stored at -70°C. • All types of specimens may be kept frozen at -20oC for up to 7 days, or at -70°C if >7 days. Samples can be preserved for extended periods. • Repeated freezing and thawing of specimens should be avoided. • Temperature should be monitored and recorded regularly to diminish risk of temperature fluctuations. Aedes mosquitoes for testing should be frozen and transported dry using standardized protocols.
Results	Diagnostic services for Zika virus are not routinely available. Contact the appropriate National authority or WHO for the assigned reference laboratory within the EDPLN.

References

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Sickle Cell Anaemia

Background

- Sickle cell anaemia is one of the most common genetic diseases worldwide. According to estimates, over 300,000 children will be born yearly with this condition. A large majority of people affected with this disease live on the African continent. Child mortality related to this disease is very high. More than 80% affected children die before the age of 5. All age groups are affected and youth from 10 to 29 years, representing 98, 2 of the sick.
- Cameroon has a sickle cell trait (SCT) prevalence of 20%, representing about 3.5 million people. Homozygotous states represent approximately 2%, representing 350 000 people considered as chronic patients. Out of all electrophoresis carried out at the Centre Pasteur of Cameroon (CPC) in 2004, sickle cell trait (SCT) prevalence represents 25.79% (749/2907) of specimen received including 8.43% of homozogotous states.
- A more recent survey conducted in 2007 by the Department for the Control of Diseases, Outbreaks and Pandemics of the Ministry of Public Health (MOH) sickle cell trait (SCT) prevalence revealed that, out of 12897 electrophoresis done in HF and laboratories in the country: SS haemoglobin was found in 6% people, or 769 cases; heterozygotous forms (HbAS) was found in 21% people, or 2738 cases; and normal haemoglobin (HbAA) in 93390 people, or 73% of the sample. These statistics show a high proportion of this disease in Cameroon.
- As for the socio-economic burden, sickle cell anaemia “SS“patients have a disrupted education due to its excess morbidity, and the frequent hospitalizations which jeopardize their education. Young “SS” adolescents and adults often find themselves having no qualification or face difficulties finding and keeping their job

Surveillance goal

- Detect sickle cell disease as quickly as possible for a good case management (comprehensive management), awareness-raising of the patient and their family;
- Describe and monitor vulnerable groups that is those at high risk of experiencing complications of the disease.
  - Early manage detected cases to reduce the morbidity and mortality rate related to this disease;
  - Plot cases by level of care.

Standard case definition of “new cases of sickle cell disease”

Suspected Case

A child aged at least 6 months with episodes of acute painful crisis in the lower and upper limbs and in the stomach, without fever and no history of trauma. Pain may be associated or not with jaundice or anaemia.

Confirmed Case

Presence of Haemoglobin “SC, SD or S Beta thalassemia” for serum haemoglobin electrophoresis.

Responding to alert threshold

If there is a suspected case of sickle cell disease

- Look at the family history to actively find similar cases (clinical history in siblings)
- Investigate personal history of blood transfusion or painful crisis in the suspected case or siblings;
- Administer the first dose of painkillers to relieve the pain.

if there is a confirmed case

- Treat the patient based on the national protocol for sickle cell management for the concerned level of care;

Sickle Cell Anaemia

- Provide psychological support to the family.

Analysing and Interpreting data

Time: Graph monthly trends of cases and deaths;
Place: Plot the distribution of case households.
Personal information:

- Analyse the age and sex distribution of cases;
- Assess the risk factors to improve disease prevention and control.

Respond to outbreak or action threshold

At the level of the community and the family, conduct information and education activities on sickle cell disease and the prevention means of this disease.

Laboratory confirmation

Haemoglobin electrophoresis must be used to confirm possible cases.

References

- *Integrated National Multisector Strategic Plan for the Control of Chronic Non-Communicable Diseases (CNCDs).*

Intestinal Helminthiasis

Background

- Intestinal Helminths are parasitic diseases caused by roundworms or nematodes living in the human intestine. The most common infections are Ascariasis (caused by Ascaris), Trichuriasis (caused by Trichocephalus), Ancylostomiasis (caused by Ancylostoma).
- Like for bilharzia, the contamination cycle is simple: the infected individual contaminates the soil and plants by defecating in open air and other persons are infected by eating contaminated foods, or by walking bare foot (in the case of ancylostomiasis).
- Indeed, intestinal helminths are common diseases. Over 10 million Cameroonians are infected by worms and the most vulnerable group affected includes children of school-going age (5 to 14 years).
- The consequences in children are stunting, decreased intellectual development, anaemia, increased risk of infection by other germs.
- Risk factor: it is a disease caused by the non-respect of hygiene rules: contaminated food, dirty hands, etc....
- This infection is preventable through respecting hygiene rules namely, washing hands with clean water and soap before and after each meal, avoiding walking barefoot and treating infected people.

Surveillance goal

- Avoid spreading the disease to risk areas.
- Map out the various areas enabling to take appropriate control measures.
- Identify the populations most at risk of intestinal helminths.
- Follow up the impact of control measures.
- Support operational research and innovation.

Standard case definitions

Suspected Case: In the case of intestinal helminths, this is any person with frequent diarrhoea and abdominal pains.

Confirmed Case: A suspected case with stool analysis in which helminths eggs or adult forms are found (Ascaris, Trichocephalus, Ancylostoma)

Responding to alert threshold

If there is a single case or a cluster of suspected or probable cases

- Collect stool specimens for laboratory confirmation.
- Diagnose and provide treatment to patients.
- Enhance surveillance to avoid spreading the transmission.

Respond to outbreak or action threshold

In addition to the response to alert threshold, if a single case is confirmed:

- Initiate the survey in living areas and the surroundings of the confirmed case.
- Conduct routine deworming of children of school-going age and risk populations if prevalence is above 10%.
- Promote hygiene and sanitation in the said areas and start social mobilisation there.

Analysing and Interpreting data

Intestinal Helminthiasis

Time: Graph the monthly curve of various cases. Draw up the curve (to monitor monthly evolution).

Place: Precisely plot the cases location and work places.

Personal information: Immediately forward information about the cases. Assess risk factors to improve prevention.

Laboratory confirmation

Specimen	Stool specimen collection
When to collect the specimen	Upon onset of the first symptoms
How to prepare, store, and transport the specimen	Collect a quantity of stool corresponding to *3 sugar lumps minimum* Transport stool specimen in the appropriate packaging (clean stool bottle) to avoid contamination.
Results	The laboratory should provide results within a few hours following the reception of the specimen.

References

Katz, N., Chaves, A., Pelligrino, J., 1972. A simple device for quantitative stool thick-smear technique in schistosomiasis mansoni. Rev. Inst. Med. Trop. Sao Paulo 14, 397–400.

World Health Organization, 1991. Basic Laboratory Methods in Medical Parasitology. World Health Organization, Geneva.

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Schistosomiasis (Bilharzia)

Background

- Schistosomiasis is a parasitic disease caused by worms called Schistosoma living around the veins around the bladder or the intestine. There are two main types: urinary bilharzia and intestinal bilharzia.
- Contamination cycle is simple: the infected person contaminates fresh water with urine or faeces. In water, eggs hatch into miracidiums which enter into vector molluscs. The latter, in their turn, emit cercarias, which will later on infect individuals in contact with the water.
- Consequences in children include stunting, decreased intellectual development, anaemia and increased risk of infection by other germs.
- Risk factors: This disease is linked to lifestyle behaviours namely, bathing or doing laundry in rivers, lakes, etc., fishing in fresh water and farming in swampy areas.
- This infection can be prevented by systematically using latrines and by respecting hygiene rules, avoid walking barefoot and wearing gloves and boots in swampy areas, treat the infected person.

Surveillance goal

- Avoid spreading the disease to risk areas.
- Map out the various areas enabling to take appropriate control measures.
- Identify the populations and areas at risk of bilharzia.
- Follow up the impact of control measures.
- Support operational research and innovation.

Standard case definitions

Intestinal Schistosomiasis

Suspected Case: A person having blood in urine WITH OR WITHOUT difficulty urinating (dysuria) or abdominal pain.

Confirmed case: suspected case with Schistosoma eggs in stool

Urinary Schistosomiasis

Suspected Case: a person with abdominal pain with presence of blood in urine

Confirmed Case: suspected case with schistosoma eggs in urine

Responding to alert threshold

If there is a single case or a cluster of suspected or probable case

- Collect stool specimens for laboratory confirmation.
- Diagnose and provide treatment to patients.
- Enhance surveillance to avoid spreading the transmission.

Respond to outbreak or action threshold

Schistosomiasis (Bilharzia)

In addition to the response to alert threshold, if a single case is confirmed:

- Initiate the survey in living areas and the surroundings of the confirmed case.
- Conduct routine deworming of children of school-going age and risk populations if prevalence is above 10%.
- Promote hygiene and sanitation in the said areas and start social mobilisation there.

Analysing and Interpreting data

Time: Graph the monthly curve of various cases. Draw up the outbreak curve (to monitor monthly evolution).

Place: Precisely plot the cases location and work places.

Personal information: Immediately forward information about the cases. Assess risk factors to improve prevention.

Laboratory confirmation

Specimen	Stool specimen collection
When to collect the specimen	Upon onset of the first symptoms
How to prepare, store, and transport the specimen	Collect a quantity of stool corresponding to *3 sugar lumps minimum* Transport stool specimen in the appropriate packaging (clean stool bottle) to avoid contamination.
Results	The laboratory should provide results within a few hours following the reception of the specimen.

References

Katz, N., Chaves, A., Pelligrino, J., 1972. A simple device for quantitative stool thick-smear technique in schistosomiasis mansoni. Rev. Inst. Med. Trop. Sao Paulo 14, 397–400.

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WHO (2000). WHO Recommended Surveillance Standards. WHO/CDS/CSR/ISR/99.2. Second Ed. - June 2000

Snakebites and other envenoming

Background

- Snakebite is a neglected public health issue in many tropical and subtropical countries. About 5.4 million snakebites occur each year, resulting in 1.8 to 2.7 million cases of envenoming (poisoning from snakebites). There are between 81 410 and 137 880 deaths and around three times as many amputations and other permanent disabilities each year.
- Most of these occur in Africa, Asia and Latin America. In Asia up to 2 million people are envenomed by snakes each year, while in Africa there are an estimated 435 000 to 580 000 snake bites annually that need treatment.
- In 2015, Cameroon started a weekly reporting of cases of snakebites in HFs. In 2018, the Cameroon Society of Epidemiology reported over 3000 victims of snakebites annually and a case fatality rate of 2.5%. Yet, this estimate is based on data provided by hospitals, exclusive of victims who could not go to consultation or who died before reaching a health structure.
- Women, children and agricultural workers in the poorest rural communities are the most affected.
- In Cameroon, 150 species of snake have been identified including more than 20% are potentially dangerous for men. Carpet Viper (*Echis ocellatus), a deadly species is found in t*he northern part of the country. The humid Equatorial climate of the Centre Region of Cameroon is host to various cytotoxic and hemotoxic vipers (*Bitis gabonica, Bitis nasicornis* and *Atheris squamigera*) whose venom can cause localized destruction of tissues likely to cause necrosis and amputation, or prevent the body from clotting, leading to haemorrhagic shock. Cobras and neurotoxic mambas (for example *Naja melanoleuca* and *Dendroaspis jamesoni*) constitute the endemic Elapides in this part of the country. Their venom can cause flaccid paralysis of cranial nerves, quickly spreading to respiratory muscles. Elapides equally include non-neurotoxic, cytotoxic and spitting cobras (for example *Naja nigricollis*). Snakes causing milder and localized cytotoxic symptoms include *Atractaspis spp,* some colubridae (*Crotaphopeltis spp*) and small vipers (*Causus spp*.).
- Bites by venomous snakes constitute medical emergencies. They can cause paralysis that may prevent breathing, bleeding disorders that can lead to a fatal haemorrhage, irreversible kidney failure and tissue damage that can cause permanent disability and limb amputation. Children often suffer more severe effects than adults, due to their smaller body mass.
- In contrast to many other serious health conditions, a highly effective treatment exists for snakebites envenoming. Most deaths and serious consequences of snakebites are entirely preventable by making antivenoms more widely available. High quality snake antivenoms are the only effective treatment to prevent or reverse most of the venomous effects of snakebites. They are included in the WHO List of essential medicines and should be part of any primary health care package where snakebites occur.
- The availability and accessibility of antivenoms, and raising the awareness of communities and health personnel on primary prevention methods are the best means to limit the serious consequences and deaths due to snakebites.
- After a snake bite, follow this procedure:
  - Immediately move away from the area where the bite occurred;
  - Remove everything that is tight around the body area bitten by the snake to avoid damages in case of swelling;
  - Reassure the victim as most venomous snake bites do not cause immediate death;
  - Completely immobilise the person and take them to the nearest HF as soon as possible;
  - Apply pressure on the bite site with a buffer (this may be appropriate on some case);
  - Avoid traditional first aid methods or herbal medicines;
  - Administer a painkiller if pain;
  - Place the person in the recovery position;
  - Closely monitor respiratory track and breathing, and be ready to resuscitate, if necessary.

Surveillance goal

- Reduce morbidity and mortality related to snake bites
- Map out the various risk area
- Identify the populations most at risk of snake bites
- Follow up the impact of control measures.
- Support operational research and innovation.

Standard case definitions

Suspected case: A person who had had a direct contact with a snake (eventually caught or filmed, if possible) and which present skin marks consistent with snake fangs, or exposed to a spraying of venom.
Confirmed case: A suspected case presenting one of the following syndromes:

- Cytotoxic (gradual oedema…)
- Neurotoxic (neurological deficiency, hypersalivation, excessive sweating, shortness of breath…)
- Hemotoxic (spontaneous bleeding, clotting disorder…).

Responding to alert threshold

If there is a single case or a cluster of suspected cases

- Establish clinical diagnostic and provide care to patients
- Enhance surveillance to avoid other bites

Respond to outbreak or action threshold

In addition to the response to alert threshold, if a single case is confirmed:

- Initiate the survey in living areas and the surroundings of the confirmed case.
- Raise awareness of risk populations
- Promote hygiene and sanitation in the said areas
- Constitute a stock of antivenom within the HDs

Analysing and Interpreting data

Time: Draw up the monthly curve of the various cases (to monitor the monthly evolution).

Place: Precisely plot the cases location and work places.

Personal information: Immediately forward information about the cases. Assess risk factors to improve prevention.

Laboratory confirmation

The diagnostic of envenoming related to snakebites is essentially clinical.

References

Snake bites: key facts. https://www.who.int/fr/news-room/fact-sheets/detail/snakebite-envenoming
Morsures de serpent https://epicentre.msf.org/portfolio/les-morsures-de-serpent
Cameroon Society of Epidemiology (CaSE), C.M.o.P.H.a.C.S.o.E., bulletin epidemiologique regional de l'Extrême-Nord. 2018
Gonwouo Nono Legrand, Lebreton Matthew, Chirio Laurent, Dzikouk G. Biogeographical distribution of snakes in Cameroon: The case of venomous snakes (PDF Download Available). ResearchGate

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Leishmaniasis

Background

- Human leishmaniasis a parasitic disease transmitted by a vector caused by an infection of the blood by an intracellular protozoan of the genus leshmania when a human being is bitten by infected sand flies (phlebotominae). It is considered as one of the main neglected tropical diseases with 1 million new cases yearly, and it is said to be the third most serious NTD after sleeping sickness and Chagas disease.
- Human leishmaniasis is mostly endemic in tropical and subtropical Regions, with approximately 94% new cases of visceral leshmaniasis occurring in India, Sudan, Brazil, South Sudan and Ethiopia only; other East African countries like Kenya have recurrent outbreaks.
- This disease mostly affects the poorest people on the planet and is associated with malnutrition, displacement of populatons (refugees, internally displaced people), poor housing conditions, weakened immune system, and lack of financial resources.
- Leishmaniasis is related to environmental changes such as deforestation, construction of dams, irrigation systems and urbanisation. Pets, such as dogs, are major vectors.
- The clinical presentation of the disease is as follows:
  - Visceral leishmaniasis (VL), also known as kala-azar is fatal if left untreated in over 95% of cases. It is characterized by irregular bouts of fever, weight loss, enlargement of the spleen and liver, and anaemia. It remains one of the top parasitic diseases with outbreak and mortality potential.
  - Cutaneous leishmaniasis (CL) is the most common form of leishmaniasis and causes skin lesions, mainly ulcers, on exposed parts of the body, leaving life-long scars and serious disability or stigma. It is estimated that between 600 000 to 1 million new cases occur worldwide annually.
  - Mucocutaneous leishmaniasis leads to partial or total destruction of mucous membranes of the nose, mouth and throat.

- Leishmaniasis is treatable and curable and requires an immunocompetent system because drugs cannot eliminate the parasite from the body hence the risk of relapse in an immunocompromised organism. All patients diagnosed with visceral leshmanisasis require rapid and comprehensive treatment.
- The main prevention strategies are:
  - Early diagnosis and a rapid and effective treatment will reduce disease prevalence and prevent disabilities and deaths
  - Vector control contribute in reducing or stopping the spread of diseases by reducing the number of phlebotominae. Control methods include insecticide spraying, use of insecticide treated mosquito nets, environmental management and personal protection.
  - Effective disease surveillance is important to monitor and act quickly during outbreaks and situations where the case fatality rate of treated cases is high.
  - The control of host animal reservoirs is complex and should be adjusted to the local situation;
  - Social mobilisation and strengthening partnerships, community mobilisation and education through effective intervention means for behaviour change must always be adjusted to the local context.

Surveillance goal

- Increase the percentage of laboratory confirmed cases
- Map out the various risk area
- Actively find cases in endemic regions through population surveys and serological screening.
- Monitor Leshmaniasis-free areas by screening out humans and animals.

Standard case definitions

Hospitalization or outpatient consultation with one of the key diagnosis of leshmaniasis;

or

A file on a medical event to be reported on leshmaniasis

Responding to alert threshold

If there is a single case or a cluster of suspected cases

- Establish a clinical diagnosis and provide care to patients;
- Strengthen surveillance to avoid an outbreak.

Respond to outbreak or action threshold

In addition to the response to alert threshold, if a single case is confirmed:

- Start a survey in the living areas and the surroundings (humans and domestic animals, namely dogs) of the confirmed case;
- Raise population awareness;
- Promote hygiene and sanitation in the said areas;
- Enhance the capacity of district laboratories in the rapid confirmation of suspected cases.

Analysing and Interpreting data

Time: Draw up the monthly curve of the various cases (to monitor the monthly evolution).

Place: Precisely graph housings.

Personal information: Immediately forward information about the cases. Assess risk factors to improve prevention.

Laboratory confirmation

Laboratory diagnosis is done by identifying amastigotes in lesions on Giemsa-stained slides in the microscope.
Specimens are then collected and isolated for molecular confirmation by PCR.

References

1. *WHO | Epidemiological situation (2019). Available at: https://www.who.int/leishmaniasis/burden/en/ (Accessed: 6 December 2019).*
2. *WHO Leishmaniasis Fact Sheets (2020). Available at: https://www.who.int/news-room/fact-sheets/detail/leishmaniasis.*
3. *Tabbabi, A. (2019) 'Review of leishmaniasis in the middle east and North Africa', African Health Sciences. Makerere University, Medical School, 19(1), pp. 1329–1337. doi: 10.4314/ahs.v19i1.4.*

APPENDICES TO SECTION 11

Appendices provide examples of specific forms for the various programmes. Some help in documenting initial results while others are designed for in-depth investigations. Refer to country surveillance programmes to obtain the right forms.

APPENDIX 11A	Adverse Events Following Immunization (AEFI) –Investigation Form
APPENDIX 11B	Acute Flaccid Paralysis – Case-based Investigation Form
APPENDIX 11C	Cholera - Case-based Investigation Form
APPENDIX 11D	Dracunculiasis – Case-based Investigation Form
APPENDIX 11E	Maternal Deaths – Reporting Form
APPENDIX 11F	Perinatal Deaths – Weekly Summary Reporting Form
APPENDIX 11G	Measles – Case-based Investigation Form
APPENDIX 11H	Meningitis - Case-based Investigation Form and Decision Tree for vaccine selection during a reactive immunization campaign
APPENDIX 11I	Neonatal Tetanus – Case-based Investigation Form
APPENDIX 11J	Respiratory diseases (including influenza) – Reporting and Investigation Form of COVID-19 in Cameroon
APPENDIX 11K	Tuberculosis (MDR and XDR) - Case-based Reporting Form
APPENDIX 11L	Viral Haemorrhagic Fever- Case-based Reporting Form
APPENDIX 11 M	Acute or Chronic Viral Hepatitis - Case-based Investigation Form
APPENDIX 11N	IDSR Descriptive List of an outbreak
APPENDIX 11O	Contact Listing Forms
APPENDIX 11P	Community-based Signals – Logbook
APPENDIX 11Q	Event-Based Surveillance– Signal Register
APPENDIX 11R	AEFI Reporting Form
APPENDIX 11 S	Aide-mémoire on AEFI Investigation

APPENDIX 11A. Adverse Events Following Immunization (AEFI) - Investigation Form

REPUBLIQUE DU CAMEROUN Paix – Travail – Patrie ---------- Minister of Public Health ----------	REPUBLIC OF CAMEROON Peace – Work - Fatherland ------------------- MINISTRY OF PUBLIC HEALTH -------------------

AEFI INVESTIGATION FORM

(Only in case of severe AEFI – Death / Permanent disability / Hospitalization / Cluster)

(in a cluster use one form for each case)

Section A - Basic Information

Case identification code

Name of investigating officer :

Date of Investigation: __ __ / __ __ / __ __ __ __

Date this form was finalized: __ __ / __ __ / __ __ __ __

Title/ Function

Fixed telephone number +237 __ __ / __ __ / __ __ __ __

Mobile telephone

E-mail address:

Patient name

Date of birth (DD/MM/YYYY): __ __ / __ __ / __ __

Sex: ☐ M ☐ F

OR *Age at onset* (DD/MM/YYYY): __ __ / __ __ / __ __ ***OR*** *Age group:* ☐ *< 1 year* ☐ *1–5 years* ☐ *>5 years*

Full and accurate address of patient (or popular neighbour):

Name of vaccine/diluent administered to patient	**Vaccination Date**	**Vaccination Hour**	**Dosage (e.g. 1st, 2nd, etc.)**	**Lot No.**	**Expiration date**
				Vaccine	Vaccine
				Diluent	Diluent
				Vaccine	Vaccine
				Diluent	Diluent

				Vaccine	Vaccine
				Diluent	Diluent
				Vaccine	Vaccine
				Diluent	Diluent
				Vaccine	Vaccine
				Diluent	Diluent

Address of Immunization Site:

☐ Public ☐ Private ☐ Mass campaign ☐ Advanced/Mobile strategy ☐ Others (specify) __________

Date of first symptom (DD/MM/YYYY): __ __ / __ __ / __ __ Time of 1st symptom (hh/mm): __ / __

Date of hospitalization (DD/MM/YYYY): __ __ / __ __ / __ __ __ __

Date of first notification to health authority (DD/MM/YYYY): __ __ / __ __ / __ __ __ __

Condition at date of investigation (✓): ☐ Death ☐ Disabled ☐ Recovering ☐ Completely recovered ☐ Unknown

If death, date and time of death (DD/MM/YYYY) : __ __ / __ __ / __ __ __ __ *(hh/mm) :* __ __ / __ __

Autopsy performed? (✓) ☐ Yes (date)________________ ☐ No ☐ Scheduled on (date)______________ time___________

Attach report (if applicable)

Section B - Patient Information Prior to Immunization

Parameter	Result	If yes, specify
History of similar events apart from vaccination	☐Yes ☐No ☐Unknown	
Pre-existing disease (30 days) / congenital pathology	☐Yes ☐No ☐Unknown	
History of allergy to any vaccine, drug or food	☐Yes ☐No ☐Unknown	
Pre-existing disease (30 days) / congenital pathology	☐Yes ☐No ☐Unknown	
History of hospitalization in the past 30 days (indicate reason)	☐Yes ☐No ☐Unknown	
Patient currently on medication? (If yes, name of drug, indication, regimen & dates of treatment)	☐Yes ☐No ☐Unknown	
Family history of any disease or allergy (related to AEFI)	☐Yes ☐No ☐Unknown	

If female of childbearing potential:

- Currently pregnant? ☐ Yes (weeks) ______________________ ☐ No ☐ Unknown
- Currently breastfeeding? ☐ Yes ☐ No

In the case of infants

Birth: ☐ At term ☐ before term ☐ after term Birth weight: ___________g

Delivery : ☐ Normal ☐ Caesarean section ☐ Assisted (forceps, vacuum...) ☐ with complications

____

Section C - First Test Details** of Severe AEFI Case

Source of Information : ☐ **Test by investigator** ☐ **Documents** ☐ **Others**__________________________

☐ **Verbal autopsy** (specify source)

____

Name of person who first examined/treated the patient:

____

Names of other persons who treated the patient:

____

Other sources of information (specify): ____

History of illness (be specific about the chronology of signs and symptoms) :

Name and contact information of the person who completed this form with detailed clinical data:	Function:	Date/time

If the patient has received medical care – attach copies of all documents (including case report form, hospital record with results of clinical, biological and imaging examinations, health records and prescriptions issued in all health centers where the patient has been and autopsy report if applicable) and write here only the information that is not available in the said documents

If the patient has not received medical care – fill in the history, examine the patient and write your observations (add additional sheets if necessary)

Specify on the basis of positive and negative arguments,

Presumptive diagnosis:

Final diagnosis:

Section D - Detailed information on vaccines provided on site in relation to AEFI on the corresponding day

Number of persons vaccinated for each antigen at the vaccination session site. Attach log if applicable	Name of vaccine									
	Number of doses									

a) Number of persons vaccinated from the relevant vial/ampoule	
b) Number of persons vaccinated with the vaccine during the same session	
c) Number of persons vaccinated with the relevant vaccine with the same lot number in other locations. Specify locations: _____________	

d) When was the patient vaccinated?

☐ During the first vaccinations of the session ☐ During the last vaccinations of the session ☐ Unknown

For multi-dose vials, was the vaccine administered

☐ Among the first doses from the vial? ☐ Among the last doses? ☐ unknown

e) Was there a prescribing error or noncompliance with the recommendations for use of this vaccine?	☐Yes ☐No	explain
f) Based on your investigation, do you believe that the vaccine(s) administered may not have been sterile?	☐Yes ☐No ☐Not assessable	explain
g) Based on your investigation, do you believe that the physical condition of the vaccine (e.g., cloudiness, presence of foreign substances, etc.) was abnormal at the time of administration?	☐Yes ☐No ☐Not assessable	explain
h) Based on your investigation, do you believe that an error in vaccine reconstitution/preparation was made by the vaccinator (e.g., wrong product, wrong diluent, improper mixing, improper syringe filling, etc.)?	☐Yes ☐No ☐Not assessable	explain

i) Based on your investigation, do you think that there was an error in the handling of the vaccine (e.g., break in the cold chain during shipment, storage, and/or vaccination session, etc.)?	☐Yes ☐No ☐Not assessable	explain
j) Based on your investigation, do you believe that the vaccine was not administered correctly (e.g., incorrect dosage, incorrect site or route of administration, incorrect injection practice, etc.?	☐Yes ☐No ☐Not assessable	explain

k) Is this case part of a cluster?	☐Yes ☐No ☐Unknown
i. If yes, how many other cases were detected in the cluster?	
a. Did all cases in the cluster receive the vaccine from the same vial?	☐Yes ☐No ☐Unknown
b. If no, number of vials used in the cluster (enter information separately)	

Section E - Vaccination practices at the place(s) where the vaccine was used

(Complete this section by interviewing and/or by observing the procedure)

Syringes and needles used:	
• Are auto-disable syringes used for vaccination?	☐Yes ☐No ☐Inc
If not, specify the type of syringes used: ☐ Glass ☐ Disposable ☐ Recycled ☐ Other ______________________________	
Reconstitution: (Complete only if relevant, otherwise tick NA (not applicable))	
• Procedure reconstitution	Status
Same syringe used for reconstitution of several vials of the same vaccine?	☐Yes ☐No ☐Inc
Same syringe used for reconstitution of several vials of different vaccines?	☐Yes ☐No ☐Inc
Different syringes used for the reconstitution of each vial of vaccine?	☐Yes ☐No ☐Inc
Different syringes used for the reconstitution of each vaccination?	☐Yes ☐No ☐Inc
• Are the vaccines and diluents used the same as those recommended by the manufacturer?	☐Yes ☐No ☐Inc
Further comments	

Section F - Cold chain and transportation

(Complete this section by interviewing and/or by observing the procedure)

Last point of vaccine storage:	
- Is the temperature of the refrigerator used to store the vaccine monitored?	☐Yes ☐No
○ If yes, was there a difference of more than 2-8°C after the vaccine was placed there?	☐Yes ☐No
○ If yes, provide separate monitoring data	
- Has the correct procedure for storing vaccines, diluent and syringes been followed?	☐Yes ☐No ☐Inc
- Was anything else (other than EPI vaccines and diluent) stored in the fridge or freezer?	☐Yes ☐No ☐Inc
- Were there partially used reconstituted vaccines in the refrigerator?	☐Yes ☐No ☐Inc
- Were there any unusable vaccines (expired, spoiled, no labelled, no control pellet (PCV) at unusable stage, frozen) in the fridge?	☐Yes ☐No ☐Inc
- Were unusable diluent (expired, spoiled, different from those recommended by the manufacturer, cracked or soiled bottles) kept?	☐Yes ☐No ☐Inc
Further comments	
Vaccines transportation	
- Type of vaccines carrier used	
- Was the vaccine carrier sent to the vaccination site on the day the vaccination took place?	☐Yes ☐No ☐Inc
- Was the vaccine carrier sent from the vaccination site on the day the vaccination took place?	☐Yes ☐No ☐Inc
- Was an insulated package used?	☐Yes ☐No ☐Inc
Further comments	

Section G - Community survey

(Please visit and interview parents/others)

Were similar events reported at the same time in the same locality? ☐Yes ☐No ☐Unknown

If yes, describe :

If yes, how many events/episodes?

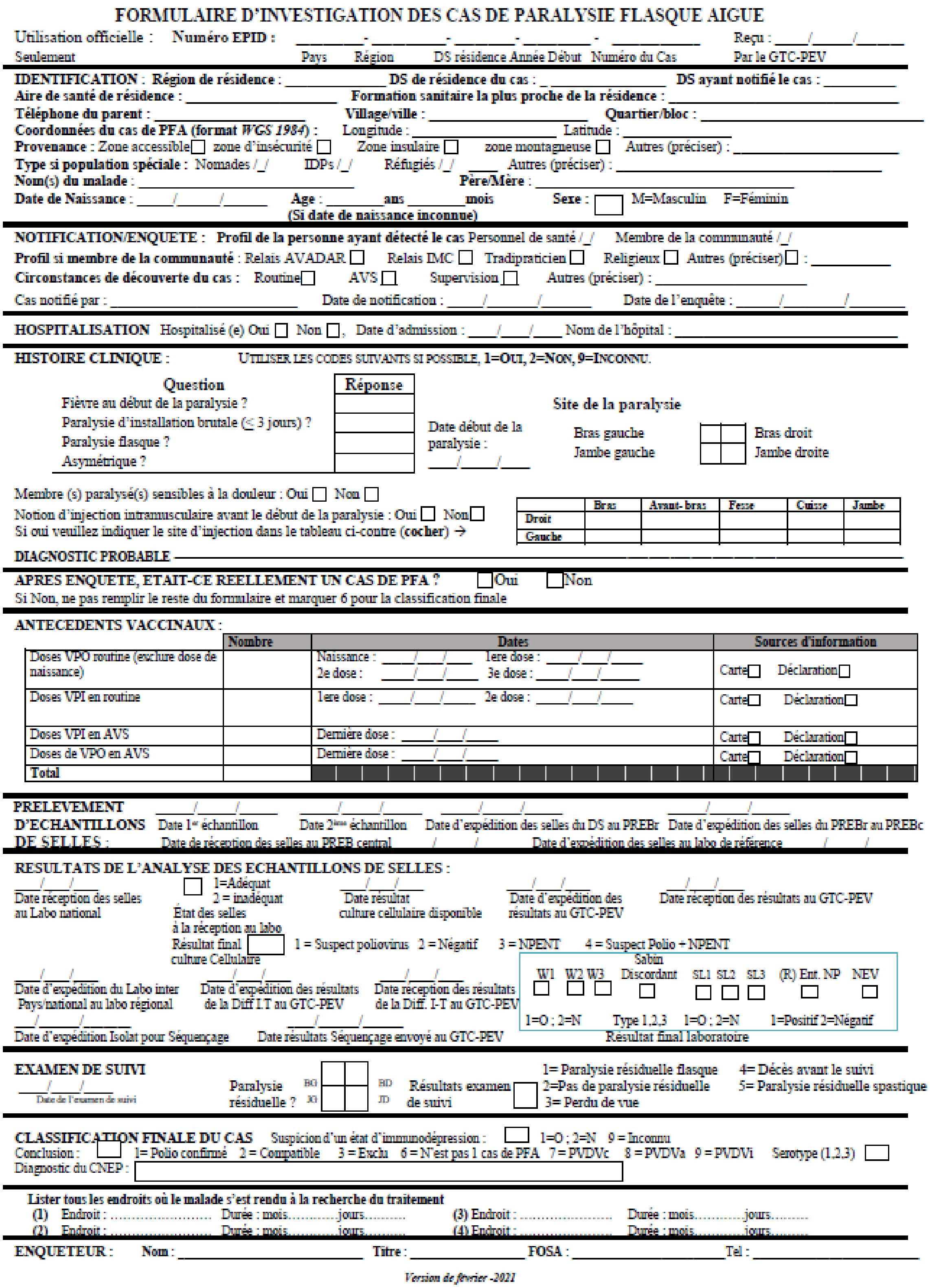
How many of those affected are:

- Vaccinated: ________________________________ (specify antigens and vaccination sites on an additional sheet)
- Non vaccinated: ____________________________
- Unknown: ___________________________________

Further comments:

Section H - Other findings/observations/comments

APPENDIX 11B. Acute Flaccid Paralysis – Case-based Investigation Form



FORMULAIRE D'INVESTIGATION DES CAS DE PARALYSIE FLASQUE AIGUE

Utilisation officielle Seulement : **Numéro EPID :** ________- _________- ________- _________ - ___________ (Pays, Région, DS résidence, Année Début, Numéro du Cas)
Reçu Par le GTC-PEV : ____/_____/______

IDENTIFICATION : Région de résidence : ______________ **DS de résidence du cas :** _______________ **DS ayant notifié le cas :** __________
Aire de santé de résidence : ____________________ **Formation sanitaire la plus proche de la résidence :** ________________________
Téléphone du parent : ________________ **Village/ville :** ________________ **Quartier/bloc :** ______________________
Coordonnées du cas de PFA (format *WGS 1984*) : Longitude : ________________ Latitude : __________
Provenance : Zone accessible ☐ zone d'insécurité ☐ Zone insulaire ☐ zone montagneuse ☐ Autres (préciser) : ________________
Type si population spéciale : Nomades /_/ IDPs /_/ Réfugiés /_/ ____ Autres (préciser) : ______________________
Nom(s) du malade : ______________________ **Père/Mère :** __________________________
Date de Naissance : _____/______/______ **Age :** ________**ans** ________**mois** (Si date de naissance inconnue) **Sexe :** ☐ M=Masculin F=Féminin

NOTIFICATION/ENQUETE : Profil de la personne ayant détecté le cas Personnel de santé /_/ Membre de la communauté /_/
Profil si membre de la communauté : Relais AVADAR ☐ Relais IMC ☐ Tradipraticien ☐ Religieux ☐ Autres (préciser) ☐ : ___________
Circonstances de découverte du cas : Routine ☐ AVS ☐ Supervision ☐ Autres (préciser) : _______________
Cas notifié par : ______________________ Date de notification : _____/_______/_______ Date de l'enquête : ______/________/________

HOSPITALISATION Hospitalisé (e) Oui ☐ Non ☐, Date d'admission : ____/____/____ Nom de l'hôpital : ______________________

HISTOIRE CLINIQUE : UTILISER LES CODES SUIVANTS SI POSSIBLE, 1=OUI, 2=NON, 9=INCONNU.

Question	Réponse
Fièvre au début de la paralysie ?	
Paralysie d'installation brutale (≤ 3 jours) ?	
Paralysie flasque ?	
Asymétrique ?	

Date début de la paralysie : ____/_____/____

Site de la paralysie

Bras gauche	☐	☐	Bras droit
Jambe gauche	☐	☐	Jambe droite

Membre (s) paralysé(s) sensibles à la douleur : Oui ☐ Non ☐
Notion d'injection intramusculaire avant le début de la paralysie : Oui ☐ Non ☐
Si oui veuillez indiquer le site d'injection dans le tableau ci-contre (**cocher**) →

	Bras	Avant- bras	Fesse	Cuisse	Jambe
Droit					
Gauche					

DIAGNOSTIC PROBABLE ______________________________

APRES ENQUETE, ETAIT-CE REELLEMENT UN CAS DE PFA ? ☐ Oui ☐ Non
Si Non, ne pas remplir le reste du formulaire et marquer 6 pour la classification finale

ANTECEDENTS VACCINAUX :

	Nombre	Dates	Sources d'information
Doses VPO routine (exclure dose de naissance)		Naissance : ____/___/____ 1ere dose : ____/___/____ 2e dose : ____/___/____ 3e dose : ____/___/____	Carte ☐ Déclaration ☐
Doses VPI en routine		1ere dose : ____/___/____ 2e dose : ____/___/____	Carte ☐ Déclaration ☐
Doses VPI en AVS		Dernière dose : ____/___/____	Carte ☐ Déclaration ☐
Doses de VPO en AVS		Dernière dose : ____/___/____	Carte ☐ Déclaration ☐
Total			

PRELEVEMENT D'ECHANTILLONS DE SELLES :
_____/_____/_____ Date 1er échantillon
_____/_____/_____ Date 2ème échantillon
_____/_____/_____ Date d'expédition des selles du DS au PREBr
_____/_____/_____ Date d'expédition des selles du PREBr au PREBc
Date de réception des selles au PREB central / /
Date d'expédition des selles au labo de référence / /

RESULTATS DE L'ANALYSE DES ECHANTILLONS DE SELLES :
____/____/____ Date réception des selles au Labo national
☐ État des selles à la réception au labo 1=Adéquat 2 = inadéquat
____/____/____ Date résultat culture cellulaire disponible
____/____/____ Date d'expédition des résultats au GTC-PEV
____/____/____ Date réception des résultats au GTC-PEV
Résultat final culture Cellulaire ☐ 1 = Suspect poliovirus 2 = Négatif 3 = NPENT 4 = Suspect Polio + NPENT
____/____/____ Date d'expédition du Labo inter Pays/national au labo régional
____/____/____ Date d'expédition des résultats de la Diff I.T au GTC-PEV
____/____/____ Date réception des résultats de la Diff. I-T au GTC-PEV
___/______/______ Date d'expédition Isolat pour Séquençage
____/______/______ Date résultats Séquençage envoyé au GTC-PEV

Résultat final laboratoire

W1	W2	W3	Sabin Discordant	SL1	SL2	SL3	(R) Ent. NP	NEV
☐	☐	☐	☐	☐	☐	☐	☐	☐
1=O ; 2=N			Type 1,2,3	1=O ; 2=N			1=Positif 2=Négatif	

EXAMEN DE SUIVI
____/____/____ Date de l'examen de suivi
Paralysie résiduelle ? BG ☐ ☐ BD / JG ☐ ☐ JD
Résultats examen de suivi ☐ 1= Paralysie résiduelle flasque 2=Pas de paralysie résiduelle 3= Perdu de vue 4= Décès avant le suivi 5= Paralysie résiduelle spastique

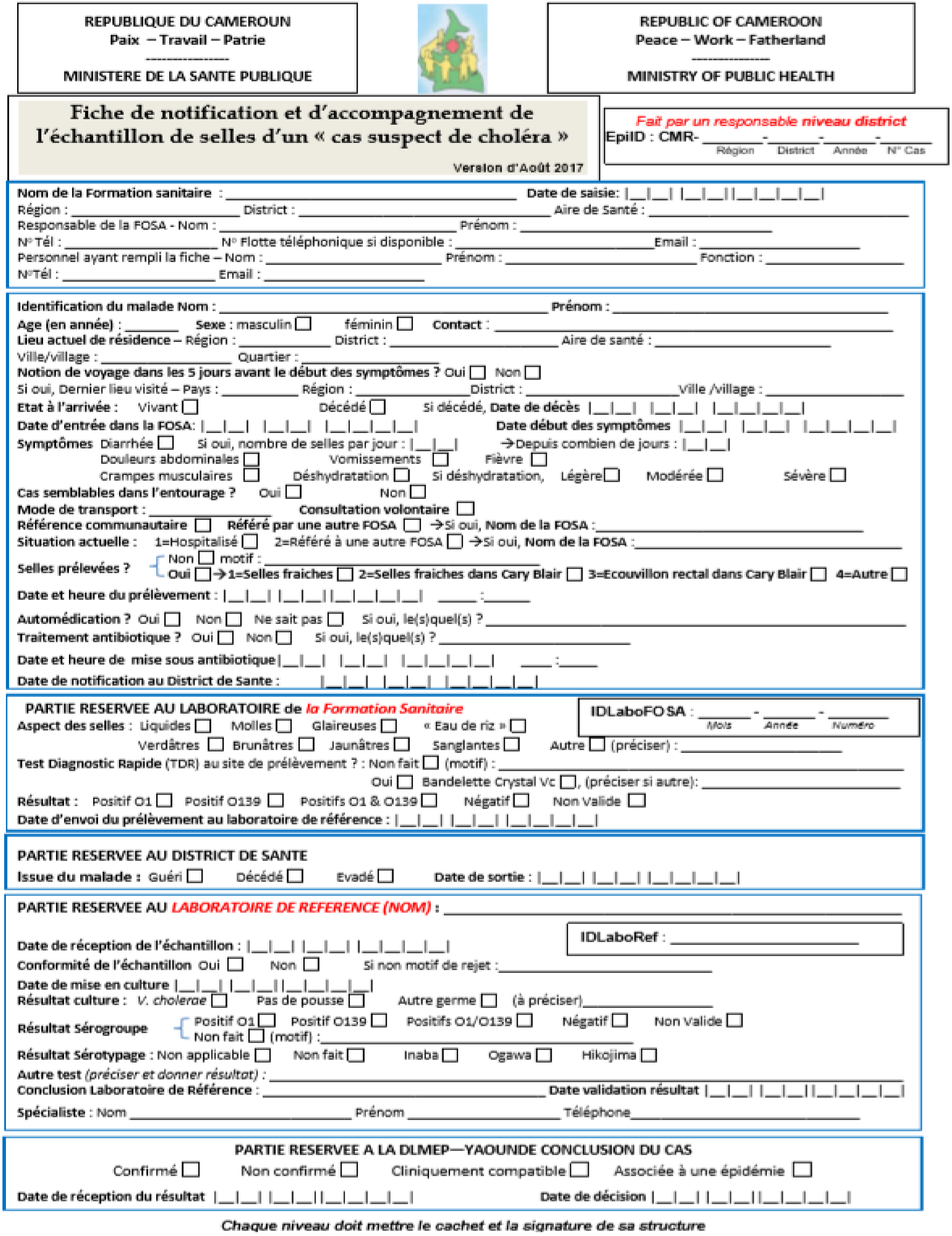
CLASSIFICATION FINALE DU CAS Suspicion d'un état d'immunodépression : ☐ 1=O ; 2=N 9 = Inconnu
Conclusion : ☐ 1= Polio confirmé 2 = Compatible 3 = Exclu 6 = N'est pas 1 cas de PFA 7 = PVDVc 8 = PVDVa 9 = PVDVi Serotype (1,2,3) ☐
Diagnostic du CNEP : ☐

Lister tous les endroits où le malade s'est rendu à la recherche du traitement
(1) Endroit : .... Durée : mois............jours..........
(2) Endroit : .... Durée : mois............jours..........
(3) Endroit : .... Durée : mois............jours..........
(4) Endroit : .... Durée : mois............jours..........

ENQUETEUR : Nom : ______________________ **Titre :** ______________ **FOSA :** __________________ **Tel :** ____________________

Version de février -2021

APPENDIX 11C. Cholera – Case-based Investigation Form



REPUBLIQUE DU CAMEROUN
Paix – Travail – Patrie

MINISTERE DE LA SANTE PUBLIQUE

REPUBLIC OF CAMEROON
Peace – Work – Fatherland

MINISTRY OF PUBLIC HEALTH

Fiche de notification et d'accompagnement de l'échantillon de selles d'un « cas suspect de choléra »

Version d'Août 2017

Fait par un responsable niveau district
EpiID : CMR- ________-________-________-________
Région District Année N° Cas

Nom de la Formation sanitaire : ____________________ **Date de saisie:** |__|__| |__|__||__|__|__|__|
Région : ____________ District : ____________ Aire de Santé : ____________
Responsable de la FOSA - Nom : ____________ Prénom : ____________
N° Tél : ____________ N° Flotte téléphonique si disponible : ____________Email : ____________
Personnel ayant rempli la fiche – Nom : ____________ Prénom : ____________ Fonction : ____________
N°Tél : ____________ Email : ____________

Identification du malade Nom : ____________________ **Prénom :** ____________
Age (en année) : ________ **Sexe** : masculin ☐ féminin ☐ **Contact** : ____________
Lieu actuel de résidence – Région : ____________ District : ____________ Aire de santé : ____________
Ville/village : ____________ Quartier : ____________
Notion de voyage dans les 5 jours avant le début des symptômes ? Oui ☐ Non ☐
Si oui, Dernier lieu visité – Pays : ________ Région : ____________District : ____________Ville /village : ____________
Etat à l'arrivée : Vivant ☐ Décédé ☐ Si décédé, **Date de décès** |__|__| |__|__| |__|__|__|__|
Date d'entrée dans la FOSA: |__|__| |__|__| |__|__|__|__| **Date début des symptômes** |__|__| |__|__| |__|__|__|__|
Symptômes Diarrhée ☐ Si oui, nombre de selles par jour : |__|__| →Depuis combien de jours : |__|__|
Douleurs abdominales ☐ Vomissements ☐ Fièvre ☐
Crampes musculaires ☐ Déshydratation ☐ Si déshydratation, Légère☐ Modérée ☐ Sévère ☐
Cas semblables dans l'entourage ? Oui ☐ Non ☐
Mode de transport : ____________ **Consultation volontaire** ☐
Référence communautaire ☐ **Référé par une autre FOSA** ☐ →Si oui, **Nom de la FOSA** :____________
Situation actuelle : 1=Hospitalisé ☐ 2=Référé à une autre FOSA ☐ →Si oui, **Nom de la FOSA** :____________
Selles prélevées ? Non ☐ motif : ____________
Oui ☐→**1=Selles fraiches** ☐ **2=Selles fraiches dans Cary Blair** ☐ **3=Ecouvillon rectal dans Cary Blair** ☐ **4=Autre** ☐
Date et heure du prélèvement : |__|__| |__|__||__|__|__|__| _____ :______
Automédication ? Oui ☐ Non ☐ Ne sait pas ☐ Si oui, le(s)quel(s) ? ____________
Traitement antibiotique ? Oui ☐ Non ☐ Si oui, le(s)quel(s) ? ____________
Date et heure de mise sous antibiotique |__|__| |__|__| |__|__|__|__| ____ :_____
Date de notification au District de Sante : |__|__| |__|__| |__|__|__|__|

PARTIE RESERVEE AU LABORATOIRE de *la Formation Sanitaire*

IDLaboFOSA : ________ - ________ - ________
Mois Année Numéro

Aspect des selles : Liquides ☐ Molles ☐ Glaireuses ☐ « Eau de riz » ☐
Verdâtres ☐ Brunâtres ☐ Jaunâtres ☐ Sanglantes ☐ Autre ☐ (préciser) : ____________
Test Diagnostic Rapide (TDR) au site de prélèvement ? : Non fait ☐ (motif) : ____________
Oui ☐ Bandelette Crystal Vc ☐, (préciser si autre): ____________
Résultat : Positif O1 ☐ Positif O139 ☐ Positifs O1 & O139 ☐ Négatif ☐ Non Valide ☐
Date d'envoi du prélèvement au laboratoire de référence : |__|__| |__|__| |__|__|__|__|

PARTIE RESERVEE AU DISTRICT DE SANTE
Issue du malade : Guéri ☐ Décédé ☐ Evadé ☐ **Date de sortie :** |__|__| |__|__| |__|__|__|__|

PARTIE RESERVEE AU *LABORATOIRE DE REFERENCE (NOM)* : ____________________

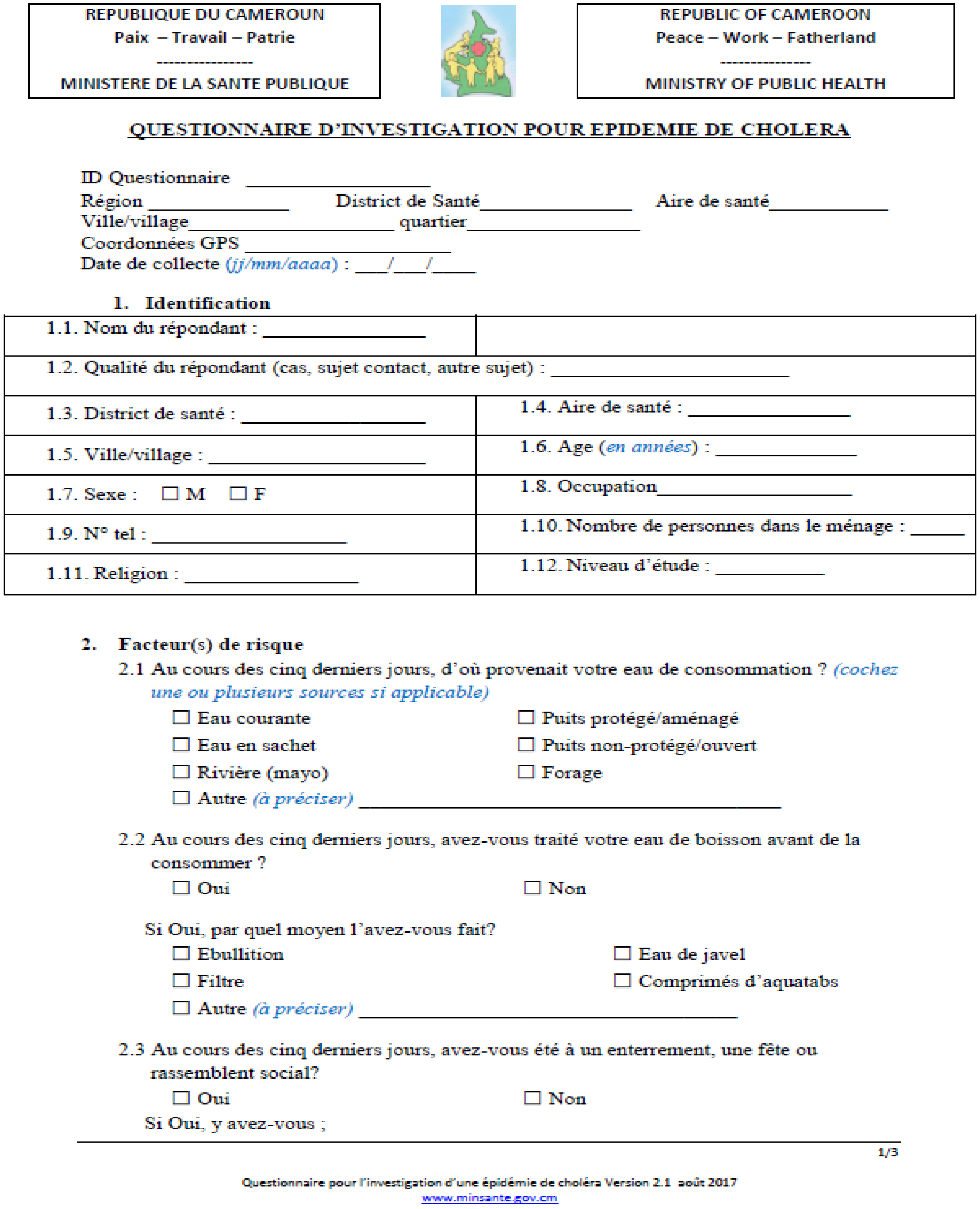
IDLaboRef : ____________

Date de réception de l'échantillon : |__|__| |__|__| |__|__|__|__|
Conformité de l'échantillon Oui ☐ Non ☐ Si non motif de rejet :____________
Date de mise en culture |__|__| |__|__||__|__|__|__|
Résultat culture : *V. cholerae* ☐ Pas de pousse ☐ Autre germe ☐ (à préciser)____________
Résultat Sérogroupe Positif O1 ☐ Positif O139 ☐ Positifs O1/O139 ☐ Négatif ☐ Non Valide ☐
Non fait ☐ (motif) :____________
Résultat Sérotypage : Non applicable ☐ Non fait ☐ Inaba ☐ Ogawa ☐ Hikojima ☐
Autre test *(préciser et donner résultat)* : ____________________
Conclusion Laboratoire de Référence : ____________________ **Date validation résultat** |__|__| |__|__||__|__|__|__|
Spécialiste : Nom ____________ Prénom ____________ Téléphone____________

PARTIE RESERVEE A LA DLMEP—YAOUNDE CONCLUSION DU CAS
Confirmé ☐ Non confirmé ☐ Cliniquement compatible ☐ Associée à une épidémie ☐
Date de réception du résultat |__|__| |__|__||__|__|__|__| **Date de décision** |__|__| |__|__||__|__|__|__|

Chaque niveau doit mettre le cachet et la signature de sa structure

Part B: Cholera outbreak Invetigation Form



REPUBLIQUE DU CAMEROUN
Paix – Travail – Patrie

MINISTERE DE LA SANTE PUBLIQUE

REPUBLIC OF CAMEROON
Peace – Work – Fatherland

MINISTRY OF PUBLIC HEALTH

QUESTIONNAIRE D'INVESTIGATION POUR EPIDEMIE DE CHOLERA

ID Questionnaire ____________________
Région _______________ District de Santé________________ Aire de santé_____________
Ville/village______________________ quartier___________________
Coordonnées GPS ______________________
Date de collecte *(jj/mm/aaaa)* : ____/____/_____

1. Identification

1.1. Nom du répondant : __________________	
1.2. Qualité du répondant (cas, sujet contact, autre sujet) : __________________________	
1.3. District de santé : ____________________	1.4. Aire de santé : __________________
1.5. Ville/village : ________________________	1.6. Age (*en années*) : ________________
1.7. Sexe : ☐ M ☐ F	1.8. Occupation_____________________
1.9. N° tel : ______________________	1.10. Nombre de personnes dans le ménage : ______
1.11. Religion : ___________________	1.12. Niveau d'étude : ____________

2. Facteur(s) de risque

2.1 Au cours des cinq derniers jours, d'où provenait votre eau de consommation ? *(cochez une ou plusieurs sources si applicable)*

☐ Eau courante
☐ Eau en sachet
☐ Rivière (mayo)
☐ Autre *(à préciser)* ____
☐ Puits protégé/aménagé
☐ Puits non-protégé/ouvert
☐ Forage

2.2 Au cours des cinq derniers jours, avez-vous traité votre eau de boisson avant de la consommer ?

☐ Oui ☐ Non

Si Oui, par quel moyen l'avez-vous fait?

☐ Ebullition
☐ Filtre
☐ Autre *(à préciser)* ______________________________________
☐ Eau de javel
☐ Comprimés d'aquatabs

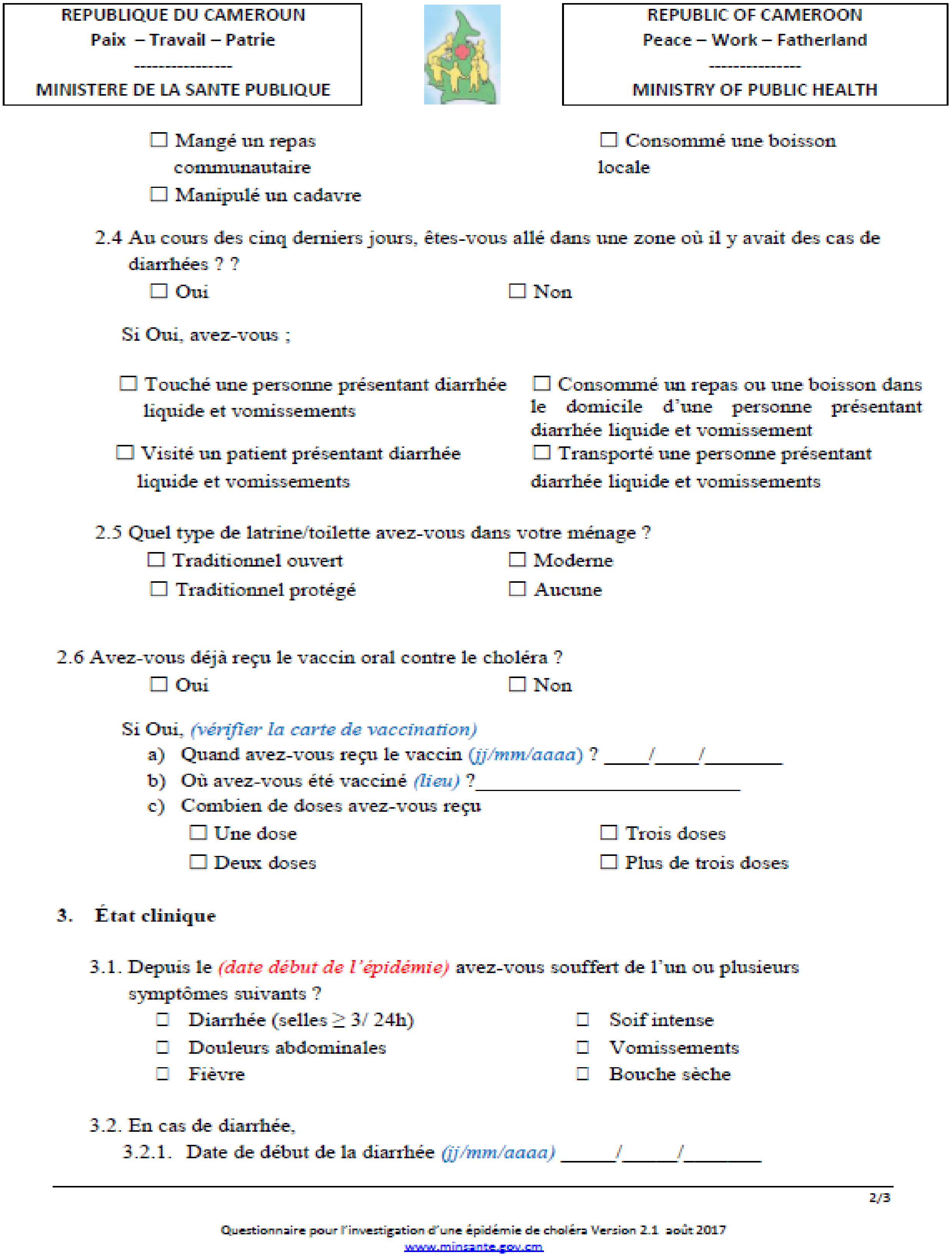
2.3 Au cours des cinq derniers jours, avez-vous été à un enterrement, une fête ou rassemblent social?

☐ Oui ☐ Non

Si Oui, y avez-vous ;

1/3

Questionnaire pour l'investigation d'une épidémie de choléra Version 2.1 août 2017
www.minsante.gov.cm



REPUBLIQUE DU CAMEROUN
Paix – Travail – Patrie

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Peace – Work – Fatherland

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☐ Mangé un repas communautaire
☐ Manipulé un cadavre
☐ Consommé une boisson locale

2.4 Au cours des cinq derniers jours, êtes-vous allé dans une zone où il y avait des cas de diarrhées ? ?

☐ Oui ☐ Non

Si Oui, avez-vous ;

☐ Touché une personne présentant diarrhée liquide et vomissements
☐ Visité un patient présentant diarrhée liquide et vomissements
☐ Consommé un repas ou une boisson dans le domicile d'une personne présentant diarrhée liquide et vomissement
☐ Transporté une personne présentant diarrhée liquide et vomissements

2.5 Quel type de latrine/toilette avez-vous dans votre ménage ?

☐ Traditionnel ouvert
☐ Traditionnel protégé
☐ Moderne
☐ Aucune

2.6 Avez-vous déjà reçu le vaccin oral contre le choléra ?

☐ Oui ☐ Non

Si Oui, *(vérifier la carte de vaccination)*

a) Quand avez-vous reçu le vaccin *(jj/mm/aaaa)* ? ____/____/________
b) Où avez-vous été vacciné *(lieu)* ?______________________________
c) Combien de doses avez-vous reçu

☐ Une dose
☐ Deux doses
☐ Trois doses
☐ Plus de trois doses

3. État clinique

3.1. Depuis le *(date début de l'épidémie)* avez-vous souffert de l'un ou plusieurs symptômes suivants ?

☐ Diarrhée (selles ≥ 3/ 24h)
☐ Douleurs abdominales
☐ Fièvre
☐ Soif intense
☐ Vomissements
☐ Bouche sèche

3.2. En cas de diarrhée,

3.2.1. Date de début de la diarrhée *(jj/mm/aaaa)* _____/_____/________

2/3

Questionnaire pour l'investigation d'une épidémie de choléra Version 2.1 août 2017
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3.2.2. Avez-vous été consulté dans une formation sanitaire pour des soins?

☐ Oui ☐ Non

Si oui :
Nom de la FOSA____
Combien de temps après le début de la diarrhée êtes-vous allé consulter ?
☐ Dans l'heure ☐ Dans les12h
☐ Dans les 24h ☐ Plus de 24h

3.2.3. Avez-vous été référé ?

☐ Oui ☐ Non

Si oui, référence communautaire FOSA précisez___________________________

3.2.4. Quel traitement avez-vous reçu à la formation sanitaire?
☐ SRO ☐Antibiotiques
☐ Perfusion ☐Autre _________________

3.2.5. Avez-vous été hospitalisé?
☐ Oui ☐ Non

Si Oui, pendant combien de temps ? (en jours) :_________________

Je vous remercie pour votre disponibilité

APPENDIX 11D. Dracunculiasis – Case-based Investigation Form

[COUNTRY NAME] DRACUNCULIASIS ERADICATION PROGRAMME

Nº Epid:_ _ _-_ _ _ -_ _ _-_ _-_ _ _

INDIVIDUAL INVESTIGATION FORM

To be completed in three copies

COUNTRY-REGION-DISTRICT-YEAR-

I. Information on notification et investigation

Village notificator:________________Area:________________District:________________Region: ________________

II. Information about the patient their place of residence

Name: ________________ Father /lodger's name: ________________

Age: ________Gender:________Occupation:________Ethnic group: ________

Permanent address: Village: ________________ Area: ________________

Sub-district:________________ District: ________________Region:________________

Area: Urban/Rural________________ Reference Points: ________________

III. Place of residence during the previous 10-14 months if different from the one indicated above.

Village: ________________ Area________________Sub-District: ________________

District: ________________Reion:________________Country:________________

IV. Travels by the patient during the previous 10-14 months

Departure date:	Arrival date:	Village:	Sub-district	District:	Region:

Water sources where the patient could have been contaminated – give precise details about its location and GPS indications:

Name	Latitude	Longitude	Type	Source	Tick in case of treatment date

V. Signs and symptoms

What was the first sign/symptom before the worm emerged? Blister/Itching/Tumefaction/Others, Specify________________

Emergence of Guinea worm: YES/NO Nbr of worms: is it the first time this year that a Guinea worm has emerged?

VI. Final classification of the case

Final classification: ________________ (1-local case 2-imported case 3- no cases related to Guinea worm)

If **the disease is not due to Guinea worm,** specify the final diagnosis:

____

If it is an IMPORTED case, indicate the type importation: **LOCAL/INTERNATIONAL.** If it is an imported case. Cross-reporting

Epid No: _ _ _-_ _ _ -_ _ _-_ _-_ _ _ _ *To be completed in three copies*

COUNTRY-REGION-DISTRICT-YEAR-

VII. Case isolation measure and Dracunculiasis register

Has the patient received health education: **YES/NO** Did the patient enter a water source: **YES/NO**

Place of care: **Country/Home/Health centre/Hospital**

Name of health facility/Health centres/Other health centres if the patient has been hospitalised: ________________________________

Date of admission: ___/____/______ ` Date of discharge: ___/____/______

SN.NO.	Location of worm	Date of detecting worm	Date of emergence of Guinea worm	Date of confirmation by the supervisor: Complete	Date of removal of worm	Bandage normal	Extract
_____	________________	___/____/______	___/____/______	___/____/______	___/____/______	☐	☐
_____	________________	___/____/______	___/____/______	___/____/______	___/____/______	☐	☐
_____	________________	___/____/______	___/____/______	___/____/______	___/____/______	☐	☐
_____	________________	___/____/______	___/____/______	___/____/______	___/____/______	☐	☐
_____	________________	___/____/______	___/____/______	___/____/______	___/____/______	☐	☐
_____	________________	___/____/______	___/____/______	___/____/______	___/____/______	☐	☐

VIII. Handling of specimens

Was a sample (of worm) taken and stored in alcohol? **YES/NO** If no, why? ________________________________

Dispatch date to the region: ______________________ Received by: ______________________Date of receipt by: ______________________

Dispatch date to the national level: ____________________Received by: ______________________Date of receipt by: ______________________

Only for use by the National Secretariat:

Has it been sent for confirmation? Yes/No Dispatch date: ___________________________Sent to:__________________________________

Receipt date of the results____________________

IX. Other information

Use of a fabric filter YES/NO Frequency of filter replacement 1-rarely; 2-sometimes; 3-always; 4-never

Notes: ____

The person who completed this form

____________________________________ ________________ _______________ __________

NAME FUNCTION MOBILE N° SIGNATURE

Person responsible for disease control or surveillance:

APPENDIX 11E Maternal deaths – Reporting Form

RÉPUBLIQUE DU CAMEROUN
Paix-Travail-Patrie

MINISTERE DE LA SANTE PUBLIQUE

SECRETARIAT GENERAL

DIRECTION DE LA SANTE FAMILIALE

REPUBLIC OF CAMEROON
Peace-Work-Fatherland

MINISTRY OF PUBLIC HEALTH

SECRETARIAT GENERAL

DEPARTMENT OF FAMILY HEALTH

INVESTIGATION FORM OF MATERNAL DEATHS

Region: _______________ HD: _____________ Health facility: _______________

Unique Identification Number of the Case ___ /____ /____ / _____ /______

INSTRUCTIONS:

1. ***This form should be completed for any death of women related to pregnancy, delivery and up to 42 days after delivery, including abortions and ectopic pregnancies***
2. ***The form should be completed in triplicate: One copy remains in the HF(service archives, review and surveillance committee) and the other copies are sent to the SDMPR committee of the appropriate HD of the HFand through hierarchical channels (SSD, DRSP, DSF). The medical head of the HFis responsible for transmitting these documents in electronic and/or paper form (weekly, Monday).***
3. ***These forms are confidential and anonymous.***

SECTION 1: IDENTIFICATION

1.1 Age		
1.2 Residence:	Rural	Urban
1.3 Occupation		

1.4 Marital Status								
1.5 Educational Level	Out-of-school		Primary		Secondary		Tertiary	
1.6 Occupation:								
1.7 Spouse/partner occupation:								

SECTION 2: DEATH

2. 1 Place of occurrence	HF		Type of structure	Public		Community	
				Faith-based			
				Private			
	Name of the Health Facility:						
2.2 Period of death	During pregnancy		During delivery		Postpartum		
	Post abortion/GEU		Number of days after delivery		After GEU		
2.3 Transfer	Yes		**2.4 Date and hour of admission to hospital**				
	No		**2.5 Date and hour of occurrence**				

2.6 If referred, from : _______________ duration and kind of conveyance: __________________________________

2.7 Condition on arrival	**Stable**		**Critical**		**Dead**	

Section 3: Development of pregnancy and care

3.1 **Obstetrical history**								
	All pregnancies (Gravida)	All births (Parity)	Total number of live births	Dead	Stillbirths	Neonatal deaths	Abortions	
							Spontaneous	

								Induced		
3.2 Type pregnancy	Simple		Twin		Triple or more					
3.3 Number of antenatal visits	4 or more		3		2		1		No visits	
3.4 Malaria prophylaxis	TPI 3+		TPI 2		TPI 1		Non received			
3.5 Anti-tetanus vaccination	VAT 2+		VAT 1		Non received		Don't know			
3.6 HIV Serology	HIV-negative		HIV-positive		Not done		Don't know			
3.6.1 Action to be taken in case of a HIV-positive status		TARV (yes/no)	if yes, specify the protocol		Not done					
3.7 Syphilis testing	**Negative**		Positive		Not done		Don't know			
3.8 Blood group Rhesus		3.**9 Haemoglobin electrophoresis**								
3.10 Was the patient ill during her pregnancy? date		If so, what was she suffering from? ______________________ ____								
3.11 Was she suffering from Malaria (Yes/No)		3.11 Was she suffering from hypertension**(Yes/No)**								
3.11 Was she suffering from diabetes **(Yes/No)**		3.11 Was she suffering from anaemia **(Yes/No)**								
3.15 Was she suffering from foetus malpresentation (Yes/No)		3.16 Has she previously underwent any caesarean section **(Yes /No)** how many? Year(s):								

Section 4: Delivery, Puerperium and neonatal information

4.1 Place of delivery	Health facility		Home		Street		Don't know	
4.2 Who assisted the deceased during delivery?	Obstetrician - Gynaecologist		Doctor		Midwife		Other, specify	
	Female (male) nurse		Care assistants		Traditional birth attendant			
4.3 How long (hours) did the labour last?								
4.4 Mode of delivery	Unassisted vaginal delivery		Assisted vaginal delivery		Caesarean section			

			vacuum/forceps					
4.5 Start of labour	Spontaneous		Induced				Caesarean section before the start	
4.6 Gestational age in weeks/months at the time of delivery or death if not delivered								
4.7 Pregnancy outcome	Living child		Is he still alive?					
			Did it die within 28 days of birth?					
			Dead after 28 days of life?					
	Stillbirth?		Spontaneous		abortion		Ectopic pregnancy	
					Induced			
4.8 Has the deceased been referred to any health centre of hospital?	Yes		No				Don't know	
4.8.1 If yes, how long did it take to go to the hospital? (hours)								

Section 5: CARE MANAGEMENT OF THE WOMAN ON ARRIVAL AT THE RECEIVING HEALTH FACILITY

5.1 Time taken to get to the HF	Less than 30 min		30-60 min		61-120 min		>120 min	
5.2 Care received on arrival at the health facility:								
Injectable antibiotics		Injectable anti-convulsant (Valium, magnesia sulphate)						
Oxytocin by infusion		Misoprostol:	Intra-vaginal		Intra -rectal		Oral	
Curettage		Extraction of products retained in the uterus (MVA)						
Oxygen		General, spinal, epidural or local anaesthesia						
Hysterectomy		Laparotomy						
Manual vacuum aspiration		Anti-malarial drugs			others			
Normal vaginal delivery					Manual delivery			
Instrumental vaginal delivery					Newborn reanimation			
Caesarean section					antihypertensive			
Management intercurrent disease (specify)					___________________________________			
In case of delivery								
Use of a partograph	Yes				No			
AMTSL practise		Yes			No			
Postpartum surveillance:								
For 2h in the delivery room	Yes				No			

For 6h in the postpartum period	Yes			No	
Surgery indicated		Yes		No	
If no, drugs not available		Surgeon/anaesthetist not available			
Operating room not available				Operating room not functional	
Blood transfusion	**Yes (done)**			No	
	Not indicated			Required blood not available	

Other care to be specified: ____

____

____

5.3 Cause maternal deaths

Eclampsia		Bleeding		Infections	
CIVD		Functional dystocia		Complicated abortion	
Malaria		HIV		Anaemia	
Uterine rupture					

Others: ____

____

____

APPENDIX 11F. Perinatal deaths - Weekly Summary Reporting form

RÉPUBLIQUE DU CAMEROUN	**REPUBLIC OF CAMEROON**
Paix-Travail-Patrie	Peace-Work-Fatherland
------------	-------------
MINISTERE DE LA SANTE PUBLIQUE	**MINISTRY OF PUBLIC HEALTH**
---------------	------------
SECRETARIAT GENERAL	**SECRETARIAT GENERAL**
------------	--------------
DIRECTION DE LA SANTE FAMILIALE	**DEPARTMENT OF FAMILY HEALTH**

STILLBIRTH AND PERINATAL DEATH INVESTIGATION FORM

Region ________________ **District** ______________ **Health Area** ________________

Unique Case Identification Number___ /____ /____ / _____ /______

INSTRUCTIONS:

1. ***This form must be completed for all deaths of newborns from birth to 28 days of life.***
2. **At each perinatal death the care team meets within 48 hours to analyse the case and complete the form**
3. ***The form must be completed in duplicate. One copy remains in the HF (service archive, review and monitoring committee) and another copy is sent through the chain of command (DHS, RDPH, FHD). The director of the HFis responsible for these mailings in electronic and/or paper form (Weekly, Monday).***
4. ***These forms are confidential and anonymous.***

PART ONE:

To be filled in by the major or team leader of the NEONATOLOGY OR PEDIATRIC DEPARTMENT of the HFwhere the death of the newborn occurred

SECTION 1: DEATH

2. Place of Occurrence	HF		Type of Structure	Public		Community	
				Religious			
				Private			
	Name of the Health Facility:						

2.2 Type de Death	Stillbirth Antepartum		First 7 days of life (early neonatal death)		Stillbirth, unknown time	
	Stillbirth Per-partum		8 to 28 days of life (late neonatal death)			

1.3 **Date and time of death**	

1.4 Transfer	Yes		**1.4.1 Date and time of arrival at the hospital:**
	Name		

1.5 If referred, origin : _______________ duration of transfer ___________________ newborn transfer medium : ___________

Carer:	Health care staff		Family		Other, specify	

1.6 Condition at arrival	**Stable**		**Critical**		**Deceased**	

SECTION 2: Pregnancy Evolution and care

1.1 Age of the mother								
1.2 Residence of the mother:	Rural				Urban			
1.3 Mother's occupation								
1.4 Matrimonial status of the mother								
1.5 Mother's level of education	Out of school		Primary		Secondary		Tertiary	
1.6 Mother's occupation								
1.7 Spouse's/partner's occupation :								

1.8 **Obstetrical history**	All the pregnancies (Gravida)	All the births (Parity)	Total number of alive births	Deceased	Stillbirths	Neonatal deaths	Abortions			
							Spontaneous			
							Provoked			
1.9 Type of pregnancy	Simple		Twins		Triple or more					
1.10 Number of prenatal consultations	4 or more		3		2		1		No visit	
1.11 Anti-malarial prophylaxis	TPI 3+		TPI 2		TPI 1		Not received			
1.12 Anti-tetanus vaccination	VAT 2+		VAT 1		Not received		Does not know			
1.13 HIV serology	HIV-negative		HIV-positive		Not done		Does not know			
1.13.1 Action in case of HIV-positive status			NVP		TARV		Not done			
1.14 Syphilis screening	**Negative**		Positive		Not done		Does not know			
1.15 Rhesus blood group		3.**9 Haemoglobin electrophoresis**								
1.16 Was the mother ill during the pregnancy?		If so, what was she suffering from? ______________________ ____								
1.17 Did she have malaria (Yes/No)		1.18 Did she have high blood pressure (Yes/No)								
1.19 Did she suffer from diabetes (Yes/No)		1.20 Did she suffer from anaemia (Yes/No)								

1.21 Was she suffering from an abnormal presentation of the foetus (Yes/No)		1.22 Has she previously had a caesarean section (Yes/No)	

SECTION 3: Labour and Birth

3.1 Date of last menstrual period	: :	Does not know							
3.2 Date of birth	: :								
3.2.1 Time of birth									
3.3 Gestational age	(in weeks)	Does not know							
3.3.1 Method of determination	Certain date of last period		Uncertain date of last period						
	Early pregnancy ultrasound		Late pregnancy ultrasound						
3.4 Staff present at the delivery	Midwife		Nurse		Doctor		Traditional midwife	Person	
	Does not know								
3.5 Beginning of labour	Spontaneous		Provoked		Pregnancy ultrasound before the beginning		Does not know		
3.6 Heart rate on admission	No		Yes Frequency:		Does not know				
3.7 Use of a partograph	No		Yes		Does not know				
3.8 Mode of delivery	Normal		Normal assisted		Pregnancy ultrasound		Does not know		
3.9 Child's gender	Male		Female						
3.10 Birth weight	(g)	≥2500 g	1500 – 2499 g	1000 – 1499 g	<1000 g				
			Low weight	Very low weight	Extremely low weight				
3.11 Height of the child	(cm)	≥ 35 cm		< 35 cm					

SECTION 4: CARE OF THE NEWBORN ON ARRIVAL IN THE RECEIVING HF

4.1 Deadline for the care at the HF	< 30 min.		30 - 60 min.		>60 min.	
4.2 Did the newborn breathe or cry at birth?	**Yes**		No			
4.3 Apgar Score	1 min.=		5 min.=	10min		
4.4 Resuscitation	Not required		Mask and balloon			
	Not practised		Other, specify ____			

4.4.1 place of resuscitation	equipment used	
4.4.2 Duration of resuscitation (in minutes) progress		
4.4.3 Qualification of resuscitation personnel		

4.5 Care received on arrival in the HF :					
Prevention of hypothermia		Airway clearance		Oxygen therapy/ventilation	
Injectable anticonvulsants (phenobarbital, diazepam)		Injectable antibiotics		Venous route	

4.6 Kangaroo maternity care for premature and low birth weight babies?	Yes		No		Does not know	
4.7 Type of feeding	Breast milk		Breast milk substitute		Others	
4.8 Placement of an orogastric or nasogastric tube	Yes					

4.9 Umbilical cord care	Yes		No		**4.10 Phototherapy**	Yes		No	
Were any products applied used on the cord?	Yes		No		If yes, specify the name of the product				
4.11 Exsanguination	Yes		No		**4.12Blood transfusion**	indicated	Yes		
							No		
						Carried out	Yes		
							No		
					If not carried out, reasons :				

4.13 Management of an intercurrent disease (specify) :				
4.14 Known post-natal visit?	Yes			

4.15 Post-natal visit carried out (date)	Yes		No	
4.16 Neonatal surgical condition indicated?	Yes		No	
4.16.1 Time for diagnosis	birth?		Number of days of life	

Operation performed?	Yes		No		If not, why?	
					Non-functional operating theatre	
					Financial means	

4.17 Recognisable malformation at birth?	Yes		No	

If yes, specify malformation or the organ affected

Other conditions?

____

____

____

Other care to be specified:

____

____

____

4.18 Cause of death

1- Complications of prematurity						
2-Neonatal infections, etc.	Septicaemia		Pneumonia		Meningitis	Other, specify if applicable
	Diarrhoea		Tetanus		syphilis	
3- low birth weight						
4- Perinatal and neonatal asphyxia						
5- Congenital condition (type or organ)						
6- Neonatal surgical emergency						
7- Nosocomial infections						
8- Others						

Note: Attach copies of the Partogram and surveillance sheets

ENDIX 11G. Measles – Case-based Investigation Form

Ministère de la Santé Publique **OMS** **Centre Pasteur du Cameroun**

FORMULAIRE DE NOTIFICATION DES CAS DE ROUGEOLE
NOTIFICATION OBLIGATOIRE

Prière de remplir toutes les cases

Utilisation officielle seulement	N0. EPID	__________	__________	__________	_________	_________	Reçu le
		pays	Région	District Santé	Année	Nbre cas	___________

IDENTIFICATION

District de Santé : ______________________ Région : ________________ Nom de la Formation Sanitaire la plus proche________

Village/Quartier __________________ Ville : ________________ ☐ Urbaine / Rurale
1 = Urbaine
2 = Rurale

Nom du patient : _______________________________ Nom du Père / Mère ____________________________________

Adresse du patient____

Date de naissance du patient : _____/_____/______ Age : ________________ Sexe ☐ M / F

NOTIFICATION

Date cas vu à la formation Sanitaire _______/______/___________ Date de la notification au District ______/_____/_______

HISTORIQUE DE LA MALADIE

Date de début De l'éruption (rash) ___/_____/______

☐ Evolution du malade
1 = Oui, décédé
2 = Non, vivant
9 = Inconnu

Nbre de dose valide du vaccin antirougeoleux ☐

______/______/____ Date de la dernière vaccination antirougeoleuse

Quelles sont les trois (03) dernières structures sanitaires ou tradipraticiens visités par le malade (nom et localisation)?

1 : ………………………………………………………; **2 :**……………………………………………………; **3 :**…………………………………………………

ECHANTILLON DE SANG (PLASMA)

Date de prélèvement _____/____/____ Date d'expédition du prélèvement Vers le labo _______/_______/________

Date de réception au labo CPC _____/____/____ Date de réception des Résultats au PEV ______/______/_____

Résultats de IgM Indirect de Rougeole ☐
1 = Positif
2 = Négatif
3 = Indéterminé

Résultat de IgM Rubéole ☐
1 = Positif
2 = Négatif
3 = Indéterminé

Autres résultats _______________

CLASSIFICATION FINALE DU CAS

☐
1 = Cas confirmé par labo ou lien épidémiologique avec un autre cas confirmé
2 = Clinique (compatible/suspect). Test de labo non réalisés
3 = Exclu

Date d'expédition des résultats du labo Au clinicien/district qui a envoyé le sang _______/_______/_______

INVESTIGATION DANS LA COMMUNAUTE

Source de l'infection identifiée : ☐
1 = Oui
2 = Non

Si le test de confirmation IgM de rougeole positif Investigation communautaire faite ? ☐
1 = Oui
2 = Non

Si oui, décrire le résultat de l'investigation : ____

INFORMATION SUR LA PERSONNE QUI A NOTIFIE

Nom ____Titre ______________________________________

Unité : _______________________________Adresse ____________________________Tél : __________________

Cas Détecté en premier par : ☐ Personnel de santé ; ☐ Tradipraticien /Lieu de Prière; ☐Relais communautaire ; ☐ Autres :…………………..

APPENDIX 11H. Bacterial Meningitis - Case-based Investigation Form and Decision Tree

Bacterial meningitis – Case-based investigation form

MINISTRY OF HEALTH **I GENERIC CASE- REPORTING FORM** ***Name of country***

HF______________________ Health area :_______________________________HD : __________________

Region : _________________

☐ **Cholera** ☐ Disease 2 **......** ☐ **Meningitis** ☐ Other (please specify) **:_______________**

EPID NUMBER : / _ _ _ / _ _ _ / _ _ _ / _ _ / _ _ _ / _ _ _ _ /

(To fill at the level of the district) Country *Region* *HD* *Year* *Disease*

Case No.

PATIENT IDENTIFICATION

Patient’s name __________________________ Patient’s surname : _____________________________

Date of birth: ____/____/______

Or age in years: ____ **or** age in months (if <12 months) ____ **or** age in days (if<1 month) ____

Sex : ☐ Female ☐ Male Occupation (enter child if <5 years) : ____

Patient’s place of residence

HD of residence: ______________ City/Village : ______________ Neighbourhood/Area : _________ ☐ Urban / ☐ Rural

Name of the father/the mother/legal guardian : ______________________ father’s/Legal guardian’s phone number _____________

Date of consultation _____ / ____ / _____ Date of onset of symptoms : : ____ / ____ / _____

☐ inpatient/in observation ☐ Outpatient

Outcome: ☐ Cured ☐ Died ☐ under treatment ☐ unknown

VACCINATED PATIENT ☐ YES ☐ NO ☐ UNKNOWN

If not a case of meningitis:

Type of vaccine: _______________ Number of doses : ____ ☐ Date of last vaccination unknown : ___ / ___ / ____

If suspected meningitis, vaccines received:	*Source of information on vaccination :*

MenAC ☐ Yes, Date : ____ / ____ / ______ ☐ Name ☐ Unknown	☐ booklet ☐ vaccination register ☐ verbal ☐ Unknown
MenACW ☐ Yes, Date : ____ / ____ / ______ ☐ Name ☐ Unknown	☐ booklet ☐ vaccination register ☐ verbal ☐ Unknown
MenACWY ☐ Yes, Date : ____ / ____ / ______ ☐ Name ☐ Unknown	☐ booklet ☐ vaccination register ☐ verbal ☐ Unknown
Conjugate A ☐ Yes, Date : ____ / ____ / ______ ☐ Name ☐ Unknown	☐ booklet ☐ vaccination register ☐ verbal ☐ Unknown
PCV13- 1 ☐ Yes, Date : ____ / ____ / ______ ☐ Name ☐ Unknown	☐ booklet ☐ vaccination register ☐ verbal ☐ Unknown
PCV13- 2 ☐ Yes, Date : ____ / ____ / ______ ☐ Name ☐ Unknown	☐ booklet ☐ vaccination register ☐ verbal ☐ Unknown
PCV13-3 ☐ Yes, Date : ____ / ____ / ______ ☐ Name ☐ Unknown	☐ booklet ☐ vaccination register ☐ verbal ☐ Unknown
Hib 1 ☐ Yes, Date : ____ / ____ / ______ ☐ Name ☐ Unknown	☐ booklet ☐ vaccination register ☐ verbal ☐ Unknown
Hib 2 ☐ Yes, Date : ____ / ____ / ______ ☐ Name ☐ Unknown	☐ booklet ☐ vaccination register ☐ verbal ☐ Unknown
Hib 3 ☐ Yes, Date : ____ / ____ / ______ ☐ Name ☐ Unknown	☐ booklet ☐ vaccination register ☐ verbal ☐ Unknown

SAMPLES COLLECTED : ☐ YES ☐ NO(Note: *IF NO, complete the form and send it to the District CISSE)*

IF THE ANSWER IS NO : Why: ☐ No kit ☐ Patient condition ☐ Other : ____________________

IF THE ANSWER IS YES :

Date of sample collection ____/ ____/ ______ Time of collection : / __/__/ HH__/ __/ Min

Source of sample: ☐ Stool ☐ Blood ☐ CSF ☐ Other: _______________________________________

Appearance of the sample : Cerebrospinal fluid (CSF): ☐ Clear ☐ Cloudy ☐ Hematic ☐ Xanthochromic ☐ Citrine ☐ Opaque ☐ Purulent

Stool: ☐ **Aqueous** ☐ **Glairy** ☐ **Glairo- bloody** ☐ **Bloody**

Date and time of inoculation into the transport medium: on___/ ____/ ______ at /__/__/HH __/__/ Min

Sample(s) sent to the laboratory: ☐ Yes ☐ No If no, why? __________________________________

Packaging: ☐ Dry tube ☐ Trans-Isolate ☐ Cryotube ☐ Cary blair ☐ Other: ____________________________

RDT practised: ☐ Cholera ☐ Meningitis ☐ Other (please specify): _________________ Results: ___________________

Date sample sent to laboratory : ____/ ____/ ______ Name of the laboratory : ____________________

Date of reporting to higher level : __/ __/ __ Person who filled the form : ___________________ Tel : _______

Date form sent to HD : ____/ ____/ ______Date of receipt of the form by the HD: ____/____/______

Date form sent to the Region : ____/ ____/ ______Date of receipt of the form by the Region: ___ / ___ / ____

Date form sent to the national level : ____/____/______

LABORATORY : ____

Date of reception : ____/ ____/ ____ Time: ___/ H ____/ Min No. in the laboratory register : ___________________

Sample(s) received: ☐ Dry tube ☐ Trans-Isolate ☐ Cryotube ☐ Cary blair ☐ Other (please specify):

____

Conditions for transportation of the sample(s) : ☐ Adequate ☐ Not adequate

Appearance of the sample : Cerebrospinal fluid (CSF): ☐ Clear ☐ Cloudy ☐ Hematic ☐ Xanthochromic ☐ Citrine ☐ Opaque ☐ Purulent

Stool: ☐ **Aqueous** ☐ **Glairy** ☐ **Glairo- bloody** ☐ **Bloody**

Type of analyses performed : ☐ Cytology ☐ Fresh ☐ Gram ☐ Latex ☐ ROT ☐ Other (please specify): ______________

Cytology: Leucocytes / ___/ ___/ ___/ ___/ ___/ mm3 NP / ___/ ___/% LYMPH / ___/ ___/%

Gram : ☐ DGP ☐DGN ☐BGP ☐ BGN ☐ Other pathogens ☐ Negative

RDT practised: ☐ Cholera ☐ Meningitis ☐ Other (please specify): ____________ Resultts :

____

Latex : ☐ NmA ☐ NmC ☐ NmW/Y ☐ NmB ☐ S. pneumonia ☐ Hib ☐ Negative

Other analysis (please specify type and results) :

____

Date sample sent to the reference laboratory : ____/____/______

REGIONAL LABORATORY OF: ____

Date of reception : ____/ ____/ ____ Time: ___/ H ____/ Min No. in the laboratory register : ___________________

Sample(s) received: ☐ Dry tube ☐ Trans-Isolate ☐ Cryotube ☐ Cary blair ☐ Other (please specify): __________________

Conditions for transportation of the sample(s) : ☐ Adequate ☐ Not adequate

Appearance of the sample : Cerebrospinal fluid (CSF): ☐ Clear ☐ Cloudy ☐ Hematic ☐ Xanthochromic ☐ Citrine ☐ Opaque ☐ Purulent

Stool: ☐ **Aqueous** ☐ **Glairy** ☐ **Glairo- bloody** ☐ **Bloody**

Type of analyses performed : ☐ Cytology ☐ Fresh ☐ Gram ☐ Latex ☐ ROT ☐ Other (please specify): _______________

Cytology: Leucocytes / ___/ ___/ ___/ ___/ ___/ mm3 NP / ___/ ___/% LYMPH / ___/ ___/%

Gram : ☐ DGP ☐DGN ☐BGP ☐ BGN ☐ Other pathogens ☐ Negative

TOR practiced: ☐ Cholera ☐ Meningitis ☐ Other (please specify): Results

____

Latex : ☐ NmA ☐ NmC ☐ NmW/Y ☐ NmB ☐ S. pneumonia ☐ Hib ☐ Negative

Culture : ☐ NmA ☐ NmC ☐ NmW ☐ NmB ☐ NmX ☐ Nm Unknown ☐ {ut11 }S. Pneumonia

☐ Hib ☐ H. influenza Indeterminate ☐ StrepB ☐ Other pathogens (please specify) :

____

☐ Contaminated ☐ Negative

Other analysis (please specify type and results) :

____

Antibiogram : Ceftriaxone : ☐ Sensitive ☐ Resistant ☐ Intermediate Not performed

Penicillin G: ☐ Sensitive ☐ Resistant ☐ Intermediate Not performed

Oxacillin : ☐ Sensitive ☐ Resistant ☐ Intermediate Not performed

Other _______________:☐ Sensitive ☐ Resistant ☐ Intermediate Not performed

Date sample sent to the reference laboratory : ____/____/______

REFERENCE LABORATORY ____

Date of reception: __/ ___/ ___ **Time:** ___H ___ **No. EPID on the tube?** ☐ YES ☐ NO No. In the laboratory register : ________

Sample(s) received: ☐ Dry tube ☐ Trans-Isolate ☐ Cryotube ☐ Cary blair ☐ Other (please specify): ____________________

Conditions for transportation of the sample (s) : ☐ Adequate ☐ Not adequate

Appearance of the sample : Cerebrospinal fluid (CSF): ☐ Clear ☐ Cloudy ☐ Hematic ☐ Xanthochromic ☐ Citrine ☐ Opaque ☐ Purulent

Stool: ☐ **Aqueous** ☐ **Glairy** ☐ **Glairo- bloody** ☐ **Bloody**

Type of analyses performed : ☐ Cytology ☐ Fresh ☐ Gram ☐ Latex ☐ RDT/reagent strip ☐ Other (please specify): _______________

Cytology: Leucocytes / ___/ ___/ ___/ ___/ ___/ mm3 NP / ___/ ___/% LYMPH / ___/ ___/%

Gram : ☐ DGP ☐DGN ☐BGP ☐ BGN ☐ Other pathogens ☐ Negative

Results of the rapid diagnostic test (RDT/reagent strip) : ☐ NmA ☐ NmC ☐ NmW ☐ NmY ☐ Negative

Latex : ☐ NmA ☐ NmC ☐ NmW/Y ☐ NmB ☐ S. pneumonia ☐ Hib ☐ Negative

Culture : ☐ NmA ☐ NmC ☐ NmW ☐ NmB ☐ NmX ☐ Nm Unknown ☐ {ut11 }S. Pneumoniae

☐ Hib ☐ H. influenzae Indeterminate ☐ StrepB ☐ Other pathogens (please specify) : ____________________

☐ Contaminated ☐ Negative

PCR : date of PCR : ____/ ____/ ______ **Type of PCR** : ☐ Real-time ☐ Conventional

☐ NmA ☐ NmC ☐ NmW ☐ NmY ☐ NmB ☐ NmX ☐ Nm Unknown ☐ S. pneumoniae

☐ Hib ☐ H. influenzae Indeterminate ☐ StrepB ☐ Other pathogens (please specify) : ____________________

☐ Contaminated ☐ Negative

Serotype : / ____/ ____/ other analysis (please specify the type and results obtained): ____

Final laboratory result :

☐ NmA ☐ NmC ☐ NmW ☐ NmY ☐ NmB ☐ NmX ☐ Nm Unknown ☐ S. pneumoniae

☐ Hib ☐ H. influenzae Indeterminate ☐ StrepB ☐ Other pathogens (please specify) : ____________

Antibiogram :	Ceftriaxone :	☐ Sensitive ☐ Resistant ☐ Intermediate Not performed
	Penicillin G:	☐ Sensitive ☐ Resistant ☐ Intermediate Not performed
	Oxacillin :	☐ Sensitive ☐ Resistant ☐ Intermediate Not performed
	Other _______________:	☐ Sensitive ☐ Resistant ☐ Intermediate Not performed

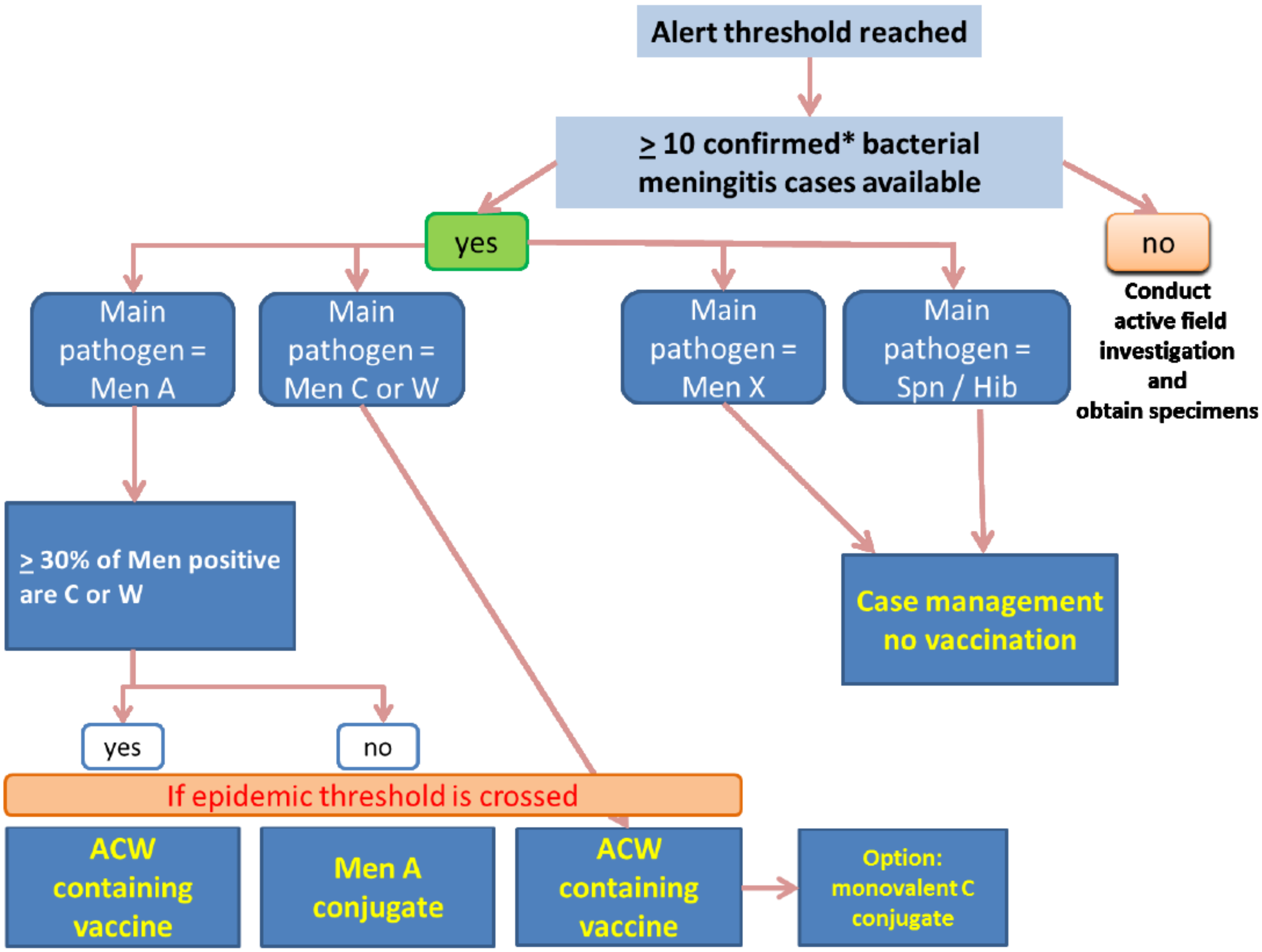
Observations:

____

Submission date of results to the Surveillance Service of the Ministry of Health : ___ / ___ / ____

Decision Tree for vaccine selection during a reactive immunization campaign

Decisional tree for Meningitis Vaccine Choice in a Reactive Vaccination Campaign


Alert threshold reached
> 10 confirmed* bacterial meningitis cases available
yes
no
Conduct active field investigation and obtain specimens
Main pathogen = Men A
Main pathogen = Men C or W
Main pathogen = Men X
Main pathogen = Spn / Hib
> 30% of Men positive are C or W
yes
no
Case management no vaccination
If epidemic threshold is crossed
ACW containing vaccine
Men A conjugate
ACW containing vaccine
Option: monovalent C conjugate

APPENDIX 11I Neonatal Tetanus – Case based investigation Form

Ministère de la Santé Publique OMS **Centre Pasteur du Cameroun**

☐

FORMULAIRE D'ENQUETE DES CAS DE TETANOS NEONATAL

NOTIFICATION OBLIGATOIRE. PRIERE DE REMPLIR CETTE FICHE POUR CHAQUE CAS

A REMPLIR AU NIVEAU CENTRAL

Numéro EPID ________ ________ ________ ________ ________ Reçu le ______/______/_____

Pays Province District Année Début N° cas

IDENTIFICATION

District : _______________ Province : ____________________ Nom de la formation Sanitaire la plus proche : ____________________

Adresse : ____________________ Village : ________________________________ Ville : ________________________________

Nom(s) du malade : ______________________________________ Mère : ____

Sexe : ☐ M = masculin F = féminin Père : ________________________________

NOTIFICATION / ENQUETE

Cas notifié Par : _________________________ Date de notification _________________ Date de l'enquête : _______________________________

ANTECEDENTS VACCINAUX DE LA MERE

Mère vaccinée avec le VAT : ☐ 1=0 2=N 9=Inconnu

Carte ? ☐ 1=0 2=N 9=Inc

Nombre de doses ☐ _____/______/_____ Date de la dernière dose

Statut vaccinal de la mère lors de l'accouchement ☐ 1=à jour 2=non à jour 9=inconnu

NAISSANCE DE L'ENFANT

Date de naissance :_____/______/______

Lieu de Naissance ☐ 1=hôpital 2=centre de santé 3=domicile, personne formée 4=domicile, personne non formée 5=domicile, pas d'AT 9=inconnu

Cordon coupé avec une lame de rasoir stérile ☐

Cordon coupé avec tout autre instrument ☐

Pansement du cordon ☐

la mère a-t-elle reçu des soins prénataux ☐ 1=O 2=N 9=Inconnu

Où ? ________________________________ Nom de l'hôpital : _______________________

Combien de visites ? ☐

Accouchement pratiqué Par médecin/infirmier Sage-femme ☐ 1=0 2=N 9=Inconnu

Nom de AT/Sage-femme ___________________________________

Accouchement pratiqué par AT ? ☐ 1=0 2=N 9=Inc

1. ________________
2. ________________
3. ________________

ANTECEDENTS CLINIQUES

Enfant normal à la Naissance ☐ 1=0 2=N 9=Inc

Raideur ? ☐

Enfant décédé ☐ 1=0 2=n 9= Inc

Date de début De symptômes _____/____/_____

Enfant a bien crié et tété les 2 premiers jours ☐ 1=0 2=N 9=Inc

Spasmes ou convulsions ? ☐

Age au début de La maladie (jours) ☐

A-t-il stoppé de têter ensuite ? ☐ 1=0 2=N 9=Inc

Position opisthotonos ☐ 1=0 2=N 3=Inc

Age au moment du décès (jours) ☐

Quelles sont les trois (03) dernières structures sanitaires ou tradipraticiens visités par le malade (nom et localisation)?

1 : ……………………………………………………; **2 :**……………………..………………………; **3 :**………………………………………………

TRAITEMENT

Hospitalisé ☐ 1=0 2=N 9=Inc

Date d'admission ________/_______/_____

Numéro du dossier clinique : _______________ Hôpital (Adresse) : ______________________________

COMMENTAIRES

____

____

CLASSIFICATION FINALE DU CAS Tétanos Néonatal ☐ 1= O 2= N 9= Inconnu

ENQUETEUR

Nom : ______________________________________ Titre ________________________________

Unité : ____________________ Adresse : ________________________ Tél : ________________ Signature______________

Cas Détecté en premier par : ☐ Personnel de santé ; ☐ Tradipraticien /Lieu de Prière; ☐ Relais communautaire ; ☐ Autres :…………………..

REPONSE MISE EN ŒUVRE

La mère du cas de TN a-t-elle reçu une dose protective de VAT durant les 3 mois suivant la déclaration du cas ☐ 1=0 2=N 9=Inc

Date de réponse : ________/________/_________

Détails de réponse : ___________________________

Activités supplémentaires de vaccination Dans la localité où le cas est survenu ☐ 1=0 2=N 9=Inc

PRIERE D'ENVOYER UNE COPIE DU FORMULAIRE REMPLI IMMEDIATEMENT A _____________________________________

SI VOUS DESIREZ AVOIR DES INFORMATIONS COMPLEMENTAIRES. VEUILLEZ VOUS ADRESSER A

INSTRUCTIONS FOR COMPLETING THE NEONATAL TETANUS CASE INVESTIGATION REPORT FORM

These instructions are given to assist the nurse, the doctor to complete the form.

EPID No.: completion is reserved for the national Level

IDENTIFICATION: completion by **the investigator**

- District: mark the name of the HD in which the NT case resides
- Region: mark the name of the administrative Region
- Name of the nearest health facility: mark the name of the HFnearest to the patient's place of residence
- Village/neighbourhood name: place of residence of the patient
- Patient's address: address of the patient or relatives where the patient can be reached in the community (PO Box, telephone, or other means)
- Village: name of the village where the patient lives.
- City: name of the city where the patient lives.
- Patient's name: mark the patient's name and surname
- Mother: Name of the patient’s mother
- Sex: mark in the box M, if male, F if female
- Father: mark the name of the patient's father:

REPORTING/INVESTIGATION

- Case reported by: Name of the person who reported to the district health service.
- Date of reporting: date on which the case was reported to the district health service (day, month, year)
- Date of Investigation: date on which the health staff in charge of the case investigated the case.

IMMUNISATION HISTORY OF THE MOTHER

- Mother vaccinated with ATT: Mark the corresponding number.

- Number of doses: Mark the number of VAT doses.

-Date of last dose: Mark the date on which she received the last dose.

-Vaccination status of the mother at the time of delivery: Mark the corresponding number

CHILD'S BIRTH:

Date of birth: Mark the child's date of birth (day, month, year)

Place of birth: Mark the corresponding number

Cord cut with a sterile razor blade: Mark a cross in the corresponding box

Cord cut with any other instrument: Mark a cross in the corresponding box

Dressing of the cord: Mark a cross in the corresponding box

Did the mother receive antenatal care? Mark the corresponding number

Where: Did the mother receive antenatal care?

Name of investigator: Name of hospital:

Indicate the name of the hospital. How many visits:

Delivery by doctor/nurse/midwife: Mark the corresponding number

Delivery by TBA: Mark the corresponding number

Name of TBA/midwife: Mark name

CLINICAL HISTORY

Date of onset of symptoms: Date of symptoms onset Mark date (day, month, year)

Age at the onset of illness (days): Indicate age of child at the onset of illness

Normal child at birth: Mark the corresponding number

Child has been crying and suckling well for the first 2 days: Mark the corresponding number

Did he stop suckling afterwards? Mark the corresponding number

Stiffness: Mark the corresponding number

Spasms or convulsions? Mark the corresponding number

Opisthotonos position: Mark the corresponding number

Deceased child: Mark the corresponding number

Age at death (days): Mark the age at death.

Normal child at birth:

Treatment

Date of admission: Mark the date of admission of the case (day, month, year)
Inpatient: Mark the corresponding number
Clinical record number Mark clinical record number
Hospital/address: Mark the name and address of the hospital

COMMENTS: Write some comments about the case

FINAL CASE CLASSIFICATION: Write the corresponding number

INVESTIGATOR:

Name: Mark name of the investigator:

Write the title, unit, address and telephone number of the investigator.

APPENDIX 11 J Respiratory diseases (including influenza) – Reporting and Investigation Form of COVID-19 in Cameroon

INVESTIGATION AND REPORTING FORM OF COVID-19 CASES IN CAMEROON

EPI ID : CO-CMR-/__/__/__/-/__/__/__/-/__/__/__/__/__/ *Code Région (3) Code district (3) Numéro d'ordre (5)*	**Date of filling**: ___/____/20__/

Region: _________ **District:** __________________ **Health Area:** __________________ Place: ___________________________

1. Socio-demographic Data

Case Status: ☐ Alive ☐ Dead if dead, date of death /______/________/20______/

Case Name and Surname: ____

Sex: ☐ M ☐ F **Age**: ____ ☐ Years ☐ Months **Date of birth**: /______/_______/________/

Country of Residence: ____________________ Town: _____________________ **Nationality**: __________________

Tel.: ________________________ **Profession**: _____________________ **Health Professional**: ☐Yes ☐No

2. Signs and symptoms

☐ Asymptomatic Date of onset of symptoms /______/_______/20_____/

Fever (t°≥38°C)	☐ Yes ☐ No ☐ NSP	Rash	☐ Yes ☐ No ☐ NSP
Chills	☐ Yes ☐ No ☐ NSP	Conjounctivitis	☐ Yes ☐ No ☐ NSP
Cough	☐ Yes ☐ No ☐ NSP	Shortness of breath	☐ Yes ☐ No ☐ NSP
Sore Throat	☐ Yes ☐ No ☐ NSP	Muscles Pain	☐ Yes ☐ No ☐ NSP
Nasal Discharge	☐ Yes ☐ No ☐ NSP	Difficulty breathing	☐ Yes ☐ No ☐ NSP
Vomiting	☐ Yes ☐ No ☐ NSP	Headaches	☐ Yes ☐ No ☐ NSP
Diarrhoea	☐ Yes ☐ No ☐ NSP	Loss of taste	☐ Yes ☐ No ☐ NSP
Loss of smelling	☐ Yes ☐ No ☐ NSP	Intense Fatigue	☐ Yes ☐ No ☐ NSP

Others, specify ____

3. Comorbidites

Cancer	☐Yes ☐No ☐ NSP	**HIV**	☐Yes ☐Nn ☐ NSP
Diabèetes	☐Yes ☐No ☐ NSP	**Heart Disease**	☐Yes ☐No ☐ NSP
HBP	☐Yes ☐No ☐ NSP	**Kidney failure**	☐Yes ☐No ☐ NSP
Obesity	☐Yes ☐No ☐ NSP	**Sickle Cell Anemia**	☐Yes ☐No ☐ NSP

Chronic Pulmonary Disease ☐Yes ☐No ☐ NSP

Amenorrhea/Pregnancy (Only for females) ☐Yes ☐No ☐ NSP if yes, indicate the term: __________________

☐ **Others, specify**: ______________________________

4. History

Did you receive the COVID-19 vaccine? ☐ Yes ☐ No Si yes, vaccination date: _____/________/20_____

Name of vaccine received:_________________________________ Number of doses received:_______

Were you diagnosed COVID-19 positive? ☐Yes ☐No, If yes, were you declared healed? ☐Yes ☐No

5. Test

Type of test: ☐RDT Ag ☐ rdt Ac ☐ PCR

Specimen Type: ☐ Nasopharyngeal ☐Others **Specimen collection Date:** _________/____________/____________

Indication on specimen collection: ☐ Voluntary ☐Control ☐ ☐Others

Results: ☐ SARS Cov-Ag positive ☐ SARS Cov-Ag negative ☐Unknown

Conclusion: ☐Positive ☐Negative ☐Unknown

Tester: ____

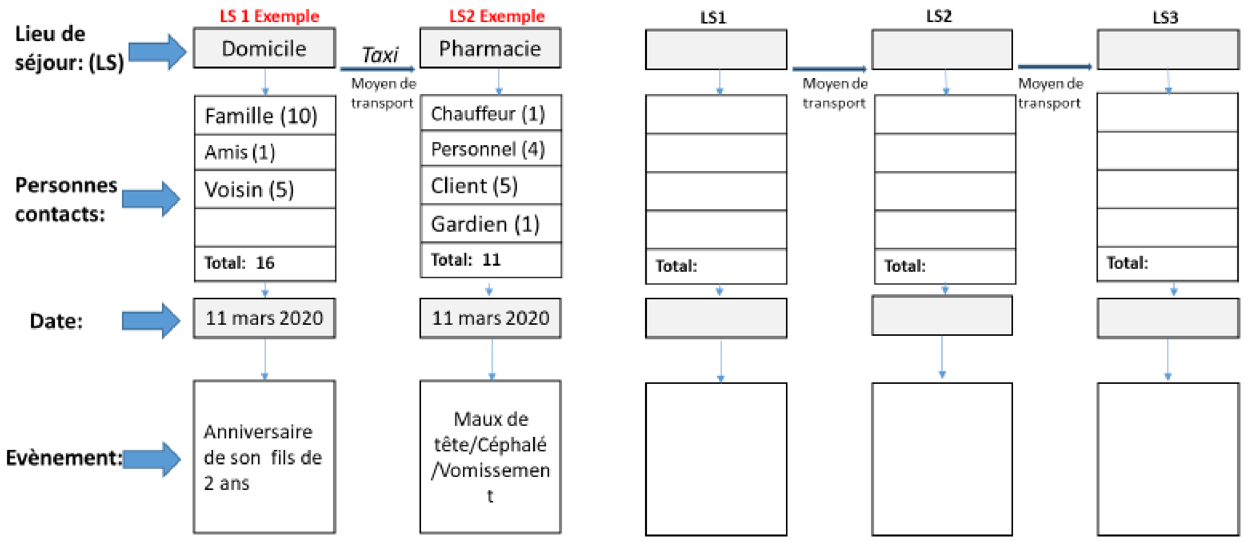
Telephone of Tester: ____

Investigator(s) Name(s): ____

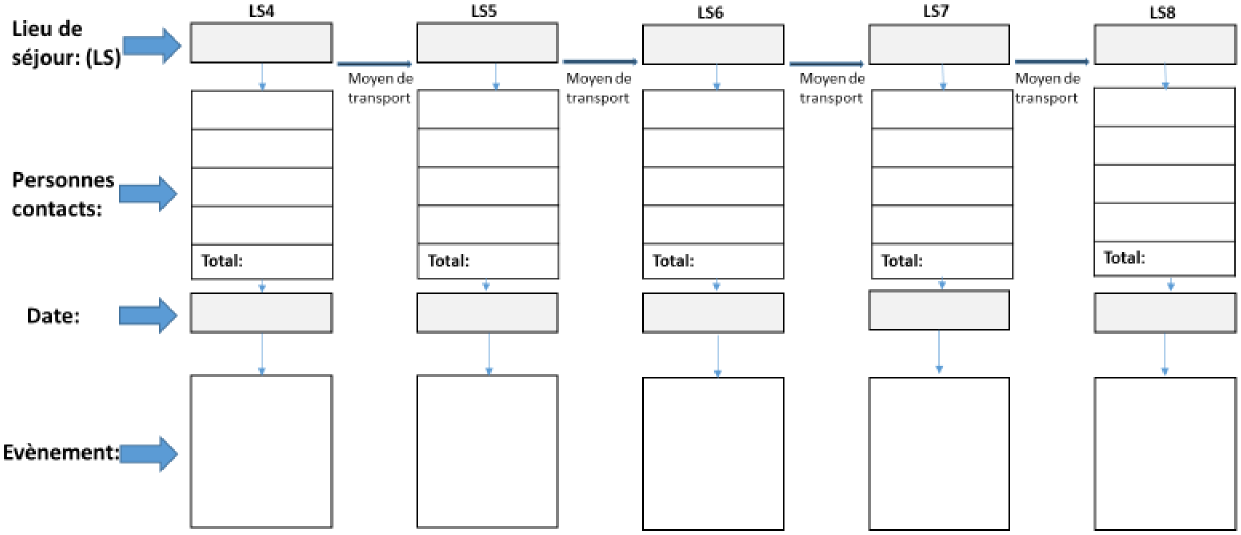
Telephone number: ____

E-mail Adress: ____

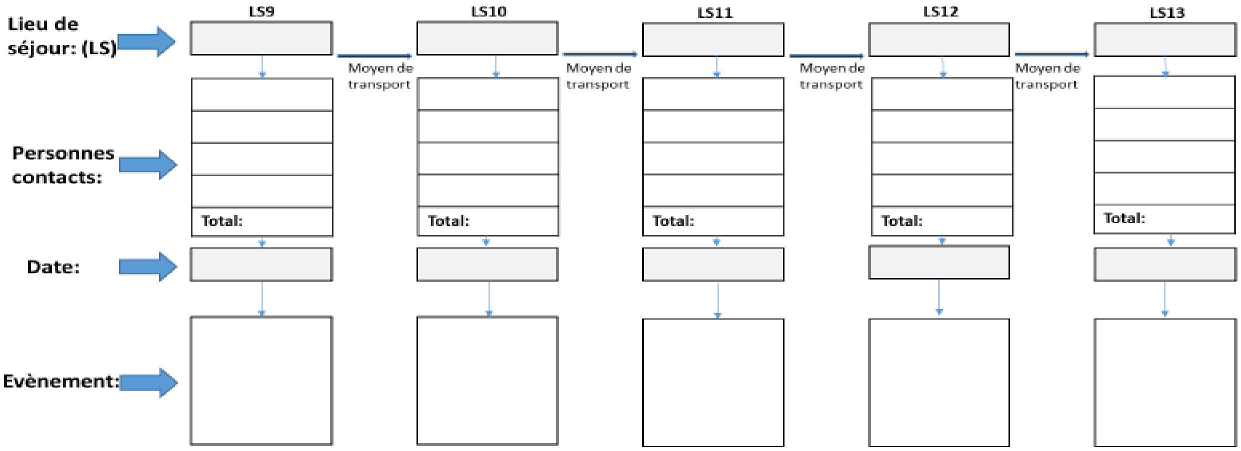
MAPPING CAS INVESTIGUE 1/3


Lieu de séjour: (LS)
Personnes contacts:
Date:
Evènement:
LS 1 Exemple
Domicile
Famille (10)
Amis (1)
Voisin (5)
Total: 16
11 mars 2020
Anniversaire de son fils de 2 ans
Taxi
Moyen de transport
LS2 Exemple
Pharmacie
Chauffeur (1)
Personnel (4)
Client (5)
Gardien (1)
Total: 11
11 mars 2020
Maux de tête/Céphalé /Vomissement
LS1
Total:
Moyen de transport
LS2
Total:
Moyen de transport
LS3
Total:

MAPPING CAS INVESTIGUE 2/3


LS4
LS5
LS6
LS7
LS8
Lieu de séjour: (LS)
Moyen de transport
Moyen de transport
Moyen de transport
Moyen de transport
Personnes contacts:
Total:
Total:
Total:
Total:
Total:
Date:
Evènement:

MAPPING CAS INVESTIGUE 3/3


LS9
LS10
LS11
LS12
LS13
Lieu de séjour: (LS)
Moyen de transport
Moyen de transport
Moyen de transport
Moyen de transport
Personnes contacts:
Total:
Total:
Total:
Total:
Total:
Date:
Evènement:

CONTACT LISTING FORM OF A COVID-19 CASE IN CAMEROON

Date of filling ___/___/___	Form Reception Date ___/___/___	Form filled by (name and surname): …………………………………………	Tel.: ………………………

Identification of case

EPI ID : CO-CMR-/___/___/___/-/___/___/___/-
/___/___/___/___/___/
Region Code (3) *District Code (3)* *Number (5)*

Case Status: ☐ Probable ☐ Confirmed ☐ Dead ☐ Alive

Date of last contact ________/_________/_________

Identification of contact cases

No.	Name	Surname	Age (years)	Sex (M/F)	Profession	Telephone	Relationship with the case	Village/Quarter (with precisions)	Residence Health Area	Residence HD	Types of contact*	Circumstan of contact
1												
2												
3												
4												
5												
6												
7												
8												
9												
10												
11												
12												
13												

Name ____________________________ Surname ___________________________ Village/Neighbourhood of the case__________________________

HA of residence of the case ___________________________ HD of residence of the case __________________________

Region of Residence of the case_____________________________

* **Type of contact**: 1 having had a contact (1 meter radius) with a confirmed case, 2 = living in the same household, 3 = personnel having provided direct individual or clinical care, 4= more than 1m radius, or open space

** **Circumstances of contact**: 1=House/ Household ; 2= Hospital / Health care ; 3= Work place ; 4=Group of visit ; 5=Other, indicate

Formulaire de suivi individuel des contacts

(à remplir par les traceurs)

Nom du traceur
Identifiant équipe
Nom du superviseur

Identifiant cas source (EPI ID) : CO-CMR-/___/___/___/-/___/___/___/-/___/___/___/___/___/

Code région (3) *Code district (3)* *Numéro d'ordre (5)*

Quartier/village : ……………………………..………… Aire de santé : …………………………………… District : …………………………………… Région : ……………………………………………………….……

Nom et téléphone chef du quartier/village : ………………………….…………………………………………

Identifiant contact : ……………………………………

Nom et prénom contact : …………………………………… Âge : ………………… Sexe : ………………… Téléphone : …………………………………… Adresse : ……………………………………………..

Date de dernier contact avec le cas (JJ/mm/aaaa) : _____/ _____/ 2020

Date du premier jour de suivi (JJ/mm/aaaa) : _____/ _____/ 2020 **Date prévisionnelle du dernier jour de suivi (JJ/mm/aaaa) : _____/_____/ 2020**

Jour de suivi	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**
Température														
Toux														
Asthénie														
Difficultés respiratoires														
Douleurs musculaires														
Diarrhées														
Céphalées														
Maux de gorges														

Yeux rouges														
Écoulement nasal														
Autres														
Prélèvement (Yes/non)														
Status du contact														

Indications Status contact : x = vu 0 = non vu 1 = devenu suspect 2 = décédé D = déplacé RS = refus de suivi R = recyclé

Indications symptômes à suivre : YES = présence de signes NON = absence de signes

APPENDIX 11K. Tuberculosis (MDR and XDR) – Case-based Reporting Form (in addition to TB, STIs, Hepatitis, HIV)

Case-based Reporting Form of multiresistant and ultraresistant tuberculosis																												
Country:						**Year:**																						
Quarter:						**Month:**			**Results of the drugs sensitivity tests** **(S=Sensitive ; R=Resistant ; I=Intermediate ; U=Unknown)**																			
Case Identifier (year of detection-country code – number in the TB register	**Sex** (F/M)	**Age** (in years)	**Diagnostic Date** (dd/mm/yyyy)	**Type of Notification** (MDR TB * or XDR TB **)	**TB Site** (Pulmonary or extra Pulmonary)	**Type of TB cases** (New /Relapse /After defaulting /After 1st treatm	**Patient Status** (On treatment /not on treatment /Unknown)	**HIV Status** (positive /negative /Unknwon)	**H**	**R**	**E**	**Z**	**S**	**Th**	**Am**	**Km**	**Cm**	**Cfx**	**Ofx**	**Lfx**	**Mfx**	**Gfx**	**Gfx**	**Pto**	**Eto**	**Cs**	**PAS**	**Other (Indicate)**

*** Multiresistant TB =** resistance at least to isoniazid and rifampicine

**** Ultraresistant TB =** MDR TB in addition with: resistance to fluoroquinolones (e.g. Ciprofloxacine, Oxfloxacine, etc), and resistance to at least one of three second line injectable anti-TB (Capreomycine, Kanamycine and Amikacine).

First line treatment: H = Isoniazid R = Rifampicine E = Ethambutol Z = Pyrazinamid S = Streptomycine Th = Thioacetazone

Second line treatment: Am=Amikacine Km=Kanamycine Cm=Capreomycine Cfx=Ciprofloxacine Ofx=Ofloxacin Lfx=Levofloxacine Mfx=Moxifloxacine Gfx=Gatifloxacine Pto=Protionamide Eto=Ethionamide Cs=Cyclosérine PAS=acide P-aminosalicylique

National Tuberculosis Control Programme **Republic of Cameroon**

Quarterly report on tuberculosis screening

Name of Tuberculosis diagnosis on treatment centre Patients recorded during the _____ quarter of 20___	Name of worker in charge of __________________________________ report Date _____________ Signature

CASES RECORDED DURING THE QUARTER *

Bacteriologically confirmed pulmonary tuberculosis (BPT+) **				Bacteriologically-negative pulmonary tuberculosis (BPT-)	Extra pulmonary tuberculosis	**TOTAL**
New cases	Relapses	Failures	Recoveries Treatment			

HIV test	**Number done**		**Positive number**		**Placed under ARVs**		**Placed under CTX**	
	0 -4 years old	**> 5 years old**	**0 -4 years old**	**> 5 years old**	**0 -4 years old**	**> 5 years old**	**0 -4 years old**	**> 5 years old**
BPT + new cases								
Total TB								

Children 0 -5 years old INH prophylaxis					
Sex:					
Male					
Female					

* Patients “coming from” are not included in this report

** BPT+= Bacteriologically confirmed pulmonary tuberculosis (Microscopy, molecular test, culture);

BPT- =Bacteriologically-negative pulmonary tuberculosis

Other risky clusters   Research of RR in the re-treatment

Refugees	Prisoners	Internally displaced person	Health worker

Expert		RR	
P	N	D	ND

Community activities and involvement of private facilities

Number of TB patients (NC all forms and relapses) referred by CHWs/RACs	Number of TB patients (NC all forms and relapses) referred by private facilities	Number of TB patients (NC all forms and relapses) followed by CHWs/RACs

All cases (bacteriologically confirmed or not) during the quarter, by age group and sex

	Age groups										**TOTAL**
Sex:	0-4	5-9	10-14	15-19	20-24	25-34	35-44	45-54	55-64	65+	
Male											
Female											

1st quarter: 1st January-31 March 2nd quarter: 1st April-31 June

3rd quarter: 1st July -31 September 4th quarter 1st October - 31 December

P Positive, N: Negative, D: Detected, ND: Non-Detected

National Tuberculosis Control Programme **Republic of Cameroon**

Quarterly report on treatment outcome of pulmonary tuberculosis case

Table 1: All TB cases recorded during the quarter (except those transferred to the second-line treatment register)

	Sex:		**Therapeutic results**

Types of tuberculosis patient		***Number of cases recorded**	**Healed**	**Completed treatments**	**Failures**	**Deaths**	**Lost to follow-up**	**Non-assessed**
BPT + new cases and relapses	M							
	F							
Re-treatment (excluding relapses)	M							
	F							
**All other cases	M		■		■			
	F		■		■			
HIV positive (new cases and relapses)	M							
	F							
Followed by CHWs/RACs (new cases all types)	M							
	F							

* According to “Quarterly report on tuberculosis screening”

**Negative-Bacteriologically Pulmonary Tuberculosis (BPT-), extrapulmonary, others (excepted “non assessed”)

Table 2: Activities against tuberculosis co-infection/HIV (all tuberculosis cases recorded during quarter)

Number of HIV+ patients (new cases and relapses)	Number of patients under cotrimoxazole	Number of patients under antivirals

1st trimester: 1st January-31 March 2nd quarter: 1st April-30 June

3rd trimester: 1st July -31 September 4th quarter 1st October - 31 December

CASE DEFINITION OF TUBERCULOSIS

SUSPECTED CASE

ANY PERSON WITH:

1. **A COUGH OF ANY DURATION**
2. **FEVER**
3. **WEIGHT LOSS**
4. **NIGHT SWEATS**

CONFIRMED CASE:

A PATIENT WITH A DIAGNOSIS OF **TUBERCULOSIS** (WHETHER OR NOT CONFIRMED BY BACTERIOLOGICAL ANALYSIS) MADE BY A CLINICIAN.

APPENDIX 11L. Viral Haemorrhagic Fever – Case-based Reporting Form

Viral haemorrhagic fever
IDSR reporting form

Variables/ questions		Answers
1	Day of Detection (dd/mm/yyyy)	
2	Place of detection (Health facility)	
3	Patient identification (year-week-country-region-district-reporting site no.)	
4	Name of patient	
5	Surname (s) of patient	
6	Age (years)	
7	Sex (F/M)	
8	Number of persons in the same house	
9	Number of other contacts	
10	Patient's home address	
11	Village/City	
12	Neighbourhood	
13	District	
14	Region	
15	Country	
16	Date of symptom onset (DD/MM/YYYY)	
17	Clinical signs and symptoms observed	
18	Was the patient exposed to known risk factors for this disease? (Yes/No)	
19	If yes, specify risk factor (s)	
20	Laboratory results	
21	Final classification (non case, presumed, probable, laboratory-confirmed, epidemiologically confirmed,	
22	Outcome of illness (death, cured, unknown)	
23	Date of last vaccine traced contact (dd/mm/yyyy)	
24	Other notes and observations	
25	Date of last form updating (dd/mm/yyyy)	

Date of Case Detection /__ /___	**Case identifier**
This case has been reported by (*check the correct answer and specify*)	Date received: ___/___/___
Mobile team _____________________________ Health training:	**Country:**
__________________________Others: ___________________________	
Form completed by (name and surname): __________________________	
Information provided by (name and surname)	

Patient identity

Name:_______________________ Surname (s) :____

For young children, son/daughter of (father's name)_____________________________ _____

Date of birth: ___/___/___ Age (years)________ Sex M/F_______

Permanent address Head of household (name and surname) ______________________

Village/Neighbourhood______ Country ______________GPS lat ________________ long _____________

Nationality: ________________ Ethnic group ______________________

Occupation: (tick the correct answer)

☐ Health worker, Name of HF_______________ ~~DS~~__________________ Quality:____________

☐ Minor ☐ Housewife ☐ Hunter/venison trade ☐ Farmer

☐ Pupil/Student ☐Child ☐Other __________________________

Patient Status

Patient status at the time of detection: Alive ☐ Deceased ☐

If deceased, specify date of death: ___/___/___

Place of death: ☐ Community, name of village/neighbourhood:

☐ HF:___________________HA____________________HD :___________________________

History of the disease

Date of symptoms onset ___/___/___

Name of the village/neighbourhood where the patient became ill:

__________________HA :_____________HD :___________

Did the patient travel during illness: Yes ☐ ☐ No ☐ DK

If yes, indicate location: ____

Has the patient experienced or exhibited the following symptoms (*check the appropriate boxes*)

Headaches	☐ Yes ☐ No ☐	Rash	☐ Yes☐ No ☐ NSP
vomiting/nausea	☐ Yes ☐ No ☐	Bleeding at injection sites	☐ Yes☐ No ☐ NSP
Anorexia/loss of appetite	☐ Yes ☐ No ☐	Bleeding gums	☐ Yes☐ No ☐ NSP
Diarrhoea	☐ Yes ☐ No ☐	Eye bleeding (red eyes	☐ Yes☐ No ☐ NSP
Intense fatigue	☐ Yes ☐ No ☐ NSP	Black or bloody stools	☐ Yes☐ No ☐ NSP
Abdominal pain	☐ Yes ☐ No ☐ NSP	Vomiting Blood	☐ Yes☐ No ☐ NSP

Muscle or joint pain	☐ Yes ☐ No ☐ NSP	Nose bleeding	☐ Yes☐ No ☐ NSP
Difficulty swallowing	☐Yes ☐ No ☐NSP	Vaginal bleeding	☐ Yes☐ No ☐ NSP
Difficulty breathing	☐Yes ☐ No ☐NSP	Hiccup	☐ Yes☐ No ☐ NSP
Fever	☐ Yes ☐ No ☐ NSP		

Risks of exposure

- Was the patient hospitalised or did he visit someone in a HF in the past three weeks before the onset of disease?
  Yes☐ No☐ NSP☐ ; If yes, in which HF:________________
  Between (dates)___/___/__ and__/___/______
- Did the patient consult or visit a traditional healer in the past three weeks before the onset of disease or during his disease? Yes ☐ ☐ No ☐NSP ; If yes, where?________________________and when? __/___/20____
- Did the patient receive any traditional treatment? ☐ Yes ☐ No ☐ NSP
  If yes, which
  one____
- Did the patient partake to funerals in the past three weeks before the onset of disease? Yes ☐No ☐NSP
- Did the patient travel in the past three weeks before the onset of disease? ☐Yes ☐ No ☐ NSP.
  If yes, where?______________________________and when? ___/___/20____
- Was the patient exposed to a known/identified suspected case in the past three weeks before the onset of symptoms?
  Yes No NSP ☐ If ☐yes, Na☐me of the suspected case:____
- During contact, was the suspected case: ☐Alive ☐ Deceased. If dead, date of death: ___/___/20___
- Was the patient exposed to a dead or sick wild animal (primate or other) in the bush, or with an animal with abnormal behaviour, in the past three weeks before the onset of disease? Yes ☐No NS☐P ;
  If yes, what type of animal Place____________________ date ___/___/20___
  Were specimens collected? ☐Yes ☐ No ☐ NSP; If yes, what specimens? ☐ Blood☐ Urine☐ Saliva☐ Skin Biopsy Sampling Date ___/___/20___
- Was the patient referred to a HF? ☐ Yes ☐ No
- Was the patient admited to an isolation ward? Yes ☐ No ☐
  If yes, which HF?______________________________ Admission Date ___/___/20____

PPENDIX 11M. Acute or Chronic Viral Hepatitis – Case-based Investigation Form (updated version)

Acute or chronic viral hepatitis – Case-based investigation form		
No	Variable/Description	Response
General information - identification		
1	Outbreak Number (e.g. Country Code-RRR-DDD-YY-NNN)	Country Code _ _ _ - _ _ _ - _ _ - _ _ _
2	GPS coordinates: Latitude; Longitude	
3	Region	

Acute or chronic viral hepatitis – Case-based investigation form		
No	Variable/Description	Response
4	HD	
5	Name of the health facility	
6	Patient code	
7	Date of consultation (dd/mm/yyyy)	/__ __/__ __/__ __ __ __/
8	Date of notification (dd/mm/yyyy)	/__ __/__ __/__ __ __ __/
9	Patient’s family name	
10		
11	Name of mother/father/legal guardian if the child is ≤12 years old	
12	Date of birth (dd/mm/yyyy)	/__ __/__ __/__ __ __ __/
13	Country of birth	
14	Age (years, month, days completed)	☐ Years ☐ Months ☐
15	Sex: M=Male F=Female	
16a	Place of residence (Village/neighbourhood)	
16b	Telephone number	
16c	profession	
16d	Place of work	
17	Urban/rural area	
18		
19	District of residence	
20		
21	Country of residence	
Clinical characteristics and analysis background		
22	Clinical diagnosis	Acute ☐ Chronic ☐
23	acute phase	Yes ☐ No ☐
24	If acute phase, date of onset (first symptoms) (dd/mm/yyyy)	
25	If chronic phase, answer 25a and 25b below.	
25a	Systematic tests (screening)	Yes ☐ No ☐
25b	Screening for chronic liver disease (e.g. cirrhosis and/or liver	Yes ☐ No ☐
27	Inpatient or outpatient?	
28	If inpatient, date of hospitalization (dd/mm/yyyy)	
29	Clinical signs and symptoms	Icterus: Yes ☐ No ☐ Others
Previous diagnosis and treatments		
30a	Previously diagnosed chronic HB virus infection	Yes ☐ No ☐ unknown ☐
30b	Previously diagnosed chronic HC virus infection	Yes ☐ No ☐ unknown ☐
31a	Patient under specific antiviral treatment against hepatitis B	Yes ☐ No ☐ unknown ☐
31 b	Patient on specific antiviral treatment for hepatitis C	Yes ☐ No ☐ unknown ☐

Acute or chronic viral hepatitis – Case-based investigation form		
No	Variable/Description	Response
Hepatitis vaccination history		
32	Has the patient ever received even one dose of hepatitis A vaccine?	Yes ☐ (_____doses) N ☐
33a	Has the patient received the birth dose of hepatitis B vaccine?	Yes ☐ (_____doses) No ☐
33b	Has the patient ever received even one dose of hepatitis B vaccine?	Yes ☐ (_____doses) No ☐
34	Has the patient ever received even one dose of hepatitis E vaccine?	Yes ☐ (_____doses) N ☐
35	Date of last vaccine received (dd/mm/yyyy)	/__ __/__ __/__ __ __ __/
General risk of exposure		
36	Is the patient a health worker who has been exposed to the blood of patients under care?	Yes ☐ No ☐ unknown ☐
37	Does the patient have homosexual intercourses?	Yes ☐ No ☐ unknown ☐
38	Is the patient on chronic haemodialysis?	Yes ☐ No ☐ unknown ☐
39	Does the patient use recreational drugs injections?	Yes ☐ No ☐ unknown ☐
40	Is the patient involved in a declared, identified outbreak?	Yes ☐ No ☐ unknown ☐
Possible exposure within 2-6 weeks prior to illness (acute hepatitis only)		
41	Has the patient been in contact with people with similar symptoms?	Yes ☐ No ☐ unknown ☐
42	Did the patient drink water from a well or any other unsafe	Yes ☐ No ☐ unknown ☐
43	Did the patient eat unsafe foods, such as raw shellfish	Yes ☐ No ☐ unknown ☐
44	Is the patient one of the children or supervisors of a day-care	Yes ☐ No ☐ unknown ☐
45	Has the patient travelled to a high hepatitis A endemic area?	Yes ☐ No ☐ unknown ☐
Possible exposure factors in the 1 to 6 months preceding the illness (acute hepatitis only)		
46	Was the patient injected in a health care facility?	Yes ☐ No ☐ unknown ☐
47	Has the patient been hospitalized?	Yes ☐ No ☐ unknown ☐
48	Has the patient been operated upon?	Yes ☐ No ☐ unknown ☐
49	Has the patient received a blood transfusion?	Yes ☐ No ☐ unknown ☐
50	Has the patient consulted a dentist?	Yes ☐ No ☐ unknown ☐
51	Has there been sexual contact with a person with hepatitis B?	Yes ☐ No ☐ unknown ☐
52	Has there been household contact with a person with heptitis B?	Yes ☐ No ☐ unknown ☐
53	Has there been unprotected sexual intercourse with one or more unusual partners?	Yes ☐ No ☐ unknown ☐
54	Piercing, tattoos?	Yes ☐ No ☐ unknown ☐
55a	Outcome (1=Alive; 2=Dead; 3=Unknown)	
55b	If dead, Date of death (dd/mm/yyyy)	/__ __/__ __/__ __ __ __/

Acute or chronic viral hepatitis – Case-based investigation form		
No	Variable/Description	Response
56	Final classification (1= Confirmed by the laboratory; 2=Confirmed by epidemiological link; 3=Excluded (negative lab result);	
57	Date the form was sent to the district (dd/mm/yyyy)	/__ __/__ __/__ __ __ __/
58	Date the form was received by the district (dd/mm/yyyy)	/__ __/__ __/__ __ __ __/
59	Person who filled in the form: Name, title, Phone number. e-mail address Signature Name, phone number and e-mail address of the person in charge of the health care facility	
Viral Hepatitis - Laboratory Notification Form		
Part I. The health agent must fill in this form and send a copy of it along with the specimen collected to the laboratory		
	Variable	Response
1	Sampling date (dd/mm/yyyy)	/__ __/__ __/__ __ __ __/
2	Date the specimen was sent to the laboratory (dd/mm/yyyy)	/__ __/__ __/__ __ __ __/
3	Type of sampling (please specify)	
4	Date the specimen was received by the laboratory (dd/mm/yyyy)	/__ __/__ __/__ __ __ __/
5	Epid. number (e.g. GHA-GAR-DDD-YY-NNN) **	GHA- _ _ _ - _ _ _ - _ _ - _ _ _
6	Patient's name(s)	
7	Sex: (M=Male F=Female)	
8	Age (years, month, days completed)	☐ Years ☐ Months ☐
9	Specimen sender: Name, title, Phone number, e-mail address	
Part II. The laboratory manager should fill in this part and return the form to the district and to the clinician		
	Laboratory name and address	
10	Specimen condition 1=adequate (good) 2=Not adequate (bad)	

Acute or chronic viral hepatitis – Case-based investigation form

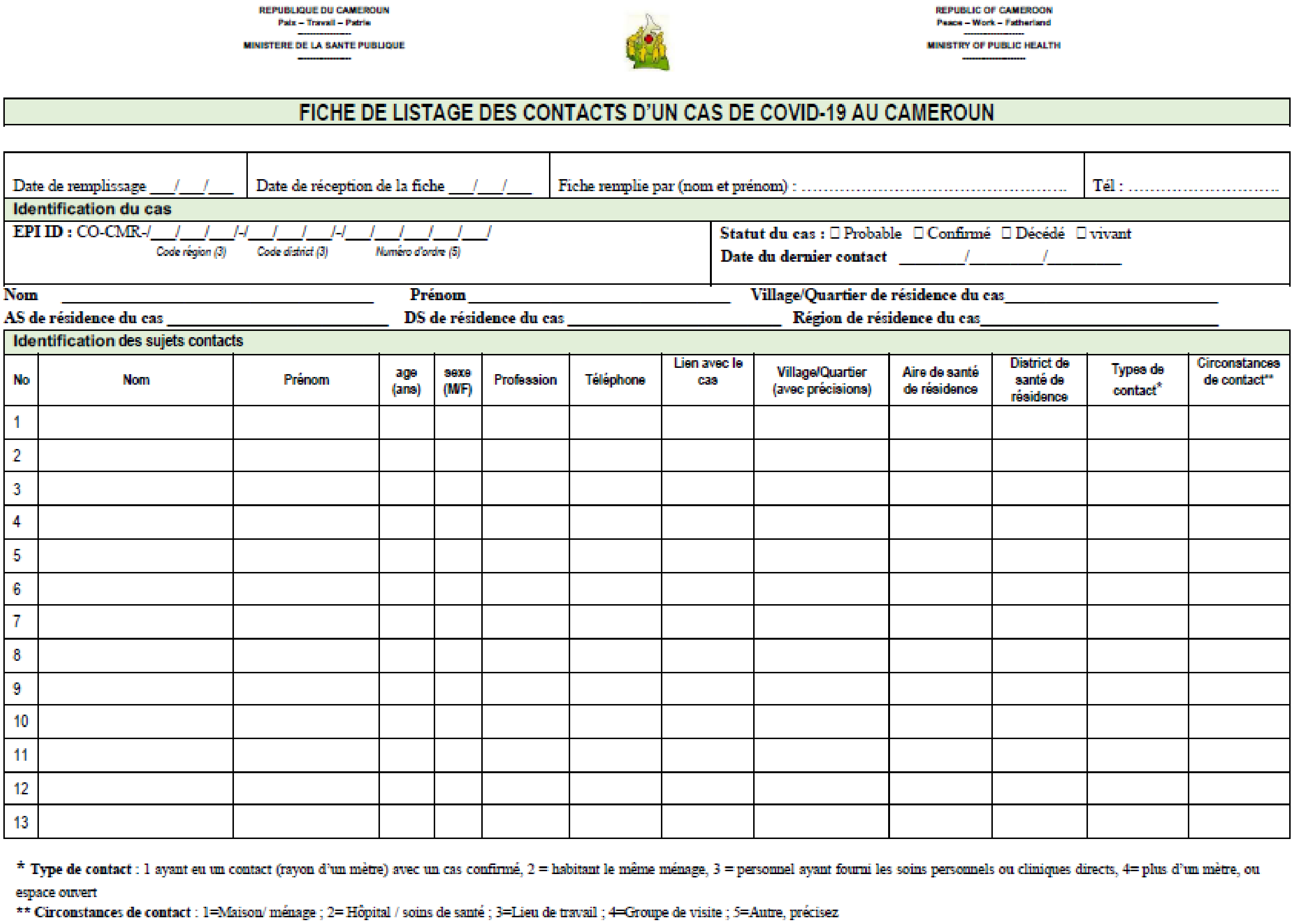
No	Variable/Description	Response
11	Laboratory results Hepatitis A IgM antibodies to HA virus Hepatitis B HB Ag or HB IgM antibodies Hepatitis C HC virus antibodies Hepatitis D HB Ag or HB IgM antibodies plus HD virus antibodies Hepatitis E IgM HE virus and/or IgG antibodies against HE virus	IgM HA virus antibodies Pos Neg unknown IgM HB antibodies Pos Neg unknown Ag HB Pos Neg unknown HC virus antibodies Pos Neg unknown RNA HC virus Pos Neg unknown HC capsidic ag Pos Neg unknown HC virus genotype
12	Other laboratory results	
13	Date the laboratory sent the results to the clinician (dd/mm/yyyy)	/__ __/__ __/__ __ __ __/
14	Date the laboratory sent the results to the district (dd/mm/yyyy)	/__ __/__ __/__ __ __ __/
15	Date the district received the lab results (dd/mm/yyyy)	/__ __/__ __/__ __ __ __/
16	Name of the laboratory personnel who filled in the form. Telephone Signature e-mail address Date	

APPENDIX 11N. IDSR Line List of an outbreak

A line list gives an account of key information on each reported case for analysis and intervention. Describe each case and related information enabling to produce the necessary data to assess cases characteristics and guide response activities. It is an important tool to gather information and quickly analyse it.

During an outbreak, the descriptive list should be established and used ad the main data collection tool. Columns in the IDSR line list should be modified based on the context. Information on each reported case should be written on one column only. This form should be systematically integrated in databases which are systematically communicated as part of IDSR to facilitate comprehensive analysis and daily and weekly reporting to the higher level. (See Line List template in appendix 4E of 4)

APPENDIX 11 O. Contact listing Forms



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Paix – Travail – Patrie

MINISTERE DE LA SANTE PUBLIQUE

REPUBLIC OF CAMEROON
Peace – Work – Fatherland

MINISTRY OF PUBLIC HEALTH

FICHE DE LISTAGE DES CONTACTS D'UN CAS DE COVID-19 AU CAMEROUN

Date de remplissage ___/___/___ | Date de réception de la fiche ___/___/___ | Fiche remplie par (nom et prénom) : ………………………………………… | Tél : ………………………

Identification du cas

EPI ID : CO-CMR-/___/___/___/-/___/___/___/-/___/___/___/___/___/
Code région (3) *Code district (3)* *Numéro d'ordre (5)*

Statut du cas : ☐ Probable ☐ Confirmé ☐ Décédé ☐ vivant
Date du dernier contact ________/________/________

Nom ______________________________ **Prénom** ______________________________ **Village/Quartier de résidence du cas**______________________________
AS de résidence du cas ______________________________ **DS de résidence du cas** ______________________________ **Région de résidence du cas**______________________________

Identification des sujets contacts

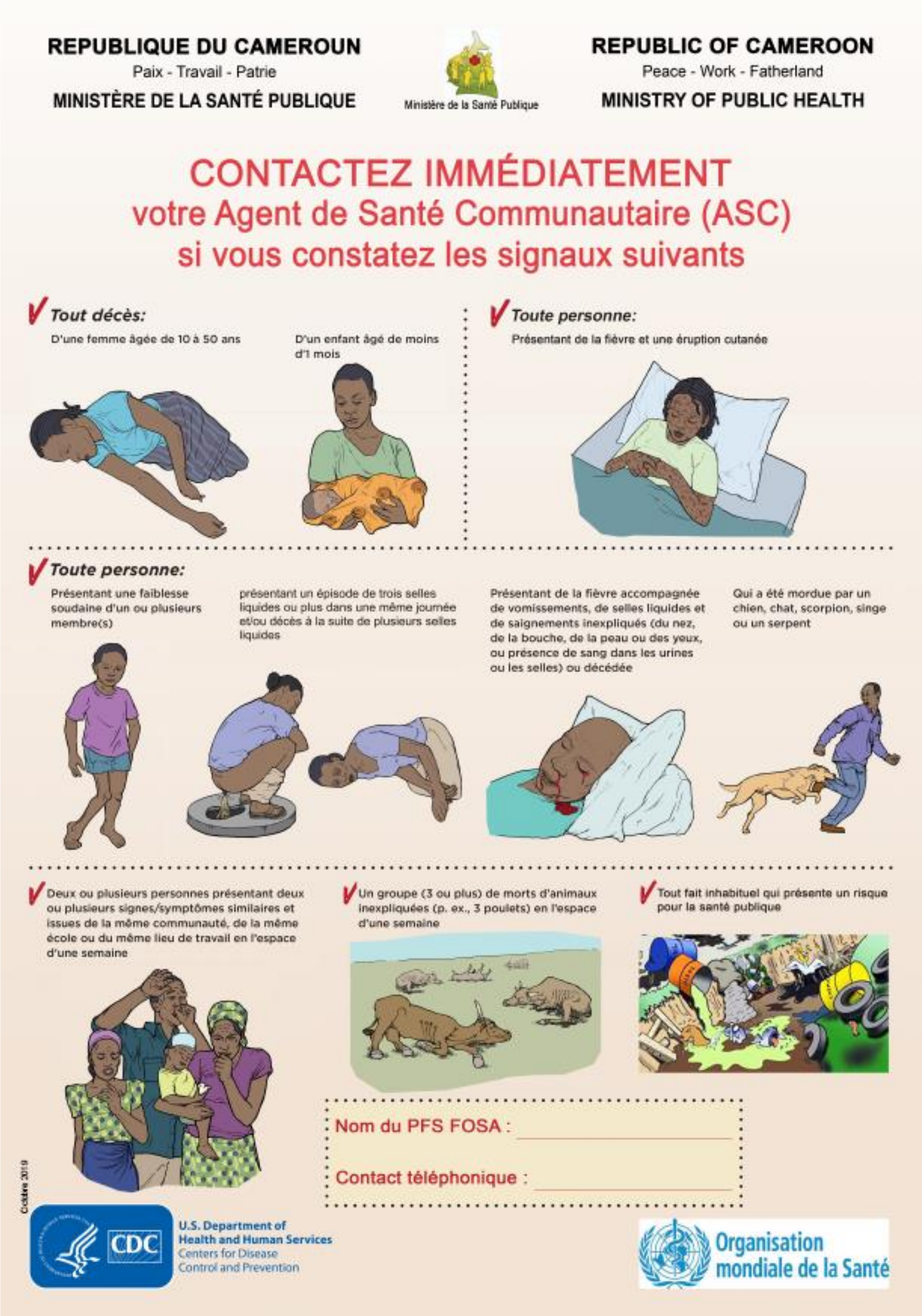
No	Nom	Prénom	age (ans)	sexe (M/F)	Profession	Téléphone	Lien avec le cas	Village/Quartier (avec précisions)	Aire de santé de résidence	District de santé de résidence	Types de contact*	Circonstances de contact**
1												
2												
3												
4												
5												
6												
7												
8												
9												
10												
11												
12												
13												

* **Type de contact** : 1 ayant eu un contact (rayon d'un mètre) avec un cas confirmé, 2 = habitant le même ménage, 3 = personnel ayant fourni les soins personnels ou cliniques directs, 4= plus d'un mètre, ou espace ouvert

** **Circonstances de contact** : 1=Maison/ ménage ; 2= Hôpital / soins de santé ; 3=Lieu de travail ; 4=Groupe de visite ; 5=Autre, précisez

APPENDIX 11P. Community-based Signals - Logbook

Introductions: *This form should be filled in by the community health worker (CHW) and immediately forwarded to the nearest health facility/health area surveillance focal point.*



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MINISTRY OF PUBLIC HEALTH

CONTACTEZ IMMÉDIATEMENT
votre Agent de Santé Communautaire (ASC) si vous constatez les signaux suivants

✓ ***Tout décès:***

D'une femme âgée de 10 à 50 ans

D'un enfant âgé de moins d'1 mois

✓ ***Toute personne:***

Présentant de la fièvre et une éruption cutanée

✓ ***Toute personne:***

Présentant une faiblesse soudaine d'un ou plusieurs membre(s)

présentant un épisode de trois selles liquides ou plus dans une même journée et/ou décès à la suite de plusieurs selles liquides

Présentant de la fièvre accompagnée de vomissements, de selles liquides et de saignements inexpliqués (du nez, de la bouche, de la peau ou des yeux, ou présence de sang dans les urines ou les selles) ou décédée

Qui a été mordue par un chien, chat, scorpion, singe ou un serpent

✓ Deux ou plusieurs personnes présentant deux ou plusieurs signes/symptômes similaires et issues de la même communauté, de la même école ou du même lieu de travail en l'espace d'une semaine

✓ Un groupe (3 ou plus) de morts d'animaux inexpliquées (p. ex., 3 poulets) en l'espace d'une semaine

✓ Tout fait inhabituel qui présente un risque pour la santé publique

Nom du PFS FOSA : ____________________

Contact téléphonique : ____________________

Octobre 2019

CDC
U.S. Department of Health and Human Services
Centers for Disease Control and Prevention

Organisation mondiale de la Santé

RÉPUBLIQUE DU CAMEROUN

Paix-Travail-Patrie

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REPUBLIC OF CAMEROON

Peace-Work-Fatherland

*****

MINISTRY OF PUBLIC HEALTH

*****

Community Health Worker Logbook

General Information

Name of the CHW ____ Telephone: ________________

Health Area ___________________________________HA SFP Phone number: _________________

Any death:

- Of a woman between 10 and 50 years old
- Of a child of less than one month

Anyone with:

- Fever and skin rash

Anyone:

- With sudden weakness of one or more limbs;
- with an episode of three or more loose stools in one day and/or death following several loose stools;
- with fever coupled with vomiting, loose stools and unexplained bleeding (from the nose, mouth, skin or eyes, or blood in the urine or stool) or who has died
- who has been bitten by a dog, cat, scorpion, monkey or snake

Two or more people with two or more similar signs/symptoms from the same community, school, or workplace within one week

A group (3 or more) of unexplained animal deaths (e.g., 3 chickens) within a week

Any event that presents a public health risk, including natural disasters, or rarely observed symptoms

Instructions

When you detect one or more community signals, the signal should be immediately reported to the Health Area Focal Point by phone, SMS message or personally. The following information must be communicated:

- *Date the signal began.*
- *Date and time of signal detection*
- *Description of the signal*
- *Location of the signal*

Date du début du signal : ____

Date et heure de détection du signal : _____________________________________

____

Localisation du signal : ____

____

Description du signal : ____

____

____

____

____

____

____

____

____

____

____

____

Date du début du signal : ____

Date et heure de détection du signal : _____________________________________

____

Localisation du signal : ____

____

Description du signal : ____

____

____

____

____

____

____

____

____

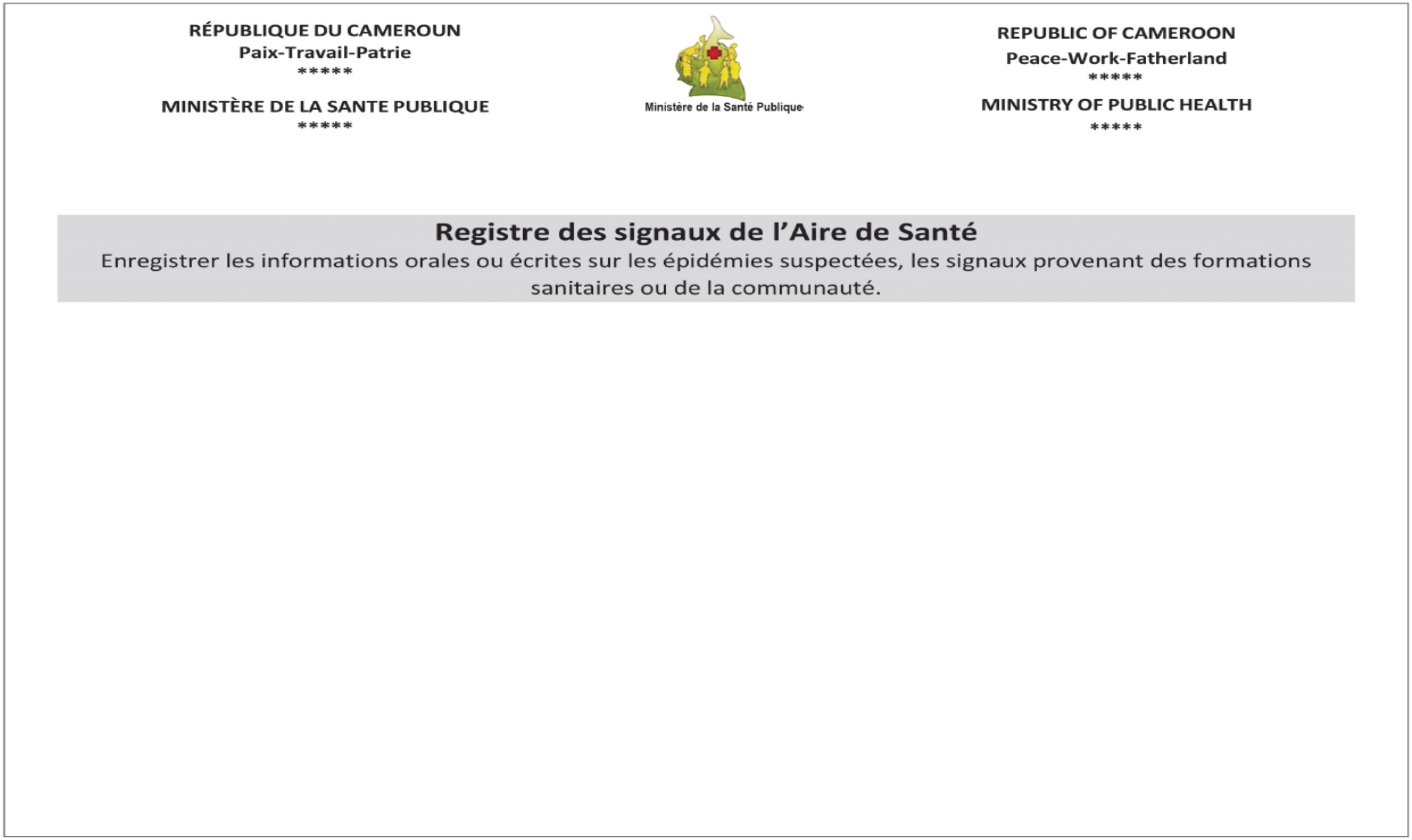
____

____

____

APPENDIX 11Q Event-Based Surveillance - Signal Register

This registry summarizes all signals recorded during the week. It is filled in by the head of the health area or that of the HFand sent to the district surveillance focal point every week.



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Registre des signaux de l'Aire de Santé

Enregistrer les informations orales ou écrites sur les épidémies suspectées, les signaux provenant des formations sanitaires ou de la communauté.

Informations générales

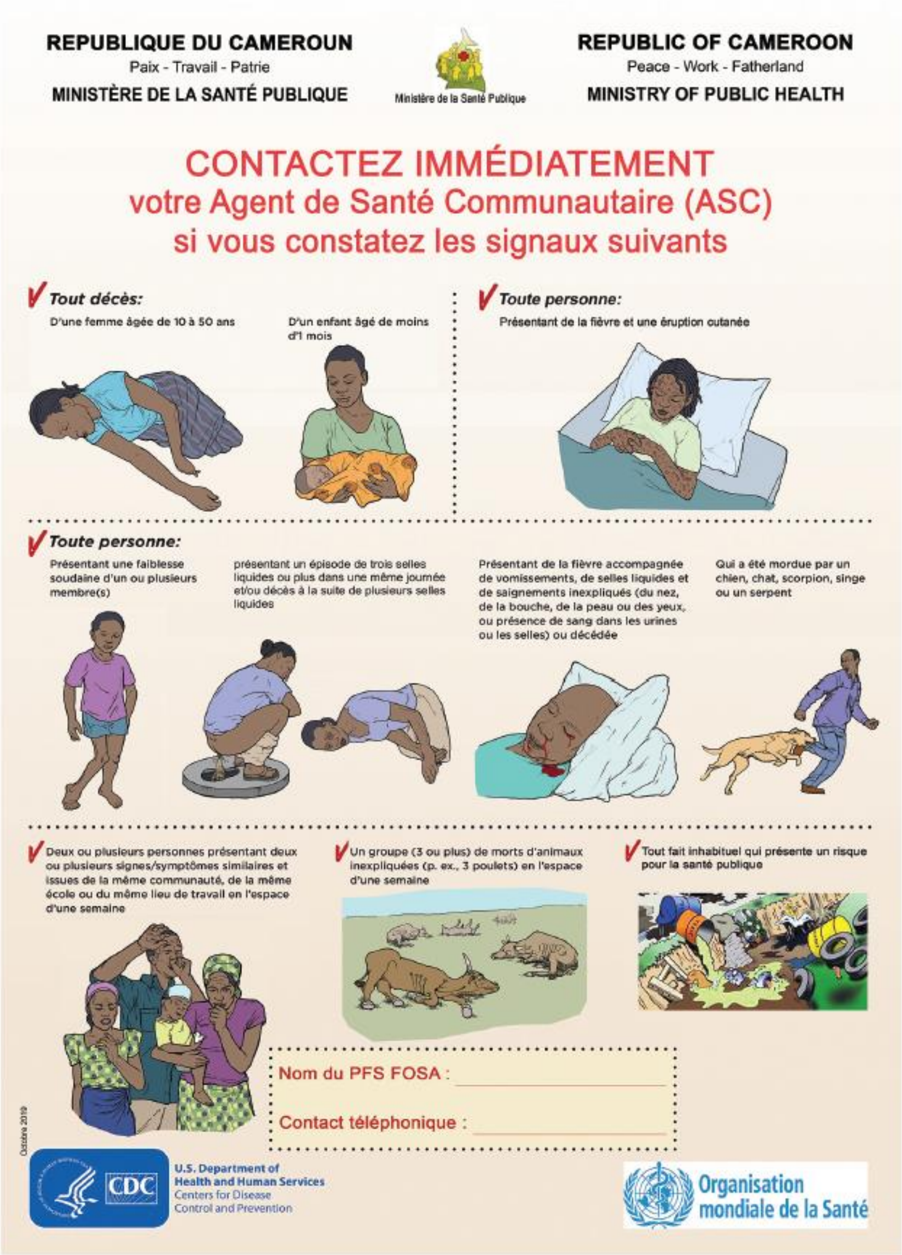
Région : ____

District de santé : ____

Aire de Santé : ____

Nom du Point Focal Surveillance de l’Aire de Santé : ______________________________

Téléphone du Point Focal Surveillance de l’Aire de Santé : _________________________



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MINISTRY OF PUBLIC HEALTH

CONTACTEZ IMMÉDIATEMENT votre Agent de Santé Communautaire (ASC) si vous constatez les signaux suivants

✓ ***Tout décès:***

D'une femme âgée de 10 à 50 ans

D'un enfant âgé de moins d'1 mois

✓ ***Toute personne:***

Présentant de la fièvre et une éruption cutanée

✓ ***Toute personne:***

Présentant une faiblesse soudaine d'un ou plusieurs membre(s)

présentant un épisode de trois selles liquides ou plus dans une même journée et/ou décès à la suite de plusieurs selles liquides

Présentant de la fièvre accompagnée de vomissements, de selles liquides et de saignements inexpliqués (du nez, de la bouche, de la peau ou des yeux, ou présence de sang dans les urines ou les selles) ou décédée

Qui a été mordue par un chien, chat, scorpion, singe ou un serpent

✓ Deux ou plusieurs personnes présentant deux ou plusieurs signes/symptômes similaires et issues de la même communauté, de la même école ou du même lieu de travail en l'espace d'une semaine

✓ Un groupe (3 ou plus) de morts d'animaux inexpliquées (p. ex., 3 poulets) en l'espace d'une semaine

✓ Tout fait inhabituel qui présente un risque pour la santé publique

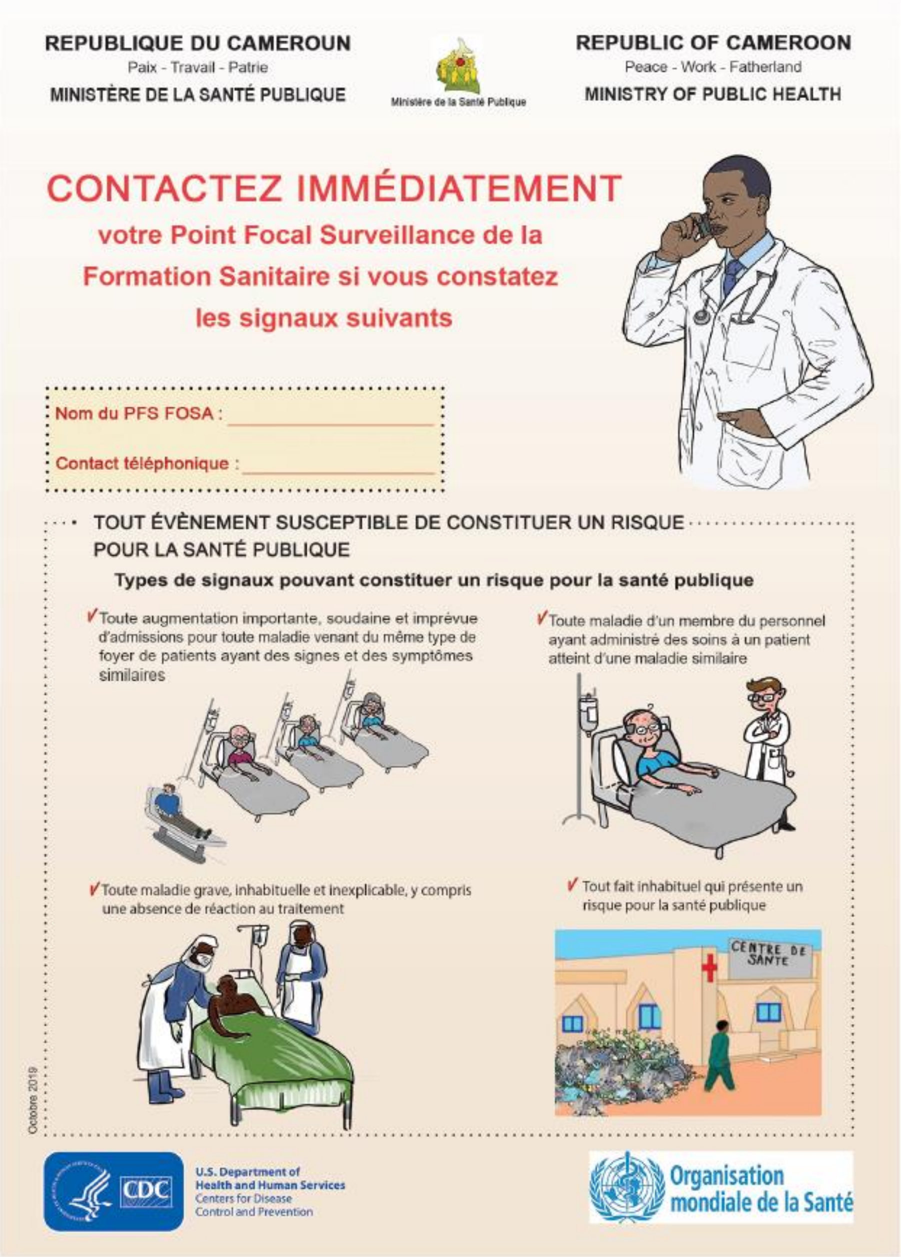
Nom du PFS FOSA : ____________________

Contact téléphonique : ____________________

Octobre 2019

U.S. Department of Health and Human Services
Centers for Disease Control and Prevention

Organisation mondiale de la Santé



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MINISTRY OF PUBLIC HEALTH

CONTACTEZ IMMÉDIATEMENT votre Point Focal Surveillance de la Formation Sanitaire si vous constatez les signaux suivants

Nom du PFS FOSA : ____________________

Contact téléphonique : ____________________

- TOUT ÉVÈNEMENT SUSCEPTIBLE DE CONSTITUER UN RISQUE POUR LA SANTÉ PUBLIQUE

Types de signaux pouvant constituer un risque pour la santé publique

✓ Toute augmentation importante, soudaine et imprévue d'admissions pour toute maladie venant du même type de foyer de patients ayant des signes et des symptômes similaires

✓ Toute maladie d'un membre du personnel ayant administré des soins à un patient atteint d'une maladie similaire

✓ Toute maladie grave, inhabituelle et inexplicable, y compris une absence de réaction au traitement

✓ Tout fait inhabituel qui présente un risque pour la santé publique

Octobre 2019

U.S. Department of Health and Human Services
Centers for Disease Control and Prevention

Organisation mondiale de la Santé

Registre des signaux du l'Aire de Santé

Enregistrer les informations orales ou écrites sur les épidémies suspectées, les signaux provenant des formations sanitaires ou de la communauté.

Source d'information (ASC, IC, FOSA, COM) [1]	Nom et contact tél. de la personne qui notifie à l'aire de santé [2]	signal rapporté [3]	Lieu du signal (Village, Quartier) [4]	Date de détection du signal[5]	Date du début du signal (1er cas)[6]	Date de notification à l'AS [7]	Nombre de cas initialement notifiés [8]	Résultat de Triage (validé, monitoré ou rejeté)[9]	Résultat de vérification (mettre le nom de l'évènement)[10]

1= *il s'agit de la «source» du rapport initial du signa ; de la personne ayant rapportée le signal à l'ASC ou autre responsable de santé (Tradipraticien, Pharmacien, etc).*

2= *Le nom et contact tél de la personne qui rapporte le signal à l'Aire de Santé*

3= *Mentionner le maladie ou signal détecté de façon brève.* Exemple: mort d'une femme.

4= *Indiquer l'emplacement où le signal s'est produit*

5= *La date (jj/mm/aaaa) à laquelle le signal a été capté par l'ASC ou une autre source*

6 = *Est la date au début du signal. Si le signal implique plus d'un cas, mentionnez juste la date du premier cas.*

7= *La date (jj/mm/aaaa) à laquelle le Chef de l'AS a eu connaissance du signal.* Lorsque le signal a été directement signalé à l'Aire de Santé par un ASC ou une autre source.

8= *C'est le nombre de cas dans ce cluster, tel qu'estimé par les informations les plus récentes obtenues par la personne, déclarant le signal*

9 = *Si vous déterminez pendant le triage que le signal est vrai écrire "validé", faux écrivez "rejeté", si elle nécessite une surveillance supplémentaire, écrivez "monitoré*

10 = *Si le signal est vérifié comme un événement, mettez le nom de l'événement sinon laissez-le vide.*

APPENDIX 11R : Reporting Form on Adverse Effects Following Immunization (AEFI)

REPUBLIQUE DU CAMEROUN
Paix – Travail - Patrie

MINISTERE DE LA SANTE PUBLIQUE

REPUBLIC OF CAMEROON
Peace – Work - Fatherland

MINISTRY OF PUBLIC HEALTH

REPORTING FORM OF ADVERSE EFFECTS POSSIBLY DUE TO A MEDICINE OR OTHER HEALTH PRODUCT

To be returned to the Pharmacovigilance Centre and the Department of Pharmacy, Drugs and Laboratories

Tel. : xxx ; Fleet : xxx

E-mail : pharmacovigilance_cam@yahoo.fr; dpharmedilab.cnr@gmail.com

Form No. ....

Patient

Name: ………………………………………………….. Surname: ……………………………………..

Sex : F M Age : /___/ year /___/ month Weight : /________/ Height: /_______/

Address (Tel.) : …………………. Region :…………………HD :……………………HF :……………..

If newborn, Product taken:
By the patient: /____/
Through breastfeeding: /____/
By the mother during pregnancy: /____/
(indicate the term) : /____/

History/Risks factors: Pregnancy /___/ Alcohol /___/ Hepatopathy /___/ Allergy /___/ Nephropathy /___/ Tobacco smoking /___/

Others (Indicate) ……..

SUSPECTED/ASSOCIATED PRODUCTS (including vaccines, solvents and médicinal plants):

No.	**Name**	**Manufacturer**	**Batch No.**	**Expiry Date**	**Administration Route**	**Dosage**	**Treatment Indication/ Reason**	**Treatment Duration**	
								Start Date	**End Date**
1									
2									
3									
4									
5									

Were one or many products stopped?: Yes /___/ No /___/ No information /___/
If yes, which one(s):
Did the reaction stop after treatment interruption? Yes /___/ No /___/ No information /___/

Were one or many products reintroduced? Yes /___/ No /___/ No information /___/
If yes, did the reaction reappear? Yes /___/ No /___/ No information /___/

Adverse Event(s):

Nature and description of event ……..…………………..…………………………………………………………………………. ……………………………… …
…….
……..

Date of intake/vaccination: /____/____/____/ Date of onset: /____/____/____/

If no, onset delay following intake/vaccination: Minutes /____/ Hours /____/ Days /____/ Months /____/

SEVERITY

EVOLUTION

/____/ Hospitalisation or prolonged hospitalisation /____/ Permanente incapacity or disability /____/ life-threatening /____/ Death	/____/ Healing without sequaele /____/ Death related to the event /____/ Death not related to the event /____/ Patient not having yet recovered /____/ Healing with sequaele /____/ Death possibly caused by the event /____/ Unknown

Treatment received: …….
……
…….. …..

For all products: indicate the place of acquisition Pharmacy /____/ HF/____/ Street /____/ Others /____/

For vaccines: indicate: the vaccination place /______________________________/ injection site (e.g.: left hand = LH) /________________/

Medicinal plants: indicate the part used: root /____/ bark /____/ leaf /____/ flower /____/

NOTIFIER :

Name and Surname: ………
/___/ Medical Doctor /___/ Pharmacist /___/ Dentist /___/ Midwife /___/ Nurse Others (indicate): …………………………………………………………
Speciality (indicate)………
Telephone: ……………………………………………. E-mail: ………………………………………………………………… Date: ………………………………
Signature and stamp:

REPUBLIQUE DU CAMEROUN
Paix – Travail – Patrie

MINISTERE DE LA SANTE PUBLIQUE

SECRETARIAT GENERAL

DIRECTION DE LA SANTE FAMILIALE

PROGRAMME ELARGI DE VACCINATION

GROUPE TECHNIQUE CENTRAL

REPUBLIC OF CAMEROON
Peace – Work – Fatherland

MINISTRY OF PUBLIC HEALTH

SECRETARIAT GENERAL

DIRECTORATE OF FAMILY HEALTH

EXPANDED PROGRAMME ON IMMUNIZATION

CENTRAL TECHNICAL GROUP

PREVENTIVE VACCINATION CAMPAIGN AGAINST MEASLES

Investigation Form of Severe Adverse Events Following Immunization (AEFI)

Fill in this summary form at the end of the investigation; record it with the other AEFI Reporting Forms

Notification Number: ………………………………………………………………………… (Number of consultation registre)

Presumptive Diagnostic:………………………………………………………………………………………………..

Date of investigation : ……….../…..……./…………(DD/MM/YY)

Place(s) of investigation :……………………………… ……………………………………………………….…..

DEMOGRAPHIC DATA

Name: …………………………………	**Surname(s):** ……………………………	**Age :**………Ans…………Mois	**Sex** : ☐ **F** ☐ **M**
Region ………………………….**District:** ……………………		**HF** :……………………………………………………………	
Village/Neighborhood: ………..……………………………		**Address/Tel.:**………………………………………………	
Vaccination Date: ______/_______/________	**Manufacturer and Vaccine batch:** ………………………………	**Manufacturer and solvent batch:** ……………………………….	**Date of AEFI onset:** _____/_____/________

TYPE OF SEVERE AEFI *(TICK)*

☐ Encephalopathy
☐ Febrile Convulsions
☐ Septicaemia
☐ Respiratory failure
☐ Bleeding
☐ Heart Failure
☐ Shock/ Anaphylactic reaction
☐ Others:………………………………………….

☐ Guillain Barré Syndrome ☐ Death

1. **Medical History:**…………………………………………………………………………………………………
………

2. **Clinical Examination:**………………..………………………………………………………………………….
………………..……………………………………………..………………………………………………………
……………..………………………………………..…………………………………………………………………
……………………………………..…………………………………………………………………………………
…………………………………..……………………………………………………………………………………
……………………………..…………………………………………………………………

3. **Résultats des examens complémentaires**

Tests généraux pratiqués	**SpecificTests performed:**
…………………………………………………….……………….	……………………………………………………….…….…….
……………………………………………………..……………	………………………………………………………….……...
……………………………………………………..……………	………………………………………………………………
……………………………………………………..……………	………………………………………………………………
…………………………………………………………………	………………………………………………………………..

4. **Results of tests/ biological analyses perormed before investigation:**

……
…………………………………………………………………………………..…………………………………………
………………………………………………………………………………..………………………

5. **Patient situation:** ☐ Alive ☐ Comateous ☐ Healed ☐Ran away ☐ Dead Date**:**___**/**___**/**___(DD/MM/YYYY)

6. **Treatment (s) received before investigation:**

……
……………..……
……………………………………………………………………………………..…………………

7. Biological specimens /vaccines/consumables and shipment date

Specimen Collection Date	Specimen Collection Time (hour and minutes)	Shipment Date:
☐ **Blood 1**………/………/……… (DD/MM/YY)	……...….............a.m./p.m.	………/………/………
☐ **Blood 2** ………/………/……… (DD/MM/YY)	……...…............. a.m./p.m	………/………/………
☐ **CSF 1** ………/………/……… (DD/MM/YY)	……...…............. a.m./p.m	………/………/………
☐ **CSF 2** ………/………/……… (DD/MM/YY)	……...…............. a.m./p.m	………/………/………
☐ **Urine 1** ………/………/……… (DD/MM/YY)	……...…............. a.m./p.m	………/………/………
☐ **Urines 2** ………/………/……… (DD/MM/YY)	……...…............. a.m./p.m	………/………/………
☐ **Stool 1** ………/………/……… (DD/MM/YY)	……...…............. a.m./p.m	………/………/………
☐ **Stool 2** ………/………/……… (DD/MM/YY)	……...…............. a.m./p.m	………/………/………
☐**Tissues :** …………………………………Date:……/………/……	Time: ……...…............... a.m./p.m	
………………………………… …Date:……/………/……	Time: ……...…............... a.m./p.m	
……………………………………Date:……/………/……	Time: ……...…............... a.m./p.m	
☐ **Vaccine, batch**: ……………………………Date: ……/………/……	Time: ……...….................a.m./p.m	
☐ **Solvent, batch**: ……………………………Date: …..../………/……..	Time: ……...…............... a.m./p.m	

8. Case investigation outcome (considering informations available after investigation

Diagnostic:

……………..………………………………………………………………………..……………………………………………………

……

On behalf of the Investigation Team:

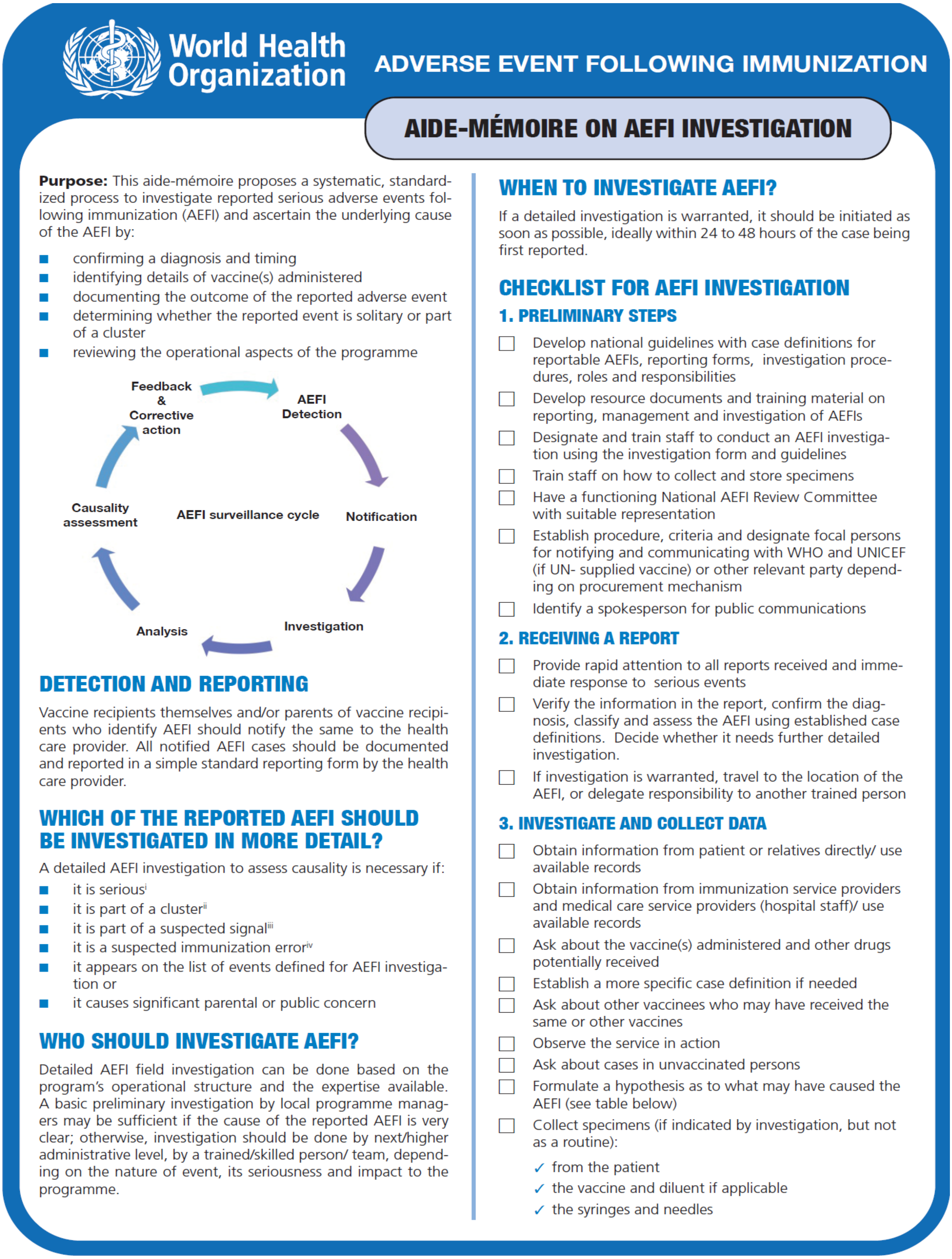
(Name, Surname and signature)

TO BE COMPLETED BY THE AEFI EXPERTS COMMITTEE	
9. Classification:	
☐ **Vaccine-related AEFI**	☐ No ☐ Yes If **Yes,** indicate *imputability level* ☐ Suspected ☐ Probable ☐ Confirmed
☐ **Programmatic AEFI**	☐ No ☐ Yes ***If Yes, indicate the reason*** *(tick)*

	☐ Defective Cold Chain ☐ Vaccine Reconstitution Error ☐ Wrong Injection Technique ☐ Non sterile Practices
☐ **Coincidental AEFI**	☐ No ☐ Yes
☐ **Other Causes of AEFI**	☐ No ☐ Yes
☐ **Undetermined AEFI**	☐ No ☐ Yes

On behalf of the Experts Committee: (Name, Surname and signature)

APPENDIX 11S : Aide-Mémoire on AEFI Investigation



World Health Organization

ADVERSE EVENT FOLLOWING IMMUNIZATION

AIDE-MÉMOIRE ON AEFI INVESTIGATION

Purpose: This aide-mémoire proposes a systematic, standardized process to investigate reported serious adverse events following immunization (AEFI) and ascertain the underlying cause of the AEFI by:

- confirming a diagnosis and timing
- identifying details of vaccine(s) administered
- documenting the outcome of the reported adverse event
- determining whether the reported event is solitary or part of a cluster
- reviewing the operational aspects of the programme

DETECTION AND REPORTING

Vaccine recipients themselves and/or parents of vaccine recipients who identify AEFI should notify the same to the health care provider. All notified AEFI cases should be documented and reported in a simple standard reporting form by the health care provider.

WHICH OF THE REPORTED AEFI SHOULD BE INVESTIGATED IN MORE DETAIL?

A detailed AEFI investigation to assess causality is necessary if:

- it is serious[i]
- it is part of a cluster[ii]
- it is part of a suspected signal[iii]
- it is a suspected immunization error[iv]
- it appears on the list of events defined for AEFI investigation or
- it causes significant parental or public concern

WHO SHOULD INVESTIGATE AEFI?

Detailed AEFI field investigation can be done based on the program's operational structure and the expertise available. A basic preliminary investigation by local programme managers may be sufficient if the cause of the reported AEFI is very clear; otherwise, investigation should be done by next/higher administrative level, by a trained/skilled person/ team, depending on the nature of event, its seriousness and impact to the programme.

WHEN TO INVESTIGATE AEFI?

If a detailed investigation is warranted, it should be initiated as soon as possible, ideally within 24 to 48 hours of the case being first reported.

CHECKLIST FOR AEFI INVESTIGATION

1. PRELIMINARY STEPS

☐ Develop national guidelines with case definitions for reportable AEFIs, reporting forms, investigation procedures, roles and responsibilities
☐ Develop resource documents and training material on reporting, management and investigation of AEFIs
☐ Designate and train staff to conduct an AEFI investigation using the investigation form and guidelines
☐ Train staff on how to collect and store specimens
☐ Have a functioning National AEFI Review Committee with suitable representation
☐ Establish procedure, criteria and designate focal persons for notifying and communicating with WHO and UNICEF (if UN- supplied vaccine) or other relevant party depending on procurement mechanism
☐ Identify a spokesperson for public communications

2. RECEIVING A REPORT

☐ Provide rapid attention to all reports received and immediate response to serious events
☐ Verify the information in the report, confirm the diagnosis, classify and assess the AEFI using established case definitions. Decide whether it needs further detailed investigation.
☐ If investigation is warranted, travel to the location of the AEFI, or delegate responsibility to another trained person

3. INVESTIGATE AND COLLECT DATA

☐ Obtain information from patient or relatives directly/ use available records
☐ Obtain information from immunization service providers and medical care service providers (hospital staff)/ use available records
☐ Ask about the vaccine(s) administered and other drugs potentially received
☐ Establish a more specific case definition if needed
☐ Ask about other vaccinees who may have received the same or other vaccines
☐ Observe the service in action
☐ Ask about cases in unvaccinated persons
☐ Formulate a hypothesis as to what may have caused the AEFI (see table below)
☐ Collect specimens (if indicated by investigation, but not as a routine):
  ✓ from the patient
  ✓ the vaccine and diluent if applicable
  ✓ the syringes and needles